AN INTERACTING FIELD STRENGTH IN THREE-DIMENSIONAL QUANTUM ELECTRODYNAMICS

Abstract. At a suitable positive charge, lattice quantum electrodynamics gives a non-Gaussian probability law supported on real tempered two-forms on $\mathbb{R}^3$ that are closed. Its tensor Schwinger functions satisfy the Osterwalder–Schrader axioms, including clustering. We first send the volume to infinity, then the lattice spacing to zero, and select an ergodic component. Coordinate and diagonal reflection positivity, together with Gaussian bounds, give the continuum symmetries. A common bare mass on two reference tori permits partition comparison. A positive fourth response and bounds on separated quadratic observables yield a non-Gaussian component.

Contents

1. Introduction 1
2. Reflection positivity and Gaussian domination in lattice QED 6
3. Block averaging in Maxwell's theory 47
4. Block averaging with fermions 69
5. The renormalization group: one step 92
6. The renormalization group: full trajectory 138
7. The renormalization group: large fields 147
8. Ultraviolet stability of the partition function 149
9. Infrared stability of the partition function 157
10. The fermion determinant: quartic response 159
11. The interacting theory: quartic response 166
12. The continuum limit and the Osterwalder–Schrader axioms 201
13. An interacting Osterwalder–Schrader theory 211
Appendix A. Consequences of extra reflection positivity 217
Appendix B. Rotation invariance from reflection positivity 218
Appendix C. Clustering from reflection positivity 221
Appendix D. Polymer geometry 221
Appendix E. Notation 224
References 254

1. Introduction

A field strength is a two-form F on $\mathbb{R}^3$ whose coefficients are real tempered distributions. We evaluate it on a Schwartz two-form f, writing $F(f)$, and call it closed if $dF = 0$ in the distributional sense, where d is the exterior derivative. For a probability law λ of such fields, its Schwinger functions are its moments,

$$S_n(f_1 \otimes \cdots \otimes f_n) = \int \prod_{j=1}^{n} F(f_j)\, \mathrm{d}\lambda(F), \qquad S_0 = 1.$$

We extend field evaluation complex-linearly. The moments will extend from products to all tensor tests in $\mathcal{S}_n = \mathcal{S}(\mathbb{R}^{3n}; (\Lambda^2\mathbb{C}^3)^{\otimes n})$. A translation-invariant law is ergodic if its translation-invariant events have probability zero or one. Its ergodic components are the conditional laws given the translation-invariant events.

Theorem 0. At a suitable positive charge, the lattice QED3 construction has a limit obtained by first taking infinite volume and then sending the lattice spacing to zero, along subsequences. An ergodic component λ of this limit is non-Gaussian and is supported on real tempered two-forms that are closed. It satisfies

$$\int e^{F(f)}\,\mathrm{d}\lambda(F) \le e^{\|f\|_2^2/2} \qquad (f \text{ a real Schwartz two-form}).$$

Its Schwinger functions have the following properties.

(E0′) *Regularity and growth.* Each S_n is a continuous linear functional on $\mathcal{S}_n$, with

$$|S_n(\Psi)| \le 2\,66^n\sqrt{n!}\,q_{6n}(\Psi), \qquad n \ge 1,$$

where

$$q_m(\Psi) = \max_{|\alpha|\le m, I} \sup_{y\in\mathbb{R}^{3n}} (1+|y|^2)^{m/2}|\partial^\alpha \Psi_I(y)|$$

and I ranges over the tensor components.

(E1) *Euclidean tensor covariance.* For every $g(x) = Qx + a$, $Q \in O(3)$, $a \in \mathbb{R}^3$,

$$S_n(g\cdot\Psi) = S_n(\Psi), \qquad (g\cdot\Psi)(x_1,\dots,x_n) = (\Lambda^2 Q)^{\otimes n}\Psi(g^{-1}x_1,\dots,g^{-1}x_n).$$

Here $\Lambda^2 Q$ is the natural action of Q on two-form indices.

(E2) *Reflection positivity.* Let r reflect in a plane through the origin and let H be either closed half-space bounded by that plane. For every finite family $\Psi_k \in \mathcal{S}_k$, $1 \le k \le M$, with $\operatorname{supp}\Psi_k \subset H^k$, and $\Psi_0 \in \mathbb{C}$,

$$\sum_{k,l=0}^{M} S_{k+l}(\Theta_k\Psi_k \otimes \Psi_l) \ge 0, \qquad \Theta_k\Psi_k = \overline{r\cdot\Psi_k}, \quad \Theta_0 c = \overline{c}.$$

(E3) *Permutation symmetry.* $S_n(P_\pi\Psi) = S_n(\Psi)$ for every permutation π, where P_π is the continuous extension of

$$P_\pi(f_1\otimes\cdots\otimes f_n) = f_{\pi^{-1}(1)}\otimes\cdots\otimes f_{\pi^{-1}(n)}.$$

Thus the arguments and their two-form indices are permuted together.

(E4) *Clustering.* For $\Psi \in \mathcal{S}_k$ and $\Phi \in \mathcal{S}_l$,

$$S_{k+l}(\Psi\otimes\tau_a^{\otimes l}\Phi) \longrightarrow S_k(\Psi)S_l(\Phi) \qquad \text{as } |a|\longrightarrow\infty,$$

where $(\tau_a^{\otimes l}\Phi)(x_1,\dots,x_l) = \Phi(x_1-a,\dots,x_l-a)$.

Moreover, for some real $f \in C_c^\infty(\mathbb{R}^3;\Lambda^2)$,

$$S_4(f^{\otimes 4}) - 3S_2(f^{\otimes 2})^2 > 0.$$

■□

The precise regulator, mass choice, limit topology and constants are specified in Thm. A.

The proof must recover continuous Euclidean symmetry from a lattice regulator and retain a nonzero fourth cumulant after taking limits and selecting an ergodic component. Reflection positivity and Gaussian domination give the symmetries and compactness. The fourth cumulant requires quantitative estimates from the renormalization group.

The lattice fermions are placed in the chambers cut out by the coordinate and diagonal mirrors. A chamber is one of the simplices in a unit cube described in §2.1.2. Reflected neighboring chambers have a positive crossing expansion. This proves reflection positivity: an observable supported on one side of a mirror has nonnegative expectation when multiplied by its reflected complex conjugate. Repeated reflection compares observables in a large torus with reflected copies in a smaller one. For diagonal mirrors we first work on tori adapted to their directions. Comparison estimates transfer the resulting forms to the infinite-volume limit of cubic tori, giving three coordinate and six diagonal reflection families.

A Gaussian bound controls the limiting field. In its continuum form it says

$$\mathbb{E}e^{F(f)} \le \exp(\|f\|_2^2/2) \qquad (f \text{ a real Schwartz two-form}).$$

It supplies tightness and all moment bounds needed to pass to the limit. The diagonal reflection forms provide information absent from cubic symmetry: in the reflected Hilbert space, energy dominates each transverse momentum. Zero energy therefore forces zero momentum. This identifies observables invariant in one coordinate direction with those invariant under all translations, and yields clustering in every direction for an ergodic law. A general reflection-positive rotation theorem gives covariance for tests with disjoint supports. Closedness of the field removes the remaining terms supported where arguments coincide.

To distinguish the field from a Gaussian one, we integrate fields one lattice scale at a time. Photon averaging splits the quadratic energy exactly into a minimizing background and independent fluctuations. Fermion integration gives the corresponding coarse Dirac operator and determinant. Bounds on these operators allow the local interaction and mass to be followed through successive scales. One bare mass must control two reference tori at once; otherwise their partition functions and response coefficients would describe different cutoff theories. Compatibility of the local integration on covers provides this common choice.

At the reference volumes, a compactly supported probe has a strictly positive fourth-order free response. After normalization by its leading power of the charge, the interacting response differs from the free coefficient by an error bounded by a positive power of the charge. Partition comparison then transfers these estimates to arbitrarily large volumes, with an error depending only on the two reference partition functions.

A positive fourth cumulant need not survive passage to an ergodic component. To see the obstruction, write the parent law as $\mu = \int \lambda_\xi \, \mathrm{d}\varpi(\xi)$. For a fixed real test f, put $m_2(\xi) = \int F(f)^2 \, \mathrm{d}\lambda_\xi(F)$. Gaussian domination gives $\int F(f) \, \mathrm{d}\lambda_\xi(F) = 0$ for almost every ξ. For a centered real variable X, write $\kappa_4(X) = \mathbb{E}X^4 - 3(\mathbb{E}X^2)^2$. The mixture identity is

$$\int \kappa_4^{\lambda_\xi}(f) \, \mathrm{d}\varpi(\xi) = \kappa_4^{\mu}(f) - 3 \operatorname{Var}_\varpi(m_2).$$

Variation of the component variances can therefore account for a positive parent cumulant, even if every component is Gaussian. We bound this variation by estimating products of quadratic observables with their reference values subtracted, in separated cells. Averaging their translates controls the translation-invariant part of $F(f)^2$. The fourth-order estimate then leaves the displayed integral positive; see Lem. 13.2. A positive-measure set of components has positive fourth cumulant. Since the reflection and Gaussian bounds pass to almost every component, one component has all the required properties.

The construction proceeds through the following tasks. §2 constructs the lattice law, proves Gaussian domination and obtains the nine reflection families. The photon and fermion block constructions are developed in §§3 and 4, respectively. §5 proves the one-step renormalization identity and estimates; §6 iterates them with a common choice of bare mass on the reference tori.

The large-field bounds in §7 control the complement of the fluctuation domains. §8 bounds the reference partition functions uniformly in the cutoff, and §9 transfers the reference estimates to arbitrary volumes. §10 constructs a probe with positive free fourth response; §11 compares it with the interacting response and controls separated quadratic observables. §12 takes the volume and continuum limits and proves the structural axioms. §13 makes the parameter choices and selects a non-Gaussian ergodic component.

The general reflection results are proved in §§A, B, and C: the reflected Hilbert space and spectral cone, rotation invariance, and clustering. The remaining appendices give polymer geometry and the index of notation and conventions.

The Osterwalder–Schrader framework is due to [1, 2]; lattice gauge reflection positivity and chessboard estimates were developed in [3, 4]. The block analysis follows the constructive renormalization

framework of [7]. We give the chamber, comparison and response arguments here. The general spectral-cone and disjoint-support rotation results are taken from [11], with their hypotheses and tensor conventions checked where they are used.

1.1. **Conventions.** The components of a two-form are indexed by $1 \le i < j \le 3$; their L^2 inner product is the sum of the three component inner products. We complexify test spaces when needed, while the random field is real. The adjoint d^* is taken in this Euclidean inner product. The matrices $\gamma_i = \mathrm{diag}(\sigma_i, -\sigma_i)$ use the Pauli matrices, and satisfy $\gamma_i\gamma_j + \gamma_j\gamma_i = 2\delta_{ij}$.

For a Euclidean motion $g(x) = Qx + a$, its action on a one-argument test is $(g \cdot f)(x) = (\Lambda^2 Q) f(g^{-1}x)$ and on a distribution is $(gF)(f) = F(g^{-1} \cdot f)$. Translations follow this convention: $(\tau_a f)(x) = f(x - a)$. Observable reflection includes complex conjugation. The Hodge dual of a two-form transforms as an axial vector, by $\det(Q)Q$; this convention will also be used in the rotation argument.

Notation and references to its definitions are collected in App. E. Fig. I maps the construction; its boxes link to the corresponding sections and appendices.

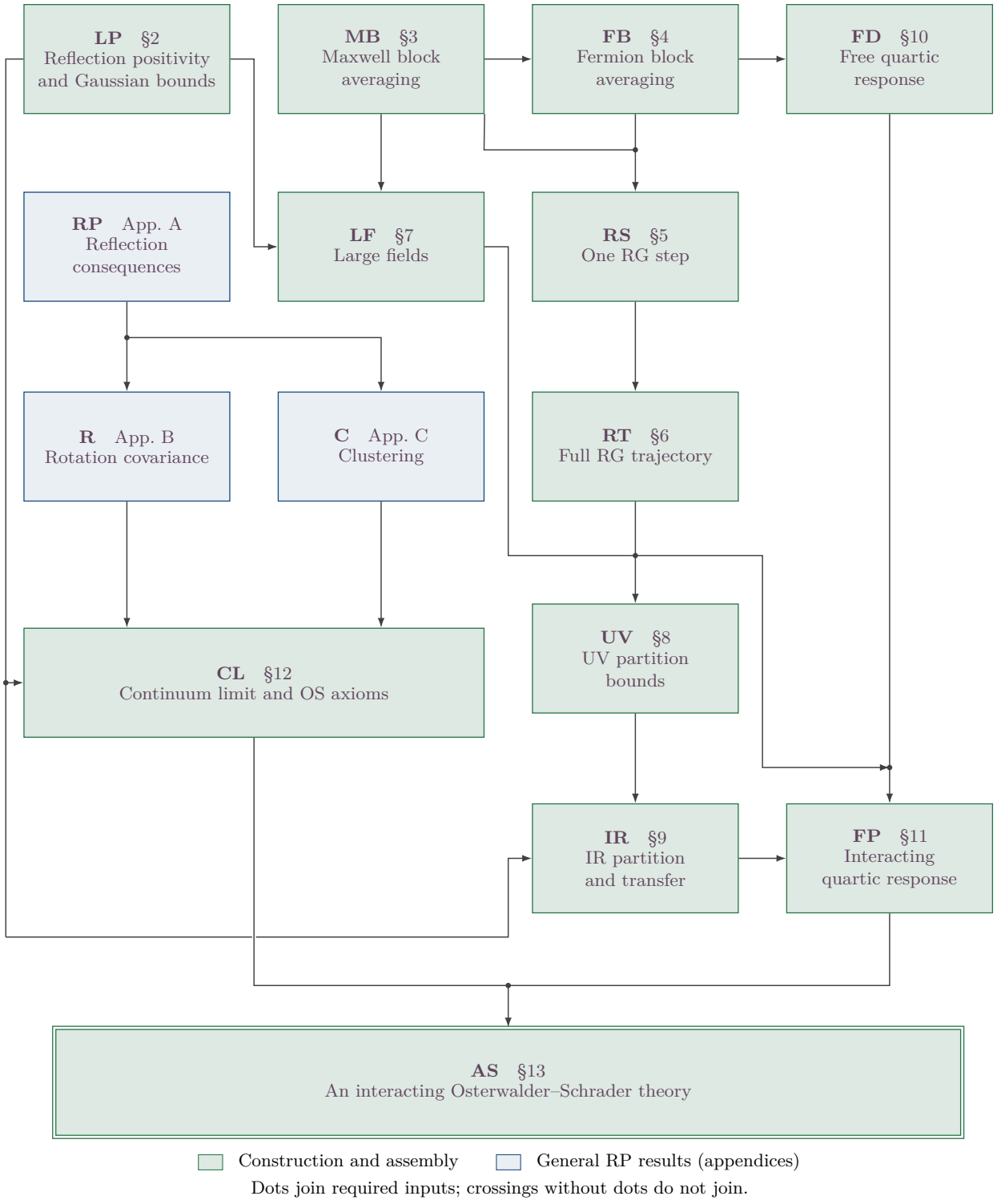


FIGURE I. Proof structure. Codes and references link to the corresponding sections or appendices.

2. Reflection positivity and Gaussian domination in lattice QED

2.1. **The chamber regularization.** We use the lattice gauge reflection-positivity method of [3], with the chamber crossing action constructed below. We first construct a finite-volume curvature law with reflection positivity and Gaussian bounds. The photon is a real one-cochain on the cubic lattice. The coordinate and diagonal reflection planes partition space into chambers; one fermion site is placed in each chamber, and neighboring sites are joined across their common wall. Each such bond joins reflected points orthogonally. This geometry makes the crossing action a sum of reflected Grassmann products with positive coefficients. A nonzero determinant coefficient, independent of the mass, will also give strict positivity of the partition function when the bare mass has undetermined sign.

Throughout, $0 < \rho < 1$, $q > 0$, $M \in \mathbb{R}$ and $w_0, \dots, w_3 > 0$; the continuum construction uses the standard hopping weights.

2.1.1. *The photon field and the flux sum.* We orient each link (x, μ) from x to $x + e_\mu$ and each plaquette $(x; \mu\nu)$, $\mu < \nu$, by the ordered pair (e_μ, e_ν). The real k-cochains are denoted by C^k, and the integer cochains by $C^k(\mathbb{Z})$. The coboundary d and its adjoint d^* use these orientations. Thus

$$(dA)_{\mu\nu}(x) = A_\mu(x) + A_\nu(x + e_\mu) - A_\mu(x + e_\nu) - A_\nu(x). \tag{2.i}$$

Write also $A(y, x) = -A(x, y)$ and $A(x, x) = 0$. Pullback includes cell orientation and commutes with d; coordinate reflection therefore negates curvature components containing the normal and preserves tangential ones.

Let $\Gamma \subset \mathbb{Z}^3$ be a rank-three subgroup with

$$\min\{|g|_\infty : g \in \Gamma \setminus \{0\}\} \geq 4.$$

We write $C^k(\Gamma)$ for the cochains on $\mathbb{Z}^3/\Gamma$ and identify them with their periodic extensions. The torus has $V = [\mathbb{Z}^3 : \Gamma]$ photon sites, $3V$ links and $3V$ plaquettes. The inner product on real cochains is the sum over positively oriented cells, and $\|\cdot\|$ denotes its norm.

A photon configuration is a class in

$$\mathrm{Conf}(\Gamma) = \{(A, z) : A \in C^1(\Gamma),\ z \in C^2(\Gamma; \mathbb{Z}),\ dz = 0\}/\!\sim, \qquad (A, z) \sim \left(A + \frac{2\pi}{q}k, z - dk\right), \tag{2.ii}$$

where $k \in C^1(\Gamma; \mathbb{Z})$. Its curvature and link phases are

$$F = dA + \frac{2\pi}{q}z, \qquad u_b = \exp(iqA(b)). \tag{2.iii}$$

These are representative-independent, $dF = 0$, and the closed integer cochain has unrestricted flux; the curvature is real and unbounded.

Put $r_q = 2\pi/q$. Each class has a unique representative with $A(b) \in [0, r_q)$ at every link. The measure on the quotient is defined by

$$\int_{\mathrm{Conf}(\Gamma)} \Phi(A, z)\, \mathrm{d}(A, z) = \sum_{\substack{z \in C^2(\Gamma;\mathbb{Z}) \\ dz = 0}} \int_{[0, r_q)^{3V}} \Phi(A, z)\, \mathrm{d}A. \tag{2.iv}$$

The formula holds for nonnegative or absolutely integrable functions on any measurable fundamental domain, since integer changes of representative preserve Lebesgue and counting measure.

We will repeatedly use the finite number

$$C_q(\varepsilon) = \sup_{y \in \mathbb{R}} \sum_{j \in \mathbb{Z}} e^{-\varepsilon(y + r_q j)^2}, \qquad \varepsilon > 0. \tag{2.v}$$

It satisfies

$$C_q(\varepsilon) \leq 2 + \frac{q}{2\pi}\sqrt{\frac{\pi}{\varepsilon}}.$$

Retain the two terms nearest the origin and bound the others by r_q^{-1} times the Gaussian integral over the disjoint adjacent intervals closer to the origin. Write $C_q = C_q(1/2)$. Dropping the constraint $dz = 0$ in the flux sum gives

$$0 < \int_{\mathrm{Conf}(\Gamma)} e^{-\|F\|^2/2}\, \mathrm{d}(A, z) \le (r_q C_q)^{3V}. \tag{2.vi}$$

Retaining $z = 0$ gives positivity; dropping closedness and summing each z_σ gives the upper bound.

The periodic extensions of the curvatures lie in

$$\Omega = \{F \in C^2 : dF = 0\},$$

with the topology of pointwise convergence on plaquettes. This is a closed subspace of a countable product of copies of $\mathbb{R}$. Local observables depend on finitely many plaquette values. For a finitely supported real two-cochain β, the character $\exp(i\langle \beta, F\rangle)$ is bounded and continuous on Ω.

2.1.2. *Chambers and the fermion graph.* The lattice mirrors are

$$\{x_\mu = c\}, \quad c \in \tfrac{1}{2}\mathbb{Z}, \qquad \{x_\mu \pm x_\nu = c\}, \quad c \in \mathbb{Z}, \quad \mu < \nu. \tag{2.vii}$$

They form three coordinate families and six diagonal families. Let $\mathcal{L}$ be the corresponding linear forms ℓ and write $\ell(x) = n_\ell \cdot x$. The admissible positions are $\mathrm{Adm}(x_\mu) = \frac{1}{2}\mathbb{Z}$ and $\mathrm{Adm}(x_\mu \pm x_\nu) = \mathbb{Z}$. The reflection in a mirror $\Pi = (\ell, c)$ is

$$\theta_\Pi x = x - \frac{2(\ell(x) - c)}{|n_\ell|^2} n_\ell. \tag{2.viii}$$

Its upward normal is $n_\ell/|n_\ell|$. Coordinate mirrors at integer positions are called site planes, and those at half-integer positions are called link planes.

The group generated by these reflections is

$$W = \mathbb{Z}^3 \rtimes O_{\mathrm{cub}}, \qquad O_{\mathrm{cub}} = O(3) \cap \mathrm{GL}(3, \mathbb{Z}).$$

Here O_{cub} consists of the 48 signed permutation matrices. A fundamental chamber is

$$C_0 = \{0 < x_3 < x_2 < x_1 < \tfrac{1}{2}\}.$$

Its wall reflections are

$$\begin{aligned} s_0(x) &= (x_1, x_2, -x_3), & s_1(x) &= (x_1, x_3, x_2), \\ s_2(x) &= (x_2, x_1, x_3), & s_3(x) &= (1 - x_1, x_2, x_3). \end{aligned}$$

Denote the walls by Π_i and their inward unit normals by

$$n_0 = e_3, \quad n_1 = (e_2 - e_3)/\sqrt{2}, \quad n_2 = (e_1 - e_2)/\sqrt{2}, \quad n_3 = -e_1. \tag{2.ix}$$

Fix $v_0 = (v_1, v_2, v_3) \in C_0$. The fermion sites and their neighbors are

$$\mathcal{X} = W v_0, \qquad (g v_0)^{(i)} = g s_i v_0. \tag{2.x}$$

The unordered bond $\{x, x^{(i)}\}$ has type i. A site has the unique form $x = k + r v_0$, with $k \in \mathbb{Z}^3$ and $r \in O_{\mathrm{cub}}$; put $\pi(x) = k$ and $y(x) = r v_0$. Thus $\pi(x)$ is its nearest photon site.

Proposition 2.1. The chambers are precisely gC_0, $g \in W$, and W acts on them simply transitively. No fermion site lies on a mirror. Every bond crosses exactly one mirror, its chamber wall, and crosses it orthogonally at its midpoint. Consequently a bond crossing Π is $\{x, \theta_\Pi x\}$.

The projection π is W-equivariant. Bonds of types $0, 1, 2$ have the same photon site at both ends; a type-3 bond joins two nearest photon sites. Exactly eight fermion bonds lie over each photon link. If a mirror does not cut the link and its open segment lies on one side, all eight bonds lie on that side. Each site has one neighbor of each type. ∎

Proof. Coordinate reflections and coordinate transpositions generate O_{cub}, and $\theta_{x_\mu,1/2}\theta_{x_\mu,0} = \tau_{e_\mu}$ gives the translations. Conversely, eq. 2.viii is an affine signed permutation with integer translation, so the mirror group is W.

For a point on no mirror, subtract the unique nearest lattice point. The resulting three coordinates are nonzero, have distinct absolute values, and lie in $(-1/2, 1/2)$. There is exactly one signed permutation making them positive and decreasing. This places the point in exactly one translate of C_0. These translates are disjoint open convex sets, meet no mirror, and cover its complement. They are therefore the chambers, with the asserted free transitive action.

For the four bonds from v_0, the midpoint is, respectively,

$$(v_1, v_2, 0), \quad (v_1, (v_2+v_3)/2, (v_2+v_3)/2), \quad ((v_1+v_2)/2, (v_1+v_2)/2, v_3), \quad (1/2, v_2, v_3).$$

Each belongs to the relative interior of its wall and to no other mirror. Each half of the bond, except its midpoint, lies in one of the two adjacent chambers. Transport by $g \in W$ proves the bond assertion at every site.

In the coordinates $x = k + rv_0$, the first three neighbors are $k + rs_i v_0$, whereas the fourth is $k + re_1 + rr_{e_1}v_0$, where r_{e_1} changes the sign of the first coordinate. This proves the statements about π and its equivariance. Over (k, μ) the lower ends are exactly the eight sites for which $y_\mu = v_1$: the remaining two absolute values can be ordered in two ways, with four choices of their signs. Their type-3 neighbors are the upper ends of these eight bonds.

The bond over $b = (k, \mu)$ has the segment $k + te_\mu + y_\perp$, $v_1 \le t \le 1 - v_1$, where the other two coordinates of $y_\perp$ have absolute values v_2, v_3. For a diagonal height involving x_μ, its varying part is $t \pm y_\nu \in (0, 1)$, the same interval between integers as on the open photon link. For heights not involving x_μ, the changes are y_ν or $y_\nu \pm y_{\nu'}$, whose absolute values are smaller than $1/2$ or 1, respectively. Away from a mirror containing b these preserve its side. The coordinate height x_μ gives the same side except at the unique link plane through the midpoint, which cuts b. This proves the assertion about photon links.

□

The geometric location is useful for reflection, while the coordinates (k, r) are useful for the renormalization analysis. If $d_i = \mathrm{dist}(v_0, \Pi_i)$, then

$$x^{(i)} - x = -2d_i\, rn_i, \qquad \frac{x^{(i)} - x}{|x^{(i)} - x|} = -rn_i.$$

The directions, bond types and projected endpoints in these coordinates do not depend on v_0. Hence neither the operator below nor its determinant depends on v_0 after this identification. The standard weights are

$$w_i = 48|\Pi_i \cap \overline{C}_0|, \qquad (w_0, w_1, w_2, w_3) = (6, 6\sqrt{2}, 6\sqrt{2}, 6). \tag{2.xi}$$

Here the bars denote two-dimensional Euclidean area. The inward normals n_i satisfy $\sum_i w_i n_i = 0$.

2.1.3. *The chamber operator and the finite-volume law.* Let σ_μ be the Pauli matrices and

$$\gamma_\mu = \mathrm{diag}(\sigma_\mu, -\sigma_\mu), \qquad \gamma_v = \sum_\mu v_\mu \gamma_\mu.$$

They form a Hermitian Clifford triple. For a unit vector n, put $P_n^{\pm} = (1 \pm \gamma_n)/2$. Both projections have rank two, and

$$\gamma_n \gamma_v \gamma_n = -\gamma_{r_n v}, \qquad r_n v = v - 2(v \cdot n)n. \tag{2.xii}$$

This follows by multiplying $\gamma_n\gamma_v + \gamma_v\gamma_n = 2n \cdot v$ by γ_n.

For an oriented bond (x, x') of type i define

$$U_A(x, x') = \exp(iqA(\pi(x), \pi(x'))), \qquad w(x, x') = \frac{w_i}{2}\left(\rho - \gamma_{(x'-x)/|x'-x|}\right). \tag{2.xiii}$$

The chamber operator is

$$(\mathcal{D}_A\psi)(x) = M\psi(x) - \sum_{i=0}^{3} w(x,x^{(i)})U_A(x,x^{(i)})\psi(x^{(i)}). \tag{2.xiv}$$

Its lattice mass is

$$m = M - \frac{\rho}{2}\sum_i w_i. \tag{2.xv}$$

For the standard weights, $\sum_i w_i n_i = 0$, so the Clifford part of the diagonal coefficient cancels. Thus eq. 2.xiv, in coordinates (k,r), is $D^{\mathrm{ch}}[qA] + m$, where $D^{\mathrm{ch}}[B]$ is the chamber matrix at unit mesh and zero mass, with external hopping phase $\exp(iB(b))$.

A fermion boundary condition is a character $\chi : \Gamma \to \{1,-1\}$. Write $\mathbb{T} = (\Gamma,\chi)$. Its spinors satisfy $\psi(x+g) = \chi(g)\psi(x)$ for $g \in \Gamma$. The restriction $\mathcal{D}_A^{\mathbb{T}}$ acts on a space of dimension $192V$, with the inner product obtained by summing over representatives of $\mathcal{X}/\Gamma$. For each site introduce generators $\psi_\alpha(x), \bar\psi_\alpha(x)$, $1 \le \alpha \le 4$, obeying the same boundary conditions. Their Grassmann algebra is denoted by $\mathrm{Gr}_{\mathbb{T}}$. Its Berezin integral is normalized by

$$\int_{\mathbb{T}} \prod_{x\in\mathcal{X}/\Gamma}\prod_{\alpha=1}^{4} \psi_\alpha(x)\bar\psi_\alpha(x) = 1. \tag{2.xvi}$$

Products of pairs are even, so their ordering is immaterial. With $S_{\mathbb{T}}(A) = \sum_x \bar\psi(x)(\mathcal{D}_A\psi)(x)$, this normalization gives $\int_{\mathbb{T}} e^{-S_{\mathbb{T}}(A)} = \det \mathcal{D}_A^{\mathbb{T}}$.

The unnormalized functional is

$$Z_{\mathbb{T}}[G] = \int_{\mathrm{Conf}(\Gamma)} e^{-\|F\|^2/2} \int_{\mathbb{T}} e^{-S_{\mathbb{T}}(A)} G(A,z)\,\mathrm{d}(A,z). \tag{2.xvii}$$

For a scalar G the inner integral is $G(A,z)\det\mathcal{D}_A^{\mathbb{T}}$. For a Grassmann-valued integrand, absolute convergence is understood coefficient by coefficient. Bounded measurable coefficients suffice: $e^{-S_{\mathbb{T}}(A)}$ has finitely many coefficients, each a polynomial in the link phases, and eq. 2.vi supplies an integrable majorant.

Proposition 2.2. For every real A and M,

$$\det\mathcal{D}_A^{\mathbb{T}} \ge 0, \qquad \|\mathcal{D}_A^{\mathbb{T}}\| \le |M| + \frac{1+\rho}{2}\sum_i w_i.$$

Moreover $\Re\mathcal{D}_A^{\mathbb{T}} = m + \frac{\rho}{2}\nabla_A^*\nabla_A$, where

$$(\nabla_A\psi)(x,i) = \sqrt{w_i/2}\,\big(\psi(x) - U_A(x,x^{(i)})\psi(x^{(i)})\big).$$

The determinant is not identically zero as a function of the link phases, for every real mass parameter M. ∎

Proof. Write

$$H_A\psi(x) = \sum_i w_i U_A(x,x^{(i)})\psi(x^{(i)}), \qquad J_A\psi(x) = \sum_i w_i \gamma_{(x^{(i)}-x)/|x^{(i)}-x|} U_A(x,x^{(i)})\psi(x^{(i)}).$$

Reversing a bond conjugates its phase and reverses its direction. Reindexing by the involution $x \mapsto x^{(i)}$ therefore gives $H_A^* = H_A$ and $J_A^* = -J_A$. Hence

$$\mathcal{D}_A = M - \frac{\rho}{2}H_A + \frac{1}{2}J_A, \qquad \nabla_A^*\nabla_A = \sum_i w_i - H_A.$$

This proves the real-part identity. In the splitting $\mathbb{C}^4 = \mathbb{C}^2 \oplus \mathbb{C}^2$, the two blocks of H_A coincide, and the two blocks of J_A have opposite signs. It follows that

$$\mathcal{D}_A^{\mathbb{T}} = \mathrm{diag}(\check{\mathcal{D}}_A^{\mathbb{T}}, (\check{\mathcal{D}}_A^{\mathbb{T}})^*), \qquad \det\mathcal{D}_A^{\mathbb{T}} = |\det\check{\mathcal{D}}_A^{\mathbb{T}}|^2. \tag{2.xviii}$$

Each hopping matrix has norm $w_i(1+\rho)/2$, while the associated phase-weighted permutation of sites is unitary. Their sum gives the norm bound.

We also need the determinant to be nonzero for some connection, whatever the real mass M. View it as a Laurent polynomial $P_{\mathbb{T}}$ in the independent link phases u_b, and select an extreme coefficient in a fixed coordinate direction. Only type-3 bonds contain a variable, with u_b at the lower end and u_b^{-1} at the upper end. The selected phase powers force hopping entries in the determinant permutation and exclude the mass entries. Fix a coordinate direction μ. The coefficient of

$$\prod_{b \| e_\mu} u_b^{32} \tag{2.xix}$$

will be computed without using M.

Partition the sites modulo Γ into six layers according to the increasing values

$$y_\mu = -v_1, -v_2, -v_3, v_3, v_2, v_1.$$

Each layer has $8V$ sites. Each site has one upward neighbor, one downward neighbor, and two neighbors in its own layer. The upward map is a permutation of the sites and visits successive layers modulo six. Its bond types are

$$(i_1, i_2, i_3, i_4, i_5, i_6) = (2, 1, 0, 1, 2, 3). \tag{2.xx}$$

For layers $1, \dots, 5$ the upward phase is one; in layer six it is u_b for the link $(\pi(x), \mu)$. These facts follow by changing the smallest sign or exchanging successive absolute values in the neighbor formulas above. The sixth step changes v_1 to $-v_1$ and increases the photon coordinate by one.

Each photon link has eight lower and eight upper ends, with four rows each. Exponent 32 therefore uses every upward type-3 row and excludes every reverse row. Layer six fills all layer-one columns, forcing layer-one rows upward. Inductively, occupied columns in the current and preceding layers force layers two through five upward as well. No other link variables occur.

The remaining freedom is a permutation of the four spin indices at each site. The site permutation, repeated four times, has sign one. Summing over the spin permutations therefore gives the product of the 4×4 determinants of the selected hopping blocks. The boundary signs have fourth power one, and

$$\det w(x, x^{\uparrow}) = (w_{i_k}/2)^4 (1-\rho^2)^2$$

at a site in layer k. Consequently the coefficient of eq. 2.xix is C_*^V, where

$$C_* = (w_0/2)^{32} (w_1/2)^{64} (w_2/2)^{64} (w_3/2)^{32} (1-\rho^2)^{96} > 0. \tag{2.xxi}$$

This coefficient is independent of M. In particular the determinant polynomial, and its restriction to the phase torus, are nonzero. □

The coefficient bound gives a positive determinant integral for every real M, without a sign assumption on m.

Proposition 2.3. The partition function satisfies $0 < Z_{\mathbb{T}}[1] < \infty$. More precisely,

$$r_q^{3V} e^{-6r_q^2 V} C_*^V \le Z_{\mathbb{T}}[1] \le (r_q C_q)^{3V} \left(|M| + \frac{1+\rho}{2} \sum_i w_i \right)^{192V}. \tag{2.xxii}$$

Thus $\mathbb{E}_{\mathbb{T}}[G] = Z_{\mathbb{T}}[G]/Z_{\mathbb{T}}[1]$ defines a probability law on scalar observables. Its curvature law $\mu_{\mathbb{T}}$ is the law of the periodic extension $\widetilde{F}$ on Ω. ■

Proof. For a nonnegative Laurent polynomial P on the phase torus, every Fourier coefficient satisfies

$$|\widehat{P}(\lambda)| = \left| \int P(u) \overline{u^\lambda} \, \mathrm{d}u \right| \le \int P(u) \, \mathrm{d}u,$$

where Haar measure is normalized. By the preceding proposition and eq. 2.xxi,

$$\int_{[0,r_q)^{3V}} \det \mathcal{D}_A^{\mathbb{T}} \, \mathrm{d}A \geq r_q^{3V} C_*^V .$$

On this box $|(dA)_\sigma| \leq 2r_q$ and $\|dA\|^2 \leq 12Vr_q^2$. The $z = 0$ term in eq. 2.xvii gives the lower bound in eq. 2.xxii. For the upper bound use $|\det \mathcal{D}_A^{\mathbb{T}}| \leq \|\mathcal{D}_A^{\mathbb{T}}\|^{192V}$ and eq. 2.vi. Normalizing the nonnegative density $e^{-\|F\|^2/2} \det \mathcal{D}_A^{\mathbb{T}}$ and pushing it forward by $(A, z) \mapsto \widetilde{F}$ gives the asserted law. □

To state the continuum result, specialize the finite-volume law to cubic tori. For an integer $n \geq 4$, take $\Gamma = n\mathbb{Z}^3$ and $\chi(nk) = (-1)^{k_1+k_2+k_3}$. Thus the photon cochains are periodic and the fermions are antiperiodic. For the continuum theorem, fix the Wilson parameter $\rho = 1/2$, the standard chamber weights $(6, 6\sqrt{2}, 6\sqrt{2}, 6)$ of eq. 2.xi, and the physical reference mass $\bar{m} = 1$. The integers $N, T \geq 0$ index the ultraviolet cutoff and the volume, respectively; $\boldsymbol{\mu}_N$ will denote the lattice mass m in eq. 2.xv at cutoff N. The following definitions specify the physical field and the topology of its limiting laws.

Let $\eta > 0$ and $R > 0$, with $n = R/\eta$ an integer. A plaquette of the unit covering lattice is $\sigma = (x; \mu\nu)$, with $\mu < \nu$ and center $y_\sigma = x + (e_\mu + e_\nu)/2$. Its averaging cube is

$$Q_\eta(\sigma) = \eta(y_\sigma + [-1/2, 1/2)^3).$$

For each fixed orientation these cubes partition $\mathbb{R}^3$. If u is a real lattice curvature, its physical step field is $\eta^{-3/2} u(\sigma)$ on $Q_\eta(\sigma)$ in component $\mu\nu$. The cutoff laws in the theorem below are the images of $\mu_{\mathbb{T}}$ under this map.

For a physical charge $e > 0$ and a mass shift $m_{0,N}(e)$, the continuum cutoff uses

$$m = \bar{m}\eta + m_{0,N}(e), \qquad M = m + \frac{\rho}{2} \sum_i w_i .$$

Let $A = 1 + |x|^2 - \Delta$ and let $h_\kappa = h_k e_{ij}$, $k \in \mathbb{Z}^3_{\geq 0}$, $i < j$, be the real Hermite basis of $L^2(\mathbb{R}^3; \Lambda^2)$. Its eigenvalues are $\lambda_\kappa = 2|k|_1 + 4$.

The Hilbert scale used for the limit laws is

$$\|F\|_{-s}^2 = \sum_\kappa \lambda_\kappa^{-2s} |F(h_\kappa)|^2, \qquad H_{-s} = \{F \in \mathcal{S}' : \|F\|_{-s} < \infty\}.$$

A volume-first limit means a weak limit of infinite-volume limits on H_{-4}: first let the torus side tend to infinity at fixed spacing, then let the spacing tend to zero. Compactness in this topology and its compatibility with tempered distributions are proved in §§12.1 and 12.2.

In the continuum formulas below, d is the exterior derivative on $\mathbb{R}^3$, d^* its Euclidean adjoint, and ∂ the coordinate gradient. The matrices σ_j are the Pauli matrices. The cutoff law uses the lattice coboundaries of eq. 2.i.

On $L^2(\mathbb{R}^3; \mathbb{C}^4)$, let $S_{\mathrm{D}} = (1 + \gamma \cdot \partial)^{-1}$ be the mass-one Dirac resolvent. For a real $\beta \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$, $i\gamma \cdot d^*\beta$ denotes multiplication by this matrix-valued function. The trace below is the operator trace on this Hilbert space; the trace-class property is proved in Prop. 10.1.

Theorem A (Existential field-strength theorem)**.** There are an odd integer $L \geq 5$, a charge $e > 0$, and real mass shifts $m_{0,N}$, independent of volume and with $|m_{0,N}| < 1/8$, such that the all-flux chamber laws with periodic photons, antiperiodic fermions, and

$$\eta = L^{-N}, \qquad R = 8L^T, \qquad q = e\eta^{1/2}, \qquad \boldsymbol{\mu}_N = \eta + m_{0,N} \tag{2.xxiii}$$

have a volume-first limit μ. There is a translation-ergodic component λ of μ supported on real tempered two-forms that are closed. It obeys Gaussian domination

$$\int \exp(F(g)) \, \mathrm{d}\lambda(F) \leq \exp(\|g\|_2^2/2) \qquad (g \in \mathcal{S}(\mathbb{R}^3; \Lambda^2) \text{ real}). \tag{2.xxiv}$$

Its moments, defined on products by $S_n(g_1 \otimes \cdots \otimes g_n) = \int \prod_{j=1}^n F(g_j)\,\mathrm{d}\lambda(F)$, extend to the tensor Schwartz spaces $\mathcal{S}_n = \mathcal{S}(\mathbb{R}^{3n}; (\Lambda^2\mathbb{C}^3)^{\otimes n})$. Writing these moments as S_n and setting $S_0 = 1$, they have the following properties.

(E0′) For $n \geq 1$ and $\Psi \in \mathcal{S}_n$,

$$|S_n(\Psi)| \leq 2\,66^n\sqrt{n!}\,q_{6n}(\Psi), \qquad q_m(\Psi) = \max_{|\alpha|\leq m} \sup_y (1+|y|^2)^{m/2}|\partial^\alpha \Psi(y)|, \tag{2.xxv}$$

where the maximum is also over tensor components.

(E1) For every Euclidean motion $g(x) = Qx + a$, $Q \in O(3)$, $S_n(g\cdot\Psi) = S_n(\Psi)$, where

$$(g\cdot\Psi)(x_1,\ldots,x_n) = (\Lambda^2 Q)^{\otimes n}\Psi(g^{-1}x_1,\ldots,g^{-1}x_n).$$

(E2) Let r be reflection in a plane through the origin and H either closed half-space bounded by that plane. If $\Psi_0 \in \mathbb{C}$ and $\Psi_k \in \mathcal{S}_k$, $1 \leq k \leq M$, have $\operatorname{supp}\Psi_k \subset H^k$, then

$$\sum_{k,l=0}^{M} S_{k+l}(\Theta_k\Psi_k \otimes \Psi_l) \geq 0, \qquad \Theta_k\Psi_k = \overline{r\cdot\Psi_k}, \quad \Theta_0 c = \overline{c}. \tag{2.xxvi}$$

(E3) S_n is unchanged by simultaneous permutation of the arguments and their tensor indices.

(E4) For $\Psi \in \mathcal{S}_k$ and $\Phi \in \mathcal{S}_l$,

$$S_{k+l}(\Psi \otimes \tau_a^{\otimes l}\Phi) \longrightarrow S_k(\Psi)S_l(\Phi) \quad \text{as } |a| \longrightarrow \infty, \tag{2.xxvii}$$

where

$$(\tau_a^{\otimes l}\Phi)(x_1,\ldots,x_l) = \Phi(x_1-a,\ldots,x_l-a).$$

There is a real $\beta \in C_c^\infty(\mathbb{R}^3;\Lambda^2)$ with $f = dd^*\beta$ such that

$$b(f) = -6\,\mathrm{tr}[(S_{\mathrm{D}} i\gamma\cdot d^*\beta)^4] > 0, \qquad S_{\mathrm{D}} = (1+\gamma\cdot\partial)^{-1}, \quad \gamma_j = \mathrm{diag}(\sigma_j, -\sigma_j), \tag{2.xxviii}$$

and

$$\kappa_4^\lambda(f) = S_4(f^{\otimes 4}) - 3S_2(f^{\otimes 2})^2 > 0. \tag{2.xxix}$$

In particular λ is non-Gaussian. ■□

The two-form action in (E1) is also the action in (E2): reflection in $x_1 = 0$ changes the signs of F_{12}, F_{13} and preserves F_{23}. The estimates concern full tensor test spaces, including coincident arguments.

2.1.4. *Symmetries and reflected Grassmann products.* The field strength transforms as a two-cochain, with the orientation of each plaquette included. For an antiperiodic cubic torus the law is invariant under W and under $F \mapsto -F$. We record the corresponding fermionic identities, since reflection positivity uses their signs.

Gauge covariance follows directly from

$$U_{A-d\lambda}(x,x') = e^{iq\lambda(\pi(x))}U_A(x,x')e^{-iq\lambda(\pi(x'))}.$$

For lattice covariance set $\gamma_5 = \left(\begin{smallmatrix} 0 & 1_2 \\ 1_2 & 0 \end{smallmatrix}\right)$. It anticommutes with the Clifford triple, and

$$(\gamma_n\gamma_5)\gamma_v(\gamma_n\gamma_5)^{-1} = \gamma_{r_n v}.$$

If the linear part of $g \in W$ is $r_{n_1}\cdots r_{n_k}$, the corresponding product of these spin matrices, followed by the permutation $x \mapsto g^{-1}x$, intertwines $\mathcal{D}_A^{\mathbb{T}}$ and $\mathcal{D}^{g\mathbb{T}}_{(g^{-1})^*A}$, where $g\mathbb{T} = (r\Gamma, \chi\circ r^{-1})$ and r is the linear part of g. Taking determinants and changing variables in eq. 2.xvii proves

$$\mu_{g\mathbb{T}} = ((g^{-1})^*)_*\mu_{\mathbb{T}}.$$

The configuration measure is preserved because pullback permutes cochain coordinates up to signs and maps one fundamental domain to another. All translations preserve $\mathbb{T}$, and all elements of W preserve an antiperiodic cubic torus, proving the stated invariance.

For charge conjugation let $C = \sigma_2 \oplus \sigma_2$. The Pauli identity $C\gamma_v^\top C^{-1} = -\gamma_v$, together with reversal of the oriented bond, gives

$$C(\mathcal{D}_A^{\mathbb{T}})^\top C^{-1} = \mathcal{D}_{-A}^{\mathbb{T}}.$$

Thus the determinant is unchanged by $A \mapsto -A$. The substitution $(A, z) \mapsto (-A, -z)$ preserves the configuration measure and sends F to $-F$, proving charge symmetry of the curvature law.

At a mirror Π with upward normal n, the Grassmann reflection Θ_Π is antilinear and reverses products. In components it is

$$\Theta_\Pi \psi_\alpha(x) = -\sum_\beta \bar\psi_\beta(\theta_\Pi x)(\gamma_n)_{\beta\alpha}, \qquad \Theta_\Pi \bar\psi_\alpha(x) = -\sum_\beta (\gamma_n)_{\alpha\beta}\psi_\beta(\theta_\Pi x). \tag{2.xxx}$$

On scalar coefficient functions it acts as $\Phi(A, z) \mapsto \overline{\Phi(\theta_\Pi^* A, \theta_\Pi^* z)}$. The Clifford identities give $\Theta_\Pi^2 = 1$ and

$$\Theta_\Pi(\bar\psi(x) Y \psi(x')) = \bar\psi(\theta_\Pi x')\gamma_n Y^* \gamma_n \psi(\theta_\Pi x). \tag{2.xxxi}$$

For a hopping term, $\gamma_n w(x, x')\gamma_n = w(\theta_\Pi x', \theta_\Pi x)$, by eq. 2.xii. Reflection therefore permutes the oriented terms of the action, including the real mass terms.

If Y is a finite set of sites, then

$$\int_{\theta_\Pi Y} \Theta_\Pi \xi = \overline{\int_Y \xi}. \tag{2.xxxii}$$

Only the top-degree coefficient needs to be checked. At one site reflection maps $\prod_\alpha \psi_\alpha \bar\psi_\alpha$ to the same product at the reflected site, multiplied by $(\det \gamma_n)^2 = 1$. Products from different sites are even. This proves eq. 2.xxxii, and a measure-preserving change of photon variables gives

$$Z_{\theta_\Pi \mathbb{T}}[\Theta_\Pi G] = \overline{Z_{\mathbb{T}}[G]} \tag{2.xxxiii}$$

for bounded coefficient integrands, and for the integrable Gaussian-growth classes considered below.

The reflections act on even integrands through W: two words for the same isometry differ by an even word with spin matrix $\mathrm{diag}(u, u)$. Here u is an even product of Pauli matrices, so has determinant one; trivial geometry forces it to commute with all Pauli matrices. Thus $u = \pm 1_2$, which acts trivially on even monomials. Write this action as G^g; for real scalar observables, $G^g(F) = G(g^* F)$.

2.1.5. *The fermion crossing at a mirror.* Let a bond of type j cross Π, with lower endpoint x and upper endpoint $x' = \theta_\Pi x$. Its direction is the upward normal n. The two hopping matrices are

$$\begin{aligned} w(x, x') &= \tfrac{w_j}{2}((1+\rho)P_n^- - (1-\rho)P_n^+), \\ w(x', x) &= \tfrac{w_j}{2}((1+\rho)P_n^+ - (1-\rho)P_n^-). \end{aligned} \tag{2.xxxiv}$$

Choose orthonormal bases $e_{n,1}^\pm, e_{n,2}^\pm$ of the two projected spaces. For each $k = 1, 2$ introduce four crossing modes. Their coefficients and phases are displayed together:

	ζ	ω	u
1	$(e_{n,k}^-)^* \psi(x')$	$\frac{w_j}{2}(1+\rho)$	$U_A(x, x')$
2	$(e_{n,k}^+)^* \psi(x')$	$\frac{w_j}{2}(1-\rho)$	$U_A(x, x')$
3	$\psi(x') e_{n,k}^+$	$\frac{w_j}{2}(1+\rho)$	$\overline{U_A(x, x')}$
4	$\bar\psi(x') e_{n,k}^-$	$\frac{w_j}{2}(1-\rho)$	$\overline{U_A(x, x')}$

(2.xxxv)

All coefficients are positive in the range $0 < \rho < 1$.

Proposition 2.4. Let J be a finite collection of bonds crossing Π, and let $S_\Pi^\times(J)$ be their part of the action. Index all their crossing modes by a finite set $\mathcal{I}$. Then

$$-S_\Pi^\times(J) = \sum_{i \in \mathcal{I}} \omega_i u_i \Theta_\Pi(\zeta_i)\zeta_i. \tag{2.xxxvi}$$

Choose an order on $\mathcal{I}$ and, for $I \subset \mathcal{I}$, put $\zeta_I = \prod_{i\in I} \zeta_i$, $\omega_I = \prod_{i\in I} \omega_i$ and $u_I = \prod_{i\in I} u_i$. Then

$$e^{-S^{\times}_{\Pi}(J)} = \sum_{I\subset\mathcal{I}} \omega_I u_I \Theta_\Pi(\zeta_I)\zeta_I. \tag{2.xxxvii}$$

At site and diagonal mirrors all u_i are one. At a link plane, groups one and two have charge $+1$, groups three and four charge -1, and their phases are $e^{iq\mathfrak{q}(i)A(b)}$ at the cut photon link b. ■

Proof. The reflections of the four modes in eq. 2.xxxv are, respectively,

$$\bar\psi(x)e^-_{n,k}, \quad -\bar\psi(x)e^+_{n,k}, \quad -(e^+_{n,k})^*\psi(x), \quad (e^-_{n,k})^*\psi(x).$$

Multiplying each reflected mode by the original mode and using anticommutation gives the four projected terms in eq. 2.xxxiv, with the indicated phases. Summation over k uses $P^\pm_n = \sum_k e^\pm_{n,k}(e^\pm_{n,k})^*$ and gives the negative of the two bond terms. Summation over bonds proves eq. 2.xxxvi.

The commuting even elements $\Theta_\Pi(\zeta_i)\zeta_i$ square to zero; reversal of order under Θ_Π gives

$$\prod_{i\in I} \Theta_\Pi(\zeta_i)\zeta_i = \Theta_\Pi(\zeta_I)\zeta_I,$$

with no remaining sign. Expanding the product of the finite exponentials proves eq. 2.xxxvii. Only type-3 bonds carry a photon phase, and they cross link planes. Their projected endpoints are the oriented endpoints of the cut link, giving the last assertion. □

2.1.6. *Adapted tori and the second mirror.* An adapted torus for $\ell \in \mathcal{L}$ has the form

$$\Gamma = \Gamma_\parallel \oplus \mathbb{Z}v, \quad \Gamma_\parallel = \Gamma \cap \{\ell = 0\}, \quad v = \frac{p}{|n_\ell|^2} n_\ell \in \mathbb{Z}^3, \quad \chi(v) = -1, \tag{2.xxxviii}$$

where $p = \ell(v) > 0$ is the height period. For an admissible c set

$$c' = c + p/2, \quad \Pi = (\ell, c), \quad \Pi' = (\ell, c'), \quad U_+ = \{c < \ell < c'\}, \quad U_- = \{c - p/2 < \ell < c\}.$$

Write $\theta = \theta_\Pi$ and $\Theta = \Theta_\Pi$. The two mirrors are admissible: for coordinate families $p \geq 4$, while for diagonal families p is even and $p \geq 8$. Their reflections satisfy $\theta_{\Pi'} = \tau_v \theta_\Pi$ and induce the same geometric reflection on the torus, exchanging its two halves. An antiperiodic coordinate torus is adapted to each coordinate family. The torus with periods $ae_1 - ae_2$, $be_1 + be_2$ and ce_3, antiperiodic on these periods, is adapted to the corresponding two diagonal families and the third coordinate family. Cubic images give adapted tori for all nine families.

2.1.7. *The photon crossing and Gauss' law.* The photon crossing must retain the integer closedness condition $dz = 0$. We impose that condition at the cut cubes by Fourier multipliers and apply Poisson summation at the cut plaquettes. The upper and lower variables then separate into reflected factors. At a cut link, integrating its phase imposes Gauss' law, again with a nonnegative coefficient. The cell classification below determines which variables enter each factor.

Let $\mathcal{M}$ be the two boundary mirrors, understood periodically. Classify a cell by its relative interior: it is upper or lower if that interior is in the respective half, lies on the mirrors if the whole cell does, and is cut if its relative interior meets a mirror without being contained in it. Write $B_+, B_-, B_0, B_\times$ for the four sets of links, $P_+, P_-, P_0, P_\times$ for the plaquettes, and $Q_+, Q_-, Q_\times$ for the cubes.

Reflection exchanges upper and lower cells. A cell meeting a mirror in its relative interior is fixed, since distinct relative interiors are disjoint. Its affine hull is fixed pointwise for a mirror cell, giving orientation sign $+1$, and reflected for a cut cell, giving -1. No cell meets both boundary mirrors: its height variation is at most $|n_\ell|^2 < p/2$.

An upper cell has only upper faces or faces lying on the mirrors; a cut cell has only upper, lower or cut faces. Moreover a plaquette containing a cut link is cut, and a cube containing a cut plaquette is cut. An upper face can touch the boundary at a relative interior point only when its affine height is constant. A face of a cut cell cannot lie in the mirror: reflection in the supporting plane of a face

would not preserve that cell. Finally, the height takes both signs on a cut face, hence on the relative interior of its coface. In particular, the incidences at a cut plaquette and a cut cube give

$$(dA)_\sigma = (d_+A)_\sigma - (d_+\theta^*A)_\sigma + (d_\times A)_\sigma, \qquad (dz)(Q) = (d_+z)(Q) - (d_+\theta^*z)(Q) + (d_\times z)(Q), \tag{2.xxxix}$$

where d_+ and $d_\times$ retain the incidences from upper and cut faces, respectively. For example, $(d_+A)_\sigma = \sum_{b\in B_+}[\sigma : b]A(b)$. The minus sign in the lower term follows from orientation reversal of the cut cell and the identity $d\theta^* = \theta^* d$.

Use representatives

$$A = (A_+, A_-, A_0, a), \qquad z = (z_+, z_-, z_0, z_\times).$$

Choose the interval $[0, r_q)$ for upper and mirror links, its signed reflection for lower links, and $[-r_q/2, r_q/2)$ for cut links. This is a fundamental domain. Reflection maps the lower variables measure-preservingly to an independent copy of the upper variables, fixes (A_0, z_0), and changes the sign of a.

At a cut cube impose integer closedness by the Fourier multiplier

$$1_{\{(dz)(Q)=0\}} = \int_0^{2\pi} e^{i\alpha_Q (dz)(Q)} \frac{\mathrm{d}\alpha_Q}{2\pi}.$$

For $\alpha \in [0, 2\pi)^{Q_\times}$ and $k \in \mathbb{Z}^{P_\times}$ define

$$\begin{aligned} \alpha'_\sigma &= \sum_{Q\in Q_\times} [Q : \sigma]\alpha_Q, \quad \beta_\sigma = \frac{q}{2\pi}(\alpha'_\sigma - 2\pi k_\sigma), \\ u_{\alpha,k} &= \exp\left(i\sum_Q \alpha_Q (d_+z)(Q) - i\sum_\sigma \beta_\sigma (d_+A)_\sigma \right). \end{aligned} \tag{2.xl}$$

The function $u_{\alpha,k}$ depends only on the upper variables and has modulus one.

Poisson summation, in the normalization needed here, reads

$$\sum_{j\in\mathbb{Z}} e^{-(y+r_q j)^2/2 + ij\alpha} = \frac{q}{\sqrt{2\pi}} \sum_{k\in\mathbb{Z}} e^{-\beta_k^2/2 - i\beta_k y}, \qquad \beta_k = \frac{q}{2\pi}(\alpha - 2\pi k). \tag{2.xli}$$

It follows by Fourier transforming the Gaussian under $t = y + r_q s$. Insert the cube multipliers and apply this formula at each cut plaquette. Splitting dA and dz by eq. 2.xxxix gives the exact identity

$$\begin{aligned} &\sum_{z_\times} \prod_{\sigma\in P_\times} e^{-F_\sigma^2/2} \prod_{Q\in Q_\times} 1_{\{(dz)(Q)=0\}} \\ &= \int \prod_{Q\in Q_\times} \frac{\mathrm{d}\alpha_Q}{2\pi} \sum_{k\in\mathbb{Z}^{P_\times}} \prod_{\sigma\in P_\times} \frac{q}{\sqrt{2\pi}} e^{-\beta_\sigma^2/2}\, u_{\alpha,k}\Theta(u_{\alpha,k}) \exp\left(iq \sum_{b\in B_\times} a_b (d^*k)(b) \right). \end{aligned} \tag{2.xlii}$$

Here k is extended by zero outside $P_\times$. To obtain the final factor, observe that at a cut link b,

$$\sum_{\sigma\in P_\times} [\sigma : b]\alpha'_\sigma = \sum_{Q\in Q_\times} \alpha_Q \sum_\sigma [Q : \sigma][\sigma : b] = 0.$$

The restriction to cut plaquettes may be dropped because any plaquette containing a cut link is cut; the last sum is zero by $d^2 = 0$. Thus the part of β involving α' cancels from the cut-link factor.

All the weights before the two reflected factors on the right of eq. 2.xlii are positive. Their total integral is at most $C_q^{|P_\times|}$. Indeed, with $y = 0$ in eq. 2.xli,

$$\frac{q}{\sqrt{2\pi}} \sum_k e^{-\beta_k^2/2} \le \sum_j e^{-(r_q j)^2/2} \le C_q$$

at each cut plaquette. The original Gaussian sum also has an absolute bound $C_q^{|P_\times|}$, justifying insertion of the Fourier multipliers and the exchanges of their sums and integrals.

Cut links occur only at coordinate link planes. A link fixed by a mirror reflection but not contained in its plane has its endpoints exchanged; hence it is perpendicular to the mirror, whose normal is a coordinate vector. A diagonal mirror has cut plaquettes and cubes, but no cut links.

For a crossing-mode subset I, let $\mathfrak{q}(I)(b)$ be the sum of its charges at modes over b. By eq. 2.xxxvii, their photon phase is

$$u_I = \exp\left(iq \sum_{b \in B_\times} a_b \mathfrak{q}(I)(b) \right).$$

Integration of the cut links over one period gives

$$\int_{[-r_q/2, r_q/2)^{B_\times}} e^{iq \sum_b a_b ((d^* k)(b) + \mathfrak{q}(I)(b))} \, \mathrm{d}a = r_q^{|B_\times|} 1_{\{d^* k + \mathfrak{q}(I) = 0 \text{ on } B_\times\}}. \tag{2.xliii}$$

The coefficients $(d^*k)(b) + \mathfrak{q}(I)(b)$ are integers. This is precisely Gauss' law at the cut links, with a nonnegative multiplier. When there are no cut links the condition is empty.

2.2. **Slab operators and finite-volume comparison.** We represent the finite-volume functional as a trace of operators associated with slabs between parallel reflection planes. Reflecting a slab takes its operator to an adjoint, so repeating a reflected pair expresses its partition function as a Schatten norm. The trace inequality then compares tori obtained by changing the normal closing vector or its fermion character. Three such comparisons connect the cubic and diagonal-adapted geometries at equal volume; the resulting pressure error will vanish in the volume limit.

A second trace representation retains every integer flux sector and proves Gaussian domination. Here an external two-cochain shifts the electric and magnetic curvatures, meaning the components containing the chosen time direction and those tangent to a slice. The electric shift enters through unitary operators and can be removed by the trace inequality. Repeating the argument with another time direction removes the magnetic shift.

Keep $q > 0$, $0 < \rho < 1$, $w_i > 0$ and $M \in \mathbb{R}$ fixed during the volume limit, allowing mesh dependence. We use $r_q = 2\pi/q$ and counting Berezin integration eq. 2.xvi. Its agreement with the RG convention follows from the $192V$ fermion modes: with the pairing

$$\langle \bar{\Psi}, O\Psi \rangle_{\mathrm{ch}} = \frac{1}{48} \sum_x \bar{\Psi}(x)(O\Psi)(x), \qquad \int \mathrm{d}\Psi := 48^{192V} \int_{\mathbb{T}},$$

one has

$$\int \mathrm{d}\Psi \, e^{-\langle \bar{\Psi}, O\Psi \rangle_{\mathrm{ch}}} = \det O.$$

The factor 48^{192V} cancels the weighted determinant factor, so both conventions give the same curvature law. Slab modes use counting integration.

2.2.1. *Local observables and their lifts.* Local observables may contain fermion variables and coefficients depending on a connection. To evaluate the same observable in every flux sector, we shall lift the curvature and link phases to a real connection on its support. Gauge invariance will make the value independent of that lift; a strict Gaussian margin will give integrability. Let $B = y + [-r, r]^3$, and write $B^{(j)}$ for the oriented lattice j-cells contained in B. The fermion sites assigned to B are

$$\mathcal{X}(B) = \{x \in \mathcal{X} : \pi(x) + [-\tfrac{1}{2}, \tfrac{1}{2}]^3 \subset B\}.$$

This support condition includes the photon site at which the fermion transforms under gauge transformations.

Let $\mathcal{G}(B)$ consist of the Borel maps $G : \mathbb{R}^{B^{(1)}} \to \mathrm{Gr}(\mathcal{X}(B))$ with gauge-invariant values and with the following growth bound. If $G(\widehat{A}) = \sum_\Psi a_\Psi(\widehat{A})\Psi$, then, for some $C < \infty$ and $\varepsilon > 0$,

$$|a_\Psi(\widehat{A})| \le C \exp\left((\tfrac{1}{2} - \varepsilon) \sum_{\sigma \in B^{(2)}} (d\widehat{A})_\sigma^2 \right). \tag{2.xliv}$$

Here gauge invariance means

$$a_\Psi(\widehat{A} - d\lambda)e^{iq\langle\lambda,\mathfrak{q}(\Psi)\rangle} = a_\Psi(\widehat{A}) \tag{2.xlv}$$

for every real function λ on the photon sites of B; $\mathfrak{q}(\Psi)$ counts the ψ generators minus the $\bar{\psi}$ generators at fermion sites projecting to each photon site. We denote the even subspace by $\mathcal{G}^{\mathrm{ev}}(B)$.

A local lift of a configuration (A, z) on B is a real one-cochain $\widehat{A}$ on $B^{(1)}$ satisfying

$$d\widehat{A} = F \quad \text{on } B^{(2)}, \qquad e^{iq\widehat{A}(b)} = e^{iqA(b)} \quad (b \in B^{(1)}). \tag{2.xlvi}$$

We evaluate G at this lift and then impose the fermion identifications of the torus or cylinder. We now verify that this procedure is Borel and independent of the lift.

Replace B by its integer box $\prod_\mu [a_\mu, b_\mu]$, where $a_\mu = \lceil y_\mu - r \rceil$ and $b_\mu = \lfloor y_\mu + r \rfloor$. If this box has no sites, use the unique empty lift. Otherwise its axial spanning tree consists of all links in direction 3, the links in direction 2 on $x_3 = a_3$, and the links in direction 1 on $x_2 = a_2$, $x_3 = a_3$. Prescribe a one-cochain on this tree. Its values on the remaining links are determined by

$$\begin{aligned}
\widehat{A}_1(x + e_2) &= \widehat{A}_1(x) + \widehat{A}_2(x + e_1) - \widehat{A}_2(x) - F_{12}(x), && x_3 = a_3, \\
\widehat{A}_1(x + e_3) &= \widehat{A}_1(x) + \widehat{A}_3(x + e_1) - \widehat{A}_3(x) - F_{13}(x), \\
\widehat{A}_2(x + e_3) &= \widehat{A}_2(x) + \widehat{A}_3(x + e_2) - \widehat{A}_3(x) - F_{23}(x).
\end{aligned} \tag{2.xlvii}$$

First use the first line in increasing x_2, then the last two lines in increasing x_3. Each link outside the tree occurs as an unknown once. The construction is linear in F and the tree values. It gives the prescribed 13- and 23-curvatures and the 12-curvature on the bottom face. If $dF = 0$, induction over the cubes gives the remaining 12-curvatures: on each cube the two closed two-cochains F and $d\widehat{A}$ agree on five faces, hence on the sixth.

On the tree choose $\widehat{A}(b) = \arg_q(e^{iqA(b)})$, where $\arg_q$ takes values in $[0, r_q)$. This is a Borel choice. Subtract the recursion for A from eq. 2.xlvii. The tree differences lie in $r_q\mathbb{Z}$, and every subsequent difference is a signed sum of earlier ones and of a value of $r_q z$. Thus $\widehat{A} - A$ is integer-valued in units of r_q on every link. This proves eq. 2.xlvi, with Borel dependence on F and the tree phases.

Two local lifts differ by $r_q k$, where k is an integer closed one-cochain. Sum k along the tree paths from its bottom corner to define an integer function λ. Then $d\lambda = k$ on the tree. The zero-curvature recursion shows that $k - d\lambda = 0$ on every link. eq. 2.xlv, applied with $-r_q\lambda$, gives $a_\Psi(\widehat{A} + r_q d\lambda) = a_\Psi(\widehat{A})$, since $\langle\lambda, \mathfrak{q}(\Psi)\rangle$ is integer. The value of G is therefore independent of the lift.

For an open set U, let $\mathcal{A}(U)$ be the linear span of the maps

$$\widehat{A} \longmapsto \Phi(d\widehat{A}) \prod_{j=1}^{r} G_j(\widehat{A}|_{B_j^{(1)}}), \tag{2.xlviii}$$

where the boxes $B_j \subset U$ are pairwise disjoint, $G_j \in \mathcal{G}^{\mathrm{ev}}(B_j)$, and Φ is a bounded Borel function of finitely many curvature values at plaquettes contained in U. A box B embeds in a period group Γ when the sets $B + [-1, 1]^3 + \gamma$, $\gamma \in \Gamma$, are pairwise disjoint. Boxes $B_1, \dots, B_r$ embed disjointly in Γ when each embeds and all sets $B_j + \gamma$, indexed by distinct pairs (j, γ), are pairwise disjoint. We evaluate a product on the quotient only when its boxes embed disjointly.

It remains to check that equal maps eq. 2.xlviii have equal values on configurations. This requires one lift for every box in a representation. On $\mathbb{Z}^3$, a closed integer two-cochain z has the integer primitive

$$\begin{aligned}
k_1(x) &= 0, \\
k_2(x) &= \sum_{s=0}^{x_1 - 1} z_{12}(s, x_2, x_3),
\end{aligned}$$

$$k_3(x) = \sum_{s=0}^{x_1-1} z_{13}(s, x_2, x_3) + \sum_{s=0}^{x_2-1} z_{23}(0, s, x_3).$$

The sums are oriented for negative endpoints. The first two components of $dk = z$ follow by taking differences. In the third, $dz = 0$ changes the summand into a difference of consecutive z_{23} values, so the sum telescopes. Therefore $\widehat{A} = A + r_q k$ is a lift on the whole lattice. It need not be periodic, even when (A, z) is periodic.

The same conclusion holds on a closed slab. Choose a coordinate μ_0 absent from the normal n_ℓ. The slab cell complex is a planar strip times $\mathbb{Z}$ in direction μ_0. Let $\mu < \nu$ be the other coordinates, denoted by (a, r). Write $Q(a, r)$ for the unit square with lower corner (a, r) and I_r for the interval of a such that $[(a, r), (a, r + 1)]$ lies in the strip. When I_r is nonempty choose $a_r \in I_r$. On the base sheet set the μ-link values of k to zero and put

$$k([(a, r), (a, r + 1)] \times \{0\}) = \sum_{a_r \le s < a} z(Q(s, r) \times \{0\}).$$

All squares in this oriented sum lie in the strip because I_r is an interval. Set $k = 0$ on the μ_0-links. For a horizontal link b in direction $\nu' \in \{\mu, \nu\}$, extend by

$$k(b \times \{j\}) = k(b \times \{0\}) + s_{\nu'} \sum_{0 \le j' < j} z(b \times [j', j' + 1]), \qquad s_{\nu'} = \begin{cases} -1, & \nu' < \mu_0, \\ +1, & \nu' > \mu_0. \end{cases}$$

The sums with reversed endpoints have the opposite sign. The resulting primitive has $dk = z$ on all vertical plaquettes and on the base sheet. The closed two-cochain $z - dk$ then has vanishing vertical components; closedness makes its horizontal component constant between sheets. It vanishes on the base sheet and hence everywhere. Again the primitive is taken on the lifted strip, not on its periodic quotient.

Evaluation at this common lift, followed by the quotient homomorphism on fermion generators, agrees with the separate local evaluations. It is linear in the map G, so it is independent of any representation of G as a sum of products. Moreover, different box plaquettes have different representatives on the quotient. Multiplying eq. 2.xliv by the Maxwell density therefore leaves

$$C \exp(-\varepsilon_G \|F\|^2), \qquad \varepsilon_G = \min(\tfrac{1}{2}, \varepsilon_1, \dots, \varepsilon_r) > 0. \tag{2.xlix}$$

The finite Grassmann expansion of the action has bounded coefficients for real A. The Gaussian sum eq. 2.v now proves absolute integrability, coefficient by coefficient. This also proves the required Borel measurability on the quotient.

2.2.2. *Boundary spaces and slab kernels.* The slab operator must retain enough boundary data to glue both fields. Fermion crossing modes supply a finite exterior space. Fourier variables for the closed integer flux and the remaining mirror photon values will supply a positive boundary measure. The crossing identities then become operator composition.

Fix a mirror family ℓ and a cross-section $\Xi = (\Gamma_\parallel, \chi_\parallel)$, where $\Gamma_\parallel \subset \mathbb{Z}^3 \cap \{\ell = 0\}$ has rank two and systole at least four. Here and below, the systole of a period lattice Γ is $\min_{\gamma \in \Gamma \setminus \{0\}} |\gamma|_\infty$. Work on the cylinder $\mathbb{Z}^3/\Gamma_\parallel$. The admissible mirror positions are $c \in \frac{1}{2}|n_\ell|^2\mathbb{Z}$. Write $\mathcal{M}^j(c)$ for the j-cells contained in $\{\ell = c\}$, $\mathcal{X}^j(c)$ for those cut through their relative interiors by this plane, and $\mathcal{B}^j(c, c')$ for the j-cells contained in $\{c \le \ell \le c'\}$ and in neither bounding plane. In particular a slab cell may touch either boundary. All these sets are finite on a bounded slab.

For a chain of admissible mirrors, the cylinder cells partition into the cells in its mirrors, those cut by them, and those lying between consecutive mirrors. A cell cut by c has its height range $[c - |n_\ell|^2/2, c + |n_\ell|^2/2]$. Its non-cut faces lie on one side of the plane; the other faces are cut by the same plane. This partition applies also when two mirrors are consecutive. A fermion bond crosses at most one mirror of such a chain, and no fermion site lies in a mirror.

A closed-slab configuration $\varpi = (A, z)$ assigns real link values and integer plaquette values to all cells of the closed slab, with $dz = 0$ on its cubes. We integrate its interior values using $A(b) \in [0, r_q)$ and counting measure for z. Its interior photon weight is

$$W_{c,c'}(\varpi) = \prod_{\sigma \in \mathcal{B}^2(c,c')} e^{-F_\sigma^2/2} \prod_{Q \in \mathcal{B}^3(c,c')} \mathbf{1}[(dz)(Q) = 0]. \tag{2.li}$$

We use the larger class $\mathcal{I}_\Xi[c, c']$ of Borel Grassmann-valued functions of closed-slab configurations, with fermion support $c < \ell < c'$, invariant under $(A, z) \mapsto (A + r_q k, z - dk)$, whose coefficients obey

$$|a_\Psi(\varpi)| \le C_G \exp\left((\tfrac{1}{2} - \varepsilon_G) \sum_{\sigma \in \mathcal{B}^2(c,c')} F_\sigma^2 \right), \qquad 0 < \varepsilon_G \le \tfrac{1}{2}. \tag{2.lii}$$

The lift construction puts open-slab local products in this class. Consecutive products retain the smaller ε; reflection and translation preserve the bound.

At the mirror c, let $\mathcal{J}(c)$ be the ordered set of crossing modes from eq. 2.xxxv, taken modulo $\Gamma_\parallel$. Let $\mathcal{F}(c)$ be the exterior space with orthonormal basis e_I, $I \subset \mathcal{J}(c)$. Put $\omega_c e_I = \omega_I e_I$ and $U_{\mathrm{f}} e_I = (-1)^{|I|} e_I$. Translation of the modes includes both the character $\chi_\parallel$ and the sign required to reorder their product:

$$U_v e_I = s_v(I) e_{r_v(I)}, \qquad \tau_v \zeta_I^{(c)} = s_v(I) \zeta_{r_v(I)}^{(c+\ell(v))}.$$

The U_v are unitary, satisfy the translation law, and commute with parity. For an integer translation parallel to the normal, choose the orders compatibly so that $s_v(I) = 1$.

For a Grassmann element H supported in the open slab define

$$N_{c,c'}[H]_{I',I} = \int_{(c,c')} \zeta_I^{(c)} H \Theta_{c'}(\zeta_{I'}^{(c')}). \tag{2.liii}$$

An operator Υ on $\mathcal{F}(c)$ is represented at the mirror by

$$\mathrm{Ins}_c[\Upsilon] = \sum_{I,J} \Upsilon_{IJ} \Theta_c(\zeta_J^{(c)}) \zeta_I^{(c)}.$$

The particular insertion $\mathrm{Ins}_c[\omega_c \operatorname{diag}(u_I(A))]$ is the crossing exponential eq. 2.xxxvii. The ordering in these definitions gives

$$N_{c,c''}[H \,\mathrm{Ins}_{c'}[\Upsilon] H'] = N_{c',c''}[H'] \Upsilon N_{c,c'}[H]. \tag{2.liv}$$

Indeed, an entry of the left side is a sum of products $[\zeta_I H \Theta_{c'} \zeta_J][\zeta_K H' \Theta_{c''} \zeta_{I''}]$ in disjoint sets of slab generators. Berezin factorization gives the entries on the right without moving any of these factors. Reflecting the integrand reverses their order and conjugates the Berezin coefficient; with $v = 2(c' - c) n_\ell / |n_\ell|^2$ this gives

$$N_{c',2c'-c}[\Theta_{c'} H] = U_v N_{c,c'}[H]^*. \tag{2.lv}$$

We next include the photon crossing in the boundary data. The cut incidence maps d_c and d_c^+ are the restrictions of the coboundary to cut faces and to upper faces, as in eq. 2.xxxix. Introduce one angle ϑ_Q for each cut cube and one integer k_σ for each cut plaquette, and set

$$\begin{aligned}
\varkappa_\sigma &= r_q^{-1}((d_c^* \vartheta)_\sigma - 2\pi k_\sigma), \\
\mathrm{d}\mu_c^\times &= \prod_{Q \in \mathcal{X}^3(c)} \frac{\mathrm{d}\vartheta_Q}{2\pi} \prod_{\sigma \in \mathcal{X}^2(c)} \frac{q}{\sqrt{2\pi}} e^{-\varkappa_\sigma^2/2}, \\
e_c((\vartheta, k); \varpi) &= e^{i\langle \vartheta, d_c^+ z \rangle - i\langle \varkappa, d_c^+ A \rangle}, \\
D_c(k, I) &= r_q^{|\mathcal{X}^1(c)|} \mathbf{1}[d_c^* k + \mathfrak{q}(I) = 0].
\end{aligned} \tag{2.lvi}$$

Here $\mathfrak{q}(I)(b)$ is the number of modes of groups $1,2$ over the cut link b, minus the number of groups $3,4$ over it. There are no cut links at diagonal mirrors. At a coordinate site plane there are no cut cells at all; its Fourier space is a point.

The one-mirror photon identity proved by the Fourier and Poisson calculation of §2.1.7 is

$$\sum_{z^\times}\int \mathrm{d}A^\times \prod_{\sigma\in\mathcal{X}^2(c)} e^{-F_\sigma^2/2} \prod_{Q\in\mathcal{X}^3(c)} \mathbf{1}[(dz)(Q)=0]\, u_I(A^\times) \\ = \int \mathrm{d}\mu_c^\times\, D_c(k,I) e_c((\vartheta,k);\varpi_+)\overline{e_c((\vartheta,k);\theta_c^*\varpi_-)}. \quad (2.\mathrm{lvi})$$

The data agree on the mirror and supply the upper and reflected lower non-cut faces. Cube Fourier multipliers impose closedness, Poisson summation supplies $e^{-\varkappa^2/2}$, and cut-link integration supplies $D_c(k,I)$. Kernel construction and gluing use the two directions of this identity.

Mirror variables are $\xi=(\mathfrak{m},\vartheta,k,I)$, with $\mathfrak{m}=(A_0,z_0)\in[0,r_q)^{\mathcal{M}^1(c)}\times\mathbb{Z}^{\mathcal{M}^2(c)}$. On their space Y_c put

$$\mathrm{d}\mu_c(\xi)=\mathrm{d}A_0 \prod_{\sigma\in\mathcal{M}^2(c)} e^{-F_\sigma(\mathfrak{m})^2/2}\,\mathrm{d}\mu_c^\times(\vartheta,k) D_c(k,I), \qquad \mathcal{H}(c)=L^2(Y_c,\mu_c). \quad (2.\mathrm{lvii})$$

The integer and mode variables carry counting measure. This is a finite positive measure. Summing the mirror plaquette Gaussians gives $C_q^{|\mathcal{M}^2(c)|}$, while Poisson summation at fixed angles gives

$$\frac{q}{\sqrt{2\pi}}\sum_{k_\sigma} e^{-\varkappa_\sigma^2/2} \le \sum_{j\in\mathbb{Z}} e^{-r_q^2 j^2/2}.$$

The remaining factors are bounded by $r_q^{|\mathcal{M}^1(c)|+|\mathcal{X}^1(c)|}2^{|\mathcal{J}(c)|}$. The induced signed translations $V_v:\mathcal{H}(c)\to\mathcal{H}(c+\ell(v))$ are unitary and satisfy the group law. The parity operator multiplies the component I by $(-1)^{|I|}$.

Let $S_{(c,c')}(A)$ be the fermion action containing precisely the mass and bond terms with fermion endpoints in the open slab. For $G\in\mathcal{I}_\Xi[c,c']$, define the kernel

$$\mathfrak{S}[G](\xi',\xi)=\sqrt{\omega_{I'}\omega_I}\int \mathrm{d}\varpi^\circ\, W_{c,c'}(\varpi) e_c((\vartheta,k);\varpi)\overline{e_{c'}((\vartheta',k');\theta_{c'}^*\varpi)} \\ \cdot N_{c,c'}[G(\varpi)e^{-S_{(c,c')}(A)}]_{I',I}, \qquad \varpi=\mathfrak{m}\vee\varpi^\circ\vee\mathfrak{m}'. \quad (2.\mathrm{lviii})$$

Set the whole integrand to zero where $W_{c,c'}(\varpi)=0$; $G(\varpi)$ is evaluated only on the closed configurations in its domain. The associated integral operator maps $\mathcal{H}(c)$ to $\mathcal{H}(c')$. The square roots of the mode weights belong in the kernel; the measure eq. 2.lvii contains none of these weights. This places one full crossing weight at a glued mirror.

Theorem B. Every $\mathfrak{S}[G]$ is Hilbert–Schmidt and depends linearly on G. If G has constants C_G,ε_G in eq. 2.li, then

$$|\mathfrak{S}[G](\xi',\xi)| \le C_{c,c'}C_G C_q(\varepsilon_G)^{|\mathcal{B}^2(c,c')|}, \\ \|\mathfrak{S}[G]\|_2 \le \sqrt{\mu_c(Y_c)\mu_{c'}(Y_{c'})}\,C_{c,c'}C_G C_q(\varepsilon_G)^{|\mathcal{B}^2(c,c')|}. \quad (2.\mathrm{lix})$$

The constant depends on the fixed cross-section, mirror positions, q,M,ρ,w_i, and chamber base point. For consecutive slabs,

$$\mathfrak{S}[G']\mathfrak{S}[G]=\mathfrak{S}[GG'], \\ \mathfrak{S}[\tau_v G]=V_v\mathfrak{S}[G]V_v^{-1}, \quad (2.\mathrm{lx}) \\ \mathfrak{S}[\Theta_{c'}G]=V_{2(c'-c)n_\ell/|n_\ell|^2}\mathfrak{S}[G]^*.$$

If $\mathbb{T} = (\Gamma, \chi)$ has cross-section Ξ, $g \in \Gamma$ is a closing vector with minimal positive height $p = \ell(g)$, and $c_0 < \cdots < c_r = c_0 + p$ are admissible, $r \geq 2$, and $G_j \in \mathcal{I}_\Xi[c_{j-1}, c_j]$ for every j, then

$$Z_{\mathbb{T}}[G_1 \cdots G_r] = \operatorname{Tr}_{\mathcal{H}(c_0)} \left(V_g^{-1} U_{\mathrm{f}}^{(1+\chi(g))/2} \mathfrak{S}[G_r] \cdots \mathfrak{S}[G_1] \right). \tag{2.lxi}$$

These identities apply to arbitrary parities. For homogeneous G, $U_{\mathrm{f}} \mathfrak{S}[G] = (-1)^{\deg G} \mathfrak{S}[G] U_{\mathrm{f}}$. ■□

Proof. Each coefficient of the finite Grassmann polynomial $e^{-S_{(c,c')}(A)}$ is bounded uniformly in real A. Each coefficient in the matrix in eq. 2.lviii is consequently bounded by a fixed multiple of the majorant for G. The phases have modulus one. After multiplication by the photon weight, the absolute integrand is bounded by

$$C C_G \prod_{\sigma \in \mathcal{B}^2(c,c')} e^{-\varepsilon_G F_\sigma^2}.$$

Dropping cube constraints gives a factor $C_q(\varepsilon_G)$ per integer sum and r_q per link integral; finite mode factors enter $C_{c,c'}$. This proves eq. 2.lix. Finite boundary measures give its L^2 bound and justify the gluing integrals absolutely.

In the product kernel the middle square roots give $\omega_{I'}$. Apply eq. 2.lvi to the boundary measure and phases, then eq. 2.liii with $\Upsilon = \omega_{c'} \operatorname{diag}(u_{I'})$. These restore the cut photon weights, cube constraints, crossing phases and fermion action. The boundary measure supplies mirror plaquettes; since no cube lies in a mirror, the cell partition counts every constraint once. The result is $\mathfrak{S}[GG']$.

Translation relabels modes and oriented cells. Reflection uses eq. 2.liv, swaps the boundary phases and preserves the interior weight; integer-shift invariance permits the reflected integration interval. Its upper boundary is the lower one translated by $2(c' - c)n_\ell/|n_\ell|^2$, proving eq. 2.lx. Odd Berezin integrals vanish, giving parity.

For the trace, composition reduces the claim to two slabs. A product of two Hilbert–Schmidt kernels is trace class, and its trace is the absolutely convergent integral of their cyclic product. The closing operator identifies the upper mirror data with the translate by g of the lower data. Its contribution to this cyclic kernel is

$$(-\chi(g))^{|I_0|} s_g(I_0).$$

The relabeling gives $s_g(I_0)$; translating and reordering the lower crossing product give $\chi(g)^{|I_0|}$ and $(-1)^{|I_0|}$ in even total degree. Thus closure contributes the identity for antiperiodic character and fermion parity for periodic character.

Apply eq. 2.lvi at the two mirrors. For fixed photon variables the remaining mode sum is

$$\operatorname{tr}_{\mathcal{F}(c_0)} (U_g^{-1} U_{\mathrm{f}}^{(1+\chi(g))/2} N_2 \Upsilon_1 N_1 \Upsilon_0), \qquad \Upsilon_j = \omega_{c_j} \operatorname{diag}(u_I(A)).$$

Berezin composition and closure give the torus integral: the two slab and crossing actions count each mass and hopping term once, and the cell partition counts each link, plaquette and cube constraint once. This is eq. 2.xvii, proving eq. 2.lxi in even degree. Odd Berezin integrals vanish, and odd trace terms anticommute with parity. Decomposing into homogeneous parts proves the general case. □

2.2.3. *Reflected folds and Schatten estimates.* Repeated reflection gives the chessboard method of [4]. We implement it through the chamber slab operators. A reflected slab pair gives Y^*Y after identifying its boundary spaces. Repeating the pair therefore expresses a folded partition function as a Schatten norm, while the original torus trace is bounded by the product of these norms. For an operator Y between Hilbert spaces, write $\|Y\|_s = (\operatorname{Tr}(Y^*Y)^{s/2})^{1/s}$. We use Schatten's inequality in the following form. If $1 \leq s_j < \infty$, $Y_j : H_{j-1} \to H_j$ belongs to the Schatten class of exponent s_j, $H_r = H_0$, and $\sum_j s_j^{-1} = 1$, then the cyclic product is trace class and

$$|\operatorname{Tr}(Y_r \cdots Y_1)| \leq \prod_{j=1}^{r} \|Y_j\|_{s_j}. \tag{2.lxii}$$

Apply the usual inequality to zero extensions on $\bigoplus_j H_j$; their singular values are unchanged and their product has the displayed cyclic block. Absorb any unitary closing map into Y_r.

Let $\mathbb{T}$ have height period p in the family ℓ. Its straightened torus $\mathbb{T}^0$ has the same cross-section and the closing vector

$$g^0 = \frac{p}{|n_\ell|^2} n_\ell, \qquad \Gamma^0 = \Gamma_\parallel \oplus \mathbb{Z} g^0, \qquad \chi^0(g^0) = -1. \tag{2.lxiii}$$

We use this torus only when $g^0 \in \mathbb{Z}^3$ and its lattice has systole at least four. Its volume is the volume of $\mathbb{T}$: both lattices have the same covolume in the mirror and the same normal height period.

Divide one period into $2k$ slabs of equal height h', with all $c_j = c_0 + jh'$ admissible and $p = 2kh'$. For an integrand G in one of these slabs, form its fold by reflecting it successively in the bounding planes, taking all $2k$ copies. This is an integrand on $\mathbb{T}^0$. Its fermion boxes are disjoint: they lie strictly between different consecutive mirrors; within each slab the cross-section embedding separates their translates.

Proposition 2.5. For integrands $G_j \in \mathcal{I}_\Xi[c_{j-1}, c_j]$,

$$Z_{\mathbb{T}^0}[\mathrm{fold}_\ell\, G_j] = \|\mathfrak{S}[G_j]\|_{2k}^{2k},$$
$$|Z_{\mathbb{T}}[G_1 \cdots G_{2k}]| \le \prod_{j=1}^{2k} Z_{\mathbb{T}^0}[\mathrm{fold}_\ell\, G_j]^{1/(2k)}. \tag{2.lxiv}$$

In particular every folded factor on the right is nonnegative. ■

Proof. Fix j and write $a = c_{j-1}$. Put $Y = \mathfrak{S}[G_j] : \mathcal{H}(a) \to \mathcal{H}(a + h')$, and form the fold on $[a, a + p]$, with initial boundary a in the trace formula. Let $v = 2h' n_\ell / |n_\ell|^2$, which is integer because both mirror positions are admissible. The first reflected pair has product $V_v Y^* Y$ by eq. 2.lx. Its translate by sv has product $V_{(s+1)v} Y^* Y V_{sv}^{-1}$. Multiplying all k pairs telescopes to

$$V_{kv} (Y^* Y)^k.$$

Since $kv = g^0$ and $\chi^0(g^0) = -1$, the closing translation in eq. 2.lxi cancels V_{g^0} and introduces no parity. The trace is $\mathrm{Tr}_{\mathcal{H}(a)} (Y^* Y)^k = \|Y\|_{2k}^{2k}$. This calculation takes place at the starting boundary of each slab and proves the first identity for every G_j.

For the original torus, the operator $V_g^{-1} U_{\mathrm{f}}^{(1+\chi(g))/2}$ is unitary. Apply eq. 2.lxii to its trace formula, with every exponent equal to $2k$. The first identity identifies the resulting norms with folded partition functions. The Hilbert–Schmidt estimate ensures all operators belong to the required Schatten classes. □

Corollary 2.6. Let $\mathbb{T}$ be adapted as in eq. 2.xxxviii, and put $c' = c + p/2$. For even $G, G' \in \mathcal{I}_\Xi[c, c']$,

$$Z_{\mathbb{T}}[(\Theta_c G') G] = \mathrm{Tr}(\mathfrak{S}[G']^* \mathfrak{S}[G]). \tag{2.lxv}$$

Thus the scalar curvature law is reflection invariant and reflection positive at the boundary mirror. The same form remains positive after multiplication of the weight by $\Phi_+ \Theta_c \Phi_+$, with Φ_+ bounded and scalar in the open upper half-slab. ■□

Proof. Take $k = 1$ in eq. 2.lxiv. The straightened torus is $\mathbb{T}$ itself, and the right side is $\|\mathfrak{S}[G]\|_2^2$. Since $\theta_{c'} = \tau_v \theta_c$, the fermion boundary condition gives $\Theta_{c'} = \chi(v)^{\deg} \Theta_c$. On even integrands the two reflections therefore agree, and the fold is $(\Theta_c G) G$. Polarization gives eq. 2.lxv; all its integrals converge by Thm. B. For scalar tests divide by $Z_{\mathbb{T}}[1] > 0$; reflection invariance follows from eq. 2.xxxiii. Finally replace G, G' by $\Phi_+ G, \Phi_+ G'$. □

We next apply this inequality to cells. A frame is a triple of the mirror families with mutually orthogonal normals n_1, n_2, n_3. The two frames needed below are (x_1, x_2, x_3) and $(x_1 - x_2, x_1 + x_2, x_3)$.

Choose admissible offsets o_a and an integer height width $h > 0$. The open cells are

$$Q_\kappa = \{x : o_a + h\kappa_a < \ell^a(x) < o_a + h(\kappa_a + 1),\ a = 1, 2, 3\}, \qquad |Q| = \frac{h^3}{|n_1||n_2||n_3|}. \tag{2.lxvi}$$

The reflections in the planes $\ell^a = o_a + hj$ generate a product of three infinite dihedral groups. Exactly one element of this group carries a given open cell to any specified open cell. Its action on even Grassmann-valued functions is generated by the reflections Θ and translations used above. Denote the resulting copy of $G \in \mathcal{A}(Q_\kappa)$ in $Q_{\kappa'}$ by $G^{[\kappa']}$.

A rectangular torus in this frame has periods $\gamma_a = p_a n_a/|n_a|^2 \in \mathbb{Z}^3$, with $\chi(\gamma_a) = -1$. If each $p_a \in 2h\mathbb{Z}$, the copies fit its period lattice. Let $N = |\mathbb{T}|/|Q|$ be the number of cells, and define

$$\mathrm{Fold}_{\mathbb{T}}\, G = \prod_{\kappa' \bmod \Gamma} G^{[\kappa']}, \qquad \|G\|_{\mathbb{T}} = Z_{\mathbb{T}}[\mathrm{Fold}_{\mathbb{T}}\, G]^{1/N}. \tag{2.lxvii}$$

This is an unnormalized seminorm. Its defining partition function is nonnegative, by the first identity of eq. 2.lxiv applied to one layer of the folded array. The value is independent of the chosen representatives of the cells.

Theorem C. If a rectangular torus $\mathbb{T}$ carries the cells eq. 2.lxvi, then

$$\left| Z_{\mathbb{T}} \left[\prod_\kappa G_\kappa \right] \right| \le \prod_\kappa \|G_\kappa\|_{\mathbb{T}}, \qquad G_\kappa \in \mathcal{A}(Q_\kappa). \tag{2.lxviii}$$

Let $C(a, b, c)$ have coordinate periods ae_1, be_2, ce_3, all with character -1. Let $e_- = e_1 - e_2$, $e_+ = e_1 + e_2$, and let $D(a, b, c)$ have periods ae_-, be_+, ce_3, all with character -1. Let $D'(a, b, c)$ have the same periods and characters, except for $\chi(be_+) = +1$. For the diagonal frame and $n \in 2h\mathbb{Z}$, $n \ge 8$, the cells have representatives

$$0 \le \kappa_1 < n/h, \qquad 0 \le \kappa_2 < 2n/h, \qquad 0 \le \kappa_3 < n/h.$$

They represent the cells of $C(n, n, n)$, $D'(n/2, n, n)$, and $D(n/2, n, n)$, and

$$\left| Z_{\mathbb{T}} \left[\prod_\kappa G_\kappa \right] \right| \le \prod_\kappa \|G_\kappa\|_{D(n/2,n,n)}, \quad \mathbb{T} = C(n, n, n) \text{ or } D'(n/2, n, n). \tag{2.lxix}$$

For coordinate cells of width h, $a, c \in 2h\mathbb{Z}$, $a, c \ge 4$, one has

$$\left| Z_{D(a,2a,c)} \left[\prod_\kappa G_\kappa \right] \right| \le \prod_\kappa \|G_\kappa\|_{C(a,4a,c)}. \tag{2.lxx}$$

■□

Proof. Group the cell integrands into layers perpendicular to the first normal, and apply eq. 2.lxiv. Its right side consists of the arrays obtained by retaining one layer and reflecting it throughout the straightened torus. Next apply the same inequality to each such array in direction two, and then in direction three. The reflected array after these three steps is the full fold of the integrand from one original cell. The three exponents multiply to

$$\frac{h}{p_1}\frac{h}{p_2}\frac{h}{p_3} = \frac{1}{N}.$$

Thus the last product consists of one factor $\|G_\kappa\|_{\mathbb{T}_3}$ for each original cell, where $\mathbb{T}_3$ is the final rectangular torus.

We verify that all three arrays have consistent periods and that the same cells represent them. If Γ_0 is the initial lattice, then straightening successively in the frame gives

$$\Gamma_1 = (\Gamma_0 \cap \{\ell^1 = 0\}) \oplus \mathbb{Z}\gamma_1,$$
$$\Gamma_2 = (\Gamma_0 \cap \{\ell^1 = \ell^2 = 0\}) \oplus \mathbb{Z}\gamma_1 \oplus \mathbb{Z}\gamma_2,$$

$$\Gamma_3 = \mathbb{Z}\gamma_1 \oplus \mathbb{Z}\gamma_2 \oplus \mathbb{Z}\gamma_3, \qquad \gamma_a = p_a n_a/|n_a|^2.$$

The kernel in the second line is a rank-one group along n_3; its positive generator is γ_3, by the definition of the third height period. Characters already fixed at γ_1, γ_2 are retained in subsequent cross-sections, so all three final characters are -1. The brick $0 \le \kappa_a < p_a/h$ represents Γ_3. It represents Γ_2 as well; then its section $\kappa_2 = 0$ represents the second cross-section, and its section $\kappa_1 = 0$ represents the first one. Restoring the normal coordinate in each case gives representatives for the preceding lattice. These groups all have the same covolume.

We also check the stronger box embedding condition used in the local classes. If $B = y + [-r, r]^3$ lies in an open cell, then $2|n_a|^2 r < h$ for every a, by taking the range of ℓ^a on B. For a nonzero period γ, at least one height is nonzero and $|\ell^a(\gamma)| \ge 2h$. Consequently $|\gamma|_\infty > 4r$, and the systole also gives $|\gamma|_\infty \ge 4$. These inequalities imply $|\gamma|_\infty > 2r + 2$: use the second if $r < 1$, the first if $r \ge 1$. Thus translates of $B + [-1, 1]^3$ are disjoint. Boxes in different cell orbits cannot meet after period translations, since the open cells themselves are disjoint. It follows that every layer belongs to its slab class. This justifies the three applications.

For a rectangular torus, every straightening leaves the torus unchanged, which proves eq. 2.lxviii. For a cubic torus with even side n, the diagonal operations are

$$C(n,n,n) \xrightarrow{\ell_-} D'(n/2,n,n) \xrightarrow{\ell_+} D(n/2,n,n), \qquad D(n/2,n,n) \xrightarrow{x_1} C(n/2,2n,n). \tag{2.lxxi}$$

For the first operation the cross-section is generated by ne_+, ne_3, with characters $+1, -1$; its height period is n, and straightening replaces the closing vector ne_1 by $(n/2)e_-$. For the second, the cross-section is generated by $(n/2)e_-, ne_3$, both with character -1. Its closing vector ne_+ already points along the normal, and the operation changes its character from $+1$ to -1. For the third, the cross-section is generated by $2ne_2, ne_3$, both with character -1; the height period is $n/2$ and the straight closing vector is $(n/2)e_1$. The lattices have systole at least four.

The first two diagonal operations, followed by the unchanged x_3 operation, have height periods $(n, 2n, n)$. Hence they give eq. 2.lxix; starting at D' instead, use the order (ℓ_+, x_3, ℓ_-) to obtain the same final fold. Likewise, starting at $D(a, 2a, c)$ and using coordinate directions gives periods $(a, 4a, c)$ and ends at $C(a, 4a, c)$, proving eq. 2.lxx. □

2.2.4. *Pressure and the comparison error.* Write $C(n) = C(n,n,n)$ and

$$f(n_1,n_2,n_3) = \frac{1}{n_1 n_2 n_3} \log Z_{C(n_1,n_2,n_3)}[1].$$

The mass and charge in this function are held fixed. The following error will compare local expectations in the cubic and diagonal geometries.

Proposition 2.7. The pressure f is nonincreasing in each side, and

$$f(n_1,n_2,n_3) \longrightarrow f_\infty := \inf f \in \mathbb{R} \quad \text{as } \min_\mu n_\mu \longrightarrow \infty.$$

For every even $n \ge 8$,

$$Z_{C(n)}[1] \le Z_{D'(n/2,n,n)}[1] \le Z_{D(n/2,n,n)}[1] \le Z_{C(n/2,2n,n)}[1]. \tag{2.lxxii}$$

With $\Delta(n) = f(n/2, 2n, n) - f(n,n,n)$,

$$0 \le \Delta(n) \le f(n/2,n/2,n/2) - f_\infty \longrightarrow 0. \tag{2.lxxiii}$$

The logarithms of any two partition functions in eq. 2.lxxii differ in absolute value by at most $n^3\Delta(n)$. ■

Proof. Fix the cross-section with sides n_2, n_3 and let $Y = \mathfrak{S}[1] : \mathcal{H}(0) \to \mathcal{H}(1/2)$ be the coordinate half-step. The first identity of eq. 2.lxiv, applied to $2n_1$ half-slabs, gives

$$Z_{C(n_1,n_2,n_3)}[1] = \operatorname{Tr}(Y^*Y)^{n_1}. \tag{2.lxxiv}$$

The positive trace-class operator Y^*Y is independent of n_1 and is nonzero, since the partition function is positive. Consequently

$$f(n_1,n_2,n_3)=\frac{1}{n_2n_3}\log\|Y^*Y\|_{n_1}.$$

Schatten norms decrease as their exponent increases, so f decreases in n_1. Permuting the time direction proves the other two claims. The same trace identity shows separate convexity of $\log Z$ in the side lengths: Hölder's inequality gives $\operatorname{Tr}A^{(1-t)s+ts'}\le(\operatorname{Tr}A^s)^{1-t}(\operatorname{Tr}A^{s'})^t$ for $A=Y^*Y$.

The lower partition bound eq. 2.xxii gives the volume-independent lower bound

$$f\ge\log C_*+3\log r_q-6r_q^2.$$

The upper bound in the same equation makes f_∞ finite. For any $\delta>0$, choose a triple with $f<f_\infty+\delta$. Monotonicity gives the same bound whenever all three sides exceed that triple, proving convergence.

Apply eq. 2.lxiv with constant integrands to each of the three operations eq. 2.lxxi. A fold of the constant is the constant, so $Z_{\mathbb{T}}[1]\le Z_{\mathbb{T}^0}[1]$. This proves the chain. Each torus in it has volume n^3, so its first and last logarithms differ by $n^3\Delta(n)$. Side monotonicity gives $f(n/2,2n,n)\le f(n/2,n/2,n/2)$, while $f(n,n,n)\ge f_\infty$, proving eq. 2.lxxiii. All intermediate logarithms lie between the endpoints, which proves the last assertion. □

The partition functions can still differ by a factor as large as $e^{n^3\Delta(n)}$. For the local comparison, fix a cell Q and place $n^3/|Q|$ copies of the observable in the torus. Tilting their sum by a strength $t>0$ and dividing the logarithmic comparison by $tn^3/|Q|$ leaves the error $|Q|\Delta(n)/t$. This tends to zero when volume tends to infinity with the cell and t fixed; removing the tilt is a subsequent step.

2.2.5. *Coordinate slices and the all-flux trace.* The time-slice determinant factorization follows the transfer-matrix method of [12, 3]. We give its chamber version and retain every flux sector in the trace.

For Gaussian domination we use a slice representation with real spatial connections. Take direction 1 as time on $C_{\mathbf{n}}=C(n_1,n_2,n_3)$, $\mathbf{n}=(n_1,n_2,n_3)$, $n_\mu\ge4$, and let $\Sigma=\mathbb{Z}/n_2\mathbb{Z}\times\mathbb{Z}/n_3\mathbb{Z}$. Its links and plaquettes are $\Sigma^{(1)}$ and $\Sigma^{(2)}$, its coboundary is d_s, and $v=|\Sigma|=n_2n_3$. A connection has spatial slices A_j and temporal links a_j, and its curvature has the parts

$$F_j^{\mathrm{el}}=A_{j+1}-A_j-d_sa_j+r_qz_j^{\mathrm{el}},\qquad F_j^{\mathrm{mag}}=d_sA_j+r_qz_j^{\mathrm{mag}}.\tag{2.lxxv}$$

Closedness reads $z_{j+1}^{\mathrm{mag}}-z_j^{\mathrm{mag}}=d_sz_j^{\mathrm{el}}$.

Closedness allows us to replace the electric integer cochains by shifts of the real slice connections. The remaining integer data are a spatial magnetic flux and a closed winding vector at the last time slice. Keeping both is necessary for the trace to represent the original all-flux law. Put $k_0=0$, $k_{j+1}=k_j+z_j^{\mathrm{el}}$. Closedness is equivalent to

$$z_j^{\mathrm{mag}}=z_0^{\mathrm{mag}}+d_sk_j,\qquad d_sk_{n_1}=0.$$

The integer shift with spatial values k_j and zero temporal values removes all electric integers except the last one, k_{n_1}, and changes every spatial slice to $A_j+r_qk_j$. Summing k_j while integrating A_j over its compact box unfolds the intermediate slices to all of $\mathbb{R}^{\Sigma^{(1)}}$. The last integer remains as a winding vector

$$g=r_qk_{n_1}\in\Gamma_s,\qquad\Gamma_s=r_q(\mathbb{Z}^{\Sigma^{(1)}}\cap\ker d_s).$$

Fix a plaquette σ_0 of Σ and write $k=\sum_\sigma z_0^{\mathrm{mag}}(\sigma)$. Then $z_0^{\mathrm{mag}}-k\mathbf{1}_{\sigma_0}=d_sk'$ for an integer one-cochain k'. Integer exactness follows from the dual graph whose vertices are the plaquettes. A link of Σ is a dual edge directed toward the plaquette where its coefficient in d_s is positive. The dual graph is connected. Choose a spanning tree rooted at σ_0. Unit signed flow along the path from the root to a vertex σ has divergence $\delta_\sigma-\delta_{\sigma_0}$. Summing these paths with the integer coefficients of

a zero-sum two-cochain gives precisely that cochain as its divergence, hence as $d_s k'$. Two choices of k' differ by an integer element of $\ker d_s$.

Shift every slice by $r_q k'$. The magnetic integers now equal $k\mathbf{1}_{\sigma_0}$ in every slice. Summation over k' modulo closed integer cochains replaces the compact box for A_0 by a fundamental domain of Γ_s. We arrive at paths

$$A_0 \in \mathbb{R}^{\Sigma^{(1)}}/\Gamma_s, \qquad A_j \in \mathbb{R}^{\Sigma^{(1)}}\ (0<j<n_1), \qquad A_{n_1} = A_0 + g, \quad g \in \Gamma_s,$$

with $a_j \in [0, r_q)^{\Sigma}$ and flux $k \in \mathbb{Z}$. Their curvatures are

$$F_j^{\mathrm{el}} = A_{j+1} - A_j - d_s a_j, \qquad F_j^{\mathrm{mag}} = d_s A_j + r_q k \mathbf{1}_{\sigma_0}. \tag{2.lxxvi}$$

Thus integration over the original configuration quotient equals summation over k, g and integration over these paths. For a nonnegative integrand the identity follows by Tonelli's theorem. For an absolutely integrable complex integrand it follows by applying that identity to its modulus and then using Fubini's theorem. Each change above preserves link phases and curvature, so it applies to the determinant and to all curvature observables used here.

There are two finite Fock spaces in this representation: $\mathcal{F}'$ at a coordinate site plane and $\mathcal{F}$ at a link plane. The matrix of the open half-slab gives a map $N(A) : \mathcal{F}' \to \mathcal{F}$, and we include both crossing weights by writing

$$\widehat{N}(A) = \omega^{1/2} N(A) (\omega')^{1/2}.$$

Let $u(a)$ be the diagonal unitary on $\mathcal{F}$ with entries $e^{iq\langle a, \mathfrak{q}(I)\rangle}$. The fermion trace is

$$\det \mathcal{D}_A^{C_{\mathbf{n}}} = \mathrm{tr}_{\mathcal{F}'} \prod_{j=n_1-1,\dots,0} (\widehat{N}(A_{j+1})^* u(a_j) \widehat{N}(A_j)). \tag{2.lxxvii}$$

This formula also holds on an unfolded path, since $A_{n_1} - A_0 \in \Gamma_s$ leaves $\widehat{N}$ unchanged.

To derive eq. 2.lxxvii, split the torus at each site and link plane. The half-slab above a link plane is the reflected half-slab below it, hence its matrix is $N(A_{j+1})^*$. The crossing at a site plane has no link phase and gives ω'. At a link plane it gives $\omega^{1/2} u(a_j) \omega^{1/2}$. Compose the matrices and close antiperiodically; the closing parity is the identity. Cyclicity moves half of the final ω' to the front, giving eq. 2.lxxvii. Every entry of $\widehat{N}(A)$ is a polynomial in the link phases: the half-slab action is quadratic in finitely many generators, its exponential terminates, and Berezin integration extracts a coefficient. Consequently

$$C_N := \sup_{A \in \mathbb{R}^{\Sigma^{(1)}}} \|\widehat{N}(A)\| < \infty, \qquad \widehat{N}(A + r_q k') = \widehat{N}(A) \quad (k' \in \mathbb{Z}^{\Sigma^{(1)}}). \tag{2.lxxviii}$$

On real slice connections the two Hilbert spaces are

$$H^s = \bigoplus_{k \in \mathbb{Z}} L^2(\mathbb{R}^{\Sigma^{(1)}}/\Gamma_s; \mathcal{F}'), \qquad H^l = \bigoplus_{k \in \mathbb{Z}} L^2(\mathbb{R}^{\Sigma^{(1)}}/\Gamma_s; \mathcal{F}).$$

All quotient measures are induced by Lebesgue measure. Define commuting self-adjoint electric operators E_b by $e^{-i\langle x, E\rangle} \Phi(A) = \Phi(A - x)$. On H^l let

$$U_G(a)\Phi(A) = u(a)\Phi(A - d_s a), \qquad P_G = r_q^{-v} \int_{[0,r_q)^{\Sigma}} U_G(a)\, \mathrm{d}a, \qquad K = r_q^v e^{-|E|^2/2} P_G. \tag{2.lxxix}$$

The integer mode charges make U_G a representation of the compact gauge group $\mathbb{R}^{\Sigma}/r_q\mathbb{Z}^{\Sigma}$. Its average P_G is the orthogonal projection onto invariant vectors. It commutes with every function of E, because $u(a)$ acts on the Fock index and all spatial translations commute. Hence

$$K^{1/2} = r_q^{v/2} e^{-|E|^2/4} P_G. \tag{2.lxxx}$$

For a real slice two-cochain b, define the multiplication operator $M_b : H^s \to H^l$ by

$$(M_b \Phi)_k(A) = e^{-\|d_s A + r_q k \mathbf{1}_{\sigma_0} + b\|^2/4} \widehat{N}(A) \Phi_k(A), \qquad X_b = K^{1/2} M_b. \tag{2.lxxxi}$$

The magnetic weight is bounded and $\widehat{N}$ is bounded, so M_b is bounded. We must prove that X_b is Hilbert–Schmidt; the real quotient is not compact and there are infinitely many flux sectors.

The periodized Gaussian

$$\widetilde{g}_{1/2}([x]) = \sum_{\gamma\in\Gamma_s} \prod_{b\in\Sigma^{(1)}} \pi^{-1/2} e^{-(x_b+\gamma_b)^2}$$

has integral one on the quotient and is bounded. Indeed $\Gamma_s \subset r_q\mathbb{Z}^{\Sigma^{(1)}}$, so the series is dominated by the product of one-dimensional Gaussian sums on that larger lattice. It is therefore in $L^1 \cap L^\infty \subset L^2$. The kernel of $K^{1/2}$ is

$$K^{1/2}(A', A) = r_q^{-v/2} \int_{[0,r_q)^\Sigma} \widetilde{g}_{1/2}([A' - A - d_s a])u(a)\,\mathrm{d}a.$$

Its operator norm is bounded by the nonnegative function

$$\widehat{g}([x]) = r_q^{-v/2} \int_{[0,r_q)^\Sigma} \widetilde{g}_{1/2}([x - d_s a])\,\mathrm{d}a,$$

which has finite L^1 and L^∞ norms and hence finite L^2 norm. Integrating the square of the kernel of X_b over A' now gives

$$\|X_b\|_2^2 \le (\dim \mathcal{F}')C_N^2 \|\widehat{g}\|_2^2 \sum_{k\in\mathbb{Z}} \int_{\mathbb{R}^{\Sigma^{(1)}}/\Gamma_s} e^{-\|d_s A + r_q k \mathbf{1}_{\sigma_0} + b\|^2/2}\,\mathrm{d}A. \tag{2.lxxxii}$$

To bound this last sum, split the quotient as

$$(\ker d_s/\Gamma_s) \times (\ker d_s)^\perp.$$

The first factor is compact: $\ker d_s$ is the kernel of an integer matrix, hence has an integer basis, and Γ_s has full rank in it. On the second factor d_s is injective, so $\|d_s A\|^2$ is positive definite. Moreover $\ker d_s^*$ consists of constant plaquette functions. This follows from the two equations

$$(d_s^* b)(y,2) = b(y) - b(y - e_3), \qquad (d_s^* b)(y,3) = b(y - e_2) - b(y).$$

Thus $\operatorname{ran} d_s$ is the orthogonal complement of the constant two-cochain. The orthogonal constant component of $r_q k \mathbf{1}_{\sigma_0} + b$ has squared norm $v^{-1}(r_q k + \sum_\sigma b_\sigma)^2$. Translate the variable in $(\ker d_s)^\perp$ to remove the remaining component. We obtain

$$\sum_k \int e^{-\|d_s A + r_q k \mathbf{1}_{\sigma_0} + b\|^2/2}\,\mathrm{d}A = C_s \sum_{k\in\mathbb{Z}} e^{-(r_q k + \sum_\sigma b_\sigma)^2/(2v)} < \infty, \tag{2.lxxxiii}$$

where $C_s = \operatorname{vol}(\ker d_s/\Gamma_s) \int_{(\ker d_s)^\perp} e^{-\|d_s A\|^2/2}\,\mathrm{d}A < \infty$. This constant is independent of k and b. eq. 2.lxxxii–eq. 2.lxxxiii prove the Hilbert–Schmidt property for every real magnetic perturbation. The bounds here depend on the fixed cross-section; their purpose is to justify the trace argument.

For a real external two-cochain β on $C_{\mathbf{n}}$, put

$$Z_{C_{\mathbf{n}}}(\beta) = \int_{\mathrm{Conf}(C_{\mathbf{n}})} e^{-\|F+\beta\|^2/2} \det \mathcal{D}_A^{C_{\mathbf{n}}}\,\mathrm{d}(A, z). \tag{2.lxxxiv}$$

Let β_j^{el} and β_j^{mag} be its electric and magnetic slices. Then

$$Z_{C_{\mathbf{n}}}(\beta) = (2\pi)^{n_1 n_2 n_3} \operatorname{Tr}_{H^s} \prod_{j=n_1-1,\dots,0} (X_{j+1}^* U_j X_j), \quad X_j = X_{\beta_j^{\mathrm{mag}}}, \quad U_j = e^{i\langle \beta_j^{\mathrm{el}}, E\rangle}, \quad X_{n_1} = X_0. \tag{2.lxxxv}$$

Here is the conversion from the unfolded path integral. Write $g_t(x) = (2\pi t)^{-1/2} \exp(-x^2/(2t))$ for $t > 0$. The electric Maxwell factors are the kernels

$$e^{-|A_{j+1} - A_j - d_s a_j + \beta_j^{\mathrm{el}}|^2/2} = (2\pi)^v g_1^{\otimes 2v}(A_{j+1} - A_j - d_s a_j + \beta_j^{\mathrm{el}}).$$

After summing the winding vector they are periodized on Γ_s. Gaussian convolution with a translated kernel is $e^{i\langle\beta_j^{\mathrm{el}},E\rangle}e^{-|E|^2/2}$. Integration over a_j includes the matrix $u(a_j)$ from eq. 2.lxxvii; it is exactly the gauge average in eq. 2.lxxix. The magnetic Gaussian in slice j splits into the two quarter-Gaussians in M_j and M_j^*. The determinant half-slab matrices supply their other factors. Thus the cyclic path integral is the trace on H^l of

$$\prod_{j=n_1-1,\dots,0}(K^{1/2}U_jK^{1/2}M_jM_j^*).$$

Each time slice contributed $(2\pi)^v$, giving the constant in eq. 2.lxxxv. Move the final M_0^* to the front and regroup the products as $X_{j+1}^*U_jX_j$. The Hilbert–Schmidt property above and the absolute Gaussian path majorant justify this cyclicity and every interchange. In particular the external-field integral is finite; alternatively this follows directly by absorbing a linear factor into a smaller Gaussian.

We spell out the path and kernel identification used here. On a fundamental domain write every intermediate real slice as $A_j = A_j^\flat + g_j$, $g_j \in \Gamma_s$, and set $g_0 = 0$ and $g_{n_1} = A_{n_1} - A_0$. Each unperiodized electric kernel then has the argument $A_{j+1}^\flat - A_j^\flat + g_{j+1} - g_j - d_s a_j$. The map $(g_1,\dots,g_{n_1}) \mapsto (g_{j+1}-g_j)_{0\le j<n_1}$ is a bijection of $\Gamma_s^{n_1}$. Summing these differences independently is therefore precisely periodization of every cyclic kernel. The magnetic factors and $\widehat{N}$ are unchanged by the shifts g_j. The finite Fock trace is changed from $\mathcal{F}'$ to $\mathcal{F}$ by moving its last $\widehat{N}(A_0)^*$ to the front; periodicity identifies it with $\widehat{N}(A_{n_1})^*$.

Replace matrix factors by their norms and shifted Gaussians by constants times $e^{-|A'-A|^2/4}$ and $e^{-\|F^{\mathrm{mag}}\|^2/4}$. The nonnegative electric convolutions are bounded and in L^2; eq. 2.lxxxiii gives magnetic square integrability after the flux sum. The majorant operators are therefore Hilbert–Schmidt. Their cyclic kernel integral is finite by Hilbert–Schmidt Cauchy–Schwarz, grouping one factor against the product of the others. Tonelli for these majorants and Fubini for the original integrand justify all kernel, periodization and trace exchanges.

2.2.6. *Gaussian domination.* We prove Gaussian domination by showing that the shifted partition function is largest at zero shift. Completing the square will then give the exponential moment bound. In the all-flux trace, the electric shift occurs only in unitary factors. Schatten's inequality removes these factors and leaves shifts in the magnetic component. Choosing a second coordinate as time makes that remaining shift electric, so the same estimate removes it. This is the reflection method of [13] in the present chamber construction.

Theorem D. For every coordinate torus with sides at least four and every real two-cochain β,

$$0 < Z_{C_{\mathbf{n}}}(\beta) \le Z_{C_{\mathbf{n}}}(0). \tag{2.lxxxvi}$$

Consequently its normalized curvature law obeys

$$\mathbb{E}_{C_{\mathbf{n}}}e^{\langle\beta,F\rangle} \le e^{\|\beta\|^2/2}. \tag{2.lxxxvii}$$

For real two-cochains β_i and complex numbers z_i,

$$\left|\mathbb{E}_{C_{\mathbf{n}}}e^{\sum_i z_i\langle\beta_i,F\rangle}\right| \le \exp\left(\tfrac{1}{2}\left\|\sum_i(\operatorname{Re}z_i)\beta_i\right\|^2\right). \tag{2.lxxxviii}$$

The constants are independent of the volume, the charge, and the mass. ■□

Proof. The determinant is nonnegative and positive on a set of positive configuration measure, since $Z_{C_{\mathbf{n}}}(0) > 0$. The shifted photon density is strictly positive everywhere, proving the strict lower bound in eq. 2.lxxxvi.

In eq. 2.lxxxv move X_0^* through the trace to obtain a trace on H^l:

$$(2\pi)^{-n_1n_2n_3}Z_{C_{\mathbf{n}}}(\beta) = \operatorname{Tr}_{H^l}(U_{n_1-1}X_{n_1-1}X_{n_1-1}^*\cdots U_0X_0X_0^*).$$

Every U_j is unitary. Apply Schatten's inequality with all exponents equal to n_1; the trace on the left is nonnegative. Since $X_jX_j^*$ and $X_j^*X_j$ have the same nonzero eigenvalues, this gives

$$Z_{C_{\mathbf{n}}}(\beta) \le \prod_{j=0}^{n_1-1} \left((2\pi)^{n_1n_2n_3} \operatorname{Tr}(X_j^*X_j)^{n_1}\right)^{1/n_1} = \prod_{j=0}^{n_1-1} Z_{C_{\mathbf{n}}}(\beta^{(j)})^{1/n_1}. \tag{2.lxxxix}$$

Here $\beta^{(j)}$ has zero electric part and the same magnetic part β_j^{mag} in every time slice. The equality uses the trace formula again, with the same X_j in each slice and all unitary factors equal to the identity.

Interchanging directions $1, 2$ turns the 23-supported $\beta^{(j)}$ into a purely electric 13 perturbation on $C(n_2, n_1, n_3)$. Applying eq. 2.lxxxix there and using isometry covariance bounds $Z_{C_{\mathbf{n}}}(\beta^{(j)})$ by $Z_{C_{\mathbf{n}}}(0)$. Substitution proves eq. 2.lxxxvi.

Replacing β by $-\beta$ and completing the square gives eq. 2.lxxxvii. For complex parameters, take the absolute value inside the probability integral and apply that bound to $\sum_i (\operatorname{Re} z_i)\beta_i$, proving eq. 2.lxxxviii. □

If a finite-support lattice test embeds in the torus, its periodization has the same norm and the same pairing with the periodic curvature. Thus eq. 2.lxxxvii also applies to these local tests on the lifted field law.

2.2.7. *The finite-volume fold inequality.* We give the local form of the comparison that will be used to obtain diagonal positivity. Fix diagonal cells of even height width h, and let O be a real trigonometric polynomial in finitely many curvature values at plaquettes contained in the base cell Q_0. Thus O is bounded and depends only on this open cell. Let $n \ge 8$ be divisible by $4h$, and abbreviate $D_n = D(n/2, n, n)$. Its cell count, and that of the cubic torus, is

$$N_n = \frac{n^3}{|Q|} = \frac{2n^3}{h^3}.$$

The translations between cells form the integer lattice

$$\Lambda_h = \frac{h}{2}\mathbb{Z}e_- + \frac{h}{2}\mathbb{Z}e_+ + h\mathbb{Z}e_3.$$

On $C(n)$ form the translated copies, and on D_n the reflected copies:

$$\widetilde{O}_n(F) = \sum_{v \in \Lambda_h/n\mathbb{Z}^3} O(\tau_v^* F), \qquad O_n^{\mathrm{fold}}(F) = \sum_{\kappa \bmod \Gamma_{D_n}} O(g_\kappa^* F). \tag{2.xc}$$

Each sum has N_n terms. The first sum is evaluated on cubic-periodic curvatures, the second on D_n-periodic curvatures. Here g_κ is the reflected map from the base cell to Q_κ.

Let $\mu_{C(n)}$ and μ_{D_n} be the normalized curvature laws. For $t > 0$ define the tilted diagonal law

$$\mu_{D_n}^t(\Phi) = \frac{\mu_{D_n}(e^{tO_n^{\mathrm{fold}}}\Phi)}{\mu_{D_n}(e^{tO_n^{\mathrm{fold}}})}. \tag{2.xci}$$

The folded array is invariant under the group generated by the cell walls: that group permutes the cells of D_n and their reflected copies. Thus the tilt retains these symmetries.

Proposition 2.8. For the preceding data and every $t > 0$,

$$\mu_{C(n)}(O) \le \mu_{D_n}^t(O) + \frac{|Q|\Delta(n)}{t}. \tag{2.xcii}$$

The error is the fixed-parameter pressure error of eq. 2.lxxiii. ■

Proof. Jensen's inequality and translation invariance on $C(n)$ give

$$\log \frac{Z_{C(n)}[e^{t\widetilde{O}_n}]}{Z_{C(n)}[1]} \geq t\mu_{C(n)}(\widetilde{O}_n) = tN_n\mu_{C(n)}(O). \tag{2.xciii}$$

The diagonal chessboard inequality gives

$$Z_{C(n)}[e^{t\widetilde{O}_n}] \leq \|e^{tO}\|_{D_n}^{N_n} = Z_{D_n}[e^{tO_n^{\mathrm{fold}}}]. \tag{2.xciv}$$

To check that its N_n factors have the same seminorm, translate the base observable by $v \in \Lambda_h$. Translation permutes the cells and conjugates their reflected maps, so the fold of the translated observable is the translate of the original full fold. The law on D_n is translation invariant, hence its partition function and the seminorm are unchanged. All observables here are bounded scalar functions supported in distinct open cells, so the chessboard inequality applies directly.

Set $L(t) = \log \mu_{D_n}(e^{tO_n^{\mathrm{fold}}})$. Boundedness permits differentiation, giving $L'(t) = \mu_{D_n}^t(O_n^{\mathrm{fold}})$. Convexity and $L(0) = 0$ give $L(t) \leq tL'(t)$. Wall-group invariance of the tilted law makes the expectation of every reflected copy equal to that of O; hence

$$\log \frac{Z_{D_n}[e^{tO_n^{\mathrm{fold}}}]}{Z_{D_n}[1]} \leq tN_n\mu_{D_n}^t(O). \tag{2.xcv}$$

Combining eq. 2.xciii–eq. 2.xcv leaves, in addition, the normalization difference

$$\log Z_{D_n}[1] - \log Z_{C(n)}[1] \leq n^3\Delta(n).$$

Division by tN_n proves eq. 2.xcii. □

For fixed O, its cell geometry, and $t > 0$, the error in eq. 2.xcii tends to zero as $n \to \infty$ along the admissible sides. The estimate compares the cubic law with a tilted diagonal law. The next step must control these tilted laws as $t \downarrow 0$; the pressure estimate alone does not identify their local limits.

2.3. **Diagonal reflection positivity.** The finite fold inequality compares the cubic law with a tilted diagonal-adapted law. The latter is reflection positive at its cell walls, but the tilt preserves only the translations of the reflected array. To transfer positivity for arbitrary local tests, we must recover more translations as the tilt is removed.

We first realize local observables near a coordinate link plane as bounded-operator insertions. A second use of the chessboard estimate, in coordinate cells, controls these insertions in the tilted law. Fermion equations of motion and a contour shift then reduce general local polynomials to two adjacent fermion layers. These reductions give coordinate reflection invariance after taking volume to infinity and then removing the tilt. Products of coordinate reflections yield the even translations needed to position a test at any diagonal wall. The fold comparison applied to a negative reflected product finally transfers diagonal positivity to the cubic limit.

Fix $q > 0$, $0 < \rho < 1$, positive weights w_i and a real mass M. All volume limits in this subsection keep these parameters fixed. We use cubic sides $n_i \to \infty$ such that eight divides all sufficiently large sides and the set

$$\mathcal{D} = \{m \in \mathbb{N} : m \mid n_i \text{ for all sufficiently large } i\}$$

is infinite. The set $\mathcal{D}$ is closed under divisors and least common multiples. The cutoff sides $8hL^{T+N}$ have this property at fixed N, since $8L^j \in \mathcal{D}$ for each $j \geq 0$.

2.3.1. *The tilted law at its walls.* Fix diagonal cells with offsets (c_0, c_+, c_3), even width h, and $c_0, c_+ \in \mathbb{Z}$, $c_3 \in \mathbb{Z} + \frac{1}{2}$, and $4h \in \mathcal{D}$. Let O be the real trigonometric polynomial used in the finite fold comparison. Its support is contained in a box of radius r_O, and we require $h > 8r_O + 8$. Such widths can be chosen because $\mathcal{D}$ contains arbitrarily large multiples of eight. For large divisible sides, put $D_i = D(n_i/2, n_i, n_i)$ and $\mu_i^t = \mu_{D_i}^t$, as in eq. 2.xci. Write

$$O(F) = \sum_\beta a_\beta e^{i\langle \beta, F\rangle}, \qquad |O|_1 = \sum_\beta |a_\beta|.$$

The sum is over the distinct exponents in the chosen finite expansion. In particular $|O| \le |O|_1$. Let $V_i = O_{n_i}^{\mathrm{fold}}$ and $N_i = n_i^3/|Q|$. For an admissible Grassmann integrand we use the same notation for the normalized functional $\mu_i^t(G) = Z_{D_i}[e^{tV_i}G]/Z_{D_i}[e^{tV_i}]$. On bosonic integrands it agrees with integration against the tilted probability law. The normalizing denominator satisfies

$$Z_{D_i}[e^{tV_i}] \ge e^{-tN_i|O|_1} Z_{D_i}[1] > 0, \qquad t \ge 0. \tag{2.xcvi}$$

This uses only the positive bosonic measure; it makes no positivity assertion about a Grassmann integrand.

At a wall $\Pi = (\ell^a, o_a + jh)$, the adapted torus has height period p_a, with $(p_1, p_2, p_3) = (n_i, 2n_i, n_i)$. The number p_a/h is even. Split the cells into the two half-slabs adjacent to this wall, and define

$$\Phi_+(F) = \exp\left(t \sum_{Q_\kappa \subset \{o_a + jh < \ell^a < o_a + jh + p_a/2\}} O(g_\kappa^* F) \right),$$

with representatives in one half-period. Reflection in the wall maps these cells onto the other half-period modulo the normal closing vector. Because the reflected copies of O form one wall-invariant array, on periodic curvatures

$$e^{tV_i} = \Phi_+ \Theta_\Pi \Phi_+. \tag{2.xcvii}$$

The function Φ_+ is bounded, real, and supported strictly within the upper half-slab. The support lies there because each plaquette of a copy of O is contained in its open cell.

Apply eq. 2.lxv to $\Phi_+ G$ and $\Phi_+ G'$. For local even integrands G, G' in the upper half-slab,

$$\mu_i^t(\Theta_\Pi G'\, G) = \frac{\mathrm{Tr}(\mathfrak{S}[\Phi_+ G']^* \mathfrak{S}[\Phi_+ G])}{Z_{D_i}[e^{tV_i}]}. \tag{2.xcviii}$$

The numerator is the Hilbert–Schmidt inner product of the two slab operators. Thus, on D_i, the tilted functional is reflection positive at each cell wall for local even integrands supported in the corresponding open positive half-period slab. Wall reflection also gives $\mu_i^t(\Theta_\Pi G) = \overline{\mu_i^t(G)}$: the unperturbed functional is reflection covariant, the scalar weight e^{tV_i} is invariant, and its denominator is real and positive.

The bosonic law is invariant under the full group generated by the cell walls. Indeed, that group preserves D_i and permutes the reflected copies in V_i, while the unperturbed bosonic law is invariant under its lattice isometries. In particular the tilt is invariant under translations by he_-, he_+ and $2he_3$. These are the periods of the reflected cell array, rather than all lattice translations.

To compare weak limits we use the Polish space

$$\Omega = \{F \in \mathbb{R}^{\{\text{oriented plaquettes of } \mathbb{Z}^3\}} : dF = 0\},$$

with its product topology. The pullback by a lattice reflection or translation is continuous, so invariance passes to weak limits. Let P be a finite set of plaquettes contained in the open positive half-space of Π, and write $\Phi(F) = \phi((F(p))_{p\in P})$. For $\phi \in C_b^0(\mathbb{R}^P; \mathbb{C})$, positivity passes to weak limits once P lies in the positive half-period slab for all large volumes. After reflection invariance passes, the same holds for bounded Borel ϕ, by L^2 approximation with $C_c^0(\mathbb{R}^P; \mathbb{C})$ in the plaquette marginal. The reflected-form difference is at most

$$\|\Phi - \Phi_k\|_2 (\|\Phi\|_2 + \|\Phi_k\|_2),$$

because reflection preserves the L^2 norm. The Gaussian domination bound also passes to weak limits, by truncating $e^{\langle \beta, F\rangle}$ and then using monotone convergence. These facts will be applied after obtaining tightness of the tilted laws; their extensive weight alone gives no uniform moment estimate.

2.3.2. *Local operators at a coordinate link plane.* To place the same local operator insertion in tori of different cross-sections, its norm must remain unchanged when the electric variables are shifted by a curvature character. We construct a local operator algebra with this property. Fix a finite set S of links of $\mathbb{Z}^2$ and a nonempty finite set Λ of sites containing their endpoints. Let $\mathcal{F}_\Lambda$ be the Fock space of crossing modes over Λ, and put

$$H_S = L^2(\mathbb{R}^S) \otimes \mathcal{F}_\Lambda, \qquad E_b = -i\partial_{A(b)}.$$

The diagonal unitary $u(a)$ on $\mathcal{F}_\Lambda$ has integer mode charges. An operator Υ has charge $z \in \mathbb{Z}^\Lambda$ when $u(a)\Upsilon u(a)^{-1} = e^{iq\langle a,z\rangle}\Upsilon$.

Let $\mathfrak{A}(S)$ be the span of the functions $e^{-Q(\xi_{S'})+\langle \lambda, \xi_{S'}\rangle}$, where $S' \subset S$, Q is real and positive definite on $\mathbb{R}^{S'}$, and $\lambda \in \mathbb{C}^{S'}$. The empty subset gives the constant one. These functions are bounded and uniformly continuous. Their span is closed under products, conjugation and real translations: products add the positive quadratic forms on the union of their supports, while translation completes the same quadratic exponent.

For $\delta \in \mathbb{R}^S$, extended by zero to the full slice, require $d_s^*\delta = qz(\delta)$ with $z(\delta) \in \mathbb{Z}^\Lambda$. The local operator algebra consists of the finite sums

$$X = \sum_\delta e^{i\langle \delta, A\rangle}\Upsilon_\delta(E_S), \qquad \Upsilon_\delta \in \mathfrak{A}(S) \otimes \mathrm{End}_{z(\delta)}(\mathcal{F}_\Lambda). \tag{2.xcix}$$

We denote it by $\mathfrak{W}(S,\Lambda)$. The Weyl identity gives

$$\begin{aligned}(e^{i\delta A}\Upsilon(E_S))(e^{i\delta' A}\Upsilon'(E_S)) &= e^{i(\delta+\delta')A}\Upsilon(E_S+\delta')\Upsilon'(E_S),\\ (e^{i\delta A}\Upsilon(E_S))^* &= e^{-i\delta A}\Upsilon^*(E_S-\delta).\end{aligned} \tag{2.c}$$

Here δA abbreviates the real pairing. These formulas make $\mathfrak{W}(S,\Lambda)$ a unital $*$-algebra. The normal form is unique. In Fourier variables its terms send a function supported near ξ_0 to functions supported near $\xi_0 + \delta$; small enough neighborhoods separate the finitely many distinct exponents. If the sum vanishes as an operator, every continuous symbol vanishes on each such neighborhood and hence everywhere.

Every X commutes with the local gauge transformations $\Phi(A) \mapsto u(a)\Phi(A - d_s a)$. The phase change $e^{-i\langle \delta, d_s a\rangle}$ cancels the Fock charge phase. The total charge of $z(\delta)$ is zero, since it is the divergence of a finitely supported cochain. Each Fock term is therefore even. For $\beta \in \mathbb{R}^S$, define

$$\alpha_\beta X = e^{i\langle \beta, A\rangle} X e^{-i\langle \beta, A\rangle} = \sum_\delta e^{i\langle \delta, A\rangle}\Upsilon_\delta(E_S - \beta). \tag{2.ci}$$

This is an isometric $*$-automorphism on H_S.

For a rectangular cross-section Ξ containing the smallest integer rectangle spanned by Λ, the same formulas define X_Ξ on the real slice quotient space H^l of §2.2.5. The Fock operators are localized into the full crossing Fock space by ι. The localization is the tensor inclusion for the exterior space, with the reordering sign and cross-section character attached to each representative; it is a unital injective $*$-homomorphism. The phase $e^{i\langle \delta, A\rangle}$ is well defined on the quotient. Indeed, on the support hull an integer closed one-cochain is an integer coboundary, by the box Poincare argument. Thus for $\gamma \in \Gamma_s$, $\langle \delta, \gamma\rangle \in 2\pi\mathbb{Z}$, using $d_s^*\delta = qz(\delta)$. The local Weyl formulas therefore remain valid on the quotient. Uniqueness of the normal form makes the representation independent of its presentation. It commutes with P_G and with functions of the electric variables outside S.

Define the norm needed for characters by

$$\|X\|_{\#} = \sup_{\Xi,\beta} \|(\alpha_\beta X)_\Xi\|. \tag{2.cii}$$

The support hull must embed in the cross-section; the supremum is over such cross-sections and all $\beta \in \mathbb{R}^S$. The electric spectrum of a quotient can be smaller than $\mathbb{R}^S$, so its unshifted operator norm need not equal the local norm.

Proposition 2.9. For $X \in \mathfrak{W}(S,\Lambda)$,

$$\|X\|_{\#} = \|X\|_{\mathcal{L}(H_S)}.$$

The norm is unchanged by enlarging S, Λ. In particular it is a C^*-norm, $\|1\|_{\#} = 1$, and every quotient realization satisfies $\|(\alpha_\beta X)_\Xi\| \le \|X\|_{\#}$. ■

Proof. First remove the Fock localization. Its tensor extension is an injective $*$-homomorphism and hence isometric; we may work on $L^2(\mathbb{R}^{\Sigma^{(1)}}/\Gamma_s) \otimes \mathcal{F}_\Lambda$ with the original finite Fock operators. Call this realization X^0_Ξ. Introduce an additional variable $z \in \mathbb{R}^S$, with electric operators $E' = -i\partial_z$, on

$$L^2(\mathbb{R}^S_z) \otimes L^2(\mathbb{R}^{\Sigma^{(1)}}/\Gamma_s) \otimes \mathcal{F}_\Lambda.$$

The operator $X(z, E')$ has norm $\|X\|$, since it is X tensored with the identity. Choose a Borel representative $A^\flat$ of the quotient and use the two unitaries

$$(U_1\Phi)(z,[A]) = \Phi(z + A^\flat|_S, [A]), \qquad (U_2\Phi)(z,[A]) = \Phi(z, [A - z]),$$

where z is extended by zero outside S. The first conjugation multiplies $e^{i\delta z}$ by $e^{i\delta A}$. The second replaces E' by $E' + E_S$ and cancels the factor $e^{i\delta z}$. Hence

$$U_2U_1X(z,E')(U_2U_1)^{-1} = \sum_\delta e^{i\delta A}\Upsilon_\delta(E_S + E').$$

Fourier transform in z gives the direct integral of $T_p = (\alpha_{-p}X)^0_\Xi$. Uniform continuity of the symbols makes T_p norm continuous, so its essential supremum norm equals $\sup_p \|T_p\|$ (test a nearly maximizing vector on a small ball). Hence

$$\|X\| = \sup_p \|(\alpha_{-p}X)^0_\Xi\| = \sup_p \|(\alpha_{-p}X)_\Xi\|.$$

Taking the cross-section supremum proves the claim. Enlarging S or Λ tensors with an identity on the added L^2 or Fock factor, so preserves the norm. □

2.3.3. *Insertions and a coordinate trace bound.* Let $c = j + \frac{1}{2}$. Write Ins_c for the crossing insertion map, and ω for its diagonal mode weights. For an even matrix kernel $k(A', A)$ on $\mathbb{R}^S$, set

$$\begin{aligned} G[k](\widehat{A}) = {} & \mathrm{Ins}_c\Big[\omega^{1/2}k(\widehat{A}_{j+1}|_S, (\widehat{A}_j + d_s\widehat{a}_j)|_S)u(\widehat{a}_j)\omega^{1/2}\Big] \\ & \cdot \prod_{b\in S} g_1(F^{\mathrm{el}}_j(b))^{-1} \exp(S^\times_c(\Lambda)). \end{aligned} \tag{2.ciii}$$

Here $S^\times_c(\Lambda)$ is the part of the fermion action crossing the plane over Λ. The two factors on the second line remove the original electric and fermion crossing weights on the support of k. Thus $G[k]$ inserts k in their place. Its definition uses the infinite lattice and is independent of a torus.

We require

$$\begin{aligned} u(a)^{-1}k(A' + d_sa, A + d_sa)u(a) &= k(A', A), \\ \|k(A', A)\| &\le C\exp(-\epsilon|A' - A|^2), \qquad \epsilon > 0. \end{aligned} \tag{2.civ}$$

Choose a box containing the two copies of S, the temporal links over Λ, and the unit cubes centered at (j, y) and $(j+1, y)$, $y \in \Lambda$. In what follows we use the cube $B^\#(c, S, \Lambda)$ obtained by taking twice the largest half-width of the smallest rectangular box containing these sets, with the same center.

This choice is covariant under coordinate translations and reflections. The insertion depends only on these link variables and the crossing fields, so it is supported in $B^{\#}$.

The insertion is local, even and gauge invariant. Indeed, an even kernel has entries only between occupation vectors of the same parity, so every monomial $\Theta_c\zeta_J\,\zeta_I$ in its insertion is even. If the gauge parameter is λ, its upper and lower slice values change the matrix in eq. 2.ciii to

$$u(\lambda_{j+1})^{-1}k(\widehat{A}_{j+1},\widehat{A}_j+d_s\widehat{a}_j)u(\widehat{a}_j)u(\lambda_j).$$

The matrix entry acquires precisely the inverse of the phase of $\Theta_c\zeta_J\,\zeta_I$. The electric curvature and the crossing action are gauge invariant. Finally, the kernel estimate and $g_1(x)^{-1}=\sqrt{2\pi}e^{x^2/2}$ bound every Grassmann coefficient by

$$C'\exp\left((\tfrac12-\epsilon')\sum_{b\in S}F_j^{\mathrm{el}}(b)^2\right),\qquad \epsilon'=\min(\epsilon,\tfrac12). \tag{2.cv}$$

The other factors have bounded coefficients. This is the local integrand bound of §2.2.1, since its electric plaquettes lie in the box.

For $X\in\mathfrak{W}(S,\Lambda)$ let $k_X^{(s)}$ be the continuous kernel of $e^{-s|E_S|^2}Xe^{-s|E_S|^2}$, with $s=\frac14$ or $\frac18$, and write

$$G_X=G[k_X^{(1/4)}],\qquad G'_X=G[k_X^{(1/8)}]. \tag{2.cvi}$$

Both kernels satisfy eq. 2.civ. To see their decay, consider one normal-form term $e^{i\delta A}\Upsilon(E)$ with $\Upsilon(\xi)=e^{-Q(\xi)+\lambda\xi}\Upsilon_0$. Identify Q with its symmetric matrix, so $Q(\xi)=\langle\xi,Q\xi\rangle$, extending both Q and λ by zero to the other links. The brackets in the following exponent are the complex-bilinear extension of the real Euclidean pairing. With $Q_s=4sI+2Q$ and $\lambda_s=\lambda-2s\delta$, its kernel is

$$\begin{aligned}k_X^{(s)}(A',A)&=e^{i\delta A'}(2\pi)^{-|S|/2}(\det Q_s)^{-1/2}e^{-s|\delta|^2}\\&\quad\cdot\exp\left(\tfrac12\langle\lambda_s+i(A'-A),Q_s^{-1}(\lambda_s+i(A'-A))\rangle\right)\Upsilon_0.\end{aligned} \tag{2.cvii}$$

The real part of this exponent has a strictly negative quadratic form in $A'-A$, with a fixed linear term. It gives the required Gaussian decay after decreasing its exponent. Gauge covariance follows by conjugating the regularized operator by the gauge unitary; equality of their continuous kernels gives the first identity in eq. 2.civ. Finite sums preserve these properties. The identity operator has kernel $\prod_{b\in S}g_1(A'(b)-A(b))$ when $s=\frac14$, so $G_1=1$; unused links and modes cancel in the same way.

Reflection acts on these insertions by adjoints:

$$\Theta_cG_X=G_{X^*},\qquad \Theta_cG'_X=G'_{X^*}. \tag{2.cviii}$$

The kernel is changed to $k(A,A')^*$. At the same time the two slices exchange, $\widehat{a}_j$ changes sign, and covariance moves $u(\widehat{a}_j)$ to the required side of the kernel. This gives eq. 2.cviii, including its field signs. We also need coordinate translations and spatial reflections to preserve the operator norm, since each coordinate fold contains such images.

For a spatial reflection θ in the second or third coordinate, the spin matrix exchanges the two eigenspaces of γ_1. Its unitary matrix v on crossing modes preserves their weights and reverses their charges. On exterior products the reordering sign is $r_I=(-1)^{|I|(|I|-1)/2}$; hence the Fock unitary is $U=(\bigwedge v)r$. The anticommutation of the two field reflection maps contributes the sign $(-1)^{\deg}$, and moving the two mode products past one another contributes $(-1)^{|I||J|}$. For the parity-matched entries of an even kernel these signs leave exactly r_Ir_J. Consequently

$$\Theta_\theta\,\mathrm{Ins}_c[\Upsilon(\widehat{A})]=\mathrm{Ins}_c[U\overline{\Upsilon(\theta^*\widehat{A})}U^*].$$

The antiunitary $(J\Phi)(A)=U\overline{\Phi(\theta_s^*A)}$ gives this kernel transformation. It sends an oriented electric operator to the negative of the reflected electric operator, and sends $e^{i\delta A}$ to $e^{-i\langle\theta_s^*\delta,A\rangle}$. Thus it preserves the charge constraint, the local algebra and the heat factors $e^{-s|E|^2}$. Translations act by

the corresponding unitary relabeling. Together with adjoints at the link plane, these facts show that each translate or signed coordinate reflection of G_X is an insertion G_{X^g} with $\|X^g\| = \|X\|$. The same holds for G'_X. The box $B^{\#}$ transforms with the insertion.

An even local integrand G in a box B satisfies the coordinate bound $C \geq 0$ if, on every rectangular coordinate torus $T = (\Gamma, \chi) = C(n_1, n_2, n_3)$ with $n_\mu \geq 4$, for every nonempty finite family of images $g_a B$ under lattice translations and signed coordinate reflections that embeds disjointly in Γ, and every $\beta \in C^2(\Gamma)$,

$$\left| Z_T \left[\varphi_\beta \prod_{a=1}^{r} G^{g_a} \right] \right| \leq C^r Z_T[1], \qquad \varphi_\beta(F) = e^{i\langle \beta, F \rangle}, \tag{2.cix}$$

provided that, in each coordinate direction μ, there is a plaquette σ orthogonal to e_μ such that none of the boxes or their period translates contains any plaquette $\sigma + k e_\mu$, $k \in \mathbb{Z}$. This condition places the flux representative used in the slice trace on plaquettes outside the insertion boxes and their period translates. When the coordinate cell structure cuts T and its walls are at half-integers, boxes contained in its open cells satisfy the free-column condition. Choose a plaquette orthogonal to the chosen direction that meets a wall normal to another direction. Period translations permute the open cells, while every plaquette in this column meets that wall. Thus none of these plaquettes is contained in a box or any of its period translates.

Proposition 2.10. The insertion G_X satisfies eq. 2.cix with $C = \|X\|$. The insertion G'_X satisfies it with $C = 2^{|S|/2}\|X\|$. A bounded bosonic integrand satisfies it with $C = \|G\|_\infty$. These bounds survive box enlargement, multiplication by a curvature character, and transport by a signed permutation of the coordinates. ■

Proof. We give the trace calculation, including the support condition. Choose the time direction to be the first coordinate. By the covariance just proved, the images of the insertions have the same operator norms and are at link planes $j + \frac{1}{2}$. The free column lets us choose the magnetic flux plaquette σ_0 so that no plaquette in its column is contained in an insertion box. At a fixed step j their spatial supports and modes are disjoint. Let S_j be the union of the supports of the primed insertions at that step.

The unfolded path variables are genuine local lifts on each $[j, j+1] \times \mathrm{hull}(\Lambda_a)$. On electric plaquettes their coboundary is $A_{j+1} - A_j - d_s a_j$. On magnetic plaquettes it is $d_s A_j$; the flux correction is supported at σ_0, which is not a plaquette of the local hull. Two local lifts with these curvatures and phases differ by an integer exact cochain $\frac{2\pi}{q} d\lambda$. Gauge invariance therefore allows the insertion to be evaluated directly at the path variables. This is why the free-column hypothesis is needed.

At each crossing, the factors $e^{S_c^{\times}(\Lambda_a)}$ remove the crossing weights on the local modes. The remaining crossing factors, together with their insertions, form

$$\mathrm{Ins}_c \left[\omega^{1/2} \left(\prod_a \iota(k_a(A_{j+1}, A_j + d_s a_j)) \right) u(a_j) \omega^{1/2} \right].$$

The even localized matrices commute because their mode sets are disjoint. The factors g_1^{-1} similarly remove the electric Gaussian densities on the local links. The slice determinant factorization of §2.2.5 now gives the path product with these modified crossing kernels. Its Gaussian majorant is integrable: every modified kernel has strict decay in its electric difference, all remaining links retain g_1, and the magnetic multiplier has its Gaussian weight. Hence unfolding, periodization and the path trace are valid for this product.

Separate the curvature character into electric factors $e^{i\langle \beta_j, F_j^{\mathrm{el}} \rangle}$ and magnetic unitaries u_j. Multiplying a regularized kernel by $e^{i\langle \beta_j, A' - A \rangle}$ changes the corresponding operator to

$$e^{-s|E-\beta_j|^2} (\alpha_{\beta_j} X) e^{-s|E-\beta_j|^2}.$$

Fourier transformation of the disjoint link kernels therefore gives

$$Z_T\left[\varphi_\beta \prod_a G_a\right] = (2\pi)^{|T|}\,\mathrm{Tr}_{H^l} \prod_{j=n_1-1,\dots,0} \left(K^{1/2}_{\beta_j,S_j} B_j K^{1/2}_{\beta_j,S_j} M_0 u_j M_0^*\right), \tag{2.cx}$$

where

$$B_j = \prod_{a:j_a=j} (\alpha_{\beta_j} X_a)_\Xi, \qquad K_{\beta,S} = r_q^v e^{-|E-\beta|^2/2+|E-\beta|_S^2/4} P_G.$$

Here $r_q^v = (2\pi/q)^{|\Sigma|}$ is the gauge-volume factor in eq. 2.lxxix; the primed kernels change the electric Gaussian variance only on S. Both the kernels and the local operators commute with P_G. In particular $\|B_j\| \le \prod_{a:j_a=j} \|X_a\|$ by the norm identity above.

Set $T_j = K^{1/2}_{\beta_j,S_j} M_0$. These are Hilbert–Schmidt by the same quotient Gaussian calculation as eq. 2.lxxxii. Move the first square root cyclically to the end of the trace. The product then has alternating factors $B_j T_j u_j T_{j-1}^*$. Schatten Hölder, with $2n_1$ equal exponents, bounds its absolute trace by

$$\prod_j \|B_j\| \prod_j \|T_j\|_{2n_1}^2.$$

The slice trace applied to $(T_j^* T_j)^{n_1}$ gives exactly

$$\mathrm{Tr}(T_j^* T_j)^{n_1} = (2\pi)^{-|T|} Z_T \left[\prod_{l=0}^{n_1-1} e^{i\langle \beta_j, F_l^{\mathrm{el}}\rangle} \prod_{l=0}^{n_1-1} \prod_{b\in S_j} \sqrt{2} e^{-F_l^{\mathrm{el}}(b)^2/2}\right]. \tag{2.cxi}$$

The second product changes g_1 into $g_{1/2}$ on those links. The positive bosonic law bounds the absolute value by $2^{|S_j|n_1/2}(2\pi)^{-|T|} Z_T[1]$. Consequently

$$\prod_j \|T_j\|_{2n_1}^2 \le 2^{\frac{1}{2}\sum_j |S_j|} (2\pi)^{-|T|} Z_T[1].$$

Together with eq. 2.cx this proves the two operator bounds. For insertions in another time direction use an orientation-preserving coordinate permutation. The scalar bound follows directly from the positive law. Multiplication by a character merely changes β, and enlargement restricts the allowed placements. A signed permutation transports the torus, the free columns and the placements; conjugation of a character still has modulus one. These observations prove the remaining assertions. □

2.3.4. *Amplification in two cell structures.* A diagonal fold exchanges the first and second time directions. Its copies of a coordinate insertion therefore cannot all be put in one coordinate trace. We will place these copies in different coordinate cells and apply a second chessboard estimate. Each resulting coordinate fold contains copies of only one insertion, all with the same time direction, so eq. 2.cix applies.

The placement requires two scales depending on the support radius r, but not its center or the tilt. Choose $h'' \in h\mathbb{Z}$ and $k \in \mathbb{N}$ such that

$$h'' \ge 2r + 6h, \qquad kh \ge 16h'', \qquad kh \in h''\mathbb{Z}, \qquad 4h'', 2kh \in \mathcal{D}. \tag{2.cxii}$$

These choices exist: take $4h''$ to be the least common multiple of $4h$ and an element of $\mathcal{D}$ exceeding $4(2r+6h)$, then take $2kh$ to be the least common multiple of $4h''$ and an element exceeding $32h''$. Fix choices depending only on h, r and the side sequence. For all sufficiently large i, the needed divisibility conditions then hold simultaneously.

Here is the geometry of the placement. Let $B = y + [-r,r]^3$. Choose coarse diagonal offsets $c'_\pm \in c_\pm + h\mathbb{Z}$ (with $c_- = c_0$) and $c'_3 \in c_3 + h\mathbb{Z}$ so that

$$\ell_-(y) - c'_- \in [2h'', 2h''+h), \quad \ell_+(y) - c'_+ \in [4h'', 4h''+h), \quad y_3 - c'_3 \in [h'', h''+h). \tag{2.cxiii}$$

The box lies strictly in the coarse cell Q_0' of width kh. Every element of its wall group has the form

$$g = \tau_{kh(z_-e_-+z_+e_++2z_3e_3)}\theta_-^{a_-}\theta_+^{a_+}\theta_3^{a_3}, \qquad z \in \mathbb{Z}^3, \quad a \in \{0,1\}^3,$$

where the reflections are at the lower walls of Q_0'. From eq. 2.cxiii one obtains

$$gy = y + h''\kappa(g) + s(a), \qquad |s(a)|_\infty < 2h, \tag{2.cxiv}$$

with

$$\kappa(g) = (-2a_- - 4a_+,\, 2a_- - 4a_+,\, -2a_3) + \frac{kh}{h''}(z_- + z_+,\, z_+ - z_-,\, 2z_3).$$

The eight first vectors are distinct, and their pairwise differences have size at most six. Since $kh/h'' \geq 16$, equality of two indices first forces equality of their a's and then of their z's. Thus $g \mapsto \kappa(g)$ is injective.

Choose half-integer coordinate offsets o_μ in

$$I_\mu = (y_\mu + r + 2h - h'',\; y_\mu - r - 2h).$$

These intervals have length at least $2h$. eq. 2.cxiv places every gB strictly inside coordinate cell $\kappa(g)$ of width h''. We can also choose the offsets so that all these coordinate walls avoid every reflected copy of $\operatorname{supp} O$. Under the original diagonal wall group, the centers of the copies of a box containing O have at most four residue classes modulo h in either of the first two coordinates, and at most two in the third. Its radius r_O therefore forbids at most $4(2r_O+1) = 8r_O + 4$ half-integer classes. Since $h > 8r_O + 8$, an allowed class exists. Each I_μ contains a representative, and $h'' \in h\mathbb{Z}$ keeps every wall in the allowed class.

For large i, the adapted torus D_i carries the original and coarse diagonal structures; its period lattice lies in the translation subgroup of the coarse wall group. The coordinate structure of width h'' cuts D_i and is carried by $T_i = C(n_i/2, 2n_i, n_i)$. Formula eq. 2.cxiv also shows that coarse diagonal cells distinct modulo the periods have their boxes in coordinate cells distinct modulo those periods. Indeed a period translation changes κ by that translation divided by h''; injectivity then gives the converse. Thus the counts are

$$N_i' = \frac{n_i^3}{|Q|k^3}, \qquad N_i'' = \frac{n_i^3}{(h'')^3}. \tag{2.cxv}$$

The first is the number of coarse diagonal cells in D_i; the second is the number of coordinate cells in either torus. The shortest periods are at least $n_i/2 \geq kh \geq 16h''$, so the boxes, with the margins used for local lifts, embed. Folding one such box in the coordinate structure gives a disjoint placement with free columns, since every image stays inside its open cell.

Theorem E. Put $\Delta_i = \Delta(n_i)$, with Δ defined in eq. 2.lxxiii. It tends to zero as $i \to \infty$, with the charge, mass and weights fixed. If G satisfies the coordinate bound C in a box of radius r, then, for k as above and all sufficiently large i,

$$|\mu_i^t(G)| \leq C\gamma_{t,i}(k), \qquad \gamma_{t,i}(k) = \exp(2t|O|_1k^3 + |Q|k^3\Delta_i), \qquad t \geq 0. \tag{2.cxvi}$$

The threshold and k depend only on the side sequence, h and r. In particular they are uniform in the location of G. ■□

Proof. Each coarse diagonal cell contains k^3 original cells. Put $P = \exp(t\sum_{Q_\kappa \subset Q_0'} O^{[\kappa]})$. The tilt e^{tV_i} is exactly the product of the reflected copies of P over the coarse cells: their walls are original walls, so each reflection carries the original array to the same array. Apply the adapted chessboard estimate to the integrands PG in one coarse cell and P in the others. The fold of P is e^{tV_i}. After dividing its powers out, this gives

$$|\mu_i^t(G)|^{N_i'} \leq \frac{Z_{D_i}[e^{tV_i}\prod_{a=1}^{N_i'} G^{g_a}]}{Z_{D_i}[e^{tV_i}]}. \tag{2.cxvii}$$

The right side is nonnegative by the reflected fold form. This step uses reflection positivity of even integrands and does not require G itself to be a positive scalar function.

Expand e^{tV_i} in curvature characters. The coefficient sum is at most $\exp(tN_i|O|_1)$: the character algebra is closed under multiplication, and the exponential series is absolutely summable in its coefficient norm. Every plaquette in a character of this expansion lies strictly within a coordinate cell, because the chosen walls avoid the support of every copy of O. For each character term, collect its factors by coordinate cells. Each cell then contains a character alone or a character times one G^{g_a}; precisely N_i' cells have the latter form. Period translations can be used to place the integrands in a chosen coordinate brick without changing their value on D_i.

Use the chessboard estimate from the adapted to the coordinate frame, eq. 2.lxx. A cell with a character alone has fold bounded by $Z_{T_i}[1]$. A cell with a copy of G has a fold consisting of a character and its coordinate images; the geometry above supplies the free columns and disjoint embedding, so its fold is bounded by $C^{N_i''}Z_{T_i}[1]$. Taking the chessboard roots and multiplying the cell contributions gives, for every character term,

$$\left| Z_{D_i}\left[\varphi_\beta \prod_{a=1}^{N_i'} G^{g_a}\right]\right| \le C^{N_i'} Z_{T_i}[1].$$

The coefficient sum therefore bounds the numerator of eq. 2.cxvii by $e^{tN_i|O|_1}C^{N_i'}Z_{T_i}[1]$. The denominator bound eq. 2.xcvi contributes a second $e^{tN_i|O|_1}$. Finally, the pressure comparison gives $Z_{T_i}[1]/Z_{D_i}[1] \le e^{n_i^3\Delta_i}$. The root of order N_i' gives eq. 2.cxvi. □

We also need exponential moments, for which a coordinate operator bound is unnecessary. Repeat eq. 2.cxvii with $G = e^{\langle\beta,F\rangle}$ supported in B. It is an admissible local integrand, since $\langle\beta,F\rangle \le \|\beta\|^2 + \frac{1}{4}\sum_{\operatorname{supp}\beta} F_\sigma^2$. Its diagonal copies have disjoint supports. Remove the tilt from the numerator by its pointwise bound, and apply the second chessboard estimate to their coordinate cell integrands. The fold of one exponential has N_i'' disjoint copies of β, and hence squared norm $N_i''\|\beta\|^2$. Gaussian domination on T_i bounds its chessboard root by $e^{\|\beta\|^2/2}Z_{T_i}[1]^{1/N_i''}$. There are N_i' nonempty cells. The same denominator and pressure bounds therefore yield

$$\mu_i^t(e^{\langle\beta,F\rangle}) \le e^{\|\beta\|^2/2}\gamma_{t,i}(k). \tag{2.cxviii}$$

For each fixed t this proves tightness on Ω. Indeed, Markov's inequality applied to $\beta = \pm s1_\sigma$ gives a uniform coordinate tail $2e^{-sR+s^2/2}\gamma_{t,i}(k(h,1))$. Choose radii R_σ with these tails summable over plaquettes; the intersection of Ω with their product of compact intervals is compact. The finitely many initial laws are tight individually. Any weak limit ν_t consequently satisfies

$$\nu_t(e^{\langle\beta,F\rangle}) \le e^{\|\beta\|^2/2+2t|O|_1k^3}. \tag{2.cxix}$$

This uses lower semicontinuity of the exponential integral, as explained after eq. 2.xcviii.

2.3.5. *The two adjacent fermion layers.* The remaining comparison argument uses gauge invariant polynomials

$$P(\widehat{A}) = \sum_a b_a e^{i\langle\delta_a,\widehat{A}\rangle}\Psi_a, \qquad d^*\delta_a = q\mathfrak{q}(\Psi_a),$$

where the cochains have finite support and the Ψ_a are ordered Grassmann monomials. Denote their class by $\mathcal{P}$. The charge condition is exactly gauge invariance. It also makes each term even: the total divergence is zero, so its numbers of ψ and $\bar\psi$ factors agree. The expansion is unique after equal pairs (δ_a,Ψ_a) have been collected, because the monomials and the distinct characters are linearly independent. Each such polynomial is an admissible local integrand with bounded coefficients.

At a coordinate link plane $c = j + \frac{1}{2}$, consider polynomials depending on photon links in the planes j and $j+1$ and between them, and on fermion fields only at the two ends of bonds crossing c. In the geometric coordinates of eq. 2.x, these endpoint sets have first-coordinate heights $j + v_1$

and $j+1-v_1$. Denote them by $\mathcal{X}_{j,6}$ and $\mathcal{X}_{j+1,1}$, respectively, and call this polynomial class $\mathcal{P}_1(c)$. For a monomial $Y = e^{i\delta\widehat{A}}\Psi$, write its exponent as $(\delta_0, \delta_1, \delta_\perp)$ on the lower spatial, upper spatial and temporal links. If $\mathfrak{q}_\pm$ are the lower and upper charge densities, gauge invariance gives

$$d_s^*\delta_0 - \delta_\perp = q\mathfrak{q}_-, \qquad d_s^*\delta_1 + \delta_\perp = q\mathfrak{q}_+. \tag{2.cxx}$$

Let $S = \operatorname{supp}\delta_0 \cup \operatorname{supp}\delta_1$ and choose a nonempty Λ containing their endpoints, the support of $\delta_\perp$ and all sites over which the fields of Ψ occur.

Every such Y is a kernel insertion. To apply the operator norm bound, however, we need bounded middle operators between the heat factors. The quarter-heat representation may fail this requirement. We shall first use eighth-heat factors to obtain a bounded operator, then approximate its kernel by quarter-heat insertions with a controlled error. The support remains fixed throughout, so the amplification scales are independent of the approximation parameter. The crossing modes and their reflected modes form a basis of the two-layer generators. Since all their weights are positive when $0 < \rho < 1$, there is a unique finite matrix Υ_Y with

$$\Psi e^{-S_c^\times(\Lambda)}\Big|_{\widehat{a}_j=0} = \mathrm{Ins}_c[\omega^{1/2}\Upsilon_Y\omega^{1/2}]. \tag{2.cxxi}$$

To obtain it, expand the left side in the basis $\Theta_c\zeta_J\zeta_I$ and multiply the resulting matrix by $\omega^{-1/2}$ on both sides. Its nonzero entries have $\mathfrak{q}(I) - \mathfrak{q}(J) = \mathfrak{q}_- + \mathfrak{q}_+$. Thus Υ_Y has charge $q^{-1}d_s^*(\delta_0+\delta_1)$ and is even.

Set

$$k_Y^0(A', A) = e^{i\delta_1 A'}\prod_{b\in S} g_1(A'(b) - A(b))e^{i\delta_0 A}\Upsilon_Y. \tag{2.cxxii}$$

It is the continuous kernel of $e^{i(\delta_0+\delta_1)A}e^{-|E_S+\delta_0|^2/2}\Upsilon_Y$. We also write k_Y^0 for this integral operator. The charge identity and unchanged Gaussian difference prove its covariance. Moreover

$$Y = G[k_Y^0]. \tag{2.cxxiii}$$

Indeed, the Gaussian factors in the insertion cancel. Applying the automorphism that multiplies the lower fields by their charge phases to eq. 2.cxxi replaces its matrix by $\omega^{1/2}\Upsilon_Y u(\widehat{a}_j)\omega^{1/2}$ and its left side by $e^{-iq\langle\mathfrak{q}_-,\widehat{a}_j\rangle}\Psi e^{-S_c^\times(\Lambda)}$. The remaining temporal exponent in $G[k_Y^0]$ is $d_s^*\delta_0 - q\mathfrak{q}_- = \delta_\perp$ by eq. 2.cxx, proving the identity.

Write $E = E_S$, $\delta = \delta_0 + \delta_1$ and

$$U = e^{i\delta A}, \qquad a(E) = |E+\delta|^2 + |E|^2, \qquad b(E) = \tfrac{1}{2}|E+\delta_0|^2.$$

For $\epsilon > 0$ define

$$X'_Y = Ue^{a(E)/8 - b(E)}\Upsilon_Y, \tag{2.cxxiv}$$

$$X_{Y,\epsilon} = Ue^{(1/4-\epsilon)a(E) - b(E)}\Upsilon_Y, \tag{2.cxxv}$$

Their exponents have quadratic parts $-|E|^2/4$ and $-2\epsilon|E|^2$, respectively, so their symbols lie in $\mathfrak{A}(S)\otimes\mathrm{End}(\mathcal{F}_\Lambda)$. The charge of $U\Upsilon_Y$ is fixed by eq. 2.cxx, so both belong to $\mathfrak{W}(S,\Lambda)$. Put $H_\alpha = e^{-\alpha|E|^2}$ and $P_\epsilon = H_\epsilon$. The Weyl rule gives $H_\alpha U f(E) H_\alpha = Ue^{-\alpha a(E)}f(E)$. Using $k_Y^0 = Ue^{-b(E)}\Upsilon_Y$ therefore proves

$$\begin{aligned} H_{1/8}X'_Y H_{1/8} &= k_Y^0, \qquad Y = G'_{X'_Y}, \\ H_{1/4}X_{Y,\epsilon}H_{1/4} &= P_\epsilon k_Y^0 P_\epsilon, \\ Y - G_{X_{Y,\epsilon}} &= G'_{X'_Y - P_\epsilon X'_Y P_\epsilon}. \end{aligned} \tag{2.cxxvi}$$

The last identities use the equal continuous insertion kernels. The primed coordinate bound makes the error at most $2^{|S|/2}\xi_Y(\epsilon)$, where

$$\begin{aligned}\xi_Y(\epsilon) &= \|X'_Y - P_\epsilon X'_Y P_\epsilon\| \\ &\le \epsilon\|\Upsilon_Y\| \sup_E e^{a(E)/8-b(E)}a(E) = C_Y\epsilon.\end{aligned} \tag{2.cxxvii}$$

Indeed the difference has symbol $Ue^{a(E)/8-b(E)}(1-e^{-\epsilon a(E)})\Upsilon_Y$; use $a(E)\ge 0$, $1-e^{-s}\le s$, and the Gaussian quadratic part. All supports are independent of ϵ; only the finite norm of $X_{Y,\epsilon}$ may grow as $\epsilon\downarrow 0$.

Reflection changes $(\delta_0,\delta_1,\delta_\perp)$ to $(-\delta_1,-\delta_0,\delta_\perp)$ and Υ_Y to Υ_Y^*. The latter follows by reflecting eq. 2.cxxi; $\mathrm{Ins}_c[\omega]$ is even and invariant. Taking adjoints in the two heat identities then gives $X'_{\Theta_c Y} = (X'_Y)^*$ and $X_{\Theta_c Y,\epsilon} = X^*_{Y,\epsilon}$; the heat operator is injective with dense range, so these identities determine the bounded middle operators uniquely. All constructions extend linearly to two-layer polynomials. Coordinate permutations give the same assertions at every coordinate link plane.

Proposition 2.11. For every $Y\in\mathcal{P}_1(c)$ at a coordinate link plane,

$$\lim_{t\downarrow 0}\limsup_{i\to\infty}|\mu_i^t(\Theta_c Y) - \overline{\mu_i^t(Y)}| = 0. \tag{2.cxxviii}$$

The assertion concerns the finite-volume functionals on fermion polynomials; it does not presume that they converge to fermion laws. ■

Proof. For a fixed support, $o_i^t(X) = \mu_i^t(G_X)$ is a unital linear functional on the local operator algebra with $|o_i^t(X)|\le\gamma\|X\|$, where $\gamma=\gamma_{t,i}(k)$. A unital functional this close in norm to one is nearly Hermitian. If $X_0 = X_0^*$ and $\|X_0\|\le 1$, write $o_i^t(X_0) = u+iv$. For each real s,

$$(v+s)^2 \le |o_i^t(X_0+is)|^2 \le \gamma^2\|X_0+is\|^2 \le \gamma^2(1+s^2).$$

For $\gamma>1$ take $s = v/(\gamma^2-1)$; this gives $|v|\le\sqrt{\gamma^2-1}$. For $\gamma=1$ the same conclusion follows by letting s tend to either infinity. Decomposing a general X into its real and imaginary self-adjoint parts yields

$$|o_i^t(X^*) - \overline{o_i^t(X)}| \le 4\sqrt{\gamma^2-1}\|X\|. \tag{2.cxxix}$$

Use eq. 2.cviii to interpret this as reflection covariance of the insertion expectation.

Apply eq. 2.cxxix to $X_{Y,\epsilon}$ and use eq. 2.cxxvi for Y and its reflection. The amplification theorem controls the two errors and gives

$$\begin{aligned}|\mu_i^t(\Theta_c Y) - \overline{\mu_i^t(Y)}| &\le 4\sqrt{\gamma^2-1}\|X_{Y,\epsilon}\| \\ &\quad + 2\cdot 2^{|S|/2}\gamma\xi_Y(\epsilon).\end{aligned} \tag{2.cxxx}$$

The support box and its coarse scales do not depend on ϵ. First take $i\to\infty$, and then $t\downarrow 0$ with ϵ fixed. Since $\Delta_i\to 0$, the first term vanishes and the remaining upper bound is $2\cdot 2^{|S|/2}\xi_Y(\epsilon)$. Let $\epsilon\downarrow 0$ using eq. 2.cxxvii. Finite sums and coordinate transport give the stated result. □

2.3.6. *Removing fields from a thin layer.* The fermion graph has six thin layers in each thick layer j, at heights $j-v_1, j-v_2, j-v_3, j+v_3, j+v_2, j+v_1$. Each site has one neighbor in the next thin layer, one in the preceding one and two in its own layer. The forward matching bonds have types $2,1,0,1,2,3$ in succession. This follows directly from the chamber reflections: type zero reverses the coordinate of modulus v_3, types one and two interchange the adjacent moduli, and type three reverses the coordinate of modulus v_1 and shifts its nearest photon site by the corresponding unit vector. The bonds of types zero, one and two carry no photon phase. Type three bonds within thin layers $2,\dots,5$ carry spatial phases in plane j, while those between layer six and the next layer one carry temporal phases between j and $j+1$. These facts control the spatial support of the recursion.

For a photon site y, let $\mathcal{P}^{\pm}(y)$ consist of the same exponential monomials as $\mathcal{P}$, with $d^*\delta = q(\mathfrak{q}(\Psi) \mp 1_y)$. They transform with gauge phase $e^{\pm iq\lambda(y)}$ and are odd. Removing a $\bar\psi(x)$ from an invariant monomial leaves an element of $\mathcal{P}^+(\pi(x))$; removing a $\psi(x)$ leaves one in $\mathcal{P}^-(\pi(x))$. Multiplication by a hopping phase transports the specified charge to the other endpoint. These assertions are the identities

$$d^*1_{(y,y')} = 1_{y'} - 1_y, \qquad U_A(x,x') = e^{iq\langle 1_{(\pi(x),\pi(x'))},A\rangle}.$$

They show that every term of the following replacements stays in $\mathcal{P}$.

Berezin integration by parts, with left derivatives, gives

$$\begin{aligned} Z_T[\Phi(\bar\psi\mathcal{D}_A)_\alpha(x')G] &= -Z_T[\Phi\partial_{\psi_\alpha(x')}G], \\ Z_T[\Phi(\mathcal{D}_A\psi)_\alpha(x')G] &= Z_T[\Phi\partial_{\bar\psi_\alpha(x')}G]. \end{aligned} \tag{2.cxxxi}$$

The appropriate charge of G makes each integrand invariant. These identities first hold at fixed complex A: the derivatives of $e^{-S_T(A)}$ are respectively $(\bar\psi\mathcal{D}_A)_\alpha e^{-S_T(A)}$ and $-(\mathcal{D}_A\psi)_\alpha e^{-S_T(A)}$, and the Berezin integral annihilates a derivative. Integrating against any bounded bosonic factor $\Phi(F)$ then proves eq. 2.cxxxi. The enlarged local box must embed so that all generators involved are distinct torus generators and the phases can be evaluated on the local lift.

Let x' be the neighbor of x in the next thin layer. Its matching hopping matrix $w(x,x')$ has eigenvalues $\frac{w_i}{2}(\rho \mp 1)$ and is invertible for our fixed $0 < \rho < 1$. Solve the equation at x' for the field at x. With the replaced field moved to the left of the monomial, the two rules are

$$\begin{aligned} \bar\psi(x) &\longmapsto \left(M\bar\psi(x') - \sum_{x''\sim x',\, x''\neq x} \bar\psi(x'')w(x'',x')U_A(x'',x') + \partial_{\psi(x')} \right) w(x,x')^{-1}U_A(x',x), \\ \psi(x) &\longmapsto U_A(x,x')w(x',x)^{-1} \left(M\psi(x') - \sum_{x''\sim x',\, x''\neq x} w(x',x'')U_A(x',x'')\psi(x'') - \partial_{\bar\psi(x')} \right). \end{aligned} \tag{2.cxxxii}$$

The derivatives act on the monomial to their right. Multiplying by the inverse matching matrix and its inverse phase in eq. 2.cxxxi proves equality of the expectations before and after either replacement, for every bosonic factor Φ. This equality is independent of a choice of time direction or a symmetry of the torus.

For a thin layer $\mathbb{L}$ define a linear map $R^+_{\mathbb{L}}$ by recursion on the number of its fields in an ordered monomial. Select the first such field, move it to the front with its Grassmann sign, and apply eq. 2.cxxxii through its matching neighbor in $\mathbb{L}^+$. That neighbor has only one neighbor in $\mathbb{L}$, the field just removed. All other fields created by the rule lie in $\mathbb{L}^+$ or $\mathbb{L}^{++}$; its derivatives remove fields in $\mathbb{L}^+$. Thus the number in $\mathbb{L}$ falls by exactly one at every recursive call. The map is well-defined, preserves $\mathcal{P}$, removes all fields in $\mathbb{L}$, and creates no new fields outside its next two layers. A downward map $R^-_{\mathbb{L}}$ is obtained using the preceding matching neighbor.

If the initial polynomial is a local integrand in a box B, every polynomial produced by the one-layer recursion is supported in $B + [-1,1]^3$. The fields being removed stay in the original box, and their next neighbors and the other neighbors of those sites have photon projections equal or nearest to the original projection: among two successive different bonds at most one has type three. Hence these sites and every added phase are in the same enlarged box, irrespective of the number of recursive calls. Induction therefore gives

$$Z_T[\Phi P] = Z_T[\Phi R^{\pm}_{\mathbb{L}} P] \tag{2.cxxxiii}$$

whenever this enlarged box embeds. The new photon links are only those of type three bonds with an endpoint in $\mathbb{L}^{\pm}$. The recursion commutes with multiplication by a curvature character, since it differentiates only fields and its choices depend only on the monomial.

We first use this recursion to bound an arbitrary polynomial; here its photon links need not yet be reduced. Move its fields upwards through all thin layers below $\mathcal{X}_{j,6}$ and downwards through

those above $\mathcal{X}_{j+1,1}$. The last downward replacement can reach $\mathcal{X}_{j,6}$, and cannot reach a lower layer. We obtain, on every sufficiently large torus and with every bounded bosonic factor, an equality

$$Z_T[\Phi P] = \sum_a b_a Z_T[\Phi \varphi_{\beta_a} Y_a], \qquad Y_a \in \mathcal{P}_1(c). \tag{2.cxxxiv}$$

Here the Y_a can be chosen with integer exponents qj_a. Indeed each remaining monomial has integer charge of total zero on the two adjacent photon planes. In their connected rectangular support graph choose paths from a fixed vertex to each charged vertex; the charge-weighted sum of these paths is an integer cochain j_a with the required divergence. Then $u_a = \delta_a - qj_a$ is divergence free in the same integer box and can be written $d^*\beta_a$ there.

The filling must stay inside the support box for the polynomial bound below. In $\prod_\mu [a_\mu, a'_\mu]$, set electric components $\beta^{\mathrm{el}}_j = -\sum_{l=a_1}^{j} (u_a)_l$ for $a_1 \le j < a'_1$. Subtracting $d^*\beta$ removes all spatial links below the top plane; divergence zero then removes the temporal links. On that top plane a second partial-sum sweep in the second coordinate removes the third-direction links below its top edge. Divergence zero removes the second-direction links and leaves only a terminal segment in direction three. Its divergence equation and vanishing boundary value make it zero. The two sweeps use only plaquettes of the box. This proves the asserted filling and thus the splitting $\delta_a = d^*\beta_a + qj_a$ in eq. 2.cxxxiv.

Apply the coordinate bound to each Y_a, absorb the extra character and use amplification. For any fixed box B_0 there are constants C_P, k_P and a volume threshold with

$$\sup_{\operatorname{supp}\beta \subset B_0} |\mu_i^t(\varphi_\beta P)| \le C_P \gamma_{t,i}(k_P), \qquad t \ge 0. \tag{2.cxxxv}$$

One may take $C_P = \sum_a |b_a| 2^{|S_a|/2} \|X'_{Y_a}\|$, where the factor $2^{|S_a|/2}\|X'_{Y_a}\|$ is one for $Y_a = 1$. Such a term therefore contributes $|b_a|$. All support boxes in the finite normal form and B_0 fit in one fixed box; its radius determines k_P. The size of β does not enter the bound. Reflected polynomials have the same kind of bound, since the normal form identity can be reflected on the reflected torus and the coordinate bounds are preserved by reflection.

2.3.7. *The contour shift and an outer plane.* The field recursion alone does not remove a plane from the support: its photon phases remain. We remove these by shifting the links of the next plane in an imaginary direction. The spatial and outgoing temporal exponents then cancel together, using gauge invariance. The shift is justified in the original all-flux law: summing over exact integer fluxes unfolds the affected compact link intervals to real lines. We first prove that contour identity, then choose the shift that removes the lowest plane.

Let u be a real cochain supported on finitely many links S of one coordinate plane, and let $P \in \mathcal{P}$. Choose a box B with $P \in \mathcal{G}^{\mathrm{ev}}(B)$ that contains $b + [-1,1]^3$ for every $b \in S$. Put $P^{(u)}(\widehat{A}) = P(\widehat{A} - iu)$ by its entire exponential expansion. Define

$$\begin{aligned} Q_u = \sum_{b \in S} \sum_{(x,x') \text{ over } b} & [(e^{qu_b} - 1)\bar\psi(x) w(x,x') U_A(x,x') \psi(x') \\ & + (e^{-qu_b} - 1)\bar\psi(x') w(x',x) U_A(x',x) \psi(x)], \qquad V_u = e^{Q_u}. \end{aligned} \tag{2.cxxxvi}$$

The bonds are oriented with b. Each term is invariant and even, and the exponential is a finite polynomial by nilpotence. It depends only on the link variables with $b \in S$, $u_b \neq 0$, and on the fields at their overlying bonds. For a plane normal to e_1, those fields are in thin layers $2, \dots, 5$ of its thick layer. The change in the action is $S_T(A - iu) - S_T(A) = -Q_u$, so $e^{-S_T(A-iu)} = e^{-S_T(A)} V_u$.

If $\Phi(F)$ is entire and bounded on each tube of bounded imaginary curvature, then on every torus in which B embeds we have

$$Z_T[\Phi P] = e^{\|du\|^2/2} Z_T[\Phi(F - idu)\varphi_{du}(F) V_u P^{(u)}]. \tag{2.cxxxvii}$$

The norm here is the lattice norm, equal to the torus norm because its support embeds. We supply the contour justification. For each b choose its electric plaquette leaving the plane in the normal direction. Its only side in S is b, with coefficient ±1. Consequently d is injective on cochains supported in S, and the action $z \mapsto z + dk$, $k \in \mathbb{Z}^S$, on closed integer fluxes is free. Choose orbit representatives z_0. The compact A_S intervals and the sum over k become the integral over $\mathbb{R}^S$, since $(A, z_0 + dk)$ represents the same configuration as $(A + \frac{2\pi}{q}k, z_0)$.

For fixed z_0 and the other real links, choose an integer one-cochain k' with $dk' = z_0$ in the local box. Then $A + \frac{2\pi}{q}k'$ is a local lift affine in the complex variables $\zeta = A_S$. The Berezin coefficients of $e^{-S_T(A)}$ are polynomials in the link phases, so the full integrand extends to an entire function $f(\zeta)$. On the strip $|\Im\zeta_b| \le \|u\|_\infty$ the selected plaquettes give

$$|f(\zeta)| \le C \prod_{b\in S} e^{-(\Re\zeta_b - a_b)^2/2}. \tag{2.cxxxviii}$$

Indeed each of their real curvatures is $\pm\Re\zeta_b + a'_b$, all imaginary curvatures are bounded in this finite strip, the factor Φ is bounded there, and each remaining exponential phase is bounded there. The constant can be chosen independent of the remaining real links and the orbit representative; only the real Gaussian centers vary.

Move one variable at a time along the rectangle with real endpoints $\pm R$ and imaginary displacement $-iu_b$. The vertical sides tend to zero by eq. 2.cxxxviii. Fubini then gives $\int f(\xi)\,\mathrm{d}\xi = \int f(\xi - iu)\,\mathrm{d}\xi$. At the shifted arguments the Maxwell exponent, using the complex-bilinear cochain pairing, becomes

$$-\tfrac12\langle F - idu, F - idu\rangle = -\tfrac12\|F\|^2 + i\langle du, F\rangle + \tfrac12\|du\|^2.$$

The change in the fermion action is the polynomial V_u, and the observable becomes $P^{(u)}$. Both the initial integrand and the resulting real integrand are absolutely integrable by their local bounds. Hence we may sum the identities over z_0 and integrate the remaining links, which proves eq. 2.cxxxvii.

We now eliminate the lowest plane j in the support of P. This means that its fields have photon projection height at least j and its links have no endpoint below plane j. Apply the six upward thin-layer maps of this thick layer to obtain $P^\uparrow = \sum_a b_a e^{i\delta'_a \widehat{A}}\Psi_a$. It contains no fields of thick layer j and depends on no link variables below it; the only new links are spatial links in plane j and temporal links from j to $j+1$. At each term, gauge invariance in plane j therefore says

$$(\delta'_a)^\perp_j = d^*_s(\delta'_a)_j. \tag{2.cxxxix}$$

For this term let u_a be supported on plane $j+1$ with $(u_a)_{j+1} = (\delta'_a)_j$. Define

$$P_a = b_a e^{\|du_a\|^2/2 + \langle\delta'_a, u_a\rangle} e^{i\langle\delta'_a + d^*du_a, \widehat{A}\rangle} V_{u_a}\Psi_a. \tag{2.cxl}$$

Then eq. 2.cxxxiii and eq. 2.cxxxvii give the exact relation

$$Z_T[\Phi P] = \sum_a Z_T[\Phi(F - idu_a)P_a]. \tag{2.cxli}$$

The construction is on the infinite lattice; the same relation holds for every Φ entire in the plaquette variables and bounded on each tube $|\Im F_\sigma| \le r$ for all σ, $r > 0$, and every torus containing the fixed enlarged support box.

Here is the cancellation of the eliminated exponent. Write $v = (\delta'_a)_j$. From the slice formulas $(du)^{\mathrm{el}}_l = u_{l+1} - u_l - d_s u^\perp_l$ and $(d^*\beta)_l = \beta^{\mathrm{el}}_{l-1} - \beta^{\mathrm{el}}_l + d^*_s\beta^{\mathrm{mag}}_l$, the only nonzero components of d^*du_a are

$$\begin{aligned} (d^*du_a)_j &= -v, & (d^*du_a)^\perp_j &= -d^*_s v,\\ (d^*du_a)_{j+1} &= (2 + d^*_s d_s)v, & (d^*du_a)^\perp_{j+1} &= d^*_s v,\\ (d^*du_a)_{j+2} &= -v. \end{aligned} \tag{2.cxlii}$$

Thus the spatial exponent in plane j vanishes, and eq. 2.cxxxix cancels its outgoing temporal exponent as well. The polynomial V_{u_a} depends only on fields and link variables in thick layer $j+1$. Hence P_a has lowest plane at least $j+1$ and highest plane at most the maximum of $j+2$ and the previous highest plane. The six field recursions enlarge the box by six, and the shifted links with their plaquettes require two more. The polynomials and shifts produced by the step are supported in $B+[-8,8]^3$. Elimination from the highest plane is obtained by the downward maps and a shift in plane $j-1$. The corresponding incoming temporal exponent is $-d_s^*(\delta'_a)_j$, and the same two components cancel. The highest plane falls by one, while the lowest can fall only to $j-2$.

The source of error in the tilted law is precisely the shifted bosonic factor. For a finite cochain η, only finitely many copies in the array V_i meet its support. Their change is the fixed trigonometric polynomial

$$H_\eta(F)=\sum_{\kappa,b} a_b(e^{\langle (g_\kappa^{-1})^*\beta_b,\eta\rangle}-1)\varphi_{(g_\kappa^{-1})^*\beta_b}(F), \tag{2.cxliii}$$

where $O=\sum_b a_b\varphi_{\beta_b}$ and only contributing cells occur. At most $|\operatorname{supp}\eta|$ cells contribute; each lies in the support box of η enlarged by h. Its coefficient norm has the bound

$$|H_\eta|_1\le C(\eta):=|\operatorname{supp}\eta|\,|O|_1(e^{\max_b\|\beta_b\|_1\|\eta\|_\infty}-1). \tag{2.cxliv}$$

If that enlarged box embeds, no two contributing cells are period equivalent and the same polynomial is the torus difference $V_i(F-i\eta)-V_i(F)$. Expand its exponential in characters:

$$e^{tV_i(F-i\eta)}=e^{tV_i(F)}\left(1+\sum_\alpha b_\alpha(t)\varphi_{\beta_\alpha}(F)\right),\qquad \sum_\alpha|b_\alpha(t)|\le e^{tC(\eta)}-1. \tag{2.cxlv}$$

The coefficient norm is submultiplicative, so the exponential series proves absolute uniform convergence. All character supports lie in one fixed enlarged box, independent of i and t; sums of character exponents do not enlarge this support.

Set $P^\flat=\sum_a P_a$ in eq. 2.cxli. Insert eq. 2.cxlv with $\eta=du_a$ and divide by the positive denominator. Termwise integration is valid: at a fixed torus the functional on bounded bosonic factors multiplying P_a is continuous for their supremum norm, by the absolute integrability of its Berezin coefficients. We get

$$|\mu_i^t(P)-\mu_i^t(P^\flat)|\le(e^{Ct}-1)\sum_a\sup_{\operatorname{supp}\beta\subset B'}|\mu_i^t(\varphi_\beta P_a)|, \tag{2.cxlvi}$$

where $C=\max_a C(du_a)$ and B' is a fixed box enlarged by h. The a priori bound eq. 2.cxxxv makes this $O(t)$ uniformly for all large i and $0\le t\le 1$: each of the finitely many amplification factors is bounded once $\Delta_i\le 1$, and $e^{Ct}-1\le(e^C-1)t$ on this interval.

The same estimate applies after transporting the identity. This needs care because D_i need not be invariant under a coordinate permutation. Reflect the equality on the reflected torus and then reflect it back, using functional covariance. The conjugations change a shift η to $-\theta^*\eta$. Indeed, if $\Phi^\theta(F)=\overline{\Phi(\theta^*\overline{F})}$, then $(\Phi^\theta(\cdot-i\eta))^\theta=\Phi(\cdot+i\theta^*\eta)$. Thus eq. 2.cxli becomes an exact identity for ΘP with shifts $-\theta^*du_a$ and polynomials ΘP_a on the original torus. Iterating reflections transports it by any g, with shifts $\det(\vec g)(g^{-1})^*du_a$, where $\vec g$ is the linear part of g. Its support box is transported at the same time. Apply eq. 2.cxlv and eq. 2.cxxxv on D_i itself to obtain the transported $O(t)$ estimate. No invariance of D_i or of its tilted array under g is used.

Theorem F. For each $P\in\mathcal{P}$ and coordinate link plane c there is $Y_P\in\mathcal{P}_1(c)$, independent of the torus and tilt, such that, for every fixed lattice isometry g,

$$|\mu_i^t(P^g)-\mu_i^t((Y_P)^g)|\le C_{P,g}t\quad(0\le t\le 1, i\ge i_0(P,g)). \tag{2.cxlvii}$$

At a plane normal to either other coordinate the corresponding reduction and its reflected estimate hold as well. ■□

Proof. For $c = j_0 + \frac{1}{2}$, eliminate lowest planes until the lowest is at least j_0. Every step raises it by at least one, so this takes finitely many steps. Then eliminate highest planes until the highest is at most $j_0 + 1$. During this phase an eliminated plane $j \geq j_0 + 2$ can create links only as low as $j - 2 \geq j_0$, preserving the first reduction. The resulting polynomial depends only on the fields and link variables in the two thick layers and on the links connecting them. Apply the upward recursions in thin layers $1, \dots, 5$ of the lower thick layer, followed by the downward recursions in thin layers $6, \dots, 2$ of the upper one. New fields remain at or above the lower layer six and at or below the upper layer one. The new type three phases are spatial phases in these two planes or the temporal phases between them. Thus the result Y_P is in $\mathcal{P}_1(c)$.

Sum eq. 2.cxlvi over the finitely many outer-plane steps, using one threshold for their fixed supports and a priori bounds. The final ten thin-layer recursions are exact, proving eq. 2.cxlvii. For direction μ, apply the construction to $P^{g_\mu^{-1}}$ and transport by the cyclic permutation g_μ; the transported estimate holds on D_i itself, including its reflected version. □

2.3.8. *Comparison states and translation invariance.* A comparison state for the chosen cell structure and O is a law ν_0 obtained by the ordered limits

$$\mu_{i_m}^{t_k} \longrightarrow \nu_{t_k} \quad (m \to \infty) \text{ for each } k, \qquad \nu_{t_k} \longrightarrow \nu_0 \quad (k \to \infty), \qquad t_k > 0, \quad t_k \to 0. \tag{2.cxlviii}$$

All convergences here are weak convergences of bosonic laws on Ω, and the volume subsequence is common to all strengths. Given a prescribed subsequence of volumes, a further subsequence produces such a state. To see this, take strengths $t_k = 1/k$, extract nested convergent volume subsequences by the tightness proved above, and choose their diagonal subsequence. The limiting laws ν_{t_k} are themselves uniformly tight: applying eq. 2.cxix to $\beta = \pm 1_\sigma$ and using $t_k \leq 1$ bounds their exponential coordinate moments by a common constant. The same product-of-intervals argument gives a weakly convergent strength subsequence. This proves existence of eq. 2.cxlviii subordinate to any prescribed volume subsequence.

Wall invariance and wall reflection positivity also pass through both limits. For each local positive-side test, its support is contained in the positive half-period slab once i_m is large, so eq. 2.xcviii applies. The continuity and L^2 extension described in §2.3.1 give positivity for local tests that are bounded and Borel. In particular ν_0 is invariant and reflection positive at $(x_1 - x_2, c_0)$.

Suppose the unperturbed cubic laws $\mu_{C(n_i)}$ converge to ν on the prescribed subsequence. For a comparison state subordinate to it, the finite fold inequality gives

$$\nu(O) \leq \nu_0(O). \tag{2.cxlix}$$

Indeed fix $t_k > 0$, take the volume limit in

$$\mu_{C(n_{i_m})}(O) \leq \mu_{i_m}^{t_k}(O) + |Q|\Delta_{i_m}/t_k,$$

and use bounded continuity of O and $\Delta_{i_m} \to 0$. Then let $t_k \to 0$. The denominator t_k in the error precludes reversing this order.

We now obtain the extra invariance required to use eq. 2.cxlix for observables at arbitrary positions. Fix a coordinate link reflection θ_c and a curvature character $P = \varphi_\beta$. Its polynomial representation is $e^{i\langle d^*\beta, \widehat{A}\rangle}$; on every sufficiently large torus this equals $e^{i\langle \beta, \widetilde{F}\rangle}$ on the local lift. Apply Thm. F to P and its reflection, and then Prop. 2.11 to its two-layer reduction. The finite errors are at most two $O(t)$ terms plus a finite sum of the terms in eq. 2.cxxx. Along the common volume subsequence in eq. 2.cxlviii, take volume to infinity, then $t_k \to 0$ with the regularization fixed, and finally remove the regularization. Only the character expectations are passed to weak limits. The conclusion is

$$\nu_0(\Theta_c \varphi_\beta) = \overline{\nu_0(\varphi_\beta)}. \tag{2.cl}$$

There is no need to extract limiting fermion expectations in this argument.

For a bosonic character $\Theta_c \varphi_\beta = \varphi_{-\theta_c^*\beta}$, while positivity of the probability law gives $\overline{\nu_0(\varphi_\beta)} = \nu_0(\varphi_{-\beta})$. Thus eq. 2.cl, with $-\beta$ in place of β, is invariance under the tensor pullback $F \mapsto \theta_c^* F$.

Characters determine all finite plaquette marginals by uniqueness of their Fourier transforms; these marginals determine the law because cylinder sets generate the Borel sigma algebra of Ω. Consequently every comparison state is invariant under all coordinate link reflections, in all three directions. The product of reflections at c and $c+1$ is translation by $2e_\mu$, so

$$(\tau_v^*)_*\nu_0 = \nu_0, \qquad v \in 2\mathbb{Z}^3. \tag{2.cli}$$

We need only these even translations. The original wall periods alone would not have permitted the next placement.

2.3.9. *Direct comparison and the nine mirror families.* Fix an integer c_0. For every local trigonometric polynomial O that is real-valued and every cubic limit ν along our divisible side sequence, there is a comparison state ν_0, invariant and reflection positive at $(x_1 - x_2, c_0)$, satisfying eq. 2.cxlix. The polynomial need not initially fit in a cell having its lower wall at c_0. Let $\operatorname{supp} O \subset y_0 + [-r, r]^3$, with $r \geq 1$, and choose an admissible even width $h > 8r + 8$. Choose an even integer a in

$$(c_0 + 2r - \ell_-(y_0),\ c_0 + h - 2r - \ell_-(y_0)).$$

The interval has length greater than two. For $v = ae_1$ and $y = y_0 + v$ its defining inequalities place the translated box between the two ℓ_- walls. Choose integer c_+ and half-integer c_3 so that

$$c_+ < \ell_+(y) - 2r < \ell_+(y) + 2r < c_+ + h, \qquad c_3 < y_3 - r < y_3 + r < c_3 + h.$$

The available intervals have length greater than one. The box therefore lies in the open diagonal cell, since $|\ell_\pm(x) - \ell_\pm(y)| \leq 2r$ on it. Form the tilted law with the translated polynomial O^{τ_v}.

Choose a comparison state subordinate to the volume subsequence defining ν. It has the stated wall invariance and positivity, and the fold inequality gives $\nu(O^{\tau_v}) \leq \nu_0(O^{\tau_v})$. The original cubic laws are invariant under every lattice translation, and this invariance passes to ν. eq. 2.cli gives the same invariance under this particular v for ν_0. Hence

$$\nu(O) = \nu(O^{\tau_v}) \leq \nu_0(O^{\tau_v}) = \nu_0(O). \tag{2.clii}$$

The comparison state may depend on O. To prove positivity of a given reflected form, we choose O to be the negative real part of that form. Its expectation in a reflection-positive comparison state is nonpositive, so the comparison forces the real part of the original reflected form to be nonnegative.

Let θ reflect in $x_1 - x_2 = c_0$, and let A be a finite positive-side character sum. Put $Q_A = (\Theta_\theta A)A$, where $\Theta_\theta A(F) = \overline{A(\theta^* F)}$. Since $\Theta_\theta \varphi_\beta = \varphi_{-\theta^*\beta}$, $O_A = -\operatorname{Re} Q_A$ is a local trigonometric polynomial and is real-valued. Its comparison state ν_A in eq. 2.clii is reflection positive, so

$$-\operatorname{Re}\nu(Q_A) = \nu(O_A) \leq \nu_A(O_A) = -\nu_A(Q_A) \leq 0.$$

The original limit ν is θ-invariant, since θ is coordinate exchange followed by the integer translation $(c_0, -c_0, 0)$. Thus $\nu(Q_A)$ is real and nonnegative. Characters span a dense subspace of the positive-side L^2 space by finite-dimensional Fourier uniqueness and convergence of finite-coordinate conditional expectations. Reflection is an L^2 isometry, so Cauchy–Schwarz bounds the reflected form by $\|A\|_2\|B\|_2$. Positivity extends by density to the full space and, by reflection, to the opposite side.

Theorem G. At fixed $q > 0$, $0 < \rho < 1$, positive chamber weights and real mass, the cubic curvature laws along the divisible side sequence are tight on Ω. Every weak limit satisfies Gaussian domination, is invariant under lattice translations, signed coordinate permutations and $F \mapsto -F$, and is reflection positive at all mirrors

$$x_\mu = c, \quad c \in \tfrac{1}{2}\mathbb{Z}; \qquad x_\mu \pm x_\nu = c, \quad c \in \mathbb{Z}, \quad \mu < \nu.$$

Thus it has all nine lattice mirror families. ■□

Proof. The coordinate Gaussian domination theorem gives the uniform one-plaquette tails and hence tightness. Any finitely supported β embeds without overlap once the side is large, so its periodization has unchanged norm. Truncation passes Gaussian domination to every limit. The laws are supported on the closed subspace $dF = 0$. Finite-volume translation, signed permutation and charge-conjugation invariances pass to the limit under their continuous actions on Ω.

For a coordinate mirror $x_\mu = c$, a continuous local test supported on the positive side lies in its positive half-period slab for all large sides. The cubic torus is adapted to this mirror, and its reflected slab trace gives nonnegativity of the reflected form. Weak convergence and the L^2 extension give coordinate reflection positivity of the limit. The direct reflected-product argument above proves reflection positivity at $(x_1 - x_2, c_0)$ for every integer c_0.

A signed permutation transports this diagonal mirror and its positive half-space to any plane $x_\mu \pm x_\nu = c$. The corresponding reflection is the conjugate of the original one. Pull a positive-side test back by this isometry; invariance of ν identifies its reflected form with the already nonnegative one. This gives the six diagonal families and completes the proof. □

The pressure error and amplification scales may depend on fixed q, M, w_i. At mesh $\eta = L^{-N}$ these are $q = e\eta^{1/2}$ and $M = \bar{m}\eta + m_{0,N}(e) + \frac{\rho}{2}\sum_i w_i$ by eq. 2.xv; the side sequence has the required divisibility. This proves positivity for fixed-N volume limits, which must precede the mesh limit; it gives no estimate for simultaneous mesh and volume variation.

3. Block averaging in Maxwell's theory

The photon block analysis follows the constructive renormalization framework of [7]. We keep the coarse photon average and split the Maxwell energy into a minimizing background and Gaussian fluctuations. The two minimizations use three lattices: the fine lattice of spacing L^{-k}, the unit lattice, and the coarse lattice of spacing L. The first reconstructs a fine field from its unit-lattice average; the second fixes the next coarse average and minimizes the energy induced on the unit lattice. The reconstruction kernel and the positive covariance square root have uniform decay, which permits local block integration at every scale.

3.1. **Photon block integration.**

3.1.1. *Coarse averages and the Landau minimizer.* Let $L \geq 5$ be odd, $\eta = \eta_k = L^{-k}$, and let ℓ be a positive integer divisible by L. The photon fields are periodic cochains on $T_k = \eta\mathbb{Z}^3/\ell\mathbb{Z}^3$, with the pairing $\eta^3 \sum_c$ on each component. Distances on T_k are periodic sup distances: $d(c, c') = \min_{n \in \ell\mathbb{Z}^3} |c - c' + n|_\infty$. For bonds we use the distance between their initial points; for finite supports we take the infimum over their pairs of points. On the covering lattice, $d(c, c') = |c - c'|_\infty$.

The block step reconstructs a fine field from its average and integrates the remaining zero-average fluctuation. Adding a gradient does not change the curvature energy. For the reconstruction we require its divergence to be orthogonal to the scalar Laplacians of functions with zero block average, the restricted Landau condition. For the fluctuation we require the averages of coordinate-path integrals within each block to vanish, the axial condition. Uniform locality of the reconstruction and of the positive covariance square root will then make this Gaussian change of variables usable on local polymers, that is, connected unions of blocks. Here the differences are normalized by the mesh:

$$\partial_\mu f(x) = \eta^{-1}(f(x + \eta e_\mu) - f(x)), \qquad (d\lambda)_\mu = \partial_\mu \lambda, \qquad (d\mathscr{A})_{\mu\nu} = \partial_\mu \mathscr{A}_\nu - \partial_\nu \mathscr{A}_\mu.$$

We put $\delta = d^T$ and $\Delta_{\mathrm{sc}} = \delta d$ on scalars. Fix $0 < \alpha < 1$. The Hölder quotient $\delta_\alpha \partial$ is the difference of a first-difference kernel at two parallel bonds, divided by the distance between their initial sites to the power α; only distances in $(0, 1]$ are used. The estimate will be stated in eq. 3.xi. The unit lattice is $T_k^0 = \mathbb{Z}^3/\ell\mathbb{Z}^3$ and the unit blocks are $\Delta_z = \{c : |c - z|_\infty < 1/2\}$. Write $T_k^c = L\mathbb{Z}^3/\ell\mathbb{Z}^3$ for the coarse lattice and $\Omega^p(T)$ for the real p-cochains on a lattice T. All operators in this section act on these lattice cochains. The averages and reconstructions extend complex-linearly when complex

fields are used below. Set $N = L^k$. The cochain averages are denoted by $\mathbf{Q}_k$; they commute with the coboundary, so a fine gauge change induces the coarse gauge change and coarse curvature is the average of fine curvature. On scalars, bonds, and plaquettes they are respectively

$$\begin{aligned}(\mathbf{Q}_k\lambda)(z) &= \eta^3 \sum_{c\in\Delta_z} \lambda(c),\\ (\mathbf{Q}_k\mathscr{A})_\mu(z) &= \eta^3 \sum_{c\in\Delta_z}\sum_{j=0}^{N-1} \eta\mathscr{A}_\mu(c+j\eta e_\mu),\\ (\mathbf{Q}_k^{(2)}F)_{\mu\nu}(z) &= \eta^3 \sum_{c\in\Delta_z}\sum_{i,j=0}^{N-1} \eta^2 F_{\mu\nu}(c+i\eta e_\mu + j\eta e_\nu).\end{aligned} \tag{3.i}$$

All sums are on the periodic covering lattice. These averages preserve constants and the mean of a bond field, are onto on bond fields, and satisfy

$$\mathbf{Q}_k d = d\mathbf{Q}_k, \qquad d\mathbf{Q}_k = \mathbf{Q}_k^{(2)} d, \qquad \|\mathbf{Q}_k^{(2)}F\|_2 \le \|F\|_2. \tag{3.ii}$$

On scalars $\mathbf{Q}_k^T\lambda$ is constant with value $\lambda(z)$ on Δ_z, and $\mathbf{Q}_k\mathbf{Q}_k^T = 1$. The differential identities follow by telescoping along the segments in eq. 3.i. The plaquette average has total weight one. Jensen's inequality and summation over coarse sites count each fine plaquette with total weight η^3, proving its contraction. To see surjectivity on bonds, give the fine bond leaving a unit block in direction μ the value $NA_\mu(z)$, and give the other bonds value zero. Every segment of length one starting in Δ_z crosses exactly one such boundary bond, and its average is $A_\mu(z)$.

The prescribed average is unchanged by adding $d\lambda$ when $\mathbf{Q}_k\lambda = 0$. We fix this freedom among the energy minimizers by a condition on the divergence. For a fine field $\mathscr{A}$, write $E(\mathscr{A}) = \frac{1}{2}\|d\mathscr{A}\|_2^2$. Let $\mathcal{R}_k$ be the orthogonal projection on $\Delta_{\mathrm{sc}}(\ker \mathbf{Q}_k)$ in the scalar space. Denote by $H_k^{\mathrm{ph}}A$ the minimizer of $\|d\mathscr{A}\|_2^2$ subject to

$$\mathbf{Q}_k\mathscr{A} = A, \qquad \mathcal{R}_k\delta\mathscr{A} = 0. \tag{3.iii}$$

This is the restricted Landau gauge of [6, Section 3.5, Prop. 5]. The superscript marks the photon reconstruction; the fermion reconstruction introduced later will have a different domain. Define the coarse photon form

$$\Delta_k = (dH_k^{\mathrm{ph}})^T dH_k^{\mathrm{ph}}. \tag{3.iv}$$

Proposition 3.1. The minimizer exists uniquely. It is real and linear, and $\mathbf{Q}_k H_k^{\mathrm{ph}} = 1$. It preserves constant fields; if $dA = 0$, then $dH_k^{\mathrm{ph}}A = 0$. At $k = 0$ it is the identity. Its characterization is

$$\mathbf{Q}_k\mathscr{A} = A, \qquad \delta d\mathscr{A} \in \operatorname{ran}\mathbf{Q}_k^T \text{ on bonds}, \qquad \Delta_{\mathrm{sc}}\delta\mathscr{A} \in \operatorname{ran}\mathbf{Q}_k^T \text{ on scalars}. \tag{3.v}$$

■

Proof. Surjectivity gives a nonempty affine space of curvatures; finite-dimensional least squares gives its unique norm minimizer. Variation along $B \in \ker\mathbf{Q}_k$ gives

$$\langle \delta d\mathscr{A}, B\rangle = 0;$$

the middle condition in eq. 3.v.

For a closed periodic bond field the discrete Hodge decomposition is $B = c + d\lambda$, where c is its mean. One can see this directly in Fourier variables: at nonzero momentum, closedness says that the components of $\widehat{B}$ are proportional to the forward-difference symbol, and the proportionality factor is $|\widehat{d}|^{-2}\widehat{d}^* \cdot \widehat{B}$. At zero momentum only the constant remains. Exact fields have mean zero. Hence a closed $B \in \ker\mathbf{Q}_k$ is $d\lambda$ with $\lambda \in \ker\mathbf{Q}_k$: average the decomposition, use eq. 3.ii, and subtract the constant value of $\mathbf{Q}_k\lambda$. Thus the unconstrained minimizers are $\mathscr{A}_1 + d(\ker\mathbf{Q}_k)$.

The restriction of Δ_{sc} to $\ker \mathbf{Q}_k$ is injective: its kernel consists of constants, whose zero block average forces zero. For $\lambda \in \ker \mathbf{Q}_k$,

$$\mathcal{R}_k \delta(\mathscr{A}_1 + d\lambda) = \mathcal{R}_k \delta \mathscr{A}_1 + \Delta_{\mathrm{sc}} \lambda.$$

Since $\mathcal{R}_k$ projects onto $\Delta_{\mathrm{sc}}(\ker \mathbf{Q}_k)$, exactly one such λ imposes the Landau constraint. The equivalence $\mathcal{R}_k f = 0 \iff \Delta_{\mathrm{sc}} f \perp \ker \mathbf{Q}_k$ gives the last Euler condition. Uniqueness in this real linear system makes H_k^{ph} real and linear.

Constants satisfy the equations. If $A = c + d\lambda$ is closed, then $c + d\mathbf{Q}_k^T \lambda$ is a closed fine field with average A, so the minimum curvature energy is zero. This proves preservation of closed fields. Finally $\mathbf{Q}_0 = 1$ gives the level-zero assertion. □

The minimizing property also separates the fine fluctuation from the reconstructed field. For any fine bond field $\mathscr{A}$, put $A = \mathbf{Q}_k \mathscr{A}$ and $Z_0 = \mathscr{A} - H_k^{\mathrm{ph}} A$. Then

$$\mathbf{Q}_k Z_0 = 0, \qquad E(\mathscr{A}) = E(H_k^{\mathrm{ph}} A) + E(Z_0). \tag{3.vi}$$

Indeed, variation in the direction Z_0 gives $\langle dH_k^{\mathrm{ph}} A, dZ_0 \rangle = 0$. The Gaussian fluctuation integrated in the next block step lies on the unit lattice and also satisfies the axial condition.

3.1.2. *Axial coercivity on a block.* On the unit lattice, let $B(y) = \{z \in \mathbb{Z}^3 : |z - y|_\infty < L/2\}$, with centers in $L\mathbb{Z}^3$. The one-step scalar average is

$$(\mathbf{Q}\lambda)(y) = L^{-3} \sum_{z \in B(y)} \lambda(z).$$

The bond average and axial functional are

$$(\mathbf{Q}A)(y, y + Le_\mu) = L^{-4} \sum_{z \in B(y)} A(\Gamma_{z, z+Le_\mu}), \qquad (\tau A)(y,z) = \tfrac{1}{6} \sum_\pi A(\Gamma^\pi(y,z)).$$

Here the first path is straight, and Γ^π changes the coordinates in the order π. The sums $A(\Gamma)$ are oriented bond sums. These are the symmetrized axial paths of [6, Section 2.2, eqs. (12)–(15)]. The condition $\tau A = 0$ selects one representative without changing its coarse average or reconstructed curvature. For $x \in B(y)$, set

$$\lambda(x) = -(\tau A)(y,x) + L^{-3} \sum_{z \in B(y)} (\tau A)(y,z). \tag{3.vii}$$

The block means of λ vanish, and $\mathbf{Q}d = d\mathbf{Q}$ follows from the same path telescoping as for $\mathbf{Q}_k$. Since every path sum of $d\lambda$ telescopes to $\lambda(x) - \lambda(y)$, we have

$$\mathbf{Q}\lambda = 0, \qquad \mathbf{Q}(A + d\lambda) = \mathbf{Q}A, \qquad \tau(A + d\lambda) = 0.$$

If two scalars with zero mean on each block impose this gauge, their difference is constant on each block and hence zero. Moreover, $dH_k^{\mathrm{ph}} d\lambda = 0$ by preservation of closed fields. Thus this change of gauge preserves $E(H_k^{\mathrm{ph}} A)$. We write

$$\mathcal{Z}_k = \{Z \in \Omega^1(T_k^0) : \mathbf{Q}Z = 0,\ \tau Z = 0\}$$

for the unit-lattice fluctuation space.

The axial condition controls interior bond values by curvature. Across a block face, curvature controls differences of bond values; the coarse average then controls their mean. We give both estimates before summing them to obtain a bound independent of the number of blocks. For an interior bond $b = (z, z + e_\nu)$, subtract the two paths from y to its endpoints and append b. For each order this loop bounds the strips belonging to the directions that come after ν. Their total number of plaquettes is $\sum_{\mu \text{ after } \nu} |z_\mu - y_\mu|$. If $\tau A = 0$, averaging over the six orders gives

$$|A(b)| \le \tfrac{1}{2} \sum_{\mu \neq \nu} |z_\mu - y_\mu| \sup_{p \subset B(y)} |dA(p)| \le \tfrac{L-1}{2} \sup_{p \subset B(y)} |dA(p)|. \tag{3.viii}$$

For a face between $B(y)$ and $B(y+Le_\nu)$, let $a(z)$ be its face bond values and let p_c be the maximum curvature on plaquettes in these two blocks and across their common face. A transverse face step is one plaquette plus the difference of two interior bond values. Thus $|a(z)-a(z')| \le L|z-z'|_1 p_c$. The defining average $\mathbf{Q}A$ counts each face bond L times and $L^3(L-1)$ interior bond occurrences. Therefore its face mean $\bar{a}$ satisfies

$$|\bar{a}| \le L|(\mathbf{Q}A)(c)| + \tfrac{1}{2}(L-1)^2 p_c.$$

The average transverse distance from a face site is at most $L-1$, so $|a(z)-\bar{a}| \le L(L-1)p_c$. Combining these gives

$$|A(b)| \le \tfrac{3}{2}L^2 p_c + L|(\mathbf{Q}A)(c)|. \tag{3.ix}$$

For interior bonds use eq. 3.viii and omit the coarse-average term. In particular, on the axial fluctuation space,

$$\mathbf{Q}Z = 0, \quad \tau Z = 0 \quad \Longrightarrow \quad \|Z\|_\infty \le \tfrac{3}{2}L^2\|dZ\|_\infty, \qquad \|Z\|_2^2 \le 9L^7\|dZ\|_2^2. \tag{3.x}$$

For the second inequality, square the local bound and replace each local curvature maximum by its sum of squares. An interior plaquette belongs to the neighborhoods of at most $3L^3+3L^2 \le 4L^3$ bonds; a face plaquette to those of at most L^2 bonds. Summation gives $(9/4)L^4 \cdot 4L^3 = 9L^7$.

3.1.3. *The minimizer symbol and uniform regularity.* Changing one unit-lattice average should have an exponentially small effect on the reconstructed field far away. We need the same decay for its first differences and their Hölder quotients, uniformly as the fine mesh tends to zero. The estimates come from Bałaban's minimizer analysis; compare [5, Section 7.2, eq. (7.2.2)].

Proposition 3.2 (Uniform locality of reconstruction)**.** For every $0<\alpha<1$, there are $C_H(\alpha), \gamma_H > 0$, independent of k and ℓ, such that

$$|H_k^{\mathrm{ph}}(b,b')|, \quad |\partial H_k^{\mathrm{ph}}(b,b')|, \quad |\delta_\alpha \partial H_k^{\mathrm{ph}}((b,\widetilde{b});b')| \le C_H(\alpha)e^{-\gamma_H d(b,b')}. \tag{3.xi}$$

Here $b, \widetilde{b}$ are parallel fine bonds whose initial sites are at distance in $(0,1]$, and b' is a unit bond. ■

Proposition 3.3 (Bounds for the induced energy)**.** There are constants $C_\Delta, C_\Delta^{(2)}$, independent of k and ℓ, such that

$$|\Delta_k(b,b')| \le C_\Delta e^{-\gamma_H d(b,b')}, \qquad \|\Delta_k\|_{2\to 2} \le C_\Delta^{(2)}. \tag{3.xii}$$

In particular, $E(H_k^{\mathrm{ph}}A) \le \frac{1}{2}C_\Delta^{(2)}\|A\|_2^2$. ■

Sampling on the unit lattice identifies fine momenta that differ by $2\pi m$; we call them aliases. We first solve the three minimizer equations in each alias class. The class containing zero has an apparent pole. The prescribed-average equation forces its cancellation, after which uniform analytic bounds give a strip of fixed width. A contour shift and periodization then prove the two propositions. We use the Hermitian extensions of the inner products for real Fourier frequencies, and analytic continuation of the displayed symbols for complex frequencies.

Averaging plane waves. Fix

$$p \in [-\pi,\pi)^3 \cap \frac{2\pi}{\ell}\mathbb{Z}^3.$$

The frequencies

$$q_m = p + 2\pi m, \qquad m \in \mathbb{Z}^3, \quad |m|_\infty \le \frac{N-1}{2},$$

give the same plane wave on T_k^0, since

$$e^{iq_m\cdot y} = e^{ip\cdot y}, \qquad y \in T_k^0.$$

They are distinct frequencies on T_k. Thus one frequency on T_k^0 corresponds to N^3 frequencies on T_k.

The centered average and the segment average have factors

$$u(t) = \frac{1}{N} \sum_{r=-(N-1)/2}^{(N-1)/2} e^{itr/N}, \qquad U(q) = \prod_{j=1}^{3} u(q_j),$$

$$V_\mu(q) = \frac{1}{N} \sum_{r=0}^{N-1} e^{iq_\mu r/N}.$$

The corresponding derivative factors are

$$d_\mu(q) = N(e^{iq_\mu/N} - 1), \qquad \rho_\mu(p) = e^{ip_\mu} - 1.$$

Proposition 3.4 (Fourier multipliers of block averaging)**.** For $f(x) = e^{iq_m \cdot x}$ and $A(x) = ae^{iq_m \cdot x}$, where $a \in \mathbb{C}^3$, we have

$$(\mathbf{Q}_k f)(y) = U(q_m)e^{ip\cdot y}, \qquad (\mathbf{Q}_k A)_\mu(y) = U(q_m)V_\mu(q_m)a_\mu e^{ip\cdot y}.$$

Moreover,

$$V_\mu(q) = e^{i(1-N^{-1})q_\mu/2}u(q_\mu), \qquad V_\mu(q_m)d_\mu(q_m) = \rho_\mu(p). \tag{3.xiii}$$

■

Proof. Substituting the plane wave into the scalar average gives one factor $u(q_j)$ for each coordinate. In the μ-component of $\mathbf{Q}_k A$, the additional average along a segment of N bonds gives $V_\mu(q_m)$. The identity $e^{iq_m \cdot y} = e^{ip\cdot y}$ yields the displayed formulas.

Shifting the summation index in V_μ by $(N-1)/2$ gives its expression in terms of u. Finally,

$$\begin{aligned} V_\mu(q_m)d_\mu(q_m) &= \sum_{r=0}^{N-1} \left(e^{iq_{m,\mu}(r+1)/N} - e^{iq_{m,\mu}r/N} \right) \\ &= e^{iq_{m,\mu}} - 1 = e^{ip_\mu} - 1 = \rho_\mu(p). \end{aligned}$$

Multiplying by $U(q_m)$ gives the identity $\mathbf{Q}_k d = d\mathbf{Q}_k$ on this plane wave. □

Solving the reconstruction equations. Fix a nonzero p on the Fourier grid of the preceding calculation, and let $a \in \mathbb{C}^3$. We compute $H_k^{\mathrm{ph}} A$ for

$$A(y) = ae^{ip\cdot y}, \qquad y \in T_k^0.$$

Write

$$d_\mu^*(q) = d_\mu(-q), \qquad V_\mu^*(q) = V_\mu(-q), \qquad \rho_\mu^*(p) = \rho_\mu(-p),$$

and

$$D(q) = \sum_{\mu=1}^{3} d_\mu^*(q)d_\mu(q).$$

For real arguments the starred functions are complex conjugates. The definitions by reversal also specify their analytic extensions. We use the Hermitian extensions of the real inner products. For this fixed p, set

$$E_\mu = \sum_m \frac{U(q_m)^2 u(q_{m,\mu})^2}{D(q_m)}, \qquad G = \sum_m \frac{U(q_m)^2}{D(q_m)^2}, \qquad r = \sum_{\mu=1}^{3} \frac{\rho_\mu \rho_\mu^*}{E_\mu}.$$

Every sum over m uses the range specified in the preceding calculation.

We seek

$$F_\mu(x) = \sum_m f_\mu(q_m)e^{iq_m \cdot x}, \qquad x \in T_k,$$

with $b \in \mathbb{C}^3$ and $w \in \mathbb{C}$ to be determined, subject to

$$\sum_\mu \rho_\mu^* b_\mu = 0,$$

and with amplitudes

$$f_\mu(q_m) = \frac{U(q_m)V_\mu^*(q_m)}{D(q_m)} b_\mu + \frac{d_\mu(q_m)U(q_m)}{D(q_m)^2} w. \tag{3.xiv}$$

The solution below is linear in a; denote its coefficients by $h_{\mu\nu}(p,m)$, so that

$$f_\mu(q_m) = \sum_{\nu=1}^{3} h_{\mu\nu}(p,m) a_\nu.$$

The two terms in eq. 3.xiv will give stationarity and the restricted Landau condition. The prescribed average determines b, w: by Prop. 3.4 and $V_\mu d_\mu = \rho_\mu$,

$$(\mathbf{Q}_k F)_\mu(y) = (E_\mu b_\mu + \rho_\mu G w) e^{ip\cdot y}.$$

Thus the remaining equations are

$$E_\mu b_\mu + \rho_\mu G w = a_\mu, \qquad \sum_\mu \rho_\mu^* b_\mu = 0. \tag{3.xv}$$

Proposition 3.5 (Solution of the Fourier reconstruction equations)**.** For the fixed nonzero real momentum p, the numbers E_μ, G, r are positive. The system eq. 3.xv has the unique solution

$$\begin{aligned} w &= \frac{\sum_\nu \rho_\nu^* a_\nu / E_\nu}{Gr}, \\ b_\mu &= \sum_\nu \Lambda_{\mu\nu} a_\nu, \qquad \Lambda_{\mu\nu} = \frac{\delta_{\mu\nu}}{E_\mu} - \frac{\rho_\mu \rho_\nu^*}{E_\mu E_\nu r}. \end{aligned} \tag{3.xvi}$$

With these values, the field F defined above equals $H_k^{\mathrm{ph}} A$. ∎

Proof. First, $D(q_m) > 0$ for every m, since $p \neq 0$. For $|t| \leq \pi$, the identity

$$u(t) = \frac{\sin(t/2)}{N \sin(t/(2N))}, \qquad u(0) = 1,$$

gives $u(p_\mu) > 0$. Thus the $m = 0$ terms make $E_\mu, G > 0$. At least one ρ_μ is nonzero, so $r > 0$. Substitute $b_\mu = (a_\mu - \rho_\mu G w)/E_\mu$ into the second equation of eq. 3.xv. It gives the displayed formula for w; substitution back gives $b = \Lambda a$. Positivity of the denominators proves uniqueness.

We verify the remaining equations characterizing $H_k^{\mathrm{ph}} A$. The identity $d_\mu^* V_\mu^* = \rho_\mu^*$ makes the first term of f divergence-free. The second term is a gradient; consequently the amplitude of δF at q_m is

$$\sum_\mu d_\mu^*(q_m) f_\mu(q_m) = \frac{U(q_m)}{D(q_m)} w.$$

The amplitude of $\Delta_{\mathrm{sc}} \delta F$ is therefore $U(q_m) w$. Since scalar averaging has multiplier $U(q_m)$, this places $\Delta_{\mathrm{sc}} \delta F$ in the range of its adjoint. Hence

$$\langle \delta F, \Delta_{\mathrm{sc}} \lambda \rangle = \langle \Delta_{\mathrm{sc}} \delta F, \lambda \rangle = 0 \qquad (\lambda \in \Omega^0(T_k),\ \mathbf{Q}_k \lambda = 0),$$

which is the restricted Landau condition.

The amplitude of δdF is

$$D(q_m) f_\mu(q_m) - d_\mu(q_m) \sum_\nu d_\nu^*(q_m) f_\nu(q_m) = U(q_m) V_\mu^*(q_m) b_\mu.$$

This belongs to the range of the adjoint of averaging on Ω^1, so

$$\langle dF, dK \rangle = 0 \qquad (K \in \Omega^1(T_k),\ \mathbf{Q}_k K = 0).$$

Thus stationarity also holds. We have verified all three equations characterizing $H_k^{\mathrm{ph}}A$, so $F = H_k^{\mathrm{ph}}A$. □

At $p = 0$, constant preservation gives $H_k^{\mathrm{ph}}A = A$: the coefficient at $q_0 = 0$ is a, and all other coefficients vanish.

Removing the apparent zero-momentum singularity.

For locality, these Fourier coefficients must extend analytically through $p = 0$.

We regard the preceding finite-sum formulas as functions of complex p, wherever they are defined. The formula for $f_\mu(p)$ contains $D(p)^{-1}$ and $D(p)^{-2}$, while $D(0) = 0$. Its two terms must be combined using the relation between b and w imposed by the prescribed average:

$$E_\mu b_\mu + \rho_\mu G w = a_\mu.$$

We first use this relation to expose a cancellation, then check that the remaining denominators are nonzero.

Proposition 3.6 (Removal of the zero-momentum singularity)**.** For each fixed N, the matrices $h(p, m)$ extend analytically to a complex neighborhood of $[-\pi, \pi]^3$. At $p = 0$, they give $f(0) = a$ and $f(2\pi m) = 0$ for $m \neq 0$. ■

Proof. Write $D = D(p)$, $U = U(p)$, and $u_\mu = u(p_\mu)$. Eliminating b_μ with the prescribed-average relation and using $V_\mu^* \rho_\mu = u_\mu^2 d_\mu$ gives

$$f_\mu(p) = \frac{U V_\mu^*}{D E_\mu} a_\mu + \frac{d_\mu U}{D^2 E_\mu}(E_\mu - u_\mu^2 D G) w.$$

Separating $m = 0$ from the sums defining E_μ, G, write

$$E_\mu = \frac{U^2 u_\mu^2}{D} + X_\mu, \qquad G = \frac{U^2}{D^2} + Y,$$

where X_μ, Y are the sums with $m \neq 0$. These are analytic near the real cube, since $D(q_m) > 0$ there for $m \neq 0$. The two displayed singular contributions come from the same term $m = 0$. They cancel in the combination

$$E_\mu - u_\mu^2 D G = X_\mu - u_\mu^2 D Y.$$

To write the result with analytic numerators and denominators, put

$$N_\mu = D E_\mu = U^2 u_\mu^2 + D X_\mu, \qquad N_G = D^2 G = U^2 + D^2 Y.$$

The right sides define N_μ, N_G even when $D = 0$.

Multiplying the preceding cancellation by D gives

$$N_\mu - u_\mu^2 N_G = D(X_\mu - u_\mu^2 D Y). \tag{3.xvii}$$

Applying this cancellation to the displayed formula for $f_\mu(p)$ gives

$$f_\mu(p) = \frac{U V_\mu^*(p)}{N_\mu} a_\mu + \frac{d_\mu(p) U}{N_\mu}(X_\mu - u_\mu^2 D Y)\frac{w}{D}. \tag{3.xviii}$$

The factors D^{-1} and D^{-2} have thus combined into the single quotient w/D. We must still prove that this quotient has an analytic continuation at $D = 0$.

For that check, we rewrite the scalar denominator r in the formula for w, separating its factor D^2. Define

$$\Pi = \prod_\mu N_\mu - \sum_\mu d_\mu^*(p) d_\mu(p) X_\mu \prod_{\nu \neq \mu} N_\nu.$$

Using $\rho_\mu \rho_\mu^* = u_\mu^2 d_\mu^*(p) d_\mu(p)$ and $\sum_\mu d_\mu^*(p) d_\mu(p) = D$, the definition of r gives

$$r = D \sum_\mu \frac{d_\mu^*(p) d_\mu(p) u_\mu^2}{N_\mu} = \frac{D^2 \Pi}{U^2 \prod_\mu N_\mu}.$$

The formulas for w and b therefore become

$$\frac{w}{D} = \frac{U^2 \prod_\mu N_\mu}{N_G \Pi} \sum_\nu \frac{\rho_\nu^* a_\nu}{N_\nu}, \qquad b_\mu = \frac{D}{N_\mu} a_\mu - \frac{\rho_\mu N_G}{N_\mu} \frac{w}{D}. \tag{3.xix}$$

These identities are first obtained for real $p \neq 0$; being algebraic, they also hold wherever the complex expressions are defined. The right side of the formula for w/D will define its continuation where $D = 0$.

All remaining denominators are positive on the real cube. For N_μ, N_G this follows from $X_\mu, Y \geq 0$ and $u_\mu > 0$. For $p \neq 0$, the identity for r gives

$$\Pi = \sum_\mu \frac{d_\mu^*(p) d_\mu(p)}{D} U^2 u_\mu^2 \prod_{\nu \neq \mu} N_\nu > 0 : \tag{3.xx}$$

this is a weighted average of positive quantities. At $p = 0$, $N_\mu = N_G = \Pi = 1$. For fixed N, they therefore remain nonzero in a complex neighborhood of the real cube. The displayed formulas prove analyticity of w/D, b, and $f(p)$; for $m \neq 0$, use eq. 3.xiv. At $p = 0$, the continued quantities w/D and b both vanish, so these same formulas give the asserted values of f. □

Keeping the analytic neighborhood uniform in N.

The preceding neighborhood could shrink as N grows. We prevent this by bounding the denominators below on the real domain and above on a fixed complex rectangle, with both bounds independent of N. Analyticity then controls how rapidly they change away from the real domain.

Proposition 3.7 (Uniform analytic neighborhood)**.** There is $g > 0$, independent of N, such that N_μ, N_G, Π have reciprocals bounded independently of N on

$$|\operatorname{Re} p_\mu| < 5\pi/4 - 1/4, \qquad |\operatorname{Im} p_\mu| < g.$$

The matrices $h(p, m)$ are analytic there. ■

Proof. We first obtain the upper bounds on the larger rectangle

$$|\operatorname{Re} p_\mu| < 5\pi/4, \qquad |\operatorname{Im} p_\mu| < 1/4.$$

Although the number of terms in X_μ, Y grows with N, their decay in m gives a summable bound independent of N. Write $\langle n \rangle = \max(1, |n|)$ for an integer n. We claim that

$$\operatorname{Re} D(q_m) \geq |m|_2^2 \quad (m \neq 0), \qquad |u(q_{m,\mu})| \leq C \langle m_\mu \rangle^{-1}, \tag{3.xxi}$$

with C independent of N.

To verify these bounds, put

$$s(t) = 2N \sin(t/(2N)), \qquad \kappa = \frac{\sin(5\pi/8)}{5\pi/8}.$$

Then $D(q) = \sum_\mu s(q_\mu)^2$ and $u(t) s(t) = 2 \sin(t/2)$. In the allowed range of m, $|\operatorname{Re} q_{m,\mu}|/(2N) < 5\pi/8$ and $|\operatorname{Im} q_{m,\mu}| < 1/4$. The real sine bound $|\sin x| \geq \kappa |x|$ for $|x| \leq 5\pi/8$, together with the complex sine identity, gives

$$\operatorname{Re}(s(q_{m,\mu})^2) \geq \kappa^2 (\operatorname{Re} q_{m,\mu})^2 - \cosh^2(1/8) (\operatorname{Im} q_{m,\mu})^2.$$

If $m_\mu \neq 0$, then $|\operatorname{Re} q_{m,\mu}| \geq (3\pi/4)|m_\mu|$. Summing in μ therefore gives, for $m \neq 0$,

$$\operatorname{Re} D(q_m) \geq \kappa^2 (3\pi/4)^2 |m|_2^2 - 3 \cosh^2(1/8)/16 \geq |m|_2^2.$$

The same bound in one coordinate gives $|s(q_{m,\mu})| \geq |m_\mu|$ when $m_\mu \neq 0$. The numerator in $u = 2\sin(t/2)/s(t)$ is uniformly bounded, since adding $2\pi m_\mu$ changes only its sign. When $m_\mu = 0$, the centered finite sum defining u is uniformly bounded. This proves eq. 3.xxi.

It follows that X_μ, Y are uniformly bounded by a constant multiple of the fixed convergent series

$$\sum_{m\in\mathbb{Z}^3} \prod_{\mu=1}^{3} \langle m_\mu\rangle^{-2} < \infty.$$

The same elementary formulas bound $D(p), U(p), u(p_\mu)$, and $d_\mu(p)$ uniformly on the larger rectangle. Thus the definitions of N_μ, N_G, Π give a common upper bound B, independent of N.

On the real part of this rectangle, the sine quotient gives $u(p_\mu) \geq \kappa$. Their positive formulas therefore give

$$N_\mu \geq \kappa^8, \qquad N_G \geq \kappa^6, \qquad \Pi \geq \kappa^{24}.$$

For Π, use the weighted average in eq. 3.xx; its weights are nonnegative and sum to one. That algebraic identity also holds on this larger real cube. At $p = 0$, $\Pi = 1$. Thus $c = \kappa^{24}$ is a common positive lower bound.

We now use these two bounds to move off the real domain. Let J be any of N_1, N_2, N_3, N_G, Π, and write $p = x + iy$ with x, y real. Suppose

$$|x_\mu| < 5\pi/4 - 1/4, \qquad |y|_\infty < c/(16B).$$

For complex t with $|t| < 4B/c$, the point $x + ity$ stays in the larger rectangle. The analytic function $J(x+ity) - J(x)$ vanishes at $t = 0$ and has modulus at most $2B$. Schwarz's lemma on this disc gives

$$|J(x + iy) - J(x)| \leq \frac{2B}{4B/c} = c/2.$$

Since $J(x) \geq c$, we obtain $|J(x + iy)| \geq c/2$. Taking $g = c/(16B)$ proves the uniform nonvanishing. The formulas from the preceding calculation, together with eq. 3.xxi, give the asserted analyticity. □

Summing the reconstruction coefficients. We also need to bound the sum over m, uniformly in N. A difference in direction λ contributes $d_\lambda(q_m)$, and the Hölder estimate also requires $|q_m|_1^\alpha$, where $|q|_1 = \sum_j |q_j|$.

Proposition 3.8 (Uniform summability)**.** For every $0 < \alpha < 1$, there is $S_\alpha < \infty$, independent of N, such that

$$\sup_p \sum_m |h_{\mu\nu}(p, m)| \max\{1, |d_\lambda(q_m)|, |d_\lambda(q_m)||q_m|_1^\alpha\} \leq S_\alpha \tag{3.xxii}$$

for every μ, ν, λ. The supremum is over the strip in Prop. 3.7, and the sum uses the same range of m as the Fourier reconstruction. ■

Proof. The formulas eq. 3.xix and the uniform bounds from the preceding proof give $|b| + |w| \leq C|a|$. Here $w = D(p)(w/D)$ and $D(p)$ is uniformly bounded. The formula eq. 3.xviii also gives $|h_{\mu\nu}(p, 0)| \leq C$.

For $m \neq 0$, the estimates already proved give

$$|U(q_m)| \leq C \prod_\sigma \langle m_\sigma\rangle^{-1}, \qquad |D(q_m)|^{-1} \leq |m|_2^{-2}.$$

The identity $V_\mu^*(q) = e^{-i(1-N^{-1})q_\mu/2} u(q_\mu)$ gives $|V_\mu^*(q_m)| \leq C\langle m_\mu\rangle^{-1}$, since the imaginary parts are bounded on the strip. The definition of d_λ likewise gives

$$|d_\lambda(q_m)| \leq C\langle m_\lambda\rangle.$$

Applying these bounds to the two terms in eq. 3.xiv gives

$$|h_{\mu\nu}(p, m)| \leq C\left(\langle m_\mu\rangle^{-1}|m|_2^{-2} + \langle m_\mu\rangle|m|_2^{-4}\right) \prod_\sigma \langle m_\sigma\rangle^{-1}. \tag{3.xxiii}$$

The first term comes from UV_μ^*/D and the second from $d_\mu U/D^2$.

It remains to check summability with the requested weights. Put $R = \max_j \langle m_j\rangle$. Since $|q_m|_1 \le CR$ and $|d_\lambda(q_m)| \le CR$, all three weights are bounded by $CR^{1+\alpha}$. For $m \neq 0$, $|m|_2 \ge R \ge 1$. The first term in eq. 3.xxiii, multiplied by $R^{1+\alpha}$, is therefore bounded by

$$C\langle m_\mu\rangle^{-2} R^{-(1-\alpha)} \prod_{\sigma\neq\mu} \langle m_\sigma\rangle^{-1} \le C\langle m_\mu\rangle^{-2} \prod_{\sigma\neq\mu} \langle m_\sigma\rangle^{-(3-\alpha)/2}.$$

The last inequality uses $R^2 \ge \prod_{\sigma\neq\mu}\langle m_\sigma\rangle$. For the second term, use $\langle m_\mu\rangle \le R$ and $\alpha < 1$ to obtain

$$CR^{\alpha-2}\prod_\sigma \langle m_\sigma\rangle^{-1} \le CR^{-1}\prod_\sigma \langle m_\sigma\rangle^{-1} \le C\prod_\sigma \langle m_\sigma\rangle^{-4/3}.$$

Here $R^3 \ge \prod_\sigma \langle m_\sigma\rangle$.

Every exponent in these products exceeds one, because $(3-\alpha)/2 > 1$. Their sums over $\mathbb{Z}^3$ are finite, and bound the sums over the allowed m independently of N. The term $m = 0$ is uniformly bounded, including all three weights. This proves eq. 3.xxii. □

Turning analyticity into spatial decay.

We now use the complex neighborhood to obtain decay in x. Moving the Fourier contour introduces an exponentially decreasing factor. The summability estimate controls the remaining integral uniformly in N.

For $x \in \eta\mathbb{Z}^3$, define

$$K_{\mu\nu}(p;x) = \sum_m e^{i(p+2\pi m)\cdot x} h_{\mu\nu}(p,m), \qquad H^\infty_{\mu\nu}(x) = \frac{1}{(2\pi)^3}\int_{[-\pi,\pi]^3} K_{\mu\nu}(p;x)\,\mathrm{d}p.$$

We will recover the torus kernel by periodizing H^∞. As before, differences are normalized by the spacing:

$$\partial_\lambda H^\infty(x) = \frac{H^\infty(x+\eta e_\lambda) - H^\infty(x)}{\eta}.$$

Proposition 3.9 (Decay of H^∞)**.** Put $\gamma_H = g/2$, with g as in Prop. 3.7. For every $0 < \alpha < 1$, there is C_α, independent of N, such that

$$|H^\infty_{\mu\nu}(x)| + |\partial_\lambda H^\infty_{\mu\nu}(x)| \le C_\alpha e^{-\gamma_H|x|_1},$$

and, for $z \in \eta\mathbb{Z}^3$ with $0 < |z|_\infty \le 1$,

$$|\partial_\lambda H^\infty_{\mu\nu}(x+z) - \partial_\lambda H^\infty_{\mu\nu}(x)| \le C_\alpha |z|_\infty^\alpha e^{-\gamma_H|x|_1}.$$

These bounds hold for every $x \in \eta\mathbb{Z}^3$ and every μ,ν,λ. ■

Proof. Before moving the contour, we check that its opposite boundary faces cancel. The complete sum $K_{\mu\nu}(p;x)$ is 2π-periodic in each coordinate of p. Indeed, replacing p_j by $p_j + 2\pi$ reindexes m_j by one. At the end of its range, reducing m_j modulo N changes $q_{m,j}$ by $2\pi N$. The finite-sum formulas for u, V, d and their reversed arguments are invariant under this change; so is $e^{iq_m\cdot x}$, since $x \in \eta\mathbb{Z}^3$. Also ρ is 2π-periodic. Thus the sums defining E_μ, G, r are unchanged, and the reconstruction coefficients are permuted with their frequencies. This proves periodicity where the original formulas are defined. Their analytic continuations give the same identity across the removed singularities.

Shift each coordinate of integration to

$$p_j + i\gamma_H \operatorname{sgn}(x_j).$$

The contours remain in the analytic strip, and periodicity cancels the opposite vertical faces. Since $2\pi m$ is real, every exponential acquires the same factor:

$$e^{i(p+i\gamma_H \operatorname{sgn} x + 2\pi m)\cdot x} = e^{-\gamma_H|x|_1} e^{i(p+2\pi m)\cdot x}.$$

The unweighted part of eq. 3.xxii therefore gives

$$|H^{\infty}_{\mu\nu}(x)| \le S_\alpha e^{-\gamma_H |x|_1}.$$

A difference ∂_λ inserts $d_\lambda(q_m)$ in the integral. This factor respects the same reindexing, so the identical contour shift and the second weight in eq. 3.xxii give the first-difference bound.

For the Hölder estimate, use this same contour, chosen from x, for both terms. Their difference inserts the additional factor $e^{iq_m\cdot z}-1$. On the shifted contour,

$$|\operatorname{Im}(q_m\cdot z)| \le 3\gamma_H, \qquad |e^{iq_m\cdot z}-1| \le C\min\{1,|q_m\cdot z|\} \le C|z|_\infty^\alpha |q_m|_1^\alpha.$$

The middle inequality follows by integrating the exponential when $|q_m\cdot z|\le 1$, and by bounding its two terms otherwise. The increment factor respects the frequency reindexing because $z\in\eta\mathbb{Z}^3$. The third weight in eq. 3.xxii now gives the asserted bound, with the same factor $e^{-\gamma_H|x|_1}$. □

Returning to the torus.

The torus kernel is obtained by summing H^∞ over translates by ℓn. We first identify this sum with the original reconstruction, then show that summing the translates preserves all three estimates uniformly in ℓ.

Completion of the proof of Prop. 3.2. Let b,b' have initial sites c,z' and directions μ,ν, respectively. Choose representatives $c\in\eta\mathbb{Z}^3$, $z'\in\mathbb{Z}^3$, and put $x=c-z'$. Fourier inversion on T_k^0 and Prop. 3.5 give

$$H_k^{\mathrm{ph}}(b,b') = \frac{1}{\ell^3}\sum_{p\in[-\pi,\pi)^3\cap(2\pi/\ell)\mathbb{Z}^3} K_{\mu\nu}(p;x).$$

Here $e^{-ip\cdot z'}e^{iq_m\cdot c}=e^{iq_m\cdot(c-z')}$, since $z'\in\mathbb{Z}^3$.

For $j\in\mathbb{Z}^3$, the definition of K gives $K_{\mu\nu}(p;x-j)=e^{-ip\cdot j}K_{\mu\nu}(p;x)$. Thus its Fourier coefficient at j is $H^\infty_{\mu\nu}(x-j)$, and

$$K_{\mu\nu}(p;x)=\sum_{j\in\mathbb{Z}^3} H^\infty_{\mu\nu}(x-j)e^{ip\cdot j}.$$

This series converges absolutely by Prop. 3.9. Averaging on the displayed Fourier grid retains exactly the terms with $j\in\ell\mathbb{Z}^3$. Consequently

$$H_k^{\mathrm{ph}}(b,b') = \sum_{n\in\mathbb{Z}^3} H^\infty_{\mu\nu}(x+\ell n). \tag{3.xxiv}$$

To bound the sum, choose the representative of x with $|x_j|\le \ell/2$. Then $|x|_\infty = d(b,b')$. For any real t with $|t|\le\ell/2$, the geometric series gives

$$\sum_{n\in\mathbb{Z}} e^{-\gamma_H|t+\ell n|} = \frac{e^{-\gamma_H|t|}+e^{-\gamma_H(\ell-|t|)}}{1-e^{-\gamma_H\ell}} \le \frac{2}{1-e^{-\gamma_H}}e^{-\gamma_H|t|},$$

where we used $\ell\ge 1$. Multiplying the three one-dimensional estimates yields

$$\sum_{n\in\mathbb{Z}^3} e^{-\gamma_H|x+\ell n|_1} \le \left(\frac{2}{1-e^{-\gamma_H}}\right)^3 e^{-\gamma_H|x|_1} \le C_{\gamma_H} e^{-\gamma_H d(b,b')}.$$

Applying the bounds for H^∞ and $\partial_j H^\infty$ term by term in eq. 3.xxiv proves eq. 3.xi.

Finally, let $b,\widetilde{b}$ be parallel and put $r=d(b,\widetilde{b})\in(0,1]$. Choose $z\in\eta\mathbb{Z}^3$ such that $\widetilde{b}=b+z$ on the torus and $|z|_\infty=r$. The periodization formula gives

$$\begin{aligned}&\partial_j H_k^{\mathrm{ph}}(\widetilde{b},b') - \partial_j H_k^{\mathrm{ph}}(b,b')\\ &\quad= \sum_{n\in\mathbb{Z}^3}\left[\partial_j H^\infty_{\mu\nu}(x+z+\ell n) - \partial_j H^\infty_{\mu\nu}(x+\ell n)\right].\end{aligned}$$

The Hölder estimate from Prop. 3.9, followed by the same geometric-series bound, gives

$$|\partial_j H_k^{\mathrm{ph}}(\widetilde{b}, b') - \partial_j H_k^{\mathrm{ph}}(b, b')| \le C r^{\alpha} e^{-\gamma_H d(b,b')}.$$

This is eq. 3.xi. All constants are independent of N and ℓ, as required. □

The reconstructed energy.

The same Fourier solution identifies the quadratic energy on $\Omega^1(T_k^0)$. We first prove its uniform upper bound, then the decay of its kernel.

Operator bound in Prop. 3.3. First take $A(y) = ae^{ip\cdot y}$ at a nonzero torus frequency p, using the Hermitian extensions of the inner products. Let $F = H_k^{\mathrm{ph}} A$ have amplitudes $f(q_m)$, with the notation of eq. 3.xiv. The Euler equation computed there gives

$$\widehat{(\delta dF)}_\mu(q_m) = U(q_m) V_\mu^*(q_m) b_\mu, \qquad b = \Lambda(p) a.$$

Distinct aliases are orthogonal, and every fine plane wave has squared norm $\eta^3(\ell/\eta)^3 = \ell^3$. Since U is real at real frequencies, pairing the Euler equation with F gives

$$\begin{aligned} \|dF\|_2^2 &= \ell^3 \sum_{m,\mu} \overline{f_\mu(q_m)} U(q_m) V_\mu^*(q_m) b_\mu \\ &= \ell^3 \sum_{\mu} \overline{\sum_{m} U(q_m) V_\mu(q_m) f_\mu(q_m)} b_\mu = \ell^3 a^* \Lambda(p) a. \end{aligned}$$

The last equality uses $\mathbf{Q}_k F = A$. At $p = 0$, reconstruction preserves constants and the energy vanishes; the regularized formula gives $\Lambda(0) = 0$.

For general A, set

$$\widehat{A}(p) = \ell^{-3} \sum_{y \in T_k^0} A(y) e^{-ip\cdot y}, \qquad A(y) = \sum_p \widehat{A}(p) e^{ip\cdot y},$$

where p ranges over the unit-lattice Fourier grid. Orthogonality of distinct frequency classes yields

$$E(H_k^{\mathrm{ph}} A) = \frac{\ell^3}{2} \sum_p \widehat{A}(p)^* \Lambda(p) \widehat{A}(p). \tag{3.xxv}$$

The bound for $b = \Lambda(p)a$ in the proof of Prop. 3.8 bounds every entry of Λ by a constant A_1 independent of N. Hence $\sup_p \|\Lambda(p)\| \le 3A_1$. Parseval's identity proves the operator bound in eq. 3.xii with $C_\Delta^{(2)} = 3A_1$. □

The Fourier identity eq. 3.xxv identifies $\Lambda(p)$ as the symbol of Δ_k. Its bounded analytic continuation gives kernel decay by the same contour shift and periodization used for H_k^{ph}.

Kernel decay in Prop. 3.3. The regularized formulas for $b = \Lambda(p)a$ and Prop. 3.7 give an analytic continuation to the stated neighborhood of the real cube, with entries bounded independently of N. Shifting p by $2\pi e_j$ permutes the aliases modulo $2\pi N\mathbb{Z}^3$, leaving the sums E_μ, G, r and the factors ρ_μ unchanged. This proves periodicity where the original formulas are defined. Their analytic continuations agree on overlapping neighborhoods and therefore define an analytic matrix on the strip $|\operatorname{Im} p_j| < g$. This matrix is periodic and bounded, with both $g > 0$ and the bound independent of N.

For $x \in \mathbb{Z}^3$, define

$$\Delta_{k;\mu\nu}^{\infty}(x) = \frac{1}{(2\pi)^3} \int_{[-\pi,\pi]^3} e^{ip\cdot x} \Lambda_{\mu\nu}(p)\, \mathrm{d}p.$$

Choose $\gamma_H = g/2$. Shifting the jth contour to $p_j + i\gamma_H \operatorname{sgn}(x_j)$ gives

$$|\Delta_{k;\mu\nu}^{\infty}(x)| \le C_\Lambda e^{-\gamma_H |x|_1},$$

where C_Λ bounds the entries of Λ on the strip. The vertical faces cancel by periodicity.

For $b = (y, y + e_\mu)$ and $b' = (y', y' + e_\nu)$, finite Fourier inversion and periodization give

$$\Delta_k(b, b') = \ell^{-3} \sum_p e^{ip\cdot(y-y')} \Lambda_{\mu\nu}(p) = \sum_{n\in\mathbb{Z}^3} \Delta^\infty_{k;\mu\nu}(y - y' + \ell n),$$

where p runs over the unit-lattice Fourier grid. This is the same Fourier sampling identity used in eq. 3.xxiv. Choose the representative x of $y - y'$ with $|x_j| \le \ell/2$. The geometric-series bound used there yields

$$|\Delta_k(b, b')| \le C_\Lambda \left(\frac{2}{1 - e^{-\gamma_H}}\right)^3 e^{-\gamma_H |x|_1} \le C e^{-\gamma_H d(b,b')}.$$

□

3.1.4. *Fluctuation coordinates and the positive square root.* The fluctuations in one block integration satisfy $\mathbf{Q}Z = 0$ and $\tau Z = 0$. We first parametrize this space by vectors supported in neighboring blocks. The energy in these coordinates is uniformly positive. Its positive inverse square root then converts the Gaussian law to a product measure and has exponentially decaying entries. Thus the initial coordinate map has bounded support, while the final Gaussian change of variables is controlled by exponential decay.

The condition $\tau Z = 0$ involves only bonds inside each block $B(y)$. We first choose coordinates satisfying this condition. The equations $\mathbf{Q}Z = 0$ also involve bonds joining neighboring blocks; we will use them to determine one value on each face after the other values have been chosen.

For $y \in T^c_k$, let $\Omega^1(B(y))$ denote the real one-forms on the interior bonds of the L-block $B(y)$. Each coordinate line in this cube has L sites and $L - 1$ interior bonds; the cube may wrap around the torus. The paths defining τ remain inside $B(y)$, so they define

$$\tau_y : \Omega^1(B(y)) \longrightarrow \Omega^0(B(y)), \qquad (\tau_y Z)(x) = (\tau Z)(y, x).$$

Proposition 3.10 (Coordinates in a block)**.** For each $y \in T^c_k$,

$$\operatorname{rank} \tau_y = L^3 - 1, \qquad \dim \ker \tau_y = 2L^3 - 3L^2 + 1.$$

The kernel has an orthonormal basis of simultaneous eigenvectors of the three coordinate reflections through y. Here the inner product is the sum of products on the positively oriented interior bonds. ■

Proof. Since $(\tau_y df)(x) = f(x) - f(y)$, the range of τ_y is precisely the functions vanishing at y, and its rank is $L^3 - 1$. Subtracting this from the number $3L^2(L-1)$ of interior bonds gives the asserted kernel dimension. The coordinate reflections, including the signs from reversing bond orientation, preserve the symmetrized path sums and act on $\ker \tau_y$ by commuting orthogonal involutions. Choosing an orthonormal basis in their common eigenspaces gives the final assertion. □

Choose such a basis in $B(0)$ and translate it to every $B(y)$, writing the translated vectors as e^y_i. Then the coefficients in

$$Z\big|_{B(y)} = \sum_i z^y_i e^y_i$$

give Euclidean coordinates for the restriction of Z to the interior bonds of $B(y)$:

$$\big\|Z\big|_{B(y)}\big\|_2^2 = \sum_i |z^y_i|^2.$$

Reflections in coordinate planes through the origin act on these translated basis vectors by signs and permutations.

We now impose $\mathbf{Q}Z = 0$. For each positively oriented bond $c = (y, y + Le_\mu)$ of T^c_k, let $F(c)$ be the L^2 positively oriented bonds joining $B(y)$ to $B(y + Le_\mu)$. Reserve the central bond $b_c \in F(c)$, on the line joining the block centers. The other $L^2 - 1$ values can be chosen freely; the average equation at c will determine the value at b_c. This is the local parametrization used in [6, Section 3.7, eqs. (103)–(109)].

Represent the freely chosen values by

$$V_k = \{z \in \Omega^1(T_k^0) : \tau z = 0,\ z(b_c) = 0 \text{ for every } c\}.$$

The interior basis vectors e_i^y, extended by zero outside $B(y)$, together with the point masses δ_b on $F(c) \setminus \{b_c\}$, form an orthonormal basis of V_k. The zeros at b_c will be replaced by the values required by $\mathbf{Q}Z = 0$.

Let $I(c)$ be the interior bonds used by the straight paths in the average at c, all oriented in direction μ, and let $n_c(b)$ be the number of occurrences of b among these paths. The bonds in $I(c)$ lie in the two blocks adjacent to c. Define $Cz = z$ except at the reserved bonds, where

$$(Cz)(b_c) = -\sum_{b \in F(c)\setminus\{b_c\}} z(b) - \sum_{b \in I(c)} \frac{n_c(b)}{L} z(b). \tag{3.xxvi}$$

Proposition 3.11 (Coordinates for $\mathcal{Z}_k$)**.** The map C is a real linear bijection

$$C : V_k \longrightarrow \mathcal{Z}_k, \tag{3.xxvii}$$

and

$$\dim V_k = \dim \mathcal{Z}_k = (2 - 2L^{-3})\ell^3. \tag{3.xxviii}$$

■

Proof. For each choice of the two transverse coordinates, the average at c contains L straight paths, one for each longitudinal starting site in $B(y)$. Each path crosses the face once, using the same bond. Thus every bond in $F(c)$ occurs L times, and the normalization of $\mathbf{Q}$ gives

$$L^4(\mathbf{Q}Z)(c) = L \sum_{b \in F(c)} Z(b) + \sum_{b \in I(c)} n_c(b) Z(b).$$

Solving this equation with $(\mathbf{Q}Z)(c) = 0$ gives eq. 3.xxvi.

Changing $z(b_c)$ affects neither an axial path nor the average equation at another positively oriented bond of T_k^c. The reserved values can therefore be filled independently, giving $Cz \in \mathcal{Z}_k$. Conversely, if $Z \in \mathcal{Z}_k$, setting its reserved values to zero gives $z \in V_k$, and the same equations recover exactly the removed values. This proves bijectivity; linearity follows from the displayed formula.

Each block contributes $2L^3 - 3L^2 + 1$ interior coordinates and $L^2 - 1$ coordinates on each of its three outgoing faces. The total is $2L^3 - 2$ per block. There are $(\ell/L)^3$ blocks, which gives the stated dimension. □

We need bounds for C that do not grow with the number of blocks. The completion formula supplies them because each new value depends only on the two adjacent blocks and their common face. Write

$$z = \sum_{\Upsilon} z_\Upsilon \Upsilon$$

in the orthonormal basis of V_k just constructed.

Proposition 3.12 (Bounds for C)**.** Under the lattice hypotheses of Prop. 3.2, every $z \in V_k$ satisfies

$$\|z\|_2 \le \|Cz\|_2 \le \sqrt{5L^3}\,\|z\|_2, \qquad \|Cz\|_\infty \le 3L^{9/2} \max_{\Upsilon} |z_\Upsilon|. \tag{3.xxix}$$

For each basis vector Υ,

$$\sum_{b \text{ unit bond}} |(C\Upsilon)(b)| \le 14 L^{3/2},$$

and

$$d(b, b') \le L \qquad (b \in \operatorname{supp} C\Upsilon, \quad b' \in \operatorname{supp} \Upsilon). \tag{3.xxx}$$

The constants are independent of N and ℓ. ■

Proof. Each central value in eq. 3.xxvi is a sum of at most $2L^3$ original bond values, with coefficients of absolute value at most one. Each original bond value enters at most two such sums. Cauchy–Schwarz therefore gives

$$\sum_c |(Cz)(b_c)|^2 \le 4L^3 \|z\|_2^2.$$

Since z vanishes on the reserved bonds and $Cz = z$ elsewhere,

$$\|Cz\|_2^2 = \|z\|_2^2 + \sum_c |(Cz)(b_c)|^2.$$

This proves both bounds for the 2-norm.

For the supremum bound, normalize $\max_\Upsilon |z_\Upsilon| = 1$. Each block has at most $2L^3$ interior coordinates, so orthonormality gives $|z(b)| \le \sqrt{2L^3}$ on its interior bonds. A noncentral crossing value has absolute value at most one. The completion formula consequently gives

$$|(Cz)(b_c)| \le L^2 + 2L^3\sqrt{2L^3} \le 3L^{9/2},$$

where $L \ge 5$. The same bound covers the unchanged values.

If Υ is an interior basis vector, it has norm one and at most $3L^3$ nonzero bond values. Hence $\sum_b |\Upsilon(b)| \le \sqrt{3L^3}$. It contributes only to the six central bonds adjacent to its block, and each added value is bounded by this same sum. Thus

$$\sum_b |(C\Upsilon)(b)| \le 7\sqrt{3L^3} \le 14L^{3/2}.$$

For a face vector $\Upsilon = \delta_b$, the completion adds just the value -1 at the central bond of the same face, so the sum is two.

Finally, for an interior basis vector in $B(y)$, every original bond's initial site is within $(L-1)/2$ of y in the supremum norm. Each added central bond's initial site is within $(L+1)/2$ of y. The distance between any such sites is therefore at most L. For a crossing basis vector, the original and central bonds lie on the same face and have distance at most $(L-1)/2$. These bounds also hold when blocks wrap around the torus, since periodic distance can only decrease. This proves eq. 3.xxx. □

We use the constants

$$c_C = 3L^{9/2}, \qquad c_1 = 14L^{3/2}, \qquad d_V = 2 - 2L^{-3}$$

for these coordinate bounds and for the dimension per unit volume.

Let M denote the auxiliary block size, an integer multiple of L whose quantitative choice is made in eq. 3.xlv. Coordinate sign changes and translations by $M\mathbb{Z}^3$ permute $\{\pm\Upsilon\}$. For the interior vectors this follows from the simultaneous eigenvector choice; for face coordinates it follows from the preservation of central bonds. For a coordinate cutoff $p_{0,k} > 0$, the standard Gaussian μ_I on V_k and its normalized restriction $\widehat{\mu}_k$ to

$$\mathrm{Box}_k = \{z : \ |z(\Upsilon)| \le p_{0,k} \text{ for every } \Upsilon\}$$

are invariant under these transformations and under $z \mapsto -z$. Independence of the coordinates gives

$$\mu_I(\mathrm{Box}_k) = \exp(-\ell^3 \delta\varepsilon_k^0), \qquad \delta\varepsilon_k^0 = -d_V \log \Pr(|g| \le p_{0,k}) \le 12\exp(-p_{0,k}^2/2), \tag{3.xxxi}$$

when $p_{0,k} \ge 1$, where g is standard normal. For $p_{0,k} \ge 1$, the tail bound $\Pr(|g| > p_{0,k}) \le 2(2\pi)^{-1/2} p_{0,k}^{-1} e^{-p_{0,k}^2/2} < 1/2$, together with $d_V < 2$ and $-\log(1-t) \le 2t$, gives the stated estimate.

Set

$$\widetilde{\Delta}_k = C^T \Delta_k C.$$

We first prove a positive lower bound for this coordinate energy; its inverse will then define the fluctuation covariance. From the axial estimate, the contraction of the plaquette average, and $\mathbf{Q}_k H_k^{\mathrm{ph}} = 1$, we get

$$\|z\|_2^2 \le \|Cz\|_2^2 \le 9L^7 \|dCz\|_2^2 = 9L^7 \|\mathbf{Q}_k^{(2)} dH_k^{\mathrm{ph}} Cz\|_2^2 \le 9L^7 \langle z, \widetilde{\Delta}_k z\rangle.$$

The upper bound follows from eq. 3.xii and eq. 3.xxix. Thus, with

$$c_P = \max\{9L^7,\ 5L^3 C_\Delta^{(2)}\}, \qquad C_{\mathrm{Dt}} = c_1^2 C_\Delta \exp(2\gamma_H L),$$

we have

$$c_P^{-1} \le \widetilde{\Delta}_k \le c_P, \qquad |\widetilde{\Delta}_k(\Upsilon, \Upsilon')| \le C_{\mathrm{Dt}} \exp(-\gamma_H d(\Upsilon, \Upsilon')). \tag{3.xxxii}$$

The positive lower bound defines the fluctuation covariance $C_k = \widetilde{\Delta}_k^{-1}$. For the kernel estimate, expand $\langle C\Upsilon, \Delta_k C\Upsilon'\rangle$ as a double bond sum. Every contributing pair satisfies $d(\Upsilon, \Upsilon') \le d(b, b') + 2L$ by eq. 3.xxx; the two ℓ^1 bounds then give the asserted constant.

The distance between supports has a triangle defect at most L. We use the exponential-conjugation method of Combes–Thomas [14]. We record the weighted resolvent argument with this defect, since it is responsible for the uniform decay of the square root. Suppose that $T \ge t_0 > 0$ is a real symmetric matrix, that $|T(a,b)| \le C_T \exp(-g d(a,b))$, and that

$$d(a,c) \le d(a,b) + d(b,c) + \phi.$$

Choose

$$S \ge \exp(g\phi/2) \sup_a \sum_b (d(a,b) + \phi) \exp(-g d(a,b)/2), \qquad \theta = \min\{g/2,\ t_0/(2C_T S)\}.$$

Fix c and conjugate T by the diagonal matrix $W(a,a) = \exp(\theta d(a,c))$. The error $E = WTW^{-1} - T$ satisfies

$$|E(a,b)| \le \theta C_T \exp(g\phi/2)(d(a,b) + \phi) \exp(-g d(a,b)/2),$$

because $|d(a,c) - d(b,c)| \le d(a,b) + \phi$ and $|\exp(u) - 1| \le |u| \exp(|u|)$. The Schur test gives $\|E\| \le t_0/2$. Therefore $\|(T + x + E)^{-1}\| \le (t_0/2 + x)^{-1}$ for $x \ge 0$. Undoing the conjugation yields

$$|(T + x)^{-1}(a,c)| \le (t_0/2 + x)^{-1} \exp(-\theta d(a,c)).$$

The scalar integral $\pi^{-1} \int_0^\infty x^{-1/2}(t + x)^{-1}\, \mathrm{d}x = t^{-1/2}$ and finite-dimensional diagonalization now give

$$T^{-1/2} = \frac{1}{\pi} \int_0^\infty x^{-1/2}(T + x)^{-1}\, \mathrm{d}x, \qquad |T^{-1/2}(a,c)| \le (2/t_0)^{1/2} \exp(-\theta d(a,c)). \tag{3.xxxiii}$$

Apply this to $T = \widetilde{\Delta}_k$, $\phi = L$, $t_0 = c_P^{-1}$ and $g = \gamma_H$. The basis support count gives

$$\sup_\Upsilon \sum_{\Upsilon'} \exp(-g d(\Upsilon, \Upsilon')) \le c_{\mathrm{bas}}(g), \qquad c_{\mathrm{bas}}(g) = 2L^3 \exp(2gL)(1 + 6/g)^3.$$

For example, assign a basis vector to its block center or to an end of its face bond. At most $2L^3$ vectors are assigned to any site, and replacing support distance by point distance costs at most $2L$; the remaining lattice sum is bounded by $(1 + 6/g)^3$. Since $t \exp(-gt/4) \le 2/g$, we may take

$$S_P = \exp(\gamma_H L/2)(L + 2/\gamma_H) c_{\mathrm{bas}}(\gamma_H/4), \quad \gamma_P = \min\{\gamma_H/2,\ (2c_P C_{\mathrm{Dt}} S_P)^{-1}\}.$$

With

$$C_P^{\mathrm{ph}} = \max\{C_H(\alpha), C_{\mathrm{Dt}}, (2c_P)^{1/2}, 4\exp(2)\},$$

all the kernels in eq. 3.xi, eq. 3.xxxii, and the positive square root satisfy their respective bounds at rate γ_P with constant C_P^{ph}. In particular,

$$C_k^{1/2} = \widetilde{\Delta}_k^{-1/2}, \qquad |C_k^{1/2}(\Upsilon, \Upsilon')| \le C_P^{\mathrm{ph}} \exp(-\gamma_P d(\Upsilon, \Upsilon')). \tag{3.xxxiv}$$

At $k = 0$, $H_0^{\mathrm{ph}} = 1$; its first differences and their Hölder differences are supported within distance two and have entries of modulus at most two. Our choice $C_P^{\mathrm{ph}} \geq 4\exp(2)$ therefore includes this level as well.

3.1.5. *The exact block decomposition.* We now separate a field of prescribed next-scale average into its energy minimizer and an independent Gaussian fluctuation. This is a second minimization: the energy on the unit lattice is the one induced by the fine reconstruction, namely $\langle A, \Delta_k A\rangle = \|dH_k^{\mathrm{ph}} A\|_2^2$. For a real bond field A' on the unit lattice of the torus of side ℓ/L, put $A'_L(b) = L^{-1/2}A'(b/L)$. Let $A_k^{\min}(A')$ minimize $\langle A, \Delta_k A\rangle$ under

$$\mathbf{Q}A = A'_L, \qquad \tau A = 0.$$

This affine space is nonempty: set each face bond of the face c equal to $LA'_L(c)$, and set every interior bond to zero. The gauge correction eq. 3.vii sends every unit field with this average to the axial slice without changing its reconstructed energy. Thus minimizing on the slice gives the same minimum as imposing only the average. Its direction is $C(V_k) = \mathcal{Z}_k$, and eq. 3.xxxii makes the quadratic form positive definite on this direction. The minimizer is therefore unique and is characterized by

$$\langle A_k^{\min}(A'), \Delta_k C z\rangle = 0 \quad (z \in V_k). \tag{3.xxxv}$$

Uniqueness in this real linear system makes $A_k^{\min}$ real and linear.

Define $\mathrm{Sq}_k = CC_k^{1/2}$. It maps V_k bijectively onto the axial fluctuations, has zero block average and mean, and

$$\langle \mathrm{Sq}_k\zeta, \Delta_k \mathrm{Sq}_k\zeta'\rangle = \langle \zeta, C_k^{1/2}\widetilde{\Delta}_k C_k^{1/2}\zeta'\rangle = \langle \zeta, \zeta'\rangle.$$

Thus the exact Gaussian decomposition is

$$\langle A_k^{\min}(A') + \mathrm{Sq}_k\zeta, \Delta_k(A_k^{\min}(A') + \mathrm{Sq}_k\zeta)\rangle = \langle A_k^{\min}(A'), \Delta_k A_k^{\min}(A')\rangle + \|\zeta\|_2^2. \tag{3.xxxvi}$$

The fields with $\mathbf{Q}A = A'_L$ and $\tau A = 0$ form the affine space $A_k^{\min}(A') + \mathcal{Z}_k$. Since $\mathrm{Sq}_k : V_k \to \mathcal{Z}_k$ is a bijection, each such field has a unique representation $A = A_k^{\min}(A') + \mathrm{Sq}_k\zeta$. Equip this affine space with the volume induced by the counting inner product on unit bonds. The change of variables has a constant Jacobian $j > 0$. By eq. 3.xxxvi, its unnormalized Gaussian density becomes

$$j\exp(-\tfrac{1}{2}\langle A_k^{\min}(A'), \Delta_k A_k^{\min}(A')\rangle)e^{-\|\zeta\|_2^2/2}\,\mathrm{d}\zeta.$$

Its integral is the same constant factor times $(2\pi)^{\dim V_k/2}$. Thus the normalized measure is the image of μ_I under this affine map; see also [6, Sections 3.9–3.10, eqs. (141), (151)–(152)]. The block integration uses this standard Gaussian restricted to Box_k. The loss of normalization is exactly eq. 3.xxxi.

We also need the relation between the two backgrounds. With the scalar scaling $\lambda_L(x) = L^{1/2}\lambda(x/L)$, there is a unique real scalar $\omega_k(A')$ with $\mathbf{Q}_{k+1}\omega_k(A') = 0$ such that

$$H_k^{\mathrm{ph}} A_k^{\min}(A') = (H_{k+1}^{\mathrm{ph}} A')_L - d\omega_k(A'), \qquad \|dH_k^{\mathrm{ph}} A_k^{\min}(A')\|_2 = \|dH_{k+1}^{\mathrm{ph}} A'\|_2. \tag{3.xxxvii}$$

The harmonic part of A' is arbitrary. Here $\mathbf{Q}_{k+1} = \mathbf{Q}\mathbf{Q}_k$ is the average from the mesh-η_k lattice of side ℓ to its L-spaced lattice. The operator H_{k+1}^{ph} is formed at mesh η_{k+1} and side ℓ/L, before scaling by L.

Here is the minimizing argument. At mesh η, the scaled field $(H_{k+1}^{\mathrm{ph}} A')_L$ minimizes the curvature norm under the $(k+1)$-step constraint $\mathbf{Q}_{k+1}\mathscr{A} = A'_L$. This follows by scaling the defining problem; the curvature L^2 norm is unchanged. The field $\mathscr{B} = H_k^{\mathrm{ph}} A_k^{\min}(A')$ has the same average. If $\mathscr{A}$ is any competitor, set $A = \mathbf{Q}_k\mathscr{A}$. Choose the one-step restricted gauge

$$\mu(x) = (\tau A)(y, x) - L^{-3}\sum_{x' \in B(y)} (\tau A)(y, x'), \qquad x \in B(y).$$

Then $\mathbf{Q}\mu = 0$ and $\tau(A - d\mu) = 0$, by telescoping the exact-field path sums. Therefore

$$\|d\mathscr{A}\|_2^2 \geq \|dH_k^{\mathrm{ph}} A\|_2^2 = \|dH_k^{\mathrm{ph}}(A - d\mu)\|_2^2 \geq \|d\mathscr{B}\|_2^2.$$

Both backgrounds are minimizers of the same problem. The Landau-minimizer argument shows that their difference is the coboundary of a scalar with zero $(k+1)$-step average. This scalar is unique: a closed field with zero average is exact, and the only constant scalar with zero block average is zero. The two backgrounds are real and linear in A', hence so is ω_k. This proves eq. 3.xxxvii and identifies the first term on the right of eq. 3.xxxvi with $\|dH_{k+1}^{\mathrm{ph}} A'\|_2^2$.

3.1.6. *Curvature control in local gauges.* The inverse estimates will require a small connection in each local gauge, even when its harmonic part is large. We therefore bound the reconstructed field modulo a real gauge by its curvature. The proof uses the local strips from the axial estimate and the exponential reconstruction kernel to suppress the distant seams of a periodic gauge.

Assume from now on that $\ell/(LM)$ is odd and at least 13. For an M-cube $\square$, let $\square^\dagger$ be its enlargement by LM in each direction. Its side is $s_d = (2L+1)M \leq 3\ell/13$. For a bond field B on a set W, write

$$\|B\|_{1,W} = \|B\|_{\infty,W}, \qquad \|B\|_{2,W} = \|\partial B\|_{\infty,W}, \qquad \|B\|_{3,W} = \|\delta_\alpha \partial B\|_{\infty,W}.$$

Only bonds and differences contained in W enter these norms. They are ordinary differences of the photon coefficients; no connection occurs in their definition.

Proposition 3.13. For every real bond field A on the unit lattice, including an arbitrary constant part, and every M-cube $\square$, there is a real scalar ν on $\square^\dagger$ such that

$$\|H_k^{\mathrm{ph}} A - d\nu\|_{i,\square^\dagger} \leq C_B \|dA\|_\infty, \qquad i = 1, 2, 3. \tag{3.xxxviii}$$

The constant C_B depends on L, M, α and is independent of k, ℓ and the harmonic part of A. ■

Proof. Let a be the mean of A, put $A^0 = A - a$, and choose the center x_0 of $\square$ as base point. On the canonical lifted period cell

$$W = x_0 + \{-(\ell-1)/2, \dots, (\ell-1)/2\}^3$$

define $\lambda(x)$ to be the line sum of A^0 along the rectilinear path from x_0 to x, in coordinate order $1, 2, 3$. This defines a scalar on the torus. Put $A'' = A^0 - d\lambda$ and $p_A = \|dA\|_\infty$. The path sums of A'' vanish. For a bond whose two ends lie in W without crossing its boundary, the strip identity used in the axial estimate gives

$$|A''(x, x+e_\mu)| \leq \sum_{\nu>\mu} |x_\nu - x_{0,\nu}| p_A \leq 2d(x, x_0) p_A. \tag{3.xxxix}$$

Indeed, commuting the final bond through the later portions of the two base-point paths sweeps one strip in each direction $\nu > \mu$. Each strip contributes its plaquette sum, and both remaining path sums are zero.

A seam bond lies on a closed coordinate line with $\ell - 1$ inner bonds. Let $\mathrm{hol}(v)$ be the line sum of A^0 in that direction. Translating this line by one transverse step changes its sum by at most ℓp_A, by summing the intervening ℓ plaquettes. Its mean over the transverse two-torus is zero, since A^0 has mean zero. The mean distance in each coordinate circle is at most $\ell/4$, hence $|\mathrm{hol}(v)| \leq \ell^2 p_A/2$. The line sum of $d\lambda$ around a closed line vanishes. Removing the inner bonds using eq. 3.xxxix therefore gives

$$|A''(b')| \leq |\mathrm{hol}(v)| + (\ell-1)\ell p_A \leq \tfrac{3}{2}\ell^2 p_A \tag{3.xl}$$

for the seam value. In particular this construction also gives a global real gauge with $\|A^0 - d\lambda\|_\infty \leq 2\ell^2 \|dA\|_\infty$.

Since $H_k^{\mathrm{ph}} d\lambda$ is closed and has mean zero, it is exact, say $d\nu'$. On the nonwrapping cube $\square^\dagger$, let x denote lifted coordinates and set $\nu = a \cdot x + \nu'$. Then $H_k^{\mathrm{ph}} A - d\nu = H_k^{\mathrm{ph}} A''$ there. If $b \subset \square^\dagger$ and b' is inner, its initial point is at distance at most $d(b, b') + 2$ from an end of b. Thus eq. 3.xxxix gives

$$|A''(b')| \le (2d(b, b') + 4 + s_d) p_A.$$

For a seam bond, $d(b, b') \ge (\ell - 1 - s_d)/2$.

Put $g = \gamma_P$ and $c_{\mathrm{bd}}(g) = 3\exp(3g)(1 + 6/g)^3$, which bounds the sum of $\exp(-g d(b, b'))$ over unit bonds. Summing any of the three kernel bounds against the inner bonds gives

$$C_P^{\mathrm{ph}} p_A [(s_d + 4) c_{\mathrm{bd}}(g) + (2/g) c_{\mathrm{bd}}(g/2)],$$

using $2t \exp(-gt) \le (2/g)\exp(-gt/2)$. For the seam contribution split off half the decay. Since $s_d \le 3\ell/13$,

$$\exp(-g d(b, b')) \le \exp(g/4)\exp(-5g\ell/26)\exp(-g d(b, b')/2), \qquad \ell^2 \exp(-5g\ell/26) \le 15/g^2.$$

Together with eq. 3.xl, this bounds the seam contribution by $62 C_P^{\mathrm{ph}} p_A g^{-2} c_{\mathrm{bd}}(g/2)$. The proposition follows with

$$C_B = C_P^{\mathrm{ph}} [(s_d + 4) c_{\mathrm{bd}}(\gamma_P) + (2/\gamma_P + 62/\gamma_P^2) c_{\mathrm{bd}}(\gamma_P/2)]. \tag{3.xli}$$

□

3.1.7. *Weakened photon kernels.* To localize the later Gaussian integration, we attach auxiliary parameters to sets of blocks joining the two indices of a kernel. At parameter value one we recover the exact kernels. Setting parameters to zero will disconnect components exactly, while a polydisc larger than the unit disc must retain exponential decay. The path counts below compare the growth of each parameter product with the decay available in the exact kernel.

For $a = M$ or LM, let π_a consist of the closed cubes of half-side a centered at $a\mathbb{Z}^3/\ell\mathbb{Z}^3$. These are overlapping parameter cubes, distinct from the cubes that partition polymers. Two parameter cubes are linked when they contain a common lattice site; components below refer to chains of such links. For a set S of unit sites, let $\mathrm{Cub}_a(S)$ be all parameter cubes meeting S.

Choose the shortest rectilinear path $\mathrm{Pth}(p, p')$ in coordinate order $1, 2, 3$. Its signed coordinate offsets are uniquely determined because ℓ is odd. A site belongs to at most 27 parameter cubes. If S is connected by nearest-neighbor steps, then $\mathrm{Cub}_a(S)$ is linked: consecutive sites lie together in a cube, and all cubes containing either site are linked to this cube. Along a coordinate segment, at most nine transverse choices of centers occur. Counting the longitudinal centers for the three segments of $\mathrm{Pth}(p, p')$ gives

$$\#\mathrm{Cub}_a(\mathrm{Pth}(p, p')) \le 9(3d(p, p')/a + 9). \tag{3.xlii}$$

An L-block, and the ends of a unit bond, meet at most 27 cubes of either kind.

We smooth only in the row variable of the minimizer. Let

$$\vartheta(r) = \begin{cases} 0 & r \le 0, \\ 3r^2 - 2r^3 & 0 \le r \le 1, \\ 1 & r \ge 1, \end{cases} \qquad \varphi(t) = \vartheta(2(3/4 - |t|)), \qquad \chi_c(x) = \prod_{\nu=1}^{3} \varphi((x_\nu - c_\nu)/M).$$

The coordinates in the last expression are represented with modulus at most $\ell/2$. Put $\chi_c(b) = \chi_c(x_b)$, where x_b is the midpoint of b. The identity $\vartheta(r) + \vartheta(1 - r) = 1$ implies $\sum_{j \in \mathbb{Z}} \varphi(t - j) = 1$, and therefore $\chi_c \ge 0$ and $\sum_c \chi_c = 1$ on the torus. Moreover φ' is bounded by three and is Lipschitz with constant 24. Tensor products and the averaged derivative formula for the discrete difference give, for $\tilde{b} = b + r$ with $|r| \le 1$,

$$|\partial_\nu \chi_c(b)| \le 3/M, \qquad |\chi_c(\tilde{b}) - \chi_c(b)| \le 9|r|/M, \qquad |\partial_\nu \chi_c(\tilde{b}) - \partial_\nu \chi_c(b)| \le 42|r|/M^2. \tag{3.xliii}$$

For the last bound the derivative factor contributes $24/M^2$, and the two remaining factors contribute $2 \cdot 9/M^2$. If $\chi_c(b) \ne 0$, both ends of b belong to the cube centered at c.

For a unit bond b' with ends p'_1, p'_2, define

$$P'(c,b') = \mathrm{Cub}_M(\mathrm{Pth}(c,p'_1) \cup \mathrm{Pth}(c,p'_2)), \qquad w(b,b';s) = \sum_c \chi_c(b) \prod_{a \in P'(c,b')} s_a.$$

For an interior basis vector set $\mathrm{St}(e_i^y) = B(y)$ and $\mathrm{Pt}(e_i^y) = \{y\}$; for δ_b, both sets are the two ends of b. Put

$$P''(\Upsilon,\Upsilon') = \mathrm{Cub}_{LM}\left(\mathrm{St}(\Upsilon) \cup \mathrm{St}(\Upsilon') \cup \bigcup_{\substack{p \in \mathrm{Pt}(\Upsilon) \\ p' \in \mathrm{Pt}(\Upsilon')}} \mathrm{Pth}(p,p')\right).$$

The weakened kernels are

$$\begin{aligned} H_k^{\mathrm{ph}}(s)(b,b') &= H_k^{\mathrm{ph}}(b,b')w(b,b';s), \quad (k \geq 1), \\ C_k^{1/2}(s)(\Upsilon,\Upsilon') &= C_k^{1/2}(\Upsilon,\Upsilon') \prod_{a \in P''(\Upsilon,\Upsilon')} s_a, \quad (k \geq 0). \end{aligned} \tag{3.xliv}$$

The two families have parameter sets π_M and π_{LM}, respectively. When both occur in one expression, s denotes their pair. Each entry is a polynomial, independent of the connection and real for real parameters. At $s = 1$ the exact operators are recovered. For general complex parameters, the second family is defined by the displayed kernel; positivity or the square-root identity is used only at $s = 1$.

Fix $0 < \alpha_0 \leq 1/4$ and take

$$M = L^m \geq L^2, \qquad 2\alpha_0 n_P \log M \leq \gamma_P M, \qquad n_P = 402. \tag{3.xlv}$$

Parameters range over the closed polydisc $|s_a| \leq M^{\alpha_0}$. Such an M exists because $(\log M)/M \to 0$. If $\chi_c(\hat{b}) \neq 0$ for a translate of b by distance at most two, then $d(c,x_b) \leq M+1$; applying eq. 3.xlii to both paths gives

$$\#P'(c,b') \leq 234 + 54d(b,b')/M \leq n_P(1 + d(b,b')/M).$$

At most 27 such centers occur: in each coordinate their representatives lie in an interval of length $3M/2 + 5 < 3M$. The set P'' has two endpoint sets and at most four paths. Since their endpoints are within L of the respective supports, the same count gives

$$\#P''(\Upsilon,\Upsilon') \leq 378 + 216/M + 108d(\Upsilon,\Upsilon')/(LM) \leq n_P(1 + d(\Upsilon,\Upsilon')/M).$$

For either product, with corresponding distance D, we have

$$\left|\prod_{a \in P} s_a\right| \leq M^{\alpha_0 n_P} \exp(\alpha_0 n_P D \log M/M) \leq M^{\alpha_0 n_P} \exp(\gamma_P D/2). \tag{3.xlvi}$$

Multiplying by the unweakened kernel bound leaves decay $e^{-\gamma_P D/2}$, uniformly on the parameter polydisc.

Proposition 3.14. Under eq. 3.xlv, put $\gamma^P = \gamma_P/2$ and $C^P = 116 C_P^{\mathrm{ph}} M^{\alpha_0 n_P}$. The three kernels of $H_k^{\mathrm{ph}}(s)$ satisfy

$$|H_k^{\mathrm{ph}}(s)(b,b')|, |\partial H_k^{\mathrm{ph}}(s)(b,b')|, |\delta_\alpha \partial H_k^{\mathrm{ph}}(s)((b,\tilde{b});b')| \leq C^P \exp(-\gamma^P d(b,b')), \tag{3.xlvii}$$

and

$$|C_k^{1/2}(s)(\Upsilon,\Upsilon')| \leq C^P \exp(-\gamma^P d(\Upsilon,\Upsilon')). \tag{3.xlviii}$$

These bounds hold on the closed polydiscs. ∎

Proof. The root estimate follows at once from eq. 3.xxxiv and eq. 3.xlvi. For the minimizer, fix b' and abbreviate $H(b) = H_k^{\mathrm{ph}}(b, b')$, $w(b) = w(b, b'; s)$ and $b^+ = b + \eta e_\nu$. Put

$$\mathrm{Pm} = M^{\alpha_0 n_P} \exp(\gamma_P d(b, b')/2).$$

For $b, b^+, \tilde{b}, \tilde{b}^+$, and the intervening translates, eq. 3.xliii and the 27-center count give

$$|w| \le \mathrm{Pm}, \quad |\partial_\nu w| \le 9\mathrm{Pm}, \quad |w(\tilde{b}^+) - w(b^+)| \le 27|r|\mathrm{Pm}, \quad |\partial_\nu w(\tilde{b}) - \partial_\nu w(b)| \le 14|r|\mathrm{Pm}.$$

Here $\eta \le 1/3$ and $M \ge 9$; all translates are within two of the initial point. A rectilinear path between b and $\tilde{b} = b + r$ has at most $3|r|/\eta$ steps, at distance at most one from b. Summing the first-derivative bound for H along this path gives

$$|H(\tilde{b}) - H(b)| \le 3|r| C_P^{\mathrm{ph}} \exp(\gamma_P) \exp(-\gamma_P d(b, b')).$$

Use the discrete product rule

$$\partial_\nu (Hw)(b) = (\partial_\nu H)(b) w(b^+) + H(b) \partial_\nu w(b).$$

It bounds the zeroth and first kernels with factors one and ten. Subtracting this formula at b and $\tilde{b}$ gives four terms: the difference of $\partial_\nu H$, the difference of $w(b^+)$, the difference of H, and the difference of $\partial_\nu w$. Their bounds have factors $1, 27, 27\exp(\gamma_P), 14$, respectively, times $|r|^\alpha C_P^{\mathrm{ph}} \mathrm{Pm} \exp(-\gamma_P d(b, b'))$. We used $|r| \le |r|^\alpha$ and $\gamma_P \le 1$. Their sum is at most 116 times this expression, proving eq. 3.xlvii. □

At level zero the row smoothing is replaced by the diagonal family

$$H_0^{\mathrm{ph}}(s)(b, b') = \delta_{bb'} \prod_{a \in \pi_M : b \subset a} s_a. \tag{3.xlix}$$

Every product has at most 27 factors and at least one. The first difference has two diagonal terms; the Hölder difference has four, supported within distance two. Thus their entries are bounded by $4\exp(2\gamma^P) M^{27\alpha_0} \exp(-\gamma^P d(b, b'))$, which is at most the right side of eq. 3.xlvii by $C_P^{\mathrm{ph}} \ge 4\exp(2)$.

The weakening also separates components exactly. Suppose that $s_a = 0$ outside a set $\mathscr{Y}$ of parameter cubes, and let Y_1 be one linked component of their union. A linked path set P meeting this component either lies entirely in it, or contains a cube outside $\mathscr{Y}$ and its parameter product vanishes. Both P' and P'' are linked and contain every parameter cube meeting the respective column support; P'' also contains every cube meeting the row support. In the first family, every contributing row cube c contains the fine bond b and belongs to $P'(c, b')$. It follows that a row or column supported in Y_1 depends only on parameters in that component and has no nonzero entry at an index outside it. A nonzero row is supported in some component. The diagonal family has the same properties because all cubes containing b are linked through an end of b.

Coordinate signs and center-lattice translations preserve the ordered paths, constraints and basis up to signed permutation. Uniqueness gives covariance of the exact kernels, and the parameter products and χ_c transform accordingly. Thus the weakened H_k^{ph} is covariant under signs and M translations, and $C_k^{1/2}$ under signs and LM translations, with transformed parameters.

3.1.8. *The domain bounds for block integration.* The block step needs a domain stable under its real background and its Gaussian fluctuations. We allow a real field that is small only after a local gauge change, together with a globally small complex remainder. This permits arbitrary real harmonic fields while giving an analytic margin for the fluctuation. Take $0 < \varepsilon \le 1/200$, $e_k \in (0, 1)$, and put $t_{i,k} = e_k^{-3/4+i\varepsilon}$ for $i = 1, 2, 3$. The set R_k consists of complex fine bond fields with $\|B\|_i < t_{i,k}$ in the three norms. A field belongs to $c\widetilde{R}_k$, $0 < c \le 1$, if it has a decomposition $B = B_0 + B_1$, with $B_1 \in cR_k$ and B_0 real, such that on each $\square^\dagger$ the field B_0 is gauge equivalent by a real scalar to a real field with norms strictly below $ct_{i,k}$. The decomposition and gauges need only exist. For $\widetilde{R}_k^-$ the

local real representative has the bounds $t_{i,k}$, while the global term B_1 has the bounds $(1-e_k^{\varepsilon})t_{i,k}$. The domain Dom_k consists of the real bond fields A on the unit lattice for which

$$H_k^{\mathrm{ph}} A \in \tfrac{1}{4}\widetilde{R}_k.$$

These definitions permit every real harmonic part.

Let $C_w \geq C^P$ and $0 < \gamma_w \leq \gamma^P$, and choose c_γ to bound both sums

$$\sup_{\Upsilon} \sum_{\Upsilon'} \exp(-\gamma_w d(\Upsilon, \Upsilon')), \qquad \sup_{b} \sum_{b'} \exp(-\gamma_w d(b, b'))$$

over basis elements and unit bonds, respectively. The counts above make c_γ independent of k, ℓ. Choose fixed cutoff exponents $p_0, p > 0$ and set $p_{0,k} = (-\log e_k)^{p_0}$ and $p_k = (-\log e_k)^p$. Their later choice is part of the renormalization parameters. Assume

$$\begin{aligned} p_{0,k} &\leq e_k^{-\varepsilon/4}, \qquad 2C_w^2 c_\gamma^2 c_C e_k^{\varepsilon/4} \leq 1, \\ 8C_B p_k &\leq t_{3,k}, \qquad 8e_k^{3/4-7\varepsilon/2} \leq 1, \qquad L^{3/4} \geq 8/3. \end{aligned} \tag{3.l}$$

Since the constants are fixed before the charge, these are conditions on an upper bound $\bar{e}_0$, valid for every $0 < e_k \leq \bar{e}_0$.

Proposition 3.15. Under eq. 3.l, every real bond field on the unit lattice with $\|dA\|_\infty \leq p_k$ belongs to Dom_k. For every $\zeta \in \mathrm{Box}_k$ and every pair of parameters in the closed polydiscs, the field

$$\mathscr{Z}_k(s, \zeta) = H_k^{\mathrm{ph}}(s) C C_k^{1/2}(s) \zeta$$

has all three norms bounded by $\frac{1}{2}e_k^{-\varepsilon/2}$. If ℓ/L is also an admissible side and $e_{k+1} = L^{1/2} e_k$, then

$$A_k^{\min}(A') + \mathrm{Sq}_k \zeta \in \mathrm{Dom}_k \quad (A' \in \mathrm{Dom}_{k+1},\ \zeta \in \mathrm{Box}_k). \tag{3.li}$$

■

Proof. The first assertion follows from eq. 3.xxxviii: its local real representative has each norm at most $C_B p_k \leq t_{3,k}/8 < t_{i,k}/4$. Use $B_1 = 0$ in the domain decomposition.

The root row sum gives

$$\sup_{\Upsilon} |(C_k^{1/2}(s)\zeta)(\Upsilon)| \leq C_w c_\gamma e_k^{-\varepsilon/4}.$$

Applying the real bound for C to its row signs gives $\sum_\Upsilon |C(b, \Upsilon)| \leq c_C$, also controlling complex inputs. Each of the three reconstruction norms has row bound $C_w c_\gamma$, so all three norms of $\mathscr{Z}_k(s, \zeta)$ are at most

$$C_w^2 c_\gamma^2 c_C e_k^{-\varepsilon/4} \leq \tfrac{1}{2} e_k^{-\varepsilon/2}.$$

These estimates include the diagonal family at $k = 0$.

For eq. 3.li, write $\mathscr{A}' = H_{k+1}^{\mathrm{ph}} A'$. It is real and belongs to $\frac{1}{4}\widetilde{R}_{k+1}$. On any coarse enlarged cube there is a real gauge in which $\mathscr{A}'$ is represented by a real field A^c with i-th norm below $t_{i,k+1}/2$: add the real global summand to the local representative. For any fine M-cube $\square$, choose a coarse cube $\square'$ with $\square \subset L\square'$. Then $\square^\dagger \subset L(\square'^\dagger)$. The scaling of the three norms is $L^{-1/2}, L^{-3/2}, L^{-3/2-\alpha}$, respectively, and

$$t_{i,k+1} = L^{-3/8+i\varepsilon/2} t_{i,k}.$$

Thus $(A^c)_L$ has i-th norm strictly below $\frac{1}{2}L^{-7/8+3\varepsilon/2} t_{i,k} \leq \frac{1}{2}L^{-3/4} t_{i,k} \leq 3t_{i,k}/16$. By eq. 3.xxxvii, the new background on $\square^\dagger$ is gauge equivalent to $(A^c)_L + \mathscr{Z}_k(1, \zeta)$. The latter fluctuation is real, and its norms satisfy

$$\tfrac{1}{2} e_k^{-\varepsilon/2} \leq t_{3,k}/16 \leq t_{i,k}/16$$

by eq. 3.l. The sum has norms strictly below $t_{i,k}/4$. This is the required local representative, again with global summand zero. □

The weakened fluctuation also gives the complex radius used in the later Taylor expansions. If $A \in \widetilde{R}_k^-$ and

$$|t| \le \mathrm{rad}_1 := \tfrac{1}{2} e_k^{-3/4+9\varepsilon/2}, \tag{3.lii}$$

then $A + t\mathscr{Z}_k(s,\zeta) \in \widetilde{R}_k$. Indeed the three added norms are at most $\frac{1}{4} e_k^{-3/4+4\varepsilon} < e_k^{\varepsilon} t_{3,k}$; this fits the margin in the global summand of $\widetilde{R}_k^-$. The fluctuation is a polynomial in the two parameter families and linear in ζ.

3.1.9. *Consistency of the exact operators on covers.* Let $\ell_2 = J\ell_1$ with J odd and with both sides admissible. Write pj^* for periodic pullback, also on the coarse unit lattices and on fluctuation coordinates. The common reflection eigenbasis in each block makes the last pullback well defined. Its image is the subspace of ℓ_1-periodic fields, and

$$\langle \mathrm{pj}^* f, \mathrm{pj}^* h \rangle_{\ell_2} = J^3 \langle f, h \rangle_{\ell_1}.$$

All exact photon operators commute with this pullback:

$$O^{(2)} \mathrm{pj}^* = \mathrm{pj}^* O^{(1)}, \qquad O = H_k^{\mathrm{ph}}, \Delta_k, C, \widetilde{\Delta}_k, (\widetilde{\Delta}_k + x)^{-1}, C_k^{1/2}, \mathrm{Sq}_k. \tag{3.liii}$$

The superscript specifies the torus; $x \ge 0$. Also

$$(A_k^{\min})^{(2)} (\mathrm{pj}^* A') = \mathrm{pj}^* (A_k^{\min})^{(1)} (A'). \tag{3.liv}$$

The finite rows of $d, \delta, \Delta_{\mathrm{sc}}, \mathbf{Q}_j, \mathbf{Q}_j^T, \tau, C$ commute with pullback and with the deck translations. Pulling back eq. 3.v and using uniqueness gives the identity for H_k^{ph}. Adjoints preserve the periodic subspaces, and pairing two pullbacks cancels their common factor J^3, so the identity also holds for transposes. Products give Δ_k and $\widetilde{\Delta}_k$; inversion gives the resolvents and eq. 3.xxxiii gives $C_k^{1/2}$, hence Sq_k.

Finally let $A = (A_k^{\min})^{(1)}(A')$ and test its pullback against a larger-torus axial fluctuation $\widetilde{Z}$. Sum the translates of $\widetilde{Z}$ over $\ell_1\mathbb{Z}^3/\ell_2\mathbb{Z}^3$ to obtain $\mathrm{pj}^* Z$. Its average and axial sums vanish, so $Z \in C(V_k)$ on the smaller torus. The field $\Delta_k^{(2)} \mathrm{pj}^* A$ is periodic; consequently

$$\langle \mathrm{pj}^* A, \Delta_k^{(2)} \widetilde{Z} \rangle_{\ell_2} = J^{-3} \langle \mathrm{pj}^* \Delta_k^{(1)} A, \mathrm{pj}^* Z \rangle_{\ell_2} = \langle A, \Delta_k^{(1)} Z \rangle_{\ell_1} = 0.$$

Its average and axial constraints also pull back. The characterization eq. 3.xxxv proves eq. 3.liv.

4. Block averaging with fermions

4.1. **Chamber propagators.** The chamber fiber has one four-component subspace constant in its 48 labels. We call this the light subspace and its orthogonal complement the heavy subspace. Internal hopping has a gap on the heavy subspace. The light part retains the low-momentum Dirac behavior, which will be controlled by penalties on block averages. Together these estimates give local inverses with bounds independent of the fine mesh. We then join the local inverses by a convergent walk expansion to obtain the propagator.

4.1.1. *The cluster decomposition.* Let $a > 0$ be a mesh. A fermion site is written $x = (c, g)$, where $c \in a\mathbb{Z}^3$ and $g \in O_{\mathrm{cub}}$. In these coordinates the four neighbors are

$$x^{(i)} = (c, g s_i) \quad (0 \le i \le 2), \qquad x^{(3)} = (c + a g e_1, g r_{e_1}).$$

Here s_0 changes the sign of the third coordinate, s_1 interchanges the second and third coordinates, s_2 interchanges the first and second coordinates. The matrix r_{e_1} changes the sign of the first coordinate; it is the linear part of the affine wall reflection s_3 used in the regulator. The inward normals of the four walls are

$$n_0 = e_3, \qquad n_1 = \frac{e_2 - e_3}{\sqrt{2}}, \qquad n_2 = \frac{e_1 - e_2}{\sqrt{2}}, \qquad n_3 = -e_1.$$

Thus $\widehat{d}_i(c,g) = -gn_i$ and

$$M_i(c,g) = \frac{w_i}{2}(\rho - \gamma \cdot \widehat{d}_i(c,g)), \qquad (w_0,w_1,w_2,w_3) = (6, 6\sqrt{2}, 6\sqrt{2}, 6).$$

Throughout this subsection $0 < \rho < 1$ is fixed.

The discussion applies on the covering lattice and on a torus of side $R \in a\mathbb{N}$, with any unitary character $f(c + Re_\mu, g) = -\exp(i\theta_\mu) f(c,g)$. Pairings on the torus are formed over a period cell. The notation $d(c,c')$ denotes the sup distance modulo $R\mathbb{Z}^3$, or the covering sup distance when $R = \infty$. The chamber measure is

$$\int \mathrm{d}x\, v(x) = \frac{a^3}{48} \sum_c \sum_g v(c,g), \qquad \|f\|_2^2 = \int \mathrm{d}x\, |f(x)|_2^2.$$

On four-component spinors depending only on c, the measure is $a^3 \sum_c$. The cluster injection and its adjoint are

$$(J\xi)(c,g) = \xi(c), \qquad (J^* f)(c) = \frac{1}{48} \sum_g f(c,g).$$

Set $P = JJ^*$ and $Q = 1 - P$. The decomposition $f = J\xi + h$, with $\xi = J^* f$ and $h = Qf$, is orthogonal. In particular,

$$\|f\|_2^2 = \|\xi\|_2^2 + \|h\|_2^2. \tag{4.i}$$

The normalization by 48 makes J an isometry. Kernels on the chamber lattice are taken with respect to $\mathrm{d}x$, so the kernel of the identity is $48a^{-3}\mathbf{1}[x = x']1_4$.

Let A be a connection and write $u_A(c,c') = \exp(ie_a a A(c,c'))$ on a photon bond. The charge e_a in this notation is the charge at the level under consideration. The phases on the internal bonds are one. Put

$$R_i(A) f(x) = U_A(x, x^{(i)}) f(x^{(i)}), \qquad \mathcal{L}_A = \sum_{i=0}^{3} w_i (1 - R_i(A)).$$

The massless chamber operator is

$$D_A f(x) = -a^{-1} \sum_{i=0}^{3} M_i(x) (U_A(x, x^{(i)}) f(x^{(i)}) - f(x)). \tag{4.ii}$$

At level k, $a = \eta_k$ and dilation to the unit lattice gives

$$D_A = \eta_k^{-1} D^{\mathrm{ch}}[B_k], \qquad B_k(c/\eta_k, c'/\eta_k) = e_k \eta_k A(c,c'). \tag{4.iii}$$

The reference mass is added separately as $D_A + \mu_k$. For real A, the involutions $R_i(A)$ are unitary and self-adjoint, and the bond expansion gives

$$\mathrm{Re}(f, D_A f) = \frac{\rho}{2a}(f, \mathcal{L}_A f), \qquad D_A^* = \gamma_5 D_A \gamma_5, \qquad \gamma_5 = \begin{pmatrix} 0 & 1_2 \\ 1_2 & 0 \end{pmatrix}. \tag{4.iv}$$

Here γ_5 is Hermitian, $\gamma_5^2 = 1$, and it anticommutes with each γ_μ. Each unoriented bond contributes $a^3 w_i |U_A(x, x^{(i)}) f(x^{(i)}) - f(x)|_2^2 / 48$ to the energy.

On one cluster let

$$\mathcal{L}_{\mathrm{int}} = \sum_{i=0}^{2} w_i (1 - R_i), \qquad R_i v(g) = v(g s_i).$$

Its kernel consists of the constants: s_1, s_2 give adjacent swaps and their conjugates of s_0 give each sign change, generating O_{cub}. Denote its least positive eigenvalue by λ_{int} and set

$$g_{\mathrm{h}} = \frac{\rho \lambda_{\mathrm{int}}}{2}, \qquad c_* = \frac{\min(w_3, \lambda_{\mathrm{int}})}{18}, \qquad c_{\mathrm{lh}} = \sqrt{3}(1 + \rho).$$

No numerical evaluation of λ_{int} is needed.

Proposition 4.1. For real A and $\xi = J^*f$,

$$(f, \mathcal{L}_A f) \geq \lambda_{\mathrm{int}} \|Qf\|_2^2, \qquad a^{-2}(f, \mathcal{L}_A f) \geq c_* \|\nabla_A \xi\|_2^2, \tag{4.v}$$

where $\nabla_{A,\mu}\xi(c) = a^{-1}(u_A(c, c+ae_\mu)\xi(c+ae_\mu) - \xi(c))$. The same inequalities hold on a finite set E of photon sites, with the energy restricted to bonds whose endpoints lie over E and the photon differences restricted to bonds in E. Moreover,

$$\|QD_A J\xi\|_2 \leq c_{\mathrm{lh}} \|\nabla_A \xi\|_2, \qquad |(\xi, J^* D_A Q f)| \leq c_{\mathrm{lh}} \|\nabla_A \xi\|_2 \|f\|_2. \tag{4.vi}$$

On $\operatorname{ran} Q$, for $t > -g_{\mathrm{h}}/a$,

$$\|(QD_A Q + t)^{-1}\|_{2\to 2} \leq (t + g_{\mathrm{h}}/a)^{-1}. \tag{4.vii}$$

All constants are independent of a, R, the character, and the real connection. ■

Proof. Sum the internal gap $a^3(f(c,\cdot), \mathcal{L}_{\mathrm{int}} f(c,\cdot)) \geq \lambda_{\mathrm{int}} a^3 \|Qf(c,\cdot)\|^2$ over $c \in E$.

For the second bound, consider a photon bond $c' = c + av$. The eight labels with $ge_1 = v$ correspond to the eight chamber bonds crossing it. Write $\mathcal{O}_v(c)$ for this octet. Since their endpoints form $\mathcal{O}_{-v}(c')$, the difference of the cluster means is the sum of the mean of the eight bond differences and two means of Qf:

$$\begin{aligned} u_A(c,c')\xi(c') - \xi(c) = \frac{1}{8} \sum_{x \in \mathcal{O}_v(c)} (U_A(x, x^{(3)}) f(x^{(3)}) - f(x)) \\ - \frac{u_A(c,c')}{8} \sum_{x' \in \mathcal{O}_{-v}(c')} Qf(x') + \frac{1}{8} \sum_{x \in \mathcal{O}_v(c)} Qf(x). \end{aligned}$$

The square is bounded by $3/8$ times the sum of the squared terms in these three octets. Distinct photon bonds use distinct chamber bonds; each octet of cluster remainders occurs at most once. Consequently, with $\mathcal{E}_3$ and $\mathcal{E}_{\mathrm{int}}$ denoting the two parts of the energy,

$$a^2 \|\nabla_A \xi\|_{2,E}^2 \leq 18(w_3^{-1}\mathcal{E}_3 + \|Qf\|_{2,E}^2) \leq 18(w_3^{-1}\mathcal{E}_3 + \lambda_{\mathrm{int}}^{-1}\mathcal{E}_{\mathrm{int}}).$$

This proves the local bound and therefore the global one.

The internal hops vanish on $J\xi$. Thus

$$(D_A J\xi)(c,g) = -3(\rho - \gamma \cdot ge_1) \nabla_{A, ge_1} \xi(c).$$

Each of the six directions occurs eight times. The norm of the matrix in this formula is $3(1+\rho)$, and the two directed differences on a photon bond have equal norms. It follows that $\|D_A J\xi\|_2^2 \leq 3(1+\rho)^2 \|\nabla_A \xi\|_2^2$. Projection gives the first bound in eq. 4.vi; the second follows from eq. 4.iv, since J and Q commute with γ_5.

For $Qf = f$, eq. 4.iv gives $\operatorname{Re}(f, (QD_A Q + t)f) \geq (t + g_{\mathrm{h}}/a)\|f\|_2^2$. The adjoint has the same bound; closed range and zero adjoint kernel give surjectivity, including on the infinite lattice, and eq. 4.vii. □

4.1.2. *The free symbol and its Schur complement.* Use lattice units for the moment. Let $\mathcal{D}_m(p)$ be the symbol of $D_0 + m$, $0 \leq m \leq 1$, on $V = \ell^2(O_{\mathrm{cub}}; \mathbb{C}^4)$ with measure $48^{-1}\sum_g$:

$$(\mathcal{D}_m(p)v)(g) = mv(g) + \sum_{i=0}^{2} M_i(g)(v(g) - v(gs_i)) + M_3(g)(v(g) - \exp(ip \cdot ge_1) v(gr_{e_1})).$$

Let $D_{\mathrm{ll}}, D_{\mathrm{lh}}, D_{\mathrm{hl}}, D_{\mathrm{hh}}$ be its blocks in $V = \operatorname{ran} J \oplus \operatorname{ran} Q$. The mass is included in the diagonal blocks. Set

$$H_m(p) = D_{\mathrm{hh}}(p)^{-1}, \qquad B_m(p) = D_{\mathrm{ll}}(p) - D_{\mathrm{lh}}(p) H_m(p) D_{\mathrm{hl}}(p), \qquad t(p) = \sum_\mu (1 - \cos p_\mu).$$

The compression of the full operator is the Wilson operator:

$$D_{\mathrm{ll}}(p) = m + i\gamma \cdot \sin p + \rho t(p).$$

Solving the heavy equation and substituting it into the light equation gives the correction $-D_{\mathrm{lh}}H_m D_{\mathrm{hl}}$ in B_m. Thus B_m governs the low-momentum behavior of the full inverse, including the effect of the eliminated chamber modes. The mass enters both diagonal blocks, so it changes H_m as well as D_{ll}.

Proposition 4.2. There are positive constants $\theta_0, c_{\mathrm{e}}, \kappa_{\mathrm{strip}}$, depending on ρ only, such that H_m and B_m are holomorphic in a neighborhood of $\{|\operatorname{Im} p_\mu| \le \theta_0\}$ and $2\pi\mathbb{Z}^3$-periodic. On the real cube $[-\pi,\pi]^3$,

$$\|H_m(p)\| \le (g_{\mathrm{h}} + m)^{-1}, \tag{4.viii}$$

$$\|D_{\mathrm{lh}}(p)\| + \|D_{\mathrm{hl}}(p)\| \le 2c_{\mathrm{lh}}(2t(p))^{1/2}, \tag{4.ix}$$

$$B_m(p) = m + i\gamma\cdot\sin p + E_m(p), \qquad \|E_m(p)\| \le K_\rho t(p), \tag{4.x}$$

$$\operatorname{Re}(v, B_m(p)v) \ge (m + \rho c_* t(p))|v|_2^2, \tag{4.xi}$$

$$|B_m(p)v|_2 \ge c_{\mathrm{e}}(|p|_2^2 + m^2)^{1/2}|v|_2, \tag{4.xii}$$

where $K_\rho = \rho + 2c_{\mathrm{lh}}^2/g_{\mathrm{h}}$. For every multi-index ν,

$$\|\partial_p^\nu B_m(p)^{-1}\| \le C_\nu(\rho)(|p|_2^2 + m^2)^{-(1+|\nu|)/2}. \tag{4.xiii}$$

If $m > 0$, the inverse exists also on $|\operatorname{Im} p_\mu| \le \kappa_{\mathrm{strip}} m$, where the same bound holds with $\operatorname{Re} p$ on the right, for $\operatorname{Re} p \in [-\pi,\pi]^3$. At $m = 0$ the real derivative bound is asserted away from $p = 0$. ∎

Proof. Only the fourth hop depends on p. For $p = p' + ip''$,

$$\|\mathcal{D}_m(p) - \mathcal{D}_m(p')\| \le 3(1+\rho)(\exp(|p''|_\infty) - 1).$$

Choose θ_0 so that the right side is $g_{\mathrm{h}}/2$ when $|p''|_\infty = \theta_0$. The heavy gap then gives $\operatorname{Re} D_{\mathrm{hh}}(p) \ge m + g_{\mathrm{h}}/2$ on this strip. Invertibility holds on a slightly larger strip, which proves holomorphy up to the closed strip. For real p, the off-diagonal bounds in Prop. 4.1 give $\|D_{\mathrm{lh}}\|, \|D_{\mathrm{hl}}\| \le c_{\mathrm{lh}}(2t(p))^{1/2}$. This proves eq. 4.x.

For $u = Jv - H_m D_{\mathrm{hl}} v$, one has $J^* u = v$ and $\mathcal{D}_m u = JB_m v$, so

$$\operatorname{Re}(v, B_m v) = \operatorname{Re}(u, \mathcal{D}_m u) = m\|u\|^2 + \frac{\rho}{2}(u, \mathcal{L}_0(p)u) \ge (m + \rho c_* t(p))|v|_2^2.$$

This proves eq. 4.xi.

The ellipticity estimate uses the Dirac part near zero and the real part away from zero. Put $t_0 = (4K_\rho^2)^{-1}$. One has $K_\rho \ge 48/\lambda_{\mathrm{int}} > 1$, since $\lambda_{\mathrm{int}} \le 12 + 24\sqrt{2} < 48$. Hence $t_0 \le 1/4$. If $t(p) \le t_0$, then $|\sin p|_2^2 \ge t(p)$ and $\|E_m(p)\| \le \frac{1}{2}t(p)^{1/2}$. Since

$$|(m + i\gamma\cdot\sin p)v|_2^2 = (m^2 + |\sin p|_2^2)|v|_2^2,$$

it follows that $|B_m(p)v|_2 \ge \frac{1}{2}(m^2 + t(p))^{1/2}|v|_2$. On $[-\pi,\pi]^3$, $t(p) \ge 2|p|_2^2/\pi^2$. This proves eq. 4.xii in this region. If $t(p) \ge t_0$, then eq. 4.xi gives $|B_m(p)v|_2 \ge (m + \rho c_* t_0)|v|_2$. The inequality $(|p|_2^2 + m^2)^{1/2} \le m + \pi\sqrt{3}$ closes the estimate with, for example,

$$c_{\mathrm{e}} = \min\left(0.22, \frac{\rho c_* t_0}{\pi\sqrt{3}}\right).$$

The strip bounds give uniform bounds on the derivatives of B_m by Cauchy's formula. Differentiating $B_m B_m^{-1} = 1$ and inducting on $|\nu|$ proves eq. 4.xiii. For the complex strip, let K_s bound B_m on the fixed strip. On half that strip, $\|\partial_\mu B_m\| \le 2K_s/\theta_0$. Choose

$$\kappa_{\mathrm{strip}} = \min\left(\frac{\theta_0}{2}, \frac{c_{\mathrm{e}}\theta_0}{12K_s}\right).$$

Integration along the segment from p' to $p' + ip''$ shows that the change in B_m is at most $c_{\mathrm{e}}m/2$. The lower bound is thus at least $c_{\mathrm{e}}(|p'|_2^2 + m^2)^{1/2}/2$. Differentiating the inverse identity once more gives the asserted complex derivative bounds. □

4.1.3. *Free kernels and coercivity.* Solving the heavy equation gives the inverse symbol

$$\mathcal{D}_m^{-1} = \begin{pmatrix} B_m^{-1} & -B_m^{-1} D_{\mathrm{lh}} H_m \\ -H_m D_{\mathrm{hl}} B_m^{-1} & H_m + H_m D_{\mathrm{hl}} B_m^{-1} D_{\mathrm{lh}} H_m \end{pmatrix}. \tag{4.xiv}$$

The first term is independent of the cluster labels after insertion by J and J^*. Every other term has a weaker singularity at $p = 0$: D_{lh} and D_{hl} vanish there, whereas H_m and all its derivatives are uniformly bounded.

Proposition 4.3. On the unit covering lattice let

$$G_m(c, g; c', g') = \int_{[-\pi,\pi]^3} \frac{\mathrm{d}p}{(2\pi)^3} \exp(ip \cdot (c - c'))[\mathcal{D}_m(p)^{-1}](g, g').$$

The bracket is the kernel on V relative to the label measure $48^{-1} \sum_g$, consistently with the chamber kernel convention. It is well-defined also at $m = 0$. It decomposes as $G_m(x, x') = G_m^{\mathrm{l}}(c - c') + G_m^{\mathrm{h}}(x, x')$, where

$$|G_m^{\mathrm{l}}(n)|_2 \le C_G (1 + |n|_\infty)^{-2}, \tag{4.xv}$$

$$|G_m^{\mathrm{l}}(n + v) - G_m^{\mathrm{l}}(n)|_2 \le C_G (1 + |n|_\infty)^{-3}, \quad v \in \{\pm e_\mu\}, \tag{4.xvi}$$

$$|G_m^{\mathrm{h}}(x, x')|_2 \le C_G (1 + |c - c'|_\infty)^{-3}. \tag{4.xvii}$$

Consequently $|G_m(x, x')|_2 \le 2C_G(1 + |c - c'|_\infty)^{-2}$ and

$$|G_m(x^{(i)}, x') - G_m(x, x')|_2 \le 20 C_G (1 + |c - c'|_\infty)^{-3}. \tag{4.xviii}$$

For $m > 0$ these bounds have the additional factor $\exp(-\kappa_{\mathrm{strip}} m |c - c'|_\infty)$. The constants are uniform for $0 \le m \le 1$. ∎

Proof. If a periodic symbol T satisfies $|\partial^\nu T(p)| \le A|p|_2^{-q-|\nu|}$, where $q < 3$ and $|\nu| \le N > 3 - q$, then its Fourier coefficients are bounded by $CA(1 + |n|_\infty)^{-(3-q)}$. For $n \neq 0$, take $r = |n|_\infty^{-1}$ and a cutoff at radii $r, 2r$. The inner integral is $O(Ar^{3-q})$. On the complement, N integrations by parts in a coordinate with $|n_\mu| = |n|_\infty$ give $r^N O(Ar^{3-q-N})$, including cutoff derivatives on the annulus; periodicity cancels boundary terms.

Apply this estimate with $(q, N) = (1, 3)$ to B_m^{-1}. Multiplication by $\exp(ip \cdot v) - 1$ lowers its singularity to order zero, for which $(q, N) = (0, 4)$ applies. The mixed terms of eq. 4.xiv have order zero, the double-mixed term has order -1, and H_m has fixed-strip exponential decay. Absorb the label-measure kernel factor 48 in C_G.

For $m > 0$, shift one coordinate of the contour by $i\kappa_{\mathrm{strip}} m \operatorname{sgn}(n_\mu)$, where $|n_\mu| = |n|_\infty$. Prop. 4.2 gives the same estimates on the shifted contour and hence the exponential factor; H_m has the stronger fixed strip.

An internal hop leaves c fixed, so its difference cancels G_m^{l}. An external hop replaces c by $c + ge_1$; use the light difference bound and the two heavy bounds at these points. The inequalities $1 + |n + v|_\infty \ge (1 + |n|_\infty)/2$ and $\exp(\kappa_{\mathrm{strip}} m) < \sqrt{3}$ give the factor 20 in eq. 4.xviii. □

The Green kernel at mesh a, for $0 \le \mu a \le 1$, is $a^{-2} G_{\mu a}(x/a, x'/a)$. Thus

$$|G(x, x')|_2 \le 2C_G (a + |c - c'|_\infty)^{-2}, \qquad |G(x^{(i)}, x') - G(x, x')|_2 \le 20 C_G a (a + |c - c'|_\infty)^{-3}. \tag{4.xix}$$

Fourier multiplication gives both inverse identities on finitely supported functions. The factor 48 in the identity kernel is consistent with the measure $\mathrm{d}x$ above.

For distinct sites put $d_X(x, \widetilde{x}) = \max(|c - \widetilde{c}|_\infty, a)$. The chamber graph has paths from x to $\widetilde{x}$ of at most $39 d_X(x, \widetilde{x})/a$ bonds, with photon sites in the coordinate box spanned by $c, \widetilde{c}$. Indeed, any signed permutation can be reduced to $(1, 2, 3)$ in at most nine internal moves. Put $+1$ first in at most five moves (if negative, move it to the third position, flip its sign, then move it first), put $+2$ second in at most three moves, and correct the last sign in one move. Left translation gives this

bound between any two labels. A monotone photon path has at most $3|c-\widetilde{c}|_\infty/a$ steps. Before each step move to a label with the required external direction, and after the last step move to the final label. This uses at most $10\cdot 3|c-\widetilde{c}|_\infty/a+9\le 39d_X/a$ bonds. If d_X is less than half the distance to c', sum eq. 4.xix along such a path. In the other case use the two size bounds. The resulting interpolation gives, for $0\le\alpha\le 1$,

$$|G(x,x')-G(\widetilde{x},x')|_2\le 6240C_G d_X(x,\widetilde{x})^\alpha\left((a+|c-c'|_\infty)^{-2-\alpha}+(a+|\widetilde{c}-c'|_\infty)^{-2-\alpha}\right). \tag{4.xx}$$

The real-part estimate controls the hopping energy, but the block inverse also needs to control the deviation from a block mean. We obtain this by first bounding the hopping energy by $\|D_0 f\|_2^2$, then applying a Poincaré estimate to the cluster mean and the internal gap to its orthogonal complement.

Proposition 4.4. There is $c_b=c_b(\rho)>0$ such that

$$\|D_0 f\|_2^2\ge c_b a^{-2}(f,\mathcal{L}_0 f). \tag{4.xxi}$$

For a cube E of side ℓ_E, let P_E be the mean over $E\times O_{\mathrm{cub}}$. Then

$$\|f-P_E f\|_{2,E}^2\le C_P\ell_E^2 a^{-2}(f,\mathcal{L}_0 f)_E,\qquad C_P=\frac{1}{8c_*}+\frac{1}{\lambda_{\mathrm{int}}}. \tag{4.xxii}$$

■

Proof. The Schur estimate at zero mass implies $|B_0(p)v|\ge c_{\mathrm{l}}t(p)^{1/2}|v|$, with $c_{\mathrm{l}}=\rho c_*/(2K_\rho)$: use the Dirac part for $t(p)\le t_0$ and the real part for $t(p)\ge t_0$. Write $u=Jv+h$ in a fibre and $f=\mathcal{D}_0(p)u$. The heavy equation and then the light equation give

$$h=H_0(f_{\mathrm{h}}-D_{\mathrm{hl}}v),\qquad B_0 v=f_{\mathrm{l}}-D_{\mathrm{lh}}H_0 f_{\mathrm{h}}.$$

Since $t(p)\le 6$, these imply $(2t(p))^{1/2}|v|\le A_1\|f\|$ and $\|h\|\le A_2\|f\|$, where

$$A_1=\frac{\sqrt{2}}{c_{\mathrm{l}}}\left(1+\frac{6(1+\rho)}{g_{\mathrm{h}}}\right),\qquad A_2=g_{\mathrm{h}}^{-1}(1+c_{\mathrm{lh}}A_1).$$

The bond decomposition also gives

$$(u,\mathcal{L}_0(p)u)\le 6t(p)|v|^2+(30+24\sqrt{2})\|h\|^2.$$

For the internal bonds the light part cancels, leaving at most $2(w_0+w_1+w_2)\|h\|^2$. In an external bond difference, split off the photon difference of v and the two heavy endpoint values, and bound the square by three times their squared norms. The octet count then gives $6t(p)|v|^2+3w_3\|h\|^2$. This proves the displayed upper bound. This yields eq. 4.xxi in each fibre with $c_b=(3A_1^2+(30+24\sqrt{2})A_2^2)^{-1}$. Fourier decomposition gives the assertion for all characters on the torus and on the covering lattice, restoring the factor a^{-2}.

For eq. 4.xxii, first apply the scalar lattice Poincaré bound $\|\xi-P_E\xi\|_{2,E}^2\le\ell_E^2\|\nabla_0\xi\|_{2,E}^2/8$. In one dimension this follows by writing each difference as a sum of nearest-neighbor differences; the coefficient of a bond at position j is $j(n-j)/2\le n^2/8$. In three dimensions use the three commuting line averages, whose successive remainders are orthogonal. Now $f-P_E f=J(\xi-P_E\xi)+Qf$. Its two parts are orthogonal, and the local bounds eq. 4.v give eq. 4.xxii. The heavy term uses $\ell_E\ge a$. □

4.1.4. *Averaging penalties and the inverse at zero field.* We now add block-average penalties to the Dirac operator. The Poincaré estimate controls a field after subtracting its block mean; the penalty controls that remaining mean. This yields a uniform inverse even when the reference mass tends to zero. The auxiliary parameter y will later express the determinant through resolvents.

Fix an odd $L\ge 5$ and put $a=\eta_k=L^{-k}$. Write e_k for the charge e_a at this mesh. The level data k,μ,R,e_k are independent, with $0<e_k\le\bar{e}_0$; the flow will specify their relation later. Unit blocks have side one, and the next blocks have side L. In this and the following subsubsections a finite period R is a positive integer multiple of L. For $z\in\mathbb{Z}^3$ the unit block is $\Delta_z=\{c\in a\mathbb{Z}^3:|c-z|_\infty<1/2\}$. For $z'\in L\mathbb{Z}^3$ the L-block is $E_{z'}=\{c\in a\mathbb{Z}^3:|c-z'|_\infty<L/2\}$. These partitions are taken modulo

$R\mathbb{Z}^3$ on the torus. Since L is odd, no fine site lies on a block boundary. Their means, including the cluster mean, are denoted Π_1 and Π_L. At nonzero connection the means use the gauge phases specified below. Let

$$b = 1, \qquad b_L = b/L, \qquad b_- = b(1 - L^{-1}), \qquad b_k = \frac{b(1 - L^{-1})}{1 - L^{-k}} \quad (k \geq 1).$$

For $k \geq 1$, $y \in \mathbb{R}$, define

$$\alpha_y = b_k i\gamma_5 y(b_k + i\gamma_5 y)^{-1}, \qquad \beta_y = b_k^2 b_L (b_k + b_L + i\gamma_5 y)^{-1}(b_k + i\gamma_5 y)^{-1},$$

and set $\alpha_\infty = b_k$, $\beta_\infty = 0$. The blocked operators at zero connection are

$$F_y(0) = D_0 + \mu + \alpha_y \Pi_1 + \beta_y \Pi_L \quad (k \geq 1), \qquad F_y(0) = D_0 + \mu + b_L \Pi_L + i\gamma_5 y \quad (k = 0). \tag{4.xxiii}$$

Here $0 \leq \mu \leq 1$. In the first formula $y \in \mathbb{R} \cup \{\infty\}$; in the second $y \in \mathbb{R}$. We use $-\infty = \infty$ when an adjoint identity reverses this parameter. The notation F_y denotes an operator in this section, not the curvature.

The penalty bounds follow by diagonalizing γ_5. On its eigenspace of sign ϵ, the two eigenvalues and their sum are

$$\frac{b_k i\epsilon y}{b_k + i\epsilon y}, \quad \frac{b_k^2 b_L}{(b_k + b_L + i\epsilon y)(b_k + i\epsilon y)}, \quad \frac{b_k(b_L + i\epsilon y)}{b_k + b_L + i\epsilon y}.$$

For the first eigenvalue v_1 and the sum v_2, direct calculation gives

$$|v_1|^2 = b_k \operatorname{Re} v_1, \qquad |v_2|^2 \leq b_k \operatorname{Re} v_2, \qquad \operatorname{Re} v_2 \geq \frac{b_k b_L}{b_k + b_L} \geq \frac{b_-}{L}.$$

Also $|\alpha_y| \leq b_k \leq b$ and $|\beta_y| \leq b_L$. On an L-block the decomposition $\operatorname{Pen} = \alpha_y(\Pi_1 - \Pi_L) + (\alpha_y + \beta_y)\Pi_L$ has orthogonal ranges. The scalar inequalities on the two spin eigenspaces therefore give

$$\operatorname{Re}(f, \operatorname{Pen} f)_E \geq \frac{b_-}{L} \|P_E f\|_{2,E}^2, \qquad \|\operatorname{Pen} f\|_{2,E}^2 \leq b \operatorname{Re}(f, \operatorname{Pen} f)_E. \tag{4.xxiv}$$

The kernel on E is $\alpha_y \mathbf{1}[c, c' \text{ in the same unit block}] + \beta_y L^{-3}$, and its norm is at most $2b$.

Proposition 4.5. There are $C_I, C_5, \gamma_5' > 0$, depending on L, ρ and a fixed $\alpha \in (0, 1)$ only, such that $F_y(0)$ is invertible and

$$\|F_y(0)^{-1}\|_{2 \to 2} \leq L C_I \qquad (k \geq 1).$$

If f is supported over one unit block $\Delta_{z'}$ and $v = F_y(0)^{-1} f$, then for x over Δ_z and $0 < d_X(x, \widetilde{x}) \leq 1$,

$$|v(x)|_2 + d_X(x, \widetilde{x})^{-\alpha} |v(x) - v(\widetilde{x})|_2 \leq C_5 \exp(-\gamma_5' d(z, z')) \|f\|_\infty. \tag{4.xxv}$$

This includes pairs in one cluster. The constants are uniform in k, R, the character, μ, and y. ∎

Proof. First suppose $k \geq 1$. Put $f = F_y(0)u$ and $X = \operatorname{Re}(u, f)$. The real-part identity and eq. 4.xxiv imply

$$\sum_E \|P_E u\|_{2,E}^2 \leq \frac{L}{b_-} X, \qquad \|\operatorname{Pen} u\|_2^2 \leq bX, \qquad \|\mu u\|_2^2 \leq X.$$

Therefore $\|D_0 u\|_2 \leq \|f\|_2 + (1 + \sqrt{b}) X^{1/2}$. Use eq. 4.xxi for $D_0 u$ and eq. 4.xxii on each L-block. With $K_1 = \sqrt{L} + \sqrt{Lb}$ this gives

$$\|u\|_2^2 \leq L \left(\frac{1}{b_-} + \frac{2C_P K_1^2}{c_b} \right) \|u\|_2 \|f\|_2 + \frac{2C_P}{c_b} L^2 \|f\|_2^2.$$

Hence $\|u\|_2 \leq L C_I \|f\|_2$, where $C_I = b_-^{-1} + 2C_P K_1^2 / c_b + (2C_P/c_b)^{1/2}$. The identity $F_y(0)^* = \gamma_5 F_{-y}(0) \gamma_5$ gives the same lower bound for the adjoint. Thus $F_y(0)$ has closed, dense range and is onto, including on the infinite lattice.

To obtain decay, let ψ be real, bounded and periodic, with $|\psi(c) - \psi(c')| \leq a_0 |c - c'|_1 / L$. The mass commutes with this weight. The external hopping term gives $\|\exp(\psi)[D_0, \exp(-\psi)]\|_{2 \to 2} \leq$

$3(1+\rho)\exp(1)a_0/L$. The penalty kernel is supported in one L-block, has norm at most $2b$, and $|\psi(c)-\psi(c')|\le 3a_0$ there. Its contribution is at most $6L^4 b\exp(1)a_0/L$. Put

$$c_w=\exp(1)(3(1+\rho)+6L^4b),\qquad a_0=\min(1/3,(2C_Ic_w)^{-1}).$$

Applying the L^2 inverse bound and absorbing the commutator gives

$$\|\exp(\psi)F_y(0)^{-1}f\|_2\le 2LC_I\|\exp(\psi)f\|_2. \tag{4.xxvi}$$

Take truncated multiples of the ℓ^1 distance, divided by L, to the source L-block. Letting the truncation increase proves an exponentially weighted L^2 bound.

The passage from L^2 to the sup norm uses the free kernel. On the covering lattice choose $c^\circ\in a\mathbb{Z}^3$ and $r\ge 8a$, and put

$$B=\{|c-c^\circ|_\infty<r\},\quad B^+=\{|c-c^\circ|_\infty<r+a\},\quad B'=\{|c-c^\circ|_\infty\le r/4\}.$$

Suppose u is defined over B^+ and $(D_0+\mu)u=h$ over B, with $0\le\mu a\le 1$. Both evaluation points below lie over B', and C depends on the fixed $\alpha\in(0,1)$ and ρ. Let χ equal one on the concentric half cube and zero outside the cube, with Lipschitz constant $2/r$. Then

$$\chi u=G(\chi h)+G([D_0,\chi]u).$$

The commutator contains only the external hop. Its rows are supported where $|c-c^\circ|_\infty>r/2-a$. It satisfies $\|[D_0,\chi]u\|_1\le 45r^{1/2}\|u\|_{2,B^+}$. For a point of the quarter cube, this support is at distance at least $r/4-a\ge r/8$. The bounds eq. 4.xix and eq. 4.xx consequently give

$$\begin{aligned}|u(x)|_2&\le C(r\|h\|_{\infty,B}+r^{-3/2}\|u\|_{2,B^+}),\\ |u(x)-u(\widetilde{x})|_2&\le C\left(\frac{d_X(x,\widetilde{x})}{r}\right)^{\alpha}(r\|h\|_{\infty,B}+r^{-3/2}\|u\|_{2,B^+}).\end{aligned} \tag{4.xxvii}$$

For the first term, sum $(a+|c-c'|_\infty)^{-2}$ over the cube; the result is $O(r)$. For the difference, the analogous sum with exponent $2+\alpha$ is $O(r^{1-\alpha}/(1-\alpha))$. The cluster average in $\mathrm{d}x'$ contributes one to both sums. For the commutator term, its support separation gives the size bound; summing bond differences along the chamber path gives the difference bound.

For $u=F_y(0)^{-1}f$, use $h=f-\operatorname{Pen}u$ in eq. 4.xxvii, with $r=L/4$. The enlarged cube meets at most eight L-blocks. On each, the penalty kernel and Cauchy–Schwarz give $\|\operatorname{Pen}u\|_\infty\le 2bL^{3/2}\|u\|_{2,E}$. The weighted bound eq. 4.xxvi therefore bounds both terms in eq. 4.xxvii by $CL\exp(-a_0d_1(x,E'))\|f\|_\infty$, where $d_1(x,E')$ is the ℓ^1 distance to the source L-block divided by L. This applies for $k\ge 2$. For $k=1$, the pointwise L^2 bound costs only $\sqrt{48}L^{3/2}$; L is fixed. Larger Hölder separations are controlled by the two sup bounds. Finally, $d_1(x,E')\ge d(z,z')/L-3$, and the source unit block lies in E'. This proves eq. 4.xxv with $\gamma_5'=a_0/L$. The argument uses only periodic moduli and so applies to every unitary character. □

The first level has a different resolvent parameter. On an L-block the real-part bound and Poincaré inequality imply

$$\operatorname{Re}(u,F_y(0)u)\ge\frac{c_r}{L}\|u\|_2^2,\qquad c_r=\min(b_-,\rho/(2C_PL)).$$

The term $i\gamma_5 y$ has zero real part. Conjugation by a weight with Lipschitz constant a_z changes the hopping and penalty terms by at most $\exp(1)(6+3b)a_z$. Choose

$$a_z\le\min\bigl((3L)^{-1},c_r/(2\exp(1)L(6+3b))\bigr).$$

The weighted inverse then has L^2 norm at most $2L/c_r$. On the other hand its part other than $i\gamma_5 y$ has norm at most $60+b$. For $|y|\ge 2(60+b)$, the inverse norm is at most $2/|y|$. Combining the two ranges and converting to a kernel, with its factor 48, gives

$$|F_y(0)^{-1}(x,x')|_2\le C_{00}(1+|y|)^{-1}\exp(-a_zd(c,c')). \tag{4.xxviii}$$

This factor in y is needed when the determinant is represented by an integral of resolvents.

4.1.5. *Small connections and local inverses.* The zero-field inverse persists under a connection that is small in the sup norm. A real gauge transformation is unitary, so this perturbation estimate will also apply to connections that are small only in local real gauges. We first define the averaging phases, then prove the perturbation estimate, and finally construct inverses on the enlarged cubes used for patching. The required local inverse properties follow [17, (128), (131), (132)]; here they are realized on mirror tori by folded fields.

The covariant block means are defined by the same nested blocks. Let $\tau A(c,c')$ be the arithmetic average of the oriented bond sums over the six coordinate-order paths from c to c', and let $c=c_0,c_1,\dots,c_k=z$ be the centers of the nested blocks containing c. Put

$$\varphi_A(z,c)=\exp\left(ie_k\eta_k\sum_{j<k}\tau A(c_{j+1},c_j)\right).$$

The unit-block average on four-component spinors and its inverse injection are

$$(Q_k(A)\xi)(z)=\eta_k^3\sum_{c\in\Delta_z}\varphi_A(z,c)\xi(c),\qquad (Q_k^T(-A)v)(c)=\varphi_A(z,c)^{-1}v(z).$$

For the next average, use $U_A(z',z)=\exp(ie_k\eta_k\tau A(z',z))$, with the same six-path average of fine-bond sums between the two unit sites. On the unit lattice set

$$(Q(A)v)(z')=L^{-3}\sum_{|z-z'|_\infty<L/2}U_A(z',z)v(z),\qquad (Q^T(-A)v')(z)=U_A(z',z)^{-1}v'(z').$$

On the chamber space, for $k\ge 1$, put

$$\Pi_1(A)=JQ_k^T(-A)Q_k(A)J^*,\qquad \Pi_L(A)=JQ_k^T(-A)Q^T(-A)Q(A)Q_k(A)J^*.$$

At $k=0$ only $\Pi_L(A)=JQ^T(-A)Q(A)J^*$ is used. This defines $F_y(A)$ by eq. 4.xxiii. The formulas permit complex connections; for real A the idempotents are orthogonal projections.

For a pair of sites define

$$U_A(x,x')=\exp(ie_k\eta_k\tau A(c,c')),\qquad (\delta_{\alpha,A}v)(x,x')=\frac{U_A(x,x')v(x')-v(x)}{d_X(x,x')^\alpha}.$$

Only pairs with $0<d_X<1$ enter the local norms. If $c=c'$ then $U_A=1$, so these quotients include variations between the labels in one cluster. There are no such short pairs when $k=0$.

For an operator on the torus define

$$N_\gamma(T)=\sup_z\sum_{z'}\exp(\gamma d(z,z'))\|\mathbf{1}_{\Delta_z}T\mathbf{1}_{\Delta_{z'}}\|_{\infty\to\infty}.$$

For operators with values on pairs, the left indicator restricts the first point. Inserting the block partition between two factors and using the triangle inequality for d proves $N_\gamma(TT')\le N_\gamma(T)N_\gamma(T')$. Write $w_k(y)=1$ for $k\ge 1$ and $w_0(y)=(1+|y|)^{-1}$. Enlarge C_5,C_{00} to at least one, and choose

$$\gamma=\tfrac12\min(\gamma_5',a_z),\qquad a'=\max(C_5,C_{00})\sum_{n\in\mathbb{Z}^3}\exp(-\gamma|n|_\infty).$$

The preceding estimates imply

$$N_\gamma(F_y(0)^{-1})\le a'w_k(y),\qquad N_\gamma(\delta_{\alpha,0}F_y(0)^{-1})\le a'.\tag{4.xxix}$$

At $k=0$ the analogous bound holds in the kernel row norm N_γ^{ker}, obtained by replacing the localized operator norm with the supremum of $|T(x,x')|_2$ over the two clusters.

Proposition 4.6. Let B be a complex connection with $e_k\|B\|_\infty \le \delta$, where $3L\delta \le 1$ and $2C_V a'\delta \le 1$, with $C_V = 2\exp(1)(42+6b)$. Then $F_y(B)$ is invertible and

$$N_\gamma(F_y(B)^{-1}) \le 2a' w_k(y), \qquad N_\gamma(\delta_{\alpha,B} F_y(B)^{-1}) \le 18a'. \tag{4.xxx}$$

At $k=0$ the first bound holds also in N_γ^{ker}. ■

Proof. Write $V = F_y(B) - F_y(0)$. The internal hops, the mass, and the resolvent term at level zero cancel. For the external hop, $|u_B - 1| \le \eta_k\delta\exp(\eta_k\delta)$; its weighted row sum is at most $42\exp(\gamma)\exp(\delta)\delta$. The two averaging penalties contribute respectively $3b\delta\exp(3\delta)$ and $3b\delta\exp(\gamma(L-1))\exp(3L\delta)$. Indeed, the nested paths have total length at most $3/2$, and the path to an L-block center has length at most $3L/2$. Summing their normalized block kernels removes the numbers of sites. The bounds $\gamma L \le \log 2$ and $3L\delta \le 1$ give $N_\gamma(V) \le C_V\delta$.

Thus $N_\gamma(VF_y(0)^{-1}) \le 1/2$. The Neumann series gives

$$F_y(B)^{-1} = F_y(0)^{-1}\sum_{n\ge0}(-VF_y(0)^{-1})^n = \sum_{n\ge0}(-F_y(0)^{-1}V)^n F_y(0)^{-1}.$$

It proves the first bound in eq. 4.xxx. The first form gives $N_\gamma(\delta_{\alpha,0}F_y(B)^{-1}) \le 2a'$. For distinct photon sites in a short pair, $d_X^{-\alpha}|U_B - 1| \le 3\delta\exp(3\delta) \le 1/4$; for intracluster pairs the expression is zero. Evaluation at the second point of a short pair uses at most 27 neighboring blocks, and costs at most $27\exp(\gamma)$. Its contribution to the covariant quotient is at most $16a'$, proving the second bound. At level zero, the second form of the Neumann series keeps the kernel norm in the last factor, so the kernel estimate follows as well. □

Fix a positive exponent ε; its choice in the flow will be made with the other renormalization parameters. The connection domain used by the RG is the following. Write $\ell = R$ for the side, and assume $\ell/(LM)$ is odd and at least 13. Here M is a power of L with $M \ge L^2$. The M-cubes have centers $z_\square \in M\mathbb{Z}^3/\ell\mathbb{Z}^3$ and are the periodic images of $\{c : |c - z_\square|_\infty < M/2\}$. The differential of a scalar in a connection is normalized by $(d\lambda)(c,c') = \eta_k^{-1}(\lambda(c') - \lambda(c))$. A connection belongs to $\widetilde{\mathcal{G}}_k$ if $A = A_0 + A_1$, where A_0 is real, $\|A_1\|_\infty < e_k^{-1+\varepsilon}$, and on every $11M$ enlargement of an M-cube, $A_0 = B_0 + d\lambda$, with B_0, λ real and $\|B_0\|_\infty < e_k^{-1+\varepsilon}$. Consequently, on the $3M$ enlargement $W_\square$,

$$A = B + d\lambda, \qquad e_k\|B\|_\infty < 2e_k^\varepsilon, \qquad e_k\|\operatorname{Im} A\|_\infty < e_k^\varepsilon. \tag{4.xxxi}$$

This follows by restricting $B_0 + A_1$ and λ to that cube. The domain is open: a sufficiently small complex perturbation can be added to A_1. It is invariant under real gauge transformations.

For a lift $z_\square$ of the cube center put $W_\square = \{|c - z_\square|_\infty < 3M/2\}$ and $W_\square^- = \{|c - z_\square|_\infty < 3M/2 - \eta_k\}$. Embed $W_\square$ in the positive chamber of the periodic torus of side $6M$, whose lattice is translated by $\eta_k(1,1,1)/2$. Its level-j block centers are $L^j\eta_k(1,1,1)/2 + L^j\eta_k\mathbb{Z}^3$ for $0 \le j \le k+1$. Translation by $L(1,1,1)/2$ identifies these blocks with the unshifted nested blocks, because L^{k+1-j} is odd. The zero-field and small-connection estimates therefore apply on this torus with the same constants. Fold its photon coordinates across the planes at multiples of $3M$. Adjacent sites fold either to adjacent sites in $W_\square$ or to one site; in the latter case assign phase one. Pull back A by this fold to obtain A^W. The fold preserves the supremum norm and commutes with d on scalar functions. It also preserves the nested blocks, since $3M/(L^j\eta_k)$ is odd for $j \le k+1$.

Let ext extend a function over $W_\square$ by zero in the seven other chambers, and let res restrict to its embedded copy. Labels are unchanged in these two maps. Define

$$S_\square(A) = \operatorname{res} F_y(A^W)^{-1}\operatorname{ext}.$$

For $A = B + d\lambda$ as in eq. 4.xxxi, $A^W = B^W + d\lambda^W$, with λ^W real. Gauge conjugation by $\exp(ie_k\lambda^W)$ reduces the inverse to the small connection B^W. If $6L\bar{e}_0^\varepsilon \le 1$ and $4C_V a'\bar{e}_0^\varepsilon \le 1$, Prop. 4.6 applies to every $0 < e_k \le \bar{e}_0$. Restriction, extension, and unitary gauge factors do not increase the localized norms. Thus

$$N_\gamma(S_\square(A)) \le 2a' w_k(y), \qquad N_\gamma(\delta_{\alpha,A}S_\square(A)) \le 18a'. \tag{4.xxxii}$$

The second operator here is formed on the local cube before extension to the original torus. It does not differentiate a discontinuous extension by zero.

The local inverse has both identities needed for patching. If f is defined over $W_\square$, then

$$F_y(A)S_\square(A)f = f \quad \text{over } W_\square^-. \tag{4.xxxiii}$$

If f is supported over $W_\square^-$, then $F_y(A)f$ is supported over $W_\square$ and

$$S_\square(A)F_y(A)f = f. \tag{4.xxxiv}$$

For eq. 4.xxxiii, the rows at an interior site use only its internal neighbors, its external neighbor, and its L-block. These lie in $W_\square$, where the original and folded operators agree. For eq. 4.xxxiv, the support margin ensures that applying F_y cannot leave $W_\square$. After embedding, applying the folded operator to the extension of f agrees with extending $F_y(A)f$. Apply the inverse and restrict. For an antiperiodic spinor on the original torus, use one lift and then its signed translates; the enlarged cubes do not overlap their translates. These arguments also show that $S_\square$ depends on A only in $W_\square$ and is analytic in those connection variables.

4.1.6. *The parametrix and the walk expansion.* The local inverse and walk method originates in [15], with the fermion version in [16]. We use the organization of [17, (135)–(140)] and [18, (100)–(105)], with the local inverses just constructed.

To assemble the local inverses, we multiply them by cutoffs whose squares sum to one. The resulting operator is an approximate inverse: its error consists of commutators of the Dirac operator with the cutoffs. Their M^{-1} factor makes the error small. Expanding its inverse as a geometric series will organize the propagator into walks of overlapping cubes.

Choose the product cutoff $h_\square$ with center $z_\square$ and one-dimensional factor

$$h_0(t) = \begin{cases} 1, & |t| \le 1/4, \\ \cos(\pi(2|t| - 1/2)/2), & 1/4 \le |t| \le 3/4, \\ 0, & |t| \ge 3/4. \end{cases}$$

On a lift of the cube set

$$h_\square(c) = \prod_{\mu=1}^{3} h_0\big((c_\mu - z_{\square,\mu})/M\big), \qquad h_\square(c,g) = h_\square(c).$$

Extend this with period R and set it to zero away from the periodic images of the lifted cube neighborhood. The cutoffs satisfy $\sum_\square h_\square^2 = 1$, $\operatorname{supp} h_\square \subset \{|c - z_\square|_\infty < 3M/4\}$, and $|h_\square(c) - h_\square(c')| \le \pi|c - c'|_1/M$. Put $C_\square = -[F_y(A), h_\square]$. Internal hops commute with $h_\square$, and the penalty has range at most L. The phase bounds and the same normalized sums used in the perturbation estimate give

$$N_\gamma(C_\square) \le \pi C_V/M. \tag{4.xxxv}$$

The kernel is $(h_\square(c) - h_\square(c'))F_y(A)(x,x')$. Its two photon sites lie within $3M/4 + L$ of $z_\square$. If $M > 4L$, both lie in $W_\square^-$; multiplication by $h_{\square'}$ can give a nonzero operator only when $d(z_\square, z_{\square'}) \le M$.

Using eq. 4.xxxiii and eq. 4.xxxiv gives

$$F_y h_\square S_\square h_\square = h_\square^2 - C_\square S_\square h_\square, \tag{4.xxxvi}$$

$$h_\square S_\square h_\square F_y = h_\square^2 + h_\square S_\square C_\square. \tag{4.xxxvii}$$

Let $S^{\mathrm{p}} = \sum_\square h_\square S_\square h_\square$ and $R^{\mathrm{p}} = \sum_\square C_\square S_\square h_\square$. At most 27 cubes contribute to one row of R^{p}. Consequently,

$$N_\gamma(R^{\mathrm{p}}) \le q_0 := \frac{54\pi C_V a'}{M}.$$

Choose $M \geq 108\pi C_V a'$. eq. 4.xxxvi now gives a right inverse by a convergent series:

$$S_y(A) = F_y(A)^{-1} = S^{\mathrm{p}} \sum_{n \geq 0} (R^{\mathrm{p}})^n. \tag{4.xxxviii}$$

The finite-dimensional operator $F_y(A)$ is therefore invertible. The left identity gives the second form with $R^{\mathrm{p}\prime} = -\sum_{\square} h_{\square} S_{\square} C_{\square}$.

The terms in eq. 4.xxxviii have a local description. A walk of length n is a sequence $\omega = (\square_0, \dots, \square_n)$ with successive centers at distance at most M. Its term is

$$S_\omega = (h_0 S_0 h_0)(C_1 S_1 h_1) \cdots (C_n S_n h_n),$$

where the subscript i denotes the cube $\square_i$. Other sequences give zero because of the support of the commutator. Its connection variables lie in $\bigcup_{i=0}^{n} W_{\square_i}$.

Proposition 4.7. For $p \geq 1$ with $q(p) = 54\pi C_V a' p/M \leq 1/2$,

$$\sup_z \sum_{z'} \exp(\gamma d(z,z')) \sum_{|\omega|=n} p^{n+1} \|\mathbf{1}_{\Delta_z} S_\omega \mathbf{1}_{\Delta_{z'}}\| \leq 16 a' p w_k(y) q(p)^n, \tag{4.xxxix}$$

$$\sup_z \sum_{z'} \exp(\gamma d(z,z')) \sum_{|\omega|=n} p^{n+1} \|\mathbf{1}_{\Delta_z} \delta_{\alpha,A} S_\omega \mathbf{1}_{\Delta_{z'}}\| \leq 152 a' p q(p)^n. \tag{4.xl}$$

The norms are sup operator norms. At $k = 0$ the first bound holds also with the supremum of the normalized kernel over the two clusters. In particular, $S_y(A)$ is analytic on $\widetilde{\mathcal{G}}_k$ and

$$N_\gamma(S_y(A)) \leq 32 a' w_k(y), \qquad N_\gamma(\delta_{\alpha,A} S_y(A)) \leq 304 a'. \tag{4.xli}$$

These constants are independent of the level, the side, the running mass in $[0,1]$, and the connection in the stated domain. ∎

Proof. The first factor in a walk has norm at most $2a' w_k(y)$. Every additional factor has norm at most $2\pi C_V a'/M$ by eq. 4.xxxii and eq. 4.xxxv. For a fixed output block, at most eight initial cutoffs contribute; there are at most 27 choices at each further step. Apply submultiplicativity row by row and sum these $8 \cdot 27^n$ bounds. This proves eq. 4.xxxix.

For the covariant quotient, only the factor at the left end of the product is differentiated. The product identity is

$$\delta_{\alpha,A}(h_{\square} v)(x,x') = h_{\square}(c) \delta_{\alpha,A} v(x,x') + d_X^{-\alpha}(h_{\square}(c') - h_{\square}(c)) U_A(x,x') v(x').$$

The first contribution has localized norm at most $18a'$. The second has coefficient at most $3\pi \exp(1)/M$, and evaluation at the second point costs at most $27\exp(\gamma)$. With the stated choice of M this contribution is at most a'. Thus $N_\gamma(\delta_{\alpha,A} h_{\square} S_{\square}) \leq 19a'$. For intracluster pairs the second term vanishes, since $c = c'$. There are again at most eight possible initial cubes, including when the second point of a short pair enters the cutoff support. The count $8 \cdot 27^n$ now gives eq. 4.xl.

At level zero keep the kernel norm in the rightmost local inverse; the other factors are controlled in operator norm. The same submultiplicative row estimate proves the kernel assertion. Summing the geometric bounds with $p = 1$ gives eq. 4.xli. For a fixed torus the sum of the operator norms over all walks is finite; its tail of lengths at least j is bounded by $4a'(\ell/M)^3 2^{-j}$. Each walk term is analytic in the finitely many connection variables on which it depends. The uniform tail bound therefore proves analyticity of the sum. The local bounds in eq. 4.xxxix contain no factor depending on ℓ. □

4.2. **Local fermion integration.** We now localize the fermion covariance, coarse Dirac operator and determinant. Their polymer expansions let the next renormalization step integrate fine fermions against local interactions.

We work at mesh $\eta_k = L^{-k}$ on the torus of side ℓ; this is the side denoted by R in the local inverse construction. Throughout this subsection the mass satisfies $0 < \mu \leq 1$, the range of the renormalization data. Transposition is with respect to the bilinear lattice pairings: for $O : E \to E'$,

$\langle v, Ou\rangle_{E'} = \langle O^T v, u\rangle_E$. These pairings use the counting, chamber, or coarse lattice measure of the respective space. No complex conjugation enters this operation.

4.2.1. *Weakened walks and polymer propagators.* The resolvent parameter ranges over $Y_0 = \mathbb{R}$ at level zero and $Y_k = \mathbb{R} \cup \{\infty\}$ at $k \geq 1$, with $-\infty = \infty$ as in the preceding resolvent construction.

Let π_k be the partition into M-cubes and let $\mathcal{P}_k$ be the set of nonempty unions of these cubes connected through faces. For $X \in \mathcal{P}_k$ write $|X|_M$ for its number of cubes. A net is a connected finite union of straight segments in the continuum torus, and its length is the sum of their Euclidean lengths. Allowing all finite segment representations, as specified in App. D, set

$$M d_M(X) = \inf\{\text{length}(\mathcal{N}) : \mathcal{N} \text{ is a net meeting every closed cube of } X\}.$$

For a face-connected union of cubes, joining successive cube centers gives $d_M(X) \leq |X|_M - 1$.

Following the weakening of walk sums in [17, (153)–(158)], we attach a parameter to each cube visited by a walk and to its one-layer neighborhood. For an M-cube $\square$, write $\square^{\sim 1}$ for the union of the 3^3 cubes whose centers are within sup distance M of its center, with periodic identification. We attach parameters to all these cubes so that a surviving walk and all of its connection variables lie in one active component. For a walk $\omega = (\square_0, \dots, \square_n)$ from §4.1.6, put

$$X_\omega = \bigcup_{i=0}^{n} \square_i^{\sim 1}, \qquad s_\omega = \prod_{\square \subset X_\omega} s_\square, \qquad |s_\square| \leq M^{\alpha_0}, \quad \alpha_0 = 1/54.$$

The union X_ω is a polymer and

$$|X_\omega|_M \leq 27(n+1), \qquad d_M(X_\omega) \leq 27(n+1). \tag{4.xlii}$$

Indeed, each one-layer neighborhood contains 27 cubes. Successive centers differ by at most M in each coordinate, so their neighborhoods are joined through faces. The preceding bound on d_M applies. Every walk, including a walk of length zero, carries the parameters of all cubes in X_ω.

The left parametrix gives terms

$$S'_{y,\omega} = (h_n S_n)(-C_n h_{n-1} S_{n-1}) \cdots (-C_1 h_0 S_0) h_0.$$

Here, as for $S_{y,\omega}$, S_i is the local inverse on $\square_i$ and $C_i = -[F_y, h_i]$. Define

$$S_y(s, A) = \sum_\omega s_\omega S_{y,\omega}(A), \qquad S_y(X, A) = \sum_{X_\omega = X} S_{y,\omega}(A), \tag{4.xliii}$$
$$S'_y(s, A) = \sum_\omega s_\omega S'_{y,\omega}(A), \qquad S'_y(X, A) = \sum_{X_\omega = X} S'_{y,\omega}(A).$$

Their averages are denoted by S_y^{sy}. The superscript # below means any of these three choices. Put $\kappa' = \kappa_{\mathrm{w}} + \kappa_0$, where $\kappa_{\mathrm{w}}, \kappa_0 > 0$ are fixed polymer decay rates. Choose M large enough that

$$q(M^{27\alpha_0}) \leq \tfrac{1}{2}, \qquad q(e^{27\kappa'}) \leq \tfrac{1}{2}, \qquad q(p) = 54\pi C_V a' p/M. \tag{4.xliv}$$

Proposition 4.8. For $A \in \widetilde{\mathcal{G}}_k$, $y \in Y_k$, and $s \in \prod_{\pi_k}\{|s_\square| \leq M^{\alpha_0}\}$, the series eq. 4.xliii converge absolutely in N_γ, uniformly in (s, A), and

$$N_\gamma(S_y^{\#}(s, A)) \leq 32 a' M^{27\alpha_0} w_k(y), \tag{4.xlv}$$
$$N_\gamma(\delta_{\alpha,A} S_y^{\#}(s, A)) \leq 304 a' M^{27\alpha_0},$$
$$\|\mathbf{1}_{\Delta_z} S_y^{\#}(X, A) \mathbf{1}_{\Delta_{z'}}\| \leq 32 a' e^{27\kappa'} w_k(y) e^{-\gamma d(z,z') - \kappa' d_M(X)}. \tag{4.xlvi}$$

The norm in the last line is the sup operator norm. At $k = 0$ it can also be replaced by the supremum of the normalized kernel over the two clusters, with $w_0(y) = (1 + |y|)^{-1}$. At parameter value one,

$$S_y^{\#}(1, A) = S_y(A) = \sum_{X \in \mathcal{P}_k} S_y^{\#}(X, A). \tag{4.xlvii}$$

The weakened operators are continuous on the closed parameter polydisc, jointly analytic in its interior and in A, and analytic in A for each fixed parameter value, including boundary values. For finite real y the families are continuous in y.

The operator $S_y^{\#}(X,A)$ is supported on X in both arguments and depends on A only on bonds with endpoints in X. If parameters vanish off a union Y of M-cubes, then each component of Y is preserved by $S_y^{\#}(s,A)$. On such a component the operator depends only on its parameters and its connection. Its row at $x=(c,g)$ vanishes unless all M-cubes meeting $K_c=\{c' : |c'-c|_\infty \le 2L\}$ have active parameters; the corresponding assertion holds for columns. ∎

Proof. By eq. 4.xlii, $|s_\omega| \le p_s^{n+1}$ with $p_s = M^{27\alpha_0}$. Apply the walk row estimate with this weight. The length-n contribution is bounded by $16a'p_s w_k(y)2^{-n}$; its Hölder row estimate is bounded by $152a'p_s 2^{-n}$. Summing these bounds proves the first two estimates. For a fixed polymer use

$$\mathbf{1}[X_\omega = X] \le e^{-\kappa' d_M(X)} p_X^{n+1}, \qquad p_X = e^{27\kappa'}.$$

The same row estimate with p_X proves eq. 4.xlvi. The left walk estimates follow from the left parametrix by the same weighted product estimate; taking the average preserves all bounds.

On a fixed torus, summing the geometric row bounds over its finitely many unit blocks gives uniform absolute convergence and permits regrouping by X_ω. At $s=1$ the right and left inverse expansions give eq. 4.xlvii. Each term is polynomial in s, analytic in A and continuous in finite real y; the uniform tails, with $w_k(y) \le 1$, give all stated regularity, including boundary s.

A walk term is a product of local factors on the neighborhoods $\square_i^{\sim 1}$. Its row and column cutoffs, its commutators, and all connection variables consequently lie in X_ω. If the first row is nonzero, its photon site satisfies $|c - z_{\square_0}|_\infty < 3M/4$. Since $3M/4 + 2L < 3M/2$, every M-cube meeting K_c is in $\square_0^{\sim 1}$, and its parameter occurs in s_ω. The last cutoff gives the column assertion; the order is reversed for a left walk. Finally, a surviving walk has every cube of X_ω in Y. Since X_ω is face-connected, it lies in a single component of Y. Its local factors use only the connection in that component. This proves all the locality statements. □

The lattice symmetries used in this subsection have coordinate sign changes and translations in $M\mathbb{Z}^3$. Their spin lifts lie in the group generated by $\gamma_1,\gamma_2,\gamma_3,\gamma_5$; they act on (c,g) by $(rc+a, rg)$, where $S\gamma_\mu S^{-1} = r_{\mu\mu}\gamma_\mu$ determines r. Write $\widetilde{g} = (S,a)$ for such a symmetry. The step group restricts a to $LM\mathbb{Z}^3$. A lift commuting with γ_5 preserves y; a lift anticommuting with it sends y to $-y$. Gauge covariance follows term by term: the conjugating phases of adjacent factors cancel. A lattice symmetry permutes cubes and walks, and hence acts on the parameters by $(\widetilde{g}s)_\square = s_{\widetilde{g}^{-1}\square}$. Transposition reverses the order of the local factors and interchanges the right and left expansions. More precisely, with $\mathsf{C} = i\gamma_2$ one has $F_y(A)^T = \mathsf{C}^{-1}F_{-y}(-A)\mathsf{C}$. Therefore

$$S_y(s,A)^T = \mathsf{C}^{-1}S'_{-y}(s,-A)\mathsf{C}. \tag{4.xlviii}$$

Thus at $y=0,\infty$ for $k\ge 1$, and $y=0$ for $k=0$, the symmetrized family has the same transpose covariance as the exact inverse. This is the reason for using it in the coarse covariance and determinant.

4.2.2. *The fine and coarse resolvents.* The fine inverse gives both the reconstructed fermion field and the inverse needed to integrate the unit-scale variables. We derive their exact relation before localizing either operator. The algebra below proves invertibility of the coarse operator from the already constructed fine inverses, rather than assuming a coarse inverse in the block formula.

For $k \ge 1$ let

$$\mathcal{Q}_k = Q_k(A)J^*, \qquad \mathcal{Q}_k^\sharp = JQ_k^T(-A), \qquad P_L = Q^T(-A)Q(A).$$

Here Q_k averages the fine lattice onto the unit lattice, and Q averages the unit lattice onto the lattice of spacing L. The notation $\sharp$ denotes the displayed algebraic transpose with reversed connection; for complex A it is not a Hermitian adjoint. Here $B(z') = E_{z'} \cap \mathbb{Z}^3 = \{z \in \mathbb{Z}^3 : |z-z'|_\infty < L/2\}$ is

the unit-lattice part of the L-block in §4.1.4; the blocks are taken modulo the torus periods. To specify the phases, if $c \in \Delta_z$ and $z \in B(z')$, set

$$\varphi_A(z,c) = e^{ie_k\eta_k(\tau_k A)(z,c)}, \qquad U_A(z',z) = e^{ie_k\eta_k(\tau A)(z',z)}.$$

The path average τA is the arithmetic mean of the oriented bond sums in the six coordinate orders, and $\tau_k A$ is their sum along the nested block centers. Then

$$\begin{aligned}(\mathcal{Q}_k f)(z) &= \eta_k^3 \sum_{c\in\Delta_z} \varphi_A(z,c)\frac{1}{48}\sum_g f(c,g),\\ (\mathcal{Q}_k^\sharp v)(c,g) &= \varphi_A(z,c)^{-1}v(z),\\ (P_L v)(z) &= U_A(z',z)^{-1}L^{-3}\sum_{z''\in B(z')} U_A(z',z'')v(z'').\end{aligned}$$

The nesting of the odd blocks and cancellation of the phases give

$$\mathcal{Q}_k\mathcal{Q}_k^\sharp = 1, \quad P_L^2 = P_L, \quad \mathcal{Q}_k^\sharp\mathcal{Q}_k = \Pi_1(A), \quad \mathcal{Q}_k^\sharp P_L\mathcal{Q}_k = \Pi_L(A). \tag{4.xlix}$$

All these operators commute with constant spin matrices.

Write $S_k = S_\infty(A)$ and define maps on the unit spinors by

$$\begin{aligned} H_k &= b_k S_k \mathcal{Q}_k^\sharp, & \mathfrak{H}_k &= b_k\mathcal{Q}_k S_k,\\ D_k &= b_k - b_k^2\mathcal{Q}_k S_k\mathcal{Q}_k^\sharp, & M_k &= b_k - D_k.\end{aligned} \tag{4.l}$$

For real y put

$$\mathcal{B}_{k,y} = (b_k + i\gamma_5 y)^{-1}(1-P_L) + (b_k + b_L + i\gamma_5 y)^{-1}P_L.$$

These 4×4 inverses exist since $(t+i\gamma_5 y)^{-1} = (t - i\gamma_5 y)/(t^2+y^2)$ for $t > 0$. The idempotent decomposition and eq. 4.xlix give

$$\mathcal{B}_{k,y}^{-1} = b_k + b_L P_L + i\gamma_5 y, \qquad F_y = F_\infty - b_k^2\mathcal{Q}_k^\sharp\mathcal{B}_{k,y}\mathcal{Q}_k. \tag{4.li}$$

Proposition 4.9. For $k \geq 1$, $A \in \widetilde{\mathcal{G}}_k$, and real y, the operator $D_k + b_L P_L + i\gamma_5 y$ is invertible and its inverse is

$$\Gamma_{k,y} = \mathcal{B}_{k,y} + b_k^2\mathcal{B}_{k,y}\mathcal{Q}_k S_y \mathcal{Q}_k^\sharp\mathcal{B}_{k,y}. \tag{4.lii}$$

Moreover,

$$S_y = S_k + H_k\Gamma_{k,y}\mathfrak{H}_k. \tag{4.liii}$$

In particular, $\Gamma_k = \Gamma_{k,0}$ exists and $\delta Z_k(A) = \det(D_k + b_L P_L) \neq 0$. ■

Proof. Set $Z = b_k^2\mathcal{Q}_k^\sharp\mathcal{B}_{k,y}\mathcal{Q}_k$. The two fine inverses are constructed in §4.1.6 and recovered at parameter value one in Prop. 4.8. eq. 4.li gives both identities

$$S_y - S_k = S_k Z S_y = S_y Z S_k.$$

Since $D_k + b_L P_L + i\gamma_5 y = \mathcal{B}_{k,y}^{-1} - b_k^2\mathcal{Q}_k S_k\mathcal{Q}_k^\sharp$, multiplication by the right side of eq. 4.lii gives the identity: the additional term is $b_k^2\mathcal{Q}_k(S_y - S_k - S_k Z S_y)\mathcal{Q}_k^\sharp\mathcal{B}_{k,y} = 0$. The product in the other order uses $S_y - S_k - S_y Z S_k = 0$. This proves the inverse formula without assuming that the coarse operator is invertible. Substituting the formula in $H_k\Gamma_{k,y}\mathfrak{H}_k$ gives

$$S_k Z S_k + S_k Z S_y Z S_k = S_k Z(S_k + S_y Z S_k) = S_k Z S_y = S_y - S_k.$$

At $y = 0$ the coarse operator is invertible on the finite-dimensional space of spinors on the unit lattice, so its determinant is nonzero. □

At level zero this algebra is replaced by a single chamber integration. The operators b_0, S_∞, $\mathcal{B}_{0,y}$, and M_0 are not defined. Instead,

$$F_0 = D_A + \mu + b_L J P_L J^*, \qquad \Gamma_0 = F_0^{-1}, \qquad \delta Z_0(A) = \det F_0 \neq 0. \tag{4.liv}$$

Here Γ_0 acts on 192 components per photon site, whereas Γ_k for $k \geq 1$ acts on four components per unit site.

4.2.3. *Localized coarse covariances.* For local integration we need the exact field map and covariance together with parameter-dependent versions that separate active components. We define the latter by substituting the localized fine propagators into the exact formulas. At parameter value one they recover the exact operators; away from that value we use the stated bounds and locality properties.

The weakened families for $k \geq 1$ are

$$\begin{aligned} H_k(s,A) &= b_k S_\infty(s,A)\mathcal{Q}_k^\sharp, \\ \Gamma_k(s,A) &= \mathcal{B}_{k,0} + b_k^2 \mathcal{B}_{k,0}\mathcal{Q}_k S_0^{\mathrm{sy}}(s,A)\mathcal{Q}_k^\sharp \mathcal{B}_{k,0}, \\ M_k(X,A) &= b_k^2 \mathcal{Q}_k S_\infty^{\mathrm{sy}}(X,A)\mathcal{Q}_k^\sharp. \end{aligned} \tag{4.lv}$$

The unprimed polymer propagators $S_{k,y}(X,A) = S_y(X,A)$ are used in the determinant integral. At level zero let $\mathrm{Cub}(c)$ be the M-cubes having a lattice site in common with K_c, and set

$$\begin{aligned} (H_0(s,A)u)(c,g) &= \left(\prod_{\square \in \mathrm{Cub}(c)} s_\square\right) u(c,g), \\ \Gamma_0(s,A) &= S_0^{\mathrm{sy}}(s,A), \qquad S_{0,y}(X,A) = S_y(X,A). \end{aligned} \tag{4.lvi}$$

At $s = 1$ these are the exact operators; also $\sum_X M_k(X,A) = M_k(A)$. The exponential formulas for the averaging phases and Prop. 4.8 give analyticity on the same domains.

For use in the estimates, let $\mathfrak{i}(A) = e_k \| \operatorname{Im} A\|_\infty$. The small-connection domain and the charge bound give $\mathfrak{i}(A) \leq 1/(6L)$; reduce γ, if necessary, so that $\gamma L \leq 1/6$. The path lengths imply

$$|\varphi_A(z,c)^{\pm 1}| \leq e^{3\mathfrak{i}(A)/2}, \qquad \|\mathcal{Q}_k\|_{\infty\to\infty}, \|\mathcal{Q}_k^\sharp\|_{\infty\to\infty} \leq e^{3\mathfrak{i}(A)/2}. \tag{4.lvii}$$

The sup norms here are formed with the Euclidean spinor norm. Each unit block has chamber measure one. In particular, for a fine operator O,

$$|(\mathcal{Q}_k O \mathcal{Q}_k^\sharp)(z,z')|_2 \leq \exp(1)\, \|\mathbf{1}_{\Delta_z} O \mathbf{1}_{\Delta_{z'}}\|. \tag{4.lviii}$$

Indeed, apply the two averaging bounds to a unit spinor supported at z'; their product is $e^{3\mathfrak{i}(A)} \leq \exp(1)$.

Proposition 4.10. The families eq. 4.lv satisfy

$$\|H_k(s,A)\|_{\infty\to\infty} \leq 32\exp(1) b_k a' M^{27\alpha_0}, \tag{4.lix}$$

$$\begin{aligned} \|\delta_{\alpha,A} H_k(s,A)\|_{\infty\to\infty} &\leq 304\exp(1) b_k a' M^{27\alpha_0}, \\ |\Gamma_k(s,A)(z,z')|_2 &\leq C_\Gamma M^{27\alpha_0} e^{-\gamma d(z,z')}, \\ |M_k(X,A)(z,z')|_2 &\leq 32\exp(1) b_k^2 a' e^{27\kappa'} e^{-\gamma d(z,z') - \kappa' d_M(X)}, \end{aligned} \tag{4.lx}$$

where

$$C_\Gamma = \frac{2(1+2\exp(1))}{b_-} + 128\exp(1)(1+2\exp(1))^2 a'.$$

The Hölder estimate is on short pairs, including the distinct sites of one cluster. The families $H_k(s,A)$ and $\Gamma_k(s,A)$ preserve every component of the active M-cubes and depend there only on its parameters and connection. The row of $H_k(s,A)$ at (c,g) vanishes unless all cubes meeting K_c are active. The polymer $M_k(X,A)$ is supported on the unit sites of X and depends only on the connection in X.

At level zero the normalized kernel of $\Gamma_0(s,A)$ is bounded by $32a'M^{27\alpha_0}e^{-\gamma d(c,c')}$, and eq. 4.xlvi bounds $S_{0,y}(X,A)$ with its factor $(1+|y|)^{-1}$. The multiplier $H_0(s,A)$ has sup norm at most $M^{8\alpha_0}$ and equals the identity at $s=1$. Its row at (c,g) vanishes unless every cube meeting K_c is active. There are no short pairs of distance less than one at this level. ■

Proof. The first two estimates follow by composing the weakened fine bounds with eq. 4.lvii. The third and fourth use eq. 4.lviii. For the covariance it remains to control the two factors $\mathcal{B}_{k,0}$. Its kernel is

$$\mathcal{B}_{k,0}(z,z') = b_k^{-1}\mathbf{1}[z=z']1_4 - \frac{b_L}{b_k(b_k+b_L)}P_L(z,z').$$

It vanishes unless z,z' belong to one L-block. Since $|P_L(z,z')|_2 \le \exp(1)L^{-3}$ on that block, its row and column sums are at most $(1+2\exp(1))/b_k$. In a nonzero term of $\mathcal{B}_{k,0}G\mathcal{B}_{k,0}$, where $G = \mathcal{Q}_k S_0^{\mathrm{sy}}(s,A)\mathcal{Q}_k^\sharp$, the two intermediate sites are within L of the exterior sites. Therefore

$$e^{-\gamma d(z_1,z_2)} \le e^{2\gamma L}e^{-\gamma d(z,z')} \le 4e^{-\gamma d(z,z')}.$$

The fine bound and eq. 4.lviii give $|G(z_1,z_2)|_2 \le 32\exp(1)a'M^{27\alpha_0}e^{-\gamma d(z_1,z_2)}$. Multiplying the row and column sums cancels the factor b_k^2 and gives $128\exp(1)(1+2\exp(1))^2a'M^{27\alpha_0}$. The local term $\mathcal{B}_{k,0}$ is bounded by $2(1+2\exp(1))b_-^{-1}e^{-\gamma d(z,z')}$, proving eq. 4.lx.

The odd block partitions are nested, and every unit block and every L-block belongs to a single M-cube. Moreover, the paths entering $\varphi_A(z,c)$ lie in Δ_z, and the paths entering $U_A(z',z)$ lie in the fine block of side L centered at z': each coordinate of a rectilinear path remains between those of its endpoints. Consequently the averaging maps and $\mathcal{B}_{k,0}$ preserve every union of M-cubes and depend only on its connection. Composing them with the component-local fine families proves the locality assertions. This applies also to the parameter-independent term $\mathcal{B}_{k,0}$ in $\Gamma_k(s,A)$. For the row of H_k, the terminal average does not change the first row index of $S_\infty(s,A)$, so the stronger vanishing statement is preserved. At level zero the assertions for Γ_0 follow from the fine kernel bound and locality with $\eta_0 = 1$.

For H_0, each coordinate interval of K_c has $4L+1$ sites and meets at most two consecutive M-blocks, since $M \ge L^2$, $L \ge 5$. Its at most eight meeting cubes are face-connected and contain the cube of c. The defining parameter product therefore has degree at most eight, norm at most $M^{8\alpha_0}$, vanishes if any required cube is inactive, and otherwise depends only on parameters in that active component. It is scalar in spin and chamber labels and independent of A, giving the remaining claims. □

These families have the gauge and lattice covariance of their exact counterparts. For Γ_k and M_k, transposition follows from eq. 4.xlviii at $y=0$ and $y=\infty$, respectively, and the scalar averaging phases. The barred field map is $\bar{H}_k(s,A) = \mathsf{C}^{-1}H_k(s,-A)\mathsf{C}$. Its bounds follow from those for H_k because C is unitary and $\mathsf{C}\gamma_\mu\mathsf{C}^{-1} = -\gamma_\mu^T$. When component sup norms are used in the Grassmann substitution, the Euclidean operator bounds above acquire at most the factor two from $|v|_c \le |v|_2 \le 2|v|_c$.

4.2.4. *Weighted Grassmann integration.* We fix the integration normalization and the coefficient norm together, since the first block step uses chamber measure $1/48$ where later steps use counting measure. The formulas below distinguish an operator kernel, which includes an input measure in its action, from its matrix entries. This distinction determines both the Gaussian two-point function and the weight lost in substitution.

Let $\phi_i(t)$, $1 \le i \le p$, be independent Grassmann families on finite ordered index sets. If $\phi_i(I_i)$ is the ordered product over I_i, write

$$F = \sum_{I_1,\dots,I_p} c_F(I_1,\dots,I_p)\phi_1(I_1)\cdots\phi_p(I_p), \qquad \|F\|_{\mathsf{w}} = \sum_I |c_F(I)| \prod_i \mathsf{w}_i^{|I_i|}.$$

The positive measures $\nu_i(t)$ on the index sets enter the kernels, not this coefficient norm. Thus an antisymmetric kernel expansion

$$F = \sum_n \prod_i \frac{1}{n_i!} \int \mathrm{d}t\, K_n(t) \prod_i \prod_{j=1}^{n_i} \phi_i(t_j^i)$$

has norm

$$\|F\|_{\mathsf{w}} = \sum_n \prod_i \frac{\mathsf{w}_i^{n_i}}{n_i!} \int \mathrm{d}t\, |K_n(t)|. \tag{4.lxi}$$

For arbitrary kernels the right side is an upper bound. Products of ordered monomials vanish on repeated generators and otherwise multiply their weights, so

$$\|FG\|_{\mathsf{w}} \le \|F\|_{\mathsf{w}} \|G\|_{\mathsf{w}}, \qquad \|e^F - 1\|_{\mathsf{w}} \le e^{\|F\|_{\mathsf{w}}} - 1 \quad (F \text{ even}). \tag{4.lxii}$$

For an additional finite nonempty mode set $\mathcal{I}$ with positive weights $\nu(t)$, introduce $W(t), \bar{W}(t)$ and set

$$\langle \bar{W}, OW \rangle_\nu = \sum_{t,t'} \nu(t)\nu(t') \bar{W}(t) O(t,t') W(t'), \qquad \mathrm{Top} = \prod_{t \in \mathcal{I}} W(t)\bar{W}(t).$$

The kernel represents O with measure ν; its matrix entry is $O(t,t')\nu(t')$. Define $\int \mathrm{d}W$ as extraction of the coefficient of Top, multiplied by $\prod_t \nu(t)^{-1}$, and define

$$\int \mathrm{d}\mu_I(W) F = \int \mathrm{d}W\, e^{-\langle \bar{W}, W \rangle_\nu} F, \tag{4.lxiii}$$

$$\int \mathrm{d}\mu_C(W) F = (\det C) \int \mathrm{d}W\, e^{-\langle \bar{W}, C^{-1} W \rangle_\nu} F \quad (C \text{ invertible}).$$

These are algebra-valued functionals when other Grassmann generators are present. The order of modes in Top is immaterial because its factors are even.

Proposition 4.11. Put $w_W = \max_t \nu(t)^{-1/2}$. Then

$$\left\| \int \mathrm{d}\mu_I(W) F \right\|_{\mathsf{w}} \le \|F\|_{(\mathsf{w}, w_W)}. \tag{4.lxiv}$$

The integral factors for polynomials involving disjoint sets of modes. For an invertible covariance,

$$\int \mathrm{d}\mu_C(W) F(\bar{W}, W) = \int \mathrm{d}\mu_I(W) F(\bar{W}, CW), \tag{4.lxv}$$

$$\int \mathrm{d}\mu_C(W) W(t) \bar{W}(t') = C(t,t'),$$

$$\int \mathrm{d}W\, e^{-\langle \bar{W}, OW \rangle_\nu} = \det O \quad \text{for every } O. \tag{4.lxvi}$$

Finally, consider a linear substitution $\phi_i \mapsto \sum_j O_{ij} \theta_j$, and put

$$\|O_{ij}\|_{1,\infty} = \sup_t \sum_s |O_{ij}(t,s)| \nu_j'(s).$$

If $\sum_j \|O_{ij}\|_{1,\infty} \mathsf{w}_j' \le \mathsf{w}_i$ for each i, then its image F' obeys $\|F'\|_{\mathsf{w}'} \le \|F\|_{\mathsf{w}}$. Only rows corresponding to generators occurring in F need satisfy this bound. ∎

Proof. The identity density factors as

$$e^{-\langle \bar{W}, W \rangle_\nu} = \prod_t (1 + \nu(t) W(t) \bar{W}(t)).$$

An ordered monomial $W(J)$ contributes to its integral only if J contains both generators of every mode it meets. In that case its integral is its reordering sign times $\prod_{(t,0) \in J} \nu(t)^{-1}$; otherwise it is zero. Its modulus is therefore at most $\prod_{u \in J} \nu(u)^{-1/2} \le w_W^{|J|}$. Apply this coefficientwise to obtain eq. 4.lxiv. For disjoint mode sets a nonzero integral requires each monomial separately to be paired.

Their degrees in W are then even, so the signs from exchanging external generators cancel. The paired-mode formula gives factorization. It also shows that modes absent from the integrand may be removed.

A linear change $W \mapsto OW$ multiplies the coefficient of the top monomial by $\det O$, since the matrix of the change is $O(t,t')\nu(t')$. Applied to the identity density this proves eq. 4.lxvi, without an invertibility assumption. For invertible C, use $W \mapsto CW$, leaving $\bar{W}$ fixed, in the density with covariance C. This proves eq. 4.lxv and normalization. Its two-point function is

$$\sum_s C(t,s)\nu(s)\int \mathrm{d}\mu_I(W)W(s)\bar{W}(t') = \sum_s C(t,s)\nu(s)\frac{\mathbf{1}[s=t']}{\nu(t')} = C(t,t').$$

Thus it is the kernel, rather than the matrix entry.

Each substituted generator has norm at most

$$\sum_j \mathsf{w}'_j \sum_s |O_{ij}(t,s)|\nu'_j(s) \le \mathsf{w}_i$$

Submultiplicativity and the coefficient sum prove the substitution bound; only rows occurring in the polynomial are used. □

For linear combinations $v_1,\dots,v_{2r}$ of the $W,\bar{W}$, Wick's formula gives

$$\int \mathrm{d}\mu_I(W)v_1\cdots v_{2r} = \mathrm{Pf}\left(\int \mathrm{d}\mu_I(W)v_iv_j\right)_{i,j}. \tag{4.lxvii}$$

Odd products integrate to zero. To fix the signs, let ∂_u be the left Grassmann derivative. It satisfies $\partial_u(vG) = (\partial_u v)G - v\partial_u G$, and its Berezin integral is zero. Differentiating the factored Gaussian density gives

$$\int \mathrm{d}\mu_I(W)W(t)G = \nu(t)^{-1}\int \mathrm{d}\mu_I(W)\partial_{\bar{W}(t)}G,$$
$$\int \mathrm{d}\mu_I(W)\bar{W}(t)G = -\nu(t)^{-1}\int \mathrm{d}\mu_I(W)\partial_{W(t)}G.$$

Consequently, if $I(v) = \int \mathrm{d}\mu_I(W)v$,

$$I(v_1\cdots v_q) = \sum_{j=2}^{q}(-1)^j I(v_1v_j)I(v_2\cdots\widehat{v}_j\cdots v_q).$$

This is the expansion of the Pfaffian along its first row and proves eq. 4.lxvii by induction. Formula eq. 4.lxv gives the corresponding Wick rule for μ_C.

In the renormalization step the modes have weight one for $k \ge 1$. At $k=0$ they are (c,g,β) with weight $1/48$. Thus the integration weight is

$$w_W = 1 \quad (k\ge 1), \qquad w_W = \sqrt{48} \quad (k=0). \tag{4.lxviii}$$

In particular the first-step identity covariance has two-point function $48\mathbf{1}[(c,g,\beta)=(c',g',\beta')]$. The pair index used for a Hölder field has measure $(\eta_k^3/48)^2$. In a substitution involving $\delta_{\alpha,A}H_k$, its scalar component row sum is computed with the input measure, as in Prop. 4.11. Internal-cluster differences are therefore controlled by the Hölder bound with the same field weight as other pairs.

4.2.5. *The Gaussian block identity.* Fermion block integration produces three objects: the coarse quadratic form, a determinant, and a Gaussian fluctuation with its own covariance. Completing the square identifies all three in one identity. We use the same calculation for the first chamber step and for the later four-component steps, keeping their different averages and measures explicit.

At $k \geq 1$ let $\mathcal{D}_k = D_k$ and $\mathcal{A}_k(A) = Q(A)$; at $k = 0$ let $\mathcal{D}_0 = D_A + \mu$ and $\mathcal{A}_0(A) = Q(A)J^*$. The integration measure is respectively counting measure or the chamber measure $1/48$. The target is the four-component spinor space on the lattice of spacing L, with measure L^3. Set

$$\mathcal{A}_k^\sharp(A) = \mathcal{A}_k(-A)^T, \qquad K_k = \mathcal{D}_k + b_L \mathcal{A}_k^\sharp \mathcal{A}_k, \qquad \Gamma_k = K_k^{-1}.$$

Thus $K_k = D_k + b_L P_L$ for $k \geq 1$ and $K_0 = F_0$. Define the quadratic operator on the coarse lattice by

$$D_{k+1}^0 = b_L - b_L^2 \mathcal{A}_k \Gamma_k \mathcal{A}_k^\sharp.$$

Let $\Phi, \bar{\Phi}$ be external Grassmann fields on the coarse lattice and put

$$\Psi^c = b_L \Gamma_k \mathcal{A}_k^\sharp \Phi, \qquad \bar{\Psi}^c = b_L \Gamma_k^T \mathcal{A}_k(A)^T \bar{\Phi}.$$

For $\Psi = \Psi^c + W'$ and $\bar{\Psi} = \bar{\Psi}^c + \bar{W}'$, completion of the square is the exact identity

$$\langle \bar{\Psi}, \mathcal{D}_k \Psi \rangle + b_L \langle \bar{\Phi} - \mathcal{A}_k(-A)\bar{\Psi}, \Phi - \mathcal{A}_k(A)\Psi \rangle_L \\ = \langle \bar{\Phi}, D_{k+1}^0 \Phi \rangle_L + \langle \bar{W}', K_k W' \rangle. \quad (4.\text{lxix})$$

Put $s = b_L \mathcal{A}_k^\sharp \Phi$ and $\bar{s} = b_L \mathcal{A}_k(A)^T \bar{\Phi}$. The left side is

$$\langle \bar{\Psi}, K_k \Psi \rangle - \langle \bar{\Psi}, s \rangle - \langle \bar{s}, \Psi \rangle + b_L \langle \bar{\Phi}, \Phi \rangle_L.$$

Substitute the means $\Gamma_k s, \Gamma_k^T \bar{s}$. The cross terms cancel and the constant is $b_L \langle \bar{\Phi}, \Phi \rangle_L - \langle \bar{s}, \Gamma_k s \rangle = \langle \bar{\Phi}, D_{k+1}^0 \Phi \rangle_L$. All factors retain their displayed order; no Grassmann conjugation is used.

Translation by external odd fields preserves the top coefficient in the integration variables. Applying this to eq. 4.lxix and then using eq. 4.lxv gives, for any Grassmann polynomial F, the block integration formula

$$\int \mathrm{d}\Psi \, e^{-\langle \bar{\Psi}, \mathcal{D}_k \Psi \rangle - b_L \langle \bar{\Phi} - \mathcal{A}_k(-A)\bar{\Psi}, \Phi - \mathcal{A}_k(A)\Psi \rangle_L} F(\bar{\Psi}, \Psi) \\ = \delta Z_k(A) e^{-\langle \bar{\Phi}, D_{k+1}^0 \Phi \rangle_L} \int \mathrm{d}\mu_I(W) F(\bar{\Psi}^c + \bar{W}, \Psi^c + \Gamma_k W). \quad (4.\text{lxx})$$

At the first step this is a single integration over all chamber components. The determinant and the integration weight are those of eq. 4.liv and eq. 4.lxviii; the heavy components are included in both.

4.2.6. *An integral for the determinant.* The Gaussian block identity leaves a determinant as well as the coarse quadratic form. To write its logarithm as a sum of local terms, we first express it through the resolvents already decomposed into polymers. At a real connection, the adjoint symmetry makes $K_k \gamma_5$ Hermitian; integration of its scalar eigenvalue resolvents gives the determinant, including its constant phase. The positive levels and level zero require different subtractions at large y, which we keep separate below. The local estimates will then continue the formulas to complex connections. Put $n_k = \ell^3$ and

$$c_k^d = 4((1 - L^{-3}) \log b_k + L^{-3} \log(b_k + b_L)) \quad (k \geq 1), \qquad c_0^d = 0.$$

For real $A \in \widetilde{\mathcal{G}}_k$ and $0 < \mu \leq 1$,

$$\delta Z_k(A) = \exp\left\{ c_k^d n_k - i b_k^2 \int_0^\infty \mathrm{d}y \ \mathrm{Tr}[\gamma_5 \mathcal{B}_{k,y} \mathcal{Q}_k S_y \mathcal{Q}_k^\sharp \mathcal{B}_{k,y}] \right\} \quad (k \geq 1), \quad (4.\text{lxxi})$$

$$\delta Z_0(A) = \exp\left\{ \int_1^\infty \frac{\mathrm{d}y}{y} \mathrm{Tr}[F_0 S_y] - i \int_0^1 \mathrm{d}y \ \mathrm{Tr}[\gamma_5 S_y] \right\}. \quad (4.\text{lxxii})$$

The trace in the first line is on the unit spinor space. At level zero it is the normalized chamber trace $\mathrm{Tr}\, O = 48^{-1} \sum_{c,g} \mathrm{tr}\, O((c,g),(c,g))$.

To prove these identities, write $K = K_k$ as in the preceding subsubsection. At real A, the fine adjoint identity $D_A^* = \gamma_5 D_A \gamma_5$, the reality of μ, b_k, b_L, and the orthogonal averaging projections give $K^* = \gamma_5 K \gamma_5$. Hence $K\gamma_5$ is Hermitian and invertible. Since $\det \gamma_5 = 1$ on each four-component

spinor, its determinant is $\det K$. Let λ_j be its real nonzero eigenvalues, counted with multiplicity. There are $4n_k$ at positive levels and $192n_0$ at level zero. Then

$$\operatorname{Tr}[\gamma_5(K+i\gamma_5 y)^{-1}] = \sum_j (\lambda_j + iy)^{-1}, \qquad \operatorname{Tr}[K(K+i\gamma_5 y)^{-1}] = \sum_j \lambda_j(\lambda_j+iy)^{-1}.$$

For a real $\lambda \neq 0$, choose along $y \geq 0$ the logarithm with value $\log|\lambda| + i\pi \mathbf{1}[\lambda < 0]$ at zero. Its derivative is $i/(\lambda+iy)$ and $\operatorname{Log}(\lambda+iy) - \log y \to i\pi/2$. Integration gives

$$-i\int_0^Y \frac{\mathrm{d}y}{\lambda+iy} = \operatorname{Log}\lambda - \operatorname{Log}(\lambda+iY),$$
$$\int_1^\infty \frac{\mathrm{d}y}{y}\frac{\lambda}{\lambda+iy} - i\int_0^1 \frac{\mathrm{d}y}{\lambda+iy} = \operatorname{Log}\lambda - i\pi/2.$$

Summing the second identity proves eq. 4.lxxii: its constant phase is $-96\pi i n_0 \in 2\pi i\mathbb{Z}$.

For $k \geq 1$, the ranks of P_L and $1-P_L$ are respectively $4L^{-3}n_k$ and $4(1-L^{-3})n_k$. The explicit formula for $\mathcal{B}_{k,y}$ therefore gives

$$-i\int_0^Y \mathrm{d}y\ \operatorname{Tr}[\gamma_5 \mathcal{B}_{k,y}] = -4n_k \log Y + c_k^d n_k + o(1) \quad (Y\to\infty).$$

Insert eq. 4.lii in the first scalar integral. The terms $4n_k \log Y$ cancel. The remaining phase is $2\pi i n_k$, which disappears on exponentiation, and this proves eq. 4.lxxi. The integral converges absolutely: block compression bounds $\mathcal{Q}_k S_y \mathcal{Q}_k^\sharp$ uniformly, while each row and column sum of $\mathcal{B}_{k,y}$ is at most $(1+2\exp(1))(b_k^2+y^2)^{-1/2}$. Its trace is bounded by

$$128\exp(1)(1+2\exp(1))^2 a' n_k (b_k^2+y^2)^{-1}.$$

At level zero the spectral identities already give absolute convergence. The uniform kernel bounds below will give a bound independent of the eigenvalues.

4.2.7. *Local determinant terms.* Insert each polymer propagator separately into the determinant integral. Its decay must absorb both the number of possible trace sites in the polymer and the resolvent integration. We then average the local terms under charge and complex conjugation. The sum still gives the determinant ratio, while each term acquires the symmetries needed in the renormalization step.

Choose $\gamma_{\mathrm{w}} \leq \gamma$ and $\kappa_{\mathrm{w}} = \kappa + \kappa_0 + 3$ with $\kappa, \kappa_0 > 0$. The propagator bounds above give a fixed constant C_w such that, in component sup norms, for $y \geq 0$,

$$\|\mathbf{1}_{\Delta_z} S_{k,y}(X,A)\mathbf{1}_{\Delta_{z'}}\| \leq C_w e^{-\gamma_{\mathrm{w}} d(z,z') - \kappa_{\mathrm{w}} d_M(X)} \quad (k\geq 1),$$
$$|S_{0,y}(X,A)(x,x')|_c \leq C_w (1+y)^{-1} e^{-\gamma_{\mathrm{w}} d(c,c') - \kappa_{\mathrm{w}} d_M(X)}.$$

Here $|\cdot|_c$ is the matrix norm induced by the component sup norm. We may use the same C_w after reducing the rates from γ, κ'. For $X \in \mathcal{P}_k$ define

$$\widetilde{E}_k(X,A) = -ib_k^2 \int_0^\infty \mathrm{d}y\ \operatorname{Tr}[\gamma_5 \mathcal{B}_{k,y}\mathcal{Q}_k S_{k,y}(X,A)\mathcal{Q}_k^\sharp \mathcal{B}_{k,y}] \quad (k\geq 1), \tag{4.lxxiii}$$
$$\widetilde{E}_0(X,A) = \int_1^\infty \frac{\mathrm{d}y}{y} \operatorname{Tr}[F_0 S_{0,y}(X,A)] - i\int_0^1 \mathrm{d}y\ \operatorname{Tr}[\gamma_5 S_{0,y}(X,A)],$$
$$E_k^d(X,A) = \tfrac{1}{4}(\widetilde{E}_k(X,A) + \widetilde{E}_k(X,-A) + \overline{\widetilde{E}_k(X,\overline{A})} + \overline{\widetilde{E}_k(X,-\overline{A})}),$$
$$E_k^{\det}(X,A) = E_k^d(X,A) - E_k^d(X,0).$$

Set $c_X = 54$ and

$$C_d = 16 c_X C_w \max\left\{\tfrac{\pi}{2} b\exp(1)(1+2\exp(1)),\ 2+58\exp(1)+b\exp(1)\right\}.$$

Proposition 4.12. The integrals eq. 4.lxxiii converge absolutely for $A \in \widetilde{\mathcal{G}}_k$. They define analytic functions, depending only on bonds meeting X, with

$$|\widetilde{E}_k(X,A)|, |E_k^d(X,A)| \le C_d M^3 e^{-(\kappa+1)d_M(X)}. \tag{4.lxxiv}$$

They satisfy

$$\delta Z_k(A) = e^{c_k^d n_k + \sum_X \widetilde{E}_k(X,A)}, \qquad \frac{\delta Z_k(A)}{\delta Z_k(0)} = e^{\sum_X E_k^{\det}(X,A)}. \tag{4.lxxv}$$

The functions E_k^d are gauge invariant, even in A, and satisfy $\overline{E_k^d(X,\overline{A})} = E_k^d(X,A)$. They are covariant under the lattice step group. ∎

Proof. For $k \ge 1$ write $N_y(X) = \mathcal{Q}_k S_{k,y}(X,A)\mathcal{Q}_k^\sharp$. It is supported on the unit sites of X, and each entry is bounded by $\exp(1)C_w e^{-\kappa_{\mathrm{w}} d_M(X)}$ by normalized block compression. To estimate the trace, cycle one factor $\mathcal{B}_{k,y}$ to the left and use

$$G_y = \mathcal{B}_{k,y}\gamma_5\mathcal{B}_{k,y} = \gamma_5\big((b_k + i\gamma_5 y)^{-2}(1-P_L) + (b_k + b_L + i\gamma_5 y)^{-2}P_L\big).$$

Its column sum in Euclidean matrix norm is at most $(1+2\exp(1))/(b_k^2+y^2)$: the identity contributes one and the two projection coefficients contribute at most $2\exp(1)$. Since $|\operatorname{tr}(mm')| \le 8|m|_2 \max_{i,j}|m'_{ij}|$ for four-component matrices, the integrand is bounded by

$$8\exp(1)(1+2\exp(1))C_w M^3 |X|_M e^{-\kappa_{\mathrm{w}} d_M(X)}(b_k^2+y^2)^{-1}. \tag{4.lxxvi}$$

Integrating and multiplying by b_k^2 leaves the factor $\pi b_k/2 \le \pi b/2$.

At $k=0$, the formula for D_A and the averaging penalty give the scalar component row bound

$$\sup_{x,\beta}\sum_{\beta'}\int \mathrm{d}x' \, |F_0(A)(x,x')_{\beta\beta'}| \le R_F := 58\exp(1) + 1 + b\exp(1). \tag{4.lxxvii}$$

The diagonal and three internal hops contribute at most $41+\mu$, the external hop at most $6e^{\mathrm{i}(A)}$, and the block penalty at most $b\exp(1)$. The bound follows from $\mu \le 1$ and $e^{\mathrm{i}(A)} \le \exp(1)$. Point evaluation has normalized kernel $48\mathbf{1}[x=x']$; its factor 48 cancels the chamber measure. The two traces in the level-zero integral are consequently bounded by

$$4R_F C_w M^3 |X|_M (1+y)^{-1} e^{-\kappa_{\mathrm{w}} d_M(X)}, \qquad 4C_w M^3 |X|_M (1+y)^{-1} e^{-\kappa_{\mathrm{w}} d_M(X)}.$$

The second bound uses that γ_5 has four entries of modulus one. Their integrals contribute respectively $\int_1^\infty (y(1+y))^{-1}\,\mathrm{d}y = \log 2$ and $\int_0^1 (1+y)^{-1}\,\mathrm{d}y = \log 2$.

The geometric bound eq. D.i gives $|X|_M \le c_X(d_M(X)+1) \le c_X e^{d_M(X)}$. Applying this to the trace bounds, and using $\kappa_{\mathrm{w}} - 1 \ge \kappa + 1$, proves eq. 4.lxxiv for $\widetilde{E}_k$ and hence for its four-term mean. The majorants are independent of A and integrable in y. Each integrand is analytic in A; Cauchy's formula on a compact polydisc and the majorant permit its contour integrals to be exchanged with the y integral. The result is analytic.

Every phase in the compressed trace belongs to a unit block or an L-block meeting X, and that block is contained in X by nesting. The fine propagator is supported on X and uses only its connection. At level zero a nonzero external-hop trace uses a bond meeting X; the penalty uses the L-block of its row. This proves locality. Gauge transformations conjugate each product under the trace and hence leave it unchanged. A lattice transformation whose spin matrix commutes with γ_5 does the same. Every spatial action of the step group has such a lift: multiplication by $\gamma_1\gamma_2\gamma_3$ changes its commutation sign with γ_5 and leaves its spatial action fixed. This proves step covariance. The mean in eq. 4.lxxiii then makes E_k^d even and real in the stated analytic sense.

Summing the propagators in eq. 4.lxxi or eq. 4.lxxii proves the first quotient identity for real A. Both sides are analytic on the open star-shaped $\widetilde{\mathcal{G}}_k$; equality near real zero determines their Taylor series and hence their analytic continuation.

To justify the symmetrized quotient, put $\mathcal{L}(A) = \sum_X \widetilde{E}_k(X, A)$. The charge-conjugation transpose and the complex-conjugation identity give $\delta Z_k(-A) = \delta Z_k(A) = \overline{\delta Z_k(\overline{A})}$. For the latter, the spin matrix $\varkappa = \gamma_5 \mathsf{C}$ conjugates $\overline{K_k(\overline{A})}$ to $K_k(-A)$. Thus $\mathcal{L}(-A) - \mathcal{L}(A)$ and $\overline{\mathcal{L}(\overline{A})} - \mathcal{L}(A)$ take values in $2\pi i\mathbb{Z}$. They are continuous on the connected domain, so they are constant. Subtracting their values at zero gives

$$\sum_X (E_k^d(X, A) - E_k^d(X, 0)) = \mathcal{L}(A) - \mathcal{L}(0).$$

This proves the second identity in eq. 4.lxxv without a choice of logarithm branch for the complex determinant. □

4.2.8. *The small-field gain for the determinant.* The local determinant bound is uniform but does not yet become small with the charge. We obtain that gain in two ways. For a large polymer, one unit of its decay rate supplies $e_k^{1/4}$. For a small polymer, a real gauge makes the background small throughout a box containing its support. The determinant is then analytic in the amplitude of that background on a disc of radius proportional to $e_k^{-1/4}$, and Cauchy's formula gives the same gain. We now specify the domains and size conditions for these two cases.

Use the photon domains R_k and $\widetilde{R}_k$ from §3.1.8, with mesh-normalized differences and curl. The three bounds for R_k are $e_k^{-3/4+i\varepsilon}$, $i = 1, 2, 3$, for the field, its first differences, and their Hölder quotients, respectively. The domain $\widetilde{R}_k$ allows the same local bounds after a real gauge change on each $\square^\dagger = \square^{\sim L}$, together with a global remainder in R_k. The external connection domain is

$$\mathfrak{X}_k = \{\mathscr{C} : \|\mathscr{C}\|_\infty < e_k^{-1+2\varepsilon}, \quad \|\partial\mathscr{C}\|_\infty < e_k^{-1+3\varepsilon}\}.$$

If $e_k^{1/4} + e_k^\varepsilon < 1$, both $\mathfrak{X}_k$ and $\widetilde{R}_k + \mathfrak{X}_k$ lie in $\widetilde{\mathcal{G}}_k$: for $\mathscr{A} = A_0 + A_1 \in \widetilde{R}_k$, the real summand A_0 has on the local cubes a representative of size $e_k^{-3/4+\varepsilon}$, and the remainder $A_1 + \mathscr{C}$ has size at most $(e_k^{1/4} + e_k^\varepsilon)e_k^{-1+\varepsilon}$. For $\mathfrak{X}_k$ alone take $A_0 = A_1 = 0$.

Put $v_k = -\log e_k$. Assume

$$v_k \geq 9, \qquad e_k^\varepsilon \leq \tfrac{1}{4}, \qquad 8M(v_k/4 + 4)e_k^\varepsilon \leq 1, \qquad \ell \geq Mv_k^2. \tag{4.lxxviii}$$

The last condition follows from the stop-scale volume condition $\ell \geq Mr_k$ with $r_k \geq v_k^2$.

Proposition 4.13. For $\mathscr{A} \in \widetilde{R}_k$, $\mathscr{C} \in \mathfrak{X}_k$, and $X \in \mathcal{P}_k$, under eq. 4.lxxviii,

$$|E_k^d(X, \mathscr{A} + \mathscr{C}) - E_k^d(X, \mathscr{C})| \leq 8C_d M^3 e_k^{1/4} e^{-\kappa d_M(X)}. \tag{4.lxxix}$$

■

Proof. If $d_M(X) > v_k/4$, bound the two terms separately by eq. 4.lxxiv and use $e^{-d_M(X)} \leq e_k^{1/4}$. Suppose therefore that $d_M(X) \leq v_k/4$. The polymer lift and diameter estimates give a nonwrapping lattice box W containing every bond meeting X, with each side of length at most $s_W = M(d_M(X)+3)$. Their applicability follows from $2M(d_M(X) + 2) < \ell$ and $M(d_M(X) + 4) < \ell$, which hold by eq. 4.lxxviii.

Write $\mathscr{A} = A_0 + A_1$ as in the domain definition. On each plaquette, the local real representative gives $\|dA_0\|_\infty < 2e_k^{-3/4+2\varepsilon}$. Choose a fixed-order rectilinear path $\Gamma(o, c)$ from a site $o \in W$ and set $\lambda(c) = \eta_k A_0(\Gamma(o, c))$ on W. Then

$$\|A_0 - d\lambda\|_{\infty,W} \leq 2s_W \|dA_0\|_{\infty,W} < 4s_W e_k^{-3/4+2\varepsilon}.$$

To see the factor $2s_W$, append one bond to $\Gamma(o, c)$ and subtract $\Gamma(o, c')$. After cancelling common bonds, the resulting loop bounds at most two rectangles of width η_k. Together they contain at most $2s_W/\eta_k$ plaquettes, each contributing $\eta_k dA_0$. The paths and rectangles remain in W because it is a product of intervals. Extend the real function λ by zero outside the projected box. On every bond meeting X, the bound is unchanged. By eq. 4.lxxviii it is at most $\frac{1}{2}e_k^{-3/4+\varepsilon}$.

Let $B = A_0 - d\lambda + A_1$ on bonds meeting X and zero elsewhere. Then $\|B\|_\infty < \frac{3}{2} e_k^{-3/4+\varepsilon}$. For $|t| \le \frac{1}{3} e_k^{-1/4}$,

$$\|tB + \mathscr{C}\|_\infty < (\tfrac{1}{2} + e_k^{\varepsilon}) e_k^{-1+\varepsilon} \le \tfrac{3}{4} e_k^{-1+\varepsilon}.$$

Thus this connection belongs to the complex connection ball $\mathcal{G}_k = \{A : \|A\|_\infty < e_k^{-1+\varepsilon}\}$, which is contained in $\widetilde{\mathcal{G}}_k$, and $f(t) = E_k^d(X, tB + \mathscr{C})$ is analytic on this disk and bounded by the right side of eq. 4.lxxiv. Gauge invariance and locality identify the difference in eq. 4.lxxix with $f(1) - f(0)$. On the circle of radius $r = \frac{1}{4} e_k^{-1/4} > 2$, Cauchy's formula yields

$$f(1) - f(0) = \frac{1}{2\pi i} \oint_{|t|=r} \frac{f(t)}{t(t-1)}\, \mathrm{d}t, \qquad |f(1) - f(0)| \le \frac{\sup |f|}{r-1} \le 8 e_k^{1/4} \sup |f|.$$

This proves the claimed gain, retaining the stronger decay $e^{-(\kappa+1) d_M(X)}$ in this case. □

5. The renormalization group: one step

A block step must preserve the form of the density while reducing the interaction that remains after its field-independent and mass terms are removed. We construct the local step, prove its contraction estimate and retain the scalar normalization in the exact density. Compatibility on covers will allow the next section to iterate the same local construction on both reference tori.

5.1. **Small-field renormalization.** We use $\gamma_\mu = \sigma_3 \otimes \sigma_\mu$, $\gamma_5 = \sigma_1 \otimes 1_2$ and $\Sigma_3 = \sigma_3 \otimes 1_2$. The matrices acting on the Grassmann variables below are

$$\mathsf{C} = i\gamma_2, \qquad \varkappa = \gamma_5 \mathsf{C} = \sigma_2 \otimes \sigma_2.$$

Both are real, $\mathsf{C}^T = \mathsf{C}^{-1} = -\mathsf{C}$, and $\varkappa^T = \varkappa = \varkappa^{-1}$. The regulator matrix in the reflection-positive construction is $C = \sigma_2 \oplus \sigma_2$; the relation in these spin conventions is $\mathsf{C} = i\Sigma_3 C$.

5.1.1. *Activities, norms and fields.* At level k, the interaction supplied to the step is a family of local Grassmann polynomials, one for each polymer. We call it the incoming activity. Its norm must control both the polynomial coefficients and their dependence on the connections. Fix a level k, mesh $\eta_k = L^{-k}$ and an admissible side ℓ, with $\ell/(LM)$ odd and at least 13. Let $\mathcal{P}_k$ be the collection of face-connected polymers of M-cubes on this torus. A polymer is small when $d_M(X) \le L$. A small polymer has an injective lift, unique up to a period translation, by the diameter and lifting estimates of App. D. Write $\mathrm{Vol}(X) = M^3 |X|_{\underline{M}}$.

We use ordinary Grassmann generators $\psi, \bar\psi$ on chamber sites $x = (c, g)$ and pair generators $\chi, \bar\chi$ on ordered pairs. The latter record covariant Hölder quotients when the fields are evaluated; before evaluation all generators are independent. For distinct chamber sites on the covering lattice, put $d_X(x, x') = \max(|c - c'|_\infty, \eta_k)$ and restrict pairs to $d_X \le M$. Pairs on the torus are simultaneous period classes; their shortest lifts are unique since $2M < \ell$. A pair is admissible for X when its photon sites and all six rectilinear paths between them lie in X, or when it is $(x, x^{(3)})$ and one of its photon sites belongs to X. Thus a pair may leave X only along a bond of type 3. Pairs inside a cluster have $d_X = \eta_k$ and are admissible when their common photon site belongs to X. Ordinary generators acquire the sign $(-1)^{n_1+n_2+n_3}$ under translation by ℓn. Pair generators acquire this sign under simultaneous translation of both endpoints by ℓn. Their kernels are antiperiodic in each ordinary argument and in each simultaneous pair argument.

Let $\xi = (x, \beta, \omega)$ index an ordinary leg and $\zeta = (x, x', \beta, \omega)$ a Hölder leg, where $\omega = 0$ is unbarred and $\omega = 1$ is barred. We use the index measures

$$\nu(\xi) = \eta_k^3/48, \qquad \nu(\zeta) = (\eta_k^3/48)^2.$$

Integrals sum the spin and bar indices as well. An activity has a unique expansion with separately antisymmetric kernels,

$$\mathscr{E}(X;\mathscr{A},\mathscr{C}) = \sum_{n,m} \frac{1}{n!m!} \int \mathrm{d}\xi\, \mathrm{d}\zeta\, \mathscr{E}_{nm}(X;\mathscr{A},\mathscr{C};\xi;\zeta)\psi(\xi_1)\cdots\psi(\xi_n)\chi(\zeta_1)\cdots\chi(\zeta_m). \tag{5.i}$$

Barred generators are included in this notation. The generator weights, as distinct from the index measures, are

$$h_{k,1} = e_k^{-1/4}, \qquad h_{k,2} = e_k^{-1/4+\varepsilon}.$$

For an element F the weighted coefficient norm is

$$\|F\|_{h_k} = \sum_{n,m} \frac{h_{k,1}^n h_{k,2}^m}{n!m!} \int \mathrm{d}\xi\, \mathrm{d}\zeta\, |F_{nm}|. \tag{5.ii}$$

This is the coefficient norm already used for weighted fermion integration. In particular a generator has norm $h_{k,i}$; the weight ν enters only when coefficients are written as integral kernels.

An activity family in $K_k^{x,\mathbb{C}}$ is balanced: every monomial has as many barred as unbarred legs. Its kernels are holomorphic on $\widetilde{R}_k \times \mathfrak{X}_k$, vanish unless all ordinary legs lie over X and all pairs are admissible for X, and depend on the two connections only on bonds meeting X. The norm is

$$\|\mathscr{E}\|_k^x = \sup_{X\in\mathcal{P}_k} \exp(\kappa d_M(X)) \sup_{(\mathscr{A},\mathscr{C})\in\widetilde{R}_k\times\mathfrak{X}_k} \|\mathscr{E}(X;\mathscr{A},\mathscr{C})\|_{h_k}. \tag{5.iii}$$

We require $\kappa \geq \kappa_0$ and $\|\mathscr{E}\|_k^x < \infty$.

The required symmetries act on both the connections and the generators. For $\tilde{g} = (S, a)$ with the coordinate sign action $r(S)$ and $a \in M\mathbb{Z}^3$, put

$$\tilde{g}(c,g) = (r(S)c + a, r(S)g), \qquad a_{\tilde{g}}\psi(x) = S\psi(\tilde{g}^{-1}x), \qquad a_{\tilde{g}}\bar{\psi}(x) = (S^{-1})^T\bar{\psi}(\tilde{g}^{-1}x).$$

The same spin matrices act on pair generators after transforming both endpoints. Local gauge transformations multiply an unbarred ordinary or pair leg by $\exp(ie_k\Lambda(c))$, using the photon site of its first point, and a barred leg by the inverse phase. Write this automorphism as G_Λ. Charge conjugation and the antilinear reality map act by

$$\mathrm{Cg}\,\psi = \mathsf{C}\bar{\psi}, \qquad \mathrm{Cg}\,\bar{\psi} = -\mathsf{C}\psi, \qquad \mathbf{K}\psi = \varkappa\psi, \qquad \mathbf{K}\bar{\psi} = \varkappa\bar{\psi},$$

and by the same rules on the Hölder generators.

The covariance requirements on a family are

$$\begin{aligned} a_{\tilde{g}}\mathscr{E}(\tilde{g}X;\tilde{g}\mathscr{A},\tilde{g}\mathscr{C}) &= \mathscr{E}(X;\mathscr{A},\mathscr{C}), \\ \mathrm{Cg}\,\mathscr{E}(X;-\mathscr{A},-\mathscr{C}) &= \mathscr{E}(X;\mathscr{A},\mathscr{C}). \end{aligned} \tag{5.iv}$$

For real scalar functions λ, λ' on the photon torus and two pairs in the domain which satisfy $\mathscr{A}^1 = \mathscr{A} - d\lambda$ and $\mathscr{C}^1 = \mathscr{C} - d\lambda'$ on bonds at distance at most one from X, the local gauge requirement is

$$G_{\lambda+\lambda'}\mathscr{E}(X;\mathscr{A}^1,\mathscr{C}^1) = \mathscr{E}(X;\mathscr{A},\mathscr{C}).$$

The real subspace K_k^x has in addition

$$\mathbf{K}\mathscr{E}(X;-\overline{\mathscr{A}},-\overline{\mathscr{C}}) = \mathscr{E}(X;\mathscr{A},\mathscr{C}).$$

All requirements refer to the generators before evaluating fields. They are essential to the form of the counterterms.

The spaces $K_k^{\mathbb{C}}, K_k$ without the second connection argument have the same definitions on $\widetilde{R}_k$. Restriction $\mathscr{E}|_0(X;\mathscr{A}) = \mathscr{E}(X;\mathscr{A},0)$ preserves the real and complex spaces and has norm at most one: set the second connection and its gauge function to zero in the defining properties, and take the smaller supremum in the norm.

For $k \geq 1$ the step is defined on balls with windows

$$w_k^{\mathrm{m}} = e_k^{3/4-8\varepsilon}, \qquad w_k^{\mathrm{E}} = e_k^{1/4-7\varepsilon}, \qquad \mathrm{Ball}_k^x(c) = \{(m, \mathscr{E}) : |m| \leq c w_k^{\mathrm{m}}, \|\mathscr{E}\|_k^x \leq c w_k^{\mathrm{E}}\}. \tag{5.v}$$

At level zero the ball is instead $\{(m, 0) : |m| \leq c w_0^{\mathrm{m}}\}$. This restriction is part of the first-step construction.

When fields are evaluated, the total connection is $\mathscr{T} = \mathscr{A} + \mathscr{C}$ and

$$\psi = H_k(\mathscr{T})\Psi, \qquad \chi = \delta_{\alpha,\mathscr{T}}\psi, \qquad \bar{\psi} = \bar{H}_k(\mathscr{T})\bar{\Psi}, \qquad \bar{\chi} = \delta_{\alpha,-\mathscr{T}}\bar{\psi}.$$

For an intra-cluster pair the phase is one, so $\psi(x') - \psi(x) = \eta_k^\alpha \chi(x, x')$. Accordingly

$$(1 - JJ^*)\psi(x) = -\frac{\eta_k^\alpha}{48} \sum_{x' \neq x} \chi(x, x').$$

The sum is within the cluster of x. This identity expresses the heavy part through the same ordinary and Hölder legs. The norm of a Hölder generator remains $h_{k,2}$, including for an intra-cluster pair; the factor η_k^α belongs to this evaluated identity.

5.1.2. *The total connection and its inverse domain.* An activity keeps the photon connection $\mathscr{A}$ and the external connection $\mathscr{C}$ as separate variables, while the fermion operators depend on their sum. To apply the inverse estimates on the activity domain, we must show that this sum belongs to the inverse domain $\widetilde{G}_k = \widetilde{\mathcal{G}}_k$. We also retain bounds on its imaginary part and curvature, which control the transport phases used below.

Recall that $\widetilde{G}_k$ consists of fields $B_0 + B_1$ with B_0 real and $\|B_1\|_\infty < e_k^{-1+\varepsilon}$, where on each $\square^{\sim 5}$ the field B_0 has a real-gauge representative of supremum norm below $e_k^{-1+\varepsilon}$. The external domain is

$$\mathfrak{X}_k = \{\mathscr{C} : \|\mathscr{C}\|_\infty < e_k^{-1+2\varepsilon}, \|\partial\mathscr{C}\|_\infty < e_k^{-1+3\varepsilon}\}.$$

The sets $R_k, \widetilde{R}_k, \widetilde{R}_k^-$ were defined with the photon estimates. Their real-gauge representatives may be chosen independently on each enlarged cube.

Assume $L \geq 5$, $0 < \varepsilon \leq 1/200$ and

$$e_k^\varepsilon \leq \tfrac{1}{4}, \qquad e_k^{1/4} + e_k^\varepsilon \leq 1, \qquad 4e_k^{1/4+2\varepsilon} + 2e_k^{3\varepsilon} \leq 1. \tag{5.vi}$$

Then

$$\widetilde{R}_k + \mathfrak{X}_k \subset \widetilde{G}_k. \tag{5.vii}$$

Indeed, write $\mathscr{A} = A_0 + A_1$ with $A_1 \in R_k$ and A_0 locally real-gauge equivalent on $\square^\dagger$ to $A_\square^b$ with the three threshold bounds. Restrict the representative to $\square^{\sim 5} \subset \square^\dagger$. Its supremum norm is below $e_k^{-3/4+\varepsilon} \leq e_k^{-1+\varepsilon}$, and

$$\|A_1 + \mathscr{C}\|_\infty < e_k^{-3/4+\varepsilon} + e_k^{-1+2\varepsilon} = (e_k^{1/4} + e_k^\varepsilon) e_k^{-1+\varepsilon} \leq e_k^{-1+\varepsilon}.$$

This is the required decomposition of the total connection. Real gauge transformations can absorb any real constant part, locally on the nonwrapping cubes.

Put $\mathfrak{i}(B) = e_k \|\operatorname{Im} B\|_\infty$ and $q_d(B) = e_k \|dB\|_\infty$. The same decomposition gives

$$\mathfrak{i}(\mathscr{A} + \mathscr{C}) \leq e_k^{1/4+\varepsilon} + e_k^{2\varepsilon} \leq 1, \qquad q_d(\mathscr{A} + \mathscr{C}) \leq 4e_k^{1/4+2\varepsilon} + 2e_k^{3\varepsilon} \leq 1. \tag{5.viii}$$

For the curvature, every plaquette lies in an enlarged cube, and $dA_0 = dA_\square^b$ there. This costs at most $2\|\partial A_\square^b\|_\infty$; the global small field costs another $2\|\partial A_1\|_\infty$, and the external field costs $2\|\partial\mathscr{C}\|_\infty$. The imaginary bound uses only that A_0 is real.

The domains are open: the strict inequalities define open sets for the global summand, which can absorb a sufficiently small complex perturbation while the local real representative is kept fixed. The norm-defined sets are invariant under sign and conjugation; the local real-gauge definitions are also invariant under real gauges and real constant additions. Coordinate sign changes and M translations preserve them. Their derivative and Hölder norms are preserved, with a possible reversal of a bond

and translation of its base point. Thus eq. 5.vii applies also at the signs, conjugates and real gauges used in the step.

5.1.3. *Vacuum and mass extraction.* We record separately the terms that would grow under scaling. The constant term changes the normalization, while the zero-connection average of the ordinary quadratic kernel contributes to the mass. Extracting these terms from small polymers leaves the cancellations used in the contraction estimate. For a balanced activity, its part of bidegree $(2,0)$ can be written uniquely as

$$\mathscr{E}^{[2,0]}(X;\mathscr{A},\mathscr{C}) = \int \mathrm{d}x\,\mathrm{d}x'\,\bar\psi(x)^T K_X(\mathscr{A},\mathscr{C};x,x')\psi(x').$$

The matrix kernel is the ordinary antisymmetric kernel with the first bar index one and the second bar index zero. Antisymmetry accounts for the other ordering; in particular

$$\|\mathscr{E}^{[2,0]}(X;\mathscr{A},\mathscr{C})\|_{h_k} = h_{k,1}^2 \sum_{\beta,\beta'} \int \mathrm{d}x\,\mathrm{d}x' |K_X(\mathscr{A},\mathscr{C};x,x')_{\beta\beta'}|. \tag{5.ix}$$

The kernel is antiperiodic in each argument separately.

For a small X, choose one lift $\widetilde{X}$ and define

$$a_0(\mathscr{E},X) = \mathrm{Vol}(X)^{-1}\mathscr{E}_{00}(X;0,0), \qquad a_2(\mathscr{E},X) = \mathrm{Vol}(X)^{-1}\int_{\widetilde{X}} \mathrm{d}x \int_{\widetilde{X}} \mathrm{d}x'\, K_X(0,0;x,x'). \tag{5.x}$$

Both cluster labels are averaged with measure $\eta_k^3/48$. The second expression is independent of the lift: simultaneous translation by a period multiplies its two arguments by the same antiperiodic sign, whose product is one.

For an M-cube $\square$ put

$$\varepsilon_{\mathrm{v}}(\mathscr{E},\square) = -\sum_{\substack{X\supset\square\\ X \text{ small}}} a_0(\mathscr{E},X), \qquad M(\mathscr{E},\square) = -\sum_{\substack{X\supset\square\\ X \text{ small}}} a_2(\mathscr{E},X). \tag{5.xi}$$

Translations by $M\mathbb{Z}^3$ identify the small-polymer sets over different cubes. They leave the two extracted quantities unchanged. Thus the two sums do not depend on $\square$. Write their values as $\varepsilon_{\mathrm{v}}(\mathscr{E})$ and $M(\mathscr{E})$. Let $K_{\mathrm{pol}} = e^{\kappa_0}/36 = 72^{54}/36$ be the scalar polymer-sum constant in eq. D.iii; it is distinct from the level-zero operator K_0.

Proposition 5.1. There is a complex-linear scalar map $m(\mathscr{E})$ such that $M(\mathscr{E}) = m(\mathscr{E})1$. Both m and ε_{v} are real on K_k^x, and

$$|\varepsilon_{\mathrm{v}}(\mathscr{E})| \le K_{\mathrm{pol}} M^{-3}\|\mathscr{E}\|_k^x, \qquad |m(\mathscr{E})| \le K_{\mathrm{pol}} M^{-3} e_k^{1/2}\|\mathscr{E}\|_k^x. \tag{5.xii}$$

They commute with restriction of the second connection argument to zero. ∎

Proof. Substitution of the generator transformations in a bilinear, followed by uniqueness of its kernel, gives

$$\begin{aligned} a_2(\mathscr{E},\tilde g X) &= S a_2(\mathscr{E},X) S^{-1},\\ a_2(\mathscr{E},X) &= \mathsf{C}^{-1} a_2(\mathscr{E},X)^T \mathsf{C},\\ a_2(\mathscr{E},X) &= \varkappa^{-1}\overline{a_2(\mathscr{E},X)}\varkappa \quad (\mathscr{E}\in K_k^x). \end{aligned} \tag{5.xiii}$$

For the first relation, the two variables and their labels are transformed together, and the measure is unchanged. For charge conjugation, moving the unbarred leg past the barred leg cancels the minus sign in its transformation, and exchanges the two kernel variables. For the last relation, the reality map conjugates the coefficient and multiplies both legs by $\varkappa$; its inverse transpose is $\varkappa^{-1}$. The constant kernel has the corresponding scalar invariance and is real for a real family.

Let z be the center of $\square$ and take $S = \gamma_1, \gamma_2, \gamma_3$ or γ_5. The symmetry $(S, (1 - r(S))z)$ fixes $\square$ and permutes the small polymers containing it. The first relation in eq. 5.xiii therefore shows that $M(\mathscr{E})$

commutes with all four matrices. Their commutant is scalar. In the fixed Dirac basis, γ_3 and $\Sigma_3 = -i\gamma_1\gamma_2\gamma_3$ are diagonal with the four distinct pairs of diagonal entries $(1,1),(-1,1),(-1,-1),(1,-1)$. A commuting matrix is diagonal; commutation with γ_1 identifies its first two and last two entries, and commutation with γ_5 identifies these pairs. This proves the scalar assertion. The last relation in eq. 5.xiii makes this scalar real for a real family.

The coefficient norm gives, for each small X,

$$\begin{aligned} \mathrm{Vol}(X)|a_0(\mathscr{E},X)| &\le \|\mathscr{E}(X;0,0)\|_{h_k},\\ \mathrm{Vol}(X)\sum_{\beta,\beta'}|a_2(\mathscr{E},X)_{\beta\beta'}| &\le h_{k,1}^{-2}\|\mathscr{E}^{[2,0]}(X;0,0)\|_{h_k}. \end{aligned} \tag{5.xiv}$$

Use $h_{k,1}^{-2} = e_k^{1/2}$ and $\mathrm{Vol}(X) \ge M^3$. The individual vacuum and mass contributions are bounded by $M^{-3}\|\mathscr{E}\|_k^x$ and $M^{-3}e_k^{1/2}\|\mathscr{E}\|_k^x$, respectively, times $\exp(-\kappa d_M(X))$. The polymer sum over a fixed cube is at most K_{pol} by eq. D.iii and $\kappa \ge \kappa_0$. For the mass use one diagonal entry of the matrix sum, now known to be scalar. This proves the bounds. Linearity follows from uniqueness of the antisymmetric kernels. Restriction gives the same kernels at $(0,0)$, proving the last assertion. □

Subtract these local terms only on small polymers:

$$(R\mathscr{E})(X;\mathscr{A},\mathscr{C}) = \mathscr{E}(X;\mathscr{A},\mathscr{C}) - a_0(\mathscr{E},X)\,\mathrm{Vol}(X) - \int_X \mathrm{d}x\,\bar\psi(x)^T a_2(\mathscr{E},X)\psi(x). \tag{5.xv}$$

For large X set $R\mathscr{E} = \mathscr{E}$. Then

$$\sum_X \mathscr{E}(X;\mathscr{A},\mathscr{C}) = -\varepsilon_{\mathrm{v}}(\mathscr{E})\ell^3 - m(\mathscr{E})\langle\bar\psi,\psi\rangle + \sum_X (R\mathscr{E})(X;\mathscr{A},\mathscr{C}). \tag{5.xvi}$$

Indeed, decompose the integral and volume of each small polymer over its M-cubes, interchange the finite sums, and use eq. 5.xi on each cube. Every chamber component receives the same scalar mass coefficient:

$$m(\mathscr{E})\langle\bar\psi,\psi\rangle = m(\mathscr{E})\frac{\eta_k^3}{48}\sum_c\sum_g\sum_{\beta=1}^4 \bar\psi_\beta(c,g)\psi_\beta(c,g).$$

The diagonal bilinear of a constant matrix A on X has kernel $48\eta_k^{-3}1[x = x' \text{ over } X]A$ when both arguments are taken over the same lift. Its integral is $\mathrm{Vol}(X)A$, and its norm is

$$h_{k,1}^2\,\mathrm{Vol}(X)\sum_{\beta,\beta'}|A_{\beta\beta'}|.$$

These identities and eq. 5.xiv show that

$$\begin{aligned} (R\mathscr{E})_{00}(X;0,0) &= 0 \quad (X \text{ small}),\\ \int_{\widetilde X}\mathrm{d}x\,\mathrm{d}x'\,(R\mathscr{E})_{20}(X;0,0;x,x') &= 0 \quad (X \text{ small}),\\ \|R\mathscr{E}\|_k^x &\le 2\|\mathscr{E}\|_k^x. \end{aligned} \tag{5.xvii}$$

For the bound, the two subtracted terms have distinct bidegrees and together have norm at most $\|\mathscr{E}(X;0,0)\|_{h_k}$. Apply the triangle inequality to eq. 5.xv, then the activity norm.

The map R preserves the activity spaces and restriction to zero. The subtracted kernels are independent of the connections, have both legs over X, and are diagonal bilinears, so analyticity, locality and gauge invariance are preserved. Their lattice, charge and reality symmetries follow from eq. 5.xiii, as do those of the volume and constant term. This argument does not require an individual matrix $a_2(\mathscr{E},X)$ to be scalar.

For later comparisons, the extraction has a precise local meaning across different sides. Fix a lifted M-cube and consider all lifted small polymers containing it. Their projections are bijective onto the small polymers containing its image, and their d_M and volumes are unchanged. If two families have the same constant and quadratic kernels on every such lifted polymer at zero connections, then their

vacuum and mass matrices over this cube agree term by term in eq. 5.xi. Volume independence of a flow will require this equality of kernels from the step comparison.

5.1.4. *Reconstruction by interior and boundary legs.* We first reconstruct a coarse generator from its fine field. Splitting the reconstruction into interior and boundary terms will make this substitution local. Let $k \geq 1$, $0 < \mu \leq 1$ and $b_- \leq b_k$. With $\mathcal{Q}_k(A) = Q_k(A)J^*$ and $\mathcal{Q}_k^\sharp(A) = JQ_k^T(-A)$, set

$$T_k(A) = b_k^{-1}\mathcal{Q}_k(A)(D_A + \mu + b_k\mathcal{Q}_k^\sharp(A)\mathcal{Q}_k(A)) = \mathcal{Q}_k(A)(b_k^{-1}D_A + 1 + \mu/b_k). \tag{5.xviii}$$

The second form follows from $\mathcal{Q}_k\mathcal{Q}_k^\sharp = 1$. The fine inverse used in the definition of $H_k(A)$ exists on $\widetilde{G}_k$. Hence

$$T_k(A)H_k(A) = b_k^{-1}\mathcal{Q}_k(A)F_\infty(A)b_kF_\infty(A)^{-1}\mathcal{Q}_k^\sharp(A) = 1. \tag{5.xix}$$

The barred identity follows at $-A$, by conjugation with C. At level zero all four field and reconstruction operators are the identity on the 192 chamber components per photon site. This first step uses the unblocked chamber spinors.

For $c \in \Delta_y$, let $c_j(c)$ be the center of its block of side $L^j\eta_k$, $0 \leq j \leq k$, so that $c_0(c) = c$ and $c_k(c) = y$. The averaging phase is

$$\varphi_A(y,c) = \exp\left(ie_k\eta_k\sum_{j=0}^{k-1}(\tau A)(c_{j+1}(c), c_j(c))\right). \tag{5.xx}$$

The sum uses the average of the six rectilinear path sums. We write $(\tau_k A)(y,c)$ for this sum. All of these paths lie in Δ_y and their total length in bonds is at most $\frac{3}{2}(L^k - 1)$. Thus

$$\begin{aligned} |\varphi_A(y,c)^{\pm 1}| &\leq \exp(3\mathfrak{i}(A)/2), \\ |u_A(c,c')^{\pm 1}| &\leq \exp(\eta_k\mathfrak{i}(A)) \leq \exp(\mathfrak{i}(A)). \end{aligned} \tag{5.xxi}$$

for a photon bond, with $u_A(c,c') = \exp(ie_k\eta_kA(c,c'))$.

The row form of eq. 5.xviii is

$$\begin{aligned} (T_k(A)\psi)(y) = \int_{\Delta_y} \mathrm{d}x\, \varphi_A(y,c)\Big\{&(1+\mu/b_k)\psi(x) \\ &- (b_k\eta_k)^{-1}\sum_{i=0}^{3} M_i(x)[U_A(x,x^{(i)})\psi(x^{(i)}) - \psi(x)]\Big\}. \end{aligned} \tag{5.xxii}$$

Write $c_3 = c + \eta_k g e_1$ for the photon base point of $x^{(3)}$. Here $U_A = 1$ for $i \leq 2$ and $U_A(x, x^{(3)}) = u_A(c, c_3)$. The normalized chamber integral over Δ_y has total mass one. This row exists for every complex connection, regardless of invertibility; the inverse domain was needed only for eq. 5.xix.

The rows map antiperiodic fields to antiperiodic coarse fields: translation by a period preserves blocks, neighbors and phases, and supplies the single antiperiodic sign from the source field.

If Y is a union of M-cubes and $y \in Y$ is a unit site, then $\Delta_y \subset Y$: both M and L^k are odd, and the unit block is contained in the M-cube of its center. Define the two rows

$$\begin{aligned} (T_Y^{(1)}(A)\psi)(y) &= \int_{\Delta_y} \mathrm{d}x\, \varphi_A(y,c)\Big\{(1+\mu/b_k)\psi(x) \\ &\qquad - (b_k\eta_k)^{-1}\sum_{i:\, x^{(i)} \text{ over } Y} M_i(x)[U_A(x,x^{(i)})\psi(x^{(i)}) - \psi(x)]\Big\}, \\ (T_Y^{(2)}(A)\chi)(y) &= -b_k^{-1}\eta_k^{\alpha-1}\int_{\Delta_y} \mathrm{d}x\, 1[c_3 \notin Y]\varphi_A(y,c)M_3(x)\chi(x,x^{(3)}). \end{aligned} \tag{5.xxiii}$$

Rows outside Y are zero. The first row contains ordinary legs only over Y. The second row contains precisely the type-3 pairs leaving Y, which are admissible boundary pairs of length η_k. Their

coefficients depend on A only on bonds with an endpoint in Δ_y. If $\chi = \delta_{\alpha,A}\psi$ on these boundary pairs, then

$$T_Y^{(1)}(A)\psi + T_Y^{(2)}(A)\chi = T_k(A)\psi \quad \text{on the unit sites of } Y. \tag{5.xxiv}$$

This follows by replacing the missing type-3 difference in eq. 5.xxii by $\eta_k^\alpha \chi(x, x^{(3)})$. The intra-cluster differences remain in $T_Y^{(1)}$.

Proposition 5.2. For any complex connection, write $\mathfrak{i} = \mathfrak{i}(A)$ and $q_d = q_d(A)$. In the norm obtained by taking the supremum of the Euclidean spinor norm,

$$\begin{aligned} \|T_Y^{(1)}(A)\|_{\mathrm{su}},\ \|T_k(A)\|_{\mathrm{su}} &\le \exp(3\mathfrak{i}/2)\left[1 + b_-^{-1}(54Lq_d\exp(6q_d) + 12 + 6\exp(\mathfrak{i}) + \mu)\right], \\ \|T_Y^{(2)}(A)\|_{\mathrm{su}} &\le 6b_-^{-1}\exp(3\mathfrak{i}/2)\eta_k^\alpha. \end{aligned} \tag{5.xxv}$$

In particular, if $A \in \widetilde{R}_k + \mathfrak{X}_k$ and eq. 5.vi holds, the weighted coefficient row sums satisfy

$$|T_Y^{(i)}(A)|_{1,\infty},\ |T_k(A)|_{1,\infty}, |\bar{T}_Y^{(i)}(A)|_{1,\infty},\ |\bar{T}_k(A)|_{1,\infty} \le C_T, \tag{5.xxvi}$$

where $C_T = 2\exp(3/2)[1 + b_-^{-1}(54L\exp(6) + 12\exp(1) + 1)]$. The bound is independent of k, ℓ and Y. ∎

Proof. Put $E = \Delta_y \times O_{\mathrm{cub}}$ and divide it into

$$E_1 = \{x \in E : c_3 \in \Delta_y\}, \quad E_2 = \{x \in E : c_3 \in Y \setminus \Delta_y\}, \quad E_3 = \{x \in E : c_3 \notin Y\}.$$

For each coordinate direction $a \in \{\pm e_\mu\}$ there are eight labels with $ge_1 = a$. The face of Δ_y through which $c + \eta_k a$ leaves the unit block has η_k^{-2} photon sites. Thus

$$\#E = 48\eta_k^{-3}, \qquad \#(E_2 \cup E_3) = 48\eta_k^{-2}. \tag{5.xxvii}$$

The chamber hopping matrices satisfy

$$M_i(x^{(i)}) - M_i(x) = w_i\gamma \cdot \widehat{d}_i(x), \qquad \sum_{i=0}^{3} w_i\gamma \cdot \widehat{d}_i(x) = 0.$$

Also $\|M_3(x)\|_2 \le 6$ and $\|w_3\gamma \cdot \widehat{d}_3(x)\|_2 = 6$. Reindex each intra-cluster sum by $x \mapsto x^{(i)}$ and each type-3 sum in E_1 by $x \mapsto x^{(3)}$. Both are involutions of their summation sets. The intra-cluster sums become $\sum_{x\in E}\varphi_A(y,c)\sum_{i\le 2}(M_i(x^{(i)}) - M_i(x))\psi(x)$, which equals $-\sum_{x\in E}\varphi_A(y,c)w_3\gamma\cdot\widehat{d}_3(x)\psi(x)$. The part over E_1 cancels with the corresponding matrix difference from the reindexed type-3 sum. The resulting row is $(T_Y^{(1)}(A)\psi)(y) = R_0 + R_1 + R_2 + R_3$, with

$$\begin{aligned} R_0 &= (1 + \mu/b_k)\frac{\eta_k^3}{48}\sum_{x\in E}\varphi_A(y,c)\psi(x), \\ R_1 &= b_k^{-1}\frac{\eta_k^2}{48}\sum_{x\in E_2\cup E_3}\varphi_A(y,c)w_3\gamma\cdot\widehat{d}_3(x)\psi(x), \\ R_2 &= -b_k^{-1}\frac{\eta_k^2}{48}\sum_{x\in E_2}\varphi_A(y,c)M_3(x)[u_A(c,c_3)\psi(x^{(3)}) - \psi(x)], \\ R_3 &= -b_k^{-1}\frac{\eta_k^2}{48}\sum_{x\in E_1}[\varphi_A(y,c_3)u_A(c_3,c) - \varphi_A(y,c)]M_3(x^{(3)})\psi(x). \end{aligned} \tag{5.xxviii}$$

The face count and phase moduli bound the first three terms by

$$\exp(3\mathfrak{i}/2)\|\psi\|_{\mathrm{su}}[1 + \mu/b_k + 6b_k^{-1} + 6b_k^{-1}(1 + \exp(\mathfrak{i}))].$$

The fourth term requires the curvature estimate.

For neighboring photon sites $c, c' = c + \eta_k a$ in Δ_y, let $j(c,a)$ be the first scale at which their block centers agree. The centers at all larger scales then agree as well. For $j \ge 2$, such a c lies on a

face of a block of side $L^{j-1}\eta_k$. There are $L^{3(k-j+1)}$ such blocks inside Δ_y, with $L^{2(j-1)}$ sites on the indicated face. For $j = 1$ use all sites. We obtain

$$\#\{c : c, c+\eta_k a \in \Delta_y,\ j(c,a) = j\} \le \eta_k^{-3} L^{-(j-1)}. \tag{5.xxix}$$

Write

$$\varphi_A(y,c')u_A(c',c) - \varphi_A(y,c) = \varphi_A(y,c)(\exp(v(c,a)) - 1),$$
$$v(c,a) = ie_k\eta_k[(\tau_k A)(y,c') + A(c',c) - (\tau_k A)(y,c)].$$

The contributions above scale $j(c,a)$ cancel in this bracket. The remaining expression is the mean of the line sums over closed paths from their common scale-j center down to c', then across the bond to c, and back up the reversed paths. All these loops lie in the scale-j block, of side at most $L^j\eta_k$, and have at most $3L^j$ bonds. For a closed path of n bonds in a box of side s, the axial path estimate gives $|A(\Gamma)| \le 2ns\|dA\|_\infty$. To recall the reason, subtract the gauge obtained by line sums from one corner in coordinate order $1, 2, 3$. The difference on each bond is a sum over at most two strips of width one mesh and length at most s. Its modulus is at most $2s\|dA\|_\infty$, and the gauge sum around the loop vanishes. This argument applies to complex fields as well. Consequently, with $j = j(c,a)$,

$$|v(c,a)| \le 6L^{2(j-k)}q_d, \qquad |\exp(v(c,a)) - 1| \le 6L^{2(j-k)}q_d \exp(6q_d). \tag{5.xxx}$$

The factor $L^{2(j-k)}$ must be kept until after summing the locations of the first common block. By eq. 5.xxix,

$$\eta_k^2 \sum_{\substack{c\in\Delta_y \\ c+\eta_k a\in\Delta_y}} |\exp(v(c,a)) - 1| \le 6q_d \exp(6q_d) L^{1-k} \sum_{j=1}^{k} L^j$$
$$\le 9Lq_d \exp(6q_d).$$

The six directions, eight labels per direction, and $\|M_3\|_2 \le 6$ now give

$$|R_3|_2 \le 54Lq_d \exp(6q_d) b_k^{-1} \exp(3\mathfrak{i}/2)\|\psi\|_{\mathrm{su}}.$$

Together with the first three terms this proves the first line of eq. 5.xxv for $T_Y^{(1)}$. Take Y to be the full torus to obtain the bound for T_k.

The second row is supported on E_3. Its definition and eq. 5.xxvii give directly

$$|(T_Y^{(2)}(A)\chi)(y)|_2 \le b_k^{-1}\eta_k^{\alpha-1}\frac{\eta_k^3}{48}48\eta_k^{-2}\exp(3\mathfrak{i}/2)6\|\chi\|_{\mathrm{su}},$$

which is the second bound.

For the coefficient norm, fix a row and choose each source component to have the conjugate phase of its row coefficient. Its component supremum is at most one, and its Euclidean spinor supremum is at most two. The row coefficient sum is then at most twice the corresponding bound in eq. 5.xxv. At a total connection, eq. 5.viii gives $\mathfrak{i}, q_d \le 1$; also $\mu \le 1$ and $\eta_k^\alpha \le 1$. The resulting bound is C_T as stated. The normalized index measure is already in the row coefficients, so no extra factor 48 enters. The domain is invariant under sign, and the matrix C is monomial; conjugating $T_Y^{(i)}(-A)$ or $T_k(-A)$ by it permutes row and column moduli. This proves the barred bounds. □

The local reconstruction respects the symmetries of an activity. Telescoping the scalar gauge sum in the averaging paths gives

$$\varphi_{A-d\lambda}(y,c)\exp(ie_k\lambda(c)) = \exp(ie_k\lambda(y))\varphi_A(y,c),$$
$$u_{A-d\lambda}(c,c')\exp(ie_k\lambda(c')) = \exp(ie_k\lambda(c))u_A(c,c').$$

Insert these two identities in eq. 5.xxiii. Every term acquires the coarse factor $\exp(ie_k\lambda(y))$, where the boundary leg transforms with the phase of its first point. Thus, with λ^{u} the restriction to unit sites,

$$\begin{aligned} T_Y^{(1)}(A-d\lambda)\exp(ie_k\lambda)\psi &= \exp(ie_k\lambda^{\mathrm{u}})T_Y^{(1)}(A)\psi, \\ T_Y^{(2)}(A-d\lambda)\chi^\lambda = \exp(ie_k\lambda^{\mathrm{u}})T_Y^{(2)}(A)\chi, &\qquad \chi^\lambda(x,x') = \exp(ie_k\lambda(c))\chi(x,x'). \end{aligned} \tag{5.xxxi}$$

The identity also holds for gauges specified locally on the bonds and sites occurring in the row.

A scheme symmetry $\tilde g = (S,a)$ maps nested blocks and their centers to their images, preserves the order of each coordinate path, and permutes the 48 labels. Moreover $M_i(\tilde g x)S = SM_i(x)$ and $(\tilde g x)^{(i)} = \tilde g(x^{(i)})$. Changing variables in the two rows therefore proves

$$T^{(i)}_{\tilde g Y}(\tilde g A)\tilde g = \tilde g T_Y^{(i)}(A). \tag{5.xxxii}$$

Conjugating the phases evaluated at $\overline{A}$ gives those at $-A$, and $\overline{M_i} = \varkappa^{-1}M_i\varkappa$, giving

$$\overline{T_Y^{(i)}(\overline{A})} = \varkappa^{-1}T_Y^{(i)}(-A)\varkappa. \tag{5.xxxiii}$$

The barred rows are $\bar T_Y^{(i)}(A) = \mathsf{C}^{-1}T_Y^{(i)}(-A)\mathsf{C}$. Their gauges have the opposite sign. For the lattice covariance, $(S^{-1})^T = \sigma(S)\mathsf{C}^{-1}S\mathsf{C}$ with $\sigma(S)\in\{\pm1\}$: both matrix maps are homomorphisms, and on each generating Dirac matrix they differ by a minus sign. This same sign occurs in the source and target spin actions and cancels in the barred intertwining identity. For reality, C is real and $\varkappa\mathsf{C} = -\mathsf{C}\varkappa$; the two signs from commuting the conjugations cancel. Thus the barred versions of eq. 5.xxxii and eq. 5.xxxiii hold with the inverse transpose spin action and the same $\varkappa$.

5.1.5. *Adjustment of the quadratic kernel.* Extraction has cancelled the integral of the zero-field quadratic kernel. We now turn this integral cancellation into a vanishing ordinary quadratic kernel. Write each ordinary leg as a transported average plus covariant Hölder differences, represented by the pair generators. At zero connection, the term containing two averages is multiplied by the integral of the kernel and therefore vanishes, so every remaining quadratic term contains a pair generator. The chain identity below makes this a rewriting of the same evaluated interaction, although its polynomial in the independent generators changes.

Let X be small, choose a lift $\widetilde X$, and let $x = (c,g)$ and $\tilde x = (\tilde c,\tilde g)$ lie over this lift. Take all shortest face chains of its M-cubes between their two cubes. For a chain $(\square_0,\dots,\square_n)$ with centers z_i, use the points

$$p_0 = x, \qquad p_{i+1} = (z_i, g)\quad (0\le i\le n), \qquad p_{n+2} = \tilde x.$$

The initial label is kept at each intermediate center. Each distinct consecutive pair is admissible for X of the first kind. The paths in the first and last cube remain in those cubes; between adjacent centers all six paths are the straight segment across their common face. The last pair can have different labels, and can lie inside one cluster. For a coincident pair there is no difference term.

With $u_B(p,p') = U_B(p,p')$ and $u_B(p,p) = 1$, set

$$U_i^{\pm} = \prod_{j<i} u_B(p_j,p_{j+1})^{\pm1}.$$

The chain has no repeated cube, and $n+1\le |X|_M \le c_X(L+1)$. Hence

$$\sum_{i=0}^{n+1}|\pi p_i - \pi p_{i+1}|_1 \le (n+3)M \le 3c_XLM, \qquad \sum_{i:p_i\neq p_{i+1}} d_X(p_i,p_{i+1})^\alpha \le 3c_XLM. \tag{5.xxxiv}$$

The second estimate uses $d_X^\alpha \le \max(1,d_X) \le M$. The transport bound is $|u_B(p,p')^{\pm1}| \le \exp(e_k\|\operatorname{Im}B\|_\infty|\pi p - \pi p'|_1)$, obtained by counting the bonds in each path.

For $B = \mathscr{A} + \mathscr{C}$, the domain decomposition and $e_k^{\varepsilon} \le 1/4$ give $e_k \| \operatorname{Im} B\|_\infty \le e_k^{\varepsilon}/2$. Assume in addition

$$90 c_X L M e_k^{\varepsilon} \le 1. \tag{5.xxxv}$$

The modulus of every partial product $U_i^{\pm}$ is then at most $\exp(1)$, uniformly in the chain and mesh.

Let Mn be the mean over the shortest chains. Define the homomorphism $\Theta_{X,B}$ on ordinary generators by

$$\begin{aligned}
\Theta_{X,B}\psi(x) &= \operatorname{Vol}(X)^{-1} \int_{\widetilde{X}} \mathrm{d}\tilde{x}\ \operatorname{Mn}\left[U_{n+2}^{+}\psi(\tilde{x}) - \sum_{i : p_i \neq p_{i+1}} U_i^{+} d_X(p_i,p_{i+1})^{\alpha} \chi(p_i,p_{i+1}) \right], \\
\Theta_{X,B}\bar{\psi}(x) &= \operatorname{Vol}(X)^{-1} \int_{\widetilde{X}} \mathrm{d}\tilde{x}\ \operatorname{Mn}\left[U_{n+2}^{-}\bar{\psi}(\tilde{x}) - \sum_{i : p_i \neq p_{i+1}} U_i^{-} d_X(p_i,p_{i+1})^{\alpha} \bar{\chi}(p_i,p_{i+1}) \right].
\end{aligned} \tag{5.xxxvi}$$

The spin components are unchanged. Both displayed images are linear combinations of odd independent generators, so they anticommute and determine a unique homomorphism on the subalgebra of ordinary legs over X. Translation of the lift by a period preserves every transport and chain and multiplies each generator by the same antiperiodic sign. Thus the homomorphism is independent of this choice of lift.

If the generators are evaluated at fields satisfying their covariant Hölder relations, repeated substitution of

$$\psi(p_i) = u_B(p_i,p_{i+1})\psi(p_{i+1}) - d_X(p_i,p_{i+1})^{\alpha}\chi(p_i,p_{i+1})$$

along a chain gives exactly the first expression in eq. 5.xxxvi; the barred relation uses u_B^{-1}. Taking the two means therefore reproduces the original leg. This proves equality after evaluation for every polynomial in the ordinary legs over X.

The adjusted family is

$$\mathscr{E}^{\mathrm{nat}}(X) = \mathscr{E}(X) - \mathscr{E}^{[2,0]}(X) + \Theta_{X,\mathscr{A}+\mathscr{C}}\mathscr{E}^{[2,0]}(X) \qquad (X \text{ small}), \tag{5.xxxvii}$$

and is unchanged on large polymers. The equality just proved shows that the adjusted and original families have equal evaluated values. Only bidegrees $(2,0),(1,1),(0,2)$ can change.

At zero connection all $U_i^{\pm}$ are one. The ordinary part of an adjusted bilinear with kernel K is therefore

$$\bar{\psi}_{\mathrm{av}}^{T}\left[\int_{\widetilde{X}} \mathrm{d}x\,\mathrm{d}x'\,K(x,x')\right]\psi_{\mathrm{av}}, \qquad \psi_{\mathrm{av}} = \operatorname{Vol}(X)^{-1}\int_{\widetilde{X}} \mathrm{d}x\,\psi(x).$$

Apply this adjustment to $R\mathscr{E}$, whose quadratic kernel has zero integral. We call $E = (R\mathscr{E})^{\mathrm{nat}}$ the normalized activity. Its vacuum and ordinary quadratic kernels vanish at zero connection on every small polymer:

$$E_{00}(X;0,0) = 0, \qquad E_{20}(X;0,0;\cdot,\cdot) = 0 \quad (X \text{ small}). \tag{5.xxxviii}$$

The same chain estimate controls the norm of adjustment. Decompose $\Theta\psi = L_\psi + L_\chi$ by bidegree. For arbitrary generator weights (w_1,w_2),

$$\|L_\psi\|_w \le \exp(1) w_1, \qquad \|L_\chi\|_w \le \exp(1)(3c_X L M) w_2.$$

The normalized site and label mean has total mass one. Writing $s_K = \sum_{\beta,\beta'} \int_{\widetilde{X}} \mathrm{d}x\,\mathrm{d}x'\,|K_{\beta\beta'}(x,x')|$ for the balanced quadratic kernel and $F = \Theta\mathscr{E}^{[2,0]}$ gives the three bounds

$$\begin{aligned}
\|F^{[2,0]}\|_w &\le \exp(2) w_1^2 s_K, \\
\|F^{[1,1]}\|_w &\le 2\exp(2)(3c_X L M) w_1 w_2 s_K, \\
\|F^{[0,2]}\|_w &\le \exp(2)(3c_X L M)^2 w_2^2 s_K.
\end{aligned}$$

Since $h_{k,2} = e_k^{\varepsilon} h_{k,1}$ and $3c_X L M e_k^{\varepsilon} \le 1/30$, summing these bounds yields

$$\|\mathscr{E}^{\mathrm{nat}}(X)\|_{h_k} \le \exp(2)(1 + 3c_X L M e_k^{\varepsilon})^2 \|\mathscr{E}(X)\|_{h_k} \le 10 \|\mathscr{E}(X)\|_{h_k}. \tag{5.xxxix}$$

Subtracting the original quadratic part before adding its image uses additivity of the norm over bidegrees.

The adjustment uses only legs and chain pairs over X and transports within X, so it is local and analytic. Lattice symmetries permute the averaged shortest chains; real-gauge factors telescope to the endpoint phase; charge conjugation sends u_B to u_B^{-1}; reality uses $\overline{u_{-\overline{B}}} = u_B$. These generator identities give the four covariance relations for Θ. Restriction to $\mathscr{C} = 0$ keeps the same total connection and quadratic kernel. Thus adjustment preserves the complex and real activity spaces and commutes with restriction.

5.1.6. *Scaling and reblocking.* Throughout scaling, reblocking and the contraction proof, let both ℓ and ℓ/L be admissible sides; thus $\ell \geq 13L^2M$. Set $\bar{\varepsilon} = 1/200$ and assume

$$\begin{gathered} 0 < \varepsilon \leq \bar{\varepsilon}, \qquad 1 - \varepsilon < \alpha < 1, \qquad L^{1/8-3\bar{\varepsilon}/2} \geq 2400, \qquad e_k^{\varepsilon} \leq 1/4, \\ 160 L^{1+3\varepsilon/2} M e_k^{\varepsilon} \leq 1, \qquad 9ML^{\varepsilon} e_k^{2\varepsilon} \leq 1. \end{gathered} \tag{5.xl}$$

The adjustment condition eq. 5.xxxv is also retained. These inequalities will control the analytic disk and the phases on a small polymer.

Write the three thresholds of R_k as

$$\theta_1 = e_k^{-3/4+\varepsilon}, \qquad \theta_2 = e_k^{\varepsilon}\theta_1, \qquad \theta_3 = e_k^{\varepsilon}\theta_2,$$

and the thresholds of $\mathfrak{X}_k$ as $d_1 = e_k^{-1+2\varepsilon}$ and $d_2 = e_k^{\varepsilon} d_1$. For a level $k+1$ bond function and gauge function define

$$A_L(b) = L^{-1/2} A(b/L), \qquad \lambda_L(c) = L^{1/2}\lambda(c/L). \tag{5.xli}$$

Since $\eta_{k+1} = \eta_k/L$, the finite differences obey $(d\lambda)_L = d\lambda_L$ and

$$\|A_L\|_\infty = L^{-1/2}\|A\|_\infty, \qquad \|\partial A_L\|_\infty = L^{-3/2}\|\partial A\|_\infty, \qquad \|\delta_\alpha \partial A_L\|_\infty \leq L^{-3/2-\alpha}\|\delta_\alpha \partial A\|_\infty.$$

The last inequality uses only the coarse pairs at distance $\leq 1/L$, a subset of those in the coarse seminorm. The charge relation $e_{k+1} = L^{1/2} e_k$ therefore gives the following bounds on the fine lattice after scaling. They apply to the complex small field and to each scaled real representative supplied by a local gauge change:

$$\begin{aligned} b_1 &= L^{-7/8+\varepsilon/2}\theta_1, \\ b_2 &= L^{-15/8+\varepsilon}\theta_2, \\ b_3 &= L^{-15/8+3\varepsilon/2}\theta_3. \end{aligned} \tag{5.xlii}$$

Here the additional factor $L^{-\alpha}$ in the third bound has been discarded. For the external argument the bounds are

$$\|\mathscr{C}_L\|_\infty < L^{-1+\varepsilon} d_1, \qquad \|\partial \mathscr{C}_L\|_\infty < L^{-2+3\varepsilon/2} d_2. \tag{5.xliii}$$

The enlarged coarse cube used to define a local representative, after multiplication by L, contains the enlarged fine cube. Scaling its real gauge thus gives the corresponding fine representative. All the ratios in eq. 5.xlii and eq. 5.xliii are at most $L^{-3/4}$, by eq. 5.xl. It follows that

$$\mathscr{A}_L \in L^{-3/4}\widetilde{R}_k \subset \widetilde{R}_k^-, \qquad \mathscr{C}_L \in L^{-3/4}\mathfrak{X}_k. \tag{5.xliv}$$

For the second inclusion in the first statement use $L^{-3/4} \leq 1/2 \leq 1 - e_k^{\varepsilon}$.

The Grassmann scaling homomorphism is

$$S_L\psi(x) = L^{-1}\psi'(x/L), \qquad S_L\chi(x,x') = L^{-1-\alpha}\chi'(x/L, x'/L), \tag{5.xlv}$$

with the same powers for the barred generators. Only the photon coordinate is divided by L; the chamber label and spin index are kept. Ordinary indices are mapped bijectively onto the coarse ordinary indices. Pair indices are mapped bijectively onto the coarse pair indices with $d_X \leq M/L$, since d_X is divided by L. This includes a pair inside one cluster: $d_X = \eta_k$ becomes $d_X = \eta_{k+1}$. The antiperiodic signs agree under this bijection.

Thus distinct basis monomials have distinct images, up to their ordering signs, and every monomial of bidegree (n,m) is multiplied by $L^{-n-(1+\alpha)m}$. There is no cancellation of coefficients. For arbitrary positive weights,

$$\|S_L F\|_{w_1,w_2} = \|F\|_{L^{-1}w_1,L^{-1-\alpha}w_2}. \tag{5.xlvi}$$

As $h_{k+1,1}/h_{k,1} = L^{-1/8}$ and $h_{k+1,2}/h_{k,2} = L^{-1/8+\varepsilon/2}$, this gives

$$\|(S_L F)^{[n,m]}\|_{h_{k+1}} = L^{-p(n,m)}\|F^{[n,m]}\|_{h_k}, \qquad p(n,m) = n + (1+\alpha)m + \frac{n+m}{8} - \frac{m\varepsilon}{2}. \tag{5.xlvii}$$

The powers are identical for all admissible pair types.

In kernel notation the same calculation reads

$$(S_L F)_{nm}(\xi';\zeta') = \begin{cases} L^{3n+6m-n-(1+\alpha)m} F_{nm}(L\xi';L\zeta'), & d_X(\zeta'_j) \le M/L \text{ for every } j, \\ 0, & \text{otherwise.} \end{cases} \tag{5.xlviii}$$

Indeed, the index weights satisfy $\eta_k^3/48 = L^3\eta_{k+1}^3/48$ and hence contribute L^{3n+6m} to the kernel. Antisymmetry and antiperiodicity are preserved, so uniqueness of the kernel expansion gives eq. 5.xlviii.

Let $\bar{X}^L$ be the union of LM-cubes containing the cubes of X. Define the reblocked family by

$$(\mathcal{L}F)(Y;\mathscr{A},\mathscr{C}) = \sum_{X:\bar{X}^L = LY} S_L F(X;\mathscr{A}_L,\mathscr{C}_L). \tag{5.xlix}$$

Reblocking is local: $X \subseteq LY$, ordinary legs and the six internal pair paths scale into Y, boundary pairs satisfy $x^{(3)}/L = (x/L)^{(3)}$, and each bond meeting X on which a coefficient depends scales to a bond meeting Y. These facts give both leg support and coefficient locality.

5.1.7. *Global representatives on a small polymer.* The contraction estimate needs more than the domain inclusion for the scaled fields. The vacuum and ordinary quadratic kernels have low Grassmann degree, so their smallness must come from their zeros at zero connection. On a small polymer, a real gauge and two constant fields reduce the connections to global fields that admit an analytic disk of radius $L^{7/4}$; Cauchy's formula then uses those zeros.

Proposition 5.3. Assume eq. 5.xl. Let X be small and let $(\mathscr{A},\mathscr{C}) \in \widetilde{R}_{k+1} \times \mathfrak{X}_{k+1}$. There are a real function λ_*, constant bond functions $\mathscr{K}_1, \mathscr{K}_2$, and global bond functions $\mathscr{A}^{\mathrm{b}}, \mathscr{C}^{\mathrm{b}}$ on the fine torus such that, on every bond with an endpoint at distance at most 1 from X,

$$\mathscr{A}_L - d\lambda_* = \mathscr{A}^{\mathrm{b}} + \mathscr{K}_1, \qquad \mathscr{C}_L = \mathscr{C}^{\mathrm{b}} + \mathscr{K}_2. \tag{5.l}$$

They satisfy

$$\begin{aligned} \|\mathscr{A}^{\mathrm{b}}\|_\infty &\le 64 L^{-15/8+3\varepsilon/2}\theta_1, \\ \|\partial\mathscr{A}^{\mathrm{b}}\|_\infty &\le 137 L^{-15/8+3\varepsilon/2}\theta_2, \\ \|\delta_\alpha\partial\mathscr{A}^{\mathrm{b}}\|_\infty &\le 1382 L^{-15/8+3\varepsilon/2}\theta_3, \\ \|\mathscr{C}^{\mathrm{b}}\|_\infty < L^{-7/4}d_1, &\qquad \|\partial\mathscr{C}^{\mathrm{b}}\|_\infty < L^{-7/4}d_2, \end{aligned} \tag{5.li}$$

and

$$\|\mathscr{K}_1\|_\infty < b_1, \qquad \|\mathscr{K}_2\|_\infty < L^{-1+\varepsilon}d_1. \tag{5.lii}$$

Consequently

$$(t\mathscr{A}^{\mathrm{b}}, t\mathscr{C}^{\mathrm{b}}) \in R_k \times \mathfrak{X}_k \qquad (|t| \le L^{7/4}). \tag{5.liii}$$

The construction uses a base point of X and a decomposition and local real representative of $\mathscr{A}$, and is independent of the activity. When $\mathscr{C} = 0$, it gives $\mathscr{C}^{\mathrm{b}} = \mathscr{K}_2 = 0$. ∎

Proof. Choose a lift of X and a photon site c_* in it. Set $r = 2LM \geq 50$ and $K_{\mathrm{loc}} = K(c_*, 2r+5)$, with sup balls understood through their nonwrapping lifts. Choose the coarse M-cube containing c_*/L. If $\mathscr{A} = A_0 + A_1$ is its domain decomposition, its real representative A' and gauge λ on the enlarged cube give

$$a_0 = (A_0)_L = a_0' + d\lambda_L, \qquad a_0' = (A')_L, \qquad a_1 = (A_1)_L$$

on K_{loc}. To see that these functions are defined there, the scaled enlarged cube has side $(2L+1)LM$ and center z with $|c_* - z| < LM/2$. The inequality $4LM + 6 \leq L^2 M$ puts $K(c_*, 2r+6)$ inside this cube. Moreover $4r + 11 < \ell$, since $\ell \geq 13L^2M$. Thus the boxes and all derivative cells used below do not wrap.

We first record the cutoff estimate with its derivative support. Put

$$f(t) = \begin{cases} 1, & |t| \leq 1, \\ 1 - 2(|t|-1)^2, & 1 \leq |t| \leq 3/2, \\ 2(2-|t|)^2, & 3/2 \leq |t| \leq 2, \\ 0, & |t| \geq 2. \end{cases}$$

For a bond $(c, c + \eta_k e_\mu)$ use the product of the three functions $f((c_\nu + \eta_k \delta_{\mu\nu}/2 - c_{*,\nu})/r)$ as its cutoff multiplier. Denote this operation by cut_r. The function f is continuously differentiable, $|f'| \leq 2$, and f' is Lipschitz with constant 4. If a bond function b has local norms B_0, B_1, B_α on K_{loc}, extension by zero after multiplication gives

$$\begin{aligned} \|\operatorname{cut}_r b\|_\infty &\leq B_0, \\ \|\partial \operatorname{cut}_r b\|_\infty &\leq B_1 + 2B_0/r, \\ \|\delta_\alpha \partial \operatorname{cut}_r b\|_\infty &\leq B_\alpha + 12B_1/r + 12B_0/r^2. \end{aligned} \tag{5.liv}$$

Indeed, the multiplier has oscillation at most $6|c-c'|/r$, first difference at most $2/r$, and first-difference oscillation at most $12|c-c'|/r^2$. Apply the product rule to the first difference. In its Hölder difference the four terms are bounded by $B_\alpha |c-c'|^\alpha$, twice $6B_1|c-c'|/r$, and $12B_0|c-c'|/r^2$. Here $|c-c'| \leq 1$ implies $|c-c'| \leq |c-c'|^\alpha$. If any such term is nonzero, all its cells lie in $K(c_*, 2r+5)$: a nonzero multiplier or its difference requires a site within $2r+1$, the first-difference cell adds one mesh length, and the pair adds at most 1. The same bound covers the two-step cell at $k = 0$. Thus the local estimates suffice for the global ones. The cutoff equals one on bonds with both ends in $K(c_*, r)$.

For $b = a_0'$ or a_1, write $v = c - c_*$ and define

$$V[b]_{\nu\mu} = (\partial_\nu b_\mu)(c_*), \qquad \mathrm{Lin}[b]_\mu(c) = \sum_\nu V[b]_{\nu\mu} v_\nu, \qquad \mathrm{rem}[b] = b - b(c_*) - \mathrm{Lin}[b].$$

Its first difference is $\partial_\nu b_\mu(c) - \partial_\nu b_\mu(c_*)$, and its Hölder first-difference seminorm is at most b_3. Join the two sites by at most $2r+5$ simultaneous coordinate steps of length at most 1. Their difference is therefore at most $(2r+5)b_3 \leq 3rb_3$. A rectilinear path to c_* has length at most $3(2r+5)$, and the remainder vanishes at c_*. Keeping the sharper derivative bound along this path and using $r \geq 50$ gives

$$\|\mathrm{rem}[b]\|_{K_{\mathrm{loc}}} \leq 3(2r+5)^2 b_3 \leq 18r^2 b_3, \qquad \|\partial \mathrm{rem}[b]\|_{K_{\mathrm{loc}}} \leq 3rb_3.$$

eq. 5.liv consequently bounds the three global norms of its cutoff by $18r^2 b_3$, $39rb_3$, and $253b_3$, respectively.

Only a real gauge is available. We remove the constant and symmetric linear part of a_0', leaving its curvature and the complex linear part of a_1. Set

$$V^0 = V[a_0'], \qquad V^{\mathrm{s}} = (V^0 + (V^0)^T)/2, \qquad F = V^0 - (V^0)^T,$$

and use the real quadratic function

$$q_*(c) = \sum_{\mu} \left[a'_{0,\mu}(c_*) - \frac{\eta_k}{2} V^{\mathrm{s}}_{\mu\mu} \right] v_\mu + \frac{1}{2} \sum_{\nu,\mu} V^{\mathrm{s}}_{\nu\mu} v_\nu v_\mu. \tag{5.lv}$$

Its finite difference is exactly $(dq_*)_\mu(c) = a'_{0,\mu}(c_*) + \sum_\nu V^{\mathrm{s}}_{\nu\mu} v_\nu$. The mesh term in eq. 5.lv cancels the diagonal term from differencing the quadratic polynomial. With $\mathscr{K}_1 = a_1(c_*)$ and

$$P_\mu(c) = \sum_{\nu} \left[\frac{1}{2} F_{\nu\mu} + V[a_1]_{\nu\mu} \right] v_\nu,$$

we have the local identity

$$\mathscr{A}_L - d(\lambda_L + q_*) = \mathscr{K}_1 + P + \mathrm{rem}[a'_0] + \mathrm{rem}[a_1]. \tag{5.lvi}$$

To cut off P we need a bound that also uses the side ℓ. A periodic function with global mean zero whose oscillation between sites at distance at most 1 is at most B has sup norm at most ℓB. Any two sites can be joined by at most $\lceil \ell/2 \rceil \le \ell$ simultaneous coordinate steps of length at most 1; comparison with the mean gives the claim. Every $\partial_\nu a_{1,\mu}$ has mean zero, sup norm at most b_2, and this oscillation at most b_3. For a_0, the global curl agrees on each derivative cell with the curl of a local real representative. Its sup norm is at most $2b_2$, its oscillation at most $2b_3$, and it has mean zero. The cells of a pair at distance at most 1 lie in one enlarged cube, so that the same representative can be used for their comparison. It follows that every coefficient of P is bounded by

$$s = 2\min(b_2, \ell b_3). \tag{5.lvii}$$

Use a second radius

$$r' = \min(e_k^{-\varepsilon}, \ell/5).$$

The charge condition in eq. 5.xl gives $r' \ge r$, and $4r' + 11 < \ell$. Thus its cutoff is also defined without wrapping. The three norms of P on its support are at most $6.3r's$, s, and 0. eq. 5.liv, with r' in place of r, bounds the cutoff norms by $6.3r's$, $13.6s$, and $87.6s/r'$. If $r' = e_k^{-\varepsilon}$, use $s \le 2b_2$; in units of $(\theta_1, \theta_2, \theta_3)$ these are at most $(12.6, 27.2, 175.2)L^{-15/8+\varepsilon}$. If $r' = \ell/5 < e_k^{-\varepsilon}$, use $s \le 2\ell b_3$. Since $\ell e_k^{\varepsilon} < 5$, the corresponding three bounds are at most $(63, 136, 876)L^{-15/8+3\varepsilon/2}$.

Define the global field and real gauge by

$$\begin{aligned} \mathscr{A}^{\mathrm{b}} &= \mathrm{cut}_r \left(\mathrm{rem}[a'_0] + \mathrm{rem}[a_1] \right) + \mathrm{cut}_{r'} P, \\ \lambda_* &= \lambda_L + q_* \quad \text{on } K_{\mathrm{loc}}, \qquad \lambda_* = 0 \quad \text{elsewhere.} \end{aligned} \tag{5.lviii}$$

The two remainder cutoffs contribute $(36r^2, 78r, 506)b_3$. As $r = 2LM$, the charge condition $160L^{1+3\varepsilon/2} M e_k^{\varepsilon} \le 1$ bounds the first two by $L^{-15/8+3\varepsilon/2}\theta_1$ and $L^{-15/8+3\varepsilon/2}\theta_2$, respectively. Adding the linear contribution gives the constants 64, 137, and 1382 in eq. 5.li.

In the external argument take

$$\mathscr{K}_2 = \mathscr{C}_L(c_*), \qquad \mathscr{C}^{\mathrm{b}} = \mathrm{cut}_r(\mathscr{C}_L - \mathscr{K}_2).$$

A rectilinear path to c_* bounds the undifferentiated field on K_{loc} by $6.3r\|\partial\mathscr{C}_L\|_\infty$. The cutoff bounds therefore give

$$\|\mathscr{C}^{\mathrm{b}}\|_\infty \le 6.3r\|\partial\mathscr{C}_L\|_\infty, \qquad \|\partial\mathscr{C}^{\mathrm{b}}\|_\infty \le 13.6\|\partial\mathscr{C}_L\|_\infty.$$

Use eq. 5.xliii and $d_2 = e_k^{\varepsilon} d_1$. The first charge condition involving 160 gives the desired bound for the sup norm; the large-L condition gives $13.6L^{-2+3\varepsilon/2} \le L^{-7/4}$ for the derivative. This proves the external bounds and the constant bounds.

The diameter estimate in App. D puts X in $K(c_*, M(L+2))$. Every bond with an endpoint within distance 1 of X has both ends in $K(c_*, r)$, since $M(L+2) + 2 \le 2LM$. All cutoffs in the local identity are one there, and the gauge is $\lambda_L + q_*$ there. This proves eq. 5.l. Finally, the large-L condition implies $1382L^{-15/8+3\varepsilon/2} < L^{-7/4}$. Multiplication by any $|t| \le L^{7/4}$ consequently leaves all five norms strictly below their domain thresholds, proving eq. 5.liii. □

Proposition 5.4. Let $F(X)$ be a level-k Grassmann activity holomorphic on $\widetilde{R}_k \times \mathfrak{X}_k$, with the kernel support and coefficient locality specified in §5.1.1, and covariant under the independent real gauges of its two connection arguments. For the fields of Prop. 5.3, its kernels satisfy

$$|F_{nm}(X; \mathscr{A}_L, \mathscr{C}_L; \xi; \zeta)| \le \exp(2(n+m))|F_{nm}(X; \mathscr{A}^{\mathrm{b}}, \mathscr{C}^{\mathrm{b}}; \xi; \zeta)|. \tag{5.lix}$$

For the vacuum kernel the two values are equal. ■

Proof. For each kernel argument a let $c(a)$ be the photon site of its ordinary leg, or the first photon site of its pair leg, and let $\omega(a) \in \{0, 1\}$ distinguish unbarred and barred legs. On kernels the gauge homomorphism has the form

$$(G_\lambda F)_{nm}(\xi; \zeta) = F_{nm}(\xi; \zeta) \prod_a \exp\{(-1)^{\omega(a)} i e_k \lambda(c(a))\}. \tag{5.lx}$$

This follows by applying the homomorphism to the kernel expansion; the multiplying product is symmetric in each set of arguments, and hence preserves their antisymmetry. The gauge relation eq. 5.l first gives

$$G_{\lambda_*} F(X; \mathscr{A}^{\mathrm{b}} + \mathscr{K}_1, \mathscr{C}^{\mathrm{b}} + \mathscr{K}_2) = F(X; \mathscr{A}_L, \mathscr{C}_L).$$

Both pairs lie in the domains: a constant changes only the sup norm, and $L^{-7/4} + L^{-7/8+\varepsilon/2} < 1$ and $L^{-7/4} + L^{-1+\varepsilon} < 1$. Every phase from this real gauge has modulus one.

We remove the constant fields by continuation of a real gauge identity. Write $\mathscr{K}_j = K_j^{\mathrm{R}} + iK_j^{\mathrm{I}}$ with real constant fields. On the connected domain

$$\Omega = \{z \in \mathbb{C}^4 : |\operatorname{Im} z_1| < 2,\ |\operatorname{Im} z_2| < 2,\ |z_3| < 2,\ |z_4| < 2\},$$

set

$$A(z) = \mathscr{A}^{\mathrm{b}} + z_1 K_1^{\mathrm{R}} + z_2 K_1^{\mathrm{I}}, \qquad C(z) = \mathscr{C}^{\mathrm{b}} + z_3 K_2^{\mathrm{R}} + z_4 K_2^{\mathrm{I}},$$

and set

$$\Lambda_z(c) = -(z_1 K_1^{\mathrm{R}} + z_2 K_1^{\mathrm{I}} + z_3 K_2^{\mathrm{R}} + z_4 K_2^{\mathrm{I}}) \cdot (c - c_*) \quad (c \in K(c_*, r)),$$

with value zero elsewhere. The first connection belongs to $\widetilde{R}_k$: its complex small summand has sup norm below $(L^{-7/4} + 4L^{-7/8+\varepsilon/2})\theta_1 < \theta_1$ and derivatives those of $\mathscr{A}^{\mathrm{b}}$, and its remaining constant summand is real. Invariance under real constant additions handles its unrestricted real part. The external connection has sup norm below $(L^{-7/4} + 4L^{-1+\varepsilon})d_1 < d_1$, and derivative that of $\mathscr{C}^{\mathrm{b}}$. The large-L condition implies these two strict inequalities.

For fixed kernel arguments the function

$$H(z) = F_{nm}(X; A(z), C(z); \xi; \zeta) \prod_a \exp\{(-1)^{\omega(a)} i e_k \Lambda_z(c(a))\}$$

is holomorphic on Ω. When z is real, split Λ_z into its first two and last two terms. These are independent real gauge functions for the two connections, with differentials equal to the negatives of their respective constants on $K(c_*, r)$. Every bond entering the local gauge condition lies in this ball. The real gauge identity and eq. 5.lx therefore give $H(z) = H(0)$ for $z \in \mathbb{R}^2 \times (-2, 2)^2$. All derivatives of $H - H(0)$ at zero vanish, so its holomorphic power series is zero there. The identity theorem extends this equality to the connected domain Ω. Evaluating at $z = (1, i, 1, i)$ gives the identity for the two original complex constants.

It remains to bound the inverse phases. By locality, an ordinary argument lies over X. For a pair of the first kind its first photon site also lies in X. For a boundary pair that site has distance at most $\eta_k \le 1$ from X. Hence in all nonzero kernel terms

$$|c(a) - c_*|_1 \le 3(M(L+2)+1) \le 9LM.$$

The constant bounds and the charge conditions give

$$e_k |K_1^{\mathrm{I}} \cdot (c(a) - c_*)| \le 9LML^{-7/8+\varepsilon/2} e_k^{1/4+\varepsilon} \le 1,$$
$$e_k |K_2^{\mathrm{I}} \cdot (c(a) - c_*)| \le 9ML^{\varepsilon} e_k^{2\varepsilon} \le 1.$$

Each leg phase, and its inverse, thus has modulus at most $\exp(2)$. Combining this bound with the unit modulus phases from λ_* proves eq. 5.lix. The vacuum has no leg phases, which proves its equality. □

5.1.8. *Contraction of the normalized activity.* For a small polymer, field scaling alone leaves two bidegrees to control: the vacuum and the ordinary quadratic kernel. Normalization and charge symmetry give the vacuum a double zero at zero connection; normalization gives the quadratic kernel a zero there. Cauchy's formula on the disk of radius $L^{7/4}$ gives factors $L^{-7/2}$ and $L^{-7/4}$, respectively. Including field scaling gives a quadratic contribution of order L^{-4}, while all bidegrees are bounded by a constant times $L^{-13/4+3\varepsilon/2}$. Reblocking contributes L^3, leaving the contraction factor $L^{-1/4+3\varepsilon/2}$. For large polymers, exponential decay supplies the gain directly.

Proposition 5.5. Assume eq. 5.xxxv, eq. 5.xl, and

$$\kappa \geq 7\kappa_0 + 7, \qquad L(\kappa - 6\kappa_0 - 4) - 1 \geq \kappa. \tag{5.lxi}$$

For $\mathscr{E} \in K_k^{x,\mathbb{C}}(\ell)$ set $E = (R\mathscr{E})^{\mathrm{nat}}$. The linear map $\mathscr{E} \mapsto \mathcal{L}E$ takes $K_k^{x,\mathbb{C}}(\ell)$ into $K_{k+1}^{x,\mathbb{C}}(\ell/L)$, preserves the real subspaces, and satisfies

$$\|\mathcal{L}E\|_{k+1}^x \leq c_{\mathrm{fl}} L^{-1/4+3\varepsilon/2} \|\mathscr{E}\|_k^x \leq c_{\mathrm{fl}} L^{-1/4+2\varepsilon} \|\mathscr{E}\|_k^x, \qquad c_{\mathrm{fl}} = 61 c_{\mathrm{rb}}. \tag{5.lxii}$$

Relative to the activity windows,

$$\frac{\|\mathcal{L}E\|_{k+1}^x}{w_{k+1}^{\mathrm{E}}} \leq c_{\mathrm{fl}} L^{-3/8+11\varepsilon/2} \frac{\|\mathscr{E}\|_k^x}{w_k^{\mathrm{E}}}. \tag{5.lxiii}$$

Restriction to the zero external argument commutes with the map. ■

Proof. Write

$$N(X) = \sup_{\widetilde{R}_k \times \mathfrak{X}_k} \|E(X;\cdot,\cdot)\|_{h_k}.$$

Extraction and adjustment give

$$N(X) \leq \begin{cases} 20\|\mathscr{E}\|_k^x \exp(-\kappa d_M(X)), & X \text{ small}, \\ \|\mathscr{E}\|_k^x \exp(-\kappa d_M(X)), & X \text{ large}. \end{cases} \tag{5.lxiv}$$

For a small polymer this uses $\|R\mathscr{E}(X;\mathscr{A},\mathscr{C})\|_{h_k} \leq \|\mathscr{E}(X;\mathscr{A},\mathscr{C})\|_{h_k} + \|\mathscr{E}(X;0,0)\|_{h_k}$ and eq. 5.xxxix. For a large polymer both operations leave the activity unchanged.

Fix a small X and a coarse connection pair, and use the global representatives above. Put $\rho = L^{7/4}$. The inverse image of the open set $R_k \times \mathfrak{X}_k$ under $t \mapsto (t\mathscr{A}^{\mathrm{b}}, t\mathscr{C}^{\mathrm{b}})$ contains the closed disk $|t| \leq \rho$, hence a slightly larger disk. The functions

$$f_{nm}(t) = E_{nm}(X; t\mathscr{A}^{\mathrm{b}}, t\mathscr{C}^{\mathrm{b}})$$

are holomorphic there, with values in a finite-dimensional kernel space. Their weighted norms obey

$$\frac{h_{k,1}^n h_{k,2}^m}{n!m!} \|f_{nm}(t)\|_1 \leq N(X) \qquad (|t| \leq \rho).$$

Here $\|\cdot\|_1$ is the sum of kernel moduli with their index weights.

For the vacuum, eq. 5.xxxviii gives $f_{00}(0) = 0$. Charge conjugation fixes a scalar kernel and reverses both connections, so $f_{00}(-t) = f_{00}(t)$ and $f_{00}'(0) = 0$. Cauchy's formula gives

$$f_{00}(1) = \frac{1}{2\pi i} \oint_{|t|=\rho} \frac{f_{00}(t)}{t^2(t-1)} \, \mathrm{d}t.$$

The vacuum equality in Prop. 5.4 and $\rho \geq 2$ therefore give

$$\|(S_L E)^{[0,0]}(X;\mathscr{A}_L,\mathscr{C}_L)\|_{h_{k+1}} \leq \frac{N(X)}{\rho(\rho-1)} \leq 2L^{-7/2} N(X). \tag{5.lxv}$$

For the ordinary quadratic kernel, normalization gives $f_{20}(0) = 0$ as a full antisymmetric kernel. Its balanced matrix components vanish by eq. 5.xxxviii; their oppositely ordered components vanish by antisymmetry, and the two equally barred or equally unbarred components vanish by balance. Hence

$$f_{20}(1) = \frac{1}{2\pi i} \oint_{|t|=\rho} \frac{f_{20}(t)}{t(t-1)}\,\mathrm{d}t, \qquad \|f_{20}(1)\|_1 \le \frac{\sup_{|t|=\rho} \|f_{20}(t)\|_1}{\rho - 1}.$$

The comparison of kernels costs $\exp(4)$ in this bidegree. Its scaling exponent is $p(2,0) = 9/4$. Multiplying the last inequality by $h_{k,1}^2/2$ and using eq. 5.xlvii gives

$$\|(S_L E)^{[2,0]}(X; \mathscr{A}_L, \mathscr{C}_L)\|_{h_{k+1}} \le 2\exp(4) L^{-4} N(X). \tag{5.lxvi}$$

For all remaining bidegrees use the original scaled connection pair, which belongs to the fine domain. Evenness leaves only $(1,1)$, $(0,2)$, and total degrees at least 4. The exponents satisfy

$$p(1,1) > \frac{13}{4} - \frac{3\varepsilon}{2}, \qquad p(0,2) > \frac{13}{4} - \frac{3\varepsilon}{2}, \qquad p(n,m) \ge n + m.$$

Consequently

$$\begin{aligned}\sum_{(n,m)\neq(0,0),(2,0)} \|(S_L E)^{[n,m]}(X; \mathscr{A}_L, \mathscr{C}_L)\|_{h_{k+1}} &\le L^{-13/4+3\varepsilon/2} \sum_{n,m} \|E^{[n,m]}(X; \mathscr{A}_L, \mathscr{C}_L)\|_{h_k} \\ &\le L^{-13/4+3\varepsilon/2} N(X).\end{aligned}$$

The sum is taken at one connection pair before bounding it by $N(X)$. The large-L condition ensures $2L^{-7/2} \le L^{-13/4}$ and $2\exp(4)L^{-4} \le L^{-13/4}$. Adding the three degree contributions yields

$$\|S_L E(X; \mathscr{A}_L, \mathscr{C}_L)\|_{h_{k+1}} \le 3L^{-13/4+3\varepsilon/2} N(X). \tag{5.lxvii}$$

We now sum over the fine polymers in a coarse polymer Y. The hypotheses eq. 5.lxi imply $\kappa - 1 \ge \kappa_0$ and $L(\kappa - \kappa_0 - 2) \ge \kappa$. The reblocking bound eq. D.v at rate κ and eq. 5.lxiv therefore give

$$\sum_{\substack{X \text{ small} \\ \bar{X}^L = LY}} \|S_L E(X; \mathscr{A}_L, \mathscr{C}_L)\|_{h_{k+1}} \le 60 c_{\mathrm{rb}} L^{-1/4+3\varepsilon/2} \|\mathscr{E}\|_k^x \exp(-\kappa d_M(Y)).$$

For large X, scaling decreases the norm by eq. 5.xlvi and the ratios of the two weights. Since $d_M(X) > L$,

$$\exp(-\kappa d_M(X)) \le \exp(-L)\exp(-(\kappa-1) d_M(X)).$$

Use eq. D.v at rate $\kappa - 1$ to obtain

$$\sum_{\substack{X \text{ large} \\ \bar{X}^L = LY}} \|S_L E(X; \mathscr{A}_L, \mathscr{C}_L)\|_{h_{k+1}} \le c_{\mathrm{rb}} L^3 \exp(-L) \|\mathscr{E}\|_k^x \exp(-\kappa d_M(Y)).$$

As $L^3 \exp(-L) \le L^{-1/4}$, these estimates prove eq. 5.lxii after taking the two suprema in the activity norm. Division by $w_{k+1}^{\mathrm{E}} = L^{1/8-7\varepsilon/2} w_k^{\mathrm{E}}$ gives the relative factor $c_{\mathrm{fl}} L^{-3/8+5\varepsilon}$, which is at most the factor in eq. 5.lxiii. The additional parameter choice $c_{\mathrm{fl}} L^{-3/8+6\bar{\varepsilon}} \le 1/16$ makes its relative factor at most $1/16$.

Linearity and analyticity follow from finite sums and complex-linear connection scaling; locality was proved after eq. 5.xlix. For $g' = (S, a')$, use $g = (S, La')$: it bijects the summands, $(g'\mathscr{A})_L = g\mathscr{A}_L$, and $a_{g'} S_L = S_L a_g$, also for $\mathscr{C}$. For independent real gauges, $G_\lambda S_L = S_L G_{\lambda_L}$ since $e_k \lambda_L(c) = e_{k+1}\lambda(c/L)$; the fine bonds on which the activity kernels depend scale inside the coarse unit neighborhood. Scaling has real coefficients and preserves bars, so also commutes with charge and reality. These identities give the stated activity spaces. Extraction and adjustment commute with restriction, as does reblocking since $0_L = 0$. □

The reblocked sum retains the evaluated normalized activity. Indeed, each fine polymer belongs to exactly one reblocking class, so

$$\sum_{Y}(\mathcal{L}E)(Y;\mathscr{A},\mathscr{C}) = S_L \sum_{X} E(X;\mathscr{A}_L,\mathscr{C}_L). \tag{5.lxviii}$$

If the coarse generators are evaluated at fields with $\chi' = \delta_{\alpha,\mathscr{A}+\mathscr{C}}\psi'$ and its barred counterpart, their scaled fields satisfy the fine relations. This follows from $d_X(x/L,x'/L) = d_X(x,x')/L$ and $U_{(\mathscr{A}+\mathscr{C})_L}(x,x') = U_{\mathscr{A}+\mathscr{C}}(x/L,x'/L)$: the path bijection and $e_k\eta_k L^{-1/2} = e_{k+1}\eta_{k+1}$ give the latter. Substitution in the difference quotient produces $L^{-1-\alpha}\chi'(x/L,x'/L)$, and the barred quotient uses the inverse transport. Adjustment consequently has its original value under this evaluation. Thus the evaluation of eq. 5.lxviii is also the evaluation of

$$S_L \sum_{X}(R\mathscr{E})(X;\mathscr{A}_L,\mathscr{C}_L).$$

5.2. **The exact block density.**

5.2.1. *Integration spaces and real backgrounds.* We now combine the photon energy splitting with the fermion Gaussian identity. The result must retain the coarse quadratic density, the determinants, and the scalar normalization as well as the new interaction: the reference partition functions will later be compared across volumes. We first specify the integration spaces and then identify each factor in this exact density. Fix $k \geq 0$, $\eta_k = L^{-k}$, odd $L \geq 5$, and $M = L^m \geq L^2$, with $\ell/(LM)$ odd and at least 13. The regulator parameter satisfies $0 < \rho < 1$. Keep $b = 1$ as a letter, and write $b_L = b/L$ and $b_k = b(1-L^{-1})/(1-L^{-k})$ for $k \geq 1$. Write $\mathscr{V}_k = \mathcal{S}(1,\ell)$ for $k \geq 1$, with four spin components and counting measure, and $\mathscr{V}_0 = \mathcal{S}^{\mathrm{ch}}(1,\ell)$, with 192 components per photon site and measure $1/48$. The one-step averaging operator into $\mathcal{S}(L,\ell)$ is

$$\mathcal{A}_k(B) = \begin{cases} Q(B), & k \geq 1, \\ Q(B)J^*, & k = 0, \end{cases} \qquad \mathcal{A}_k^\sharp(B) = \mathcal{A}_k(-B)^T.$$

The target measure is L^3. On these spaces set

$$\mathcal{D}_k(B) = \begin{cases} D_k(B), & k \geq 1, \\ D_B + \mu, & k = 0, \end{cases} \qquad K_k(B) = \mathcal{D}_k(B) + b_L \mathcal{A}_k^\sharp(B)\mathcal{A}_k(B). \tag{5.lxix}$$

Thus $K_k = D_k + b_L P_L$ for $k \geq 1$ and $K_0 = F_0 = D_B + \mu + b_L J P_L J^*$. We write $D_0(B) = D_B + \mu$, so that $\mathcal{D}_k(B) = D_k(B)$ at every level. When it exists, $\Gamma_k = K_k^{-1}$, and $\delta Z_k(B) = \det K_k(B)$. The transpose refers to the displayed bilinear measures. For a real B, the phases have modulus one and $\mathcal{A}_k^\sharp = \mathcal{A}_k^*$.

The normalized averages give $Q(B)Q^T(-B) = 1$. On a coarse block $B(z')$ the product is $L^{-3}\sum_{z\in B(z')} U_B(z',z)U_B(z',z)^{-1} = 1$. At level zero the additional cluster mean contributes $48/48 = 1$. Similarly $\mathcal{Q}_k\mathcal{Q}_k^\sharp = 1$ for $k \geq 1$. In a unit block the photon count gives $\eta_k^3\eta_k^{-3} = 1$ and the cluster count again gives $48/48 = 1$. For real B the inverse phases in the embeddings are the complex conjugates of the averaging phases. The products $\Pi_1 = \mathcal{Q}_k^\sharp\mathcal{Q}_k$ and $\Pi_L = \mathcal{Q}_k^\sharp P_L \mathcal{Q}_k$ are therefore orthogonal projections; so are P_L and, at level zero, JP_LJ^*. Their scalar spin kernels commute with every constant spin matrix.

Proposition 5.6. Let B be real and $\mu > 0$. The operators $\mathcal{D}_k(B)$ and $K_k(B)$ are invertible at every level. In particular $\det D_k(0)$ and $\delta Z_k(0)$ are nonzero, including the determinants on the chamber space $\mathscr{V}_0$. ∎

Proof. Put $G = D_B + \mu$. eq. 4.iv gives $\operatorname{Re} G \geq \mu$. All spaces are finite dimensional, so strict positivity of the real part implies invertibility; in particular G is invertible. At $k = 0$, $\mathcal{D}_0 = G$ and $K_0 = G + b_L J P_L J^*$, whose real part is at least μ because JP_LJ^* is an orthogonal projection. For

$k \geq 1$, write $Q = \mathcal{Q}_k(B)$, so $Q^* = \mathcal{Q}_k^\sharp(B)$. The positive penalty implies that $G + b_k Q^* Q$ is invertible; write its inverse as S. By eq. 4.1, $D_k = b_k - b_k^2 QSQ^*$. Multiplication using $S(G + b_k Q^* Q) = 1$ gives

$$D_k(b_k^{-1} + QG^{-1}Q^*) = 1.$$

Thus D_k is invertible with the displayed right inverse. Moreover

$$\operatorname{Re} G^{-1} = (G^{-1})^*(\operatorname{Re} G)G^{-1} \geq 0, \qquad \operatorname{Re} D_k^{-1} \geq b_k^{-1}.$$

Consequently $\operatorname{Re} D_k = D_k^*(\operatorname{Re} D_k^{-1})D_k \geq b_k^{-1} D_k^* D_k > 0$. Adding $b_L P_L \geq 0$ shows that K_k is invertible. Taking $B = 0$ proves the determinant assertions. □

5.2.2. *The next quadratic density.* Assume now that ℓ/L is also admissible, and let $B = B'_L$. At the outgoing level the mass is $L\mu$ and the charge is $e_{k+1} = L^{1/2} e_k$. The following identities hold whenever $S_k(B)$ and $\Gamma_k(B)$ exist for $k \geq 1$, or $\Gamma_0(B)$ exists for $k = 0$. This invertibility domain is open and invariant under $B \mapsto -B$: all elementary entries are entire functions of the bond values, and the transpose identities are

$$\mathcal{D}_k(B)^T = \mathsf{C}^{-1}\mathcal{D}_k(-B)\mathsf{C}, \qquad K_k(B)^T = \mathsf{C}^{-1}K_k(-B)\mathsf{C}, \qquad \Gamma_k(B)^T = \mathsf{C}^{-1}\Gamma_k(-B)\mathsf{C}. \tag{5.lxx}$$

For the first identity reindex each hopping term by $x^{(i)}$, using $M_i(x^{(i)})^T = \mathsf{C}^{-1}M_i(x)\mathsf{C}$ and the reverse phase at $-B$. Transposition of the averages reverses their connection in the product P_L, and the scalar kernels commute with C. The remaining identities follow by products and inversion. Invertible matrices have analytic inverses on their open determinant domains.

Define the fine-to-L average and its embedding by

$$\mathcal{Q}_k^+(B) = \begin{cases} Q(B)\mathcal{Q}_k(B), & k \geq 1, \\ Q(B)J^*, & k = 0, \end{cases} \qquad \mathcal{Q}_k^{+,\sharp}(B) = \mathcal{Q}_k^+(-B)^T.$$

They obey $\mathcal{Q}_k^+ \mathcal{Q}_k^{+,\sharp} = 1$. Set $\bar{b}_k = b_{k+1}/L$ and

$$F_k^+(B) = D_B + \mu + \bar{b}_k \mathcal{Q}_k^{+,\sharp}\mathcal{Q}_k^+, \qquad \Sigma_k^+(B) = F_k^+(B)^{-1}. \tag{5.lxxi}$$

For $k = 0$ this is F_0 and its inverse Γ_0. For $k \geq 1$ it is the fine resolvent operator at $y = 0$, since $\bar{b}_k = b_k b_L/(b_k + b_L)$.

To verify existence and the identities needed below, write $\mathcal{B}_k = (b_k + b_L P_L)^{-1}$ for $k \geq 1$ and consider

$$\begin{pmatrix} b_k + b_L P_L & -b_k \mathcal{Q}_k \\ -b_k \mathcal{Q}_k^\sharp & F_\infty \end{pmatrix}.$$

Its lower-right Schur complement is K_k. Since P_L is idempotent and scalar in spin, $\mathcal{B}_k = b_k^{-1}(1 - P_L) + (b_k + b_L)^{-1} P_L$. The other Schur complement is

$$F_\infty - b_k^2 \mathcal{Q}_k^\sharp \mathcal{B}_k \mathcal{Q}_k = D_B + \mu + \frac{b_k b_L}{b_k + b_L} \mathcal{Q}_k^\sharp P_L \mathcal{Q}_k = F_k^+.$$

The two assumed inverses make the block matrix invertible, and hence F_k^+ is invertible. Equating the upper-left and lower-left blocks of its inverse gives

$$\Gamma_k = \mathcal{B}_k + b_k^2 \mathcal{B}_k \mathcal{Q}_k \Sigma_k^+ \mathcal{Q}_k^\sharp \mathcal{B}_k, \qquad H_k \Gamma_k = b_k \Sigma_k^+ \mathcal{Q}_k^\sharp \mathcal{B}_k. \tag{5.lxxii}$$

This is the same Schur calculation as Prop. 4.9, retaining the off-diagonal block for the mean field.

Let

$$D_{k+1}^0(B) = b_L - b_L^2 \mathcal{A}_k(B)\Gamma_k(B)\mathcal{A}_k^\sharp(B). \tag{5.lxxiii}$$

For $k \geq 1$, $QP_L = Q$ and $P_L Q^T(-B) = Q^T(-B)$ give $Q\mathcal{B}_k = (b_k + b_L)^{-1} Q$ and $\mathcal{B}_k Q^T(-B) = (b_k + b_L)^{-1} Q^T(-B)$. Insert these in eq. 5.lxxii to obtain

$$\begin{aligned} D^0_{k+1}(B) &= \bar{b}_k - \bar{b}_k^2 \mathcal{Q}^+_k \Sigma^+_k \mathcal{Q}^{+,\sharp}_k, \\ b_L H_k(B) \Gamma_k(B) \mathcal{A}^\sharp_k(B) &= \bar{b}_k \Sigma^+_k \mathcal{Q}^{+,\sharp}_k. \end{aligned} \tag{5.lxxiv}$$

For $k = 0$ both formulas follow directly from $\bar{b}_0 = b_L$, $H_0 = 1$, $\Sigma^+_0 = \Gamma_0$, and $\mathcal{Q}^+_0 = \mathcal{A}_0$.

Let R_L denote the scaling $f_L(c, g) = L^{-1} f(c/L, g)$ on chamber spinors and $v_L(z') = L^{-1} v(z'/L)$ on unit spinors. These are bijections between the corresponding incoming and outgoing spaces. Scaling preserves the label and the nested block centers, and $e_k \eta_k L^{-1/2} = e_{k+1} \eta_{k+1}$ preserves each transport phase. Consequently

$$(D_B + \mu)\mathsf{R}_L = L^{-1} \mathsf{R}_L (D_{B'} + L\mu), \qquad \mathcal{Q}^+_k(B)\mathsf{R}_L = \mathsf{R}_L \mathcal{Q}_{k+1}(B'),$$

and the corresponding embedding identity holds. For the average, its L^{-3} factor cancels the L^3 change of the chamber index measure. Thus

$$F^+_k(B)\mathsf{R}_L = L^{-1}\mathsf{R}_L F_\infty(B'), \qquad \Sigma^+_k(B)\mathsf{R}_L = L\mathsf{R}_L S_{k+1}(B'). \tag{5.lxxv}$$

In particular the outgoing inverse exists. eq. 5.lxxiv and eq. 5.lxxv now imply

$$D^0_{k+1}(B)\mathsf{R}_L = L^{-1}\mathsf{R}_L D_{k+1}(B'), \qquad b_L H_k(B)\Gamma_k(B)\mathcal{A}^\sharp_k(B)\mathsf{R}_L = \mathsf{R}_L H_{k+1}(B'). \tag{5.lxxvi}$$

All expressions in the second identity act on the outgoing four-component field.

For independent Grassmann fields at the outgoing level, define

$$\Psi^c = b_L \Gamma_k(B) \mathcal{A}^\sharp_k(B) \Psi'_L, \qquad \bar{\Psi}^c = b_L \Gamma_k(B)^T \mathcal{A}_k(B)^T \bar{\Psi}'_L. \tag{5.lxxvii}$$

Then, with $\psi' = H_{k+1}(B')\Psi'$ and $\bar{\psi}' = \bar{H}_{k+1}(B')\bar{\Psi}'$,

$$H_k(B)\Psi^c = \psi'_L, \qquad \bar{H}_k(B)\bar{\Psi}^c = \bar{\psi}'_L. \tag{5.lxxviii}$$

The first relation is eq. 5.lxxvi. For the second use eq. 5.lxx and apply the first relation at $-B$ to $\mathsf{C}\bar{\Psi}'$; the constant matrix C commutes with scaling and the averages. The covariant Hölder quotients of these induced fields are also the scaled outgoing quotients: the transports agree and d_X is divided by L, so the quotient carries the factor $L^{-1-\alpha}$. This holds for every pair, including a pair inside one cluster. Applying $T_k H_k = 1$ and its barred version shows that the means in eq. 5.lxxvii are also $T_k(B)\psi'_L$ and $\bar{T}_k(B)\bar{\psi}'_L$. At level zero these maps are identities on the chamber field.

The quadratic term in the Gaussian block formula is therefore exactly

$$\langle \bar{\Psi}'_L, D^0_{k+1}(B)\Psi'_L \rangle_L = \langle \bar{\Psi}', D_{k+1}(B')\Psi' \rangle_1,$$

since the pairings of scaled fields multiply by L. Together with eq. 4.lxix, this identifies the entire outgoing quadratic density.

Proposition 5.7. On the invertibility domain just specified,

$$\det D_{k+1}(B') = b^{4n_{k+1}} \frac{\det \mathcal{D}_k(B)}{\delta Z_k(B)}, \qquad n_k = \ell^3, \quad n_{k+1} = (\ell/L)^3. \tag{5.lxxix}$$

In particular, with $c_k = b^{-4n_{k+1}}$,

$$\frac{c_k}{\det D_k(0)} = \frac{1}{\det D_{k+1}(0)\delta Z_k(0)}. \tag{5.lxxx}$$

■

Proof. Apply Sylvester's determinant identity to $b_L \mathcal{A}_k \Gamma_k$ and $\mathcal{A}^\sharp_k$ on the two different spaces. It gives

$$b_L^{-4n_{k+1}} \det D^0_{k+1}(B) = \det(1 - b_L \mathcal{A}^\sharp_k \mathcal{A}_k \Gamma_k) = \det(\mathcal{D}_k \Gamma_k) = \frac{\det \mathcal{D}_k(B)}{\delta Z_k(B)}.$$

The incoming determinant at $k = 0$ has size $192n_0$; the exponent of b_L is the outgoing dimension $4n_1$, as in every other step. eq. 5.lxxvi gives $\det D^0_{k+1}(B) = L^{-4n_{k+1}} \det D_{k+1}(B')$. Using $Lb_L = b$ proves eq. 5.lxxix. At the zero connection all denominators are nonzero by Prop. 5.6, so eq. 5.lxxx follows. □

5.2.3. *The density with an external connection.* For this subsubsection assume $0 < \mu \le 1/L$ and $e_{k+1} = L^{1/2} e_k \le \bar{e}_0$, with the hypotheses of Props. 4.7, 4.9 and 3.15. We also impose eq. 3.l and eq. 5.vi. The inverse domain $\widetilde{G}_k$ is the domain $\widetilde{\mathcal{G}}_k$ of the propagator estimates. Let $A' \in \mathrm{Dom}_{k+1}$ be a real block field, with arbitrary harmonic part, and let $\mathscr{C}' \in \mathfrak{X}_{k+1}$. Put

$$\mathscr{A}' = H^{\mathrm{ph}}_{k+1} A', \qquad \mathscr{A} = \mathscr{A}'_L, \qquad \mathscr{C} = \mathscr{C}'_L, \qquad \mathscr{T}' = \mathscr{A}' + \mathscr{C}', \qquad \mathscr{T} = \mathscr{T}'_L.$$

The photon block objects are

$$\begin{aligned} A_k(\zeta) &= A^{\min}_k(A') + \mathrm{Sq}_k \zeta, \\ \mathscr{Z}(\zeta) &= H^{\mathrm{ph}}_k \mathrm{Sq}_k \zeta, \qquad \mathscr{B}_k = H^{\mathrm{ph}}_k A^{\min}_k(A') = \mathscr{A} - d\omega_k. \end{aligned} \tag{5.lxxxi}$$

Here ω_k is real and periodic, as in eq. 3.xxxvii. The truncated photon measure has the normalization

$$\mu_I(\mathrm{Box}_k) = \exp(-n_k \delta \varepsilon^0_k), \qquad \widehat{\mu}_k = \frac{\mathbf{1}_{\mathrm{Box}_k}}{\mu_I(\mathrm{Box}_k)} \mu_I, \tag{5.lxxxii}$$

proved in eq. 3.xxxi. Since $p_{0,k} > 0$, the box has positive Gaussian measure.

All connection arguments in the step belong to their required domains. Indeed, scaling gives $\mathscr{A} \in \frac{1}{4} L^{-3/4} \widetilde{R}_k \subset \widetilde{R}^-_k$ and $\mathscr{C} \in \mathfrak{X}_k$. Prop. 3.15 gives $A_k(\zeta) \in \mathrm{Dom}_k$ for every $\zeta \in \mathrm{Box}_k$. The real-gauge identity in eq. 5.lxxxi yields

$$H^{\mathrm{ph}}_k A_k(\zeta) = \mathscr{A} + \mathscr{Z}(\zeta) - d\omega_k.$$

Since $\frac{1}{4}\widetilde{R}_k$ is invariant under real gauges, $\mathscr{A} + \mathscr{Z}(\zeta)$ also lies in this domain. Thus both connection pairs $(\mathscr{A}, \mathscr{C})$ and $(\mathscr{A} + \mathscr{Z}, \mathscr{C})$ lie in $\widetilde{R}_k \times \mathfrak{X}_k$. By eq. 5.vii, $\mathscr{T}$, $\mathscr{T} + \mathscr{Z}$ and $\mathscr{T} + \mathscr{Z} - d\omega_k$, and their negatives, belong to $\widetilde{G}_k$. The fine and coarse inverse estimates consequently apply to them. The outgoing operators at $\mathscr{T}'$ then exist by eq. 5.lxxv.

For a total connection B let ev_B insert

$$\psi = H_k(B)\Psi, \qquad \bar{\psi} = \bar{H}_k(B)\bar{\Psi}, \qquad \chi = \delta_{\alpha,B} H_k(B)\Psi, \qquad \bar{\chi} = \delta_{\alpha,-B} \bar{H}_k(B)\bar{\Psi}.$$

At $k = 0$ the two field maps are identities. For complex $m, \varepsilon^{\mathrm{v}}$ and $\mathscr{E} \in K^{x,\mathbb{C}}_k$, with $\mathscr{E} = 0$ at $k = 0$, define

$$\begin{aligned} I^x_k[m, \mathscr{E}](A, C; \Psi) &= \mathrm{ev}_{A+C} \left(-m\langle \bar{\psi}, \psi \rangle + \sum_X \mathscr{E}(X; A, C) \right), \\ \mathrm{Act}^x_k[m, \mathscr{E}](A, C; \Psi) &= -\langle \bar{\Psi}, D_k(A + C)\Psi \rangle + I^x_k[m, \mathscr{E}](A, C; \Psi). \end{aligned} \tag{5.lxxxiii}$$

The mass pairing is always the chamber pairing $\eta_k^3 \, 48^{-1} \sum_{c,g,\beta} \bar{\psi}_\beta(c, g) \psi_\beta(c, g)$. The kinetic pairing uses the measure of $\mathscr{V}_k$. For an arbitrary finite family of scalar functions $\mathfrak{P}_k(X, C)$ on $\mathfrak{X}_k$, set

$$\mathscr{F}^x_k[\varepsilon^{\mathrm{v}}, m, \mathscr{E}](A, C; \Psi) = \det D_k(0)^{-1} \exp \left\{ \mathrm{Act}^x_k[m, \mathscr{E}](H^{\mathrm{ph}}_k A, C; \Psi) - \varepsilon^{\mathrm{v}} n_k + \sum_X \mathfrak{P}_k(X, C) \right\}. \tag{5.lxxxiv}$$

The arguments here are $A \in \mathrm{Dom}_k$ and $C \in \mathfrak{X}_k$. The fields Ψ and $\bar{\Psi}$ are independent Grassmann variables.

Define the block transformation on an even density F whose coefficients are integrable along $A_k(\zeta)$ by

$$\begin{aligned}(\mathbf{T}_k^x F)(A',\mathscr{C}';\Psi') &= c_k \int_{\mathrm{Box}_k} \mathrm{d}\mu_I(\zeta) \int \mathrm{d}\Psi\, e^{\mathcal{Q}_{\mathrm{bl}}} F(A_k(\zeta),\mathscr{C};\Psi),\\ \mathcal{Q}_{\mathrm{bl}} &= -b_L\big\langle e^{-ie_k\omega_k}\bar\Psi'_L - \mathcal{A}_k(-\mathscr{B}_k-\mathscr{C})\bar\Psi,\, e^{ie_k\omega_k}\Psi'_L - \mathcal{A}_k(\mathscr{B}_k+\mathscr{C})\Psi\big\rangle_L.\end{aligned} \tag{5.lxxxv}$$

The gauge function in the two outgoing phases is evaluated at the L-spaced sites. The Berezin functional is the weighted functional of eq. 4.lxiii; its incoming mode measure is 1 for $k \geq 1$ and $1/48$ for $k = 0$.

Its normalization is

$$\int \mathrm{d}\Psi'\,(\mathbf{T}_k^x F)(A',\mathscr{C}';\Psi') = \int_{\mathrm{Box}_k} \mathrm{d}\mu_I(\zeta) \int \mathrm{d}\Psi\, F(A_k(\zeta),\mathscr{C};\Psi). \tag{5.lxxxvi}$$

Integrating the outgoing family first removes its reciprocal gauge phases and the translation by the averaged incoming fields. The remaining exponent is $-b_L\langle\bar\Psi'_L,\Psi'_L\rangle_L = -b\langle\bar\Psi',\Psi'\rangle_1$, whose integral is $b^{4n_{k+1}} = c_k^{-1}$ by eq. 4.lxvi. The even Berezin extractions commute with each other and with the photon-box integrals, proving eq. 5.lxxxvi.

The stated operator and activity domains make all coefficients in eq. 5.lxxxiv, including those of its even Grassmann exponentials, continuous in $\zeta \in \mathrm{Box}_k$ and hence integrable.

Proposition 5.8. Let the data and domains be as above. Put

$$m' = m + m(\mathscr{E}), \qquad \mathscr{E}^r = R\mathscr{E}, \qquad m^+ = Lm', \qquad \varepsilon^{\mathrm{v},+} = L^3(\varepsilon^{\mathrm{v}} + \varepsilon_{\mathrm{v}}(\mathscr{E}) + \delta\varepsilon_k^0).$$

At $k = 0$ the activity and its extracted terms vanish. Define the outgoing inserted fields $\psi' = H_{k+1}(\mathscr{T}')\Psi'$, $\bar\psi' = \bar H_{k+1}(\mathscr{T}')\bar\Psi'$ and their covariant Hölder quotients, and let $\psi^{0,\#}$ denote their scaled fine fields, including the barred ones. Let the means $\Psi^c, \bar\Psi^c$ be those of eq. 5.lxxvii at $B = \mathscr{T}$, and put

$$\Psi^f = \Psi^c + \Gamma_k(\mathscr{T})W, \qquad \bar\Psi^f = \bar\Psi^c + \bar W.$$

With $\mathscr{Z} = H_k^{\mathrm{ph}}\mathrm{Sq}_k\zeta$, set

$$\begin{aligned}V_k^x &= \mathrm{Act}_k^x[m',\mathscr{E}^r](\mathscr{A}+\mathscr{Z},\mathscr{C};\Psi^f) - \mathrm{Act}_k^x[m',\mathscr{E}^r](\mathscr{A},\mathscr{C};\Psi^f)\\ &\quad + I_k^x[m',\mathscr{E}^r](\mathscr{A},\mathscr{C};\Psi^f) - I_k^x[m',\mathscr{E}^r](\mathscr{A},\mathscr{C};\Psi^c),\\ \Xi_k^x &= \int \mathrm{d}\mu_I(W)\,\mathrm{d}\hat\mu_k(\zeta)\, e^{V_k^x}.\end{aligned} \tag{5.lxxxvii}$$

Then the exact identity is

$$\begin{aligned}&(\mathbf{T}_k^x\mathscr{F}_k^x[\varepsilon^{\mathrm{v}},m,\mathscr{E}])(A',\mathscr{C}';\Psi')\\ &\quad= \det D_{k+1}(0)^{-1}\exp\Big\{ -\langle\bar\Psi', D_{k+1}(\mathscr{T}')\Psi'\rangle_1 - m^+\langle\bar\psi',\psi'\rangle - \varepsilon^{\mathrm{v},+}n_{k+1}\\ &\qquad + \sum_X \mathscr{E}^r(X;\mathscr{A},\mathscr{C};\psi^{0,\#}) + \sum_X \mathfrak{P}_k(X,\mathscr{C})\Big\}\frac{\delta Z_k(\mathscr{T})}{\delta Z_k(0)}\Xi_k^x.\end{aligned} \tag{5.lxxxviii}$$

All integrals in this formula are finite coefficient integrals. Its right side depends on A' through $H_{k+1}^{\mathrm{ph}}A'$. ∎

Proof. Extraction gives

$$I_k^x[m,\mathscr{E}](A,C;\Psi) = -\varepsilon_{\mathrm{v}}(\mathscr{E})n_k + I_k^x[m',\mathscr{E}^r](A,C;\Psi).$$

This is the image of eq. 5.xvi under the field evaluation homomorphism. At $k = 0$ it holds with zero activity and $m' = m$.

For a fixed ζ, change the integrated variables by $\Psi \mapsto e^{ie_k\omega_k}\Psi$ and $\bar\Psi \mapsto e^{-ie_k\omega_k}\bar\Psi$. This leaves the weighted Berezin functional unchanged, since each mode's factors cancel in its top monomial, also for the $192n_0$ modes of weight $1/48$. The transport identity

$$U_{B-d\omega_k}(c,c')e^{ie_k\omega_k(c')} = e^{ie_k\omega_k(c)}U_B(c,c')$$

intertwines the Dirac operator and every average. Products and inverses preserve this intertwining, so it also holds for H_k and D_k. Its covariant difference identity uses the phase at the first photon site, including for a pair inside one cluster. The barred identity has the inverse phase. Gauge covariance of the activity and invariance of the mass pairing consequently turn its first connection argument $\mathscr{A}+\mathscr{Z}-d\omega_k$ into $\mathscr{A}+\mathscr{Z}$. The same change removes the gauge from the block Gaussian factor. The integral in eq. 5.lxxxv is now

$$\int \mathrm{d}\Psi\, \exp\Big\{-b_L\langle \bar\Psi'_L - \mathcal{A}_k(-\mathscr{T})\bar\Psi,\, \Psi'_L - \mathcal{A}_k(\mathscr{T})\Psi\rangle_L + \mathrm{Act}_k^x[m',\mathscr{E}^r](\mathscr{A}+\mathscr{Z},\mathscr{C};\Psi)\Big\},$$

multiplied by $\det D_k(0)^{-1}\exp\{-(\varepsilon^{\mathrm{v}}+\varepsilon_{\mathrm{v}}(\mathscr{E}))n_k + \sum_X \mathfrak{P}_k(X,\mathscr{C})\}$.

Separate the background quadratic term by writing

$$\begin{aligned}\mathcal{R}(\Psi) &= I_k^x[m',\mathscr{E}^r](\mathscr{A},\mathscr{C};\Psi)\\ &\quad + \mathrm{Act}_k^x[m',\mathscr{E}^r](\mathscr{A}+\mathscr{Z},\mathscr{C};\Psi) - \mathrm{Act}_k^x[m',\mathscr{E}^r](\mathscr{A},\mathscr{C};\Psi).\end{aligned}$$

The last exponent above is then the background Gaussian exponent plus $\mathcal{R}(\Psi)$. Translation by the means eq. 5.lxxvii and the block formula eq. 4.lxx give

$$\delta Z_k(\mathscr{T})e^{-\langle\bar\Psi',D_{k+1}(\mathscr{T}')\Psi'\rangle_1}\int \mathrm{d}\mu_I(W)\, e^{\mathcal{R}(\Psi^f)}.$$

The change of variables in this formula uses the same mode measure for Ψ and W. At $k=0$ it is the identity

$$\int \mathrm{d}\Psi\, e^{-\langle\bar\Psi,F_0(\mathscr{T})\Psi\rangle^{\mathrm{ch}}}G(\bar\Psi,\Psi) = \det F_0(\mathscr{T})\int \mathrm{d}\mu_I(W)\, G(\bar W,\Gamma_0(\mathscr{T})W),$$

on the entire $192n_0$-mode space. The Gaussian measure is $e^{-\langle\bar W,W\rangle^{\mathrm{ch}}}\,\mathrm{d}W$, with the weight $\sqrt{48}$ in its coefficient bound.

By definition of V_k^x, $\mathcal{R}(\Psi^f) = I_k^x[m',\mathscr{E}^r](\mathscr{A},\mathscr{C};\Psi^c)+V_k^x$. Its first summand is independent of W and ζ, so its even exponential factors out. The induced-field identity eq. 5.lxxviii gives

$$I_k^x[m',\mathscr{E}^r](\mathscr{A},\mathscr{C};\Psi^c) = -m'\langle\bar\psi'_L,\psi'_L\rangle + \sum_X \mathscr{E}^r(X;\mathscr{A},\mathscr{C};\psi^{0,\#}).$$

The chamber measure changes by L^3 and each field by L^{-1}. Hence $\langle\bar\psi'_L,\psi'_L\rangle = L\langle\bar\psi',\psi'\rangle$, giving $m^+ = Lm'$. Normalizing the photon-box restriction in eq. 5.lxxxii adds $-\delta\varepsilon_k^0 n_k$ to the vacuum term. Since $n_k = L^3 n_{k+1}$, this term has exactly the stated outgoing vacuum coefficient. Finally eq. 5.lxxx turns the remaining constant factor into $\det D_{k+1}(0)^{-1}\delta Z_k(0)^{-1}$, proving eq. 5.lxxxviii.

The fluctuation exponent is even and continuous on the photon box, by the same coefficient argument as for the input density. Thus its integral is defined, and its two finite integrations commute. Every background in the final expression is formed from $\mathscr{A}'$ and $\mathscr{C}'$; the remaining photon maps and measure are independent of A'. The minimum and the real gauge have disappeared from that expression, proving the last assertion. □

5.2.4. *Local substitution of the mean field.* The reconstruction formulas let us eliminate the auxiliary coarse generators by a local substitution. For $k \geq 1$, let Y be a union of partition M-cubes and allow independent unit generators $\Psi, \bar\Psi$ over Y, together with the ordinary and Hölder generators.

Using eq. 5.xxiii, define the substitution that fixes the latter generators and sets, at the unit sites of Y,

$$\begin{aligned}\operatorname{sub}_{Y,B}\Psi &= T_Y^{(1)}(B)\psi + T_Y^{(2)}(B)\chi,\\ \operatorname{sub}_{Y,B}\bar\Psi &= \bar T_Y^{(1)}(B)\bar\psi + \bar T_Y^{(2)}(B)\bar\chi.\end{aligned} \tag{5.lxxxix}$$

The images are odd linear combinations of independent generators and therefore anticommute with one another and with the fixed generators. They determine a unique Grassmann homomorphism. Its new ordinary legs lie over Y and its new pair legs are boundary pairs of the second kind with first point over Y. Its coefficients depend on B only on bonds meeting Y, by the row support proved in §5.1.4. Since a bar index is preserved, the substitution preserves balance and parity.

If an evaluation has $\chi = \delta_{\alpha,B}\psi$ and the barred relation on these boundary pairs, then eq. 5.xxiv turns eq. 5.lxxxix into $T_k(B)\psi$ and $\bar T_k(B)\bar\psi$. If in addition $\psi = H_k(B)\Phi$ and $\bar\psi = \bar H_k(B)\bar\Phi$, the left-inverse identity gives Φ and $\bar\Phi$ over Y. Thus this local substitution represents the exact mean field under evaluation.

The row covariances yield

$$\begin{aligned} a_g \operatorname{sub}_{gY,gB} &= \operatorname{sub}_{Y,B} a_g,\\ G_\lambda \operatorname{sub}_{Y,B^1} &= \operatorname{sub}_{Y,B} G_\lambda, \qquad B^1 = B - d\lambda \text{ on bonds meeting } Y,\\ \mathrm{Cg}\operatorname{sub}_{Y,-B} &= \operatorname{sub}_{Y,B}\mathrm{Cg},\\ \mathbf{K}\operatorname{sub}_{Y,-\overline{B}} &= \operatorname{sub}_{Y,B}\mathbf{K}.\end{aligned} \tag{5.xc}$$

For the final relation μ is real. It suffices to check the new generators: lattice and local-gauge covariance follow from eq. 5.xxxii and eq. 5.xxxi, since the rows depend on the connection only on the stipulated bonds. Charge uses $T_Y^{(i)}(-B)\mathsf{C} = \mathsf{C}\bar T_Y^{(i)}(B)$ and its barred version; reality uses eq. 5.xxxiii at $-B$, $\overline{T_Y^{(i)}(-\overline{B})} = \varkappa^{-1}T_Y^{(i)}(B)\varkappa$, and the barred version.

Assume the constants and charge conditions of Prop. 5.2, including eq. 5.vi, with $0<\mu\le 1$ and $0<e_k\le \bar e_0$. For $B\in \widetilde R_k + \mathfrak{X}_k$, the row estimate gives the following norm statement. Assign weights w_1, w_2 to the ordinary and pair legs and weight w_3 to the independent unit generators:

$$\|\operatorname{sub}_{Y,B} F\|_{w_1,w_2} \le \|F\|_{w_1,w_2;w_3} \quad \text{if } C_T(w_1+w_2)\le w_3. \tag{5.xci}$$

The image of each new generator has norm at most $C_T(w_1+w_2)$, while old generators keep their weights. Apply Prop. 4.11 on the generators over Y, with row measures $\eta_k^3/48$ and $(\eta_k^3/48)^2$ and reciprocal column-weight identity kernels, exactly as in the reconstruction estimate.

5.2.5. *The first-step vertex.* At $k=0$ write the charge as q. There is no input activity, and $H_0(B)=\bar H_0(B)=1$. Consequently the exponent in eq. 5.lxxxvii is

$$V_0^x = -\langle\bar\Psi^f, (D_{\mathscr{T}+\mathscr{Z}} - D_{\mathscr{T}})\Psi^f\rangle^{\mathrm{ch}} - m\left(\langle\bar\Psi^f,\Psi^f\rangle^{\mathrm{ch}} - \langle\bar\Psi^c,\Psi^c\rangle^{\mathrm{ch}}\right). \tag{5.xcii}$$

The terms with μ and m cancel in the connection difference of the actions; the second difference of interactions gives the displayed mass term. Here $\Psi^c = \psi'_L$, $\bar\Psi^c = \bar\psi'_L$, $\Gamma_0(\mathscr{T}) = F_0(\mathscr{T})^{-1}$, and $\Psi^f = \Psi^c + \Gamma_0(\mathscr{T})W$. All these are chamber fields on $T_0^{\mathrm f}$. The identity measure in W has mode measure $1/48$, and its two-point kernel is 48 times the identity matrix of the modes.

For a bond fluctuation Z and an M-cube $\square$, define its local vertex by

$$\begin{aligned}\mathcal{V}_0(Z,\square) = \frac12\int_\square \mathrm{d}x\Big\{&\bar\psi(x)^T M_3(x)(e^{iqZ(c,c_3)}-1)[\psi(x)+\chi(x,x^{(3)})]\\ &+[\bar\psi(x)+\bar\chi(x,x^{(3)})]^T M_3(x^{(3)})(e^{-iqZ(c,c_3)}-1)\psi(x)\Big\},\end{aligned} \tag{5.xciii}$$

where $x=(c,g)$, $c_3 = c+ge_1$ and $\int_\square \mathrm{d}x = 48^{-1}\sum_{c\in\square}\sum_g$. The pair $(x,x^{(3)})$ has $d_X=1$ and is admissible for $\square$ of the second kind. Thus the vertex has ordinary legs over the cube and precisely

the boundary pair legs allowed in the activity algebra. Its coefficients are entire functions of Z on the bonds meeting the cube.

Proposition 5.9. Evaluate $\chi = \delta_{\alpha,\mathscr{T}} f$, $\bar\chi = \delta_{\alpha,-\mathscr{T}} \bar f$, $\psi = f$ and $\bar\psi = \bar f$ at any pair of antiperiodic odd chamber fields. Then

$$\operatorname{ev} \sum_{\square} \mathcal{V}_0(Z,\square) = -\langle \bar f, (D_{\mathscr{T}+Z} - D_{\mathscr{T}}) f\rangle^{\mathrm{ch}}. \tag{5.xciv}$$

The two halves have equal images after summing over cubes. The vertex has the lattice, real-gauge, charge-conjugation and reality symmetries

$$\begin{aligned} a_g \mathcal{V}_0(gZ, g\square) &= \mathcal{V}_0(Z,\square),\\ G_\lambda \mathcal{V}_0(Z,\square) &= \mathcal{V}_0(Z,\square),\\ \mathrm{Cg}\mathcal{V}_0(-Z,\square) &= \mathcal{V}_0(Z,\square),\\ \mathbf{K}\mathcal{V}_0(-\overline{Z},\square) &= \mathcal{V}_0(Z,\square). \end{aligned} \tag{5.xcv}$$

For any generator weights (w_1, w_2),

$$\|\mathcal{V}_0(Z,\square)\|_{w_1,w_2} \le 12(1+\rho) M^3 w_1(w_1+w_2) \sup_{b \text{ oriented and meeting } \square} |e^{iqZ(b)} - 1|. \tag{5.xcvi}$$

If $q \le 1$ and $q|Z(b)| \le \frac{1}{2} q^{1-\varepsilon/2}$ on these bonds, then

$$\|\mathcal{V}_0(Z,\square)\|_{h_0/2} \le 6(1+\rho) M^3 q^{1/2-\varepsilon/2} \le 12 M^3 q^{1/2-\varepsilon/2}. \tag{5.xcvii}$$

■

Proof. At a nearest-neighbor pair the six paths reduce to the bond, so the quotient relations read

$$f(x) + \chi(x, x^{(3)}) = u_{\mathscr{T}}(c, c_3) f(x^{(3)}), \qquad \bar f(x) + \bar\chi(x, x^{(3)}) = u_{\mathscr{T}}(c,c_3)^{-1} \bar f(x^{(3)}).$$

The first half of the summed vertex is therefore one half of the integral of

$$\bar f(x)^T M_3(x) (e^{iqZ(c,c_3)} - 1) u_{\mathscr{T}}(c,c_3) f(x^{(3)}).$$

Reindexing the second half by the measure-preserving involution $x \mapsto x^{(3)}$ gives the same integrand: both Z and $\mathscr{T}$ change sign on the reversed bond. Only the type-3 hop depends on the connection, and $u_{\mathscr{T}+Z} = e^{iqZ} u_{\mathscr{T}}$. The difference of the two Dirac operators is thus the negative of this hopping coefficient, proving eq. 5.xciv. Every product contains two antiperiodic fields, so the reindexing is legitimate on the torus.

For lattice symmetry use $g^{-1}(x^{(3)}) = (g^{-1}x)^{(3)}$ and $S^{-1} M_3(x) S = M_3(g^{-1}x)$ for the spin lift S. Each half transforms separately, and reindexing maps $g\square \times O_{\mathrm{cub}}$ onto $\square \times O_{\mathrm{cub}}$ with the same measure. For a real gauge, the two generators in each term have the same first photon site and opposite bar indices, so their phases cancel. Charge conjugation interchanges the two halves. Its algebraic rule on the first half gives

$$\mathrm{Cg}[\bar\psi(x)^T m(\psi(x) + \chi(x, x^{(3)}))] = (\bar\psi(x) + \bar\chi(x, x^{(3)}))^T (\mathsf{C}^{-1} m^T \mathsf{C}) \psi(x).$$

The sign from exchanging the odd factors cancels the minus sign in the image of $\bar\psi$. As

$$\mathsf{C}^{-1} M_3(x)^T \mathsf{C} = M_3(x^{(3)}),$$

the first half at $-Z$ becomes the second at Z; the reverse relation gives the other half. Finally

$$\varkappa^T \overline{M_3(x)} \varkappa = M_3(x), \qquad \overline{e^{\pm iq(-\overline{Z})} - 1} = e^{\pm iqZ} - 1.$$

The antilinear map consequently preserves each half in its reality identity. This proves eq. 5.xcv.

Put $\delta_x = e^{iqZ(c,c_3)} - 1$ and $\delta'_x = e^{-iqZ(c,c_3)} - 1$. Collecting the ordinary bilinear in eq. 5.xciii, its coefficient is $[M_3(x)\delta_x + M_3(x^{(3)})\delta'_x]/96$. The other two products have coefficients $M_3(x)\delta_x/96$ and $M_3(x^{(3)})\delta'_x/96$, and weights $w_1 w_2$. The ordinary product has weight w_1^2. The chamber sites in the

cube and the ordered pairs with these first sites are distinct, so these are exactly the coefficients, up to ordering signs. Each endpoint matrix satisfies $\sum_{\beta,\beta'} |M_3(\cdot)_{\beta\beta'}| \le 12(1+\rho)$, since $M_3 = 3(\rho \mp \gamma_\mu)$ and γ_μ has four entries of modulus one. There are $48M^3$ chamber sites in the cube. The triangle inequality for the collected bilinear therefore gives eq. 5.xcvi.

Under the smallness condition, $|e^{iqZ(b)} - 1| \le 2q|Z(b)| \le q^{1-\varepsilon/2}$. For $w = h_0/2$, $w_1(w_1+w_2) = \frac{1}{4}q^{-1/2}(1+q^{\varepsilon}) \le \frac{1}{2}q^{-1/2}$. Inserting these bounds proves eq. 5.xcvii. □

For $Z = \mathscr{Z}$ and the evaluation at $\Psi^f, \bar{\Psi}^f$, the first term of eq. 5.xcii is the image of the summed vertex. Under the additional charge condition $96M^3 q^{1/4} \le 1$, its bound becomes $\|\mathcal{V}_0(\mathscr{Z}, \square)\|_{h_0/2} \le q^{1/4-5\varepsilon}$ whenever the displayed bond smallness condition holds. These statements apply before any expansion of the finite integral.

The first-step density identity retains every vacuum and determinant factor:

$$\begin{aligned}(\mathbf{T}_0^x \mathscr{F}_0^x[\varepsilon^{\mathrm{v}}, m, 0])(A', \mathscr{C}'; \Psi') \\ = \det D_1(0)^{-1} \exp\Big\{ &- \langle \bar{\Psi}', D_1(\mathscr{T}')\Psi'\rangle_1 - Lm\langle \bar{\psi}', \psi'\rangle \\ &- L^3(\varepsilon^{\mathrm{v}} + \delta\varepsilon_0^0)n_1 + \sum_X \mathfrak{P}_0(X, \mathscr{C})\Big\} \\ \times \frac{\det F_0(\mathscr{T})}{\det F_0(0)} &\int \mathrm{d}\mu_I(W)\, \mathrm{d}\widehat{\mu}_0(\zeta)\, e^{V_0^x}.\end{aligned} \tag{5.xcviii}$$

The two determinants of F_0 have size $192n_0$. The original level-zero density, integrated before the block transformation, is

$$\int \mathrm{d}\Psi\, \mathscr{F}_0^x[\varepsilon^{\mathrm{v}}, m, 0](A, C; \Psi) = e^{-\varepsilon^{\mathrm{v}} n_0 + \sum_X \mathfrak{P}_0(X,C)} \frac{\det(D_{A+C} + \mu + m)}{\det D_0(0)}. \tag{5.xcix}$$

This follows directly from the weighted determinant identity. The numerator need not be invertible; the nonzero denominator uses $\mu > 0$.

5.3. **The nonlinear small-field step.**

5.3.1. *Contour domains and localization.* We construct the fluctuation correction in two stages. First, finite coordinate differences expand the dependence on weakening parameters between zero and one. For the families used here, a parameter in a component disconnected from the original polymer does not affect the term. Pairing its zero and one values therefore cancels that finite difference. Grouping the surviving terms by their connected union gives local activities without changing the sum when all parameters are one. A convergent connected expansion then recovers the local exponent. Keep the admissible sides, mass and charge of §5.2, and assume in addition $\ell \ge Mr_k$. Put $e = e_k$, $h = (h_{k,1}, h_{k,2})$. We use the weakened photon and fermion kernel families on the polydisc

$$\mathrm{Pd}_k = \{s = (s', s'') : |s_{\mathrm{pc}}| \le M^{\alpha_0}\}.$$

Here $s' = (s'_M, s'_{LM})$ contains the photon reconstruction and square-root parameters, of kinds π_M, π_{LM} in eq. 3.xliv; s'' contains the fermion M-cube parameters π_k of §4.2.1. We assume their analytic kernel and Hölder estimates, component restriction, the row-vanishing properties stated for each family, and coefficient support with common constants $(\alpha_0, \gamma_w, \kappa_w; C_w)$, where $\kappa_w = \kappa + \kappa_0 + 3$. The photon hypotheses are those of Prop. 3.14, eq. 3.xlix and the component restriction following them. The fermion hypotheses are those of Props. 4.8 and 4.10, including eq. 4.lv, eq. 4.lvi, and their covariance identities. At parameter value one all families equal the exact operators. Their gauge, lattice, transpose and reality actions are the actions of those exact operators with the parameters transformed accordingly. At level zero we retain the stated diagonal families and the chamber covariance with measure 1/48.

The fixed parameters satisfy eq. 5.lxi, eq. 5.xxxv and eq. 5.xl, $M > 4L$, $L \geq 4^8$, $0 < 1-\alpha < \varepsilon \leq 1/200$, $C_w, c_\gamma \geq 1$, and $C_w \geq 4\exp(2\gamma_w)M^{27\alpha_0}$. The parameter radius satisfies $M^{\alpha_0} - 1 \geq \exp(N_c\kappa + \log(6c_X))$, where $N_c = (2L+1)^3$. For later use put $c_S = 10^5 c_X^5 K_{\mathrm{pol}}^5 \exp(6\kappa)$. The running charge lies in the fixed range on which

$$\begin{aligned}
&p_{0,k} \leq e^{-\varepsilon/4}, \qquad 12\exp(-p_{0,k}^2/2) \leq e^7, \qquad 2e^{5\varepsilon/2} \leq 1, \qquad c_{\mathrm{fl}} e^{\varepsilon} \leq 1, \qquad 3Me^{\varepsilon} \leq 1,\\
&2c_C C_w^2 c_\gamma^2 e^{\varepsilon/4} \leq 1, \qquad 16 c_X c_\gamma C_w C_T^2 M^3 e^{\varepsilon/2} \leq 1, \qquad 96 M^3 e^{1/4} \leq 1,\\
&32 C_w C_T e^{3\varepsilon/2} \leq 1, \qquad 88 M^3 e^{1/4-6\varepsilon} \leq 1, \qquad 112 C_w^2 c_\gamma e^{2\varepsilon} \leq 3,\\
&c_S e^{1/4-5\varepsilon} \leq 1, \qquad c_S L^3 e^{\varepsilon} \leq 1, \qquad e^{\varepsilon} \leq \tfrac{1}{4}, \qquad 8 c_{\mathrm{rb}} L^3 C_d M^3 e^{2\varepsilon} \leq 1.
\end{aligned} \tag{5.c}$$

The density identity also uses the photon domain-preservation conditions of eq. 3.l and the determinant conditions of eq. 4.lxxviii. All are imposed by the fixed choice of the charge ceiling.

For $\zeta \in \mathrm{Box}_k$, write

$$Z(s') = H_k^{\mathrm{ph}}(s') C C_k^{1/2}(s')\zeta. \tag{5.ci}$$

In this formula C is the real photon coordinate map of eq. 3.xxvii. The diagonal photon family is used at $k = 0$. For a bond field Z set

$$|Z|_* = \max\{\|Z\|_\infty, \|\partial Z\|_\infty, \|\delta_\alpha \partial Z\|_\infty\}.$$

The three thresholds defining R_k are $\theta_i = e^{-3/4+i\varepsilon}$, $i = 1,2,3$. Thus $\widetilde{R}_k^-$ consists of the permitted real-gauge part plus a field satisfying the three strict bounds $(1-e^{\varepsilon})\theta_i$. Introduce the radii

$$r_1 = \tfrac{1}{2} e^{-3/4+9\varepsilon/2}, \qquad r_2 = e^{-3/4+7\varepsilon}, \qquad r_4 = e^{-1/4+3\varepsilon}. \tag{5.cii}$$

Proposition 5.10. For every $s \in \mathrm{Pd}_k$ and $\zeta \in \mathrm{Box}_k$,

$$|Z(s')|_* \leq \tfrac{1}{2} e^{-\varepsilon/2}.$$

The radii satisfy $2 \leq r_2, r_4 \leq r_1$ and $(r_i - 1)^{-1} \leq 2/r_i$, $i = 2,4$. If $A \in \widetilde{R}_k^-$, $\mathscr{C} \in \mathfrak{X}_k$, and $|Z|_* \leq \frac{1}{2}e^{-\varepsilon/2}$, then

$$A + tZ \in \widetilde{R}_k, \qquad A + \mathscr{C} + tZ \in \widetilde{\mathcal{G}}_k \quad (|t| \leq r_1).$$

■

Proof. The square-root kernel is bounded by $C_w \exp(-\gamma_w d(\Upsilon, \Upsilon'))$. The summability constant c_γ and the coordinate bound $|\zeta(\Upsilon')| \leq p_{0,k} \leq e^{-\varepsilon/4}$ give

$$\sup_{\Upsilon} |(C_k^{1/2}(s')\zeta)(\Upsilon)| \leq C_w c_\gamma e^{-\varepsilon/4}.$$

The coordinate map is real. Its bound $\|Cv\|_\infty \leq c_C \sup_\Upsilon |v(\Upsilon)|$ also gives the absolute row bound $\sum_\Upsilon |C(b,\Upsilon)| \leq c_C$: apply the real bound to the signs of the entries of a fixed row. Therefore

$$\|C C_k^{1/2}(s')\zeta\|_\infty \leq c_C C_w c_\gamma e^{-\varepsilon/4}.$$

Each of the reconstruction, derivative and Hölder derivative kernels has the same summable exponential bound. Consequently

$$|Z(s')|_* \leq c_C C_w^2 c_\gamma^2 e^{-\varepsilon/4} \leq \tfrac{1}{2} e^{-\varepsilon/2},$$

where the last inequality is $2c_C C_w^2 c_\gamma^2 e^{\varepsilon/4} \leq 1$. At level zero the polynomial diagonal family has the same row bounds: its constant is $4\exp(2\gamma_w)M^{27\alpha_0} \leq C_w$.

Since $\varepsilon \leq 1/200$,

$$\frac{r_2}{r_1} = 2e^{5\varepsilon/2} \leq 1, \qquad \frac{r_4}{r_1} = 2e^{1/2-3\varepsilon/2} \leq 2e^{5\varepsilon/2} \leq 1.$$

The small-charge condition $88M^3 e^{1/4-6\varepsilon} \leq 1$ gives $r_4^{-1} = e^{1/4-3\varepsilon} \leq 1/88$ and $r_2^{-1} \leq r_4^{-1}$. In particular both radii are at least 2, and their stated inverse bounds follow.

For $|t| \le r_1$ the three norms of tZ are at most $\frac{1}{4}e^{-3/4+4\varepsilon}$, strictly less than $e^{\varepsilon}\theta_3$. Adding this field fills less than the margin in each of the three defining bounds of $\widetilde{R}_k^-$. The permitted gauge part is unchanged. Hence $A + tZ \in \widetilde{R}_k$. Finally $\widetilde{R}_k + \mathfrak{X}_k \subset \widetilde{\mathcal{G}}_k$ gives the second inclusion. □

The first-step vertex estimate eq. 5.xcvii now applies to $Z(s')$ for every weakening parameter. In particular, $96M^3e^{1/4} \le 1$ gives its bound $e^{1/4-5\varepsilon}$ in the $h/2$ norm.

Let $B = A + \mathscr{C}$. The arguments of $H_k^{\#}$ are independent pairs $(\Phi, \bar{\Phi})$; in $H_k^{\#}\Phi$ the single symbol denotes this pair. Write $\psi^{\#} = (\psi, \chi; \bar{\psi}, \bar{\chi})$ for the ordinary, pair, barred ordinary and barred pair generators, respectively. Then

$$\begin{aligned} H_k^{\#}(s'', B)\Phi = (&H_k(s'', B)\Phi, \delta_{\alpha,B}H_k(s'', B)\Phi; \\ &\bar{H}_k(s'', B)\bar{\Phi}, \delta_{\alpha,-B}\bar{H}_k(s'', B)\bar{\Phi}), \\ \bar{H}_k(s'', B) &= \mathsf{C}^{-1}H_k(s'', -B)\mathsf{C}. \end{aligned}$$

Define $J_k^{\#}(s'', B, Z) = H_k^{\#}(s'', B + Z) - H_k^{\#}(s'', B)$ and

$$\widetilde{\Gamma}_k(s'', B)(W, \bar{W}) = (\Gamma_k(s'', B)W, \bar{W}), \qquad W_k^{\#}(s'', B) = H_k^{\#}(s'', B)(\Gamma_k(s'', B)W, \bar{W}).$$

The notation $J_k, \bar{J}_k$ denotes the components without Hölder quotient; $\mathrm{Wc}, \overline{\mathrm{Wc}}$ denotes the corresponding components of $W_k^{\#}$. The auxiliary generators W have weight $\mathsf{w}_W = 1$ for $k \ge 1$ and $\sqrt{48}$ for $k = 0$.

These fields are continuous in all closed polydisc and box variables and analytic in the open parameter polydisc and the connection variables. For the Hölder rows this follows directly from

$$(\delta_{\alpha,B}H_k)((x, x'), u) = d_X(x, x')^{-\alpha}(U_B(x, x')H_k(x', u) - H_k(x, u)).$$

The transport is entire in B. Prop. 5.10 places every connection on the closed contours inside the open domains of the kernel families. Thus compositions at $B + tZ(s')$ have the stated regularity also when the contour variable is on its circle.

In the following localization, cubes and polymers are sets of photon sites. Containment refers to all their lattice sites, and two cubes meet when they share a lattice site. Let $P_0(X)$ be the parameter cubes contained in X. For P outside $P_0(X)$ let $Y'(X, P)$ be the union of the M-cubes meeting X or a cube of $P_0(X) \cup P$. For a continuous analytic family $G(s, X)$ define

$$\mathcal{T}_P G(X) = \int_{[0,1]^P} \mathrm{d}s_P\, \partial_P G(s, X)\big|_{s=1 \text{ on } P_0(X),\, s=0 \text{ outside } P_0(X)\cup P}, \tag{5.ciii}$$

$$\mathrm{Loc}\, G(Y) = \sum_{(X,P):Y'(X,P)=Y} \mathcal{T}_P G(X). \tag{5.civ}$$

Repeated one-variable integration gives the useful identity

$$\mathcal{T}_P G(X) = \sum_{Q \subseteq P} (-1)^{|P|-|Q|} G(\mathbf{1}_{P_0(X)\cup Q}, X). \tag{5.cv}$$

In particular, coefficient locality can be checked at indicator families; no statement about the convergence of the later logarithm is used here.

Lemma 5.11. Suppose $\|G(s, X)\| \le B_0 \exp(-\kappa' d_M(X))$ on the parameter polydisc, where $\kappa_0 \le \kappa' \le \kappa$. Assume $\mathcal{T}_P G(X) = 0$ whenever $Y'(X, P)$ is disconnected. Then

$$\sum_Y \mathrm{Loc}\, G(Y) = \sum_X G(1, X), \tag{5.cvi}$$

$$\|\mathrm{Loc}\, G(Y)\| \le 4c_X K_{\mathrm{pol}} B_0 \exp\big(-(\kappa' - \kappa_0 - 1)d_M(Y)\big). \tag{5.cvii}$$

■

Proof. Apply the fundamental theorem of calculus in each parameter outside $P_0(X)$. It gives $G(1,X) = \sum_P \mathcal{T}_P G(X)$. The sums are finite. After removing the vanishing disconnected terms, each term belongs to exactly one polymer $Y'(X,P)$. This proves eq. 5.cvi.

Put $R = M^{\alpha_0}$. Cauchy's estimate on a polydisc of radius less than $R-1$ about each integration point, followed by the limit of these radii to $R-1$, gives

$$\|\mathcal{T}_P G(X)\| \le B_0 \exp(-\kappa' d_M(X))(R-1)^{-|P|}.$$

If $Y'(X,P) = Y$, then $X \subseteq Y$ and $|Y \setminus X|_M \le N_c|P|$, since every added M-cube meets a parameter cube and each parameter cube meets at most $N_c = (2L+1)^3$ such cubes. To compare their tree lengths, order the cubes of $Y \setminus X$ by graph distance from X. Each added cube is adjacent to a preceding one. Extend a net by a segment from a point of that preceding closed cube to their common face, of length at most M. Starting from an arbitrarily near-minimal net for X gives $d_M(Y) \le d_M(X) + |Y \setminus X|_M$. Conversely every net for Y is a net for X. Thus $0 \le d_M(Y) - d_M(X) \le N_c|P|$. Writing $a_X = \log(6c_X)$, the condition $R-1 \ge \exp(N_c\kappa + a_X)$ implies

$$\|\mathcal{T}_P G(X)\| \le B_0 \exp(-(\kappa' - \kappa_0) d_M(Y)) \exp(-\kappa_0 d_M(X)) \exp(-a_X|P|).$$

At most $3|Y|_M$ parameter cubes are contained in Y. Hence

$$\sum_P \exp(-a_X|P|) \le (1 + (6c_X)^{-1})^{3|Y|_M} \le \exp((d_M(Y)+1)/2).$$

The polymer counting and size estimates give

$$\sum_{X \subseteq Y} \exp(-\kappa_0 d_M(X)) \le K_{\mathrm{pol}}|Y|_M \le 2c_X K_{\mathrm{pol}} \exp(d_M(Y)/2).$$

The product of the last two bounds is at most $4c_X K_{\mathrm{pol}} \exp(d_M(Y))$. Summing the estimate for $\mathcal{T}_P G(X)$ proves eq. 5.cvii. □

5.3.2. *The two finite localizations.* At $k \ge 1$, the two localizations follow the order of the Gaussian substitutions. We first vary the photon background and the reconstructed mean field, while retaining the auxiliary coarse fermion generators. The local reconstruction formula then eliminates those generators in favor of ordinary and pair generators. Only after this substitution do we add the fermion fluctuation. Each change is a difference between amplitudes one and zero, which we express by a Cauchy integral.

For a function analytic near the closed disk $|t| \le r$, $r > 1$, write

$$\mathcal{C}_r f = \frac{1}{2\pi i} \int_{|t|=r} \frac{f(t)}{t(t-1)}\, \mathrm{d}t.$$

This is $f(1) - f(0)$ and satisfies $\|\mathcal{C}_r f\| \le (r-1)^{-1} \sup_{|t|=r} \|f(t)\|$. Let $(m, \mathscr{E}) \in \mathrm{Ball}_k^x(4)$ and set $m' = m + m(\mathscr{E})$, $\mathscr{E}^r = R\mathscr{E}$. The first localization separates the changes in the kinetic term, the extracted activity and the mass form caused by the photon fluctuation. For $k \ge 1$ define

$$G_1(s,X) = \mathcal{C}_{r_1} \langle \bar\Psi, M_k(X, B + tZ(s'))\Psi \rangle_1, \tag{5.cviii}$$

$$G_2(s,X) = \mathcal{C}_{r_2}\, \mathscr{E}^r(X; A + tZ(s'), \mathscr{C}, \psi^{\#} + tJ_k^{\#}(s'', B, Z(s'))\Psi), \tag{5.cix}$$

$$G_3(s,\square) = \mathcal{C}_{r_2}(-m') \langle \bar\psi + t\bar J_k(s'', B, Z(s'))\bar\Psi, \psi + tJ_k(s'', B, Z(s'))\Psi \rangle_\square. \tag{5.cx}$$

For $k \ge 1$, the kernels $M_k(X,\cdot)$ are the local terms of $b_k - D_k$. A function defined for one cube is zero on the other polymers. Put

$$E_i^*(Y) = \mathrm{Loc}\, G_i(Y), \qquad E_i(Y) = \mathrm{sub}_{Y,B}\, E_i^*(Y), \qquad E_{\le 3} = \sum_{i=1}^3 E_i.$$

The substitution is eq. 5.lxxxix. At $k = 0$ there is no Ψ and no substitution: $G_1(s,\square) = \mathcal{V}_0(Z(s'),\square)$, $G_2 = G_3 = 0$, and $E_{\le 3} = \mathrm{Loc}\, G_1$. All parameter kinds occur in this first localization.

The second localization adds the fermion fluctuation after the reconstruction substitution. It uses only the fermion parameters π_k. Let $v = W_k^{\#}(s'', B)$ and define

$$G^W_{\le 3}(s'', X) = E_{\le 3}(X; A, \mathscr{C}, \zeta, \psi^{\#} + v), \tag{5.cxi}$$

$$G_4(s'', X) = \mathcal{C}_{r_4}\mathscr{E}^r(X; A, \mathscr{C}, \psi^{\#} + tv), \tag{5.cxii}$$

$$G_5(s'', \square) = \mathcal{C}_{r_4}(-m')\langle \bar\psi + t\overline{\mathrm{Wc}}, \psi + t\mathrm{Wc}\rangle_\square, \tag{5.cxiii}$$

$$F(Y) = \mathrm{Loc}(G^W_{\le 3} + G_4 + G_5)(Y). \tag{5.cxiv}$$

At $k = 0$ the term G_4 is zero and $m' = m$.

Proposition 5.12. All finite expressions just defined exist on $\widetilde{R}_k^- \times \mathfrak{X}_k \times \mathrm{Box}_k$. The element $F(Y)$ is even, has ordinary legs over Y and pair legs admissible for Y, and has fluctuation generators only at unit indices in Y. Its coefficients depend on $A, \mathscr{C}$ only on bonds meeting Y and on ζ only through the coordinates $\zeta(\Upsilon)$ with $\mathrm{supp}\,\Upsilon \subset Y$. They are continuous in these arguments and analytic in the connection variables. The same assertions hold for $E_{\le 3}(Y)$, which has no fluctuation generators.

The disconnected terms in both localizations vanish, and

$$\sum_Y E_i^*(Y) = \sum_X G_i(1, X), \qquad \sum_Y F(Y) = \sum_X (G^W_{\le 3} + G_4 + G_5)(1, X). \tag{5.cxv}$$

The finite expressions on Y depend on the datum only through m' and the $\mathscr{E}^r(X)$ with $X \subseteq Y$. ■

Proof. We give the component argument which is responsible for locality. Let $\mathrm{Ps} = P_0(X) \cup P$, let $Y' = Y'(X, P)$ and let Y_1' be its component containing X. Set the weakening parameters to zero outside Ps. For each photon parameter kind, take components linked by shared lattice sites; for the fermion kind π_k, take face-connected components of the active partition M-cubes. In either case, a component Lnk of Ps is contained in the polymer formed by the M-cubes meeting it. This polymer lies in one component of Y'. If a site of Y_1' lies in Lnk, or in the enlarged parameter cube of a photon kind, that polymer has a cube in common with Y_1', and hence lies entirely in Y_1'.

The row-vanishing property of the weakened photon families and their component restriction give, on any fine bond b meeting Y_1',

$$Z(s')(b) = \sum_{b', \Upsilon, \Upsilon' \subset Y_1'} H_k^{\mathrm{ph}}(s')(b, b') C(b', \Upsilon) C_k^{1/2}(s')(\Upsilon, \Upsilon')\zeta(\Upsilon'). \tag{5.cxvi}$$

Indeed, the reconstruction row is zero unless b lies in one link-component; its columns lie in that component. The coordinate map has columns supported in the L-blocks of the ends of b', and each block lies in the M-cube of the corresponding end. The square-root row then has columns in its own link-component. Each component involved is attached to Y_1' by the preceding argument. These rows are unchanged if the parameters outside Y_1' are changed while their common zero set outside Ps is kept.

For a chamber site $x = (c, g)$ at distance at most 1 from Y_1', the fermion row is zero unless every parameter cube meeting K_c is present. These partition cubes form a face-connected set. When the row is nonzero, their active face-connected component contains c and a site of Y_1', so it lies in Y_1'. The row of $H_k(s'', B)$, and its barred counterpart, then has columns in that component and depends on B only on bonds meeting it. At a column of this row, the corresponding row of $\Gamma_k(s'', B)$ has columns in the same component. These conclusions also hold at level zero: the diagonal factor is

$$\prod_{\mathrm{pc} \in \pi_0 : \mathrm{pc} \cap K_c \neq \varnothing} s''_{\mathrm{pc}},$$

and the kernel of that multiplication operator is $48\mathbf{1}_{x=x'}$ times this factor. Its composition with Γ_0 uses the measure $1/48$.

An endpoint of a pair admissible for X is at distance at most $\eta_k \le 1$ from X. Its transport depends only on bonds contained in X, or the one type-3 bond meeting X. Thus the preceding row

argument applies to every row used in $J_k^{\#}$ and $W_k^{\#}$. The Hölder row has only the two columns x, x' and the entire transport coefficient displayed above. Finite compositions preserve this dependence. The columns of O_2O_1 are contained in the union of the columns of O_1 at the finitely many columns of O_2. Its connection dependence is confined to the union of the bond sets for the contributing rows of both factors. Entries of the transport and reconstruction rows are entire in the connections, and finite compositions of these algebraic rows retain this property. Compositions involving the weakened kernel families are holomorphic on the domains stated above. These are the finite-row properties used here; the weakened operators themselves are not asserted to have finite range.

For G_1 at $k \geq 1$, the kernel $M_k(X, \cdot)$ has both indices in X and depends on the connection only on bonds meeting X. For G_2, G_3, the incoming activity or mass form has ordinary legs in X and admissible pair legs; the substitution by $J_k^{\#}\Psi$ introduces Ψ only in Y_1'. At $k = 0$ the vertex has precisely these ordinary and type-3 pair legs and depends on Z only on bonds meeting its cube. eq. 5.cxvi therefore proves the same component dependence for every first-localization function.

If Y' is disconnected, choose an M-cube outside Y_1'. It cannot meet X or a parameter cube contained in X, so it meets a cube of P outside Y_1'. The integrand is independent of that parameter. Pairing the subsets in eq. 5.cv according to whether they contain this cube proves that the term is zero. For a term assigned to a connected Y, all parameter cubes in $P_0(X) \cup P$ and X lie in Y. eq. 5.cv now gives the claimed locality of E_i^*.

The reconstruction substitution adds only ordinary legs over Y and boundary pairs $(x, x^{(3)})$ admissible for Y. Its finite rows at a unit site y depend on the connection only on bonds contained in Δ_y or type-3 bonds from this block. These bonds meet Y since $\Delta_y \subset Y$. The barred rows have the same support. Thus it preserves the claimed coefficient locality and removes Ψ. For the second localization, the row argument for $H_k^{\#}\widetilde{\Gamma}_k W$ gives fluctuation generators only in Y_1'. It applies to each of the three functions in eq. 5.cxiv. The same disconnected parameter cancellation proves the second assertion. Lem. 5.11 gives both sum identities.

The contour domains, regularity of the kernel families and finite operations give continuity and analyticity; contour and parameter integrals preserve them by locally uniform Riemann sums. Substituting odd linear fields preserves evenness. Finally G_1 is fixed, G_2, G_4 use $\mathscr{E}^r(X)$ and G_3, G_5 use m'; the localizations involve only $X \subseteq Y$. □

For the next comparison, polymers are incompatible when their union is connected through faces; a polymer is incompatible with itself. The integrated Mayer activities $H(Y)$ and the connected-logarithm activities $E^{\#}(Y)$ are those of eq. 5.cxxvi and eq. 5.cxxvii. Their definitions and convergence estimates are given in §5.3.4 below.

Lemma 5.13. For two constructions satisfying Prop. 5.12, identify the generators and their weights over a connected U by an isometric algebra map preserving bar status and the pairings of ordered products for the identity covariance. Suppose it preserves local subpolymers, parameter cubes, P_0, Y', reconstruction memberships and incompatibility. Use the same m', contour radii and normalized one-coordinate fluctuation laws under this identification. If their local incoming activities and the kernel entries used in that proposition's proof of locality agree, then their finite expressions, integrated Mayer activities, and absolutely convergent connected logarithms agree. ■

Proof. The entries required for this comparison are the restricted photon composition eq. 5.cxvi; the rows of $H_k, \bar{H}_k, \Gamma_k$ used by ordinary fields and admissible pair endpoints, with columns in U; the local $M_k(X, \cdot)$ for $X \subseteq U$; and the finite reconstruction and pair-difference rows. Parameters vanish outside U, and equality is required at all admitted contour connections. At $k = 0$ use the first-step vertex and diagonal mean field, omit M_0, Ψ, and retain the chamber weight $1/48$.

The restricted composition first identifies Z on every bond on which the remaining rows depend. The remaining rows then identify $J_k^{\#}$ and $W_k^{\#}$ as polynomials in common generators. Substitution commutes with the algebra map because it agrees on each odd generator, so the mass form and

incoming activity give identical contour integrands and integrals. Each localization term is the same mixed derivative integrated over $[0,1]^P$, with $s=1$ on $P_0(X)$, $s=u$ on P, and $s=0$ elsewhere. The correspondence preserves this prescription and $Y'(X,P)$. Since $M^{\alpha_0}>1$, coefficient differentiation and integration occur inside the analytic polydisc and commute with the identification. This proves equality through both localizations and the intervening reconstruction.

Wick moments use the same index weights and ordering signs, including the 48 factors at zero; unused modes integrate to one. The photon product marginals also agree. Thus the finite Mayer products and sums agree. Their Ursell coefficients depend only on the common incompatibility graphs. Absolute weighted convergence and the isometry pass equality to the connected logarithms. □

5.3.3. *Bounds before fluctuation integration.* We reserve part of the Grassmann weight for each substitution: the estimates pass from h to $h/2$ and then to $h/4$. Set

$$\kappa_w=\kappa+\kappa_0+3,\qquad a_1=\kappa-\kappa_0-1,\qquad a_2=\kappa-2\kappa_0-2,$$
$$C_W=28C_w^2c_\gamma,\qquad H_{\mathrm{cl}}=36c_X^2K_{\mathrm{pol}}^2e^{1/4-5\varepsilon}.$$

The notation H_{cl} is a scalar cluster bound. The bounds for the kernel families imply

$$\|\langle\bar\Psi,M_k(X,B)\Psi\rangle_1\|_{w_3}\le 4c_Xc_\gamma C_wM^3w_3^2\exp(-(\kappa_w-1)d_M(X)),\quad k\ge1, \tag{5.cxvii}$$
$$|H_k^{\#}(s'',B)|_{1,\infty}\le 4C_w, \tag{5.cxviii}$$
$$|J_k^{\#}(s'',B,Z)|_{1,\infty}\le 16C_we^{3/4-9\varepsilon/2}, \tag{5.cxix}$$
$$\mathsf{w}_W|H_k^{\#}(s'',B)\widetilde\Gamma_k(s'',B)|_{1,\infty}\le C_W. \tag{5.cxx}$$

Here $B=A+\mathscr{C}$ and $|Z|_*\le\frac12e^{-\varepsilon/2}$. To see the first bound, count $4M^3|X|_M$ rows and sum the exponential kernel in its second index, giving $c_\gamma C_w\exp(-\kappa_wd_M(X))$ per row. Then $|X|_M\le c_X\exp(d_M(X))$ gives eq. 5.cxvii. For $H_k^{\#}$, Prop. 4.10 bounds the ordinary and short-pair rows by C_w. A long pair, $d_X\ge1$, has its Hölder quotient bounded by four times the ordinary sup norm. At level zero every pair is long. The monomial matrices $\mathsf{C}^{\pm1}$ preserve row norms, so the barred rows have the same bounds. Apply $\mathcal{C}_{r_1}$ to $H_k^{\#}(s'',B+tZ)$ to obtain the bound for $J_k^{\#}$. Finally $|\Gamma_k|_{1,\infty}\le C_wc_\gamma$ by kernel summation, using $1/48$ at level zero. Submultiplicativity and $\mathsf{w}_W\le7$ give eq. 5.cxx.

Proposition 5.14. For the data of Prop. 5.12,

$$\|E_{\le3}(Y)\|_{h/2}\le 8c_XK_{\mathrm{pol}}e^{1/4-5\varepsilon}\exp(-a_1d_M(Y)), \tag{5.cxxi}$$
$$\|F(Y)\|_{(h/4;\mathsf{w}_W)}\le H_{\mathrm{cl}}\exp(-a_2d_M(Y)). \tag{5.cxxii}$$

The element $F(Y)$ is the sum of a term independent of $(m,\mathscr{E})$ and a term complex linear in these data. ■

Proof. Extraction gives

$$\|\mathscr{E}^r(X)\|_h\le2\|\mathscr{E}\|_k^x\exp(-\kappa d_M(X))\le12e^{1/4-7\varepsilon}\exp(-\kappa d_M(X)),\qquad|m'|\le8e^{3/4-8\varepsilon}.$$

The mass bound uses $c_{\mathfrak{fl}}e^{\varepsilon}\le1$. With chamber measure $\eta_k^3/48$, direct counting gives

$$\|\langle\bar\psi,\psi\rangle_\square\|_{(w_1,w_2)}=4M^3w_1^2.$$

The number of monomials is $4\cdot48\eta_k^{-3}M^3$ and their common coefficient is $\eta_k^3/48$.

In the first localization give Ψ the weight $w_3=C_Th_{k,1}$ and the old generators the weights $h/2$. The contour bound and eq. 5.cxvii give

$$\|G_1(s,X)\|\le16c_Xc_\gamma C_wC_T^2M^3e^{1/4-9\varepsilon/2}\exp(-(\kappa_w-1)d_M(X))\le e^{1/4-5\varepsilon}\exp(-\kappa d_M(X)).$$

The last step uses

$$16c_Xc_\gamma C_wC_T^2M^3e^{\varepsilon/2}\le1.$$

At $k = 0$ the weighted vertex instead gives $12M^3 e^{1/2-\varepsilon/2} \le e^{1/4-5\varepsilon}$. Moreover,

$$r_2 |J_k^{\#}|_{1,\infty} w_3 \le 16 C_w C_T e^{3\varepsilon/2} h_{k,2} \le \tfrac{1}{2} h_{k,2}.$$

The substitution bound therefore takes the integrand norm of G_2 to the h norm of $\mathscr{E}^r$, and that of G_3 to the h norm of the mass form. It gives

$$\|G_2(s, X)\| \le 24 e^{1-14\varepsilon} \exp(-\kappa d_M(X)), \qquad \|G_3(s, \square)\| \le 64 M^3 e^{1-15\varepsilon}.$$

The condition $88M^3 e^{1/4-6\varepsilon} \le 1$ implies

$$24 e^{3/4-9\varepsilon} + 64 M^3 e^{3/4-10\varepsilon} \le 1.$$

Thus $\sum_{i\le 3} G_i$ is bounded by $2e^{1/4-5\varepsilon} \exp(-\kappa d_M(X))$. Apply Lem. 5.11, using the disconnected cancellation in Prop. 5.12. The reconstruction substitution costs no norm, since $C_T(h_{k,1} + h_{k,2})/2 \le w_3$. This proves eq. 5.cxxi.

For the second localization, the condition $112 C_w^2 c_\gamma e^{2\varepsilon} \le 3$ gives

$$r_4 C_W \le \tfrac{3}{4} h_{k,2}, \qquad C_W \le \tfrac{1}{4} h_{k,2}.$$

Consequently addition of $W_k^{\#}$ maps the $h/4$ norm to the $h/2$ norm at $t = 1$ and to the h norm on the contour $|t| = r_4$. The three input bounds are

$$\begin{aligned} \|G_{\le 3}^W(s'', X)\| &\le 8 c_X K_{\mathrm{pol}} e^{1/4-5\varepsilon} \exp(-a_1 d_M(X)), \\ \|G_4(s'', X)\| &\le 24 e^{1/2-10\varepsilon} \exp(-\kappa d_M(X)), \\ \|G_5(s'', \square)\| &\le 64 M^3 e^{1/2-11\varepsilon}. \end{aligned}$$

Their sum is at most $9 c_X K_{\mathrm{pol}} e^{1/4-5\varepsilon} \exp(-a_1 d_M(X))$, again by $88M^3 e^{1/4-6\varepsilon} \le 1$. Since $\kappa_0 \le a_1 \le \kappa$, resummation gives eq. 5.cxxii. At level zero $G_4 = 0$; all the other inequalities include the factor w_W already, so they apply unchanged. Linearity follows from the definitions: G_1 is fixed, G_2, G_4 are linear in $R\mathscr{E}$, G_3, G_5 in m', and localization, the reconstruction substitution and the contour functionals are linear. □

5.3.4. *Connected polymers and the logarithm.* After localization, separated components use disjoint fluctuation coordinates, so their integrals factor. Call two polymers incompatible when their union is connected through faces; a polymer is incompatible with itself. Expanding the exponential gives a sum over compatible collections of connected polymers. Its logarithm retains the collections connected by incompatibility; these are the terms of the new local exponent. The connected-graph identity below expresses this cancellation, and the tree bound controls the surviving sums. For an ordered tuple $(X_1, \dots, X_n)$ define $u^T(X_1) = 1$ and

$$u^T(X_1, \dots, X_n) = \sum_{G \text{ connected}} (-1)^{|G|}, \quad n \ge 2,$$

where G ranges over spanning subgraphs of the incompatibility graph of the tuple. The coefficient is zero when that graph is disconnected.

We will use the formal connected-graph identity

$$\sum_{\{X_i\} \text{ compatible}} \prod_i z_{X_i} = \exp\left\{ \sum_{n\ge 1} \frac{1}{n!} \sum_{(X_1,\dots,X_n)} u^T(X_1, \dots, X_n) \prod_i z_{X_i} \right\} \tag{5.cxxiii}$$

and the tree bound obtained by Penrose's graph partition [19]:

$$|u^T(X_1, \dots, X_n)| \le \#\{\text{spanning trees of the incompatibility graph}\}. \tag{5.cxxiv}$$

Here the identity is one of formal series in commuting variables, before substitution into a normed algebra. For its proof, express the left side as an ordered sum with factor $1/n!$ and multiply by $\prod_{i<j}(1 - \mathbf{1}_{\{X_i, X_j\} \text{ incompatible}})$. Expanding this product sums over all graphs. A graph is uniquely

the union of its connected components. Grouping by their vertex sets gives the exponential on the right: an ordered list of component sizes $n_1, \dots, n_r$ has $n!/(n_1! \cdots n_r!)$ vertex assignments, and division by $r!$ removes the order of its components. Only finitely many terms contribute to a fixed degree.

For eq. 5.cxxiv, let $\mathbf{G}$ be the incompatibility graph and suppose it is connected. For a connected spanning subgraph G, root at 1 and, for $i \neq 1$, choose as its parent the least labelled neighbor at distance $d_G(i) - 1$ from the root. These edges form a tree $T(G)$ with the same root distances as G. Given a spanning tree τ, let $R(\tau)$ contain its edges, all edges of $\mathbf{G}$ within one distance level, and all edges from i to the preceding level whose other endpoint exceeds the parent of i in τ. Then

$$T(G) = \tau \quad \Longleftrightarrow \quad \tau \subseteq G \subseteq R(\tau).$$

In one direction the distances and least parents force every edge of G to belong to $R(\tau)$. Conversely, along any edge of $R(\tau)$ the tree distance changes by at most one. Thus a graph between τ and $R(\tau)$ has the tree's distances, and the least neighbor in the preceding level is its tree parent. The connected subgraphs are therefore partitioned into these intervals, and their signed sum is

$$\sum_{\tau} (-1)^{n-1} \sum_{A \subseteq R(\tau) \setminus \tau} (-1)^{|A|} = (-1)^{n-1} \#\{\tau : R(\tau) = \tau\}.$$

This proves the tree bound, including the cases of a disconnected graph and $n = 1$.

Proposition 5.15. Let $V(X)$ be even elements in the old and fluctuation generators, continuous on Box_k, with

$$\|V(X)\|_{(w;\mathsf{w}_W)} \leq H_{\mathrm{cl}} \exp(-a_2 d_M(X)).$$

Assume that $V(X)$ has fluctuation indices only in X and depends on ζ only through coordinates supported in X. For any union Y of M-cubes put

$$K(Y) = \sum_{\substack{\{X_i\} \text{ distinct} \\ \bigcup_i X_i = Y}} \prod_i (\exp V(X_i) - 1), \qquad K(\varnothing) = 1, \tag{5.cxxv}$$

$$H(Y) = \int \mathrm{d}\mu_I(W)\, \mathrm{d}\widehat{\mu}(\zeta)\, K(Y). \tag{5.cxxvi}$$

For a connected Y define

$$E^{\#}(Y) = \sum_{n \geq 1} \frac{1}{n!} \sum_{\bigcup_i Y_i = Y} u^T(Y_1, \dots, Y_n) \prod_i H(Y_i). \tag{5.cxxvii}$$

The last series converges absolutely in the w norm and

$$\int \mathrm{d}\mu_I(W)\, \mathrm{d}\widehat{\mu}(\zeta) \exp\left\{\sum_X V(X)\right\} = \exp\left\{\sum_Y E^{\#}(Y)\right\}, \tag{5.cxxviii}$$

$$\|E^{\#}(Y)\|_w \leq c'_S H_{\mathrm{cl}} \exp(-a_3 d_M(Y)). \tag{5.cxxix}$$

Here $a_3 = \kappa - 5\kappa_0 - 4$ and $c'_S = 14 c_X^2 K_{\mathrm{pol}}^2 \exp(3\kappa)$. The sum of the norms of the terms with $n > N$ in eq. 5.cxxvii is at most 2^{-N} times the right side of eq. 5.cxxix. ∎

Proof. Set

$$a_4 = \kappa - 3\kappa_0 - 3, \quad a_5 = \kappa - 3\kappa_0 - 2, \quad a_6 = \kappa - 5\kappa_0 - 3,$$
$$H_1 = 7 c_X K_{\mathrm{pol}} \exp(3a_5) H_{\mathrm{cl}}, \qquad z_S = 28 H_1 \exp(\kappa_0) K_{\mathrm{pol}} \exp(3a_6).$$

The fixed small-charge choice

$$c_S e^{1/4 - 5\varepsilon} \leq 1, \qquad c_S = 10^5 c_X^5 K_{\mathrm{pol}}^5 \exp(6\kappa)$$

implies

$$H_{\mathrm{cl}} \leq 1, \qquad 4 H_{\mathrm{cl}} K_{\mathrm{pol}} c_X \exp(3a_5) \leq 1, \qquad z_S \leq \tfrac{1}{2}. \tag{5.cxxx}$$

Indeed the second left side is $144c_X^3K_{\mathrm{pol}}^3\exp(3a_5)e^{1/4-5\varepsilon}$. For the third,

$$2z_S = 14112c_X^3K_{\mathrm{pol}}^4\exp(3a_5+3a_6+\kappa_0)e^{1/4-5\varepsilon},$$

and $3a_5+3a_6+\kappa_0 \le 6\kappa$.

Even elements commute, so expanding their exponential gives $\exp(\sum_X V(X)) = \sum_Y K(Y)$. Moreover $K(Y)$ factors over the connected components of Y: every polymer in its defining set belongs to exactly one component. If distinct polymers $X_1,\dots,X_n$ have connected union Y, their incompatibility graph is connected. Otherwise two nonempty groups would partition their union; connectedness of Y would give coincident or adjacent cubes in different groups and hence an edge between them. For each incompatibility-tree edge, choose coincident or face-adjacent cubes in the two polymers and points where their nets meet these cubes. The two points lie in a box of side lengths $M, M, 2M$, so they can be joined by a segment of length at most $\sqrt{6}M < 3M$. Joining near-minimal nets along the tree proves

$$d_M(Y) \le \sum_i d_M(X_i) + 3(n-1).$$

Using $\|\exp V - 1\| \le 2\|V\|$ and $a_2 = a_5+\kappa_0$ therefore bounds the product by

$$\exp(-a_5 d_M(Y))\bigl(2H_{\mathrm{cl}}\exp(3a_5)\bigr)^n \prod_i \exp(-\kappa_0 d_M(X_i)).$$

Replace distinct sets by ordered tuples with factor $1/n!$, and use the counting bound for subpolymers of Y. The result is

$$\|K(Y)\| \le \exp(-a_5 d_M(Y))x_Y\exp(x_Y), \qquad x_Y = 2H_{\mathrm{cl}}K_{\mathrm{pol}}\exp(3a_5)|Y|_M.$$

By eq. 5.cxxx,

$$x_Y \le (d_M(Y)+1)/2,$$
$$x_Y \le 4H_{\mathrm{cl}}K_{\mathrm{pol}}c_X\exp(3a_5)\exp(d_M(Y)/2).$$

Since $4\exp(1/2) \le 7$ and $a_5 - 1 = a_4$, we obtain

$$\|K(Y)\|_{(w;\mathsf{w}_W)} \le H_1\exp(-a_4 d_M(Y)) =: \mu(Y). \tag{5.cxxxi}$$

Weighted Berezin integration costs no norm here: each paired moment at level zero is $\pm 48^{|J|/2}$, which is exactly covered by the weight $(\sqrt{48})^{|J|}$ of its generators. The corresponding moment is ± 1 at positive levels. The photon measure is a probability measure on its compact box. Hence $\|H(Y)\|_w \le \mu(Y)$. These integrals produce even elements, since a nonzero Berezin moment uses an even number of generators. For compatible polymers the fluctuation modes and supported photon coordinates are disjoint. The two product measures therefore factor, with the external old generators left as coefficients. It follows that

$$\int \mathrm{d}\mu_I(W)\,\mathrm{d}\widehat{\mu}(\zeta)e^{\sum_X V(X)} = \sum_{\{Y_i\}\ \text{compatible}} \prod_i H(Y_i). \tag{5.cxxxii}$$

We now bound the infinite connected series. Let

$$S_n(Y) = \sum_{\bigcup_i Y_i = Y} |u^T(Y_1,\dots,Y_n)| \prod_i \mu(Y_i).$$

For $n \ge 2$, apply eq. 5.cxxiv. For every contributing tree the union estimate and $a_4 = a_6 + 2\kappa_0$ give

$$\prod_i \exp(-a_4 d_M(Y_i)) \le \exp(-a_6 d_M(Y))\exp(3a_6(n-1))\prod_i \exp(-2\kappa_0 d_M(Y_i)).$$

Drop the prescribed union, retaining $Y_i \subseteq Y$, and root the tree at 1. Write q_j for the number of children at j. A polymer incompatible with Y' contains one of at most $7|Y'|_M$ cubes consisting of its

cubes and their face neighbors. Since $|Z|_M \le c_X(d_M(Z)+1) \le \kappa_0(d_M(Z)+1)$ and $a^q \le q!\exp(a)$, the counting estimate gives

$$\sum_{Z \text{ incompatible with } Y'} \exp(-2\kappa_0 d_M(Z))|Z|_M^q \le 7q!\exp(\kappa_0)K_{\mathrm{pol}}|Y'|_M,$$

$$\sum_{Z\subseteq Y} \exp(-2\kappa_0 d_M(Z))|Z|_M^q \le q!\exp(\kappa_0)K_{\mathrm{pol}}|Y|_M.$$

Summing from leaves to root thus costs at most $(7\exp(\kappa_0)K_{\mathrm{pol}})^n|Y|_M\prod_j q_j!$. For prescribed children counts, Cayley's degree formula [9, Prop. 4] counts $(n-2)!/((q_1-1)!\prod_{j\ge 2}q_j!)$ trees, where $q_1 \ge 1$ and $\sum_j q_j = n-1$. Consequently

$$\sum_{\text{trees}}\prod_j q_j! = (n-2)!\sum_{(q_j)} q_1 \le (n-1)!\binom{2n-2}{n-1} \le (n-1)!4^{n-1}.$$

Collecting the factors, also for $n=1$, gives

$$\frac{S_n(Y)}{n!} \le 7H_1\exp(\kappa_0)K_{\mathrm{pol}}|Y|_M\exp(-a_6 d_M(Y))z_S^{n-1}. \tag{5.cxxxiii}$$

Since $z_S \le 1/2$, sum the geometric series and use $|Y|_M \le c_X\exp(d_M(Y))$ and $a_6 - 1 = a_3$. The bound becomes

$$98c_X^2K_{\mathrm{pol}}^2\exp(\kappa_0+3a_5)H_{\mathrm{cl}}\exp(-a_3 d_M(Y)) \le c_S'H_{\mathrm{cl}}\exp(-a_3 d_M(Y)),$$

because $\kappa_0 + 3a_5 = 3\kappa - 8\kappa_0 - 6$. Summing only $n > N$ gives the additional factor 2^{-N}. This proves absolute convergence and the claimed estimates.

It remains to justify the logarithm identity after substitution in eq. 5.cxxiii. The even subalgebra with the coefficient norm is a commutative Banach algebra, and the polymer set is finite. By eq. 5.cxxxiii, the connected norm sum is finite, and the norm sum of products in its exponential is at most its exponential. Absolute regrouping by the formal z_Y monomials is therefore valid. Their coefficients are those of eq. 5.cxxiii; substitute $z_Y = H(Y)$ to obtain eq. 5.cxxxii, then group by union to obtain eq. 5.cxxviii. □

There are also polynomial approximants to this logarithm which converge uniformly for all families satisfying the preceding bound, at fixed side. Replace $\exp V - 1$ by $p_N(V) = \sum_{1\le j\le N} V^j/j!$ in K, H, and stop the connected series at $n = N$; denote the result $E_N^{\#}$. It is a polynomial in integrals of products of $V(X_i)$, $X_i \subseteq Y$, and has the same bound eq. 5.cxxix. To verify uniform convergence, write $D = |\mathcal{P}_k|$. Since $\|V(X)\| \le 1$,

$$\|\exp V(X) - 1 - p_N(V(X))\| \le \frac{\exp(1)}{(N+1)!}.$$

Telescoping each finite product in K gives

$$\|H(Y) - H_N(Y)\| \le \frac{4^D D\exp(1)}{(N+1)!}.$$

Thus this difference is at most $\delta_N\mu(Y)$, where

$$\delta_N = \frac{4^D D\exp(1)}{(N+1)!\min_{Y\in\mathcal{P}_k}\mu(Y)} \longrightarrow 0.$$

Replacing the n factors in a connected term costs at most $n\delta_N\prod_i\mu(Y_i)$. Summing eq. 5.cxxxiii with this factor, using $\sum_{n\ge 1} n2^{-(n-1)} = 4$, and adding the tail yields

$$\sup_Y \|E^{\#}(Y) - E_N^{\#}(Y)\| \le 28\delta_N H_1\exp(\kappa_0)K_{\mathrm{pol}}(\ell/M)^3 + 2^{-N}c_S'H_{\mathrm{cl}}.$$

This convergence is uniform in the connections and the allowed data. The displayed error depends on the side; no volume-independent approximation is claimed.

5.3.5. *Symmetries of the local logarithm.* Apply Prop. 5.15 with $V = F$, $w = h/4$, and denote its output by $E_k^{\#}(m, \mathscr{E})$. Products on subpolymers of Y preserve the leg supports, local connection dependence, and dependence only on m' and $\mathscr{E}^r(X)$ for $X \subseteq Y$. The norm bound and uniform polynomial approximants give analytic connection coefficients. We next carry the symmetries through these finite operations, integration and the convergent logarithm.

Let $\mathbb{G}_k^L$ denote the chamber symmetry group with translations in $LM\mathbb{Z}^3$, so that it preserves every parameter partition. Its actions and the gauge, charge-conjugation and reality actions are those of §5.1.1.

Proposition 5.16. The local logarithm satisfies

$$a_{\tilde g} E_k^{\#}(\tilde g Y; \tilde g A, \tilde g \mathscr{C}) = E_k^{\#}(Y; A, \mathscr{C}), \qquad \tilde g \in \mathbb{G}_k^L, \tag{5.cxxxiv}$$

$$G_{\lambda+\lambda'} E_k^{\#}(Y; A^1, \mathscr{C}^1) = E_k^{\#}(Y; A, \mathscr{C}), \tag{5.cxxxv}$$

$$\mathrm{Cg}\, E_k^{\#}(Y; -A, -\mathscr{C}) = E_k^{\#}(Y; A, \mathscr{C}). \tag{5.cxxxvi}$$

In the gauge identity both pairs belong to the domain and $A^1 = A - d\lambda$, $\mathscr{C}^1 = \mathscr{C} - d\lambda'$ on all bonds at distance at most 1 from Y, with real λ, λ'. Here $d\lambda$ denotes the lattice gradient, as in the preceding gauge conventions. For real m and $\mathscr{E} \in K_k^x$ one also has

$$\mathbf{K} E_k^{\#}(Y; -\overline{A}, -\overline{\mathscr{C}}) = E_k^{\#}(Y; A, \mathscr{C}). \tag{5.cxxxvii}$$

Every monomial is balanced. ∎

Proof. The four actions are continuous ring automorphisms, linear except for $\mathbf{K}$. Equivariant substitutions commute with them on generators, hence on the algebra. Real finite differences eq. 5.cv also commute. Lattice actions permute parameter cubes and localization terms, preserving connectedness, incidences and Ursell coefficients.

The contour functional commutes with the linear actions. For the antilinear action, parametrize $t = r\exp(i\theta)$:

$$\mathcal{C}_r f = \frac{1}{2\pi} \int_0^{2\pi} \frac{f(r\exp(i\theta))}{r\exp(i\theta) - 1}\, \mathrm{d}\theta.$$

Conjugating coefficients and replacing θ by $-\theta$ leaves this contour functional unchanged at the transformed arguments. Wick pairings are real and invariant under $W \mapsto UW$, $\bar W \mapsto (U^{-1})^T \bar W$, so integration commutes with these linear or antilinear changes. This covers lattice and gauge actions, $\mathbf{K}$ with $U = \varkappa$, and charge extended as the identity on W; the last case needs the separate covariance argument below.

For a lattice symmetry, covariance of the kernel families gives

$$Z(\tilde g s')[\tilde g \zeta] = \tilde g Z(s')[\zeta], \qquad H_k(\tilde g s'', \tilde g B)\tilde g = \tilde g H_k(s'', B), \qquad \Gamma_k(\tilde g s'', \tilde g B)\tilde g = \tilde g \Gamma_k(s'', B).$$

The barred action follows from $(S^{-1})^T = \pm \mathsf{C}^{-1} S \mathsf{C}$. Transported endpoints give covariance of the Hölder quotients, hence of $J_k^{\#}\Psi, W_k^{\#}$. The relation $M_k(\tilde g X, \tilde g B) = \tilde g M_k(X, B)\tilde g^{-1}$, change of chamber/spin indices, incoming symmetry, and eq. 5.xc give invariance of the bilinear, mass, activity and reconstruction terms. The photon box measure is invariant. The preceding commutation rules through both localizations, integration and the connected series prove eq. 5.cxxxiv.

For the gauge identity set $B^1 = A^1 + \mathscr{C}^1$ and $\Lambda = \lambda + \lambda'$. In a summand assigned to Y, all parameters outside Y vanish. By the row-dependence argument in the proof of Prop. 5.12, B^1 may be replaced by $B - d\Lambda$ in every row used in that summand. The latter field is still in the operator domain, since real gauge shifts preserve it. Telescoping the gradient along each path gives

$$U_{B-d\Lambda}(x, x') = U_B(x, x') \exp\big(- ie(\Lambda(c') - \Lambda(c))\big).$$

The ordinary row of H_k obeys

$$H_k(s'', B^1)(x, \cdot) = \exp(ie\Lambda(c))H_k(s'', B)(x, \cdot)\exp(-ie\Lambda^u),$$

when it is nonzero, where Λ^u is the function on the unit indices. The barred row has the inverse phases. In the Hölder row the phase at its second point cancels that from the transport, leaving exactly $\exp(ie\Lambda(c))$, the phase of the first point. The row of Γ_k has the corresponding two-sided gauge covariance. Its composition with $H_k^{\#}$ thus gives the same covariance for $W_k^{\#}$, including the rows which vanish outside Y. The bilinear with kernel M_k and the local mass form are gauge invariant, and so is $R\mathscr{E}$ under the separate shifts of its two connection arguments. The reconstruction substitution also commutes with this local gauge action. The photon fields and photon measure are unchanged. The finite operations and integration therefore prove eq. 5.cxxxv.

For reality, the photon kernel families are real on real parameters. Their analytic dependence extends this identity to $\overline{O(\bar{s}')} = O(s')$ on the full polydisc. Since C is real,

$$Z(\bar{s}')[-\zeta] = -\overline{Z(s')[\zeta]}.$$

Transform $s \mapsto \bar{s}$, $t \mapsto \bar{t}$, $\zeta \mapsto -\zeta$ and $(A, \mathscr{C}) \mapsto (-\bar{A}, -\overline{\mathscr{C}})$. The fermion identities are

$$\overline{H_k(\bar{s}'', -\bar{B})} = \varkappa^{-1}H_k(s'', B)\varkappa, \qquad \overline{\Gamma_k(\bar{s}'', -\bar{B})} = \varkappa^{-1}\Gamma_k(s'', B)\varkappa.$$

For the barred field use $\varkappa\mathsf{C} = -\mathsf{C}\varkappa$. The transport relation $\overline{U_{-\bar{B}}(x, x')} = U_B(x, x')$ then gives the same identities for the Hölder rows. The kernel $M_k(X, B)$ obeys the corresponding identity, and $\varkappa^T\varkappa = 1$ preserves the mass form. The number m' is real for the present data, and $R\mathscr{E}$ and the reconstruction substitution have the reality symmetry. The contour argument just given, invariance of the box under sign reversal, and weighted integration prove eq. 5.cxxxvii.

Charge conjugation gives the first-localization symmetry by

$$H_k(s'', -B)\mathsf{C} = \mathsf{C}\bar{H}_k(s'', B), \qquad \bar{H}_k(s'', -B)\mathsf{C} = \mathsf{C}H_k(s'', B),$$
$$M_k(X, -B)^T = \mathsf{C}^{-1}M_k(X, B)\mathsf{C}.$$

Anticommuting the two transformed legs verifies the bilinear and mass-form identities. The first-step vertex has this symmetry by its explicit two-half formula. It follows that

$$\mathrm{Cg}\, E_{\le 3}(X; -A, -\mathscr{C}, -\zeta) = E_{\le 3}(X; A, \mathscr{C}, \zeta).$$

Direct covariance of the second substitution is not available, since its unbarred and barred fluctuations use different factors. We instead compare their Wick covariances.

Introduce independent generators $\phi^{\#}$ indexed like $\psi^{\#}$, with their own charge-conjugation action, and let ad add them to the old generators. Put

$$\begin{aligned}\mathcal{V}(X) = {} & \mathrm{ad}\, E_{\le 3}(X) + \mathrm{ad}\, \mathscr{E}^r(X) - \mathscr{E}^r(X) \\ & - \mathbf{1}_{X=\square} m'\big(\mathrm{ad}\langle\bar{\psi}, \psi\rangle_\square - \langle\bar{\psi}, \psi\rangle_\square\big).\end{aligned}$$

It has charge-conjugation symmetry under $(A, \mathscr{C}, \zeta) \mapsto (-A, -\mathscr{C}, -\zeta)$. Let $I_{s'',B}$ fix the old generators and replace $\phi^{\#}$ by $W_k^{\#}(s'', B)$. The second-localization input is exactly $I_{s'',B}\mathcal{V}(X)$, since the polynomial contours in G_4, G_5 give their value at 1 minus their value at 0. For any union U, interpolation restricted to cubes of U, with the disconnected terms removed, gives

$$\sum_{Y \subseteq U} F(Y) = I_{\mathbf{1}_U, B} \sum_{X \subseteq U} \mathcal{V}(X). \tag{5.cxxxviii}$$

At $s'' = \mathbf{1}_U$ define

$$\phi_\pm = H_k(\mathbf{1}_U, \pm B)\Gamma_k(\mathbf{1}_U, \pm B)W, \qquad \bar{\phi}_\pm = \bar{H}_k(\mathbf{1}_U, \pm B)\bar{W}, \qquad \phi' = \mathsf{C}\bar{\phi}_-, \quad \bar{\phi}' = \mathsf{C}^{-1}\phi_-.$$

The substitution by $(\phi', \bar{\phi}')$ and their Hölder quotients, denoted I', satisfies $\mathrm{Cg}\, I_{\mathbf{1}_U,-B} = I'\mathrm{Cg}$ on every generator, when Cg fixes W. The only nonzero ordinary pair expectations for the positive substitution are the kernel of

$$\mathcal{G}_U(B) = H_k(\mathbf{1}_U, B)\Gamma_k(\mathbf{1}_U, B)\bar{H}_k(\mathbf{1}_U, B)^T.$$

The identity moment has kernel $\mathbf{1}_{u=u'}/\nu(u)$, with $\nu = 1/48$ at zero. This precisely cancels one of the two measures in the kernel composition, so the same formula holds at every level. The transpose identities for the fermion kernels give

$$\mathcal{G}_U(-B)^T = -\mathsf{C}^{-1}\mathcal{G}_U(B)\mathsf{C}^{-1}.$$

Anticommuting $\bar{\phi}_-$ and ϕ_- therefore gives

$$\int \mathrm{d}\mu_I(W)\, \phi'(x)\bar{\phi}'(x')^T = \mathsf{C}[-\mathcal{G}_U(-B)^T(x,x')]\mathsf{C} = \mathcal{G}_U(B)(x,x').$$

Same-bar pair expectations vanish in both substitutions. The pair generators are obtained by the identical linear Hölder maps at $B, -B$, so their pair expectations also agree. Wick's theorem now proves $\int \mathrm{d}\mu_I(W) I'Q = \int \mathrm{d}\mu_I(W) I_{\mathbf{1}_U,B}Q$ for every polynomial Q in the old and $\phi^\#$ generators: each monomial integral is the Pfaffian of its pair expectations, with old generators as coefficients.

Apply this equality to the exponential of $\sum_{X\subseteq U}\mathcal{V}(X)$ and use $\zeta \mapsto -\zeta$. By eq. 5.cxxxviii, the integrated subvolume expression $\Xi_U = \int \mathrm{d}\mu_I(W)\,\mathrm{d}\widehat{\mu}(\zeta)\exp(\sum_{Y\subseteq U}F(Y))$ has charge-conjugation symmetry. But $\Xi_U = \sum_{Z\subseteq U}H(Z)$, by the finite Mayer expansion. Inverting this finite subset sum gives

$$H(Z) = \sum_{U\subseteq Z}(-1)^{|Z\setminus U|_M}\Xi_U.$$

Thus each $H(Z)$ has the symmetry, and the real coefficients and absolute convergence of the connected series prove eq. 5.cxxxvi.

Finally take a constant λ and $\lambda' = 0$ in eq. 5.cxxxv. A monomial with n_u unbarred and n_b barred legs acquires the factor $\exp(ie\lambda(n_u - n_b))$. Invariance for every real λ and independence of the basis monomials force its coefficient to vanish unless $n_u = n_b$. □

5.3.6. *The scaled correction.* Let S_L be the field scaling of eq. 5.xlv. The nonlinear correction is

$$E_k^*(m,\mathscr{E})(Y;\mathscr{A},\mathscr{C}) = \sum_{\bar{X}^L = LY} S_L E_k^\#(m,\mathscr{E})(X;\mathscr{A}_L,\mathscr{C}_L). \tag{5.cxxxix}$$

The correction E_k^* is obtained by field scaling and reblocking $E_k^\#$. The separate term $\mathcal{L}(R\mathscr{E})^{\mathrm{nat}}$ in the full step carries the quadratic adjustment of the input activity. The domain inclusions eq. 5.xliv ensure $(\mathscr{A}_L,\mathscr{C}_L) \in \widetilde{R}_k^- \times \mathfrak{X}_k$.

Proposition 5.17. For $(m,\mathscr{E}) \in \mathrm{Ball}_k^x(4)$,

$$E_k^*(m,\mathscr{E}) \in K_{k+1}^{x,\mathbb{C}}, \qquad \|E_k^*(m,\mathscr{E})\|_{k+1}^x \le e^{1/4-6\varepsilon}.$$

For real data its value belongs to K_{k+1}^x. For two data in $\mathrm{Ball}_k^x(1)$,

$$\|E_k^*(m_1,\mathscr{E}_1) - E_k^*(m_2,\mathscr{E}_2)\|_{k+1}^x \le e^{1/4-6\varepsilon}\left(\frac{|m_1-m_2|}{w_k^{\mathrm{m}}} + \frac{\|\mathscr{E}_1 - \mathscr{E}_2\|_k^x}{w_k^{\mathrm{E}}}\right). \tag{5.cxl}$$

The second quotient is absent at $k = 0$. Restriction to $\mathscr{C} = 0$ gives the one-connection correction formed from $(m, \mathscr{E}|_0)$. ■

Proof. The scaling identity gives

$$\|S_L Q\|_{h_{k+1}} = \|Q\|_{(L^{-1}h_{k+1,1}, L^{-1-\alpha}h_{k+1,2})} \le \|Q\|_{h_k/4},$$

since $h_{k+1,1} = L^{-1/8} h_{k,1}$, $h_{k+1,2} = L^{-1/8+\varepsilon/2} h_{k,2}$ and $L \geq 4$. The polymer sum in eq. 5.cxxxix uses $L d_M(Y) \leq d_M(X)$ and $X \subseteq LY$. Splitting $a_3 = (a_3 - \kappa_0) + \kappa_0$ therefore yields

$$\begin{aligned}\sum_{\bar{X}^L = LY} \exp(-a_3 d_M(X)) &\leq \exp(-L(a_3 - \kappa_0) d_M(Y)) \sum_{X \subseteq LY} \exp(-\kappa_0 d_M(X)) \\ &\leq c_X K_{\mathrm{pol}} L^3 \exp(-[L(\kappa - 6\kappa_0 - 4) - 1] d_M(Y)) \\ &\leq c_X K_{\mathrm{pol}} L^3 \exp(-\kappa d_M(Y)).\end{aligned}$$

Together with eq. 5.cxxix this gives

$$\|E_k^*\|_{k+1}^x \leq c_X K_{\mathrm{pol}} L^3 c_S' H_{\mathrm{cl}} = 504 c_X^5 K_{\mathrm{pol}}^5 \exp(3\kappa) L^3 e^{1/4 - 5\varepsilon} \leq e^{1/4 - 6\varepsilon},$$

using $c_S L^3 e^{\varepsilon} \leq 1$.

Ordinary sites scale into Y, pair paths to scaled paths, and $(x, x^{(3)})$ to $(x/L, (x/L)^{(3)})$. This preserves admissibility, including intracluster pairs. The bonds meeting X on which the coefficients depend scale to bonds meeting Y; the scaled-domain inclusion gives analytic kernels. The map $(S, a) \mapsto (S, a/L)$ identifies the lattice actions. For gauges, $\Lambda_L(c) = L^{1/2} \Lambda(c/L)$ and $(d\Lambda)_L = d\Lambda_L$, and the scaled neighborhoods on which the kernels depend are contained in the stipulated unit neighborhood. Charge and reality commute with the real scaling. Prop. 5.16 therefore gives $K_{k+1}^{x,\mathbb{C}}$ membership and its real subspace for real data.

For data u_1, u_2 in $\mathrm{Ball}_k^x(1)$, put $B = e^{1/4 - 6\varepsilon}$ and

$$D = \max\left\{ \frac{|m_1 - m_2|}{w_k^{\mathrm{m}}}, \frac{\|\mathscr{E}_1 - \mathscr{E}_2\|_k^x}{w_k^{\mathrm{E}}} \right\},$$

omitting the second entry at $k = 0$. If $D = 0$ the outputs agree. Otherwise $D \leq 2$; set $r = 3/D$, $u(t) = u_2 + t(u_1 - u_2)$ and $f(t) = E_k^*(u(t))$. For $|t| \leq r$ the data norm is at most $1 + rD = 4$, so $\|f(t)\|_{k+1}^x \leq B$. Since F is affine in the data, the previously constructed polynomial approximants, reblocked by eq. 5.cxxxix, are polynomials $f_N(t)$ converging uniformly in activity norm on this disk. Interpolation at finitely many distinct points gives their coefficients finite activity norm. Coefficientwise Cauchy and Tonelli therefore give

$$f_N(1) - f_N(0) = \frac{1}{2\pi i} \int_{|t| = r} \frac{f_N(t)}{t(t-1)} \, \mathrm{d}t, \qquad \|f_N(1) - f_N(0)\|_{k+1}^x \leq \frac{\sup_{|t|=r} \|f_N(t)\|_{k+1}^x}{r - 1}.$$

Passing to the uniform norm limit yields

$$\|E_k^*(u_1) - E_k^*(u_2)\|_{k+1}^x \leq \frac{B}{r-1} = \frac{BD}{3 - D} \leq BD \leq B \left(\frac{|m_1 - m_2|}{w_k^{\mathrm{m}}} + \frac{\|\mathscr{E}_1 - \mathscr{E}_2\|_k^x}{w_k^{\mathrm{E}}} \right).$$

This is eq. 5.cxl, with the activity quotient again omitted at $k = 0$.

Restriction commutes with extraction, $m(\mathscr{E}) = m(\mathscr{E}|_0)$ and $(R\mathscr{E})|_0 = R(\mathscr{E}|_0)$. Fermions use $A + \mathscr{C}$, while photons and measures do not use $\mathscr{C}$. Thus setting $\mathscr{C} = 0$ gives exactly the finite expressions, connected series and scaled correction for the activity $\mathscr{E}|_0$, analytic in A alone. □

The correction introduces only scaled incoming pair indices and scaled type-3 pairs: G_2, G_4 only add fields at existing pair indices, the mass contours have none, and reconstruction and the first vertex introduce only $(x, x^{(3)})$. Sums, products and integrals preserve this list. At zero only the latter pairs occur, of length η_1.

5.3.7. *The step identity and bounds.* For the local determinant terms of Prop. 4.12, put

$$\delta E_k^{\det}(X; \mathscr{A}, \mathscr{C}) = E_k^d(X, \mathscr{A} + \mathscr{C}) - E_k^d(X, \mathscr{C}), \qquad E_k^{\det}(X, \mathscr{C}) = E_k^d(X, \mathscr{C}) - E_k^d(X, 0).$$

These are scalar activities. With reblocking $\mathcal{L}$ from eq. 5.xlix, the new activity has three contributions: the normalized input, the determinant correction, and the fluctuation correction E_k^*. Define

$$\mathscr{E}^+ = \mathcal{L}(R\mathscr{E})^{\mathrm{nat}} + \mathcal{L}\delta E_k^{\det} + E_k^*(m, \mathscr{E}), \tag{5.cxli}$$

$$m^+ = L(m + m(\mathscr{E})), \tag{5.cxlii}$$

$$\varepsilon^{\mathrm{v},+} = L^3(\varepsilon^{\mathrm{v}} + \varepsilon_{\mathrm{v}}(\mathscr{E}) + \delta\varepsilon_k^0), \tag{5.cxliii}$$

$$\mathfrak{P}_{k+1}(Y, \mathscr{C}') = \sum_{\bar{X}^L = LY} (\mathfrak{P}_k(X, \mathscr{C}'_L) + E_k^{\mathrm{det}}(X, \mathscr{C}'_L)). \tag{5.cxliv}$$

Here $\mathfrak{P}_k$ is any given family of scalar functions of the external connection $\mathscr{C}$.

The reblocked determinant term belongs to K^x_{k+1}. Indeed the scaled arguments lie in its inverse domain, and each summand is analytic and depends on the fine connection only on bonds meeting X. These scale to bonds meeting Y. There are no legs. The lattice symmetry permutes the reblocking summands; local gauge shifts of the two connection arguments become the corresponding real shifts on these fine bonds, where determinant locality and gauge invariance apply. Evenness of E^d_k gives charge conjugation, and its conjugation identity gives reality. The pointwise bound eq. 4.lxxix and eq. D.v show

$$\|\mathcal{L}\delta E_k^{\mathrm{det}}\|^x_{k+1} \le 8c_{\mathrm{rb}} L^3 C_d M^3 e_k^{1/4} \le e_k^{1/4-2\varepsilon} \le e_{k+1}^{1/4-2\varepsilon}. \tag{5.cxlv}$$

Here $L(\kappa - \kappa_0 - 1) \ge \kappa$ controls the remaining exponential weight, and the fixed charge condition is $8c_{\mathrm{rb}} L^3 C_d M^3 e_k^{2\varepsilon} \le 1$. Restriction to $\mathscr{C} = 0$ gives the same result for $\mathcal{L}E_k^{\mathrm{det}}$.

Theorem H. Let $\varepsilon^{\mathrm{v}}, m$ be real, $\mathscr{E} \in K^x_k$, and $(m, \mathscr{E}) \in \mathrm{Ball}^x_k(1)$, with $\mathscr{E} = 0$ at $k = 0$. For $A' \in \mathrm{Dom}_{k+1}$ with arbitrary harmonic part and $\mathscr{C}' \in \mathfrak{X}_{k+1}$, the exact density satisfies

$$\mathbf{T}^x_k \mathscr{F}^x_k[\varepsilon^{\mathrm{v}}, m, \mathscr{E}] = \mathscr{F}^x_{k+1}[\varepsilon^{\mathrm{v},+}, m^+, \mathscr{E}^+], \tag{5.cxlvi}$$

with external scalar families $\mathfrak{P}_k$ and $\mathfrak{P}_{k+1}$ on the respective sides. Its integral satisfies

$$y^x_{k+1}(A'; \mathscr{C}') = \int_{\mathrm{Box}_k} \mathrm{d}\mu_I(\zeta)\, y^x_k(A^{\mathrm{min}}_k(A') + \mathrm{Sq}_k\zeta; \mathscr{C}'_L), \tag{5.cxlvii}$$

where $y^x_j = \int \mathrm{d}\Psi\, \mathscr{F}^x_j$ with the data and scalar family of that level. Moreover $\mathscr{E}^+ \in K^x_{k+1}$ and

$$\|\mathscr{E}^+\|^x_{k+1} \le c_{\mathfrak{fl}} L^{-1/4+2\varepsilon} \|\mathscr{E}\|^x_k + e_{k+1}^{1/4-2\varepsilon} + e_k^{1/4-6\varepsilon}, \tag{5.cxlviii}$$

$$|m^+ - Lm| \le L c_{\mathfrak{fl}} e_k^{1/2} \|\mathscr{E}\|^x_k, \tag{5.cxlix}$$

$$|\varepsilon^{\mathrm{v},+} - L^3\varepsilon^{\mathrm{v}}| \le L^3(c_{\mathfrak{fl}} \|\mathscr{E}\|^x_k + e_k^7). \tag{5.cl}$$

At level zero this formula integrates all $192n_0$ incoming pairs of generators, with chamber measure $1/48$, and leaves $4n_1$ outgoing pairs. Its data are

$$\mathscr{E}^+ = \mathcal{L}\delta E_0^{\mathrm{det}} + E_0^*(m, 0), \quad m^+ = Lm, \quad \varepsilon^{\mathrm{v},+} = L^3(\varepsilon^{\mathrm{v}} + \delta\varepsilon_0^0).$$

In particular $0 \le \varepsilon^{\mathrm{v},+} - L^3\varepsilon^{\mathrm{v}} \le L^3 q^7$. ■□

Proof. Use the notation of Prop. 5.8, with $A = \mathscr{A}'_L$, $B = A + \mathscr{C}$ and $Z = Z(1)$ the full-strength photon fluctuation. The kernel families recover their exact operators at full strength: $H_k(1, B) = H_k(B)$, $\Gamma_k(1, B) = \Gamma_k(B)$, and $\sum_X M_k(X, B) = b_k - D_k(B)$ for $k \ge 1$. The induced fields obey

$$\psi^{0,\#} = H_k^{\#}(B)\Psi^c, \qquad H_k^{\#}(B)\Psi^f = \psi^{0,\#} + W_k^{\#}(1, B).$$

We check that the localized activities reproduce the two differences in the exact density: changing the photon background at fixed Ψ^f, and then changing the fermion field from its mean to Ψ^f. The reconstruction left inverse connects these calculations. Define three evaluation homomorphisms. Let ev send old generators to $\psi^{0,\#}$ and fix W; let ev_1 send them to $H_k^{\#}(B)\Psi^f$; for $k \ge 1$, let ev_f additionally send $\Psi, \bar{\Psi}$ to $\Psi^f, \bar{\Psi}^f$. At zero take $\mathrm{ev}_f = \mathrm{ev}_1$. These maps exist since their images are odd and have the required antiperiodicity. They commute with the finite coefficient integrals; ev also commutes with Berezin integration, since it fixes W.

On the first contours $\mathcal{C}_r f = f(1) - f(0)$. For G_1, summing $M_k(X,\cdot)$ over X gives the Dirac difference $-\langle\bar\Psi, (D_k(B+Z) - D_k(B))\Psi\rangle_1$. For G_2, G_3, evaluating $\psi^\# + J_k^\#(1,B,Z)\Psi$ gives $H_k^\#(B+Z)\Psi^f$. The evaluations of their contour differences therefore give

$$\mathrm{ev}_f \sum_X \sum_{i\le 3} G_i(1,X) = \mathrm{Act}_k^x[m',\mathscr{E}^r](A+Z,\mathscr{C};\Psi^f) - \mathrm{Act}_k^x[m',\mathscr{E}^r](A,\mathscr{C};\Psi^f). \tag{5.cli}$$

At zero the explicit first vertex gives this same identity with the chamber pairing: $D_0(B) = D_B + \mu$ and the mass form has no connection dependence.

The reconstruction left inverse gives, on every local first expression, $\mathrm{ev}_1 \,\mathrm{sub}_{Y,B} = \mathrm{ev}_f$. This follows on the Ψ generators from $T_k(B)H_k(B) = 1$ and on the old generators from their definition. Adding $W_k^\#(1,B)$ to $\psi^\#$ in a polynomial and then applying ev gives ev_1 applied to that polynomial. Apply these identities to eq. 5.cxv. The evaluated $G_{\le 3}^W$ sum is eq. 5.cli. The evaluated polynomial contours G_4, G_5 give

$$I_k^x[m',\mathscr{E}^r](A,\mathscr{C};\Psi^f) - I_k^x[m',\mathscr{E}^r](A,\mathscr{C};\Psi^c).$$

Consequently

$$\mathrm{ev} \sum_Y F(Y) = V_k^x, \qquad \Xi_k^x = \exp\left\{\sum_X \mathrm{ev}\, E_k^\#(m,\mathscr{E})(X;A,\mathscr{C})\right\}.$$

The second equality is eq. 5.cxxviii evaluated by ev; its evaluation commutes with the integrals as just noted.

The other remaining factor is the determinant quotient. eq. 4.lxxv gives

$$\frac{\delta Z_k(B)}{\delta Z_k(0)} = \exp\left\{\sum_X \delta E_k^{\det}(X;A,\mathscr{C}) + \sum_X E_k^{\det}(X,\mathscr{C})\right\}.$$

On inserted fields adjustment changes no value: $\mathrm{ev}\, R\mathscr{E}(X) = \mathrm{ev}(R\mathscr{E})^{\mathrm{nat}}(X)$. The equality is the telescoping reconstruction identity for a small polymer and the definition for a large one. Finally $\mathrm{ev} = \mathrm{ev}'\, S_L$, where ev' inserts the outgoing fields. Each fine polymer has exactly one outgoing reblocking polymer. Thus the three activity sums combine into the outgoing $\mathscr{E}^+$, and the two scalar external sums combine into eq. 5.cxliv. Insert these identities in eq. 5.lxxxviii. All exponents are even and commute, so their sums give eq. 5.cxlvi, with exactly its previously derived mass, vacuum and determinant normalization.

Integrating the outgoing generators and using eq. 5.lxxxvi gives eq. 5.cxlvii. Prop. 3.15 admits the whole affine photon box; the analytic density coefficients are continuous there and thus integrable. The external scalar is fixed and no harmonic-part restriction is imposed.

The three terms of eq. 5.cxli have the stated real activity spaces and norm bounds by Prop. 5.5, eq. 5.cxlv and Prop. 5.17; the first has the stronger $3\varepsilon/2$ gain from eq. 5.lxii. Extraction gives the mass/vacuum bounds, using

$$0 \le \delta\varepsilon_k^0 \le 12\exp(-p_{0,k}^2/2) \le e_k^7.$$

At zero all incoming extracted terms vanish. The exact algebra uses all $192n_0$ pairs with the weight $1/48$ and integrates to the $4n_1$-pair outgoing density. There is no removed multiplicity or additional vacuum normalization. The displayed zero-level formulas and bounds follow. □

The restriction to zero external connection commutes with all three terms of eq. 5.cxli. The determinant difference there becomes $E_k^d(X,\mathscr{A}) - E_k^d(X,0)$, and the external scalar map preserves $\mathfrak{P}_k(\cdot,0) = 0$. Thus eq. 5.cxlvi and eq. 5.cxlvii also give the step for $\mathscr{C} = 0$, with incoming activity $\mathscr{E}|_0$.

The norm estimates hold on each torus with the same fixed constants, independently of its side and harmonic part. Agreement of the actual outputs requires a separate argument: we must compare the local kernel formulas on the different tori.

5.4. **Covers and comparison of local steps.**

5.4.1. *One system and its projection.* The next section needs one bare mass on both reference tori. Since the mass is extracted from local quadratic coefficients, we must identify those coefficients under the cover. We choose the photon and fermion kernel families on the larger torus, project them to the smaller one, and compare the local step on neighborhoods where the projection is injective. Fix a scale $k \geq 0$, $0 < \mu \leq 1$ and an allowed charge e_k. Use the photon families eq. 3.xliv and the fermion families eq. 4.lv on that torus, taking fermion polymer decay $\kappa' = \kappa_w + \kappa_0$. With the constants of their proved estimates, put

$$\alpha_0 = 1/54, \qquad \gamma_w = \min(\gamma^F, \gamma^P), \quad \gamma^P = \gamma_P/2,$$
$$C^P = 116 C_P^{\mathrm{ph}} M^{402\alpha_0}, \qquad C_w^{(1)} = \max\{C^F, C^P, 4\exp(2\gamma_w) M^{27\alpha_0}\}, \qquad C_w = 64 K_{\mathrm{pol}} C_w^{(1)}. \tag{5.clii}$$

Here C^F, γ^F are the fermion constants at the stated polymer rate. Impose the inverse and walk conditions of §4.2, eq. 3.xlv and $\gamma_w M \geq 1$. Props. 3.14, 4.8 and 4.10, including their component and covariance proofs, verify the hypotheses on kernels in §5.3.1 with $(\alpha_0, \gamma_w, \kappa_w + \kappa_0; C_w^{(1)})$: lowering a decay rate and increasing a bound preserve each estimate. Since $64K_{\mathrm{pol}} \geq 1$, the top system also obeys the weaker $(\alpha_0, \gamma_w, \kappa_w; C_w)$ bounds used below. At $k = 0$ use the diagonal families eq. 3.xlix and eq. 4.lvi, the full chamber covariance and the local resolvent with its factor $(1+y)^{-1}$. There is no M_0. Their diagonal bounds are included above, and the block measure is $(\eta_k^3/48)(48\eta_k^{-3}) = 1$.

Let $T_i = \mathbb{L}_k/\ell_i\mathbb{Z}^3$ be the two tori, $\ell_2 = J_c\ell_1$, with J_c odd and $\ell_i/(LM)$ odd and at least 13. Write $p : T_2 \to T_1$. Photon fields and parameters have period character $\chi = 1$; spinors and simultaneously translated pair fields have $\chi(w) = (-1)^{w_1+w_2+w_3}$. Since $\chi(J_c w) = \chi(w)$, pullback preserves these period characters. On the spaces with one spatial index (photon, fluctuation-coordinate, unit and chamber indices), it identifies smaller fields with the corresponding character subspace and

$$\|p^* f\|_\infty = \|f\|_\infty, \qquad \langle p^* g, p^* f\rangle_{T_2} = J_c^3 \langle g, f\rangle_{T_1}. \tag{5.cliii}$$

Indeed the larger fundamental set is the disjoint union of J_c^3 smaller ones, with cancelling characters and unchanged index weights, including $\eta_k^3/48$ on chamber columns.

Pullback preserves the parameter polydisc and the connection domains. For the inverse domain, pull back its decomposition $A = A_0 + A_1$ and the real gauge on each neighborhood of side $11M < \ell_1$; its local gradient identity and all sup bounds remain unchanged. The chamber stencils, nested block averages and transports commute with pullback. Products and inverses preserve this identity, since $O_2 p^* = p^* O_1$ implies $O_2^{-1} p^* = p^* O_1^{-1}$ when both inverses exist. The exact fermion operators therefore intertwine, as do the photon operators by eq. 3.liii. Weighted transposes intertwine by eq. 5.cliii and nondegeneracy of the pairings.

For a parameter family set $\widetilde{O}(s, A) = O_2(p^* s, p^* A)$; for a polymer family set

$$\widetilde{O}(X, A) = \sum_{pX' = X} O_2(X', p^* A).$$

Use the latter for $M_k(X, A)$, $k \geq 1$, and $S_{k,y}(X, A)$ at every level. These operators commute with the smaller period translations: parameter covariance proves this in the first case, and translations permute the polymer summands in the second. Their restriction to the character subspace defines the projection, with kernel

$$O_1(\cdot)(t, t') = \sum_{w \in (\mathbb{Z}/J_c\mathbb{Z})^3} \chi(w) \widetilde{O}(\cdot)(t, t' + \ell_1 w). \tag{5.cliv}$$

This follows by grouping the column sum into the smaller fundamental sets; no index weight changes. Replacing w by $w + J_c v$ multiplies both its character and its kernel by $\chi(v)$, so the summand is well defined. Project directly from the chosen top system, without iterating this construction through intermediate quotients.

Proposition 5.18. The projected kernel families satisfy the hypotheses on kernels in Thm. H, with $(\alpha_0, \gamma_w, \kappa_w; C_w)$ independent of the cover degree and scale. At full strength its operators are the exact smaller-torus operators. At zero its local resolvent retains $(1+y)^{-1}$. ∎

Proof. At full strength intertwining gives the exact parameter operators; the polymer sum identities give the exact polymer operators, since each X' occurs once under $X = pX'$. There are two summations to bound. For row and column supports of diameter at most L,

$$\sum_w e^{-\gamma_w d_2(t,t'+\ell_1 w)} \le 64 e^{-\gamma_w d_1(t,t')}. \tag{5.clv}$$

Choose a nearest column lift, with distance $d_1 \le \ell_1/2$, and distinct representatives realizing each larger-torus distance. For $n = |w|_\infty \ge 2$ their distance is at least $d_1 + (\ell_1 - 2L)(n-1)$; the 27 representatives of norm at most one have distance at least d_1. Since $\gamma_w(\ell_1 - 2L) \ge 6$, the quotient by $e^{-\gamma_w d_1}$ is at most $27 + \sum_{n\ge 2} 26 n^2 e^{-6(n-1)} < 64$. Projection of a covering net also proves $d_M(pX') \le d_M(X')$. Fix the row's partition cube $\square_t$. Polymer support and eq. D.iii give

$$\sum_{\substack{pX'=X \\ X' \supseteq \square_t}} e^{-(\kappa_w+\kappa_0) d_M(X')} \le K_{\mathrm{pol}} e^{-\kappa_w d_M(X)}. \tag{5.clvi}$$

Since $d_M(pX') \le d_M(X')$, the factor $e^{-\kappa_w d_M(X')}$ is at most $e^{-\kappa_w d_M(X)}$; the remaining factor $e^{-\kappa_0 d_M(X')}$ is summable over polymers containing $\square_t$ by eq. D.iii.

Apply these bounds to eq. 5.cliv. Row derivatives and short lifted differences commute with its sums. For the block norms of the resolvent at positive levels, the indicator of a column block pulls back to the disjoint sum of its translated indicators; the triangle inequality and eq. 5.clv give the same 64 bound. In the polymer sum, the fixed unit or chamber row selects $\square_t$, so eq. 5.clvi applies. The H_k sup and short-pair estimates instead follow from the isometric pullback of fields, with the same six short paths and intracluster difference. These prove all kernel, block and Hölder bounds with C_w. At zero $(1+y)^{-1}$ is common to every summand.

For component restrictions, lift a support and its meeting parameter cubes together; their diameter is below $5LM < \ell_1$. All photon parameter cubes meeting that support are linked. Indeed their half-side is $a = M$ or LM, their centers lie in $a\mathbb{Z}^3$, and the support diameter is less than a. Thus their center differences are coordinatewise less than $3a$, hence at most $2a$; their intersection contains a lattice site (use a center coordinate or, for difference $2a$, its midpoint). Fermion sites belong to one partition M-cube. A lifted parameter component maps into one smaller component, since shared sites and common faces project. Original row and column restrictions therefore confine each image summand to that component and allow its outside parameters and connection to be changed. The row-vanishing conditions pass to the projection as well: any missing cube meeting K_c has a missing lift. The parameter-independent $\mathcal{B}_{k,0}$ term of Γ_k preserves every union of partition cubes by the proof of Prop. 4.10; it causes no exception to these component assertions. Polymer support and local connection dependence descend because both original indices lie in X' and bonds meeting X' project to bonds meeting X.

Finite sums and linear pullback preserve continuity, holomorphy and Borel resolvent dependence. Gauge and lattice actions, conjugation, C, $\varkappa$ and weighted transpose commute with pullback and the image sums; period changes give cancelling source/target signs. The zero-level diagonal parameter products are unchanged, since cubes meeting their supports have unique corresponding lifts. Their bounds are included in $C_w^{(1)}$. This verifies the remaining kernel hypotheses without a new chamber multiplicity. □

5.4.2. *A common local algebra.* Assume now $\ell_1 > (31L+16)M$, and put $d = 2L+1$. For a nonempty covering union $\widetilde{X}$ of partition M-cubes, let $\operatorname{diam}\widetilde{X}$ be the largest side of its smallest rectangular cube union. Set

$$N(\widetilde{X}) = \operatorname{box}(\widetilde{X})^{\sim 6d}.$$

Call $\widetilde{X}$ admissible if $\operatorname{diam}\widetilde{X} \le \ell_1 - 14dM$; a polymer is flat if it is the image of a connected admissible lift.

Lemma 5.19. Projection identifies all local indices and incidences in $N(\widetilde{X})$, including shortest pair lifts and their six paths. The covering generators identify the local Grassmann algebras, preserving degrees, bars and coefficient norms at fixed corresponding connections and generator weights. At $k = 0$ use chamber W and omit the auxiliary Ψ. ■

Proof. Each neighborhood side is at most $\ell_1 - 2dM$. Its sites satisfy, for $w \neq 0$ and $i = 1, 2$,

$$|u - u' - \ell_i w|_\infty \ge 2dM + \eta_k. \tag{5.clvii}$$

Thus projection is injective. To lift an index or incidence, place one support point in the chosen lift. Each other point has its prescribed representative there, and a different period would contradict this separation. All relevant incidences have diameter at most $4LM < 2dM$: neighbors, short pairs and paths, cube and block membership, linked parameter cubes and coordinate supports. Short pair distances and paths consequently agree. Independent covering generators, their ordering signs and their measures are the same, proving the stated coefficient-norm isometry at the stated connections and weights. □

For a local family, agreement means equality under this identification whenever the two admissible connection pairs agree on $N(\widetilde{X})$. It suffices to compare a smaller connection pair with its global pullback: coefficient locality then changes the latter to any other admissible pair agreeing on the bonds on which the coefficients depend. Ordinary legs over $\widetilde{X}$ and admissible pair legs lie in $\widetilde{X}^{\sim 1}$, including the outward type-3 boundary pair. Every small polymer is flat: its connected nonwrapping lift has diameter at most $(L+3)M$, and $(L+3)+14d < 31L+16$. The small-lift result in App. D and the preceding incidence comparison give identical small-polymer sets over a fixed lifted cube, with the same tree length, volume and intrinsic shortest cube chains. Subpolymers, P_0, Y', incompatibility and reblocking membership also correspond: these are defined by the common cube and meeting relations. Their reblocking closures lie in the $(L-1)$-layer enlargement, still inside $N(\widetilde{X})$.

Lemma 5.20. Give parameter cubes in an admissible $\widetilde{Y}$ common values and set the other parameters to zero on each torus. For $0 \le j \le 5d$, a row of one of these kernels with support in $\widetilde{Y}^{\sim j}$ has columns in $\widetilde{Y}^{\sim(j+2L)}$ and depends on the connections only on bonds in $\widetilde{Y}^{\sim(j+d)}$. At corresponding connections it is the same covering row. Likewise $M_k(\widetilde{X}, \cdot)$, $k \ge 1$, and $S_{k,y}(\widetilde{X}, \cdot)$ agree for connected $\widetilde{X} \subseteq \widetilde{Y}$ and connections agreeing on bonds meeting $\widetilde{X}$. ■

Proof. We first justify the component restriction for a row t. Write $\square(t)$ for the partition cubes meeting its support. A parameter cube containing that support lies in $\square(t)^{\sim 2L}$: for a site use its partition cube, for a fine bond the radius-M parameter cube at an endpoint's cube center, and for a coordinate support the radius-LM parameter cube at the LM-block center of one support point. In the last case its support diameter is at most L, and its partition-cube centers are less than $(3L/2+1)M \le (2L+1)M$ from that point's partition-cube center; being separated by integer multiples of M, they lie within $2L$ layers. If $\square(t)^{\sim 2L}$ lies in a face-component Z_1 of a partition-cube union Z, use all parameter cubes contained in Z, including any whose values happen to be zero. Their component containing the chosen cube lies in Z_1: its meeting partition cubes are face-connected, since linked photon cubes share lattice sites, and all are contained in Z. The original component restrictions therefore confine the row to Z_1 and its connection dependence to bonds meeting Z_1. The block-local term of Γ_k preserves the same union.

Take $\widetilde{B} = \widetilde{Y}^{\sim(j+2L)}$ and its component $\widetilde{B}_1$ containing the row neighborhood. The row support meets one partition cube or two adjacent ones, so that neighborhood is connected. Its one-layer enlargement lies in $N(\widetilde{Y})$. Separation eq. 5.clvii says that its image is a component both below and in the full inverse image above: an extra adjacent cube would have a lift in this enlargement and

hence could not belong to another period copy. The component restriction just proved confines the row there and its connection dependence to the one-layer enlargement, giving the two radii. The global pullback parameters and the single-copy parameters agree in this component, as do the connections on which these rows depend, so their top rows coincide. In eq. 5.cliv any surviving column image must belong to the same copy; separation leaves only the original column, with sign one.

For polymer kernels, a connected X' projecting onto the image of $\widetilde{X}$ lies in one period copy: adjacent cubes cannot cross the separated copies, and connectedness propagates that copy choice. Surjectivity onto the image makes it the entire copy. In the polymer sum and column image sum, row support selects that copy and column support selects the original image. Local connection dependence permits replacement of the global pullback by an admissible connection agreeing on the bonds on which this kernel depends. The barred row follows at $-A$ by conjugation with C. At zero the diagonal mean field has kernel $48\delta_{x,x'}$ times the common parameter product; the same argument uses the full chamber covariance and resolvent, without an M_0 family. □

5.4.3. *Agreement of one step.*

Theorem I. Suppose the two sides at levels k and $k{+}1$ are admissible and satisfy $\ell_1/L > (31L{+}16)M$. Use the top and projected systems above, with all hypotheses on kernels, domains and numerical parameters in Thm. H on both tori. Let the real masses be equal with $|m| \le w_k^{\mathrm{m}}$, and let the real incoming activities $\mathscr{E}_i \in K_k^x(T_i)$ have $\|\mathscr{E}_i\|_k^x \le w_k^{\mathrm{E}}$ and agree on flat polymers, with $\mathscr{E}_i = 0$ at $k = 0$. Their extracted mass and vacuum contributions are equal and their outgoing activities agree on flat polymers. If the external scalar families agree on the admitted external domains at the incoming level, this agreement is preserved by eq. 5.cxliv. The same conclusions hold after restriction to $\mathscr{C} = 0$. ■□

Proof. First compare a connection pair with its global pullback. We verify the local data of Lem. 5.13; the lemma then identifies both finite localizations, their integrals and the convergent connected logarithms.

Finite formulas and extraction. The coordinate map eq. 3.xxvi is independent of the connection and stays in the two endpoint L-blocks. A pair-difference row has its two endpoints and the common six paths (or outward bond), with coefficient $d_X^{-\alpha}$; an intracluster row is independent of the connection. The reconstruction formulas eq. 5.xxiii involve only Δ_y, its type-3 bonds, and the outward boundary pair. Their membership indicators agree by common-lift separation. The rows of $\mathcal{Q}_k, \mathcal{Q}_k^\sharp$ involve only one Δ_y; the defining spin-matrix combination of I and P_L for $\mathcal{B}_{k,y}$ involves only one L-block. All these are identical finite formulas on the lift, with identical column measures, hence identical kernels. At zero use the full four-neighbor chamber F_0 and its L-block penalty, whose kernel is 48 times its stencil coefficient; the first vertex eq. 5.xciii retains $1/96$, both boundary transports and its outward-pair generator.

On small polymers the common constant and quadratic coefficients, volume and measure identify eq. 5.x. The common small-polymer sets over a cube then identify the mass and vacuum sums eq. 5.xi, so m' is common. The intrinsic shortest chains and all their transports identify eq. 5.xxxvi; thus subtraction and eq. 5.xxxvii preserve agreement.

Determinants. For $k \ge 1$ cycle the local determinant trace into $\mathrm{Tr}[(\mathcal{B}_{k,y}\gamma_5\mathcal{B}_{k,y})\mathcal{Q}_k S_{k,y}(X,B)\mathcal{Q}_k^\sharp]$. The second factor has both unit indices in X, and the first stays in their L-blocks, which lie in the same partition cubes. All trace indices therefore have common lifts. The preceding finite rows and the local polymer lemma identify every factor, and the two spinor period signs cancel in a trace. At zero the traces $\mathrm{Tr}[F_0 S_{0,y}(X,B)]$ and $\mathrm{Tr}[\gamma_5 S_{0,y}(X,B)]$ similarly use only X, with all 48 labels and measure $1/48$. The integrable majorants in the proof of Prop. 4.12 apply with the common resolvent bounds: $(b_k^2 + y^2)^{-1}$ for $k \ge 1$, $(1+y)^{-1}\,\mathrm{d}y/y$ on $[1,\infty)$ and $(1+y)^{-1}\,\mathrm{d}y$ on $[0,1]$ at zero. Thus the determinant integrals agree, as do their four-term symmetrization and subtractions at $\mathscr{C}$ or zero.

Kernel coefficients in the local calculus. Let $\widetilde{U}$ be connected and admissible. Give its common photon coordinates values in the prescribed box, its contained parameter cubes common values in their polydiscs, and all others zero. For the photon composition $H_k^{\mathrm{ph}}(s')CC_k^{1/2}(s')\zeta$, at a row in $\widetilde{U}^{\sim(1+d)}$ the supported-row lemma confines the intermediate bonds to $\widetilde{U}^{\sim 2d}$. The coordinate row stays in $\widetilde{U}^{\sim(1+2d)}$, and the square-root row reaches only $\widetilde{U}^{\sim 3d}$. All lie in $N(\widetilde{U})$, so the entire composition agrees there.

Write $B_i(t) = A_i + \mathscr{C}_i + tZ_i(s')$. Prop. 5.10 admits all these connections for $|t| \le r_1$, as well as $A_i + tZ_i \in \widetilde{R}_k$. The supported-row lemma identifies $H_k, \bar{H}_k, \Gamma_k$ at all rows used by ordinary fields and admissible pair endpoints, which are at distance at most $\eta_k \le 1$ from U_i. It identifies every $M_k(X, B_i(t))$, $X \subseteq U_i$. The component proof in Prop. 5.12 confines the surviving fermion columns to U_i itself, including the block-local $\mathcal{B}_{k,0}$ term. Thus these are polynomials in just the identified generators. The finite pair rows give the same pair fields; $t = 0, 1$ give the same $J_k^{\#}$ since $r_1 > 1$. Incoming agreement and coefficient locality identify their values at $(A_i + tZ_i, \mathscr{C}_i)$ on every $X \subseteq U_i$. The same rows at $B_i(0)$ identify $W_k^{\#}$. At $k = 0$ the diagonal multiplier vanishes outside U_i, its composition with Γ_0 retains the measure $1/48$, and the common photon field identifies the vertex factors $\exp(\pm iqZ(c, c_3)) - 1$; there is no M_0 or Ψ.

Common lifts preserve the subpolymers, parameter sets, P_0, Y', reconstruction memberships and incompatibility. The contour radii, coefficient weights and pairings of ordered products for the identity covariance agree. Each common photon coordinate has the same normalized truncated Gaussian law; unused coordinates and modes integrate to one. At $k = 0$ a paired chamber mode contributes 48, the inverse of its weight $1/48$. These verify every hypothesis of Lem. 5.13, including both localizations and their intervening reconstruction. Prop. 5.15 supplies the required weighted absolute convergence on both tori. The lemma's logarithm clause identifies $E_k^{\#}$ on each flat polymer. The omitted photon normalization is the same scalar on both tori:

$$\delta\varepsilon_k^0 = -(2 - 2L^{-3}) \log \left((2\pi)^{-1/2} \int_{-p_{0,k}}^{p_{0,k}} e^{-z^2/2} \, \mathrm{d}z \right). \tag{5.clviii}$$

Reblocking. For an admissible outgoing $\widetilde{U}$, $L\widetilde{U}$ is admissible at the incoming scale and $N(L\widetilde{U}) \subset LN(\widetilde{U})$. The outgoing connection pair scales into $\widetilde{R}_k^- \times \mathfrak{X}_k$ by eq. 5.xliv; scaling and pullback commute. Each polymer in the reblocking sum has its unique connected lift $\widetilde{X} \subset L\widetilde{U}$. Its LM-closure lies in the $(L-1)$-layer enlargement, where projection remains injective; hence $\bar{X}^L = LU_i$ is equivalent to $\overline{\widetilde{X}}^L = L\widetilde{U}$. The reblocking index sets are therefore identical. Scaling preserves the common labels, period signs and the rules $\psi \mapsto L^{-1}\psi$, $\chi \mapsto L^{-1-\alpha}\chi$. Each of the three terms in eq. 5.cxli agrees before scaling, by extraction and adjustment, determinant comparison, and the local-calculus conclusion. Their outgoing sums agree. The same index set and determinant comparison identify eq. 5.cxliv. Outgoing coefficient locality extends equality from global pullbacks to all admitted connection pairs with the stated common local lifts. Restriction to $\mathscr{C} = 0$ commutes with each operation and gives the assertion for the restricted activities $\mathscr{E}_i|_0$. All zero-level terms were included above with the chamber measures and the stipulated zero incoming activity. □

6. The renormalization group: full trajectory

We choose a single bare mass for the two reference tori by solving the mass and interaction recursions together. The terminal integral retains the photon fluctuation coordinates: their Gaussian tails will control the fields excluded from the small-field calculation.

6.1. **The tuned flow.**

6.1.1. *Physical scales and stopping levels.* Fix $\bar m \in (0,1]$ and put $h = L^{\lfloor \log_L(1/\bar m)\rfloor}$. For a cutoff $N \ge 0$, define

$$\eta = L^{-N}, \qquad K = N + \log_L h, \qquad a_j = L^{j-N}, \qquad e_j = e\sqrt{a_j}, \qquad \mu_j = \bar m a_j. \tag{6.i}$$

Thus a_j is the physical length of one unit at level j, and $a_K = h$. At zero the charge is written $q = e\sqrt{\eta}$; the fixed charge ceilings $\bar e_0, \bar e_1$ retain their earlier meanings. A formula with variable index j uses e_j for the running charge, including q when $j = 0$. For $0 \le j \le K$ set

$$v_j = -\log e_j, \qquad r_j = \min\{L^i : i \in \mathbb{N} \cup \{0\},\ L^i \ge v_j^r\}, \qquad p_j = v_j^p, \quad p_{0,j} = v_j^{p_0}.$$

The integers r, p, p_0 and the constants of the preceding estimates are fixed. Here $r \ge 2$, L is odd with $L \ge 13$, and M is a power of L with $M \ge L^2$. We assume the hypotheses of the kernel and step estimates, and use a ceiling $\bar e_1 \le \min(\bar e_0, 1/3)$ such that, for all $v \ge -\log \bar e_1$,

$$\begin{gathered} v \ge 8, \qquad \left(1 + \frac{3\log L}{2v}\right)^r < L, \qquad v^r > 31L + 16, \\ (c_{\mathrm{fl}} + 2)\exp(-\varepsilon v) \le \frac{1}{16}(1 - L^{-\varepsilon/2}). \end{gathered} \tag{6.ii}$$

In addition $0 < \varepsilon \le 1/200$, $c_{\mathrm{fl}} \ge 1$, $\bar e_0^{\varepsilon} \le 1/4$, $c_{\mathrm{fl}} L^{-3/8+6/200} \le 1/16$, and $L^{-5/8} \le 1/4$. The construction below applies whenever $0 < e\sqrt{h} \le \bar e_1$.

The physical top side and reference side are

$$R_{\mathrm{top}} = hMr_K, \qquad R^* = R_{\mathrm{top}}/L = hL^{T^*(e)}, \qquad T^*(e) = \log_L(Mr_K) - 1. \tag{6.iii}$$

They are independent of N, since $v_K = -\log(e\sqrt{h})$. The admissible physical sides for the flow are

$$\mathcal{T}(e) = \{R^*(e), R_{\mathrm{top}}(e)\}.$$

For $R \in \mathcal{T}(e)$ put $\ell_j(R) = R/a_j$, $T_j(R) = \mathbb{L}_j/\ell_j(R)\mathbb{Z}^3$ and $\mathfrak{n}_j = \log_L(\ell_j/(Mr_j))$. The allowed unit sides are $\mathrm{Cl} = \{\ell : \ell/(LM)$ is an odd integer at least $13\}$. Their occurrence at every required scale is part of the following arithmetic argument.

Proposition 6.1. For both $R \in \mathcal{T}(e)$ choose the common sufficient mesh threshold $N_{\min}(R, e) = \max\{0, 3 - \log_L h\}$. When $N \ge N_{\min}(R, e)$, the first level with $\mathfrak{n}_j = 0$ exists and satisfies $1 \le k_*(R) \le K$. Its physical quantities $a_*, e_*, \mu_*, \ell_*, r_*$ are independent of N. For $j \le k_*(R)$,

$$\begin{gathered} \ell_j \in \mathrm{Cl}, \qquad \ell_j \ge Mr_j, \qquad r_j \ge r_* > 31L + 16, \\ v_j \ge v_* \ge v_K, \qquad 0 < e_j \le e_* \le e_K \le \bar e_1, \qquad 0 < \mu_j \le \mu_* \le \bar m h \le 1. \end{gathered} \tag{6.iv}$$

For $j < k_*(R)$ one has $\ell_j \ge LMr_j$, $\ell_{j+1} = \ell_j/L \in \mathrm{Cl}$, $e_{j+1} = \sqrt{L}e_j \le \bar e_0$ and $0 < \mu_j = \mu_{j+1}/L \le 1/L$. At the stop,

$$\ell_* = Mr_*, \qquad Mv_*^r \le \ell_* < LMv_*^r, \qquad \mu_*^{-1} < L\ell_*, \qquad e_*^2\ell_* = e^2 R. \tag{6.v}$$

The thresholds are equal on the two sides, and $k_*(R^*) \le k_*(R_{\mathrm{top}})$. ∎

Proof. Write $R = R_{\mathrm{top}}/L^p$, $p \in \{0, 1\}$, and use depth $s = K - j$:

$$\widehat v_s = v_K + \tfrac{s}{2}\log L, \qquad \widehat r_s = \min\{L^i : i \ge 0,\ L^i \ge \widehat v_s^r\}, \qquad \widehat n_s(R) = s - p - \log_L(\widehat r_s/r_K).$$

These are $v_j, r_j, \mathfrak{n}_j$ at that depth, with $a_j = hL^{-s}$. Condition eq. 6.ii gives $\widehat r_{s+1}/\widehat r_s \in \{1, L\}$ for every $s \ge 0$, since $\widehat v_{s+1}^r < L\widehat v_s^r \le L\widehat r_s$. Hence $\widehat n_s$ is nondecreasing with increments zero or one. The stronger three-depth condition at v_K gives $\widehat r_s \in \{r_K, Lr_K\}$ for $s = 0, 1, 2, 3$. For $p = 0$, the values at $s = 0, 1, 2$ are respectively 0, an element of $\{0, 1\}$ and an element of $\{1, 2\}$. Thus its last zero $s_*(R_{\mathrm{top}}, e)$ is 0 or 1. For $p = 1$, the values at $s = 0, 1, 2, 3$ are -1, an element of $\{-1, 0\}$, an element of $\{0, 1\}$ and an element of $\{1, 2\}$; its last zero is 1 or 2. Monotonicity excludes every later zero in either case.

Our threshold makes $K \geq 3$, so $\widehat{n}_K \geq 1$ and the first zero encountered as j increases has

$$N_{\min}(R,e) = \max\{0, 3 - \log_L h\}, \qquad k_*(R) = K - s_*(R,e). \tag{6.vi}$$

In particular $1 \leq k_*(R) \leq K$. Since $v_K = -\log(e\sqrt{h})$ and both s_* depend only on fixed physical parameters, all terminal physical quantities are independent of N. The two displayed depth ranges also give the stated ordering of stops.

For $j \leq k_*$, depth is at least s_*, giving all monotone ranges; before the stop $\mathfrak{n}_j \geq 1$. The rounded-power inequality and $\bar{m}h \in (1/L, 1]$ give eq. 6.v, since $\mu_*\ell_* = \bar{m}R \geq \bar{m}h > 1/L$ and $e_*^2 = e^2 a_*$. Moreover $r_* \geq v_*^r > 31L + 16 > L$ implies $r_* \geq L^2$. Thus $\ell_j/(LM) = r_j L^{\mathfrak{n}_j - 1}$ is a positive power of the odd integer L, at least $L \geq 13$, proving admissibility. The incoming mass and charge bounds follow at $j+1$. □

The common reference mesh threshold is

$$N_{\mathrm{ref}}(e) = N_{\min}(R^*(e), e) = \max\{0, 3 - \log_L h\}. \tag{6.vii}$$

The depth calculation gives

$$\begin{gathered} k_*(R^*) \in \{K-1, K-2\}, \qquad a_* \in \{h/L, h/L^2\}, \qquad r_* \in \{r_K, Lr_K\}, \\ k_{\mathrm{top}} \in \{K, K-1\}, \qquad L^{-3} < \mu_* \leq L^{-1} \quad \text{on } R^*. \end{gathered} \tag{6.viii}$$

These physical terminal values are independent of $N \geq N_{\mathrm{ref}}(e)$.

The scale proposition verifies every side, mass and charge hypothesis of the step theorem on R_{top} and its reference quotient $R^* = R_{\mathrm{top}}/L$. At each level use the top system of §5.4.1; on a smaller torus use its projection from Prop. 5.18. These are the actual mean-field, covariance, local resolvent and photon families constructed earlier, with the same common estimates. The ratio R_{top}/R is 1 or the odd integer L, and $k_*(R) \leq k_{\mathrm{top}}$. Thus the required top families exist whenever a smaller-volume step is used. Furthermore $\ell_{j+1}(R) \geq M r_{j+1} > (31L+16)M$; the next-side hypothesis of the cover comparison holds at every such step.

6.1.2. *Continuation within the mass range.* Write the mass and activity bounds as

$$w_j^{\mathrm{m}} = e_j^{3/4-8\varepsilon}, \qquad w_j^{\mathrm{E}} = e_j^{1/4-7\varepsilon}.$$

For a real pair $(m_{\mathrm{in}}, \varepsilon_{\mathrm{in}}^{\mathrm{v}})$, the initial vacuum is $\varepsilon_{\mathrm{in}}^{\mathrm{v}}$, the initial member of the mass sequence is m_{in}, and $E_0 = \mathscr{E}_0 = \mathfrak{P}_0 = 0$. The variable-index notation m_j includes this initial member. The recursion is

$$m_{j+1} = L(m_j + m(E_j)), \qquad \varepsilon_{j+1}^{\mathrm{v}} = L^3(\varepsilon_j^{\mathrm{v}} + \varepsilon_{\mathrm{v}}(E_j) + \delta\varepsilon_j^0), \tag{6.ix}$$

$$E_{j+1} = \mathcal{L}(RE_j)^{\mathrm{nat}} + \mathcal{L}E_j^{\mathrm{det}} + E_j^*(m_j, E_j), \tag{6.x}$$

$$\mathscr{E}_{j+1} = \mathcal{L}(R\mathscr{E}_j)^{\mathrm{nat}} + \mathcal{L}\delta E_j^{\mathrm{det}} + E_j^*(m_j, \mathscr{E}_j). \tag{6.xi}$$

The scalar family $\mathfrak{P}_j$ follows eq. 5.cxliv. In the last line the correction acts on $\mathscr{E}_j$ and depends on $\mathscr{C}$; in eq. 6.x it acts on $E_j = \mathscr{E}_j|_0$. These are the corresponding maps proved earlier. The trajectory exists up to a level if every preceding $(m_j, \mathscr{E}_j)$ lies in $\mathrm{Ball}_j^x(4)$. It is controlled when it exists to $k_*(R)$ and $|m_j| \leq w_j^{\mathrm{m}}$ throughout, including the stop.

Denote the right sides of eq. 6.x and eq. 6.xi by $\Phi_j(m,E)$ and $\Phi_j^x(m,\mathscr{E})$. For $j < k_*(R)$, take real data (m,E) and $(\widetilde{m},\widetilde{E})$ in the unit ball and put $d_m = |m - \widetilde{m}|/w_j^{\mathrm{m}}$ and $d_E = \|E - \widetilde{E}\|_j/w_j^{\mathrm{E}}$. The normalized bounds are

$$\begin{gathered} L^{-1}w_{j+1}^{\mathrm{m}}/w_j^{\mathrm{m}} = L^{-5/8-4\varepsilon} \leq \tfrac{1}{4}, \qquad |m(E) - m(\widetilde{E})|/w_j^{\mathrm{m}} \leq c_{\mathrm{fl}} e_j^{\varepsilon} d_E, \\ \|\Phi_j(m,E)\|_{j+1}/w_{j+1}^{\mathrm{E}} \leq \tfrac{1}{16}\|E\|_j/w_j^{\mathrm{E}} + 2e_{j+1}^{\varepsilon}, \\ \|\Phi_j(m,E) - \Phi_j(\widetilde{m},\widetilde{E})\|_{j+1}/w_{j+1}^{\mathrm{E}} \leq \tfrac{1}{16}d_E + e_{j+1}^{\varepsilon}(d_m + d_E). \end{gathered} \tag{6.xii}$$

The size bound holds also in the norm $\|\cdot\|^x_{j+1}$. Indeed eq. 5.xii and $K_{\mathrm{pol}}M^{-3} \le c_{\mathfrak{fl}}$ give the extraction bound because $e_j^{1/2} w_j^{\mathrm{E}}/w_j^{\mathrm{m}} = e_j^{\varepsilon}$. The linear activity factor is $c_{\mathfrak{fl}} L^{-3/8+11\varepsilon/2} \le 1/16$ by eq. 5.lxiii. The two source ratios in eq. 5.cxlviii are $e_{j+1}^{5\varepsilon}$ and $L^{-1/8+3\varepsilon} e_{j+1}^{\varepsilon}$, each at most e_{j+1}^{ε}. For differences the determinant term cancels and eq. 5.cxl has the latter ratio. These arguments include $j = 0$ with $E = \tilde{E} = 0$, using the chamber case of Thm. H.

The charge condition also gives

$$\sum_{j=0}^{k_*} (c_{\mathfrak{fl}} + 2) e_j^{\varepsilon} \le (c_{\mathfrak{fl}} + 2) e_*^{\varepsilon} (1 - L^{-\varepsilon/2})^{-1} \le \tfrac{1}{16}. \tag{6.xiii}$$

In particular $c_{\mathfrak{fl}} e_j^{\varepsilon} \le 1/16$, $2e_j^{\varepsilon} \le 1/16$, and $e_j \le e_j^{\varepsilon} \le 1/48$.

Proposition 6.2. If the recursively produced masses satisfy $|m_i| \le w_i^{\mathrm{m}}$ for $i < j \le k_*(R)$, then the trajectory is defined to j and

$$\|E_j\|_j \le \|\mathscr{E}_j\|_j^x \le \frac{32}{15} e_j^{\varepsilon} w_j^{\mathrm{E}} \le 4 e_j^{1/4-6\varepsilon} \le w_j^{\mathrm{E}}. \tag{6.xiv}$$

The data satisfy $E_j = \mathscr{E}_j|_{\mathscr{C}=0}$, $\mathfrak{P}_j(X, 0) = 0$, and belong to their real mass, vacuum and activity spaces. ■

Proof. Put $A_i = \|\mathscr{E}_i\|_i^x / w_i^{\mathrm{E}}$. Starting at $A_0 = 0$, the mass hypothesis and $A_i \le 1$ allow us to apply Thm. H. Its normalized size bound gives

$$A_{i+1} \le A_i/16 + 2e_{i+1}^{\varepsilon}, \qquad A_j \le 2 \sum_{i=1}^{j} 16^{-(j-i)} e_i^{\varepsilon} \le (32/15) e_j^{\varepsilon} \le 4 e_j^{\varepsilon} \le 1.$$

Thus induction justifies every preceding application of the step and proves eq. 6.xiv. The same step theorem preserves reality and restriction, which does not increase the norm. The finite external scalar recursion eq. 5.cxliv is defined on the rescaled external-connection domain and vanishes at zero since $E_i^{\det}(X, 0) = 0$. This proves the remaining assertions. □

6.1.3. *The terminal conditions and the fixed point.* The mass recursion magnifies an error in m_j by the factor L. Prescribing $m_{k_*} = 0$ turns it into the backward equation

$$m_j = L^{-1} m_{j+1} - m(E_j).$$

This equation contracts in the weighted mass norm, while the activity equation contracts forwards. We solve the two equations simultaneously.

Theorem J. On any $R \in \mathcal{T}(e)$ with $N \ge N_{\min}(R, e)$, there is a real controlled pair whose trajectory has $m_{k_*} = 0$ and $\varepsilon_{k_*}^{\mathrm{v}} = 0$. Along its trajectory,

$$|m_j| \le \tfrac{1}{8} w_j^{\mathrm{m}}, \qquad \|E_j\|_j \le \|\mathscr{E}_j\|_j^x \le 4 e_j^{1/4-6\varepsilon}, \qquad |\varepsilon_j^{\mathrm{v}}| \le 5 c_{\mathfrak{fl}} e_j^{1/4-6\varepsilon}. \tag{6.xv}$$

The vacuum at every level is

$$\varepsilon_j^{\mathrm{v}} = -\sum_{i=j}^{k_*-1} L^{-3(i-j)} \big(\varepsilon_{\mathrm{v}}(E_i) + \delta\varepsilon_i^0\big). \tag{6.xvi}$$

■□

Proof. We solve the mass recursion backwards from $m_t = 0$ and the activity recursion forwards from $E_0 = 0$ in a single product space. On the product of the real mass coordinates and the activity spaces K_j, with $\mathscr{C} = 0$, use the norm

$$\|\mathbf{z}\|_{\mathfrak{fl}} = \max_{0 \le j \le t} \max \left\{ |m_j|/w_j^{\mathrm{m}}, \|E_j\|_j / w_j^{\mathrm{E}} \right\}, \qquad t = k_*(R).$$

Its closed unit subset $\mathcal{B}$ with $E_0 = 0$ is nonempty and complete. Use the actual maps to define

$$\begin{aligned} &\widehat{m}_j = L^{-1} m_{j+1} - m(E_j) \quad (j < t), && \widehat{m}_t = 0, \\ &\widehat{E}_0 = 0, && \widehat{E}_{j+1} = \Phi_j(m_j, E_j) \quad (j < t). \end{aligned} \tag{6.xvii}$$

eq. 6.xii and eq. 6.xiii bound both the size on $\mathcal{B}$ and the Lipschitz constant of the mass coordinates by $1/4 + c_{\mathrm{fl}} e_j^{\varepsilon} \le 5/16$, and those of the activity coordinates by $1/16 + 2e_{j+1}^{\varepsilon} \le 1/8$. The map therefore preserves $\mathcal{B}$ and contracts by $5/16$. Banach's theorem [10, Thm. 8] gives its unique fixed point, satisfying the mass/activity recursions and $m_t = 0$.

Choose its initial mass and the initial vacuum given by eq. 6.xvi at $j = 0$. Prop. 6.2 identifies this fixed-point sequence with the restriction of the full trajectory: inductively, the identified mass admits the next full step, whose restriction is the prescribed activity. Direct substitution in the scalar recursion proves eq. 6.xvi at every level and $\varepsilon_t^{\mathrm{v}} = 0$.

Put $b = 1/4 - 6\varepsilon > 0$, $B_j = |m_j|/w_j^{\mathrm{m}}$ and $V_j = |\varepsilon_j^{\mathrm{v}}|/e_j^b$. Extraction, continuation and the backward scalar recursions give, for $j < t$,

$$\begin{aligned} B_j &\le B_{j+1}/4 + 1/16, \\ V_j &\le 4c_{\mathrm{fl}} + e_j^{7-b} + L^{-23/8-3\varepsilon} V_{j+1} \le 4c_{\mathrm{fl}} + 1/48 + V_{j+1}/16. \end{aligned}$$

Here eq. 5.xii gives $|\varepsilon_{\mathrm{v}}(E_j)| \le 4c_{\mathrm{fl}} e_j^b$, the step theorem gives $|\delta\varepsilon_j^0| \le e_j^7$, and $7 - b > 1$, $e_j \le 1/48$, $L \ge 13$ give the last inequality. Since $B_t = V_t = 0$, backward induction preserves $B_j \le 1/8$ and $V_j \le 5c_{\mathrm{fl}}$: $1/32 + 1/16 < 1/8$ and $4c_{\mathrm{fl}} + 1/48 + 5c_{\mathrm{fl}}/16 \le (13/3)c_{\mathrm{fl}} < 5c_{\mathrm{fl}}$. Together with continuation's activity estimate these are eq. 6.xv. □

6.1.4. *One pair on the reference tori.* For $N \ge N_{\min}(R_{\mathrm{top}}, e)$ define

$$P_N(e) = (m_{0,N}(e), \varepsilon_{0,N}^{\mathrm{v}}(e))$$

to be the pair constructed in the proof of Thm. J on the prescribed side $R_{\mathrm{top}}(e)$. The dependence on the fixed $\bar{m}$ and fixed construction parameters is suppressed. For lower cutoffs set $m_{0,N}(e) = 0$.

Theorem K. The pair $P_N(e)$ is controlled on every $R \in \mathcal{T}(e)$ with $N \ge N_{\min}(R, e)$. Up to its stop, its mass and vacuum sequence agree with those on the top torus; its activity and external scalar families agree there on flat polymers. For every such level,

$$|m_j| \le \tfrac{1}{8} e_j^{3/4-8\varepsilon}, \qquad \|E_j\|_j \le \|\mathscr{E}_j\|_j^x \le 4e_j^{1/4-6\varepsilon}, \qquad |\varepsilon_j^{\mathrm{v}}| \le 5c_{\mathrm{fl}} e_j^{1/4-6\varepsilon}. \tag{6.xviii}$$

The real mass sequence is independent of the side datum and of the twist. At all $N \ge 0$,

$$|m_{0,N}(e)| \le \tfrac{1}{8} q^{3/4-8\varepsilon} < \tfrac{1}{8}, \qquad q = eL^{-N/2}. \tag{6.xix}$$

The total lattice mass is $\boldsymbol{\mu}_N = \bar{m}\eta + m_{0,N}(e)$. ■□

Proof. For $R = R_{\mathrm{top}}$ use Thm. J. For the reference torus take $R = R^*$. The common threshold and $k_*(R) \le k_{\mathrm{top}}$ make the pair $P_N(e)$ and the kernel families on R_{top} available through the reference stop. Both trajectories start with that scalar pair and zero activities. Inductively suppose their scalar sequences agree, as do their activity and scalar families on flat polymers, through $j < k_*(R)$. The top mass bound $|m_j| \le w_j^{\mathrm{m}}/8$ and the reference torus's own Prop. 6.2 admit its next step with real data in $\mathrm{Ball}_j^x(1)$. The scale proposition supplies the domains and strict outgoing margin; Thm. I identifies the extracted mass, vacuum and box contribution, and the outgoing activity and external scalar families on flat polymers. The scalar recursions then give the same $m_{j+1}, \varepsilon_{j+1}^{\mathrm{v}}$, closing the induction. Restriction to zero preserves this agreement. The scalar bounds are inherited from the top trajectory; the full activity norm on the reference torus is bounded by its own continuation estimate; agreement on flat polymers does not bound this norm. This proves eq. 6.xviii.

The fixed-point problem defining $P_N(e)$ on R_{top} depends only on $(N, e, \bar{m})$ and the fixed construction, including its kernel families. The smaller side is absent; twists are subsequent harmonic

insertions in the density, and the fixed-point map uses $\mathscr{C} = 0$. Thus a twist, external connection or probe does not enter the pair. For $N \geq N_{\min}(R_{\mathrm{top}}, e)$, the top initial mass bound gives eq. 6.xix; at lower cutoffs the assigned mass is zero. Both cases are real, and $q \leq e \leq \bar{e}_1 \leq 1/3$ makes the positive power in that bound less than one. □

On the reference torus, for $N \geq N_{\mathrm{ref}}(e)$, the outputs needed at the terminal scale are

$$a_* \in \{h/L, h/L^2\}, \qquad |m_*| \leq \tfrac{1}{8} e_*^{3/4-8\varepsilon}, \qquad \|E_*\|_* \leq \|\mathscr{E}_*\|_*^x \leq 4 e_*^{1/4-6\varepsilon}. \tag{6.xx}$$

The kernel estimates have the same fixed constants on every torus and at every cutoff. We now integrate the resulting terminal fields and compare the partition functions on the two reference tori.

6.2. **The terminal integral representation.** We first put the curvature Gaussian in coordinates adapted to all block steps. The energy splits into the terminal photon energy and a sum of quadratic fluctuation terms. Under this Gaussian law, the fluctuation coordinates are independent standard Gaussians. Iterating the exact density identity in these coordinates then gives the terminal fermion integral on the small-field set.

6.2.1. *The downward photon coordinates.* Fix a flow side and cutoff as in Prop. 6.1. Put $k = k_*(R)$ and $n = R/\eta = L^k \ell_*$. On the unit torus of side $\ell_j = n/L^j$ use the fluctuation space V_j and its fixed orthonormal basis, and the maps $A_j^{\min}, \mathrm{Sq}_j$ of the photon construction. For a real bond field A on the terminal unit lattice and $\zeta = (\zeta_j)_{j<k}$ define

$$A_k^{\downarrow} = A, \qquad A_j^{\downarrow} = A_j^{\min}(A_{j+1}^{\downarrow}) + \mathrm{Sq}_j \zeta_j, \qquad j = k-1, \dots, 0. \tag{6.xxi}$$

The last field is denoted $A_0^{\downarrow}(A, \zeta)$. These fields are linear in their data.

For any field A_0 on the original unit lattice, define its block fields by $A_j = (\mathbf{Q}^{(j)} A_0)^{[j]}$, where $\mathbf{Q}^{(j)}$ is the j-fold photon average on the original unit torus and $B^{[j]}(b) = L^{j/2} B(L^j b)$. The identities for block averaging and scaling give $(A_{j+1})_L = \mathbf{Q} A_j$. Let Ax_k be the fields A_0 for which $\tau A_j = 0$ for $j < k$. For each unit torus put $V_{\mathrm{t}} = \delta C^2$, the real coclosed bond fields with mean zero. On the fine torus let $F_N = dC^1(n)$ with its counting inner product and standard Gaussian γ_N. The slice is

$$\mathrm{Sl}_k = \{A_0 \in \mathrm{Ax}_k : A_k \in V_{\mathrm{t}}(T_k^0)\}.$$

We will use the elementary cochain identities

$$C^1 = dC^0 \oplus \{\text{constant bond fields}\} \oplus \delta C^2, \qquad \ker d = dC^0 \oplus \{\text{constant bond fields}\}. \tag{6.xxii}$$

The sum is orthogonal. Indeed $dd = 0$, so dC^0 and δC^2 are orthogonal, and their common orthogonal complement is $\ker(d\delta + \delta d)$. Commuting lattice differences make this operator the scalar Laplacian in each component. Its kernel consists of constants: the energy identity forces every nearest-neighbor difference to vanish. This proves eq. 6.xxii. Consequently d is injective on V_{t} and, on a unit torus with ℓ^3 sites, $\dim V_{\mathrm{t}} = \dim dC^1 = 2\ell^3 - 2$.

Proposition 6.3. The map $(A, \zeta) \mapsto A_0^{\downarrow}(A, \zeta)$ is a linear bijection from $C^1(T_k^0) \times \prod_{j<k} V_j$ to Ax_k. The block fields of its image are exactly $A_j^{\downarrow}$. The coboundary is a linear bijection $d : \mathrm{Sl}_k \to F_N$. Thus curvature has coordinates

$$\mathcal{J}_{\mathrm{ph}} : F \longmapsto (A_*; \zeta_0, \dots, \zeta_{k-1}), \qquad F = dA_0^{\downarrow}(A_*, \zeta), \tag{6.xxiii}$$

where $A_* \in V_{\mathrm{t}}(T_k^0)$. They satisfy the exact energy identity

$$\|F\|^2 = \|dH_k^{\mathrm{ph}} A_*\|_2^2 + \sum_{j<k} \|\zeta_j\|^2. \tag{6.xxiv}$$

The image of γ_N under $\mathcal{J}_{\mathrm{ph}}$ is

$$\mathsf{dm}_*(A_*) \prod_{j<k} \mathrm{d}\mu_I(\zeta_j), \tag{6.xxv}$$

where the terminal centered Gaussian on $V_{\mathrm{t}}(T_k^0)$ is nondegenerate and has density proportional to

$$\exp(-\|dH_k^{\mathrm{ph}}A\|_2^2/2).$$

There are orthonormal vectors $\phi_{j,\Upsilon} \in F_N$ such that $\zeta_j(\Upsilon) = \langle \phi_{j,\Upsilon}, F\rangle$. For each terminal unit plaquette p there is $\phi_p \in F_N$ with

$$(dA_*)(p) = \langle \phi_p, F\rangle, \qquad \|\phi_p\| \le 1. \tag{6.xxvi}$$

There are $3\ell_*^3$ such plaquettes, and

$$\dim V_j = (2 - 2L^{-3})\ell_j^3 \le 2\ell_*^3 L^{3(k-j)}. \tag{6.xxvii}$$

■

Proof. The one-step minimizer and fluctuation maps obey $\mathbf{Q}A_j^{\min}(A') = A'_L$, $\tau A_j^{\min}(A') = 0$, $\mathbf{Q}\mathrm{Sq}_j = 0$ and $\tau \mathrm{Sq}_j = 0$. Apply $\mathbf{Q}$ to eq. 6.xxi; the injective scaling map then identifies the next block field with $A_{j+1}^{\downarrow}$. Starting from $A_0 = A_0^{\downarrow}$ proves this at all levels, including $\tau A_j = 0$. Conversely, for $A_0 \in \mathrm{Ax}_k$ its difference $A_j - A_j^{\min}(A_{j+1})$ is annihilated by $\mathbf{Q}$ and τ. The coordinate bijection eq. 3.xxvii and the invertible square root give exactly one ζ_j with this difference equal to $\mathrm{Sq}_j\zeta_j$. Downward recursion recovers A_0. This proves the first bijection and its inverse.

The orthogonality eq. 3.xxxv and the Gaussian fluctuation decomposition give at each level

$$\|dH_j^{\mathrm{ph}}A_j\|_2^2 = \|dH_{j+1}^{\mathrm{ph}}A_{j+1}\|_2^2 + \|\zeta_j\|^2.$$

Summing, and using $H_0^{\mathrm{ph}} = I$, proves eq. 6.xxiv. The dimension of the slice is

$$2\ell_k^3 - 2 + \sum_{j<k}(2\ell_j^3 - 2\ell_{j+1}^3) = 2n^3 - 2 = \dim F_N.$$

If $dA_0 = 0$ on the slice, the energy identity forces every $\zeta_j = 0$ and $dH_k^{\mathrm{ph}}A_k = 0$. Since $\mathbf{Q}_k H_k^{\mathrm{ph}} = I$ and $d\mathbf{Q}_k = \mathbf{Q}_k^{(2)}d$, this implies $dA_k = 0$. The injectivity of d on V_{t} gives $A_k = 0$, whence $A_0 = 0$ by the downward bijection. Equality of dimensions therefore proves that $d : \mathrm{Sl}_k \to F_N$ is bijective and gives eq. 6.xxiii.

Equip the coordinate space with the scalar product

$$\langle (A,\zeta), (A',\zeta')\rangle_{\mathrm{ph}} = \langle A, \Delta_k A'\rangle + \sum_{j<k}\langle \zeta_j, \zeta'_j\rangle.$$

Its terminal part is positive definite: $\langle A, \Delta_k A\rangle = \|dH_k^{\mathrm{ph}}A\|_2^2 \ge \|dA\|_2^2 > 0$ for nonzero $A \in V_{\mathrm{t}}$. Here the block curvature contraction follows from eq. 3.ii. The energy identity and polarization make $\mathcal{J}_{\mathrm{ph}}$ an isometry for this scalar product. It therefore sends the standard Gaussian to the product of the normalized Gaussians on the orthogonal summands, proving eq. 6.xxv. The terminal volume measure differs from ordinary Lebesgue measure on V_{t} by a constant factor, so its Gaussian has the stated density. Positivity of its quadratic form gives nondegeneracy.

For a basis vector Υ in V_j, let $\mathbf{e}_{j,\Upsilon}$ be the coordinate vector with that single nonzero entry. Its inverse image under the isometry is $\phi_{j,\Upsilon}$. The coordinate vectors are orthonormal, proving orthonormality and the required coordinate formula. The terminal curvature functional satisfies

$$|(dA_*)(p)| \le \|dA_*\|_2 = \|\mathbf{Q}_k^{(2)}dH_k^{\mathrm{ph}}A_*\|_2 \le \|dH_k^{\mathrm{ph}}A_*\|_2 \le \|F\|.$$

It is consequently represented by a vector of norm at most 1 in the finite Euclidean space F_N, giving eq. 6.xxvi. Finally eq. 3.xxviii and $\ell_j = L^{k-j}\ell_*$ give eq. 6.xxvii. □

If a terminal field A^c is closed, linearity gives the difference of downward fields as $A_0^{\downarrow}(A^c, 0)$. For a closed input the background identity eq. 3.xxxvii gives zero curvature energy at the preceding level. Applying the average shows that its block minimizer is closed. The identity $\mathbf{Q}A_j^{\min}(A') = A'_L$

and preservation of means by the block average give $\operatorname{mean}(A_j^{\min}(A')) = L^{-1/2}\operatorname{mean}(A')$. Its final mean is $L^{-k/2}\operatorname{mean}(A^c)$. eq. 6.xxii therefore gives

$$A_0^{\downarrow}(A + A^c, \zeta) = A_0^{\downarrow}(A, \zeta) + L^{-k/2}\operatorname{mean}(A^c) + d\lambda. \tag{6.xxviii}$$

In particular adding a coarse gradient adds a fine gradient. For the harmonic twist $c_{\theta,\mu} = \theta_\mu/(e_*\ell_*)$, $L^{-k/2}c_\theta = H_\theta$, where $H_{\theta,\mu} = \theta_\mu/(qn)$. Thus its downward shift is $H_\theta + d\lambda$.

6.2.2. *The terminal density with an external connection.* Let $(m_{\mathrm{in}}, \varepsilon_{\mathrm{in}}^{\mathrm{v}})$ be a controlled real pair on R, with the trajectory of §6.1. An external connection is a complex bond field a on the original unit lattice satisfying $a^{[j]} \in \mathfrak{X}_j$ for every $j \le k$; here $a^{[j]}(b) = L^{j/2}a(L^j b)$. Write $\mathscr{C}_* = a^{[k]}$. For $A \in \mathrm{Dom}_k$ put $\mathscr{A} = H_k^{\mathrm{ph}}A$, $B = \mathscr{A} + \mathscr{C}_*$, and insert the fermion fields $H_k^{\#}(B)\Psi$ and their barred partners. The terminal function is

$$\begin{aligned} Y_*^x(A; \mathscr{C}_*) = {} & \det D_k(0)^{-1} \exp\left(-\varepsilon_k^{\mathrm{v}}\ell_*^3 + \sum_X \mathfrak{P}_k(X, \mathscr{C}_*)\right) \\ & \times \int \mathrm{d}\Psi \, \exp\left(-\langle\bar\Psi, D_k(B)\Psi\rangle - m_k\langle\bar\psi, \psi\rangle + \sum_X \mathscr{E}_k(X; \mathscr{A}, \mathscr{C}_*, \psi, \chi)\right). \end{aligned} \tag{6.xxix}$$

Since $k \ge 1$, this Berezin integral has $4\ell_*^3$ pairs with weight 1. The inserted chamber fields and kernels retain their fine weight $\eta_k^3/48$. The inverse and activity domains in the preceding sections make every term well defined, and $\det D_k(0) \ne 0$. At $\mathscr{C}_* = 0$, $\mathfrak{P}_k(X, 0) = 0$ and $\mathscr{E}_k|_0 = E_k$; we write $Y_*(A) = Y_*^x(A; 0)$.

For a field on the original unit lattice, let

$$w_N(B) = \exp(-\varepsilon_{\mathrm{in}}^{\mathrm{v}}n^3)\frac{\det(D^{\mathrm{ch}}[qB] + \bar m\eta + m_{\mathrm{in}})}{\det(D^{\mathrm{ch}}[0] + \bar m\eta)}. \tag{6.xxx}$$

These are $192n^3$-dimensional matrices with phase argument qB. Coercivity of the free matrix at positive mass makes the denominator nonzero. The weighted Gaussian identity for Berezin integration identifies the initial integrated density with $w_N(A_0 + a)$, including its vacuum and measure factors.

The fine weight is entire and gauge invariant by eq. 11.xii: multiplication by $\exp(-iq\lambda(c))$ cancels $d\lambda$ in the external hop, while internal hops remain over c.

The same stencil bounds every row sum at $B_0 + a$, with B_0 real, by $|\boldsymbol{\mu}| + 41 + 6\exp(q\|\operatorname{Im} a\|_\infty)$, where $\boldsymbol{\mu} = \bar m\eta + m_{\mathrm{in}}$. The diagonal and internal hops contribute less than $|\boldsymbol{\mu}| + 41$, and the external hop at most $6\exp(q\|\operatorname{Im} a\|_\infty)$. The determinant permutation sum is bounded by the product of row sums, giving

$$|w_N(B_0 + a)| \le B_N(a) := \frac{\exp(-\varepsilon_{\mathrm{in}}^{\mathrm{v}}n^3)}{|\det(D^{\mathrm{ch}}[0] + \bar m\eta)|}\left(|\boldsymbol{\mu}| + 41 + 6\exp(q\|\operatorname{Im} a\|_\infty)\right)^{192n^3} < \infty. \tag{6.xxxi}$$

This bound is used for integrability at a fixed cutoff.

Let $A(F) \in V_{\mathrm{t}}$ be the transverse fine potential of $F \in F_N$, and define the finite complex measure

$$\mathrm{d}\nu^{\theta,a}(F) = w_N(A(F) + H_\theta + a)\,\mathrm{d}\gamma_N(F).$$

The small-field event and the terminal indicator are

$$\mathfrak{G} = \{F : |\zeta_j(\Upsilon)| \le p_{0,j} \ (j < k), \ \|dA_*\|_\infty \le p_*\}, \qquad \chi_*(A) = \mathbf{1}[\|dA\|_\infty \le p_*]. \tag{6.xxxii}$$

Here A_*, ζ are the actual coordinates eq. 6.xxiii. Products with χ_* are extended by zero off its support.

Theorem L. For every $A \in \mathrm{Dom}_k$,

$$Y_*^x(A;\mathscr{C}_*) = \int \prod_{j<k} (\mathbf{1}_{\mathrm{Box}_j}(\zeta_j)\,\mathrm{d}\mu_I(\zeta_j)) w_N(A_0^{\downarrow}(A,\zeta)+a). \tag{6.xxxiii}$$

The integrand is continuous on the compact product of boxes. If $A, A+d\lambda \in \mathrm{Dom}_k$, with λ real, their terminal functions agree. For every bounded Borel function Φ on $V_{\mathrm{t}}(T_k^0)$ and every real twist θ,

$$\nu^{\theta,a}(\mathbf{1}_{\mathfrak{G}}\Phi(A_*)) = \int \mathsf{dm}_*(A)\,\Phi(A)\chi_*(A)Y_*^x(A+c_\theta;\mathscr{C}_*). \tag{6.xxxiv}$$

Its right-hand density is bounded and Borel, and $Y_*^x(A+c_\theta;\mathscr{C}_*)$ is jointly continuous in (θ,A) on $\{\chi_*=1\}$. ■□

Proof. The fixed charge choice gives eq. 3.l at every scale. Prop. 3.15 admits real fields of curvature at most p_j and preserves Dom_j under each downward step with a box fluctuation. Induction therefore validates eq. 6.xxi in every domain for the stated top field and boxes.

Let $y_j^x(A;a^{[j]})$ be the integrated density at level j along this trajectory. The exact integrated step gives

$$y_{j+1}^x(A';a^{[j+1]}) = \int_{\mathrm{Box}_j} \mathrm{d}\mu_I(\zeta_j)\, y_j^x(A_j^{\min}(A')+\mathrm{Sq}_j\zeta_j;a^{[j]}).$$

Its external connection has the required scaling because $(a^{[j+1]})_L = a^{[j]}$, directly from the two scaling factors. Its coupling maps are the actual trajectory maps, and its vacuum includes the box normalization. The first such identity integrates the full $192n^3$ weighted chamber pairs into $4\ell_1^3$ coarse pairs.

On a compact product of fluctuation boxes, the downward composition is continuous and bounded by eq. 6.xxxi. Starting from $y_0^x(A_0;a) = w_N(A_0+a)$, iterate the displayed step with Fubini. This gives eq. 6.xxxiii, retaining the unnormalized factors $\mathbf{1}_{\mathrm{Box}_j}\,\mathrm{d}\mu_I$, and proves continuity. eq. 6.xxviii turns a coarse real gradient into a fine gradient, so gauge invariance of w_N proves that of Y_*^x.

To integrate against an arbitrary bounded Borel observable, first express the fine density in the curvature coordinates. The slice potential $A_0^{\downarrow}(A_*,\zeta)$ has mean zero: its block average has the mean of the fine field, and its top field has mean zero. Its difference from $A(F)$ is closed and has mean zero, hence is a gradient by eq. 6.xxii. The harmonic shift identity gives

$$w_N(A(F)+H_\theta+a) = w_N(A_0^{\downarrow}(A_*+c_\theta,\zeta)+a).$$

Under the tower isometry the Gaussian becomes eq. 6.xxv, and the event $\mathfrak{G}$ has indicator

$$\chi_*(A_*)\prod_{j<k}\mathbf{1}_{\mathrm{Box}_j}(\zeta_j).$$

The integrand is a bounded Borel function with bound $\|\Phi\|_\infty B_N(a)$. The change-of-variables identity for its image measure and Fubini therefore give

$$\nu^{\theta,a}(\mathbf{1}_{\mathfrak{G}}\Phi(A_*)) = \int \mathsf{dm}_*(A)\,\Phi(A)\chi_*(A)\int\prod_{j<k}(\mathbf{1}_{\mathrm{Box}_j}\,\mathrm{d}\mu_I(\zeta_j))w_N(A_0^{\downarrow}(A+c_\theta,\zeta)+a).$$

Where $\chi_*(A)=1$, the field $A+c_\theta$ has the same curvature bound and belongs to Dom_k. The inner integral is $Y_*^x(A+c_\theta;\mathscr{C}_*)$ by eq. 6.xxxiii, proving eq. 6.xxxiv.

For all real (θ,A) the inner integral is bounded by $B_N(a)$. Its continuous integrand is uniformly continuous on each compact parameter neighborhood times the compact box product; integration therefore gives joint continuity. Multiplication by the indicator of the closed set $\{\chi_*=1\}$ gives the stated bounded Borel density.

□

7. The renormalization group: large fields

The small-field integral excludes photon configurations outside the prescribed fluctuation domains. We bound their probability uniformly in the cutoff. These estimates concern the normalized zero-flux law, so the all-flux Gaussian bound alone is insufficient: the sector projection must remain in the slice trace. We use the photon coordinates of §6.2.1, the terminal conventions of §11.3, and the zero-flux normalization of §11.2.4.

7.1. **Gaussian bounds in the zero-flux sector.** Average the measures $\nu^{\theta,0}$ of §6.2.2 over the harmonic twist:

$$\bar{\nu} = \int_{[-\pi,\pi]^3} \nu^{\theta,0} \frac{\mathrm{d}\theta}{(2\pi)^3}, \qquad \bar{\mu} = \bar{\nu}/\bar{\nu}(1).$$

For a real lattice mass, the two spin blocks of the chamber operator are adjoints, so $\nu^{\theta,0}$ is positive. The zero-sector identity eq. 11.xxxiv gives $0 < \bar{\nu}(1) < \infty$ and identifies $\bar{\mu}$ with the zero-flux curvature law. To prove Gaussian domination for this conditional law, we retain the sector projection in the coordinate-slice trace.

On $C(n_1, n_2, n_3)$ use the spaces H^s, H^l, P_G, E, and M_0 of eq. 2.lxxix and eq. 2.lxxxi. Let P_k project onto the magnetic-flux summand k. Choose coordinate slice cycles S_2, S_3, through a fixed slice site, and for $\xi \in [0,1]^2$ set $b_\xi = q(\xi_2 \mathbf{1}_{S_2} + \xi_3 \mathbf{1}_{S_3})$. For bounded magnetic factors u_j and electric factors v_j with sub-Gaussian growth, let K_j^ξ have the periodized convolution kernel

$$v_j(A' - A) e^{i\langle b_\xi, A' - A\rangle} \prod_b g_1(A'(b) - A(b)).$$

The zero-flux trace formula is

$$Z^{(0)}\left[\prod_j u_j(F_j^{\mathrm{mag}}) v_j(F_j^{\mathrm{el}})\right] = (2\pi)^{n_1 n_2 n_3} \int_{[0,1]^2} \operatorname{tr}\left(P_0 \prod_{j=n_1-1,\dots,0} r_q^{n_2 n_3} K_j^\xi P_G M_0 u_j(F^{\mathrm{mag}}) M_0^*\right) \mathrm{d}\xi. \tag{7.i}$$

To derive it, sum the electric curvature along S_2, S_3 and over time. The sums are $r_q k_{12}$ and $r_q k_{13}$. The added electric phases are therefore $e^{2\pi i(\xi_2 k_{12} + \xi_3 k_{13})}$. Restricting the first magnetic slice to flux k_{23} inserts $P_{k_{23}}$ in the word trace, since the sum of $d_s A$ on the slice vanishes. All operators preserve that summand. The flux sum is absolutely convergent by the curvature Gaussian bound on the modulus of the integrand. Set $k_{23} = 0$ and integrate over ξ; Fourier orthogonality keeps precisely $k_{12} = k_{13} = 0$. This proves eq. 7.i.

Proposition 7.1. For the zero-flux law $\bar{\mu}$, averaged over harmonic connections, at every real lattice mass, every coordinate side $n_\mu \geq 4$, and every real two-cochain g,

$$\bar{\mu}(e^{\langle g, F^{\mathrm{u}}\rangle}) \leq e^{\|\Pr_{F_N} g\|^2/2}. \tag{7.ii}$$

In particular, for $\phi \in F_N$ with $\|\phi\| \leq 1$ and every $a \geq 0$, $\bar{\mu}\{|\langle \phi, F^{\mathrm{u}}\rangle| > a\} \leq 2e^{-a^2/2}$. ∎

Proof. Put $\mathcal{Z}(\beta) = Z^{(0)}[e^{-\langle \beta, F\rangle - \|\beta\|^2/2}]$. It is finite by the all-flux bound and is positive because the sector contains the $z = 0$ term, which has positive integral; the shifted photon weight is strictly positive. For time direction 1, split β into electric parts β_j^{el} and magnetic parts β_j^{mag}. Take the corresponding exponential u_j and v_j in eq. 7.i, replacing u_j first by $u_j^\delta = \min(u_j, \delta^{-1})$. The bounded magnetic factors make the word trace applicable.

Completing the electric Gaussian gives

$$K_j^\xi = z_j U_j e^{-|E - b_\xi|^2/2}, \qquad U_j = e^{i\langle \beta_j^{\mathrm{el}}, E\rangle}, \qquad |z_j| = 1.$$

Let $K_\xi = r_q^{n_2 n_3} e^{-|E-b_\xi|^2/2} P_G$. It is positive, and its square root is $r_q^{n_2 n_3/2} e^{-|E-b_\xi|^2/4} P_G$. The operator $K_\xi^{1/2} M_0$ is Hilbert–Schmidt. The periodized square-root Gaussian has the additional phase

$e^{i\langle b_\xi, A'-A\rangle}$; its modulus is bounded by the kernel majorant used in eq. 2.lxxxii. The same real-slice integral and flux sum therefore prove this assertion uniformly in ξ. Multiplication by bounded $(u_j^\delta)^{1/2}$ preserves it.

Put $Y_j = K_\xi^{1/2} M_0 (u_j^\delta)^{1/2}$. Cyclicity turns the trace into $\prod_j z_j$ times the closed chain

$$\operatorname{tr} \prod_j U_j P_0 Y_j (P_0 Y_j)^*.$$

All factors preserve the magnetic-flux summand; U_j is unitary. Schatten Hölder with $2n_1$ factors of exponent $2n_1$ bounds this trace by

$$\prod_j \left(\operatorname{tr}_{H^s} P_0 (Y_j^* Y_j)^{n_1} \right)^{1/n_1}.$$

Hölder in ξ and the Bloch trace formula identify its jth integrated factor with the zero-flux functional having the same magnetic factor at every slice and no electric factor. Let $\beta^{(j)}$ denote that purely magnetic two-cochain. Monotone convergence as $\delta \downarrow 0$ gives

$$\mathcal{Z}(\beta) \le \prod_j \mathcal{Z}(\beta^{(j)})^{1/n_1}.$$

Rotate the torus by an orientation-preserving cyclic coordinate permutation. Zero flux and the pairing are preserved, and $\beta^{(j)}$ becomes purely electric for the new time direction. The same inequality then bounds $\mathcal{Z}(\beta^{(j)})$ by $\mathcal{Z}(0)$. Hence $\mathcal{Z}(\beta) \le Z^{(0)}[1]$.

The zero-sector identity eq. 11.xxviii identifies its normalized curvature law with $\bar\mu$. Since this law is supported on F_N, the pairing with g equals that with $\Pr_{F_N} g$. Apply the preceding inequality with the negative of that projection and divide by the positive zero-sector partition function. This proves eq. 7.ii. Exponential Markov applied to the two signs and optimized at parameter a gives the stated tail bound. □

7.2. **The large-field complement.** At zero external connection write $\nu^\theta = \nu^{\theta,0}$.

Fix $C_0 \ge 1$ to dominate the photon reconstruction row sums, the fermion field-map row sums, and the coefficient bound for the terminal Dirac quadratic form. Require also $C_0^2 \ge C_K$, where C_K is the first-derivative constant for the terminal coarse Dirac operator in eq. 11.lxi. The common calibration eq. 13.iv satisfies these requirements using only the fixed kernel constants. Set

$$\varepsilon_3 = 6\ell_*^3 \left(e^{-p_{0,*}/2} + e^{-p_*^2/(2C_0^2)} \right), \qquad \varepsilon_4 = 6\ell_*^3 e^{-p_*^2/(2C_0^2)}. \tag{7.iii}$$

Proposition 7.2. Suppose $v_* \ge 8$, the integer $p_0 \ge 1$ satisfies $p_0 v_*^{p_0-1} \ge 28$, and $L \ge 5$. Then, at every real initial pair,

$$\bar\mu(\mathfrak{G}^c) \le \varepsilon_3, \qquad \mathsf{m}_*(\chi_* = 0) \le \varepsilon_4. \tag{7.iv}$$

For fixed $(\bar m, e, R)$, these bounds are independent of the ultraviolet cutoff and of the initial pair. ■

Proof. The zero-flux domination estimate in Prop. 7.1 and the rows of eq. 6.xxvi give

$$\bar\mu\{|\zeta_j(\Upsilon)| > p_{0,j}\} \le 2e^{-p_{0,j}^2/2}, \qquad \bar\mu\{|(dA_*)(p)| > p_*\} \le 2e^{-p_*^2/2}.$$

Using the coordinate counts in eq. 6.xxvii, the union bound is

$$\bar\mu(\mathfrak{G}^c) \le 4\ell_*^3 \sum_{i=1}^{k} L^{3i} e^{-(v_* + i\log L/2)^{2p_0}/2} + 6\ell_*^3 e^{-p_*^2/2}. \tag{7.v}$$

Here $i = k - j$ and $v_j = v_* + i\log L/2$. The first two terms of the binomial expansion give

$$p_{0,j} = (v_* + i\log L/2)^{p_0} \ge p_{0,*} + 28 i \log L/2.$$

Since $p_{0,j} \ge 1$, its square is at least $p_{0,j}$. It follows that

$$L^{3i} e^{-p_{0,j}^2/2} \le e^{-p_{0,*}/2} L^{-4i}.$$

The Gaussian tail thus overcomes the L^{3i} growth in the number of coordinates: after counting them, the bound decays as L^{-4i}. Its geometric sum is $(L^4-1)^{-1} \le 1$, so the right-hand side of eq. 7.v is at most $4\ell_*^3 e^{-p_{0,*}/2} + 6\ell_*^3 e^{-p_*^2/2}$. This is at most ε_3, since $C_0 \ge 1$.

The image of γ_N under $F \mapsto A_*$ is m_* by eq. 6.xxv. For each plaquette the variable $\langle \phi_p, F\rangle$ is Gaussian with variance $\|\phi_p\|^2 \le 1$. Its two-sided exponential Markov bound is $2e^{-p_*^2/2}$. Summing over the $3\ell_*^3$ plaquettes proves $\mathsf{m}_*(\chi_* = 0) \le 6\ell_*^3 e^{-p_*^2/2} \le \varepsilon_4$. Finally, ℓ_* and e_* are independent of N after the mesh threshold, by Prop. 6.1; hence the displayed errors have the asserted dependence. □

8. Ultraviolet stability of the partition function

We combine the common trajectory with the large-field estimates to bound the reference partition functions uniformly in the ultraviolet cutoff. Free-pressure and twist estimates isolate the normalization terms; the terminal integral bounds the remaining interaction. Their cancellation gives the reference-volume comparison used in the following infrared step. The terminal estimates use the notation and norms of §11.3 and the density identity of §6.2.2. The reference partition defect is defined in §9.1.

8.1. **Free pressure and twist dependence.** The partition comparison requires an error in the free pressure which depends on the physical size, rather than on the number of lattice sites. It also requires a twist estimate for the operator with a block penalty. The latter penalty removes the small-mass singularity before the fine mesh is taken to zero.

For $0 < m \le 1$ let $\mathfrak{d}_m(p) = \det \mathcal{D}_m(p)$, where $\mathcal{D}_m$ is the 192-dimensional chamber symbol in §4.1.2, and set

$$p^0_{\mathrm{ch}}(m) = \int_{[-\pi,\pi]^3} \log \mathfrak{d}_m(p) \frac{\mathrm{d}p}{(2\pi)^3}. \tag{8.i}$$

The spin-block identity and positive real part imply $\mathfrak{d}_m(p) > 0$ for real p. In this subsection $|z| = |z|_\infty$ for $z \in \mathbb{Z}^3$. Write $\kappa_{(0)} = \kappa_{\mathrm{strip}}$ from Prop. 4.2.

Proposition 8.1. There is a finite C_6, depending only on ρ, such that, with

$$\varepsilon_5(x) = C_6 \log \frac{1}{1 - e^{-\kappa_{(0)} x}} \qquad (x > 0), \tag{8.ii}$$

the free twisted determinant on a unit torus of side $n \ge 1$ satisfies

$$\left| \log \det(D^{\mathrm{ch}}[qH_\theta] + m) - n^3 p^0_{\mathrm{ch}}(m) \right| \le \varepsilon_5(mn). \tag{8.iii}$$

Here $H_{\theta,\mu} = \theta_\mu/(qn)$ and $\theta \in \mathbb{R}^3$. For $m = \bar{m}\eta$ and $n = R/\eta$ define

$$z^{\mathrm{free}}_{r,n} = \int_{[-\pi,\pi]^3} \det(D^{\mathrm{ch}}[qH_\theta] + \bar{m}\eta) \frac{\mathrm{d}\theta}{(2\pi)^3}.$$

Then $z^{\mathrm{free}}_{r,n} > 0$ and

$$\left| \log z^{\mathrm{free}}_{r,n} - n^3 p^0_{\mathrm{ch}}(\bar{m}\eta) \right| \le \varepsilon_5(\bar{m}R). \tag{8.iv}$$

■

Proof. Put $P_{n,\theta} = \{(2\pi j + \pi \mathbf{1} + \theta)/n : j \in \{0,\dots,n-1\}^3\}$. The antiperiodic plane waves have momenta $p \in P_{n,0}$ and diagonalize the translation part of the chamber operator. The external phase in the fourth hop evaluates its symbol at $p + \theta/n \in P_{n,\theta}$, whereas the internal hops are unchanged. Thus

$$\det(D^{\mathrm{ch}}[qH_\theta] + m) = \prod_{p \in P_{n,\theta}} \mathfrak{d}_m(p) > 0. \tag{8.v}$$

The phases contain $qH_\theta = \theta/n$, so the determinant is independent of q.

Factor the determinant by the light–heavy decomposition:

$$\log \mathfrak{d}_m(p) = f_H(p) + f_S(p), \qquad f_H(p) = \log\det D_{\mathrm{hh}}(p), \quad f_S(p) = \log\det B_m(p).$$

Both logarithms are real and smooth on real momenta. Their derivatives, which suffice for contour shifting, are

$$\partial_\mu f_H = \operatorname{tr}(D_{\mathrm{hh}}^{-1}\partial_\mu D_{\mathrm{hh}}), \qquad \partial_\mu f_S = \operatorname{tr}(B_m^{-1}\partial_\mu B_m).$$

The heavy block has dimension 188. On the strip of width θ_0 its inverse has norm at most $2/g_{\mathrm{h}}$, and the fourth-hop derivative has norm at most $3(1+\rho)e^{\theta_0} = 3(1+\rho) + g_{\mathrm{h}}/2$. Hence

$$|\partial_\mu f_H| \le C_{\mathrm{lH}} := 188\left(1 + \frac{6(1+\rho)}{g_{\mathrm{h}}}\right).$$

For $z \neq 0$ choose $|z_\mu| = |z|$. One integration by parts and a shift of that contour by $-\mathrm{i}\theta_0 \operatorname{sgn}(z_\mu)$ give

$$|\widehat{f}_H(z)| \le C_{\mathrm{lH}}|z|^{-1}e^{-\theta_0|z|}. \tag{8.vi}$$

The derivative trace is holomorphic in a neighborhood of the closed strip. Its periodicity cancels the vertical sides of the contour rectangle; no choice of a complex logarithm is needed.

For the light factor put $r_m(p') = (|p'|_2^2 + m^2)^{1/2}$. The inverse derivative bounds eq. 4.xiii hold on $|\operatorname{Im} p_\nu| \le \kappa_{(0)}m$. The bound K_s for B_m on the strip of width θ_0 and Cauchy's formula on polydiscs of radius $\theta_0/2$ give $\|\partial^\beta B_m\| \le 24K_s(2/\theta_0)^{|\beta|}$ for $1 \le |\beta| \le 4$. Leibniz's rule in the derivative trace therefore gives

$$|\partial^\alpha\partial_\mu f_S(p' + \mathrm{i}p'')| \le C_{\mathrm{lS}}\, r_m(p')^{-1-|\alpha|} \le C_{\mathrm{lS}}|p'|_2^{-1-|\alpha|} \quad (|\alpha| \le 3,\ p' \neq 0), \tag{8.vii}$$

where one may take

$$C_{\mathrm{lS}} = \max_{|\alpha|\le 3} 96K_s \sum_{\beta\le\alpha} \binom{\alpha}{\beta} C_\beta(\rho)(2/\theta_0)^{|\alpha|-|\beta|+1}(3\pi^2+1)^{(|\alpha|-|\beta|)/2}.$$

This constant is independent of m. Shift the differentiated Fourier integral by $-\mathrm{i}\kappa_{(0)}m\operatorname{sgn}(z_\mu)e_\mu$. The periodic Fourier estimate proved in Prop. 4.3, with singular order 1 and three derivatives, bounds this integral by $c_{\mathrm{F}}C_{\mathrm{lS}}(1+|z|)^{-2}$, where c_{F} is its constant at those orders. Thus

$$|\widehat{f}_S(z)| \le c_{\mathrm{F}}C_{\mathrm{lS}}|z|^{-1}(1+|z|)^{-2}e^{-\kappa_{(0)}m|z|}. \tag{8.viii}$$

The strip is contained in the heavy strip since $\kappa_{(0)}m \le \theta_0/2$. The inverse is holomorphic near the closed strip, so the contour shift is legitimate.

The Fourier coefficients of $f = f_H + f_S$ are summable. For a summable periodic Fourier series, summing over $P_{n,\theta}$ keeps precisely the indices $z = nw$:

$$\sum_{p\in P_{n,\theta}} f(p) = n^3 \sum_{w\in\mathbb{Z}^3} \widehat{f}(nw)(-1)^{w_1+w_2+w_3}e^{\mathrm{i}\theta\cdot w}. \tag{8.ix}$$

This follows by summing the geometric progression in each coordinate; absolute convergence permits the interchange. Its $w = 0$ term is $n^3 p^0_{\mathrm{ch}}(m)$. There are $24t^2 + 2 \le 26t^2$ indices with $|w| = t$. The contribution of eq. 8.viii is at most

$$26c_{\mathrm{F}}C_{\mathrm{lS}} \sum_{t\ge 1} t^{-1}e^{-\kappa_{(0)}mnt}.$$

For the heavy term use $\kappa_{(0)}m \le \theta_0/2$ and $x^2e^{-\theta_0 x/2} \le 16/(\exp(1)\theta_0)^2$. Its contribution is at most

$$\frac{416C_{\mathrm{lH}}}{(\exp(1)\theta_0)^2} \sum_{t\ge 1} t^{-1}e^{-\kappa_{(0)}mnt}.$$

Since $\sum_{t\ge1} e^{-xt}/t = \log(1-e^{-x})^{-1}$, eq. 8.iii follows with $C_6 = 26[c_{\mathrm{F}}C_{\mathrm{lS}} + 16C_{\mathrm{lH}}/(\exp(1)\theta_0)^2]$. Finally every twisted determinant lies between $\exp(n^3 p^0_{\mathrm{ch}}(m) \pm \varepsilon_5(mn))$. Its harmonic average lies in the same positive interval. Taking logarithms proves eq. 8.iv. □

We compare partitions only on $R \in \{R^*, LR^*\}$ at $N \ge N_{\mathrm{ref}}(e)$, using their common mass assignment. Prop. 6.1 and $N_{\min}(LR^*, e) \le N_{\mathrm{ref}}(e)$ give $k_* \ge 1$. By eq. 6.viii and $L^{-1} < \bar{m}h \le 1$, $\mu_* = \bar{m}a_* \in (L^{-3}, 1]$ on both tori. The terminal estimates in §11.3 apply: their fine input requires $0 < \mu \le 1$, and the coarse covariance bounds require $\mu \ge \mu_- = L^{-3}$. We recall their notation at the stop $k = k_*$. The operator D_k acts on four-component unit-lattice spinors, and its connection domain is

$$\mathcal{W} = \{A_r + C : A_r \in \widetilde{R}_k \text{ real},\ C \in \mathfrak{X}_k\}.$$

The photon domain $\widetilde{R}_k$ allows a locally gauged real background and a globally small remainder; $\mathfrak{X}_k$ is the external-connection domain already used in the determinant estimate. For a complex connection direction B, put $\epsilon_B = e_*|B|_\infty$. The positive radius c_τ is chosen with $c_\tau \le \mu_-/48$ so that the fine inverse persists on

$$\mathcal{W}^+ = \{A + A_1 : A \in \mathcal{W},\ e_*|A_1|_\infty \le c_\tau\}.$$

On this enlarged domain the coarse derivative satisfies $\|\partial_B D_k(A)\|_{\mathrm{op}} \le 2C'_K\epsilon_B$. For $A_r \in \mathcal{W}$ real and $e_*|A_1|_\infty \le c_\tau$, the coarse inverse satisfies

$$\|D_k(A_r + A_1)^{-1}\|_{\mathrm{op}} \le g_1, \qquad g_1 = b_-^{-1} + 4/\mu_-, \qquad b_- = 1 - L^{-1}.$$

These are eq. 11.lxvii, eq. 11.lxviii and eq. 11.lxxii; their constants are fixed independently of the charge, mesh and side.

Proposition 8.2. At either of these stops put $c_{\theta,\nu} = \theta_\nu/(e_*\ell_*)$ and $R^{\mathrm{tw}}_\theta = \det D_k(c_\theta)/\det D_k(0)$. There are fixed $C_\Sigma < \infty$ and $0 < \gamma_\Sigma \le 1$, independent of the charge, mesh and side, such that

$$\left|\log R^{\mathrm{tw}}_\theta - \log \frac{\det(D^{\mathrm{ch}}[qH_\theta] + \bar{m}\eta)}{\det(D^{\mathrm{ch}}[0] + \bar{m}\eta)}\right| \le C_\Sigma \ell_*^2 e^{-\gamma_\Sigma \ell_*}. \tag{8.x}$$

Both ratios are positive for real θ. ■

Proof. Write $\ell = \ell_*$ and $\mu = \mu_*$. Real constant fields lie in $\mathcal{W}$, since its real domain is invariant under real constants. If $|\operatorname{Im}\theta|_\infty \le c_\tau\ell$, then $c_\theta = c_{\operatorname{Re}\theta} + A_1$ with $e_*|A_1|_\infty \le c_\tau \le \mu_-/48$. Thus eq. 11.lxvii and eq. 11.lxxii give invertibility and $\|D_k(c_\theta)^{-1}\|_{\mathrm{op}} \le g_1$ throughout that strip. For real fields the two coarse spin blocks are adjoints, and the real covariance identity makes them invertible, so their determinant is positive. The same is true of the fine free determinant.

The direction $\partial_{\theta_\nu} c_\theta$ has $\epsilon_B = 1/\ell$. eq. 11.lviii and eq. 11.lxviii therefore give $\|\partial_{\theta_\nu} D_k(c_\theta)\|_{\mathrm{op}} \le 2C'_K/\ell$. There are $4\ell^3$ coarse components. For the real periodic function $F(\theta) = \log\det D_k(c_\theta)$, its derivative extends holomorphically near the closed strip and obeys

$$|\partial_{\theta_\nu} F| = |\operatorname{tr}(D_k^{-1}\partial_{\theta_\nu} D_k)| \le 8 g_1 C'_K \ell^2.$$

Periodicity follows by conjugation with the periodic gauge factor $e^{-2\pi i c_\nu/\ell}$. In each block average every rectilinear path has the same endpoint displacement, so its added constant-field phase is $e^{2\pi i(c_\nu - y_\nu)/\ell}$, which telescopes to the fine and coarse endpoint factors. Thus the coarse operator is conjugated too. Analytic continuation gives periodicity of the derivative trace on the complex strip as well.

For $m \in \mathbb{Z}^3 \setminus \{0\}$ choose $|m_\nu| = |m|_\infty$. Integration by parts in θ_ν, followed by shifting each contour by $-ic_\tau\ell\operatorname{sgn}(m_j)$, gives

$$|\widehat{F}(m)| \le 8 g_1 C'_K \ell^2 e^{-c_\tau \ell |m|_1}.$$

Only the derivative trace is shifted, so no complex logarithm is chosen. Its periodicity cancels every vertical side of the contour. The coefficients are absolutely summable. For $\tau \geq c_\tau$,

$$\sum_{m\neq 0} e^{-\tau|m|_1} = \left(\frac{1+e^{-\tau}}{1-e^{-\tau}}\right)^3 - 1 \leq \frac{8e^{-\tau}}{(1-e^{-c_\tau})^3}.$$

Subtracting the Fourier series at θ and zero proves $|\log R_\theta^{\mathrm{tw}}| \leq C_D \ell^2 e^{-c_\tau \ell}$, where $C_D = 128 g_1 C_K'/(1-e^{-c_\tau})^3$.

The free pressure estimate eq. 8.iii at θ and zero bounds the logarithm of the fine ratio by $2\varepsilon_5(\bar{m}R) = 2\varepsilon_5(\mu\ell)$. Since $\mu \geq \mu_-$ and $\ell \geq 1$,

$$2\varepsilon_5(\mu\ell) \leq \frac{2C_6}{1-e^{-\kappa_{(0)}\mu_-}} e^{-\kappa_{(0)}\mu_-\ell}.$$

The triangle inequality gives the proposition with

$$\gamma_\Sigma = \min\{c_\tau, \kappa_{(0)}\mu_-\}, \qquad C_\Sigma = C_D + \frac{2C_6}{1-e^{-\kappa_{(0)}\mu_-}}.$$

These constants are fixed before the physical charge. □

8.2. **The terminal remainder.** After removing the vacuum exponential and the free twist factor, the terminal density must be uniformly close to one on the allowed coarse fields. This relative-error bound gives the two-sided partition comparison after integration. We first collect the conditions needed to control that error.

Use the same C_0 as in §7.2, with the calibration eq. 13.iv. Set

$$c_\kappa = \sup_{\square} \sum_{X \supset \square} e^{-\kappa d_M(X)} \leq K_{\mathrm{pol}}.$$

In addition to the kernel and flow hypotheses, impose the following inequalities at the stop:

$$\begin{aligned} 8C_0^2\ell_*^6(b_-^{-1}+\mu_*^{-1}) &\leq e_*^{-\varepsilon/2}, \\ 4(2C_0+c_\kappa+5c_{\mathrm{fl}}+6)\ell_*^3 &\leq e_*^{-\varepsilon}, \\ e_*^{1/4-9\varepsilon} \leq \tfrac{1}{8}, \qquad 4C_0\ell_*^2 p_* &< e_*^{-\varepsilon}, \\ 12\ell_*^3\left(e^{-p_{0,*}/2}+e^{-p_*^2/(2C_0^2)}\right) \leq e_*, \qquad C_\Sigma\ell_*^2 e^{-\gamma_\Sigma\ell_*} &\leq e_*. \end{aligned} \tag{8.xi}$$

These are the terminal conditions used in the proof. The choice in Prop. 13.1 is required to satisfy them: the scale estimates give $Mv_*^r \leq \ell_* < LMv_*^r$ and $\mu_*^{-1} < L\ell_*$. Substitute the latter two bounds in the first four inequalities; use the lower bound in the last exponential inequality. Thus the conditions are uniform in the cutoff.

Put

$$\varepsilon_1 = \tfrac{1}{2}e_*^{1/4-9\varepsilon}, \qquad \varepsilon_2 = C_\Sigma\ell_*^2 e^{-\gamma_\Sigma\ell_*}. \tag{8.xii}$$

Together with eq. 7.iii, eq. 8.xi gives $\varepsilon_1 \leq 1/16$, $\varepsilon_2 \leq e_*$, and $\varepsilon_4 \leq \varepsilon_3 \leq e_*/2$.

For a mean-zero real A with $\|dA\|_\infty \leq p_*$, the gauge construction in eq. 3.xxxix–eq. 3.xl gives a real periodic λ with $A^g = A - d\lambda$ and $\|A^g\|_\infty \leq 2\ell_*^2 p_*$. Both $A + c_\theta$ and $A^g + c_\theta$ lie in Dom_k by Prop. 3.15. Put

$$\mathscr{A}^0 = H_k^{\mathrm{ph}} A^g, \qquad c_{\theta,\mu} = \theta_\mu/(e_*\ell_*), \qquad \mathscr{A} = \mathscr{A}^0 + c_\theta. \tag{8.xiii}$$

Each of its three photon norms obeys $\|\mathscr{A}^0\|_{\mathrm{each}} \leq 2C_0\ell_*^2 p_* < e_*^{-\varepsilon}/2$. The definition of Dom_k puts $\mathscr{A}$ in $\frac{1}{4}\widetilde{R}_k$. This domain is star-shaped and invariant under real constant shifts, so $s\mathscr{A}^0 + c_\theta \in \frac{1}{4}\widetilde{R}_k$ for $0 \leq s \leq 1$. Thus, for $C \in \mathfrak{X}_k$, the whole segment $s\mathscr{A}^0 + c_\theta + C$ lies in $\mathcal{W}$.

We also use the summed block version of the coarse derivative bound. Let P_B be the unit blocks containing an endpoint of a bond in $\operatorname{supp} B$, and let $|O|(y,y')$ be the ℓ^∞ operator norm of the coarse 4×4 block. For $A\in\mathcal{W}$, eq. 11.lxi gives

$$\sum_{y,y'}|\partial_B D_k(A)|(y,y')\le C_K\epsilon_B|P_B|.$$

The fine lattice in these stop-scale coordinates has mesh $\eta_*=L^{-k_*}$, distinct from the physical running length a_*. Its chamber indices have weight $\eta_*^3/48$; pair indices have the squared weight, and coarse spinor indices have weight one.

Lemma 8.3. For the stated fields and segment, assume $C_0^2\ge C_K$, $|m_k|\le e_*^{3/4-8\varepsilon}$, $\|\mathscr{E}_k\|_k^x\le e_*^{1/4-7\varepsilon}$. For $C\in\mathfrak{X}_k$ put $B=\mathscr{A}+C$ and use the fine and pair substitutions $H_k^{\#}(B)\Psi$ and their barred partners. Define

$$\begin{aligned}V_C=&-\langle\bar\Psi,[D_k(B)-D_k(c_\theta+C)]\Psi\rangle-m_k\langle\bar\psi,\psi\rangle\\&+\sum_X\mathscr{E}_k(X;\mathscr{A},C,\psi,\chi).\end{aligned}\tag{8.xiv}$$

If $K_C=D_k(c_\theta+C)^{-1}$ exists, $a_2\ge 1$ and every component row of K_C has absolute sum at most a_2, then

$$\left|\int \mathrm{d}\mu_{K_C}e^{V_C}-1\right|\le e^{|V_C|_{a_2}}-1,\qquad a_2=C_0^{-1}e_*^{-1/4+\varepsilon}.\tag{8.xv}$$

The coarse Berezin measure has $4\ell_*^3$ pairs of weight one; fine and pair indices retain weights $\eta_*^3/48$ and $(\eta_*^3/48)^2$. ■

Proof. Integrate eq. 11.lxi along the segment just verified, using $|P_{\mathscr{A}^0}|\le\ell_*^3$ and $e_*\|\mathscr{A}^0\|_\infty<e_*^{1-\varepsilon}/2$. Its block-entry sum is at most $C_Ke_*^{1-\varepsilon}\ell_*^3/2$; four spin components and a_2^2 give the first row below. The other rows are eq. 11.lxxiv and eq. 11.lxxiii, summing each polymer through one of its cubes:

term	coefficient-norm bound at a_2
coarse quadratic	$2C_KC_0^{-2}e_*^{1/2+\varepsilon}\ell_*^3$
mass	$4e_*^{1/4-6\varepsilon}\ell_*^3$
activities	$c_\kappa e_*^{1/4-7\varepsilon}\ell_*^3$.

For the second row the fine measure sums to ℓ_*^3 and the four spin components give a factor 4; its 48 labels are canceled by $1/48$. The last row uses $\sum_X e^{-\kappa d_M(X)}\le c_\kappa(\ell_*/M)^3\le c_\kappa\ell_*^3$. Since $C_K\le C_0^2$ and $C_0\ge 1$, these bounds imply

$$\begin{aligned}|V_C|_{a_2}&\le(2C_0+c_\kappa+4)e_*^{1/4-8\varepsilon}\ell_*^3,\\|V_C|_{a_2}&\le(2C_K+K_{\mathrm{pol}}+4)e_*^{1/4-7\varepsilon}\ell_*^3.\end{aligned}\tag{8.xvi}$$

The one-sided substitution $\Psi\mapsto K_C\Psi$, with barred generators fixed, sends the a_2 coefficient norm into the unit norm. Normalized identity-covariance integration has norm one there. Applying this to $e^{V_C}-1$ and using submultiplicativity proves eq. 8.xv. □

Proposition 8.4. For a controlled real pair on $R\in\{R^*,LR^*\}$, every mean-zero real coarse field A with $\|dA\|_\infty\le p_*$ and every $\theta\in\mathbb{R}^3$ satisfy

$$Y_*(A+c_\theta)=e^{-\varepsilon_k^{\mathrm{v}}\ell_*^3}R_\theta^{\mathrm{tw}}(1+r_*(\theta,A)),\qquad|r_*(\theta,A)|\le\varepsilon_1.\tag{8.xvii}$$

The remainder is continuous in (θ,A) on this set. ■

Proof. Use eq. 8.xiii and gauge invariance from Thm. L. The controlled trajectory satisfies the lemma's mass and activity hypotheses by Prop. 6.2 and $4e_*^{\varepsilon} \le 1$. At $C = 0$, the real covariance formula gives $\|K_0\|_{\mathrm{op}} \le b_-^{-1} + \mu_*^{-1}$. The first stop condition yields

$$4\ell_*^3(b_-^{-1} + \mu_*^{-1}) \le \frac{e_*^{-\varepsilon/2}}{2C_0^2\ell_*^3} \le a_2.$$

Its left side is at least one, so $a_2 \ge 1$ as well. The exact weight-one Gaussian identity in eq. 6.xxix reads

$$Y_*(A + c_\theta) = e^{-\varepsilon_k^{\mathrm{v}}\ell_*^3} R_\theta^{\mathrm{tw}} \int \mathrm{d}\mu_{K_0} e^{V_0}. \tag{8.xviii}$$

The first bound in eq. 8.xvi and the second and third stop conditions give $|V_0|_{a_2} \le e_*^{1/4-9\varepsilon}/4 \le 1/32$. Applying the lemma and $e^x - 1 \le 2x$ gives the claimed error. The density is jointly continuous by Thm. L; division by the continuous positive twist determinant proves the remainder's continuity. □

8.3. **Partition bounds and comparison of torus sizes.** Let $z_{r,n}$ be the matter factor in eq. 11.xxxi, formed at the real mass $\bar{m}\eta + m_{\mathrm{in}}$. The fine normalization is

$$z_{r,n} = e^{\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3} \det(D^{\mathrm{ch}}[0] + \bar{m}\eta)\, \bar{\nu}(1) > 0. \tag{8.xix}$$

Both signs of the total lattice mass are allowed here. The free reference mass $\bar{m}\eta > 0$ is used only in the determinant normalization.

Proposition 8.5. For every controlled real pair on $R \in \{R^*, LR^*\}$, the following two-sided estimates hold:

$$e^{-\varepsilon_k^{\mathrm{v}}\ell_*^3} R_\theta^{\mathrm{tw}}(1-\varepsilon_1)(1-\varepsilon_4) \le \nu^\theta(\mathfrak{G}) \le e^{-\varepsilon_k^{\mathrm{v}}\ell_*^3} R_\theta^{\mathrm{tw}}(1+\varepsilon_1), \tag{8.xx}$$

$$(1-\varepsilon_3)\bar{\nu}(1) \le \bar{\nu}(\mathfrak{G}) \le \bar{\nu}(1). \tag{8.xxi}$$

In particular

$$\left|\log z_{r,n} - \log z_{r,n}^{\mathrm{free}} - \varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3 + \varepsilon_k^{\mathrm{v}}\ell_*^3\right| \le 2\varepsilon_1 + \varepsilon_2 + 2\varepsilon_3 + 2\varepsilon_4 \le 2e_*^{1/4-9\varepsilon}. \tag{8.xxii}$$

■

Proof. Apply eq. 6.xxxiv at $a = 0$ and $\Phi = 1$, then insert eq. 8.xvii. The result is

$$\nu^\theta(\mathfrak{G}) = e^{-\varepsilon_k^{\mathrm{v}}\ell_*^3} R_\theta^{\mathrm{tw}} \int \chi_*(A)(1 + r_*(\theta, A))\, \mathrm{d}\mathsf{m}_*(A).$$

The left side and the prefactor are real, and the prefactor is positive. The last integral therefore equals its real part. The remainder bound and $1 - \varepsilon_4 \le \mathsf{m}_*(\chi_*) \le 1$ give eq. 8.xx. The first inequality of eq. 7.iv gives $\bar{\nu}(\mathfrak{G}^c) \le \varepsilon_3\bar{\nu}(1)$. This and positivity prove eq. 8.xxi.

Multiply eq. 8.xx by $\det(D^{\mathrm{ch}}[0] + \bar{m}\eta) > 0$. The twist comparison eq. 8.x replaces this product by the fine twisted free determinant, with factors between $e^{-\varepsilon_2}$ and e^{ε_2}. Integrating in θ gives

$$\begin{aligned} &e^{-\varepsilon_k^{\mathrm{v}}\ell_*^3} z_{r,n}^{\mathrm{free}} e^{-\varepsilon_2}(1-\varepsilon_1)(1-\varepsilon_4) \\ &\qquad \le \det(D^{\mathrm{ch}}[0] + \bar{m}\eta)\bar{\nu}(\mathfrak{G}) \le e^{-\varepsilon_k^{\mathrm{v}}\ell_*^3} z_{r,n}^{\mathrm{free}} e^{\varepsilon_2}(1+\varepsilon_1). \end{aligned}$$

All integrands are continuous in θ, and the factors being compared are positive. Use eq. 8.xxi and eq. 8.xix, then take logarithms. Writing the expression on the left of eq. 8.xxii without absolute values as Q_n, we obtain

$$-\varepsilon_2 + \log(1-\varepsilon_1) + \log(1-\varepsilon_4) \le Q_n \le \varepsilon_2 + \log(1+\varepsilon_1) - \log(1-\varepsilon_3).$$

The errors lie in $[0, 1/2]$. The inequalities $-\log(1-x) \le 2x$ and $\log(1+x) \le x$ therefore give the first bound in eq. 8.xxii. Its right side is at most $e_*^{1/4-9\varepsilon} + 3e_*$. Since $e_*^{1/4} \le 1/8$, we have $3e_* \le e_*^{1/4} \le e_*^{1/4-9\varepsilon}$, which gives the second bound. □

Use the common pair $P_N(e)$ of Thm. K on the two reference tori. Their common vacuum and pressure terms cancel in the partition difference below; the remaining errors are uniform in the ultraviolet cutoff.

Theorem M. For $N \ge N_{\mathrm{ref}}(e)$, let $R_* = R_{\mathrm{top}}/L$ and $n_* = R_*/\eta$. The two tori use the same real mass and obey

$$|D_{\mathrm{mat}}| := |L^3 \log z_{r,n_*} - \log z_{r,Ln_*}| \le C_{\mathrm{mat}}, \qquad C_{\mathrm{mat}} = (L^3+1)(\varepsilon_5(1/L)+1). \tag{8.xxiii}$$

The bound is uniform in N and depends only on L, ρ and the fixed chamber weights. ■□

Proof. Both tori belong to $\mathcal{T}(e)$, and $N_{\min}(LR_*, e) \le N_{\min}(R_*, e) = N_{\mathrm{ref}}(e)$. The common pair is therefore controlled on both. Its terminal vacuum and eq. 8.xi give, at either stop,

$$|\varepsilon_k^{\mathrm{v}}|\ell_*^3 \le 5c_{\mathrm{fl}} e_*^{1/4-6\varepsilon}\ell_*^3 \le \tfrac{1}{4} e_*^{1/4-7\varepsilon} \le e_*^{1/4-9\varepsilon}.$$

Write $\alpha = 1/4 - 9\varepsilon > 0$ and, at the common mesh,

$$Q_n = \log z_{r,n} - \log z_{r,n}^{\mathrm{free}} - \varepsilon_{0,N}^{\mathrm{v}}(e) n^3, \qquad E_n = \log z_{r,n}^{\mathrm{free}} - n^3 p_{\mathrm{ch}}^0(\bar{m}\eta).$$

eq. 8.xxii gives $|Q_n| \le 3e_*^\alpha \le 3e_K^\alpha$. The positive reference mass $\bar{m}\eta$ is identical on both tori; eq. 8.iv gives $|E_n| \le \varepsilon_5(\bar{m}R)$. Both bulk terms cancel because $(Ln_*)^3 = L^3 n_*^3$, so

$$D_{\mathrm{mat}} = L^3 Q_{n_*} - Q_{Ln_*} + L^3 E_{n_*} - E_{Ln_*}.$$

Consequently

$$|D_{\mathrm{mat}}| \le (L^3+1)[\varepsilon_5(\bar{m}R_*) + 3e_K^\alpha] \le (L^3+1)[\varepsilon_5(1/L)+1].$$

Here ε_5 decreases, $\bar{m}R_* > 1/L$, and $e_K^\alpha \le 1/8$. All constants are independent of N. □

8.4. **The photon part of the defect.** The zero-flux Gaussian factor involves the determinant of the unit-lattice scalar Laplacian with its constant mode removed. Write its logarithm as

$$\mathcal{L}(n_1,n_2,n_3) = \sum_{\substack{k \in \prod_\mu \mathbb{Z}/n_\mu\mathbb{Z} \\ k \ne 0}} \log\left(\sum_{\mu=1}^{3}\left(2 - 2\cos\frac{2\pi k_\mu}{n_\mu}\right)\right), \qquad \mathcal{L}(n) = \mathcal{L}(n,n,n).$$

For $j \ge 2$ put

$$G_j = \left(1 + \frac{j}{3.2}\right)\frac{2j^2}{2.56}, \qquad K_j = (1+j+j^2)(\log j + jG_j).$$

The estimate used for the two reference volumes is

$$\frac{1}{2}(\mathcal{L}(jn) - j^3\mathcal{L}(n)) \le -(j^3-1)\log n + K_j \qquad (n, j \in \mathbb{Z},\ n, j \ge 2). \tag{8.xxiv}$$

Its logarithmic term cancels the corresponding scale dependence of the Gaussian normalization. We retain the finite-volume calculation since a bulk determinant bound would lose this term.

Sum first in the first coordinate. For a nonzero transverse momentum $k' = (k_2, k_3)$ define $\omega(k') > 0$ by

$$2\cosh\omega(k') = 2 + \sum_{\mu=2,3}\left(2 - 2\cos\frac{2\pi k_\mu}{n_\mu}\right).$$

Factoring over the n_1-th roots of unity gives

$$\prod_{k_1 \in \mathbb{Z}/n_1\mathbb{Z}}\left(2\cosh\omega - 2\cos\frac{2\pi k_1}{n_1}\right) = 4\sinh^2(n_1\omega/2).$$

For $k' = 0$, omit $k_1 = 0$; the corresponding product is n_1^2. Therefore

$$\mathcal{L}(n_1,n_2,n_3) = 2\log n_1 + 2\sum_{k' \ne 0}\log(2\sinh(n_1\omega(k')/2)). \tag{8.xxv}$$

For $z > 0$ let $g(z) = -\log(1 - e^{-2z})$. The identity $2\sinh z = e^z(1 - e^{-2z})$ shows that

$$\log(2\sinh(jz)) - j\log(2\sinh z) \le jg(z).$$

Consequently

$$\mathcal{L}(jn_1, n_2, n_3) - j\mathcal{L}(n_1, n_2, n_3) \le 2\log(jn_1) - 2j\log n_1 + 2j\sum_{k'\neq 0} g(n_1\omega(k')/2). \quad (8.xxvi)$$

In the three applications below, $n_1 = n$ and $n_2, n_3 \in \{n, jn\}$. Represent each transverse momentum by $|k_\mu| \le n_\mu/2$ and put $p_\mu = 2\pi k_\mu/n_\mu$. For $M(p) = \sum_{\mu=2,3}(2 - 2\cos p_\mu)$,

$$\frac{4}{\pi^2}|p|^2 \le M(p) \le 8, \qquad \omega = 2\,\mathrm{arcsinh}(\sqrt{M}/2) \ge 0.51|p|.$$

The last inequality follows from concavity of arcsinh on $[0, \sqrt{2}]$ and $\mathrm{arcsinh}\,\sqrt{2} > 0.81\sqrt{2}$. Thus $n\omega(k')/2 \ge c_j|k'|$, where $c_j = 1.6/j$. Since $g(z) \le e^{-2z}(1 + (2z)^{-1})$, and $|k'| \ge (|k_2| + |k_3|)/\sqrt{2}$, one obtains

$$\sum_{k'\neq 0} g(n\omega(k')/2) \le \left(1 + \frac{1}{2c_j}\right)\left[\left(\sum_{i\in\mathbb{Z}} e^{-\sqrt{2}c_j|i|}\right)^2 - 1\right]$$
$$= \left(1 + \frac{1}{2c_j}\right)\sinh(c_j/\sqrt{2})^{-2} \le \left(1 + \frac{1}{2c_j}\right)\frac{2}{c_j^2} = G_j.$$

Enlargement to all of $\mathbb{Z}^2 \setminus \{0\}$ is valid because the chosen representatives are distinct.

Finally telescope the enlargement of the three sides:

$$\mathcal{L}(jn, jn, jn) - j^3\mathcal{L}(n, n, n) = [\mathcal{L}(jn, jn, jn) - j\mathcal{L}(n, jn, jn)] + j[\mathcal{L}(jn, n, jn) - j\mathcal{L}(n, n, jn)] + j^2[\mathcal{L}(jn, n, n) - j\mathcal{L}(n, n, n)].$$

Symmetry of $\mathcal{L}$ supplies this identity. Each bracket is bounded by $2\log j + 2(1 - j)\log n + 2jG_j$ by eq. 8.xxvi. Multiplication by $1, j, j^2$ and summation prove eq. 8.xxiv.

8.5. **The photon, matter and flux contributions.** Fix $q > 0$ and a real lattice mass. On a cubic torus with $n \ge 4$ sites per side, let $Z_n^{(w)}[1]$ be the partition function in flux sector $w \in \mathbb{Z}^3$, and put $P_n(w) = Z_n^{(w)}[1]/Z_n[1]$. The flux order is $(23, 13, 12)$. Define

$$\varsigma_n = \sum_{w\neq 0} \exp\left(-\frac{2\pi^2|w|_2^2}{q^2 n}\right). \quad (8.xxvii)$$

The notation $z_{r,n}$ below is the positive matter factor of eq. 11.xxxi.

Proposition 8.6. For $n \ge 4$ and an integer $j \ge 2$,

$$Z_n[1] = \frac{Z_n^{\mathrm{ph}} z_{r,n}}{P_n(0)}, \qquad 0 < P_n(0) \le 1, \quad (8.xxviii)$$

$$P_n(w) \le \exp\left(-\frac{2\pi^2|w|_2^2}{q^2 n}\right), \qquad P_n(0) \ge 1 - \varsigma_n. \quad (8.xxix)$$

If $q^2 n \le 1$, then

$$\varsigma_n \le 7\exp\left(-\frac{2\pi^2}{q^2 n}\right) < \frac{1}{8}. \quad (8.xxx)$$

The defect D_j of eq. 9.ii decomposes as

$$D_j = D_j^{\mathrm{ph}} + D_j^{\mathrm{mat}} + D_j^{\mathrm{fl}}, \tag{8.xxxi}$$

$$D_j^{\mathrm{mat}} = j^3 \log z_{r,n} - \log z_{r,jn}, \qquad D_j^{\mathrm{fl}} = -j^3 \log P_n(0) + \log P_{jn}(0),$$

$$\begin{aligned} D_j^{\mathrm{ph}} &= (j^3-1)\log\frac{2\pi}{q^2} + \tfrac{1}{2}[\mathcal{L}(jn) - j^3\mathcal{L}(n)] \\ &\le (j^3-1)\log\frac{2\pi}{q^2 n} + K_j. \end{aligned} \tag{8.xxxii}$$

If $\varsigma_n < 1$, then

$$D_j^{\mathrm{fl}} \le -j^3 \log(1-\varsigma_n). \tag{8.xxxiii}$$

All identities use the same charge and lattice mass at the two sides; the mass can have either sign. ■

Proof. The zero-flux factorization eq. 11.xxxii gives $Z_n^{(0)}[1] = Z_n^{\mathrm{ph}} z_{r,n} > 0$. All sectors have nonnegative weights, and their absolutely convergent sum is $Z_n[1]$. Hence $0 < P_n(0) \le 1$ and eq. 8.xxviii.

Let h_w equal $r_q w_{\mu\nu}/n^2$ on orientation $\mu\nu$. Fourier Hodge decomposition gives $r_q z = dB + h$ with constant h; plane summation yields $h = h_w$. Thus $F^{\mathrm{u}} - h_w$ is exact, and periodic summation by parts gives $\langle h, F^{\mathrm{u}}\rangle = \langle h, h_w\rangle$ for every constant h on sector w.

Apply Gaussian domination eq. 2.lxxxvii with $h = h_{w'}$ and split the expectation into its sectors:

$$1 \ge \sum_w \exp(\langle h_{w'}, h_w\rangle - \|h_{w'}\|^2/2) P_n(w) \ge \exp(\|h_{w'}\|^2/2) P_n(w').$$

Since $\|h_w\|^2 = r_q^2 |w|_2^2/n$, this proves the first bound in eq. 8.xxix. The identity $\sum_w P_n(w) = 1$ proves the second.

Set $x = \exp(-2\pi^2/(q^2 n))$. Positivity permits the factorization

$$\varsigma_n = \left(\sum_{i\in\mathbb{Z}} x^{i^2}\right)^3 - 1 \le \left(1 + \frac{2x}{1-x}\right)^3 - 1.$$

For $q^2 n \le 1$, one has $x < 2^{-18}$. The bound $(1-x)^{-1} \le 1 + 2x$ gives $2x/(1-x) \le 2.001x < 10^{-5}$. Therefore $\varsigma_n \le 2.001x(3 + 3\cdot 10^{-5} + 10^{-10}) \le 7x < 1/8$. This also proves convergence of eq. 8.xxvii.

Taking logarithms in eq. 8.xxviii at n and jn gives eq. 8.xxxi. By eq. 11.xxx,

$$\log Z_n^{\mathrm{ph}} = (n^3+2)\log r_q + (n^3-1)\log(2\pi) - \mathcal{L}(n)/2.$$

Its bulk terms cancel in the defect. The remaining constant is $(j^3-1)\log(2\pi/q^2)$. eq. 8.xxiv proves eq. 8.xxxii. Finally $\log P_{jn}(0) \le 0$, and $P_n(0) \ge 1 - \varsigma_n > 0$ gives eq. 8.xxxiii. □

9. Infrared stability of the partition function

9.1. **Transfer from the reference volumes.** Chessboard bounds compare folded numerators on an arbitrary torus with those on the reference torus. To bound expectations, we also need a lower bound for the partition function in the denominator. Convexity supplies this bound, with a loss measured by the difference of the two reference partition functions. Thus only two reference volumes are needed to control observables in larger volumes.

9.1.1. *Folded observables and the partition defect.* Use unit-lattice lengths in this subsubsection. Fix the charge $q > 0$ and a real lattice mass. Choose coordinate cells of width $w \in \frac{1}{2}\mathbb{Z}_{>0}$ and offsets $o_\mu \in \frac{1}{2}\mathbb{Z}$:

$$Q_c^\circ = \{y : o_\mu + wc_\mu < y_\mu < o_\mu + w(c_\mu + 1),\ \mu = 1, 2, 3\}.$$

The walls are admissible coordinate mirrors. For even positive integers s_μ with $ws_\mu \geq 4$, let $C(ws)$ be the rectangular torus with these side lengths and antiperiodic fermions. It has $s_1 s_2 s_3$ cells. The wall-reflection group acts simply transitively on the covering cells. Write $O^{[c]}$ for the reflected copy of $O \in \mathcal{A}(Q_0^\circ)$ in Q_c°, and

$$O_{ws}^\sharp = \prod_{c \bmod s} O^{[c]}, \qquad \mathcal{Z}_s(O) = Z_{C(ws)}[O_{ws}^\sharp].$$

The copies use the antilinear reflection action, including conjugation of scalar coefficients.

The fold proof of eq. 2.lxviii applies also at half-integer w: walls remain admissible, paired reflections translate by the integer $2w$, and even cell counts give integer sides at least four. Base each layer's trace at its own lower wall. For $B = y + [-r, r]^3 \subset Q_c^\circ$, $2r < w$; nonzero periods have sup norm at least $2w > 4r$ and at least four. Thus the embedding estimate in Thm. C separates $B + [-1, 1]^3$ and puts the products in the required slab classes.

Hold two entries of s fixed and vary s_μ. The product of the copies in the layer $c_\mu = 0$ gives a Hilbert–Schmidt slab operator Y independent of s_μ. Its reflected layer pairs fill the entire folded array. eq. 2.lxiv gives

$$\mathcal{Z}_s(O) = \operatorname{Tr}(Y^*Y)^{s_\mu/2}. \tag{9.i}$$

In particular $\mathcal{Z}_s(O) \geq 0$, and $s_\mu \mapsto \mathcal{Z}_s(O)^{1/s_\mu}$ is nonincreasing. If $Y \neq 0$, the logarithm of this folded partition function is convex in s_μ. The first assertion is the monotonicity of Schatten norms, and the second follows from Hölder's inequality for the trace of positive powers. The normalization of Y here is that of the slab operators in §2.2.3.

Choose an even reference cell count $s_0 \geq 2$ with $n_r = ws_0 \geq 4$, and an integer $j \geq 2$. Put

$$N_r = s_0^3, \qquad D_j = j^3 \log Z_{C(n_r)}[1] - \log Z_{C(jn_r)}[1], \qquad \delta_j = \frac{D_j}{(j-1)^3 N_r}. \tag{9.ii}$$

For an observable in the base cell its normalized reference seminorm is

$$\|O\|_* = \left(\frac{\mathcal{Z}_{(s_0,s_0,s_0)}(O)}{Z_{C(n_r)}[1]} \right)^{1/N_r}. \tag{9.iii}$$

The partition function is positive, and the numerator is nonnegative, so this is well-defined also when it vanishes.

Proposition 9.1. One has $D_j \geq 0$. Let s be even, with either $s = s_0$ or $s \geq js_0$, and put $r = s/s_0$, $n = ws$. For every set $\mathcal{C}$ of distinct cells of $C(n)$ and $O_c \in \mathcal{A}(Q_0^\circ)$,

$$\left| \mathbb{E}_{C(n)} \prod_{c \in \mathcal{C}} O_c^{[c]} \right| \leq \prod_{c \in \mathcal{C}} e^{\delta_j} \|O_c\|_*. \tag{9.iv}$$

The right side uses only the two reference partition functions and the one-cell observables folded on $C(n_r)$. ∎

Proof. Successively decreasing the three side counts in eq. 9.i gives, for $s \geq s_0$,

$$\mathcal{Z}_{(s,s,s)}(O) \leq \mathcal{Z}_{(s_0,s_0,s_0)}(O)^{r^3}. \tag{9.v}$$

Taking $O = 1$ and $r = j$ proves $D_j \geq 0$.

We next obtain the lower bound for the normalizing partition function. Define $p(r_1, r_2, r_3) = \log Z_{C(r_1 n_r, r_2 n_r, r_3 n_r)}[1]$ on ratios with integer side lengths at least four. The pressure trace eq. 2.lxxiv shows that p is convex in each argument and that p/r_μ is nonincreasing in that argument. For $r \geq j$

put $a=(r-1)/(j-1)$. Convexity gives $p(r,\cdot,\cdot)\ge ap(j,\cdot,\cdot)-(a-1)p(1,\cdot,\cdot)$. Apply this in each argument, retaining the positive coefficient a at the next step. Bound the three subtracted terms by the side monotonicity. The result is

$$\begin{aligned}p(r,r,r)&\ge a^3p(j,j,j)\\&\quad-(a-1)(a^2j^2+ajr+r^2)p(1,1,1)\\&=r^3p(1,1,1)-a^3D_j.\end{aligned}$$

The last equality uses $aj-(a-1)=r$ three times. The negative coefficients are why upper bounds are used on the three subtracted terms; no sign condition on p is needed. Thus, including the equality case $r=1$,

$$\log Z_{C(n)}[1]\ge r^3\left(\log Z_{C(n_r)}[1]-\frac{D_j}{(j-1)^3}\right). \tag{9.vi}$$

Fill the cells outside $\mathcal{C}$ with the constant integrand. The normalized rectangular chessboard estimate then gives

$$\left|\mathbb{E}_{C(n)}\prod_{c\in\mathcal{C}}O_c^{[c]}\right|\le\prod_{c\in\mathcal{C}}\left(\frac{\mathcal{Z}_{(s,s,s)}(O_c)}{Z_{C(n)}[1]}\right)^{1/(r^3N_r)}.$$

Use eq. 9.v in the numerator and eq. 9.vi in the denominator. Each factor is at most $e^{\delta_j}\|O_c\|_*$, proving eq. 9.iv. □

10. The fermion determinant: quartic response

10.1. **A positive continuum fourth coefficient.** To detect non-Gaussianity we need a compact probe whose fourth free-fermion response is positive. Its sign becomes visible at large mass: the leading term is a positive multiple of the integral of the fourth power of the curvature. Differentiating in the mass gives a kernel with five propagators, and a local gauge reduces its leading term to a constant-field calculation. A dilation then gives a positive probe at the prescribed mass.

For a real $a\in C_c^\infty(\mathbb{R}^3;\mathbb{R}^3)$ put

$$V_a=i\gamma\cdot a,\qquad S_M=(M+\gamma\cdot\partial)^{-1},\qquad q_{4,M}(a)=-\tfrac14\operatorname{tr}[(S_MV_a)^4],\quad M>0.$$

The trace is on $L^2(\mathbb{R}^3;\mathbb{C}^4)$; V_a denotes multiplication. We use Euclidean norms in this subsection.

Proposition 10.1. The operator $(S_MV_a)^4$ is trace class and its trace is real. There is $C_a<\infty$ such that, for $M\ge1$,

$$\left|q_{4,M}(a)-\frac{1}{240\pi M^5}\int_{\mathbb{R}^3}|\operatorname{curl}a|^4\,\mathrm{d}x\right|\le C_aM^{-6}. \tag{10.i}$$

In particular, for every $\bar m>0$ there is a real $\beta\in C_c^\infty(\mathbb{R}^3;\Lambda^2)$ for which $f=dd^*\beta$ satisfies $b(f)>0$ in eq. 2.xxviii. ■

10.1.1. *Kernels and differentiation in the mass.* The kernel and Fourier multiplier of S_M are

$$S_M(z)=(M-\gamma\cdot\partial)\frac{e^{-M|z|}}{4\pi|z|}, \tag{10.ii}$$

$$\widehat S_M(p)=\frac{M-i\gamma\cdot p}{M^2+|p|^2},\qquad|\widehat S_M(p)|=(M^2+|p|^2)^{-1/2}. \tag{10.iii}$$

In particular,

$$|S_M(z)|\le\frac{e^{-M|z|}(2M|z|+1)}{4\pi|z|^2},\qquad S_M(z)=M^2S_1(Mz),\qquad\|S_M\|_{L^1}\le3/M.$$

The operator $G = S_M V_a S_M$ is Hilbert–Schmidt. Indeed, its momentum kernel, with measure $(2\pi)^{-3}\,\mathrm{d}p$, is $\widehat{S}_M(p) i\gamma\cdot\widehat{a}(p-q)\widehat{S}_M(q)$. Writing $u(p) = (M^2+|p|^2)^{-1}$, the matrix norm bound and Young's inequality give

$$\|G\|_{\mathrm{HS}}^2 \le 8(2\pi)^{-6}\int u(p)(|\widehat{a}|^2 * u)(p)\,\mathrm{d}p \le 8(2\pi)^{-3}\|u\|_2^2\|a\|_2^2 = \frac{\|a\|_2^2}{\pi M}.$$

Here $\|u\|_2^2 = \pi^2/M$. Since $(S_M V_a)^4 = G V_a G V_a$,

$$|q_{4,M}(a)| \le \frac{\|a\|_2^2\|a\|_\infty^2}{4\pi M}. \tag{10.iv}$$

The matrix $\gamma_5 = \left(\begin{smallmatrix} 0 & 1_2 \\ 1_2 & 0 \end{smallmatrix}\right)$ conjugates S_M to S_M^* and V_a to $V_a^* = -V_a$. Thus the trace is real, using cyclicity with the bounded factor V_a and the trace-class product GV_aG.

The constant-field calculation uses a linear connection. For compact smooth or linear a, the potential is Lipschitz and satisfies $|a(z)| \le c_a(1+|z|)$; the resulting kernel coefficients will be interpreted on Schwartz functions.

Choose $s(z) = K|z|^{-2}e^{-M|z|/2}$ so that $|S_M(z)|(1+|z|)^{24} \le s(z)$. Its convolution powers obey

$$s^{*2}(z) \le C|z|^{-1}e^{-M|z|/2}, \quad s^{*2}(z) \le C|z|^{-3/2}, \quad s^{*3}(z) \le C|z|^{-1/2}. \tag{10.v}$$

For example, the bound for the second convolution follows by factoring out $e^{-M|z|/2}$ and scaling $\int |z-w|^{-2}|w|^{-2}\,\mathrm{d}w = C|z|^{-1}$; the third follows by convolving $|z|^{-2}$ and $|z|^{-3/2}$. Thus $s^{*2} \in L^2$, $s^{*4} = s^{*2} * s^{*2}$ is bounded and continuous, and so is $s^{*5} = s * s^{*4}$.

Let $R_a^{(j)}(x,y)$ be the integral kernel of $(-1)^j(S_M V_a)^j S_M$, initially off the diagonal, for $0 \le j \le 4$. Distributing the factors $1+|z_l|$ along the chain of propagators gives

$$|R_a^{(j)}(x,y)| \le [c_a(1+|x|)]^j s^{*(j+1)}(x-y). \tag{10.vi}$$

These integrals act as the indicated operators on Schwartz spinors; absolute integration against such a spinor justifies every composition. The fourth kernel extends continuously to all (x,y). To see this, use the four independent increments of the first three and last propagators. Changing (x,y) then changes four Lipschitz factors and the middle propagator. On $|x|,|y|,|x'|,|y'| \le r$, the resulting difference is bounded by

$$C_{a,r}\big((|x-x'|+|y-y'|)\|s^{*5}\|_\infty + \|s^{*4}\|_\infty\|S_M - S_M(\,\cdot\,+(x'-y')-(x-y))\|_{L^1}\big). \tag{10.vii}$$

This tends to zero. Define $r_4(x;M,a) = \operatorname{tr} R_a^{(4)}(x,x)$ using this extension.

For compact a, differentiation in M now gives

$$\frac{\mathrm{d}}{\mathrm{d}M}q_{4,M}(a) = \int r_4(x;M,a)\,\mathrm{d}x, \qquad q_{4,M}(a) \longrightarrow 0 \quad (M\longrightarrow\infty). \tag{10.viii}$$

To justify this differentiation, write the trace as the absolutely convergent four-vertex integral

$$-\tfrac{1}{4}\int \operatorname{tr}\big[S_M(z_4-z_1)V_a(z_1)S_M(z_1-z_2)V_a(z_2)S_M(z_2-z_3)V_a(z_3)S_M(z_3-z_4)V_a(z_4)\big]\,\mathrm{d}z_1\cdots\mathrm{d}z_4.$$

The absolute integral is at most $4\|a\|_\infty^3\|a\|_{L^1}s^{*4}(0)$ before the factor $1/4$. On compact positive-mass intervals, S_M and $\partial_M S_M$ have a common majorant of this form. The identity $\partial_M S_M = -S_M * S_M$ and cyclicity make the four differentiated terms equal, cancelling both minus signs and the factor $1/4$. The resulting five-propagator integral has absolute integral at most $4\|a\|_\infty^3\|a\|_{L^1}s^{*5}(0)$, proving continuous differentiability and eq. 10.viii. Its limit follows from eq. 10.iv.

10.1.2. *The local large-mass expansion.* The substitution $z_l = x + M^{-1}w_l$ in the diagonal kernel gives

$$r_4(x;M,a) = M^{-2}\operatorname{tr}\int S_1(-w_1)V_a(x+M^{-1}w_1)S_1(w_1-w_2)V_a(x+M^{-1}w_2) \\ \cdot S_1(w_2-w_3)V_a(x+M^{-1}w_3)S_1(w_3-w_4)V_a(x+M^{-1}w_4)S_1(w_4)\,\mathrm{d}w. \quad (10.\text{ix})$$

The local expansion must also be integrated outside the probe support. The following spatial tail makes that passage uniform:

$$\int_{\operatorname{dist}(x,\operatorname{supp} a)\geq 1} |r_4(x;M,a)|\,\mathrm{d}x \leq C_a e^{-M}. \quad (10.\text{x})$$

Indeed, the first and last propagators each travel at least $d = \operatorname{dist}(x,\operatorname{supp} a)$; their bound is Me^{-Md} for $d \geq 1$. Integrating the three intervening propagators gives

$$|r_4(x;M,a)| \leq 108\|a\|_\infty^4 |\operatorname{supp} a| M^{-1}e^{-2Md}.$$

For $M, d \geq 1$, $2Md \geq M + d$, which proves eq. 10.x.

The reduction to field strength must be justified at the level of this diagonal kernel. Let $\lambda \in C_c^\infty(\mathbb{R}^3)$ be real. On Schwartz spinors, as a formal power series in u,

$$(M + \gamma\cdot\partial + ui\gamma\cdot(a + d\lambda))^{-1} = e^{-iu\lambda}(M+\gamma\cdot\partial + ui\gamma\cdot a)^{-1}e^{iu\lambda}.$$

All coefficients preserve Schwartz space. Comparing the coefficient of u^4 gives, almost everywhere off the diagonal,

$$R^{(4)}_{a+d\lambda}(x,y) = \sum_{j=0}^{4} R_a^{(j)}(x,y)\frac{[-i(\lambda(x)-\lambda(y))]^{4-j}}{(4-j)!}. \quad (10.\text{xi})$$

For $j = 0,1,2,3$, the bounds in eq. 10.v–eq. 10.vi show that these terms are respectively $O(|x-y|^2)$, $O(|x-y|^2)$, $O(|x-y|^{3/2})$, and $O(|x-y|)$ locally in x. Continuity of the fourth kernel therefore implies

$$r_4(x;M,a+d\lambda) = r_4(x;M,a).$$

Write $F_{\nu\mu}(x) = \partial_\nu a_\mu(x) - \partial_\mu a_\nu(x)$ for the curvature of a. Fix a real $\chi \in C_c^\infty(\mathbb{R}^3)$ equal to one near zero. For each x, choose the real compact gauge

$$\lambda_x(x+z) = -\chi(z)\left(a(x)\cdot z + \tfrac{1}{2}\sum_{\nu,\mu} z_\nu z_\mu \partial_\nu a_\mu(x)\right), \qquad a^x = a + d\lambda_x.$$

Then $a^x(x) = 0$ and $\partial_\nu a^x_\mu(x) = F_{\nu\mu}(x)/2$. The norms $\|a^x\|_{C^2}$ are uniformly bounded in x by $C_\chi\|a\|_{C^3}$. For a constant real antisymmetric matrix F, put

$$A_{F,\mu}(w) = \tfrac{1}{2}\sum_\nu w_\nu F_{\nu\mu}, \qquad \Psi(F) = r_4(0;1,A_F).$$

Taylor's formula on each segment gives, uniformly in x, w and $M \geq 1$,

$$|a^x(x+M^{-1}w)| \leq C_a M^{-1}|w|, \qquad |a^x(x+M^{-1}w) - M^{-1}A_{F(x)}(w)| \leq C_a M^{-2}|w|^2.$$

Apply eq. 10.ix to a^x. Telescoping its four connection factors bounds their difference from the four linear factors by $C_a M^{-5}\sum_l |w_l|^2\prod_{j\neq l}|w_j|$. This polynomial is bounded by $4\prod_{l=1}^4(1+|w_l - w_{l-1}|)^8$, with $w_0 = 0$. The unit-mass majorant of weight 24 in eq. 10.v absorbs these weights, and its fifth convolution at zero bounds the integral. Diagonal gauge invariance gives

$$r_4(x;M,a) = M^{-6}\Psi(F(x)) + O_a(M^{-7}), \qquad M \geq 1, \quad (10.\text{xii})$$

uniformly in x.

10.1.3. *The constant-field coefficient.* Rotation of the variables and conjugation by the spin lift in $SU(2)$ show that $\Psi(F)$ depends on the vector $B_l = \frac{1}{2}\sum_{\nu,\mu}\varepsilon_{l\nu\mu}F_{\nu\mu}$ only through $|B|$. Take $B = (0,0,b)$, $b \geq 0$, and $A = (-bz_2/2, bz_1/2, 0)$. Set $D_u = \partial + iuA$ and

$$h_u = 1 - \Delta + uW_1 + u^2 W_2, \qquad W_1 = -2iA\cdot\partial + b\sigma_3, \quad W_2 = |A|^2.$$

The Clifford relations give

$$(1-\gamma\cdot D_u)(1+\gamma\cdot D_u) = \operatorname{diag}(h_u, h_u), \qquad (1+\gamma\cdot D_u)^{-1} = (1-\gamma\cdot D_u)\operatorname{diag}(h_u^{-1}, h_u^{-1}). \tag{10.xiii}$$

These are formal identities on Schwartz spinors.

We compute the fourth coefficient of h_u^{-1} from the kernel

$$k_u(t;x,y) = (4\pi t)^{-3/2} e^{-|x-y|^2/(4t)} \frac{\tau}{\sinh\tau} \exp\left(-\frac{(\tau\coth\tau - 1)|x_p - y_p|^2}{4t} + \frac{iub}{2} x\wedge y\right) e^{-\tau\sigma_3},$$
$$\tau = ubt, \quad x_p = (x_1, x_2), \quad x\wedge y = x_1y_2 - x_2y_1. \tag{10.xiv}$$

It is analytic in u for $|u|bt < \pi$. The kernel satisfies $(\partial_t + h_u - 1)k_u = 0$. The planar verification uses $\alpha = (ub/4)\coth(ubt)$ and $\phi = -\alpha|x_p - y_p|^2 + iub(x\wedge y)/2$:

$$\frac{\check{h}e^{\phi}}{e^{\phi}} = 4\alpha - (4\alpha^2 - u^2b^2/4)|x_p - y_p|^2, \qquad \alpha' = -(4\alpha^2 - u^2b^2/4),$$

where $\check{h} = -\partial_1^2 - \partial_2^2 - iub(x_1\partial_2 - x_2\partial_1) + u^2b^2|x_p|^2/4$. Differentiation of the prefactor supplies -4α; differentiation of $e^{-\tau\sigma_3}$ supplies the spin term. The longitudinal Gaussian satisfies the one-dimensional heat equation.

We use the coefficients of eq. 10.xiv individually: the analyticity condition $|u|bt < \pi$ is imposed at each proper time t. Write $k_u = \sum_{n\geq 0} u^n k^{(n)}$. Each coefficient is the Gaussian times a finite polynomial in $t, x-y, y$, whose monomials $t^j(x-y)^\alpha y^\beta$ have $2j + |\alpha| \geq n$. The factor $e^{iub(x-y)\wedge y/2}$ has weight at least one per power of u, and the other factors have weight at least two, if t has weight two and $x-y$ weight one. Thus

$$K^{(n)}(x,y) = \int_0^\infty e^{-t} k^{(n)}(t;x,y)\,\mathrm{d}t$$

converges off the diagonal and, for $n \geq 1$, on the diagonal. Its terms are y^β times

$$m_{j,\alpha}(x-y) = \int_0^\infty e^{-t} t^j (x-y)^\alpha (4\pi t)^{-3/2} e^{-|x-y|^2/(4t)}\,\mathrm{d}t.$$

Their Fourier transforms are $(i\partial_p)^\alpha[j!(1+|p|^2)^{-j-1}]$, so they define operators preserving Schwartz space.

Against a Schwartz spinor φ, the zeroth heat coefficient tends to φ as $t \downarrow 0$, while each higher term is $O_\varphi(t^{j+|\alpha|/2})$ and tends to zero. Integrating its heat equation from ϵ to infinity and sending $\epsilon \downarrow 0$ therefore gives

$$(1-\Delta)K^{(n)} + W_1K^{(n-1)} + W_2K^{(n-2)} = \delta_{n0}1.$$

The convergence holds in tempered distributions, where the polynomial differential operators are continuous; invertibility of $1-\Delta$ on Schwartz space makes this the unique formal inverse.

For the diagonal comparison, we need that $m_{j,\alpha}$ is continuous when $2j + |\alpha| \geq 2$ and continuously differentiable when $2j + |\alpha| \geq 3$. For this, write $m_{j,\alpha}(z) = z^\alpha \mu_j(|z|)$ with

$$\mu_j(s) = \int_0^\infty e^{-t} t^j (4\pi t)^{-3/2} e^{-s^2/(4t)}\,\mathrm{d}t.$$

For $j \ge 1$, μ_j is bounded and continuous at zero and $\mu_j'(s) = -s\mu_{j-1}(s)/2$ for $s > 0$; for $j = 0$ use $\mu_0(s) = e^{-s}/(4\pi s)$. These formulas give the asserted regularity at zero. Consequently $K^{(3)}, K^{(4)}$ and $\partial_x K^{(4)}$ are continuous. From eq. 10.xiii the fourth Dirac coefficient has kernel

$$(1 - \gamma\cdot\partial_x)\operatorname{diag}(K^{(4)}, K^{(4)})(x,y) - i\gamma\cdot A(x)\operatorname{diag}(K^{(3)}, K^{(3)})(x,y).$$

It equals $R_A^{(4)}$ because both continuous kernels give the same operator on Schwartz space. At $x = y = 0$, $A(0) = 0$, and the derivative term has zero trace: the two Pauli blocks have opposite signs. It follows that

$$\Psi(F) = 2\int_0^\infty e^{-t}\operatorname{tr} k^{(4)}(t;0,0)\,\mathrm{d}t \tag{10.xv}$$

$$= -\frac{4}{45}(4\pi)^{-3/2}\Gamma(7/2)b^4 = -\frac{1}{48\pi}|B|^4. \tag{10.xvi}$$

In the second line we used $\operatorname{tr} k_u(t;0,0) = 2(4\pi t)^{-3/2}\tau\coth\tau$ and $\tau\coth\tau = 1 + \tau^2/3 - \tau^4/45 + O(\tau^6)$.

10.1.4. *Integration and dilation.* Integrate eq. 10.xii over the unit neighborhood of $\operatorname{supp} a$, and use eq. 10.x on its complement. eq. 10.viii and eq. 10.xvi give

$$\frac{\mathrm{d}}{\mathrm{d}M}q_{4,M}(a) = -\frac{1}{48\pi M^6}\int |\operatorname{curl} a|^4\,\mathrm{d}x + O_a(M^{-7}).$$

Integration from M to infinity proves eq. 10.i.

Finally let $\varphi(x) = e^{-1/(1-|x|^2)}$ for $|x| < 1$ and zero otherwise, set $\beta_{12} = \varphi$, $\beta_{13} = \beta_{23} = 0$, and $a = d^*\beta = (\partial_2\varphi, -\partial_1\varphi, 0)$. The third component of $\operatorname{curl} a$ is $-(\partial_1^2 + \partial_2^2)\varphi$ and has value $4e^{-1}$ at zero. Thus $I = \int |\operatorname{curl} a|^4\,\mathrm{d}x > 0$. For $\ell \ge 1$ put $\beta^{(\ell)}(x) = \ell^2\beta(x/\ell)$, $a_\ell(x) = \ell a(x/\ell)$, and $f_\ell = dd^*\beta^{(\ell)}$. The unitary dilation $U_\ell\psi(x) = \ell^{-3/2}\psi(x/\ell)$ gives

$$U_\ell^{-1}S_{\bar m}V_{a_\ell}U_\ell = \ell^2 S_{\bar m\ell}V_a, \qquad q_{4,\bar m}(a_\ell) = \ell^8 q_{4,\bar m\ell}(a).$$

For $\bar m\ell \ge 1$, therefore,

$$b(f_\ell) = 24q_{4,\bar m}(a_\ell) = \frac{\ell^3 I}{10\pi\bar m^5} + O_a(\ell^2\bar m^{-6}). \tag{10.xvii}$$

This is positive for sufficiently large ℓ and completes the proof of Prop. 10.1.

10.2. **The free fourth-order coefficient.**

10.2.1. *The probe and determinant.* Fix the standard chamber weights, $0 < \rho < 1$, $0 < \bar m \le 1$, and real $\beta \in C_c^\infty(\mathbb{R}^3;\Lambda^2)$. Put $a = d^*\beta$, $f = da$, and let D_β be the side of a coordinate cube Q containing $\operatorname{supp}\beta$. For $\eta > 0$ set

$$J_\eta v(y) = \eta^{-3}\int_{[-\eta/2,\eta/2]^3} v(y+z)\,\mathrm{d}z, \qquad \delta_{\eta,\mu}v(y) = \eta^{-1}[v(y+\eta e_\mu/2) - v(y-\eta e_\mu/2)].$$

Define

$$a^0_{\eta,\nu} = -\sum_{\mu\ne\nu}\delta_{\eta,\mu}J_\eta\beta_{\mu\nu}. \tag{10.xviii}$$

If $n = R/\eta \in \mathbb{N}$, its sampled periodic bond field is

$$A_{\eta,\nu}(c) = \sum_{z\in\mathbb{Z}^3} a^0_{\eta,\nu}(\eta c + \eta e_\nu/2 + Rz), \qquad c \in \mathbb{Z}^3/n\mathbb{Z}^3. \tag{10.xix}$$

On antiperiodic chamber spinors use the phase connection $B_t(c, c+e_\nu) = t\eta A_{\eta,\nu}(c)$ and the positive free mass $m = \bar m\eta$. Set

$$N_0(t) = \frac{\det(D^{\mathrm{ch}}[B_t] + m)}{\det(D^{\mathrm{ch}}[0] + m)}, \qquad c_j = N_0^{(j)}(0), \qquad b_{\mathrm{ch}}(\eta, R) = (\log N_0)^{(4)}(0), \tag{10.xx}$$

where the logarithm is zero at zero. The denominator is nonzero by the positive real part of the free matrix. In the charge convention, $q = e\sqrt{\eta}$ and $\widehat{a} = \sqrt{\eta} A_\eta$, so that $B_t = q(t/e)\widehat{a}$. Since J_η commutes with the centered differences, eq. 11.ii gives precisely this source for the probe subsequently placed in a reference cell.

The finite determinant is entire in t. With the charge-conjugation matrix $\mathsf{C} = i\gamma_2$, transposing its chamber hops and using $\mathsf{C}\gamma_\nu^T\mathsf{C}^{-1} = -\gamma_\nu$ gives

$$\mathsf{C}(D^{\mathrm{ch}}[B] + m)^T\mathsf{C}^{-1} = D^{\mathrm{ch}}[-B] + m.$$

Thus N_0 is even. For real t the two spin blocks are adjoints and the matrix has positive real part, so $N_0(t) > 0$. In particular its derivatives are real and

$$c_{2j+1} = 0, \qquad c_2 = (\log N_0)''(0), \qquad c_4 = b_{\mathrm{ch}} + 3c_2^2. \tag{10.xxi}$$

Translations and signed coordinate permutations conjugate the matrix: for a reflection with unit normal n take $S_n = (\gamma \cdot n)\gamma_5$. Anticommutation of γ_5 with the Clifford triple gives $S_n\gamma_z S_n^{-1} = \gamma_{r_n z}$. For a product of reflections take the corresponding product of the S_n. Together with the site map $(c, g) \mapsto (rc + v, rg)$, substitution in the hopping formula eq. 2.xiii gives exact conjugacy, also for complex connections. This map preserves the antiperiodic character since $(-1)^{\sum_j (rw)_j} = (-1)^{\sum_j w_j}$ for $w \in \mathbb{Z}^3$. Consequently these determinant coefficients are unchanged by lattice translations of the probe and by its reflected copies.

On the continuum antiperiodic torus $T_R = \mathbb{R}^3/R\mathbb{Z}^3$ let $S^R = (\bar{m} + \gamma \cdot \partial)^{-1}$ and $a^R = \sum_z a(\cdot + Rz)$. Write $S = (\bar{m} + \gamma \cdot \partial)^{-1}$ on $\mathbb{R}^3$ and set

$$b_R(f) = -6\,\mathrm{Tr}(S^R i\gamma \cdot a^R)^4, \qquad b(f) = -6\,\mathrm{Tr}(Si\gamma \cdot a)^4. \tag{10.xxii}$$

Proposition 10.2. For every fixed $R \geq D_\beta + 1$, along spacings with $R/\eta \in \mathbb{N}$,

$$b_{\mathrm{ch}}(\eta, R) \longrightarrow b_R(f) \quad (\eta \downarrow 0), \qquad |b_R(f) - b(f)| \leq C_{\bar{m},\beta} e^{-\bar{m}(R - D_\beta)}. \tag{10.xxiii}$$

The displayed traces exist and are real. The constant in the second estimate is independent of R and the probe position. ∎

Proof. Use the label Hilbert space $V = \ell^2(O_{\mathrm{cub}}; \mathbb{C}^4)$ with measure $1/48$ and its isometric constant-label injection J. All discrete spaces are embedded isometrically into the common Fourier space

$$\mathcal{H}_R = \ell^2(\mathcal{P}_R; V), \qquad \mathcal{P}_R = \frac{2\pi}{R}(\mathbb{Z} + \tfrac{1}{2})^3.$$

The discrete subspace consists of the n^3 frequencies in $\mathcal{B}_\eta = \{P \in \mathcal{P}_R : \eta P \in (-\pi, \pi]^3\}$. The physical site measure η^3 makes $R^{-3/2} e^{iP \cdot \eta c}$ orthonormal for odd and even n, with label measure $1/48$.

Denote its projection by P_η and extend discrete operators by zero on its complement.

The physical inverse. Let $\mathcal{D}_m$ be the chamber symbol already treated in §4.1.2. The physical inverse

$$R_\eta = \eta(D^{\mathrm{ch}}[0] + \bar{m}\eta)^{-1}$$

is the multiplier $\eta \mathcal{D}_{\bar{m}\eta}(\eta P)^{-1}$ on $\mathcal{B}_\eta$. The Schur ellipticity and block inverse formula give, for real $p \in [-\pi, \pi]^3$,

$$\|\mathcal{D}_m(p)^{-1}\| \leq C_\rho (m + |p|)^{-1}.$$

To check this bound, the heavy inverse is uniformly bounded, both cross blocks are $O(|p|)$, and the light inverse is bounded by $C(m^2 + |p|^2)^{-1/2}$. Insert these bounds in eq. 4.xiv and absorb bounded terms on the compact momentum cube. Therefore

$$\|R_\eta(P)\| \leq \frac{C_{\rho,\bar{m}}}{1 + |P|}. \tag{10.xxiv}$$

For fixed P, the light Schur block divided by η tends to $\bar{m} + i\gamma \cdot P$, because $B_{\bar{m}\eta}(\eta P) = \bar{m}\eta + i\gamma \cdot \sin(\eta P) + O(\eta^2 |P|^2)$. The scaled heavy and cross blocks of the inverse tend to zero. It follows that

$R_\eta(P) \to R_0(P)$, where $R_0 = JS^R J^*$. Since $\sum_{P\in\mathcal{P}_R}(1+|P|)^{-4} < \infty$, the finite-dimensional bound $\|M\|^4_{\mathfrak{S}_4(V)} \le 192\|M\|^4$ and dominated summation give

$$\|R_\eta - R_0\|_{\mathfrak{S}_4} \longrightarrow 0. \tag{10.xxv}$$

The sampled vertices. Define fixed label matrices by

$$(T^\pm_\nu u)(g) = 1_{\{ge_1 = \pm e_\nu\}} 3(\rho \mp \gamma_\nu) u(gr_{e_1}).$$

Let $U^\pm_\nu u(c) = u(c \pm e_\nu)$ and $A^-_{\eta,\nu}(c) = A_{\eta,\nu}(c - e_\nu)$. Exact differentiation gives

$$D_j := \partial_t^j D^{\mathrm{ch}}[B_t]\Big|_0 = -\eta^j \sum_\nu \left[i^j A^j_{\eta,\nu} T^+_\nu U^+_\nu + (-i)^j (A^-_{\eta,\nu})^j T^-_\nu U^-_\nu \right]. \tag{10.xxvi}$$

Thus $V_{j,\eta} = \eta^{-j} D_j$, $1 \le j \le 4$, have uniform operator bounds. Smooth averaging and the fundamental theorem of calculus give $A_{\eta,\nu}(c) = a^R_\nu(\eta c) + O_{R,\beta}(\eta)$ uniformly in c, and the same holds for $A^-_{\eta,\nu}$.

For finite antiperiodic Fourier polynomials u, w, the sampled matrix element is a Riemann sum with weights $\eta^3/48$ and tends to $\langle a^R_\nu u, w\rangle$. The squared norm of the sampled product with u tends to $\|a^R_\nu u\|_2^2$. Uniform operator bounds extend weak convergence from these w to all w, and weak convergence plus convergence of norms gives strong convergence. Applying this to conjugate multipliers also proves adjoint-strong convergence.

The shifts have multipliers $e^{\pm i\eta P_\nu} P_\eta$ and converge strongly with their adjoints to the identity. Bounded products preserve both convergences. Consequently

$$V_{1,\eta} \longrightarrow V_0 = -i \sum_\nu a^R_\nu (T^+_\nu - T^-_\nu) \quad \text{strongly and adjoint-strongly.}$$

The eight labels in each signed direction give $J^* T^\pm_\nu J = (\rho \mp \gamma_\nu)/2$, and hence $J^* V_0 J = i\gamma \cdot a^R$.

It follows from eq. 10.xxv that $R_\eta V_{1,\eta} \to R_0 V_0$ in $\mathfrak{S}_4$. Indeed the term $(R_\eta - R_0)V_{1,\eta}$ tends to zero by the ideal bound; for $R_0(V_{1,\eta} - V_0)$ first approximate R_0 in $\mathfrak{S}_4$ by finite-rank operators and then use adjoint-strong convergence. Relative to $\operatorname{ran} J \oplus \operatorname{ran} J^\perp$, the limit is upper triangular, with diagonal blocks $S^R i\gamma \cdot a^R$ and zero. Thus its fourth-power trace is the desired four-component trace.

Contact terms. Put $G_\eta = (D^{\mathrm{ch}}[0] + m)^{-1}$ and $Y_j = G_\eta D_j = \eta^{j-1} R_\eta V_{j,\eta}$. Expanding the finite matrix logarithm at zero, with the factorials in the derivatives D_j, gives

$$b_{\mathrm{ch}} = \operatorname{Tr} Y_4 - 4\operatorname{Tr}(Y_1 Y_3) - 3\operatorname{Tr} Y_2^2 + 12\operatorname{Tr}(Y_1^2 Y_2) - 6\operatorname{Tr} Y_1^4. \tag{10.xxvii}$$

By counting frequency shells in the fixed lattice $\mathcal{P}_R$, eq. 10.xxiv gives

$$\|R_\eta\|_{\mathfrak{S}_1} = O_R(\eta^{-2}), \qquad \|R_\eta\|^2_{\mathfrak{S}_2} = O_R(\eta^{-1}), \qquad \|R_\eta\|^3_{\mathfrak{S}_3} = O_R(1 + |\log\eta|).$$

In detail, the number of frequencies in $k \le |P| < k+1$ is at most $C_R(k+1)^2$; summing $(1+k)^{2-p}$ up to C/η proves all three bounds. Schatten Hölder therefore bounds the four contact traces, in their displayed order, by $O_R(\eta)$, $O_R(\eta)$, $O_R(\eta)$ and $O_R(\eta(1+|\log\eta|))$. They vanish. Finally

$$|\operatorname{Tr} X^4 - \operatorname{Tr} Z^4| \le 4\max(\|X\|_{\mathfrak{S}_4}, \|Z\|_{\mathfrak{S}_4})^3 \|X - Z\|_{\mathfrak{S}_4} \tag{10.xxviii}$$

follows by telescoping and Hölder. Apply it to the preceding $\mathfrak{S}_4$ convergence to obtain the fixed-R limit. Its value is real since b_{ch} is real.

Removing the torus. The massive kernel $S(x)$ in eq. 10.ii is integrable. Its antiperiodization

$$S_R(z) = \sum_{w\in\mathbb{Z}^3} (-1)^{w_1+w_2+w_3} S(z + Rw)$$

is the kernel of S^R: integrate against an antiperiodic Fourier character and unfold the sum. Absolute integration is justified by $S \in L^1$, and its Fourier multiplier is exactly $(\bar m + i\gamma\cdot P)^{-1}$. Place Q inside

a period cell and let P_Q multiply by its indicator. On $Q \times Q$ the difference between this kernel and the infinite-space kernel is the sum over $w \neq 0$. The kernel bound preceding eq. 10.iv and

$$|x - y + Rw| \geq R|w|_\infty - D_\beta \geq |w|_\infty \qquad (x, y \in Q, R \geq D_\beta + 1)$$

give, by the shell count $24k^2 + 2$, a uniform bound $C_{\bar{m}} e^{-\bar{m}(R - D_\beta)}$ for that sum. Thus

$$\|P_Q(S^R - S)P_Q\|_{\mathfrak{S}_4} \leq \|P_Q(S^R - S)P_Q\|_{\mathfrak{S}_2} \leq C_{\bar{m},\beta} e^{-\bar{m}(R - D_\beta)}.$$

These localized operators act on the common space $L^2(Q; \mathbb{C}^4)$. Their $\mathfrak{S}_4$ norms are uniformly bounded: since $S^*S = (-\Delta + \bar{m}^2)^{-1} 1_4$, its scalar kernel is $Y_{\bar{m}}(z) = e^{-\bar{m}|z|}/(4\pi|z|)$, and

$$\|SP_Q\|_{\mathfrak{S}_4}^4 = 4 \int_{Q \times Q} |Y_{\bar{m}}(x - y)|^2 \, \mathrm{d}x \, \mathrm{d}y \leq 4|Q| \|Y_{\bar{m}}\|_2^2 = \frac{|Q|}{2\pi\bar{m}}.$$

The localized torus operator inherits this uniform bound from the localized difference. The full S^R belongs to $\mathfrak{S}_4$ at each fixed R, since $\sum_{P \in \mathcal{P}_R} (\bar{m}^2 + |P|^2)^{-2} < \infty$. Multiplication by $v = i\gamma \cdot a$ preserves these estimates. For $T = Sv$ or $S^R v^R$ one has $T = TP_Q$, so $\operatorname{Tr} T^4 = \operatorname{Tr}(P_Q T P_Q)^4$ by cyclicity. Apply eq. 10.xxviii on $L^2(Q; \mathbb{C}^4)$ to prove the exponential estimate. All constants depend on the size and norms of the probe, not its position. This completes the proof. □

11. The interacting theory: quartic response

We must compare interacting curvature moments with the free response while keeping errors uniform in the lattice spacing and in products over separated cells. An external connection turns these moments into derivatives of a partition function. Subtracting the Gaussian quadratic term in the exponent produces the Wick polynomials used below. We place the probe, estimate its terminal insertions and their interactions, and finally transfer the resulting bounds to arbitrary volumes.

11.1. **Probes on the reference cells.**

We construct the discrete source for $f = dd^*\beta$ and its orthogonal cell copies.

11.1.1. *The discrete source and the smooth test.* Fix a real nonzero $\beta \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$. Let D_β retain the unbuffered support-cube side of §10.2.1, and write $D_\beta^{\mathrm{buf}} = D_\beta + 4$. Choose a cube $Q_\beta = z_\beta + [-D_\beta^{\mathrm{buf}}/2, D_\beta^{\mathrm{buf}}/2]^3$ containing $\operatorname{supp}\beta + [-2, 2]^3$. Use the centered differences $\delta_{\eta,\mu}$ and cube average J_η of that discussion. Put

$$a'_{\eta,\nu} = -\sum_{\mu \neq \nu} \delta_{\eta,\mu} \beta_{\mu\nu},$$
$$f^\eta_{\mu\nu} = \delta_{\eta,\mu} a'_{\eta,\nu} - \delta_{\eta,\nu} a'_{\eta,\mu}. \tag{11.i}$$

All these functions are smooth and real. Since each centered difference enlarges support by at most $\eta/2$ in its direction,

$$\operatorname{supp} f^\eta, \ \operatorname{supp} J_\eta a'_\eta \subset \operatorname{supp}\beta + [-\eta, \eta]^3.$$

In particular their supports lie in Q_β for $0 < \eta \leq 1$.

On the physical lattice $\eta\mathbb{Z}^3$ set

$$b_{\eta,\mu\nu}(c) = (J_\eta \beta_{\mu\nu})(c + \eta(e_\mu + e_\nu)/2), \qquad a^o_\eta = d^*_\eta b_\eta.$$

Here d_η is the mesh-normalized forward cochain differential and d^*_η its adjoint for the pairing $\eta^3 \sum$. The adjoint formula gives

$$a^o_{\eta,\nu}(c) = (J_\eta a'_{\eta,\nu})(c + \eta e_\nu/2). \tag{11.ii}$$

Thus $J_\eta a'_\eta$ is the interpolating vector field a^0_η already used in the free response. Transport to the unit lattice and define

$$\hat{\mathfrak{a}}^o_\eta(x, x + e_\nu) = \eta^{1/2} a^o_{\eta,\nu}(\eta x).$$

Its coboundary uses unscaled unit-lattice differences. For a smooth two-form f, the normalized covering test cochain is

$$f_{[\eta]}(\sigma) = \eta^{-3/2} \int_{Q_\eta(\sigma)} f_{\mu\nu}(y)\,\mathrm{d}y, \qquad \sigma = (x;\mu\nu), \quad \mu < \nu.$$

Here $Q_\eta(\sigma)$ is the plaquette-centered averaging cube defined before Thm. A. Its volume is η^3, so this cochain is $\eta^{3/2}$ times the cube average. Applying the definition to f^η, the midpoint identities yield

$$d\hat{\mathfrak{a}}^o_\eta(\sigma) = f^\eta_{[\eta]}(\sigma). \tag{11.iii}$$

The two bond values are displaced by $\pm\eta e_\mu/2$ from the plaquette center. Their difference is $\eta\delta_{\eta,\mu} J_\eta a'_{\eta,\nu}$; commutation with J_η gives $\eta^{3/2} J_\eta f^\eta$ at that center.

Periodize the cochain on a torus $C(n)$ of physical side $R = n\eta$. Formula eq. 12.ix gives the exact observable identity

$$\langle d\hat{\mathfrak{a}}_\eta, F^{\mathrm{u}}\rangle = F^\sharp(f^\eta). \tag{11.iv}$$

The source is transverse: $a^o_\eta = d^*_\eta b_\eta$ is orthogonal to every closed bond cochain, by summation by parts. This applies to both the covering cochain and its periodization.

The continuum approximation has the uniform bound

$$\|f^\eta - f\|_{H^s} \le \eta^2 \|\beta\|_{H^{s+4}} \qquad (s \in \mathbb{R},\ 0 < \eta \le 1). \tag{11.v}$$

To see this, the difference symbol is $is_\mu(p)$ with $s_\mu(p) = 2\eta^{-1}\sin(\eta p_\mu/2)$. One has $|s_\mu| \le |p_\mu|$ and $|s_\mu - p_\mu| \le \eta^2 |p_\mu|^3/24$, so

$$|s_\mu s_\nu - p_\mu p_\nu| \le \eta^2 |p|^4/12.$$

Each component of $f^\eta - f$ contains four such terms. Cauchy–Schwarz on these terms, followed by summation over the three two-form components, gives $|\widehat{f^\eta - f}|^2 \le (\eta^2/3)^2 |p|^8 |\hat\beta|^2$. Plancherel's theorem proves eq. 11.v.

11.1.2. *Reference geometry and placement.* Let $T^* \ge 1$ be the reference exponent furnished by the construction, and put

$$R^* = hL^{T^*}, \qquad n_* = R^*/\eta, \qquad p = n_*/L, \qquad w = p/2, \qquad \ell_c = \eta w = R^*/(2L).$$

Since h and η^{-1} are powers of the odd integer L, p is an odd integer, $w \in \mathbb{Z} + 1/2$, and the open unit-lattice reference cells are $Q^\circ_c = w(c + (0,1)^3)$. The reference torus $C(n_*)$ has $2L$ cells per side and $N_* = 8L^3$ in all. The second reference torus has side Ln_*. Every cutoff torus $n = 8hL^{T+N}$ with $T \ge T^* + 1$ is an integer multiple of Ln_*. These are in the range of Prop. 9.1, with $s_0 = 2L$ and $j = L$.

The probe fits the reference cells if

$$D^{\mathrm{buf}}_\beta + 3\eta \le \ell_c. \tag{11.vi}$$

A placement is a vector $v \in \eta\mathbb{Z}^3$ satisfying

$$|v + z_\beta - (\ell_c/2)(1,1,1)|_\infty \le \eta/2.$$

Such a vector always exists by coordinatewise rounding. It translates Q_β into $[\eta, \ell_c - \eta]^3$: the displacement of its center is at most $\eta/2$, and the half-side is at most $(\ell_c - 3\eta)/2$. Hence

$$\mathrm{supp}(\tau_v\beta) \subset [2 + \eta, \ell_c - 2 - \eta]^3. \tag{11.vii}$$

Form the source $\hat{\mathfrak{a}}^o$ with this translated probe.

Every endpoint of a bond in the source support is within $3\eta/2$ in sup distance of $\mathrm{supp}(\tau_v\beta)$. By eq. 11.vii, its lattice coordinates therefore lie in $[2/\eta - 1/2, w - 2/\eta + 1/2]$. If a plaquette has nonzero $d\hat{\mathfrak{a}}^o$, at least one of its boundary bonds belongs to that support. Its other two corners are one unit away. Thus the whole plaquette lies in the box

$$B_0 = (w/2)(1,1,1) + [-r_0, r_0]^3, \qquad r_0 = w/2 - 2/\eta + 3/2. \tag{11.viii}$$

Condition eq. 11.vi gives $r_0 \geq 3$. The distance of this box from each bounding wall is $2/\eta - 3/2 \geq 1/2$, so $B_0 \subset Q_0^\circ$. Its reflected copies lie strictly inside the corresponding cells. This is the support margin needed by the local source identity.

11.1.3. *Orthogonal sources and translated observables.* Let g_c be the wall-group element carrying Q_0° to Q_c°. Put $\hat{\mathfrak{a}}_c^o = (g_c)_* \hat{\mathfrak{a}}^o$ on the covering lattice and periodize it on $C(n)$, where p divides n and $n \geq n_*$, to obtain $\hat{\mathfrak{a}}_c$. The period lattice identifies cell indices modulo $(2n/p)\mathbb{Z}^3$. The coboundary commutes with these actions and periodization, and its support lies in $B_c = g_c B_0$. For cell representatives on this torus,

$$\langle d\hat{\mathfrak{a}}_c, d\hat{\mathfrak{a}}_{c'} \rangle_n = v_\eta \, \mathbf{1}_{\{c=c'\}}, \qquad v_\eta = \|d\hat{\mathfrak{a}}^o\|^2. \tag{11.ix}$$

The covering norm is the unweighted ℓ^2 norm, and the torus pairing is the counting pairing.

To prove the identity, unfold one factor:

$$\langle d\hat{\mathfrak{a}}_c, d\hat{\mathfrak{a}}_{c'} \rangle_n = \sum_\sigma \sum_{z \in \mathbb{Z}^3} d\hat{\mathfrak{a}}_c^o(\sigma) d\hat{\mathfrak{a}}_{c'}^o(\sigma + nz).$$

Disjoint open cells force $c' = c + (2n/p)z$. For distinct torus cells no such z exists; for the same cell the sole contributing translate gives the covering sum of squares. The wall action preserves that norm, proving eq. 11.ix.

Changing the placement by a vector in $\eta\mathbb{Z}^3$ translates the covering source by an integer vector. Thus v_η depends only on β and η, and is independent of the torus and the placement.

For a closed cochain u define $X_c(u) = \langle d\hat{\mathfrak{a}}_c, u \rangle_n$. Every polynomial in this observable is represented by the same local integrand on all these tori. On the cover its local expression is

$$\widehat{A} \longmapsto P\left(\sum_{\sigma \subset B_c} d\hat{\mathfrak{a}}_c^o(\sigma)(d\widehat{A})(\sigma) \right).$$

This is a scalar gauge-invariant polynomial. Cauchy–Schwarz bounds its argument by

$$\sqrt{v_\eta} \left(\sum_{\sigma \subset B_c} |d\widehat{A}(\sigma)|^2 \right)^{1/2}.$$

The polynomial is therefore bounded by

$$C_{\mathrm{pol}} \exp\left(\frac{1}{4} \sum_{\sigma \subset B_c} |d\widehat{A}(\sigma)|^2 \right)$$

for a finite constant C_{pol} depending on the polynomial and the covering source. This is the growth condition for $\mathcal{A}(Q_c^\circ)$.

The local boxes embed in the torus. Indeed, writing $p = 2m_p + 1$, condition eq. 11.vi gives $m_p \geq 7$, and

$$2r_0 + 2 = w - 4/\eta + 5 \leq w + 1 < p \leq n.$$

For each c, the translates of $B_c + [-1,1]^3$ by nz, $z \in \mathbb{Z}^3$, are consequently disjoint. On a local lift, $d\widehat{A} = F^{\mathfrak{u}}$; unfolding the finite cochain sum gives the polynomial $P(X_c)$. For a real polynomial its reflected copy from the base cell is exactly $P(X_c)$, since conjugation fixes its coefficients. This applies to the Wick polynomials used below.

The variance in eq. 11.ix approaches the continuum normal-ordering variance:

$$\left| \sqrt{v_\eta} - \|f\|_2 \right| \leq 21\eta |\beta|_{C^3} (D_\beta^{\mathrm{buf}})^{3/2} + \eta^2 \|\beta\|_{H^4}. \tag{11.x}$$

To see this, first eq. 11.iii gives $v_\eta = |(f^\eta)_\eta|_\eta^2$. Let $\Pi_\eta f^\eta$ be its componentwise cell-average step function on a period cube. Its L^2 norm is $\sqrt{v_\eta}$. Each first derivative of $f_{\mu\nu}^\eta$ is bounded by $4|\beta|_{C^3}$, from the four centered-difference terms in eq. 11.i and the identity $\delta_{\eta,\mu} g = \int_{-1/2}^{1/2} (\partial_\mu g)(\,\cdot + t\eta e_\mu)\, \mathrm{d}t$.

The pointwise error of the cell average is at most $12\eta|\beta|_{C^3}$. It vanishes outside the periodic images of the buffered cube, whose measure in one period is $(D_\beta^{\mathrm{buf}})^3$. Summing the three component squares bounds the L^2 error by $21\eta|\beta|_{C^3}(D_\beta^{\mathrm{buf}})^{3/2}$. The norm triangle inequality and eq. 11.v prove eq. 11.x.

Finally the even cell indices give pure translations. For $y \in (R^*/L)\mathbb{Z}^3 = p\eta\mathbb{Z}^3$ set $c(y) = 2y/(p\eta) \in 2\mathbb{Z}^3$. Then $g_{c(y)}$ is the unit-lattice translation by y/η, and

$$X_{c(y)} = F^\sharp(\tau_{v+y} f^\eta). \tag{11.xi}$$

Cells coincide exactly for translations congruent modulo $R\mathbb{Z}^3$, so any fixed finite set is distinct at large volume. Lattice translation invariance removes v; Gaussian domination gives polynomial integrability. Thus the cell estimates apply to the translates $\tau_y f^\eta$ in §13.2.

11.2. **Exact external-response identities.** We write the zero-flux law as a Gaussian integral over exact curvatures and a separate integral over harmonic twists. Keeping the Jacobian and vacuum factors explicit will fix the normalization of the external-source generating function.

11.2.1. *The determinant at complex connections.* Use lattice units and fix $q > 0$, $r_q = 2\pi/q$ and a real lattice mass μ_{lat}. Put $M_{\mathrm{ch}} = \mu_{\mathrm{lat}} + \frac{\rho}{2}\sum_i w_i$. For the cutoff laws of Thm. A, $\mu_{\mathrm{lat}} = \boldsymbol{\mu}_N$. The weights are the standard chamber weights, and $0 < \rho < 1$. For an integer triple $n = (n_1, n_2, n_3)$ let

$$V = |n| = n_1 n_2 n_3, \qquad B_n = \prod_{\mu=1}^{3} \{0, \dots, n_\mu - 1\}, \qquad \mathcal{L}_n = B_n \times \{1, 2, 3\}.$$

In this subsubsection each n_μ may be as small as one. Cochains are periodic functions on the oriented covering cells, so the $3V$ link orbits remain distinct even at those small sides. The chamber spinors are antiperiodic in each direction and have dimension $192V$. Write D_B for the massless chamber operator with phases $\exp(iqB)$, and $\det_n$ for its finite-dimensional determinant. In the phase convention of §10.2.1, $D_B = D^{\mathrm{ch}}[qB]$. The weight $1/48$ in the spinor pairing does not rescale this determinant.

Every matrix entry is a finite sum of exponentials of linear forms in B, so the determinant is entire in the complex link values.

The two invariances needed below hold for complex cochains:

$$\det_n(D_{B+r_q k} + \mu_{\mathrm{lat}}) = \det_n(D_B + \mu_{\mathrm{lat}}), \qquad \det_n(D_{B+d\lambda} + \mu_{\mathrm{lat}}) = \det_n(D_B + \mu_{\mathrm{lat}}), \tag{11.xii}$$

where k is integer valued and λ is a periodic complex zero-cochain. The first identity follows directly from the phases. For the second, multiplication by $\exp(iq\lambda(c))$ preserves antiperiodicity and has inverse multiplication by $\exp(-iq\lambda(c))$. On an external hop the factor

$$e^{iq(B-d\lambda)(c,c')} e^{iq\lambda(c')} = e^{iq\lambda(c)} e^{iqB(c,c')}$$

gives $D_{B-d\lambda} = e^{iq\lambda} D_B e^{-iq\lambda}$. Internal hops lie over one photon site and give the same conjugation. This proves the determinant identity.

For $s \geq 0$ define

$$K_n(s) = \left(|M_{\mathrm{ch}}| + \frac{1+\rho}{2} \left(\sum_i w_i - 6 + 6e^{qs} \right) \right)^{192V},$$

$$|\det_n(D_B + \mu_{\mathrm{lat}})| \leq K_n(\|\operatorname{Im} B\|_\infty). \tag{11.xiii}$$

Each involution permutes site representatives. Internal phases have modulus one and external phases at most $e^{q\|\operatorname{Im} B\|_\infty}$; a hop has norm $w_i(1+\rho)/2$. Their sum bounds the operator norm by the displayed base, whose $192V$th power bounds the determinant, uniformly in $\operatorname{Re} B$.

The twist connection is

$$H_\theta = \sum_\mu \frac{\theta_\mu}{q n_\mu} \mathbf{1}_\mu, \qquad \mathbf{1}_\mu = \sum_{c \in B_n} \mathbf{1}_{(c,\mu)}. \tag{11.xiv}$$

Its determinant has period 2π in each complex twist variable. For later use we need the cochain implementing this period. Set

$$k^{(\mu)}_\nu(c) = \delta_{\mu\nu} 1_{\{c_\mu = n_\mu - 1\}}, \qquad \lambda^{(\mu)}(c) = -\frac{r_q}{n_\mu} c_\mu, \qquad c \in B_n,$$

with periodic extension. On links not crossing the μ seam, $d\lambda^{(\mu)} = -r_q/n_\mu$ in direction μ and zero in the other directions. On a seam link its value is $r_q - r_q/n_\mu$. Thus

$$H_{2\pi e_\mu} = r_q k^{(\mu)} - d\lambda^{(\mu)}. \tag{11.xv}$$

The two invariances in eq. 11.xii prove the asserted twist period. For $n_\mu = 1$, $\lambda^{(\mu)} = 0$ and $k^{(\mu)} = \mathbf{1}_\mu$, so the same formula holds.

11.2.2. *Real and integer cochains on the torus.* From now on $n_\mu \geq 4$. The counting pairings on cochains are extended bilinearly when their arguments are complex. Use

$$V_{\mathrm{g}} = dC^0(n), \qquad V_{\mathrm{h}} = \mathrm{span}_{\mathbb{R}}\{\mathbf{1}_1, \mathbf{1}_2, \mathbf{1}_3\}, \qquad V_{\mathrm{t}} = (V_{\mathrm{g}} + V_{\mathrm{h}})^\perp.$$

The unfolding separates gauge, harmonic and transverse variables. Integer exactness determines the harmonic fundamental domain, while the real decomposition determines the two Jacobians.

Let

$$\mathcal{P}^{\mathrm{per}}_n = \{p : p_\mu = 2\pi j_\mu / n_\mu,\ 0 \leq j_\mu < n_\mu\}, \qquad e_p(c) = V^{-1/2} e^{ip\cdot c}, \qquad \xi_\mu(p) = e^{ip_\mu} - 1.$$

For the Fourier calculation we use the auxiliary Hermitian counting inner product, under which the characters e_p are orthonormal. The analytic cochain pairings remain complex bilinear. At each momentum p the coboundary on exterior-valued cochains is exterior multiplication by $\xi(p)$. If $\iota_{\bar\xi}$ denotes contraction by $\bar\xi$, then

$$\iota_{\bar\xi}(\xi \wedge u) + \xi \wedge \iota_{\bar\xi} u = |\xi|^2 u.$$

For $p \neq 0$, closed u equals $\xi \wedge (|\xi|^{-2} \iota_{\bar\xi} u)$; the zero modes are constant. Orthogonality of modes and taking real parts give

$$\ker(d : C^1 \to C^2) = V_{\mathrm{g}} \oplus V_{\mathrm{h}}, \qquad C^1 = V_{\mathrm{g}} \oplus V_{\mathrm{h}} \oplus V_{\mathrm{t}}, \qquad \ker(d : C^2 \to C^3) = dC^1 \oplus \mathrm{Harm}^2. \tag{11.xvi}$$

Here Harm^2 is the space of constant two-cochains. The same Fourier formula gives $d^2 = 0$.

For a closed integer two-cochain z, define $w_{\mu\nu}(z)$ as its sum over one coordinate (μ, ν) two-torus. The sum is independent of the transverse coordinate: sum $dz = 0$ over the layer of cubes between two consecutive planes. Differences in the two tangential directions cancel by periodicity; the remaining difference is that of the two plane sums. The fluxes give

$$z \in dC^1(n; \mathbb{Z}) \quad \Longleftrightarrow \quad w_{\mu\nu}(z) = 0 \text{ for all } \mu < \nu. \tag{11.xvii}$$

This is the integer cellular complex of three subdivided circles. The coordinate two-tori form a basis of $H_2(T^3; \mathbb{Z})$ and $H_1(T^3; \mathbb{Z})$ is free, so the universal coefficient theorem identifies H^2 with evaluation on that basis. The class of z therefore vanishes exactly when all $w_{\mu\nu}(z)$ vanish, which is integer exactness. [8, Thms. 3.2, 3.5 and 3B.6]

We next describe the closed integer one-cochains. If $d\lambda$ is integer valued, its sum along a path from 0 to c shows that $\lambda(c) - \lambda(0)$ is integer valued. If $d\lambda = 0$, the same path argument shows that λ is constant. It follows that

$$C^1(n; \mathbb{Z}) \cap V_{\mathrm{g}} = dC^0(n; \mathbb{Z}).$$

Each seam cochain $k^{(\mu)}$ in eq. 11.xv is closed, because its multiple by r_q lies in $V_{\mathrm{g}} + V_{\mathrm{h}}$. For an arbitrary closed integer one-cochain k, eq. 11.xvi gives $k = d\lambda + \sum_\mu t_\mu \mathbf{1}_\mu$. Its sum along a periodic μ loop is $n_\mu t_\mu$, an integer j_μ. Subtracting $\sum_\mu j_\mu k^{(\mu)}$ leaves an integer exact cochain, so

$$k = d\lambda' + \sum_\mu j_\mu k^{(\mu)}, \qquad \lambda' \in C^0(n;\mathbb{Z}), \quad j \in \mathbb{Z}^3. \tag{11.xviii}$$

The harmonic projection of k determines j, and then determines $d\lambda'$. Moreover

$$r_q k = \left(r_q d\lambda' + \sum_\mu j_\mu d\lambda^{(\mu)} \right) + H_{2\pi j}. \tag{11.xix}$$

The first term is exact and the second is harmonic. Their coupling through j will matter when choosing a fundamental domain.

The scalar Laplacian $\Delta_0 = d^*d$ has eigenvalues $|\xi(p)|^2 = \sum_\mu (2 - 2\cos p_\mu)$ and a one-dimensional kernel. Write

$$\det{}'\Delta_0 = \prod_{p \in \mathcal{P}_n^{\mathrm{per}} \setminus \{0\}} |\xi(p)|^2, \qquad \mathcal{L}(n) = \log \det{}'\Delta_0. \tag{11.xx}$$

At each nonzero momentum the transverse one-cochains are the two-dimensional space $\xi(p)^\perp$, with orthogonality in the same Hermitian inner product. For $v \in \mathbb{C}^3$,

$$\sum_{\mu<\nu} |\xi_\mu v_\nu - \xi_\nu v_\mu|^2 = |\xi|^2 |v|^2 - |\bar{\xi} \cdot v|^2.$$

Thus d^*d acts on that transverse space as multiplication by $|\xi(p)|^2$, and

$$\dim V_{\mathrm{t}} = 2V - 2, \qquad \det(d^*d|_{V_{\mathrm{t}}}) = (\det{}'\Delta_0)^2. \tag{11.xxi}$$

Diagonalizing this positive real form gives its Gaussian integral

$$\mathcal{G}_n := \int_{V_{\mathrm{t}}} e^{-\|dA\|^2/2} \, \mathrm{d}\sigma(A) = \frac{(2\pi)^{V-1}}{\det{}'\Delta_0}. \tag{11.xxii}$$

The measure σ is Euclidean Lebesgue measure for the counting pairing, not a coordinate measure chosen along a nonorthogonal basis.

There is one more determinant needed for the exact directions. The cochains $d\mathbf{1}_c$, $c \in B_n \setminus \{0\}$, form both a real basis of V_{g} and an integer basis of $dC^0(n;\mathbb{Z})$. Indeed one subtracts the value at zero from a zero-cochain; independence follows because the resulting zero-cochain has value zero there and cannot be a nonzero constant. Their Gram determinant is

$$\det\left(\langle d\mathbf{1}_c, d\mathbf{1}_{c'} \rangle \right)_{c,c' \neq 0} = \frac{\det{}'\Delta_0}{V}. \tag{11.xxiii}$$

For the site Laplacian S, the coefficient of t in $\det(t + S)$ is both $\det{}'\Delta_0$ and the sum of its V principal cofactors. Translations equate those cofactors; the cofactor at zero is the displayed Gram determinant.

11.2.3. *Unfolding the zero-flux sector.* Let Φ be a Borel function on $C^2(n)$ with $|\Phi(u)| \le C_\Phi e^{C_\Phi \|u\|}$. Let $W : C^1(n) \to \mathbb{C}$ be bounded and continuous, and invariant under addition of real exact cochains and of r_q times integer one-cochains. Then

$$\begin{aligned} &\sum_{\substack{z \in C^2(n;\mathbb{Z}),\, dz=0 \\ w(z)=0}} \int_{[0,r_q)^{\mathcal{L}_n}} \Phi(dA + r_q z) e^{-\|dA + r_q z\|^2/2} W(A) \, \mathrm{d}A \\ &\quad = \mathcal{N}_n \int_{[-\pi,\pi]^3} \frac{\mathrm{d}\theta}{(2\pi)^3} \int_{V_{\mathrm{t}}} \Phi(dA) e^{-\|dA\|^2/2} W(A + H_\theta) \, \mathrm{d}\sigma(A), \end{aligned} \tag{11.xxiv}$$

$$\mathcal{N}_n = r_q^{V+2} (\det{}'\Delta_0)^{1/2}. \tag{11.xxv}$$

Both sides converge absolutely, as the following unfolding shows.

First, the Gaussian sum eq. 2.v gives

$$\sum_{z\in C^2(n;\mathbb{Z})}\int_{[0,r_q)^{\mathcal{L}_n}} e^{-\epsilon\|dA+r_qz\|^2}\,\mathrm{d}A \le (r_qC_q(\epsilon))^{3V}.$$

Here the integer closedness and flux restrictions have been dropped. Completing the scalar square bounds $|\Phi(u)|e^{-\|u\|^2/2}$ by $C_\Phi e^{C_\Phi^2}e^{-\|u\|^2/4}$. Together with boundedness of W, this proves absolute convergence of the left side.

Let $\mathcal{Z}^1_{\mathbb{Z}} = \ker d \cap C^1(n;\mathbb{Z})$ and choose representatives $\mathcal{R}$ of $C^1(n;\mathbb{Z})/\mathcal{Z}^1_{\mathbb{Z}}$. By eq. 11.xvii, $k \mapsto dk$ is a bijection from $\mathcal{R}$ onto the closed zero-flux integer two-cochains. Put

$$P(A) = \Phi(dA)e^{-\|dA\|^2/2}W(A), \qquad \mathcal{D}_1 = \bigcup_{k\in\mathcal{R}}([0,r_q)^{\mathcal{L}_n} + r_qk).$$

The two invariances of W turn the left side of eq. 11.xxiv into $\int_{\mathcal{D}_1} P(A)\,\mathrm{d}A$. Coordinatewise division by r_q, followed by the unique quotient representative modulo $\mathcal{Z}^1_{\mathbb{Z}}$, makes these disjoint boxes a Borel fundamental domain for $r_q\mathcal{Z}^1_{\mathbb{Z}}$. The integrand P is invariant under that group.

We replace this domain by

$$\mathcal{F}_{\mathrm{g}} = \left\{ r_q\sum_{c\neq 0} t_c\, d\mathbf{1}_c : 0 \le t_c < 1 \right\}, \qquad \mathcal{F}_{\mathrm{h}} = \{H_\theta : \theta \in [-\pi,\pi)^3\}, \qquad \mathcal{D}_2 = \mathcal{F}_{\mathrm{g}} + \mathcal{F}_{\mathrm{h}} + V_{\mathrm{t}}.$$

To check that it is a fundamental domain for the same group, write $A = A_{\mathrm{g}} + A_{\mathrm{h}} + A_{\mathrm{t}}$. Formula eq. 11.xix shows that subtracting r_qk changes the harmonic part by $H_{2\pi j}$ and the exact part by $\sum_\mu j_\mu d\lambda^{(\mu)} + r_q d\lambda'$. There is exactly one j placing the harmonic part in $\mathcal{F}_{\mathrm{h}}$. For this j there is exactly one $d\lambda'$ placing the exact part in $\mathcal{F}_{\mathrm{g}}$, by the integer basis used in eq. 11.xxiii. The transverse part is unchanged. Uniqueness in eq. 11.xviii gives uniqueness of k. The orthogonal decomposition identifies $\mathcal{D}_2$ with a product of Borel sets, so it is Borel.

Partition $\mathcal{D}_1$ by its intersections with $\mathcal{D}_2 + h$, $h \in r_q\mathcal{Z}^1_{\mathbb{Z}}$, and translate each by $-h$. These pieces partition $\mathcal{D}_2$. Invariance and Tonelli give equality first for nonnegative functions; applying this to $|P|$ justifies the complex case. Thus $\int_{\mathcal{D}_1} P(A)\,\mathrm{d}A = \int_{\mathcal{D}_2} P(A)\,\mathrm{d}A$.

The two finite-dimensional volumes are

$$\mathrm{vol}(\mathcal{F}_{\mathrm{g}}) = r_q^{V-1}(\det{}'\Delta_0/V)^{1/2}, \tag{11.xxvi}$$

$$\mathrm{d}A_{\mathrm{h}} = V^{1/2}q^{-3}\,\mathrm{d}\theta = r_q^3V^{1/2}\frac{\mathrm{d}\theta}{(2\pi)^3}. \tag{11.xxvii}$$

The gauge formula follows from the Gram determinant. In the orthonormal basis $V^{-1/2}\mathbf{1}_\mu$, the map $\theta \mapsto H_\theta$ has diagonal entries $V^{1/2}/(qn_\mu)$, of product $V^{1/2}q^{-3}$. With normalized twist measure $\mathrm{d}\theta/(2\pi)^3$, the combined factor is $\mathcal{N}_n$.

Since $P(A_{\mathrm{g}} + A_{\mathrm{h}} + A_{\mathrm{t}}) = \Phi(dA_{\mathrm{t}})e^{-\|dA_{\mathrm{t}}\|^2/2}W(A_{\mathrm{t}} + A_{\mathrm{h}})$, it is independent of A_{g}. Fubini over the orthogonal decomposition and the two displayed volume formulas prove eq. 11.xxiv, including absolute convergence of its right side.

Apply this identity to $W(A) = \det_n(D_A + \mu_{\mathrm{lat}})$. Its hypotheses follow from eq. 11.xii and eq. 11.xiii. If $Z_n^{(0)}[\Phi]$ is the zero-flux part of the configuration integral, we have proved

$$Z_n^{(0)}[\Phi] = \mathcal{N}_n\int\frac{\mathrm{d}\theta}{(2\pi)^3}\int_{V_{\mathrm{t}}}\Phi(dA)e^{-\|dA\|^2/2}\det_n(D_{A+H_\theta} + \mu_{\mathrm{lat}})\,\mathrm{d}\sigma(A). \tag{11.xxviii}$$

The twist integral always runs over $[-\pi,\pi]^3$. For real A and θ the determinant is nonnegative. Also $Z_n^{(0)}[1] > 0$ at every real μ_{lat}: the lower bound in Prop. 2.3 was obtained from the single term $z = 0$, which belongs to this sector. Dividing eq. 11.xxviii by its value at $\Phi = 1$ identifies the zero-flux curvature law with the law of dA under the harmonic-averaged transverse measure.

11.2.4. *The Gaussian normalization and matter density.* Let $F_n = dC^1(n) = dV_{\mathrm{t}}$ and let γ_n be the standard Gaussian probability measure on this Euclidean space. The map $d_{\mathrm{t}} = d|_{V_{\mathrm{t}}}$ is a linear bijection onto F_n, by eq. 11.xvi; write $A(u) = d_{\mathrm{t}}^{-1}u$. Changing variables along this map gives

$$\int_{V_{\mathrm{t}}} G(dA)e^{-\|dA\|^2/2}\,\mathrm{d}\sigma(A) = \mathcal{G}_n \int G(u)\,\mathrm{d}\gamma_n(u). \tag{11.xxix}$$

Indeed the Jacobian is $|\det d_{\mathrm{t}}|^{-1}$, and $\dim F_n = 2V - 2$. The Gaussian density contributes $(2\pi)^{V-1}$; setting $G = 1$ identifies the constant with eq. 11.xxii. The formula holds for nonnegative Borel functions, and for the exponentially bounded complex functions used above by their Gaussian integrability.

Define the photon factor and normalized matter factor by

$$Z_n^{\mathrm{ph}} = \mathcal{N}_n\mathcal{G}_n = r_q^{V+2}(2\pi)^{V-1}e^{-\mathcal{L}(n)/2}, \tag{11.xxx}$$

$$z_{r,n}(q,\mu_{\mathrm{lat}}) = \int \frac{\mathrm{d}\theta}{(2\pi)^3}\int \det_n(D_{A(u)+H_\theta} + \mu_{\mathrm{lat}})\,\mathrm{d}\gamma_n(u). \tag{11.xxxi}$$

Then eq. 11.xxviii and eq. 11.xxix imply

$$Z_n^{(0)}[1] = Z_n^{\mathrm{ph}} z_{r,n}, \qquad 0 < z_{r,n} \le K_n(0). \tag{11.xxxii}$$

Positivity follows from that of $Z_n^{(0)}[1]$; the upper bound follows because both integration measures in eq. 11.xxxi are probability measures.

On a cubic torus $V = n^3$, fix a positive reference mass $\bar{m}$, mesh η and a real pair $(m_{\mathrm{in}}, \varepsilon_{\mathrm{in}}^{\mathrm{v}})$, with $\mu_{\mathrm{lat}} = \bar{m}\eta + m_{\mathrm{in}}$. The density with external cochain a is

$$\mathrm{dens}^{\theta,a}(u) = e^{-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3}\frac{\det_n(D_{A(u)+H_\theta+a} + \mu_{\mathrm{lat}})}{\det_n(D^{\mathrm{ch}}[0] + \bar{m}\eta)}. \tag{11.xxxiii}$$

The denominator is positive, since the real part of $D^{\mathrm{ch}}[0] + \bar{m}\eta$ is at least $\bar{m}\eta > 0$. At $a = 0$, the density is continuous, bounded and nonnegative. Let $\nu^{\theta,0}$ be its measure against γ_n and let $\bar{\nu}$ be the twist average. eq. 11.xxviii and eq. 11.xxix give

$$\bar{\nu}(\Phi) = \frac{e^{-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3}}{\det_n(D^{\mathrm{ch}}[0] + \bar{m}\eta)Z_n^{\mathrm{ph}}} Z_n^{(0)}[\Phi], \tag{11.xxxiv}$$

$$\bar{\nu}(1) = \frac{e^{-\varepsilon_{\mathrm{in}}^{\mathrm{v}} n^3} z_{r,n}}{\det_n(D^{\mathrm{ch}}[0] + \bar{m}\eta)} \in (0,\infty). \tag{11.xxxv}$$

Thus $\bar{\nu}/\bar{\nu}(1)$ is the normalized zero-flux curvature law; write $\mathbb{E}_n^{\mathrm{av}}$ for expectation in this law. The vacuum normalization cancels in the quotient. The quantities Z_n^{ph} and $z_{r,n}$ depend only on $(n, q, \mu_{\mathrm{lat}})$ and the fixed chamber parameters. The auxiliary split $\mu_{\mathrm{lat}} = \bar{m}\eta + m_{\mathrm{in}}$ does not change them.

11.2.5. *Holomorphic external partition functions.* Write $\mathfrak{F}(n)$ for the exponentially bounded Borel functions Φ used above. For a complex one-cochain a put

$$\mathbf{Z}_n^{\mathrm{x}}(a;\Phi) = \int \frac{\mathrm{d}\theta}{(2\pi)^3}\int_{V_{\mathrm{t}}} \Phi(dA)e^{-\|dA\|^2/2}\det_n(D_{A+H_\theta+a} + \mu_{\mathrm{lat}})\,\mathrm{d}\sigma(A). \tag{11.xxxvi}$$

Omit Φ when it equals one. The photon curvature is not shifted.

Proposition 11.1. For $\Phi \in \mathfrak{F}(n)$ this integral converges absolutely, locally uniformly in a, and is entire in the $3V$ complex link values. Moreover $\mathbf{Z}_n^{\mathrm{x}}(0) > 0$. ∎

Proof. On $\max_b |a(b)| \le s$, the strip bound gives the majorant

$$K_n(s)C_\Phi e^{C_\Phi^2}e^{-\|dA\|^2/4} \quad \text{when } |\Phi(u)| \le C_\Phi e^{C_\Phi\|u\|}.$$

It is integrable, since eq. 11.xxii gives

$$\int_{V_{\mathrm{t}}} e^{-\|dA\|^2/4}\,\mathrm{d}\sigma(A) = 2^{V-1}\mathcal{G}_n.$$

If $v(y,a)$ is jointly measurable, entire in $a \in \mathbb{C}^d$, and dominated by an integrable function on each polydisc, its integral has the same holomorphy. Fix $a^0 \in \mathbb{C}^d$ and $r > 0$, and give $\mathbb{T} = \{\zeta : |\zeta_j - a^0_j| = 2r\}$ its product orientation. For $|a - a^0|_\infty < r$, Cauchy's formula and Fubini give

$$\int v(y,a)\,\mathrm{d}\nu(y) = \frac{1}{(2\pi i)^d}\int_{\mathbb{T}} \frac{\int v(y,\zeta)\,\mathrm{d}\nu(y)}{\prod_j(\zeta_j - a_j)}\,\mathrm{d}\zeta. \tag{11.xxxvii}$$

The contour kernel and its derivatives are bounded there. The derivative Cauchy formula therefore also permits differentiation under the y integral. Apply this observation to the determinant and the majorant above. Finally eq. 11.xxviii and $\mathcal{N}_n > 0$ give the positive value at zero. □

The densities eq. 11.xxxiii define finite complex measures $\nu^{\theta,a}$ and their twist average $\bar\nu^a$ against γ_n, with

$$|\mathrm{dens}^{\theta,a}(u)| \le \frac{e^{-\varepsilon^{\mathrm{v}}_{\mathrm{in}} n^3} K_n(\|\operatorname{Im} a\|_\infty)}{\det_n(D^{\mathrm{ch}}[0] + \bar m\eta)}. \tag{11.xxxviii}$$

This is independent of (θ, u), so Fubini gives a bounded Borel density for the average. eq. 11.xxix and the exponential majorant give, for $\Phi \in \mathfrak{F}(n)$,

$$\bar\nu^a(\Phi) = \frac{e^{-\varepsilon^{\mathrm{v}}_{\mathrm{in}} n^3}}{\det_n(D^{\mathrm{ch}}[0] + \bar m\eta)\mathcal{G}_n}\mathbf{Z}^{\mathrm{x}}_n(a;\Phi). \tag{11.xxxix}$$

For every Borel $E \subset F_n$, its indicator extends to a bounded Borel function on $C^2(n)$ because F_n is closed. Thus $a \mapsto \bar\nu^a(E)$ is entire and bounded by the right side of eq. 11.xxxviii; composition with a linear map $t \mapsto a(t)$ is entire. The positive zero-source normalization is eq. 11.xxxv.

11.2.6. *The source shift and Wick derivatives.*

Proposition 11.2. For every complex one-cochain a,

$$\mathbb{E}^{\mathrm{av}}_n e^{\langle da, F\rangle} = e^{\langle da, da\rangle/2}\frac{\mathbf{Z}^{\mathrm{x}}_n(a)}{\mathbf{Z}^{\mathrm{x}}_n(0)}. \tag{11.xl}$$

The quadratic form is bilinear, without complex conjugation. ■

Proof. For real a, eq. 11.xvi writes $a = d\lambda + H_\vartheta + a_{\mathrm{t}}$ with real λ, ϑ and $a_{\mathrm{t}} \in V_{\mathrm{t}}$. Gauge invariance removes $d\lambda$ from the determinant, and its twist period eq. 11.xv allows the normalized twist integral to be translated by ϑ. Change $B = A + a_{\mathrm{t}}$ in the transverse integral; its Lebesgue Jacobian is one and $da = da_{\mathrm{t}}$. The Gaussian exponent becomes

$$-\tfrac12\|d(B - a_{\mathrm{t}})\|^2 = -\tfrac12\|dB\|^2 + \langle da, dB\rangle - \tfrac12\|da\|^2.$$

The normalized zero-flux representation eq. 11.xxviii gives eq. 11.xl for real a.

Both sides are entire. For the left side, on $\max_b |a(b)| \le s$ put $c_s = 4s\sqrt{3V}$, so $\|d\operatorname{Re} a\| \le c_s$. In its normalized transverse representation the integrand is bounded, up to the fixed normalization, by $K_n(0)e^{c_s^2}e^{-\|dB\|^2/4}$. The dominated Cauchy formula applies. The right side is entire by Prop. 11.1 and its nonzero constant denominator. Iterating the one-variable identity theorem from real cochains proves the complex identity. □

Let $a^{(1)}, \dots, a^{(r)}$ be real cochains and $X_j = \langle da^{(j)}, F\rangle$. Apply eq. 11.xl to $a = \sum_j z_j a^{(j)}$:

$$\mathbb{E}^{\mathrm{av}}_n \exp\left(\sum_j z_j X_j - \tfrac12\sum_{i,j} z_i z_j \langle da^{(i)}, da^{(j)}\rangle\right) = \frac{\mathbf{Z}^{\mathrm{x}}_n(\sum_j z_j a^{(j)})}{\mathbf{Z}^{\mathrm{x}}_n(0)}, \qquad z \in \mathbb{C}^r. \tag{11.xli}$$

Here $\mathbb{E}_n^{\mathrm{av}}$ is the zero-flux law from eq. 11.xxviii; the covariance uses the counting pairing.

Use the reference cells of §11.1 on the torus n_*. Their index set is $\mathcal{C}_* = \{0,\dots,2L-1\}^3$, and $\widehat{a}_c$ is the periodization of $\hat{\mathfrak{a}}_c^o$. These are real periodic cochains. eq. 11.ix gives

$$\langle d\widehat{a}_c, d\widehat{a}_{c'}\rangle = v_\eta\, 1_{\{c=c'\}}. \tag{11.xlii}$$

The variance $v_\eta = \|d\hat{\mathfrak{a}}_\eta^o\|^2$ is independent of the cell and the placement. Put

$$a(t) = e^{-1}\sum_{c\in\mathcal{C}_*} t_c\widehat{a}_c, \qquad \bar{Z}^{\mathrm{ext}}(t) = \int \frac{\mathrm{d}\theta}{(2\pi)^3}\int \det_{n_*}(D_{A(u)+H_\theta+a(t)}+\mu_{\mathrm{lat}})\,\mathrm{d}\gamma_{n_*}(u).$$

The reference torus has $n_* \geq 4$, and $e>0$ is the physical charge in this convention. eq. 11.xxix gives

$$\bar{Z}^{\mathrm{ext}}(t) = \mathcal{G}_{n_*}^{-1}\mathbf{Z}^{\mathrm{x}}_{n_*}(a(t)), \qquad \mathring{\Phi}(t) := \frac{\bar{Z}^{\mathrm{ext}}(t)}{\bar{Z}^{\mathrm{ext}}(0)} = \frac{\mathbf{Z}^{\mathrm{x}}_{n_*}(a(t))}{\mathbf{Z}^{\mathrm{x}}_{n_*}(0)}. \tag{11.xliii}$$

Its zero value in the denominator is positive, and $\mathring{\Phi}$ is entire. With $X_c = \langle d\widehat{a}_c, F\rangle$, eq. 11.xli becomes

$$\mathbb{E}^{\mathrm{av}}_{n_*}\exp\sum_c (z_cX_c - \tfrac{1}{2}v_\eta z_c^2) = \mathring{\Phi}(ez). \tag{11.xliv}$$

Define the Hermite polynomials with this variance by $H_j(y) = \partial_z^j e^{zy - v_\eta z^2/2}|_{z=0}$. In particular

$$H_2(y) = y^2 - v_\eta, \qquad H_4(y) = y^4 - 6v_\eta y^2 + 3v_\eta^2.$$

To differentiate eq. 11.xliv under the expectation, on a polydisc $\max_c|z_c| \leq s$ use

$$\left|\exp\sum_c (z_cX_c - \tfrac{1}{2}v_\eta z_c^2)\right| \leq \exp\left(s|\mathcal{C}_*|\sqrt{v_\eta}\|F\| + \tfrac{1}{2}|\mathcal{C}_*|s^2v_\eta\right).$$

The Gaussian majorant integrates this exponential bound, so eq. 11.xxxvii permits differentiation. Since $|X_c| \leq \sqrt{v_\eta}\|F\|$, all resulting Hermite products are bounded by a polynomial in $\|F\|$ and are integrable as well.

For $m = 2$ or 4 and a subset $S \subset \mathcal{C}_*$, differentiate in precisely those variables. The factor ez on the right supplies $e^{m|S|}$, giving

$$\mathbb{E}^{\mathrm{av}}_{n_*}\prod_{c\in S} e^{-m}H_m(X_c) = \left(\prod_{c\in S}\partial_{t_c}^m\mathring{\Phi}\right)(0).$$

Expanding the product over subsets proves, for every real c_m,

$$\mathbb{E}^{\mathrm{av}}_{n_*}\prod_{c\in\mathcal{C}_*}(e^{-m}H_m(X_c) - c_m) = \left(\prod_{c\in\mathcal{C}_*}(\partial_{t_c}^m - c_m)\mathring{\Phi}\right)(0). \tag{11.xlv}$$

Entireness makes the derivatives commute; all normalization factors are those of eq. 11.xliii.

11.3. **Terminal operators and norms.** Differentiating the source produces local insertions in the terminal Dirac operator and in the interaction. We need bounds that retain the positions of these insertions: decay between their supports controls the response in separated cells. We give the field-map and Grassmann estimates in the coordinates of the stopping scale.

11.3.1. *Reference scales and the probe.* We use the reference torus of the tuned flow. Fix $\bar m \in (0,1]$, put $h = L^{\lfloor \log_L(1/\bar m)\rfloor}$, and assume $0 < e\sqrt{h} \le \bar e_1$ and $N \ge N_{\mathrm{ref}}(e)$. By eq. 6.i and eq. 6.vii,

$$\eta = L^{-N}, \qquad K = N + \log_L h, \qquad a_j = L^{j-N}, \\ e_j = e\sqrt{a_j}, \qquad \eta_j = L^{-j} = \eta/a_j, \qquad \ell_j = R^*/a_j. \tag{11.xlvi}$$

Thus a point with level-j coordinates c has physical coordinates $a_j c$, and $T_j = \eta_j\mathbb{Z}^3/\ell_j\mathbb{Z}^3$. At $j=0$ the charge is $q = e\sqrt{\eta}$. The symbols $\bar e_0, \bar e_1$ without a running index retain their meanings as charge ceilings. For a bond cochain a on the fine unit torus, its rescaling is

$$a^{[j]}(b) = L^{j/2} a(L^j b). \tag{11.xlvii}$$

On the reference torus, eq. 6.viii and eq. 6.iv give

$$k_* \in \{K-1, K-2\}, \qquad a_* \in \{h/L, h/L^2\}, \qquad \ell_* = Mr_*, \qquad r_* > 31L + 16.$$

The offset $K - k_*$ and the quantities a_*, ℓ_*, r_* are independent of $N \ge N_{\mathrm{ref}}(e)$; the index k_* depends on N through K. Write r_{loc} for the localization power in §6.1.

Let $\beta \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$ be real and nonzero. Use the cube $Q_\beta = z_\beta + [-D_\beta^{\mathrm{buf}}/2, D_\beta^{\mathrm{buf}}/2]^3$ of §11.1, where $D_\beta^{\mathrm{buf}} = D_\beta + 4$. In this subsection a placement $v \in \eta\mathbb{Z}^3$ is admitted if, in physical coordinates,

$$v + Q_\beta \subset [\ell_c/8, 7\ell_c/8]^3, \qquad \ell_c = R^*/(2L). \tag{11.xlviii}$$

A centered placement satisfies $|v + z_\beta - (\ell_c/2)(1,1,1)|_\infty \le \eta/2$. Adding this displacement to the cube's half-side shows that it is admitted when $D_\beta^{\mathrm{buf}}/2 + \eta/2 \le 3\ell_c/8$.

Form $\widehat{\mathfrak{a}}$ with $\tau_v\beta$ as in eq. 11.ii, and periodize it on $C(n_*)$, $n_* = R^*/\eta$. For $c \in \mathcal{C}_* = \{0, \dots, 2L-1\}^3$, write $\widehat{\mathfrak{a}}_c$ for its cell copy. In physical coordinates the wall map carrying the first cell to cell c acts in coordinate μ by translation through $2i\ell_c$ if $c_\mu = 2i$, and by reflection $y_\mu \mapsto (2i+2)\ell_c - y_\mu$ if $c_\mu = 2i+1$. Define the marked connections by

$$a(t) = e^{-1}\sum_c t_c \widehat{\mathfrak{a}}_c, \qquad \widehat{c}_c = e^{-1}\widehat{\mathfrak{a}}_c^{[k_*]}, \qquad \mathscr{C}(t) = a(t)^{[k_*]} = \sum_c t_c \widehat{c}_c. \tag{11.xlix}$$

For a level-j bond function A, ∂A uses forward differences divided by η_j, and

$$|\delta_\alpha \partial A|_\infty = \sup_{\mu,\nu} \sup_{0<d(x,x')\le 1} \frac{|\partial_\nu A_\mu(x') - \partial_\nu A_\mu(x)|}{d(x,x')^\alpha}.$$

The external norm of a fine connection is $|a|_{\mathrm{ext}} = |a^{[k_*]}|_\infty + |\partial a^{[k_*]}|_\infty + |\delta_\alpha \partial a^{[k_*]}|_\infty$.

For an integer $j \ge 0$, use the component derivative norm

$$|\beta|_{C^j} = \max_{\mu<\nu,\, |\alpha|\le j} \sup_{x\in\mathbb{R}^3} |\partial^\alpha \beta_{\mu\nu}(x)|.$$

For $0 < \eta' \le 1$, $a_{\mathrm{s}} \in \{h/L, h/L^2\}$ and a coordinate reflection r', set

$$g_\nu = -\sum_{\mu\ne\nu} \delta_{\eta',\mu} J_{\eta'}(U_{r'}\beta)_{\mu\nu},$$

$$G_\nu(c) = \sqrt{a_{\mathrm{s}}}\, g_\nu(a_{\mathrm{s}} c + \eta' e_\nu/2), \qquad W_\nu(x) = \sqrt{\eta'}\, g_\nu(\eta' x + \eta' e_\nu/2),$$

where G has mesh η'/a_{s} and $U_{r'}\beta(x) = (\wedge^2 r')\beta(r'^{-1}x)$. Choose the explicit constants

$$C_\beta = 2\sqrt{h}(1 + h + 3h^2)|\beta|_{C^3}, \qquad \mathbf{D} = 48|\beta|_{C^2}^2 (D_\beta^{\mathrm{buf}} + 3)^3. \tag{11.l}$$

They are independent of charge, mesh, placement and torus, and satisfy

$$|G|_\infty + |\partial G|_\infty + |\delta_\alpha \partial G|_\infty \le C_\beta, \qquad \|dW\|_{\ell^2}^2 \le \mathbf{D}. \tag{11.li}$$

Indeed cube averaging has norm one on derivative sup norms, and a centered difference is an average of the corresponding derivative. Thus $|D^i g_\nu|_\infty \le 2|\beta|_{C^{i+1}}$, $i = 0,1,2$. Rescaling gives bounds $2a_{\mathrm{s}}^{1/2}|\beta|_{C^1}$ and $2a_{\mathrm{s}}^{3/2}|\beta|_{C^2}$ for the first two norms of G. The mean-value formula in three

coordinates bounds the difference of its first-difference averages by $6a_{\mathrm{s}}^{5/2}|\beta|_{C^3}|c-c'|_\infty$. Dividing by $|c-c'|_\infty^\alpha$, for $0<|c-c'|_\infty\le 1$, and using $a_{\mathrm{s}}\le h$ proves the first bound. Each component of dW is at most $4(\eta')^{3/2}|\beta|_{C^2}$. Its supporting bond endpoints lie in $r'Q_\beta$, since averaging, centered differences and endpoint displacement enlarge the original support by at most $3\eta'/2\le 3/2$ per coordinate. A supporting plaquette base lies in that cube enlarged by η' in the negative directions. There are at most $3(D_\beta^{\mathrm{buf}}/\eta'+2)^3$ oriented plaquettes. Squaring the component bound gives $48|\beta|_{C^2}^2(D_\beta^{\mathrm{buf}}+2\eta')^3\le\mathbf{D}$.

At the reference stop use $\eta'=\eta$, $a_{\mathrm{s}}=a_*$ and $r'=1$. The midpoint formula and eq. 11.xlvii give

$$\widehat{\mathfrak{a}}_\nu^{[k_*]}(c)=\sum_{z\in\mathbb{Z}^3}G_\nu(c-v/a_*+\ell_* z).$$

Supporting bond endpoints lie in $a_*^{-1}(v+Q_\beta)$. Distinct periodic copies of that cube have distance at least $\ell_*-3\ell_*/(8L)>\ell_*/2>1+2\eta_*$. Each of the three norms compares bonds whose endpoints are at distance at most $1+2\eta_*$, so it involves at most one periodic copy. Its supremum is therefore the corresponding norm of G. Translations and coordinate reflections preserve these norms, including the one-bond shift that reverses an oriented component; the same argument applies to the cell copies.

The image of $v+Q_\beta$ under each wall map lies in $\prod_\mu[c_\mu\ell_c+\ell_c/8,(c_\mu+1)\ell_c-\ell_c/8]$. For distinct cells, a separating coordinate on the periodic circle gives distance at least $\ell_c/4$: any joining arc contains the two margins of size $\ell_c/8$. Thus the supporting endpoints of distinct $\widehat{c}_c$ are separated, in coordinates at the stopping scale, by at least $\ell_*/(8L)$, which is larger than $1+2\eta_*$. Again each evaluation defining the external norm involves at most one copy, so

$$|a(t)|_{\mathrm{ext}}\le C_\beta|t|_\infty/e,\qquad v_\eta=\|d\widehat{\mathfrak{a}}\|^2\le\mathbf{D}. \tag{11.lii}$$

For the second bound, the covering curvature dW is supported in the cube of side $D_\beta^{\mathrm{buf}}/\eta+2<n_*$ obtained by enlarging $\eta^{-1}(v+Q_\beta)$. Its periodic translates do not overlap, so periodization preserves its counting norm. Distinct cell curvatures also have disjoint plaquette supports: a common plaquette would have vertices in both endpoint sets at physical distance at most $\eta\le 1$, whereas $\ell_c/4>1$. This gives eq. 11.ix for every admitted placement.

Let P_c be the unit-block centers whose blocks contain an endpoint of a supporting bond of $\widehat{c}_c$, and let P_c^+ be their $(M+1)$-neighborhood. For $y\in P_c$, $y'\in P_{c'}$ choose the corresponding endpoints in $\Delta_y,\Delta_{y'}$. Each endpoint is at distance less than $1/2$ from its center. Consequently

$$d_{\mathrm{sep}}:=\min_{c\ne c'}d(P_c,P_{c'})\ge\ell_*/(8L)-1=Mr_*/(8L)-1, \tag{11.liii}$$

$$|P_c^+|\le p_\beta:=(D_\beta^{\mathrm{buf}}/a_*+2M+4)^3. \tag{11.liv}$$

For the count, lift the endpoints to a cube of side $D_\beta^{\mathrm{buf}}/a_*$. Passing to block centers enlarges each coordinate bound by less than $1/2$, and taking the neighborhood adds $M+1$. The resulting open interval has length $D_\beta^{\mathrm{buf}}/a_*+2M+3$ and contains fewer than $D_\beta^{\mathrm{buf}}/a_*+2M+4$ integers. Taking the product proves the count. If the lattice probe vanishes, these sets are empty and $d_{\mathrm{sep}}=\infty$.

11.3.2. *Terminal operators and local insertions.* At a terminal level on either R^* or LR^*, write $\eta_*=L^{-k_*}$, $T_*=\eta_*\mathbb{Z}^3/\ell_*\mathbb{Z}^3$ and $T_*^0=\mathbb{Z}^3/\ell_*\mathbb{Z}^3$. The fine chamber spinors have counting measure $\eta_*^3\,48^{-1}\sum_{c,g}$, and the coarse spinors have four components per unit site and weight one. We use

$$k_*\ge 1,\qquad \ell_*>2M+2,\qquad \gamma^F M\ge 16,$$
$$1-L^{-1}=:b_-\le b_*\le 1,\qquad L^{-3}=:\mu_-<\mu_*\le 1.$$

The running mass μ_* is positive; the bare mass coupling is treated separately. Let

$$F_*(A)=D_A+\mu_*+b_*JP_*(A)J^*,\qquad S_*(A)=F_*(A)^{-1},$$

$$H_*(A) = b_* S_*(A) J Q_*^T(-A), \qquad D_*(A) = b_* - b_* Q_*(A) J^* H_*(A). \tag{11.lv}$$

These are the blocked operators already constructed in eq. 4.xlix and eq. 4.l, at $k = k_*$. The symbol F_* here is an operator on spinors, distinct from the curvature variable in the external partition function.

For a unit site y let Δ_y be its fine block. The averaging phases have the form $\varphi_A(y,c) = e^{i\phi_A(y,c)}$, where $\phi_A = e_* \eta_* \tau_{k_*} A$ is linear in A. The nested block centers give

$$|\phi_B(y,c)| \le \tfrac{3}{2} e_* |B|_\infty, \qquad |\upsilon_B(c,c')| \le 3|c-c'| e_* |B|_\infty. \tag{11.lvi}$$

The first phase uses only bonds inside Δ_y; the second, which defines $U_A(c,c') = e^{i\upsilon_A(c,c')}$, uses only bonds in the coordinate box between its endpoints. For the first bound, the path lengths at block scales $L^j \eta_*$ add up to $\frac{3}{2}(L-1)\eta_* \sum_{j<k_*} L^j \le \frac{3}{2}$. Each of the six coordinate-order paths between c and c' has length at most $3|c-c'|$, which gives the second bound.

For a complex direction B, let P_B be the unit sites y whose block Δ_y contains an endpoint of a bond in $\operatorname{supp} B$. Define

$$\epsilon_B = e_* |B|_\infty, \qquad P_B^+ = \{y : d(y, P_B) \le M+1\}.$$

The distance is the periodic sup distance on the unit lattice. For an operator O between fine, coarse or pair spaces, localize each fine spinor in its unit block and each pair at its first point, and put

$$|O|(y,y') = \sup_{\|v\|_\infty \le 1} \|\mathbf{1}_y O \mathbf{1}_{y'} v\|_\infty,$$

$$\|O\|_\gamma = \max\left\{ \sup_y \sum_{y'} |O|(y,y') e^{\gamma d(y,y')}, \sup_{y'} \sum_y |O|(y,y') e^{\gamma d(y,y')} \right\}. \tag{11.lvii}$$

The spinor sup norm uses the maximum of component moduli. Inserting the unit-block partition between two factors gives $|O_1 O_2| \le |O_1||O_2|$ and $\|O_1 O_2\|_\gamma \le \|O_1\|_\gamma \|O_2\|_\gamma$. Write $\|\cdot\|_{\mathrm{op}}$ for the Hilbert-space operator norm and $\|\cdot\|_{\mathfrak{S}_1}$ for the trace norm. For coarse spinor operators,

$$\|O\|_{\mathrm{op}} \le 2\|O\|_0, \qquad \|O\|_{\mathfrak{S}_1} \le 4 \sum_{y,y'} |O|(y,y'). \tag{11.lviii}$$

The row and column sums are bounded by $\|O\|_0$ and $4\|O\|_0$, so Schur gives the first bound. Summing entry moduli gives the trace norm bound, with at most four times the block's maximum row sum.

Use the calibrated constants $a' \ge 1$, $0 < \gamma^F \le 1$ and

$$C^F \ge 608 \exp(1) \max(1,b) a', \qquad C^F \ge 64 \exp(1) b^2 a', \qquad b = 1.$$

On $\widetilde{G}_*$, the conditions $6L e_*^\varepsilon \le 1$ and $3M e_*^\varepsilon \le 1$, together with eq. 11.lvi, bound $|\varphi_A^{\pm 1}|$ and $|U_A^{\pm 1}|$ by $\exp(1)$. The propagator estimates eq. 4.xli, at $y = \infty$, then give the following Euclidean spinor constants; the block lift $J Q_*^T(-A)$ contributes only its phase bound.

$$S_* : 32a', \qquad \delta_{\alpha,A} S_* \text{ (short)} : 304 a', \qquad H_* : 32 \exp(1) b a',$$
$$\delta_{\alpha,A} H_* \text{ (short)} : 304 \exp(1) b a', \qquad \delta_{\alpha,A} H_* \text{ (long)} : 128 \exp(1) b a'.$$

Here the long-pair entry follows from $d_X^{-\alpha} \le 1$ and $|\delta_{\alpha,A} v(x,x')|_2 \le \exp(1)|v(x')|_2 + |v(x)|_2$; $(1 + \exp(1))32\exp(1) b a' \le 128 \exp(1) b a'$. For the first four entries the same estimates hold after localization with the factor $E(y,y') = e^{-\gamma^F d(y,y')}$; for a long pair use the sum of the two endpoint decays. The conversion $|v|_c \le |v|_2 \le 2|v|_c$ and the calibration of C^F give

$$|S_*|, \quad |\delta_{\alpha,A} S_*|, \quad |H_*|, \quad |\delta_{\alpha,A} H_*| \le C^F E, \tag{11.lix}$$

where the two pair bounds concern $d_X < 1$, including intracluster pairs. All maps are analytic by the preceding inverse construction.

The domain

$$\mathcal{W} = \{A_r + \mathscr{C} : A_r \in \widetilde{R}_* \text{ real}, \ \mathscr{C} \in \mathfrak{X}_*\}$$

lies in $\widetilde{G}_*$, is invariant under sign reversal, and has $\mathfrak{i} = e_*|\operatorname{Im} A|_\infty < 1/16$ and $3M\mathfrak{i} < 1/4$. The unweighted versions of the table imply

$$|H_*^{\#}(A)|_{1,\infty} \le C^F, \qquad |\bar{H}_*^{\#}(A)|_{1,\infty} \le C^F. \tag{11.lx}$$

Indeed a map from coarse component ℓ^∞ has norm equal to its largest row sum: prescribe the conjugate phases of one row's entries independently on a fundamental domain. Coarse sites have weight one, and antiperiodic extension does not change their moduli. The barred map is $\mathsf{C}^{-1}H_*(-A)\mathsf{C}$, with C a signed component permutation commuting with scalar quotient phases, so the same estimates apply. Localizing the long-pair argument also gives the coefficient bound $C^F e^{-\gamma^F \min\{d([x],y'),d([x'],y')\}}$.

Put

$$\mathcal{S} = C^F(24/\gamma^F)^3, \qquad h_{\mathrm{L}} = 6\mathcal{S}, \qquad C_H = 203M(C^F)^2, \qquad C_K = 226\mathcal{S}^2.$$

The lattice sum $\sum_{y'} e^{-\gamma d(y,y')} \le (12/\gamma)^3$, $0 < \gamma \le 6$, follows by choosing nearest periodic representatives and summing $e^{-\gamma|z|_\infty} \le e^{-\gamma|z|_1/3}$ over $\mathbb{Z}^3$. The one-dimensional sum is $\coth(\gamma/6) \le 12/\gamma$.

For a fine spinor or Hölder-pair index u, let $\operatorname{Leg}_A(u; y', \beta')$ be the coefficient of $\Psi_{\beta'}(y')$ after substituting $H_*^{\#}(A)\Psi$ in that generator. Barred generators use the corresponding barred coefficient. The set $\operatorname{pos}(u)$ contains the unit block of the first endpoint and, for a pair with $d_X(x, x') \ge 1$, also the block of the second endpoint. It has at most two elements. Differentiating an inverse inserts a local operator between two propagators. For separated directions, the mixed derivative therefore gives ordered exponential chains through both marked sets. In the mixed bounds below, the middle factor $E(z, z')$ retains the distance between the insertions. The two chains visit their supports in the two possible orders.

Proposition 11.3. For $A \in \mathcal{W}$, the blocked field maps and their Hölder maps are analytic in A. The first coarse-operator derivative satisfies

$$\begin{aligned} |\partial_B D_*(A)|(y, y') &\le 226(C^F)^2\epsilon_B \sum_{z\in P_B} E(y, z)E(z, y'), \\ \|\partial_B D_*(A)\|_{\gamma^F/2} &\le C_K\epsilon_B, \qquad \sum_{y,y'} |\partial_B D_*(A)|(y, y') \le C_K\epsilon_B|P_B|. \end{aligned} \tag{11.lxi}$$

If $d(P_B, P_{B'}) \ge M + 2$, then

$$\begin{aligned} |\partial_B\partial_{B'} D_*(A)|(y, y') \le C_H^2\epsilon_B\epsilon_{B'} \sum_{z\in P_B, z'\in P_{B'}} &\big(E(y, z)E(z, z')E(z', y') \\ &+ E(y, z')E(z', z)E(z, y')\big). \end{aligned} \tag{11.lxii}$$

The corresponding fine-field legs satisfy

$$\begin{aligned} |\operatorname{Leg}_A|(u; y') &\le 3C^F \sum_{w\in\operatorname{pos}(u)} E(w, y'), \\ |\partial_B \operatorname{Leg}_A|(u; y') &\le C_H\epsilon_B \sum_{w\in\operatorname{pos}(u), z\in P_B^+} E(w, z)E(z, y'), \end{aligned} \tag{11.lxiii}$$

$$\begin{aligned} |\partial_B\partial_{B'} \operatorname{Leg}_A|(u; y') \le C_H^2\epsilon_B\epsilon_{B'} \sum_{\substack{w\in\operatorname{pos}(u) \\ z\in P_B^+, z'\in P_{B'}^+}} &\big(E(w, z)E(z, z')E(z', y') \\ &+ E(w, z')E(z', z)E(z, y')\big). \end{aligned} \tag{11.lxiv}$$

The last bound has the same separation hypothesis. These estimates hold for barred and unbarred legs. ∎

Proof. The block averages and their transposes are diagonal in unit blocks. From eq. 11.lvi and $\mathrm{i}(A) \le 1/4$ their majorants are at most $3/2$. Their directional derivatives are bounded by

$$(9/4)\epsilon_B 1_{\{y\in P_B\}}.$$

Differentiating the only phased Dirac hop gives

$$(\partial_B D_A v)(x) = -ie_* B(c,c_3) M_3(x) u_A(c,c_3) v(x^{(3)}).$$

Its majorant is at most $8\epsilon_B$, vanishes unless the unit blocks coincide or are neighbors, and vanishes unless both belong to P_B. The derivative of $JP_*(A)J^*$ has diagonal majorant at most $7\epsilon_B$. Thus for $F_B = \partial_B F_*(A)$,

$$|F_B|(z,z') \le \epsilon_B(15\, 1_{\{z=z'\}} + 8\, 1_{\{z,z' \text{ neighbors}\}}) 1_{\{z,z'\in P_B\}}.$$

Summing the six neighboring blocks gives $\|F_B\|_\gamma \le 146\epsilon_B$ for $0 \le \gamma \le 1$. Also, since $E(z',y') \le \mathrm{e}E(z,y')$ for neighboring z,z',

$$\sum_{z,z'} E(y,z)|F_B|(z,z')E(z',y') \le 146\epsilon_B \sum_{z\in P_B} E(y,z)E(z,y'). \tag{11.lxv}$$

If the supports are separated by $M+2$, the mixed derivatives of D_A, the averages, JP_*J^*, F_* and $\delta_{\alpha,A}$ vanish. Each entry is an exponential of a linear phase supported in a unit block, a single external bond or a coordinate box of diameter at most M. Two separated directions cannot both occur in one phase. For JP_*J^* the remaining product terms have disjoint unit-block supports.

The inverse derivative and the blocked-field definition give

$$\partial_B H_* = -S_* F_B H_* + b_* S_* \partial_B (JQ_*^T(-A)).$$

Use eq. 11.lix and eq. 11.lxv in the first term. The diagonal derivative in the second is absorbed as the summand $z = y'$. The combined coefficient is at most $149(C^F)^2$:

$$|\partial_B H_*|(y,y') \le 149(C^F)^2\epsilon_B \sum_{z\in P_B} E(y,z)E(z,y'). \tag{11.lxvi}$$

The same estimate holds for $\delta_{\alpha,A}\partial_B H_*$ on short pairs, since the first factor S_* can be replaced by $\delta_{\alpha,A}S_*$.

For separated directions differentiating once more leaves four terms:

$$\begin{aligned}\partial_B\partial_{B'} H_* = {} & S_* F_B S_* F_{B'} H_* + S_* F_{B'} S_* F_B H_* \\ & - b_* S_* F_B S_* \partial_{B'}(JQ_*^T(-A)) - b_* S_* F_{B'} S_* \partial_B (JQ_*^T(-A)).\end{aligned}$$

Apply eq. 11.lxv twice to the first two terms. Each gives an ordered three-link exponential chain with coefficient $146^2(C^F)^3$. The diagonal last factors in the other two terms give the same chains, with coefficient $(657/2)(C^F)^2$. Each ordered chain therefore has coefficient at most $21645(C^F)^3$. Again the first factor may be replaced by the short-pair Hölder inverse.

Differentiate $D_* = b_* - b_* Q_* J^* H_*$. The derivative of the average contributes $(9/4)C^F$ to the first exponential chain and the derivative of H_* contributes $(3/2)149(C^F)^2$. Their sum is at most $226(C^F)^2$, proving the first line of eq. 11.lxi. Weighted row and column sums give two factors $(24/\gamma^F)^3$, hence C_K; without weights the same sum retains one factor $|P_B|$. For the mixed derivative the three product terms contain one derivative on each factor, or the mixed derivative of H_*. Each chain has coefficient at most $(1341/4)(C^F)^2 + (64935/2)(C^F)^3 \le 32803(C^F)^3 \le C_H^2$. This proves eq. 11.lxii.

A leg is a component row of H_* or of $\delta_{\alpha,A}H_*$. For a short pair its position is its first unit block; for a long pair its position set contains both endpoint blocks. The sum over spin components of the row is bounded by the relevant majorant. For long pairs use $|\delta_{\alpha,A}v| \le \mathrm{e}|v(x')| + |v(x)|$; for

short pairs use the Hölder input directly. This gives the first line of eq. 11.lxiii. In particular, with $d(y', \mathrm{pos}(u)) = \min_{w\in\mathrm{pos}(u)} d(y', w)$,

$$\sum_{y',\beta'} |\operatorname{Leg}_A(u; y', \beta')| e^{(\gamma^F/2)d(y',\mathrm{pos}(u))} \le 3C^F |\operatorname{pos}(u)| (24/\gamma^F)^3 \le h_{\mathrm{L}}.$$

Indeed, multiplying each endpoint decay by this weight leaves decay rate at least $\gamma^F/2$, and there are at most two endpoints.

The derivative of a Hölder leg has two terms: $(\partial_B \delta_{\alpha,A}) H_* + \delta_{\alpha,A} \partial_B H_*$. The first is bounded by $4d_X^{1-\alpha}\epsilon_B |H_* v(x')|$, and vanishes unless the pair is within M of P_B. It vanishes for pairs inside a cluster, since their phase is one. For short pairs this adds at most $11C^F$ to $149(C^F)^2$; for long pairs it adds $4MC^F$ to $(4/3)149(C^F)^2$. Both are at most C_H, after enlarging P_B to P_B^+. This proves the derivative bound in eq. 11.lxiii.

For its mixed derivative the second derivative of the Hölder phase vanishes by separation. There remain $(\partial_B \delta_{\alpha,A}) \partial_{B'} H_*$ and its interchange, and $\delta_{\alpha,A} \partial_B \partial_{B'} H_*$. On short pairs each ordered chain has coefficient at most $1639(C^F)^2 + 21645(C^F)^3$. On long pairs the coefficient at the second endpoint is at most $596M(C^F)^2 + 28860(C^F)^3$; the first endpoint has the smaller coefficient $21645(C^F)^3$. All are at most C_H^2, proving eq. 11.lxiv. Barred legs follow by the monomial charge conjugation matrix and the replacement $A \mapsto -A$, which preserve component row sums; the domain is invariant under that replacement. □

11.3.3. *Complex neighborhoods and the real-field covariance.* We need an open complex neighborhood for Cauchy estimates and a decaying covariance for the later tree expansion. A Neumann series enlarges the small-field inverse domain. At real fields, accretivity gives the coarse inverse and its decay without a small-field restriction. Choose

$$c_\tau = \min\{1/8, \mu_-/48, (292\mathcal{S})^{-1}\}, \qquad C'_K = 1328\mathcal{S}^2, \qquad \mathcal{W}^+ = \{A + A_1 : A \in \mathcal{W},\ e_*|A_1|_\infty \le c_\tau\}.$$

Every point of $\mathcal{W}^+$ has imaginary part $\mathfrak{i} < 1/4$. The elementary averages and F_* therefore satisfy the same local derivative bounds as above, although the exponential propagator input is initially given only on $\mathcal{W}$. The enlargement preserves invertibility:

$$\|S_*(A + A_1)\|_{\gamma^F/2} \le 2\mathcal{S}. \tag{11.lxvii}$$

To prove this, integrate $\partial_{A_1} F_*$ along the segment to get $\|F_*(A + A_1) - F_*(A)\|_{\gamma^F/2} \le 146 c_\tau$. Multiplying by $S_*(A)$ gives norm at most $146 c_\tau \mathcal{S} \le 1/2$. The Neumann inverse in the submultiplicative majorant norm proves eq. 11.lxvii. Since its finite-dimensional determinant is nonzero, S_* is analytic in a neighborhood of each point of $\mathcal{W}^+$, as is D_*.

Differentiate $D_* = b_* - b_*^2 Q_* J^* S_* J Q_*^T(-A)$. The three terms have majorant-norm coefficients

$$\tfrac{9}{4}(2\mathcal{S})\tfrac{3}{2}, \qquad \tfrac{3}{2}(2\mathcal{S})146(2\mathcal{S})\tfrac{3}{2}, \qquad \tfrac{3}{2}(2\mathcal{S})\tfrac{9}{4}.$$

Their sum is $\frac{27}{2}\mathcal{S} + 1314\mathcal{S}^2 \le C'_K$. Keeping the support indicator of either the average derivative or F_B gives, respectively,

$$\|\partial_B D_*(A)\|_{\gamma^F/2} \le C'_K \epsilon_B, \qquad \sum_{y,y'} |\partial_B D_*(A)|(y, y') \le C'_K \epsilon_B |P_B|, \qquad A \in \mathcal{W}^+. \tag{11.lxviii}$$

In the middle term, sum first the two outer S_* factors using their row and column bounds $2\mathcal{S}$, and then sum the local derivative over its first endpoint in P_B. Integrating eq. 11.lxviii along any segment inside $\mathcal{W}^+$ gives the same two bounds for $D_*(A + B) - D_*(A)$.

The covariance at a real field requires no small-field hypothesis. If A_r is real, $R = D_{A_r} + \mu_*$ has real part at least μ_*. The averaging map $Q = Q_*(A_r) J^*$ satisfies $Q^* = J Q_*^T(-A_r)$ and $QQ^* = 1$, because its phases have modulus one and every block has fine measure one. Thus $F_*(A_r) = R + b_* Q^* Q$ also

has real part at least μ_*. Set $C = QR^{-1}Q^*$ and $Y = (b_*^{-1} + C)^{-1}$. The inverse R^{-1} is accretive, so $b_*^{-1} + C$ is invertible. Multiplication verifies

$$S_* = R^{-1} - R^{-1}Q^*YQR^{-1}, \qquad QS_*Q^* = b_*^{-1}(1 - b_*^{-1}Y).$$

The definition of D_* therefore gives

$$D_*(A_r)^{-1} = b_*^{-1} + QR^{-1}Q^*, \qquad \|D_*(A_r)^{-1}\|_{\mathrm{op}} \le b_*^{-1} + \mu_*^{-1}. \tag{11.lxix}$$

Its block decay follows by conjugating the fine inverse with a weight. Fix y' and put $\psi(c) = \vartheta\, d(c, \Delta_{y'})$ on the periodic fine lattice, with $\vartheta = \mu_*/(12\mathrm{e}) \le 1$. This weight is zero on $\Delta_{y'}$, is ϑ-Lipschitz, and on Δ_y is at least $\vartheta(d(y, y') - 1)$. Only the external Dirac hop changes under conjugation by e^{ψ}, since the internal hops stay over one photon site. Consequently

$$\|e^{\psi} D_{A_r} e^{-\psi} - D_{A_r}\|_{\mathrm{op}} \le 6\eta_*^{-1}(e^{\vartheta\eta_*} - 1) \le 6\mathrm{e}\vartheta = \mu_*/2.$$

The conjugated R has real part at least $\mu_*/2$, so its inverse has norm at most $2/\mu_*$. Apply it between Q^* times unit spinors at y and y'; these have norm one and supports in their respective blocks. Moving the weight across the scalar product gives

$$\|D_*(A_r)^{-1}(y, y')\|_{\mathrm{op}} \le b_*^{-1} 1_{\{y=y'\}} + \frac{2\mathrm{e}}{\mu_*} e^{-\mu_* d(y,y')/(12\mathrm{e})}. \tag{11.lxx}$$

The factor e accounts for the unit block diameter in the lower bound for ψ.

Set

$$g = (b_-^{-1} + \mu_-^{-1})^{1/2}, \qquad h_G = 1 + g, \qquad g_1 = b_-^{-1} + 4/\mu_-, \qquad \theta_{\mathrm{CT}} = \mu_-/(12\mathrm{e}),$$
$$\vartheta_0 = \min\{\theta_{\mathrm{CT}}/2, \gamma^F/8, 1/M\}, \qquad \alpha_G = 2b_-^{-1} + 4\mathrm{e}\mu_-^{-1}(24/\theta_{\mathrm{CT}})^3.$$

A component row or column of a 4×4 matrix has sum at most twice its Hilbert-space norm. Using eq. 11.lxx and the lattice exponential sum, every row and column of $G = D_*(A_r)^{-1}$ therefore satisfies

$$\sum_{u'} |G(u, u')| e^{\vartheta_0 d(u,u')} \le \alpha_G, \qquad \sum_{u} |G(u, u')| e^{\vartheta_0 d(u,u')} \le \alpha_G, \qquad \|G\|_{\mathrm{op}} \le g^2. \tag{11.lxxi}$$

These constants depend on L, C^F, γ^F, M, not on the size of the real connection.

For later complex twists we also need the coarse inverse away from real fields. Suppose $A = A_r + A_1$, with A_r real, $e_*|A_1|_\infty \le \mu_*/24$, and $F_*(A)$ invertible. The external-hop phase difference gives $\|D_A - D_{A_r}\|_{\mathrm{op}} \le 12 e_* |A_1|_\infty \le \mu_*/2$. Hence $D_A + \mu_*$ is accretive with inverse norm at most $2/\mu_*$. Its two average maps have norm at most $\sqrt{2}$, because $|\varphi_A^{\pm 1}| \le e^{1/16} \le \sqrt{2}$. The same finite-rank inversion calculation then gives

$$D_*(A)^{-1} = b_*^{-1} + Q_*(A) J^* (D_A + \mu_*)^{-1} J Q_*^T(-A), \qquad \|D_*(A)^{-1}\|_{\mathrm{op}} \le g_1. \tag{11.lxxii}$$

Here invertibility of $b_*^{-1} + QR^{-1}Q'$ follows from that of $F_*(A)$ and R: $\det(1 + b_* R^{-1} Q' Q) = \det(1 + b_* Q R^{-1} Q')$. This determinant identity follows by the two block-triangular factorizations of $\left(\begin{smallmatrix} 1 & -b_* R^{-1} Q' \\ Q & 1 \end{smallmatrix}\right)$. The inverse formula is then checked by multiplication exactly as above. Each covariance entry is bounded by g_1, so each row has sum at most $4\ell_*^3 g_1$.

11.3.4. *The terminal Grassmann norms.* The terminal integral has four coarse components per unit site and Grassmann weight one. The weights below make substitution of the fine fields contractive and control the terminal mass and Dirac forms in the coefficient norm. Set

$$a_1 = e_*^{-1/4}, \qquad a_2 = C_0^{-1} e_*^{-1/4+\varepsilon}, \qquad h_* = (e_*^{-1/4}, e_*^{-1/4+\varepsilon}).$$

By eq. 11.lx, the substitution $H_*^{\#}(A)\Psi$ has row norm at most $C^F \le C_0$, for both spinor and Hölder legs, including all long and cluster pairs. The coarse kernel $b_* - D_*(A) = b_*^2 Q_* J^* S_* J Q_*^T(-A)$ is bounded by $C^F E(y, y')$: block compression gives Euclidean constant $32\exp(1) b^2 a'$ by eq. 4.lviii, and conversion to the component norm doubles it. The calibration of C^F applies.

The weighted Grassmann substitution estimate applies with $C_0 a_2 = h_{*,2} \le h_{*,1}$. Therefore

$$|F(H_*^{\#}(A)\Psi)|_{a_2} \le \|F\|_{h_*}. \tag{11.lxxiii}$$

For the mass form before substitution, the fine measure sums to ℓ_*^3 and there are four spin components. Its norm at weight $C_0 a_2$ is exactly $4\ell_*^3 (C_0 a_2)^2$. In particular

$$|\langle \bar\psi, \psi\rangle|_{a_2} \le 4 e_*^{-1/2+2\varepsilon} \ell_*^3. \tag{11.lxxiv}$$

The 48 labels are canceled by the fine measure $1/48$; there is no factor 192 in this estimate. Similarly the coarse bilinear form has norm a_1^2 times the sum of its matrix entry moduli. Its diagonal term contributes $4b_*$ per unit site, and its decaying kernel contributes at most $4C^F \sum_{y'} E(y,y')$. For $c(\gamma) = \sum_{z\in\mathbb{Z}^3} e^{-\gamma|z|_\infty}$, the earlier lattice-sum estimate gives $c(\gamma) \le (12/\gamma)^3$ when $0 < \gamma \le 6$. The chosen $C_0 \ge 4 + 16 C^F c(\gamma^F)$ therefore gives

$$|\langle \bar\Psi, D_*(A)\Psi\rangle|_{a_1} \le C_0 e_*^{-1/2} \ell_*^3. \tag{11.lxxv}$$

These estimates hold on $\widetilde G_*$. For the external activity evaluated at $\mathscr{T} = \mathscr{A} + \mathscr{C}$, the same substitution applies because $\widetilde R_* + \mathfrak{X}_* \subset \widetilde G_*$.

11.4. **Interacting response on the reference torus.** The target is a product of Wick polynomials with their free reference values subtracted, one in each reference cell. We compare each cell with its free response, use separation to bound interactions between distinct cells, and control the excluded photon fields. Specifically, we estimate the normalized interacting Wick products

$$\Theta_m^{(0)} = \left| \left(\prod_{c\in\mathcal{C}_*} (\partial_{t_c}^m - c_m) \mathring\Phi \right)(0) \right|^{1/N_*}, \qquad m = 2, 4, \qquad N_* = |\mathcal{C}_*| = 8L^3, \tag{11.lxxvi}$$

where $c_m = N_0^{(m)}(0)$. The exact relation to Wick products is eq. 11.xlv.

11.4.1. *The terminal determinant ratio.* For the marked connections of eq. 11.xlix, set

$$C_b = 1 + \sqrt{h}\, C_\beta, \qquad t_x = \frac{e_*^{3\varepsilon}}{C_b}, \qquad t_x^+ = \frac{e_*^{3\varepsilon}}{\sqrt{h}\, C_\beta}, \qquad r = t_x/4. \tag{11.lxxvii}$$

If $C_\beta = 0$, interpret $t_x^+ = \infty$. For t in the polydisc $\mathbb{D}(t_x^+) = \{|t_c| < t_x^+ \text{ for every } c\}$, all the rescaled connections $a(t)^{[j]}$ belong to $\mathfrak{X}_j$ and $e_* |\mathscr{C}(t)|_\infty < e_*^{3\varepsilon}$. Indeed, eq. 11.lii and $e_* \le e\sqrt{h}$ give $|a(t)|_{\mathrm{ext}} < e_*^{-1+3\varepsilon}$. This bounds the stop sup and first-difference norms by the defining thresholds of $\mathfrak{X}_*$. For $i = k_* - j$ the rescaling multiplies the sup norm by $L^{-i/2}$ and the first differences by $L^{-3i/2}$, while $e_j = L^{-i/2} e_*$. Since $\varepsilon < 1/3$, the rescaled bounds are strictly below $e_j^{-1+2\varepsilon}$ and $e_j^{-1+3\varepsilon}$, respectively.

We first control the determinant activities accumulated along the flow. Recall the local determinant function $E_j^d(X, A)$ of Prop. 4.12 and set $E_j^{\det}(X, A) = E_j^d(X, A) - E_j^d(X, 0)$. Its analyticity, locality, real gauge invariance, and evenness will all be used below; the relevant bound is

$$|E_j^d(X, A)| \le C_d M^3 e^{-(\kappa+1) d_M(X)} \qquad (A \in \widetilde G_j). \tag{11.lxxviii}$$

Let $\mathfrak{P}_0 = 0$ and

$$\mathfrak{P}_{j+1}(Y, C) = \sum_{\overline{X}^{\,L} = LY} \big(\mathfrak{P}_j(X, C_L) + E_j^{\det}(X, C_L)\big). \tag{11.lxxix}$$

Then $\mathfrak{P}_j(X, \cdot)$ is analytic on $\mathfrak{X}_j$, vanishes at zero, and depends only on bonds meeting X. To see locality under reblocking, a bond meeting X has, after division by L, an endpoint in Y whenever $\overline{X}^{\,L} = LY$. Analyticity follows from $C_L \in \mathfrak{X}_j$.

For $d_M(Y) \geq 1$, the reblocking estimate gives

$$\sum_{X^L = LY} e^{-\kappa d_M(X)} \leq c_{\mathrm{rb}} L^3 e^{-L(\kappa-\kappa_0-1)d_M(Y)}.$$

With $L(\kappa - \kappa_0 - 1) \geq \kappa$ and $c_{\mathrm{rb}} L^3 e^{-[L(\kappa-\kappa_0-1)-\kappa]} \leq 1/2$, the weighted supremum q_j of $\mathfrak{P}_j$ over $d_M(X) \geq 1$ satisfies $q_{j+1} \leq (q_j + 2C_d M^3)/2$. Since $q_0 = 0$,

$$|\mathfrak{P}_j(X, C)| \leq 2C_d M^3 e^{-\kappa d_M(X)} \qquad (d_M(X) \geq 1). \tag{11.lxxx}$$

At $j = 0$ the determinant activity includes all 192 chamber components. The local determinant bound and reblocking estimate have the same constants at that first step.

The determinant identity eq. 5.lxxix gives

$$\frac{\det(D^{\mathrm{ch}}[qa] + \bar{m}\eta)}{\det(D^{\mathrm{ch}}[0] + \bar{m}\eta)} = \exp\left(\sum_X \mathfrak{P}_*(X, a^{[k_*]})\right) \frac{\det D_*(a^{[k_*]})}{\det D_*(0)}, \tag{11.lxxxi}$$

$$\sum_X \mathfrak{P}_*(X, a^{[k_*]}) = \sum_{j<k_*} \sum_X E_j^{\det}(X, a^{[j]}). \tag{11.lxxxii}$$

Here $a^{[j]} \in \mathfrak{X}_j$ for every j. The fine determinant uses the phase-argument convention of eq. 10.xx: its connection argument is $qa(t) = \sqrt{\eta} \sum_c t_c \widehat{\mathfrak{a}}_c$. For a single cell this is precisely the phase connection B_t in eq. 10.xx. The one-step identity reads

$$\det D_{j+1}(a^{[j+1]}) = b^{4n_{j+1}} \det D_j(a^{[j]})/\delta Z_j(a^{[j]}).$$

Divide it by its zero-field version and use eq. 4.lxxv. The resulting ratio acquires the factor

$$\exp\left(-\sum_X E_j^{\det}(X, a^{[j]})\right)$$

at each step, which proves the first identity by iteration. Summing eq. 11.lxxix over Y proves the second: each fine polymer has exactly one reblocked image. The zero-field determinants are nonzero by the real covariance formula. At $j = 0$ the incoming determinant has dimension $192n_0$, the outgoing one dimension $4n_1$; the factor b^{4n_1} cancels in the ratio.

Fix $A \in V_{\mathrm{t}}(T_*^0)$ with $\chi_*(A) = 1$ and a real twist θ. Use the gauge and fields of eq. 8.xiii.

The flow bound eq. 6.xx gives an activity with even Grassmann kernels, with equal barred and unbarred degrees, and

$$\|\mathscr{E}_*\|_*^x \leq 4e_*^{1/4-6\varepsilon}, \qquad |m_*| \leq e_*^{3/4-8\varepsilon}/8. \tag{11.lxxxiii}$$

The coefficient norm includes the weights $\eta_*^3/48$ for a fine spinor index and $(\eta_*^3/48)^2$ for a pair index, with all admissible pairs, including pairs inside a cluster. Its polymer factor is $e^{\kappa d_M(X)}$. For $(A, C) \in \widetilde{R}_* \times \mathfrak{X}_*$ define

$$Z_{\mathrm{in}}(A, C) = \int \mathrm{d}\Psi \, \exp\left\{-\langle\bar{\Psi}, D_*(A + C)\Psi\rangle - m_*\langle\bar{\psi}, \psi\rangle + \sum_X \mathscr{E}_*(X; A, C, \psi, \chi)\right\}, \tag{11.lxxxiv}$$

where $\psi, \bar{\psi}$ are substituted using $H_*^{\#}(A + C)\Psi$ and their barred partners, and $\chi, \bar{\chi}$ using the corresponding covariant Hölder quotients. The inclusion $\widetilde{R}_* + \mathfrak{X}_* \subset \widetilde{G}_*$ makes all these maps analytic; the activity kernels are analytic in (A, C). Berezin integration extracts finitely many coefficients. The integrand is a scalar exponential times a finite polynomial in its positive-degree coefficients, so Z_{in} is analytic on the two-field domain. Set

$$Z_{\mathrm{tm}}(t) = Z_{\mathrm{in}}(\mathscr{A}, \mathscr{C}(t)), \tag{11.lxxxv}$$

$$f_{\mathrm{int}}(t) = \frac{Z_{\mathrm{tm}}(t)}{\det D_*(c_\theta + \mathscr{C}(t))}. \tag{11.lxxxvi}$$

The Berezin measure has one pair for each of the $4\ell_*^3$ coarse spinor coordinates. The two-field analyticity gives analyticity of Z_{tm} on $\mathbb{D}(t_x^+)$.

Proposition 11.4. Suppose

$$4C_0\ell_*^3 g_1 \le e_*^{-1/4+\varepsilon}, \qquad 4\ell_*^3(2C_K + K_{\mathrm{pol}} + 4) \le e_*^{-2\varepsilon}, \qquad e_*^{1/4-9\varepsilon} \le 1/8, \tag{11.lxxxvii}$$

$$48e_*^{2\varepsilon} \le \mu_-. \tag{11.lxxxviii}$$

Then f_{int} is analytic and

$$|f_{\mathrm{int}}(t) - 1| \le \frac{1}{2} e_*^{1/4-9\varepsilon} \le \frac{1}{16}. \tag{11.lxxxix}$$

In particular Z_{tm} has no zero on the polydisc. The normalized terminal ratio

$$R_{\mathrm{tm}}(t;\theta,A) = \frac{Y_*^{\mathrm{x}}(A + c_\theta; \mathscr{C}(t))}{Y_*^{\mathrm{x}}(A + c_\theta; 0)} = \exp\left(\sum_X \mathfrak{P}_*(X, \mathscr{C}(t))\right) \frac{Z_{\mathrm{tm}}(t)}{Z_{\mathrm{tm}}(0)} \tag{11.xc}$$

is nonzero, continuous in (t,θ,A) on $\mathbb{D}(t_x^+) \times \mathbb{R}^3 \times \{A \in V_{\mathrm{t}}(T_*^0) : \chi_*(A) = 1\}$, and analytic in t. It has a unique continuous logarithm $\mathcal{G}(t;\theta,A)$ that vanishes at zero, and this logarithm is analytic in t. ■

Proof. For $C = \mathscr{C}(t)$, eq. 11.lxxii gives K_C with $\|K_C\|_{\mathrm{op}} \le g_1$, because $e_*|C|_\infty < e_*^{3\varepsilon} < \mu_*/24$. The first factor condition gives $4\ell_*^3 g_1 \le a_2$ and $a_2 \ge 1$. eq. 11.lxxxiii and $4e_*^{\varepsilon} \le 1$ give the remaining hypotheses of Lem. 8.3. The normalized weight-one Gaussian identity gives $f_{\mathrm{int}}(t) = \int \mathrm{d}\mu_{K_C} e^{V_C}$. The second bound in eq. 8.xvi and the factor conditions give $|V_C|_{a_2} \le e_*^{1/4-9\varepsilon}/4 \le 1/32$. The lemma and $e^x - 1 \le 2x$ prove eq. 11.lxxxix. Its denominator is analytic and nonzero by the same inverse formula, so f_{int} is analytic.

By eq. 6.xxix the terminal density has the form

$$Y_*^{\mathrm{x}}(A'; C) = (\det D_*(0))^{-1} \exp\left(-\varepsilon_*^{\mathrm{v}}\ell_*^3 + \sum_X \mathfrak{P}_*(X, C)\right) Z_{\mathrm{in}}(H_*^{\mathrm{ph}} A', C).$$

Thm. L permits replacing A by A^g, and the photon minimizer preserves real constants. Division at $C = \mathscr{C}(t)$ and $C = 0$ gives eq. 11.xc, independently of the gauge representative. For continuity use the ungauged linear map $A + c_\theta \mapsto H_*^{\mathrm{ph}}(A + c_\theta)$ in the displayed formula: the two-field Grassmann integral is analytic and the zero-mark denominator is nonzero.

The required logarithm is

$$\mathcal{G}(t;\theta,A) = \int_0^1 \sum_c t_c \frac{\partial_{t_c} R_{\mathrm{tm}}(st;\theta,A)}{R_{\mathrm{tm}}(st;\theta,A)}\, \mathrm{d}s. \tag{11.xci}$$

Local Cauchy formulas give joint continuity and analyticity of the integral in t. Along each ray, the derivative of $R_{\mathrm{tm}} e^{-\mathcal{G}}$ is zero and its value at the origin is one, so $e^{\mathcal{G}} = R_{\mathrm{tm}}$. Two continuous logarithms differ by a constant in $2\pi i\mathbb{Z}$ on the connected polydisc; their value at zero proves uniqueness.

□

11.4.2. *The one-cell free logarithm and twists.* The free determinant ratio N_0 must be compared with the interacting response of one marked cell. Its logarithm is the sum of the scalar determinant terms accumulated along the trajectory and a terminal coarse log determinant. Subtracting this free logarithm from the interacting logarithm gives the one-cell error. Its bound supplies the single-cell terms in the later estimate for products over separated cells.

For a fixed cell c and real twist θ, let

$$g_D^\theta(t) = \mathrm{tr}\log\left(1 + D_*(c_\theta)^{-1}[D_*(c_\theta + t\widehat{c}_c) - D_*(c_\theta)]\right). \tag{11.xcii}$$

The logarithm here is the matrix power series. Set

$$C_{\mathrm{fr}} = 2592 C_d K_{\mathrm{pol}} C_\beta^2 \left(M^2 (D_\beta^{\mathrm{buf}})^3 L^4 h^{-2} + 8 M^5 h \right),$$
$$B_{\mathrm{wr}} = 16 \exp(6) C_d M^3 K_{\mathrm{pol}} p_\beta e^{-r_*}, \qquad B_1 = 16 g_1 C'_K p_\beta e_*^{2\varepsilon},$$
$$B_0 = C_{\mathrm{fr}} e_*^{4\varepsilon} + B_{\mathrm{wr}} + B_1. \tag{11.xciii}$$

Proposition 11.5. Suppose eq. 11.lxxxvii–eq. 11.lxxxviii hold, $r_* > 31L + 16$, and

$$2 g_1 C'_K e_*^{2\varepsilon} \le 1/2, \quad B_0 \le 1/4, \quad \tau_s := c_\tau \ell_* \ge 1, \quad 52 B_1 e^{-\tau_s} \le e_*^{1/4 - 9\varepsilon}. \tag{11.xciv}$$

The free ratio N_0 has the logarithm

$$\ell_0(t) = \sum_X \mathfrak{P}_*(X, t\widehat{c}_c) + g_D^0(t), \qquad N_0(t) = e^{\ell_0(t)}, \tag{11.xcv}$$

analytic on $|t| < t_x^+$. It is independent of c and of the admitted placement, is even and real on the real axis, and satisfies

$$\left| \sum_X \mathfrak{P}_*(X, t\widehat{c}_c) \right| \le C_{\mathrm{fr}} e_*^{-2\varepsilon} |t|^2 + B_{\mathrm{wr}}, \tag{11.xcvi}$$
$$|\ell_0(t)| \le B_0 \qquad (|t| \le t_x). \tag{11.xcvii}$$

The one-cell interacting error $w_c(t) = \mathcal{G}(te_c; \theta, A) - \ell_0(t)$ is analytic, vanishes at zero, and obeys

$$|w_c(t)| \le 3 e_*^{1/4 - 9\varepsilon} \qquad (|t| < t_x^+). \tag{11.xcviii}$$

■

Proof. Write $K = D_*(c_\theta + t\widehat{c}_c) - D_*(c_\theta)$. The real covariance and the integrated insertion bounds give

$$\| D_*(c_\theta)^{-1} K \|_{\mathrm{op}} \le 2 g_1 C_K e_*^{2\varepsilon} \le 1/2, \qquad \| D_*(c_\theta)^{-1} K \|_{\mathfrak{S}_1} \le 4 g_1 C_K p_\beta e_*^{2\varepsilon}.$$

Thus the series in eq. 11.xcii converges, is analytic, vanishes at zero, and has modulus at most B_1. Triangularizing the matrix identifies its exponential with $\det D_*(c_\theta + t\widehat{c}_c) / \det D_*(c_\theta)$. Combine this with eq. 11.lxxxi. The lattice symmetries carrying one cell probe to another preserve the fine determinant, so the resulting fine ratio is N_0.

To prove eq. 11.xcvi, we remove the locally constant part of the connection before summing the levels. Gauge invariance leaves a bound in terms of its derivative, and evenness supplies a double zero in the mark. The polymers that do not fit in a lifted cube are controlled by their extra exponential decay. Put $i = k_* - j$ and $B_j = e^{-1} \widehat{\mathfrak{a}}_c^{[j]}$. Rescaling gives

$$e_j |\partial B_j|_\infty \le a_j^2 a_*^{-3/2} C_\beta. \tag{11.xcix}$$

Let $\mathcal{Q}_j$ be the M-cubes containing endpoints of the supporting bonds of B_j. The buffered support cube yields

$$|\mathcal{Q}_j| \le 4 \left((D_\beta^{\mathrm{buf}})^3 M^{-3} a_j^{-3} + 8 \right), \qquad |\mathcal{Q}_j| \le L^{3i} p_\beta. \tag{11.c}$$

By locality only polymers containing a cube of $\mathcal{Q}_j$ contribute to eq. 11.lxxxii.

For such a polymer X, put $d = d_M(X)$ and $s_X = M(d+3)$. Suppose first that $2s_X < \ell_j$. All bonds meeting X lie in a coordinate cube of radius $M(d+2)+1$ around a chosen site $\tilde{c}$ of X. The strict inequality $2[M(d+2)+1] + \eta_j < 2s_X < \ell_j$ makes this cube a lift to the lattice. Define a real function there by $\lambda(c) = \sum_\nu B_{j,\nu}(\tilde{c})(c - \tilde{c})_\nu$, and extend it by zero outside. On bonds meeting X, its gradient is the constant value $B_j(\tilde{c})$. If $[\,\cdot\,]_X$ restricts a connection to those bonds, then

$$B_X = [B_j - d\lambda]_X, \qquad |B_X|_\infty \le 3 s_X |\partial B_j|_\infty.$$

The last inequality follows by integrating the first differences along coordinate paths inside the lifted cube.

Real gauge invariance must be used before continuing to the complex gauge parameter. On the convex open set where $|[tB_j - z d\lambda]_X|_\infty < e_j^{-1+\varepsilon}$, the analytic function $z \mapsto E_j^d(X, [tB_j - z d\lambda]_X)$ equals $E_j^d(X, tB_j)$ for real z near zero: the unrestricted connection is related to tB_j by a real gauge transformation, and locality permits the restriction. The identity theorem gives this equality throughout that set. In particular it holds at $z = t$ whenever $|t| < \rho_X$, where $\rho_X = e_j^{-1+\varepsilon}/(3 s_X |\partial B_j|_\infty)$.

The function $z \mapsto E_j^d(X, zB_X) - E_j^d(X, 0)$ is even, bounded by $2C_d M^3 e^{-(\kappa+1)d}$ on $|z| < \rho_X$, and has a double zero at zero. Apply the maximum principle after dividing by z^2. For $|t| \geq \rho_X$, the local determinant bound eq. 4.lxxiv gives the same inequality with the factor $(|t|/\rho_X)^2 \geq 1$. Thus, in both cases,

$$|E_j^{\det}(X, tB_j)| \leq 18 C_d M^5 C_\beta^2 (d+3)^2 e^{-(\kappa+1)d} |t|^2 e_j^{-2\varepsilon} a_j^4 a_*^{-3}. \tag{11.ci}$$

When $\partial B_j = 0$, $B_X = 0$ and this contribution vanishes. This proves the cancellation also at the chamber first step.

Since $(d+3)^2 e^{-(\kappa+1)d} \leq 9 e^{-\kappa_0 d}$, summing eq. 11.ci over cubes and polymers gives at level j the bound

$$648 C_d K_{\mathrm{pol}} C_\beta^2 |t|^2 e_j^{-2\varepsilon} a_*^{-3} \left(M^2 (D_\beta^{\mathrm{buf}})^3 a_j + 8 M^5 a_j^4 \right).$$

For the remaining polymers $2 s_X \geq \ell_j$, $d \geq r_* L^i/2 - 3$. The extra exponential in eq. 4.lxxiv then gives at level j

$$2 \exp(6) C_d M^3 K_{\mathrm{pol}} p_\beta L^{3i} e^{-r_* L^i}.$$

Sum over $i \geq 1$, using $a_j = L^{-i} a_*$, $e_j^{-2\varepsilon} = e_*^{-2\varepsilon} L^{i\varepsilon}$, $\sum_i L^{-i(1-\varepsilon)} \leq 1$, and $a_*^{-2} \leq L^4 h^{-2}$, $a_* \leq h$. For $r_* > 31L + 16$, the second sum is bounded by a constant times e^{-r_*}. The stated definitions of C_{fr} and B_{wr} give eq. 11.xcvi. Together with $|g_D^0| \leq B_1$ this proves eq. 11.xcvii.

Evenness and reality follow from the corresponding fine determinant symmetries. Both $\ell_0(t)$ and $\ell_0(-t)$ vanish at zero and have exponential $N_0(t)$, so they coincide. For real t, the nonzero real ratio $N_0(t)$ is positive by continuity from one at zero; the normalized logarithm is therefore real. Cell and placement independence follows in the same way from the determinant translation and reflection symmetries.

It remains to control the twist. For $|\operatorname{Im} \theta_\mu| \leq \tau_s$ the connections $c_\theta + s t \widehat{c}_c$ lie in $\mathcal{W}^+$, and $\|D_*(c_\theta)^{-1}\|_{\mathrm{op}} \leq g_1$. There is a positive strip margin $\ell_*(e_*^{2\varepsilon} - e_*^{3\varepsilon})$: split the imaginary constant connection into a part bounded by c_τ/e_* and a remaining part absorbed into $\mathfrak{X}_*$. The latter has vanishing first differences. The mass condition bounds its total imaginary part by $\mu_*/24$. On this larger open strip, eq. 11.lxviii gives the same matrix power-series bound $|g_D^\theta| \leq B_1$. The function is holomorphic there and 2π-periodic in each twist variable. For real twists, periodicity is conjugation by the periodic gauge factor $\exp(-2\pi i c_\mu/\ell_*)$; the identity theorem extends it to the strip.

A 2π-periodic holomorphic function g, bounded by B on a closed strip of width $\tau \geq 1$, satisfies $|\widehat{g}(m)| \leq B e^{-\tau |m|_1}$. Shift each Fourier contour towards $-i\tau \operatorname{sgn}(m_\mu)$; periodicity cancels the vertical sides. First use an interior width and then take its limit. The nonconstant Fourier sum is bounded by

$$B \left[\left(\frac{1 + e^{-\tau}}{1 - e^{-\tau}} \right)^3 - 1 \right] \leq 25 B e^{-\tau}.$$

Indeed, the bracket divided by $e^{-\tau}$ has a power series with positive coefficients, so its maximum occurs at $e^{-\tau} = e^{-1}$. The difference between two real values is therefore at most $52 B e^{-\tau}$. Apply this to $g_D^\theta(t)$ to obtain

$$|g_D^\theta(t) - g_D^0(t)| \leq 52 B_1 e^{-\tau_s} \leq e_*^{1/4 - 9\varepsilon}. \tag{11.cii}$$

Finally the normalized logarithm identity and eq. 11.lxxxvi give

$$w_c(t) = g_D^\theta(t) - g_D^0(t) + \log f_{\mathrm{int}}(t e_c) - \log f_{\mathrm{int}}(0).$$

The principal logarithm of f_{int} is analytic and has modulus at most $e_*^{1/4-9\varepsilon}$. Both sides vanish at zero and exponentiate to the same ratio; uniqueness of the logarithm justifies the identity. eq. 11.cii proves eq. 11.xcviii. □

For a one-variable polynomial $P(t) = \sum_j P_j t^j$, write $\|P\|_r = \sum_j |P_j| r^j$. Let $\phi_0(t) = \sum_{i=0}^4 N_0^{(i)}(0) t^i/i!$ be the fourth jet of N_0, that is, its Taylor polynomial through degree four. Cauchy's estimate applied to $|N_0(t) - 1| \le 2B_0$ on $|t| = t_x = 4r$ gives

$$|c_2| \le r^{-2}/16, \qquad |c_4| \le r^{-4}/16, \qquad \|\phi_0 - 1\|_r \le 1/6, \qquad \|\phi_0\|_r \le 2. \tag{11.ciii}$$

The odd coefficients vanish. Expanding $\log N_0$ through degree four gives $\ell_0^{(4)}(0) = c_4 - 3c_2^2$; hence this is the chamber fourth coefficient estimated in the free-response subsection.

11.4.3. *Derivatives in separated cells.* To estimate mixed logarithmic derivatives, we use the real covariance $G = D_*(\mathscr{A})^{-1}$. The two derivatives can act on activity kernels, blocked-field legs or the coarse Dirac operator. Their insertion chains, together with the decay of G, will control the connection between separated cells. On the coarse Grassmann algebra, write an even element without constant term as $F = \sum_{p\ge2}(p!)^{-1}\sum F_p(u_1,\dots,u_p)\Psi(u_1)\cdots\Psi(u_p)$ with antisymmetric kernels. For a symmetric weight $w \ge 1$, define

$$\rho_G(F;w) = \sum_{p\ge2} \frac{h_G^p}{(p-1)!} \sup_{u_1} \sum_{u_2,\dots,u_p} |F_p(u_1,\dots,u_p)| w,$$

$$N_G(F;w) = \sum_{p\ge2} \frac{h_G^p}{(p-1)!} \sum_{u_1,\dots,u_p} |F_p(u_1,\dots,u_p)| w. \tag{11.civ}$$

The supremum in ρ_G is taken separately at each degree. The indices include site, spin, and barred or unbarred status; their distance is the distance between the sites. We use

$$w_\delta(u_1,\dots,u_p) = e^{\vartheta_0 \max_{i,j} d(u_i,u_j)}, \qquad w_P(u_1,\dots,u_p) = e^{\vartheta_0 \max_i d(u_i,P)}.$$

The constants $h_G, \vartheta_0, \alpha_G$ are those of eq. 11.lxxi. Antisymmetrizing a kernel that is not antisymmetric bounds N_G by the same expression for that kernel. For ρ_G, replace the supremum at the first argument by the largest supremum over all argument positions.

In the fine algebra absorb the terminal mass into each single-cube activity:

$$E^a(X;t) = \mathscr{E}_*(X;\mathscr{A},\mathscr{C}(t)) - \mathbf{1}_{\{X \text{ is an } M\text{-cube}\}} m_* \langle\bar\psi,\psi\rangle_X.$$

Let $E^a_{nm}(X;t)$ denote its fine spinor/pair kernels. The antisymmetric mass kernel has norm $8M^3$: its two fine-index weights multiply $48\eta_*^{-3}$, its diagonal Kronecker factors leave $48(M/\eta_*)^3$ chamber sites, and there are four spin components and two orders. It follows from eq. 11.lxxxiii that, when $a_w \le e_*^{-1/4+\varepsilon}$ and $(4 + M^3/2)e_*^{\varepsilon} \le 1$,

$$\sum_{n,m} \frac{a_w^{n+m}}{n!m!} \|E^a_{nm}(X;t)\| \le w_*^{\mathrm{E}} e^{-\kappa d_M(X)}, \qquad w_*^{\mathrm{E}} = e_*^{1/4-7\varepsilon} \quad (|t|_\infty \le t_x). \tag{11.cv}$$

Cauchy's formula on the circles of radius $t_x/2$ in one or two distinct mark variables gives, on $|t|_\infty \le t_x/2$, the same bound for $\partial_c E^a_{nm}$ and $\partial_c\partial_{c'} E^a_{nm}$ multiplied respectively by $q_x = 2/t_x$ and q_x^2. The circle integrals precede the finite weighted kernel sums, so the coefficient norm is bounded without replacing it by a supremum of individual coefficients. Locality gives $\partial_c E^a(X;t) = 0$ unless $X \cap P_c \ne \varnothing$.

After substituting the terminal field maps, put

$$V(t) = -\langle\bar\Psi, K(t)\Psi\rangle + \sum_X E^a(X;t)[H_*^{\#}(\mathscr{A}+\mathscr{C}(t))\Psi],$$

$$K(t) = D_*(\mathscr{A}+\mathscr{C}(t)) - D_*(\mathscr{A}), \qquad V_0(t) = \sum_X E^a_{00}(X;t), \qquad V_+(t) = V(t) - V_0(t). \tag{11.cvi}$$

It is even and analytic, and V_+ has no constant term. Gaussian integration gives

$$Z_{\mathrm{tm}}(t) = \det D_*(\mathscr{A}) e^{V_0(t)} \int \mathrm{d}\mu_G \, e^{V_+(t)}. \tag{11.cvii}$$

The real covariance has the weighted row and column bounds eq. 11.lxxi. These bounds let us sum propagators along trees while retaining exponential decay between the marked supports.

Proposition 11.6. Suppose

$$5h_{\mathrm{L}}h_G \le e_*^{-1/4+\varepsilon}, \qquad (4+M^3/2)e_*^{\varepsilon} \le 1, \qquad \alpha_G \bar{r} \le 1/2, \quad \bar{r} = 4h_G^2 C_K e_*^{2\varepsilon} + 650 K_{\mathrm{pol}} e_*^{1/4-7\varepsilon}. \tag{11.cviii}$$

Let $d_{\mathrm{sep}} = \min_{c \neq c'} d(P_c, P_{c'})$ and suppose $d_{\mathrm{sep}} \ge 6M+8$. With $s_F = (24/\gamma^F)^3$ and $C_V^{\mathrm{ins}} = 10^3 C_H^2 s_F^4 K_{\mathrm{pol}}$, we have

$$\begin{aligned} \rho_G(V_+(t); w_\delta) &\le \bar{r} && (|t|_\infty \le t_x), \\ N_G(\partial_c V_+(t); w_{P_c}) &\le q_x C_V^{\mathrm{ins}} h_G^2 p_\beta, \quad \rho_G(\partial_c V_+(t); w_{P_c}) \le q_x C_V^{\mathrm{ins}} h_G^2, \\ N_G(\partial_c \partial_{c'} V_+(t); 1) + |\partial_c \partial_{c'} V_0(t)| &\le q_x^2 C_V^{\mathrm{ins}} h_G^2 p_\beta e^{-\vartheta_0(d_{\mathrm{sep}}-4M-6)}. \end{aligned} \tag{11.cix}$$

The last two lines hold on $|t|_\infty \le t_x/2$, with $c \neq c'$ in the last line. If the lattice probe vanishes, all marked derivatives vanish. ∎

Proof. We give the kernel summation which preserves the marked supports. For a polymer X, let X^+ be its unit-site one-neighborhood. Every fine index in an activity supported on X has its position set in X^+. This includes both chamber sites in an internal pair and the first site of an external nearest-neighbor pair whose other endpoint lies over X. The polymer diameter estimate gives

$$\vartheta_0 \operatorname{diam}(X^+) \le d_M(X) + 3. \tag{11.cx}$$

The weighted kernel after substituting the field maps is, at degree $p = n+m$,

$$K_{nm}^X(u_1, \dots, u_p) = \int E_{nm}^a(X; t; v_1, \dots, v_p) \prod_{i=1}^p \mathrm{Leg}(v_i; u_i) \, \mathrm{d}v.$$

Here $\mathrm{d}v$ includes the fine weights already specified. Passing from the bidegree expression to a degree-p kernel produces the factor $p!/(n!m!)$; in the tree norms its remaining factor is $ph_G^p/(n!m!)$.

Suppose a weight is bounded by $c_w \prod_i e^{\vartheta_0 d(u_i, \mathrm{pos}(v_i))}$. Each summed leg contributes at most h_{L}. If one argument u is fixed, its leg contributes at most $h_{\mathrm{L}} e^{-(\gamma^F/4)d(u,X^+)}$. Thus the full kernel sum is at most $c_w h_{\mathrm{L}}^p \|E_{nm}^a(X;t)\|$, and the sum with one fixed argument retains the extra exponential in $d(u, X^+)$. For w_δ, the triangle inequality between the two farthest arguments and their position sets, followed by eq. 11.cx, permits $c_w = e^{d_M(X)+3}$.

The needed rooted polymer sum is

$$\sum_X e^{-(\gamma^F/4)d(u,X^+)} e^{-(\kappa-1)d_M(X)} \le 32 K_{\mathrm{pol}}. \tag{11.cxi}$$

Indeed the sum of $e^{-(\gamma^F/4)d(u,\square^+)}$ over M-cubes is at most 32. The neighboring 27 cubes contribute at most 27; a cube j layers away has distance at least $(j-1)M$ and there are at most $(2j+1)^3$ such cubes. Use $\gamma^F M/4 \ge 4$ to sum the remaining geometric tail. Assigning a cube of X and applying the anchored polymer sum K_{pol} proves eq. 11.cxi.

Apply eq. 11.cv with $a_w = 5h_{\mathrm{L}}h_G$. The factor 5^{-p} absorbs the polynomial degree factors. The preceding full and rooted sums give

$$\rho_G\left(\sum_X E^a(X;t)[H_*^{\#}\Psi]_+; w_\delta\right) \le 650 K_{\mathrm{pol}} w_*^{\mathrm{E}}.$$

More generally, for kernels bounded by $\mathcal{B}e^{-\kappa d_M(X)}$ at that weight, these sums give $650K_{\mathrm{pol}}\mathcal{B}$ in ρ_G and $\exp(3)\mathcal{B}\sum_X e^{-(\kappa-1)d_M(X)}$ in N_G. This follows by summing $\sum_{p\geq 2} p5^{-p} = 9/80$ in the rooted case and using $p \leq 5^p$ in the full case. The quadratic term contributes at most $4h_G^2 C_K e_*^{2\varepsilon}$, since $\vartheta_0 \leq \gamma^F/2$. This proves the first line of eq. 11.cix.

For the first derivative, there are three types of term: the coarse quadratic derivative, the activity-kernel derivative, and a derivative of one leg. The first has respective bounds $8h_G^2 C_K \epsilon_c p_\beta$ and $4h_G^2 C_K \epsilon_c$ in N_G and ρ_G, with $\epsilon_c = e_*|\widehat{c}_c|_\infty \leq q_x$. In its chain through $z \in P_c$, the weight w_{P_c} is absorbed by the exponential distances from the two coarse endpoints to z.

For the kernel derivative, X must meet P_c. The same weight comparison uses that meeting point and eq. 11.cx. The Cauchy bound multiplies $\mathcal{B} = w_*^{\mathrm{E}}$ by q_x, and the sum of polymers meeting P_c is at most $K_{\mathrm{pol}} p_\beta$. Its N_G bound is $\exp(3)K_{\mathrm{pol}} p_\beta q_x w_*^{\mathrm{E}}$ and its ρ_G bound is $650K_{\mathrm{pol}} q_x w_*^{\mathrm{E}}$.

For a differentiated leg at position j, the insertion estimate has a chain $w \in \mathrm{pos}(v_j) \to z \in P_c^+ \to u_j$. The weight obeys

$$w_{P_c} \leq e^{d_M(X)+5} e^{\vartheta_0[d(w,z)+d(z,u_j)]} \prod_{i\neq j} e^{\vartheta_0 d(u_i,\mathrm{pos}(v_i))}.$$

The extra two in the exponent accounts for $d(P_c^+, P_c) \leq M+1$. Summing the differentiated leg and the other legs leaves, before the degree sum,

$$2e^{d_M(X)+5} h_{\mathrm{L}}^{p-1} C_H \epsilon_c s_F \sum_{z\in P_c^+} e^{-(\gamma^F/2)d(z,X^+)} \|E^a_{nm}(X;t)\|.$$

There are p choices of the differentiated leg. The kernel norm contributes one further factor p, and $p^2 \leq 5^p$ absorbs both. Use eq. 11.cxi with $\gamma^F/2$ in place of $\gamma^F/4$. The result is bounded by $9500 C_H s_F K_{\mathrm{pol}} w_*^{\mathrm{E}} p_\beta \epsilon_c$ in N_G.

For ρ_G, if the fixed argument lies on the differentiated leg, split its two-edge exponent as $7\gamma^F/8 = 3\gamma^F/8 + \gamma^F/2$. Summing the intermediate site leaves $s_F e^{-(\gamma^F/4)d(u,X^+)}$. If the fixed argument lies on another leg, that leg supplies the same rooted exponential, and the differentiated leg contributes $2C_H\epsilon_c s_F^2$. Both cases therefore have the bound $9500 C_H s_F^2 K_{\mathrm{pol}} w_*^{\mathrm{E}} \epsilon_c$. Here the degree sum is $\sum_{p\geq 2} p^2 5^{-p} = 43/160$. Adding the three types of term, using $w_*^{\mathrm{E}} \leq 1/8$, $C_K \leq C_H s_F^2$, and $C_H \geq 2$, proves the second line with $C_V^{\mathrm{ins}} = 10^3 C_H^2 s_F^4 K_{\mathrm{pol}}$.

For the mixed derivative, put

$$d_0 = d(P_c^+, P_{c'}^+), \quad d_1 = d(X^+, P_c^+), \quad d_2 = d(X^+, P_{c'}^+).$$

Then $d_0 \geq d_{\mathrm{sep}} - 2M - 2$. The differentiated coarse quadratic term is bounded in N_G by

$$16 h_G^2 C_H^2 s_F^3 p_\beta q_x^2 e^{-(\gamma^F/2)d_{\mathrm{sep}}}.$$

For an activity kernel with p legs, there are $(p+1)^2$ placements of the two derivatives. They may act on two legs, the same leg, a kernel and a leg, or both on the kernel. The summed first differentiated legs are bounded by $2C_H\epsilon_c s_F^2 e^{-(\gamma^F/2)d_1}$ and its c' version. The mixed leg is bounded by

$$2C_H^2 \epsilon_c \epsilon_{c'} s_F^3 e^{-(\gamma^F/2)d_0}\big(e^{-(\gamma^F/2)d_1} + e^{-(\gamma^F/2)d_2}\big).$$

If a derivative acts on the kernel, its corresponding distance d_i is zero by locality. Normalize the first kernel derivatives by q_x and the mixed one by q_x^2. Each of the four resulting kernel types satisfies eq. 11.cv. Every product-rule term is consequently at most $4q_x^2 C_H^2 s_F^4 h_{\mathrm{L}}^p \|E_{\mathrm{type}}\| \mathcal{D}(X)$, where

$$\mathcal{D}(X) = e^{-(\gamma^F/2)(d_1+d_2)} + e^{-(\gamma^F/2)d_0}\big(e^{-(\gamma^F/2)d_1} + e^{-(\gamma^F/2)d_2}\big).$$

The triangle inequality through X^+ gives $d_0 \leq d_1 + d_2 + M d_M(X) + 2M + 2$. Use $\vartheta_0 \leq \gamma^F/8$ and $\vartheta_0 M \leq 1$ to obtain

$$\mathcal{D}(X) e^{-\kappa d_M(X)} \leq 2e^{-\vartheta_0(d_0-2M-2)}\big(e^{-(\gamma^F/4)d_1} + e^{-(\gamma^F/4)d_2}\big) e^{-(\kappa-1)d_M(X)}.$$

Sum the two distance factors over the sites of P_c^+ and $P_{c'}^+$, then use eq. 11.cxi. The degree factor $p(p+1)^2$ is at most 5^p for $p \geq 2$. At $p = 0$ there is one kernel term with no legs, and the same bound controls $\partial_c \partial_{c'} V_0$. The four kernel types contribute in total at most $256 C_H^2 s_F^4 K_{\mathrm{pol}} p_\beta q_x^2 e^{-\vartheta_0(d_{\mathrm{sep}}-4M-4)}$. Adding the quadratic term proves the last line of eq. 11.cix. If the lattice probe is zero, $\mathscr{C}(t) = 0$ and there is no mark dependence. □

We now pass from differentiated interaction kernels to derivatives of the logarithm. We use the Brydges–Kennedy forest identity in the polynomial form of [20, Thm. III.1] to express the logarithm as a connected tree sum. On the path between the two marked vertices, the weighted covariance bounds retain decay in the distance between their supports.

For n copies of the generators, let $\Delta^{ij} = \sum_{x,y} G(x,y) \partial_{\bar\Psi^{(j)}(y)} \partial_{\Psi^{(i)}(x)}$ and $\Delta_{\{i,j\}} = \Delta^{ij} + \Delta^{ji}$. The derivatives are left Grassmann derivatives. For a forest T and edge parameters $\tau \in [0,1]^T$, let $s_{ij}^T(\tau)$ be the minimum parameter on the path from i to j, zero if there is no path, and $s_{ii}^T = 1$. Write $\Delta[s] = \sum_{i,j} s_{ij} \Delta^{ij}$ and let ϵ_0 extract the constant term. For even elements $F_1, \dots, F_n$ without constant term, define

$$T_n(F_1, \dots, F_n) = \sum_{\text{trees } T} \int_{[0,1]^T} \epsilon_0 \left\{ \prod_{\ell \in T} \Delta_\ell e^{\Delta[s^T(\tau)]} \prod_i F_i(\Psi^{(i)}) \right\} \mathrm{d}\tau. \tag{11.cxii}$$

For $n = 1$ this is the Gaussian expectation.

Proposition 11.7. If $V_+(t)$ satisfies eq. 11.cviii and is analytic, then

$$W(t) = V_0(t) + \sum_{n \geq 1} \frac{1}{n!} T_n(V_+(t), \dots, V_+(t))$$

is analytic on $\mathbb{D}(t_x)$ and $Z_{\mathrm{tm}}(t) = \det D_*(\mathscr{A}) e^{W(t)}$. For $c \neq c'$ and nonempty unit-site sets P, P',

$$\begin{aligned} |\partial_c \partial_{c'} (W - V_0)| &\leq 2 N_G(\partial_c \partial_{c'} V_+; 1) \\ &\quad + 2 \alpha_G N_G(\partial_c V_+; w_P) \rho_G(\partial_{c'} V_+; w_{P'}) e^{-\vartheta_0 d(P,P')}. \end{aligned} \tag{11.cxiii}$$

■

Proof. The forest interpolation formula for a polynomial H is

$$H(\mathbf{1}) = \sum_F \int \left(\prod_{\ell \in F} \partial_{s_\ell} H \right) (s^F(\tau)) \, \mathrm{d}\tau.$$

For a partition π of the vertices, keep fixed entries $s_{\mathrm{in}}(i,j)$ inside its blocks and put $X_\pi(u)_{ij} = u$ on pairs joining distinct blocks. The chain rule and the fundamental theorem of calculus give

$$G(X_\pi(u)) = G(X_\pi(0)) + \sum_{\substack{\ell = (i,j) \\ i,j \text{ in distinct blocks of } \pi}} \int_0^u (\partial_{s_\ell} G)(X_\pi(w)) \, \mathrm{d}w. \tag{11.cxiv}$$

When ℓ joins two blocks, merge them and assign the value w to every pair between those two blocks. The interpolation point is unchanged. Begin with singleton blocks, $G = H$ and $u = 1$, and iterate this identity on each integrand. Each derivative merges two components, so after at most $n-1$ steps the procedure ends. The successive parameters are ordered decreasingly; when two vertices merge, their common value is the minimum parameter on their eventual path. For a fixed forest, its edge orderings divide the parameter cube into these ordered simplices. Summing them gives the formula.

The matrix s^F is a Gram matrix of unit vectors: at $u \in [0,1]$, assign to vertex i the basis vector of its component among the edges with $\tau_\ell > u$, and integrate their scalar products in u. This scalar product is the length of the interval on which the two vertices remain connected, namely s_{ij}^F. Wick's formula and the Gram inequality therefore bound a residual monomial of degree p under $e^{\Delta[s^F]}$ by

g^p, where $\|G\|_{\mathrm{op}} \le g^2$. Indeed, its determinant entries are $\langle \omega_i \otimes \delta_x, \omega_j \otimes G\delta_y \rangle$. These facts apply whether or not the interpolated covariance is invertible.

Let vertex i of a tree have degree k_i and let the chosen homogeneous part of F_i have degree p_i. The edge derivatives leave the factorial $(p_i - k_i)!^{-1}$ and $p_i - k_i$ uncontracted fields at that vertex; terms with $p_i < k_i$ vanish. Wick's Gram bound supplies $g^{p_i - k_i}$. Sum contracted arguments from the leaves to vertex 1, using the weighted row and column bound α_G for every edge. If nonnegative symmetric weights w_i and a distance D satisfy

$$e^{\vartheta_0 D} \le \prod_i w_i(\text{arguments at } i) \prod_{\ell=(i,j)} e^{\vartheta_0 d(\text{edge arguments})},$$

this yields $e^{-\vartheta_0 D} \alpha_G^{n-1}$ times the full kernel sum at vertex 1 and the rooted sums at the other vertices. The number of trees with degrees $k_1, \dots, k_n$ is $(n-2)!/\prod_i (k_i - 1)!$. Enlarge the sum over degrees and use

$$\sum_{k=1}^{p} \frac{g^{p-k}}{(k-1)!(p-k)!} = \frac{(1+g)^{p-1}}{(p-1)!} \le \frac{h_G^p}{(p-1)!}.$$

We obtain, for $n \ge 2$,

$$|T_n(F_1, \dots, F_n)| \le e^{-\vartheta_0 D} (n-2)! \alpha_G^{n-1} N_G(F_1; w_1) \prod_{i=2}^{n} \rho_G(F_i; w_i). \tag{11.cxv}$$

For $n = 1$ the bound is $N_G(F_1; 1)$. This also explains why the tree norms use $(p-1)!$ rather than $p!$.

Apply the forest formula to the replicated Gaussian expectation of V_+^n. Different forest components contain disjoint generator sets and their even factors commute, so their expectations factor. Grouping by component partitions gives

$$\int \mathrm{d}\mu_G \, V_+^n = \sum_{\{B_1, \dots, B_q\}} \prod_{a=1}^{q} T_{|B_a|}(V_+, \dots, V_+).$$

The exponential formula for partitions identifies the Gaussian integral with the exponential of the connected tree sum. The identity first holds as a formal series in a scalar multiplying V_+. Bound eq. 11.cxv and $\alpha_G \rho_G(V_+; 1) \le 1/2$ make the connected series absolutely convergent up to that scalar equal to one, so the formal identity is an analytic identity there. The tree terms are polynomial in the activity coefficients, and their series converges locally uniformly in t. Thus W is analytic and its derivatives are obtained term by term.

Two derivatives of the symmetric multilinear tree expression either act on the same vertex or on two different vertices. The first terms, including $n = 1$, sum to at most $2N_G(\partial_c \partial_{c'} V_+; 1)$. For the second terms put the two derivatives at vertices 1 and 2. The weights there are $w_P, w_{P'}$, and those at all other vertices are w_δ. Along the unique path from 1 to 2, the distance $d(P, P')$ is at most the sum of the two endpoint distances, all edge distances, and the argument diameters at intermediate vertices. Hence the weight condition for eq. 11.cxv holds with $D = d(P, P')$. The tree sum leaves $\alpha_G N_G(\partial_c V_+; w_P) \rho_G(\partial_{c'} V_+; w_{P'}) e^{-\vartheta_0 d(P,P')}$ times $\sum_{n \ge 2} (\alpha_G \rho_G(V_+; w_\delta))^{n-2} \le 2$. This proves eq. 11.cxiii. □

For a function f of the marks, let t^T keep the coordinates indexed by T and set all others to zero, and define its component with variable set S by

$$f_S(t) = \sum_{T \subset S} (-1)^{|S \setminus T|} f(t^T).$$

Then $f = \sum_S f_S$ and $f_\varnothing = f(0)$. Pairing terms with and without a variable shows that f_S vanishes when any variable of S vanishes; a variable absent from f therefore makes every component

containing it zero. Representing two coordinate differences as integrals of a mixed derivative gives

$$\sup_{|t|_\infty \le R} |f_S(t)| \le 2^{|S|-2} R^2 \sup_{|t|_\infty \le R} |\partial_c \partial_{c'} f(t)| \qquad (c, c' \in S,\ c \ne c'). \tag{11.cxvi}$$

Proposition 11.8. Assume the factor and mass conditions eq. 11.lxxxvii–eq. 11.lxxxviii, the tree conditions eq. 11.cviii, and $d_{\mathrm{sep}} \ge 6M + 8$. For $|S| \ge 2$, the normalized logarithm satisfies

$$\sup_{|t|_\infty \le t_x/2} |\mathcal{G}_S(t; \theta, A)| \le \varpi_{\mathrm{sep}} \le \varpi^+_{\mathrm{sep}}, \tag{11.cxvii}$$

where

$$\begin{aligned} C_W^{\mathrm{mix}} &= 3C_V^{\mathrm{ins}} h_G^2 + 2\alpha_G (C_V^{\mathrm{ins}})^2 h_G^4 + 4\exp(6) C_d M^3 K_{\mathrm{pol}}, \\ \varpi_{\mathrm{sep}} &= 2^{N_*} C_W^{\mathrm{mix}} p_\beta e^{-\vartheta_0 (d_{\mathrm{sep}} - 4M - 6)}, \\ \varpi^+_{\mathrm{sep}} &= 2^{N_*} C_W^{\mathrm{mix}} p_\beta e^{-\vartheta_0 (M r_* / (8L) - 4M - 7)}. \end{aligned} \tag{11.cxviii}$$

■

Proof. The tree representation and uniqueness of the normalized logarithm give

$$\mathcal{G}(t) = \sum_X \mathfrak{P}_*(X, \mathscr{C}(t)) + W(t) - W(0).$$

The component of a determinant activity is zero unless X meets every P_c, $c \in S$. Such a polymer has $d_M(X) \ge d_{\mathrm{sep}}/M - 2 \ge 1$. Use eq. 11.lxxx, sum the polymers anchored at one of at most p_β cubes, and keep $e^{-2d_M(X)}$ from $\kappa - \kappa_0 \ge 2$. Its total contribution is at most $2^{|S|} 2\exp(4) C_d M^3 K_{\mathrm{pol}} p_\beta e^{-\vartheta_0 d_{\mathrm{sep}}}$.

Apply eq. 11.cxiii with $P = P_c$, $P' = P_{c'}$ and eq. 11.cix. The bound for $\partial_c \partial_{c'} W$ is

$$q_x^2 p_\beta \left[2C_V^{\mathrm{ins}} h_G^2 e^{-\vartheta_0 (d_{\mathrm{sep}} - 4M - 6)} + 2\alpha_G (C_V^{\mathrm{ins}})^2 h_G^4 e^{-\vartheta_0 d_{\mathrm{sep}}} \right].$$

In eq. 11.cxvi, take $R = t_x/2$ and use $q_x R = 1$. Add the determinant contribution and $|S| \le N_*$ to obtain the first inequality. The placement estimate $d_{\mathrm{sep}} \ge M r_*/(8L) - 1$ proves the second. When the lattice probe is zero, $R_{\mathrm{tm}} = 1$, $\mathcal{G} = 0$ and the first separation error is interpreted as zero. □

11.4.4. *The small-field representation and its complement.* By Prop. 6.3, the coordinates $(A_*; \zeta_j)_{j < k_*}$ identify the exact-curvature space F_N. Each $\zeta_j(\Upsilon) = \langle \phi_{j,\Upsilon}, F^{\mathrm{u}} \rangle$ has an orthonormal coefficient vector; each coarse plaquette $(dA_*)(p) = \langle \phi_p, F^{\mathrm{u}} \rangle$ has $\|\phi_p\| \le 1$. There are at most $2\ell_*^3 L^{3(k_* - j)}$ coordinates at level j and $3\ell_*^3$ coarse plaquettes, by eq. 6.xxvi–eq. 6.xxvii. By eq. 6.xxv, the Gaussian measure m_* at the stopping scale is nondegenerate on the space of real coarse fields that are mean-zero and coclosed. These row norms and counts allow a union bound for all cutoffs in the small-field event.

Let $\mathfrak{G}$ require $|\zeta_j(\Upsilon)| \le p_{0,j}$ and $|dA_*|_\infty \le p_*$, and let $\mathfrak{G}^-$ decrease each threshold by one. Put

$$\begin{aligned} \varpi_0 &= 18\ell_*^3 e^{-p_{0*}/2} + 6\ell_*^3 e^{-(p_* - 1)^2/(2C_0^2)}, \\ t_2 &= e \min\left\{ N_*^{-1/2},\ C_\beta^{-1},\ (2\sqrt{N_* \mathbf{D}})^{-1} \right\}, \\ \varpi_L &= \sqrt{\varpi_0}\, e^{\mathbf{D}/2}, \qquad M_L = 2 + e^{\mathbf{D}/2}, \\ \lambda_L &= (\varpi_L^{1/3} M_L^{2/3})^{1/N_*} 5 (2N_* r / t_2)^4. \end{aligned} \tag{11.cxix}$$

A reciprocal with zero denominator is interpreted as $+\infty$. The number $\mathbf{D}$ is the uniform probe energy bound, so $v_\eta \le \mathbf{D}$.

Proposition 11.9. Suppose the preceding conditions hold, with $v_* \ge 8$ and $p_0 v_*^{p_0 - 1} \ge 28$. For a fixed $0 < s < 1$, set $\bar\kappa_1 = 6 e_*^{1/4 - 9\varepsilon}$. Assume

$$2\varpi^+_{\mathrm{sep}} \le \bar\kappa_1^{N_*}, \quad N_* \bar\kappa_1^{1-s} \le 1/2, \quad \varpi_0 \le 1/4, \quad 8N_* t_2 \le t_x. \tag{11.cxx}$$

There is a finite positive measure v_{sf} on $[-\pi,\pi]^3 \times \{A \in V_{\mathrm{t}}(T_*^0) : \chi_*(A) = 1\}$, of mass in $[1-\varpi_0, 1]$, such that

$$\frac{\bar\nu^{a(t)}(\mathfrak{G})}{\bar\nu(F_N)} = \int e^{\mathcal{G}(t;\theta,A)}\, v_{\mathrm{sf}}(\,\mathrm{d}\theta\,\mathrm{d}A). \tag{11.cxxi}$$

The normalized complement $L(t) = \bar\nu^{a(t)}(F_N \setminus \mathfrak{G})/\bar\nu(F_N)$ is entire. Its fourth jet, obtained by retaining Taylor exponents at most four in every t_c, has coefficient norm at radius r at most $\lambda_L^{N_*}$. ■

Proof. First, the union bound and eq. 7.ii give

$$\bar\mu(F_N \setminus \mathfrak{G}^-) \le \varpi_0. \tag{11.cxxii}$$

For $i = k_* - j \ge 1$, write $v_j = v_* + i(\log L)/2$. The scale conditions $v_* \ge 8$, $p_0 v_*^{p_0-1} \ge 28$ give $p_{0,j} \ge p_{0*} + 14i\log L$. Since $(p_{0,j}-1)^2 \ge p_{0,j}$, the individual tail is at most $2e^{-p_{0*}/2}L^{-7i}$. Multiply by the coordinate count $2\ell_*^3 L^{3i}$ and sum L^{-4i}. The $3\ell_*^3$ coarse plaquettes give the second term in eq. 11.cxix. These estimates yield eq. 11.cxxii, with room in the coefficient 18.

The exact density identity defines

$$v_{\mathrm{sf}}(\,\mathrm{d}\theta\,\mathrm{d}A) = \frac{\chi_*(A) Y_*^{\mathrm{x}}(A + c_\theta; 0)}{\bar\nu(F_N)} \frac{\mathrm{d}\theta}{(2\pi)^3}\, \mathsf{dm}_*(A). \tag{11.cxxiii}$$

We check positivity of this density, which does not follow from the formal Grassmann representation. For real marks, the original fine law $\nu^{\theta,a(t)}$ is nonnegative. The identity eq. 6.xxxiv says, for every nonnegative bounded Borel Φ,

$$\int \Phi(A)\chi_*(A) Y_*^{\mathrm{x}}(A + c_\theta; \mathscr{C}(t))\, \mathsf{dm}_*(A) = \nu^{\theta,a(t)}(\mathbf{1}_{\mathfrak{G}} \Phi(A_*)) \ge 0.$$

The terminal density is continuous. On the interior, relative to $V_{\mathrm{t}}(T_*^0)$, of the compact convex set $\{A \in V_{\mathrm{t}}(T_*^0) : |dA|_\infty \le p_*\}$, a nonzero imaginary part or negative real part would persist on a ball of positive Gaussian measure, contradicting this identity. At the boundary, approach along uA, $u \uparrow 1$; zero is an interior point. Thus the density is nonnegative everywhere on this set. Prop. 11.4 makes it nonzero there, hence strictly positive. This reasoning also proves $\mathcal{G}(t;\theta,A) \in \mathbb{R}$ for real t: its exponential is a positive ratio and its continuous value at zero is zero.

Integration of the exact terminal identity over θ, followed by eq. 11.xc, proves eq. 11.cxxi. At $t = 0$ the mass is $\bar\mu(\mathfrak{G})$, which lies in $[1-\varpi_0, 1]$. All integrals here are well-defined Borel integrals: the density is continuous on the compact support and zero elsewhere.

For real $t = ez$ with $|t|_\infty \le t_2$, put $G_z = \sum_c z_c d\widehat{\mathfrak{a}}_c$. Disjoint supports imply $\|G_z\|^2 = v_\eta |z|_2^2 \le 1/4$. The real source shift gives

$$L(ez) = e^{-v_\eta |z|_2^2/2} \bar\mu\left(\mathbf{1}_{(F_N\setminus\mathfrak{G})+G_z} e^{\langle G_z, F^{\mathrm{u}}\rangle}\right).$$

Every photon coordinate changes by at most $\|G_z\| \le 1$ under the shift. Thus the shifted complement is contained in $F_N \setminus \mathfrak{G}^-$. Cauchy–Schwarz, eq. 11.cxxii, and Gaussian domination at $2G_z$ give

$$0 \le L(t) \le \sqrt{\varpi_0}\, e^{v_\eta |z|_2^2/2} \le \varpi_L. \tag{11.cxxiv}$$

This is the only place where the lowered thresholds of $\mathfrak{G}^-$ are needed.

The source-shift identity also gives, for complex marks,

$$|\mathring{\Phi}(t)| \le \exp\left(\frac{v_\eta}{2e^2} \sum_c (\operatorname{Im} t_c)^2\right).$$

It follows directly by bounding the expectation of the real exponential and canceling its real quadratic normalization. On $|t|_\infty \le t_2$ this is at most $e^{\mathbf{D}/2}$. The small-field integral has modulus at most 2. Indeed, evenness and the double zero of ℓ_0 give $|\ell_0(t)| \le B_0 |t|^2/t_x^2$, while the zero of w_c gives $|w_c(t)| \le (3/8)|t|/t_x$. Their sum over cells at $t_2 \le t_x/(8N_*)$ is at most $1/(256N_*) + 3/64$. The remaining logarithm components sum to at most $2^{N_*}\varpi_{\mathrm{sep}}^{+} \le (2\bar\kappa_1)^{N_*}/2 \le N_*^{-N_*}/2$, by eq. 11.cxx.

Thus $|\mathcal{G}| \le 1/2$ and $|e^{\mathcal{G}}| \le 2$. Since L is the difference between the full generating function and the small-field integral, $|L| \le M_L$ on that polydisc. It is entire because the finite determinant density is entire and is bounded on compact mark sets uniformly in the real photon field and real twist. The modulus bound in the external-response subsection supplies the required Gaussian integrable majorant also for either Borel event.

To use Cauchy's coefficient estimate, we need a small bound on a complex polydisc. We combine the small real-mark bound with the uniform complex bound by the following two-constants estimate. Let $n \ge 1$ be an integer. If a holomorphic function on a polydisc in $\mathbb{C}^n$ of radius R has modulus at most M, and at most $\delta \le M$ on its real points, then its modulus is at most $\delta^{1/3} M^{2/3}$ on the polydisc of radius $R/(2n)$. On the upper half of the unit disc the harmonic function $\omega(z) = \frac{2}{\pi} \arg[(1+z)/(1-z)]$ is zero on the diameter and one on the semicircle. Comparing $\log|f(R'z)|$ with $(1-\omega)\log\delta + \omega \log M$, for $R' < R$, the maximum principle gives the same inequality in the half disc. To handle its endpoints, subtract $\epsilon \log[4/|1-z^2|]$ before applying the principle, then let $\epsilon \downarrow 0$. Its harmonic conjugate gives a holomorphic function to which the modulus principle applies. On $|z| \le q$, $\omega(z) \le (4/\pi)\arctan q$; reflection treats the lower half disc. Let $R' \uparrow R$. Apply this one-variable estimate successively with $q = 1/(2n)$. The remaining exponent of δ is at least $(1 - \frac{4}{\pi}\arctan(1/(2n)))^n \ge 1 - 2/\pi > 1/3$, proving the assertion. The case $\delta = 0$ follows by the identity theorem, and $M = 0$ is immediate.

Apply it to L, with $n = N_*$ and $R = t_2$. Cauchy's coefficient bound gives

$$|L_\alpha| \le \varpi_L^{1/3} M_L^{2/3} (2N_*/t_2)^{|\alpha|}.$$

In the fourth jet, every exponent is at most four. Since $y = 2N_* r/t_2 \ge 1$,

$$\sum_{\alpha_c \le 4} |L_\alpha| r^{|\alpha|} \le \varpi_L^{1/3} M_L^{2/3} (1 + y + \cdots + y^4)^{N_*} \le \lambda_L^{N_*}.$$

Reality on real marks gives real jet coefficients. □

11.4.5. *The fourth-jet error.* Centering annihilates the free response in any cell untouched by an error term. A surviving product must therefore contain errors that meet every cell. Their bounds give a power of the small error for each cell, before the N_*-th root in $\Theta_m^{(0)}$ is taken.

Only derivatives through order four in each variable are needed. We keep their Taylor coefficients in $\mathcal{J} = \mathbb{C}[t_c : c \in \mathcal{C}_*]/(t_c^5 : c \in \mathcal{C}_*)$, with norm $\|P\|_r = \sum_{\alpha_c \le 4} |P_\alpha| r^{|\alpha|}$. The jet map is a ring homomorphism, and the norm is submultiplicative: discarding a monomial with any exponent at least five decreases its coefficient sum.

For $S = \{c\}$ set $k_S = e^{w_c} - 1$ in this ring, and for $|S| \ge 2$ set $k_S = e^{\mathcal{G}_S} - 1$. Every monomial of k_S uses every variable of S and no other variable.

For a component f_S bounded on a polydisc of radius R, Cauchy's estimate at a smaller radius $R' \le R/2$ gives

$$\|f_S\|_{R'} \le \sup|f_S| \left(\sum_{j=1}^{4} (R'/R)^j \right)^{|S|} \le \sup|f_S|.$$

Its monomials contain each variable of S because it vanishes on every corresponding coordinate hyperplane. Use this with $R = t_x/2$, $R' = r$, and use the one-cell estimate on radius t_x, to obtain

$$\|k_{\{c\}}\|_r \le \bar\kappa_1, \qquad \|k_S\|_r \le 2\varpi_{\mathrm{sep}}^{+} \le \bar\kappa_1^{N_*} \quad (|S| \ge 2). \tag{11.cxxv}$$

We used $\|w_c\|_r \le 3e_*^{1/4-9\varepsilon} \le 3/8$ and $\|e^P - 1\|_r \le 2\|P\|_r$ for $\|P\|_r \le 1/2$. The coefficients are continuous in (θ, A): express the logarithm coefficients by Cauchy integrals on a fixed small torus, and then take the finite polynomial expressions in those coefficients. They are real because the logarithm is real at real marks.

In $\mathcal{J}$ the normalized generating function has the exact representation

$$\mathring{\Phi} = \prod_{c} \phi_0(t_c) \int \prod_{\varnothing \neq S \subset \mathcal{C}_*} (1 + k_S)\, v_{\mathrm{sf}}(\,\mathrm{d}\theta\,\mathrm{d}A) + L^{[4]}. \tag{11.cxxvi}$$

To justify taking the jet inside the integral, use joint continuity, holomorphy, and $|e^{\mathcal{G}}| \le 2$ on a fixed polydisc of radius at most t_2. Cauchy's coefficient formulas and Fubini then commute every required coefficient with the finite measure integral. Decompose $\mathcal{G}$ into its variable-set components; its singleton components are $\ell_0(t_c) + w_c(t_c)$. Exponentiation, the jet homomorphism, and $e^{\ell_0} = N_0$ prove the representation.

Theorem N. Assume the terminal inputs and conditions above, and $\lambda_L \le \bar{\kappa}_1^s$. For $m = 2, 4$,

$$\Theta_m^{(0)} \le m! r^{-m} (2\exp(1)\bar{\kappa}_1^s + \lambda_L) \le 8 m! r^{-m} \bar{\kappa}_1^s. \tag{11.cxxvii}$$

With $\mathfrak{e} = s(1/4 - 9\varepsilon) - 12\varepsilon > 0$,

$$\begin{aligned} \Theta_4^{(0)} &\le 3 \cdot 10^5 C_b^4 e_*^{\mathfrak{e}} \le 3 \cdot 10^5 C_b^4 h^{1/8} e^{\mathfrak{e}}, \\ (1 + |c_2|)\Theta_2^{(0)} &\le 3 \cdot 10^4 C_b^4 e_*^{\mathfrak{e}} \le 3 \cdot 10^4 C_b^4 h^{1/8} e^{\mathfrak{e}}. \end{aligned} \tag{11.cxxviii}$$

The constants are independent of the ultraviolet cutoff, the twist, and the admitted placement. ■□

Proof. On the one-variable jet space define $\ell_c(t_c^j) = m! \mathbf{1}_{\{j=m\}} - c_m \mathbf{1}_{\{j=0\}}$ and let $\ell = \bigotimes_c \ell_c$. Then $(\Theta_m^{(0)})^{N_*} = |\ell(\mathring{\Phi})|$. The free jet satisfies $\ell_c(\phi_0) = 0$. From eq. 11.ciii, $|\ell(P)| \le (m! r^{-m})^{N_*} \|P\|_r$. These statements follow by evaluating the functionals on monomials with every exponent in $\{0, m\}$.

Expand the product in eq. 11.cxxvi over families $\mathcal{Y}$ of nonempty cell sets. If $\mathcal{Y}$ fails to cover a cell c, its error factors do not contain t_c; the factor $\phi_0(t_c)$ is therefore annihilated by ℓ_c. Only covering families contribute. Assign cost one to each singleton and cost N_* to each larger set, and let $a(\mathcal{Y})$ be the sum of these costs. A covering family has $a(\mathcal{Y}) \ge N_*$: either it contains a larger set or it contains every singleton. By eq. 11.cxxv its coefficient norm is at most $\bar{\kappa}_1^{a(\mathcal{Y})}$. The free factors have norm at most 2^{N_*}. Keep $\bar{\kappa}_1^{sN_*}$ from the cost and sum the rest over all families. This sum is

$$(1 + \bar{\kappa}_1^{1-s})^{N_*} (1 + \bar{\kappa}_1^{(1-s)N_*})^{2^{N_*} - 1 - N_*} \le \exp\left(N_* \bar{\kappa}_1^{1-s} + 2^{N_*} \bar{\kappa}_1^{(1-s)N_*} \right) \le \exp(1).$$

The last step uses $N_* \bar{\kappa}_1^{1-s} \le 1/2$. The positive small-field measure has mass at most one. Its total contribution is thus bounded by $(m! r^{-m})^{N_*} \exp(1) (2\bar{\kappa}_1^s)^{N_*}$. The complement contributes at most $(m! r^{-m} \lambda_L)^{N_*}$. Taking the N_*th root proves eq. 11.cxxvii.

Since $r^{-4} = 256 C_b^4 e_*^{-12\varepsilon}$ and $\bar{\kappa}_1^s \le 6 e_*^{s(1/4 - 9\varepsilon)}$, the $m = 4$ bound has coefficient at most $8 \cdot 24 \cdot 1536 < 3 \cdot 10^5$. For $m = 2$, $1 + |c_2| \le (17/16) r^{-2}$, so its coefficient is at most $17 \cdot 1536 < 3 \cdot 10^4$. Finally $0 < \mathfrak{e} < 1/4$ and $e_* \le e\sqrt{h}$ imply $e_*^{\mathfrak{e}} \le h^{1/8} e^{\mathfrak{e}}$. The scale and probe bounds entering this proof are uniform in the stated choices. □

11.4.6. *A positive charge threshold.* We now verify that the response conditions admit a common interval of charges, independent of the ultraviolet cutoff. In addition to eq. 11.lxxxvii, eq. 11.lxxxviii, eq. 11.xciv, eq. 11.cviii, and eq. 11.cxx, require $r_* > 31L + 16$, $v_* \ge 8$, $p_0 v_*^{p_0 - 1} \ge 28$, and

$$M r_*/(8L) - 1 \ge 6M + 8, \qquad \lambda_L \le \bar{\kappa}_1^s, \tag{11.cxxix}$$

$$D_\beta^{\mathrm{buf}}/2 + 1/2 \le 3\ell_c/8, \qquad D_\beta^{\mathrm{buf}} + 3 \le \ell_c. \tag{11.cxxx}$$

The construction of §6.1 fixes an odd $L \ge 13$, a power M of L with $M \ge L^2$, a localization power $r_{\mathrm{loc}} \ge 2$, and cutoff powers $p_0 \ge 2$, $p > 2p_0$. It also fixes $0 < \varepsilon \le 1/200$ and $0 < s < 1$ such that

$$\mathfrak{e} = s(1/4 - 9\varepsilon) - 12\varepsilon, \qquad \sigma = \mathfrak{e} - \frac{2a_L}{N_*} > 0, \qquad a_L = \frac{L^3 - 1}{(L-1)^3}. \tag{11.cxxxi}$$

Only $\mathfrak{e}$ enters the terminal estimate; the loss $2a_L/N_*$ belongs to the subsequent finite-volume transfer.

Proposition 11.10. For the fixed construction constants, $\bar{m}$, and probe β, there is $e_\sharp(\beta) > 0$ such that all response conditions hold for $0 < e \le e_\sharp(\beta)$ and every $N \ge N_{\mathrm{ref}}(e)$. The threshold depends on the probe only through $C_\beta, \mathbf{D}, D_\beta^{\mathrm{buf}}$. It is independent of N, the twist, and the admitted placement. All centered placements are admitted in this range. ■

Proof. Write $v_K = -\log(e\sqrt{h})$ and $r_K = \min\{L^i : L^i \ge v_K^{r_{\mathrm{loc}}}\}$. Then $R^*(e) = hMr_K/L$ and $\ell_c = R^*(e)/(2L)$. Prop. 6.1 and eq. 6.vii give a finite $N_{\mathrm{ref}}(e)$.

For $N \ge N_{\mathrm{ref}}(e)$ the stop level is $K-1$ or $K-2$ by eq. 6.viii. In particular its scales are independent of N and satisfy

$$\begin{gathered} v_K \le v_*, \quad e_* = e^{-v_*}, \quad a_* \in \{h/L, h/L^2\}, \\ v_*^{r_{\mathrm{loc}}} \le r_*, \quad Mv_*^{r_{\mathrm{loc}}} \le \ell_* < LMv_*^{r_{\mathrm{loc}}}, \quad p_{0*} = v_*^{p_0}, \quad p_* = v_*^p, \\ p_\beta \le p_\beta^+ := (D_\beta^{\mathrm{buf}} L^2/h + 2M + 4)^3, \qquad \ell_c \ge hMv_K^{r_{\mathrm{loc}}}/(2L^2). \end{gathered} \tag{11.cxxxii}$$

The strict wrapping condition $r_* > 31L + 16$ already follows from eq. 6.iv whenever $e\sqrt{h} \le \bar{e}_1$. The two tail conditions hold on a sufficiently large half-line because $p_0 \ge 2$. Thus every response condition is a condition on e alone once the construction and probe are fixed.

The remaining conditions reduce to polynomial factors times decaying exponentials. For $A, d \ge 0$ and $\gamma > 0$, split the exponential and use $v^d e^{-\gamma v/2} \le (2d/(\gamma \exp(1)))^d$ to make $Av^d e^{-\gamma v}$ uniformly small on a sufficiently large half-line. Terms with quadratic exponential decay are handled by $cv^2 \ge bv + \log^+ A$ for sufficiently large v, where $c > 0$. We apply these bounds to the following groups of conditions.

The local tree smallness and terminal factor conditions follow from exponential decay with positive rates ε, 2ε, $1/4 - \varepsilon$, and $1/4 - 9\varepsilon$; the volume factors are bounded by $(LM)^3 v_*^{3r_{\mathrm{loc}}}$. For $\bar{r}$, use $1/4 - 7\varepsilon > 2\varepsilon$. The complex mass and coarse determinant conditions similarly have rate 2ε. The support count is bounded by p_β^+. All three summands of B_0 tend to zero: their rates are 4ε, 2ε, and the quadratic decay $e^{-r_*} \le e^{-v_*^2}$. Also $\tau_s \ge c_\tau M v_*^2$, so both $\tau_s \ge 1$ and $52B_1 e^{-\tau_s} \le e_*^{1/4-9\varepsilon}$ hold once v_* is large.

The separation condition follows from $r_* \ge v_*^2$. Its error obeys

$$2\varpi_{\mathrm{sep}}^+ \le 2^{N_*+1} C_W^{\mathrm{mix}} p_\beta^+ e^{\vartheta_0(4M+7)} e^{-\vartheta_0 M v_*^2/(8L)}.$$

Since $\bar{\kappa}_1 \ge e^{-v_*/4}$, quadratic decay makes this at most $\bar{\kappa}_1^{N_*}$ for all sufficiently large v_*. The family summation condition $N_* \bar{\kappa}_1^{1-s} \le 1/2$ uses the positive rate $(1-s)(1/4 - 9\varepsilon)$.

For the photon complement, $p > 2p_0$ implies $(p_* - 1)^2 \ge C_0^2 p_{0*}$ once v_* is large. Therefore

$$\varpi_0 \le 24(LM)^3 v_*^{3r_{\mathrm{loc}}} e^{-v_*^2/2}. \tag{11.cxxxiii}$$

This is eventually at most $1/4$. Moreover t_2/e is fixed and positive, while e is between $(L/h)^{1/2} e_*$ and $Lh^{-1/2} e_*$. Hence $8N_* t_2/t_x$ is bounded by a fixed constant times $e_*^{1-3\varepsilon}$, which tends to zero.

The potentially largest loss is the fourth derivative radius in λ_L. From eq. 11.cxxxiii, the definition of λ_L, and $r/t_2 \le Ce^{v_*}$, we obtain

$$\lambda_L \le A_8 \exp\left(-\frac{v_*^2}{12N_*} + \left(\frac{r_{\mathrm{loc}}}{2N_*} + 4 \right) v_* \right) \tag{11.cxxxiv}$$

for a finite $A_8 > 0$ depending only on the fixed constants and probe bounds. For clarity one can take

$$A_8 = [24(LM)^3]^{1/(6N_*)} e^{\mathbf{D}/(6N_*)} M_L^{2/(3N_*)} 5 \left[\frac{N_* \sqrt{h}}{2C_b \sqrt{L}} (t_2/e)^{-1} \right]^4.$$

If $v_*^2/(12N_*) \ge (r_{\mathrm{loc}}/(2N_*) + 17/4)v_* + \log^+ A_8$, then $\lambda_L \le e^{-v_*/4} \le \bar{\kappa}_1^s$. Thus the complex continuation and the fourth derivatives preserve a positive small-charge interval. Finally the two placement inequalities hold when $\ell_c \ge (4/3) D_\beta^{\mathrm{buf}} + 3$; this follows from eq. 11.cxxxii as $v_K \to \infty$.

Choose V beyond those half-lines and set $e_V = h^{-1/2}\min(\bar e_1, e^{-V}) > 0$. For $e \le e_V$, $v_* \ge v_K \ge V$, so every condition holds. Set $e_\sharp = e_V$. The preceding half-line estimates give every response condition throughout $(0, e_\sharp]$.

The scale identities depend on e, h and the fixed construction constants; the probe enters through $C_\beta, \mathbf{D}, D_\beta^{\mathrm{buf}}$. They do not involve the twist or placement, and they are unchanged by the coordinate reflections used to form the cell probes. Centered placements satisfy the admission inequality by eq. 11.cxxx, since their centers differ from the cell center by at most $\eta/2 \le 1/2$. The same conditions give $D_\beta^{\mathrm{buf}} + 3\eta \le \ell_c$ and $R^*(e) \ge D_\beta + 1$. □

11.5. **Wick products on arbitrary cutoff tori.** Use the cell sources of §11.1 and write

$$N_* = 8L^3, \qquad a_L = \frac{L^3-1}{(L-1)^3}, \qquad D^{\mathrm{mat}} = L^3 \log z_{r,n_*} - \log z_{r,Ln_*}, \qquad \delta = \frac{D_L}{(L-1)^3 N_*}.$$

Fix $0 < e \le 1/3$, with $q = e\sqrt{\eta}$ and $q^2 n_* = e^2 R^*$, and assume

$$6N_* e^2 R^* \log(1/e) \le \pi^2. \tag{11.cxxxv}$$

This condition makes the nonzero-flux contribution small after taking the N_*-th root; the scale choice ensuring it is verified below. Let $\mathcal{C}_* = \{0, \dots, 2L-1\}^3$ and use the probe variance v_η of eq. 11.ix. For real c_2, c_4 put

$$H_2(x;v) = x^2 - v, \qquad H_4(x;v) = x^4 - 6vx^2 + 3v^2,$$

$$O_m^{[c]} = e^{-m} H_m(X_c; v_\eta) - c_m, \qquad m = 2, 4, \tag{11.cxxxvi}$$

$$\Theta_m^{(0)} = \left| \left[\prod_{c \in \mathcal{C}_*} (\partial_{t_c}^m - c_m) \right] \mathring{\Phi}(0) \right|^{1/N_*}. \tag{11.cxxxvii}$$

$$M_4 = 600 N_*^2 v_\eta^2 e^{-4} + |c_4|, \qquad M_2 = 12 N_* v_\eta e^{-2} + |c_2|,$$

$$T_m = \Theta_m^{(0)} + 2e^6 M_m, \qquad u_m = \exp(\delta) T_m. \tag{11.cxxxviii}$$

Here $\mathring{\Phi}$ is the normalized external partition function eq. 11.xliii. The notation e^{-m} denotes an inverse power of the charge.

Proposition 11.11. Assume the probe placement condition eq. 11.vi. For $n = n_*$, or $n \in p\mathbb{Z}$ with $n \ge Ln_*$, the all-flux law with the fixed real lattice mass obeys

$$0 \le \delta \le \frac{a_L}{N_*} \log \frac{2\pi}{e^2 R^*} + \frac{K_L + D^{\mathrm{mat}} + 1}{(L-1)^3 N_*}, \tag{11.cxxxix}$$

$$\left| \mathbb{E}_n^{\mathrm{af}} \prod_{c \in \mathcal{C}} O_m^{[c]} \right| \le u_m^{|\mathcal{C}|} \tag{11.cxl}$$

for every set $\mathcal{C}$ of distinct cells and $m = 2, 4$.

The bounds are uniform in n. Their right sides are formed on the two reference tori; $\Theta_m^{(0)}$, and therefore u_m, may depend on the placement. ■

Proof. By eq. 11.cxxxv, using $N_* \ge 1000$ and $\log(1/e) > 1$, we have $e^2 R^* < 1$. For $x = \exp(-2\pi^2/(e^2 R^*))$ they also give

$$x \le e^{12N_*}, \qquad x^{1/(2N_*)} \le e^6.$$

Apply Prop. 8.6 at n_* and $j = L$. Since $\varsigma_{n_*} \le 7x < 1/8$,

$$D_L^{\mathrm{fl}} \le -L^3 \log(1 - \varsigma_{n_*}) \le 8L^3 x = N_* x \le N_* 3^{-12N_*} \le 1.$$

The photon estimate and the defect decomposition now prove the upper bound in eq. 11.cxxxix; its lower bound is Prop. 9.1.

By §11.1, these real cell polynomials are the same local integrands on every carrying torus. Put $O_m^\sharp = \prod_{c\in\mathcal{C}_*} O_m^{[c]}$. Its all-flux expectation is nonnegative by eq. 9.i. We split it into zero and nonzero flux: the first part is the terminal response, and the second is bounded by its flux probability and Gaussian moments.

First, Gaussian domination and the moment bound give, for every positive integer i,

$$\mathbb{E}^{\mathrm{af}}_{n_*} X_c^{2i} \le (2i\exp(1)v_\eta)^i.$$

Minkowski's inequality at exponent $2N_*$ yields

$$\|H_4(X_c;v_\eta)\|_{2N_*} \le [64\exp(2)N_*^2 + 24\exp(1)N_* + 3]v_\eta^2 \le 600N_*^2 v_\eta^2,$$
$$\|H_2(X_c;v_\eta)\|_{2N_*} \le [4\exp(1)N_* + 1]v_\eta \le 12N_* v_\eta.$$

Thus $\|O_m^{[c]}\|_{2N_*} \le M_m$. The finite moments justify all uses of Hölder and Cauchy–Schwarz below, also when $v_\eta = 0$.

On the zero-flux sector, eq. 11.xlv gives

$$\left|\mathbb{E}^{\mathrm{af}}_{n_*}[O_m^\sharp \mathbf{1}_{\{\mathrm{flux}=0\}}]\right| = P_{n_*}(0)\left|\mathbb{E}^{\mathrm{av}}_{n_*} O_m^\sharp\right| \le (\Theta_m^{(0)})^{N_*}.$$

On the remaining sectors, Cauchy–Schwarz followed by Hölder with N_* factors gives

$$\begin{aligned}\left|\mathbb{E}^{\mathrm{af}}_{n_*}[O_m^\sharp \mathbf{1}_{\{\mathrm{flux}\neq 0\}}]\right| &\le \varsigma_{n_*}^{1/2}\big(\mathbb{E}^{\mathrm{af}}_{n_*}|O_m^\sharp|^2\big)^{1/2}\\ &\le \varsigma_{n_*}^{1/2}\prod_{c\in\mathcal{C}_*}\|O_m^{[c]}\|_{2N_*}\\ &\le \big(\varsigma_{n_*}^{1/(2N_*)} M_m\big)^{N_*}.\end{aligned}$$

Since $\varsigma_{n_*}^{1/(2N_*)} \le 2e^6$, the nonnegative N_*-th root satisfies

$$\|O_m\|_* = \big(\mathbb{E}^{\mathrm{af}}_{n_*} O_m^\sharp\big)^{1/N_*} \le \Theta_m^{(0)} + 2e^6 M_m = T_m.$$

For nonnegative a, b we used $(a^{N_*} + b^{N_*})^{1/N_*} \le a + b$. Prop. 9.1, with $s_0 = 2L$ and $j = L$, proves eq. 11.cxl.

□

11.6. **Reference scales and the loss of charge exponent.** The reference side is chosen from the logarithmic charge scale. Fix a power M of L with $M \ge L^2$ and an integer $r \ge 2$. With the scale h defined in §6.1, set

$$e_K = e\sqrt{h}, \qquad v_K = -\log e_K, \qquad r_K = \min\{L^i : L^i \ge v_K^r,\ i \in \mathbb{Z}_{\ge 0}\}, \qquad R^* = \frac{hMr_K}{L}. \tag{11.cxli}$$

Assume $v_K \ge 8$, $e \le 1/3$, $e_K \log h \le 1$, and

$$6N_* M v_K^r \exp(-v_K)\left(v_K\exp(-v_K) + \tfrac{1}{2}\right) \le \pi^2. \tag{11.cxlii}$$

These conditions hold on a sufficiently small charge interval: as $e \to 0$, $v_K \to \infty$ and the polynomial-exponential tail tends to zero.

Minimality gives $v_K^r \le r_K < L v_K^r$. Consequently $T^* = \log_L(Mr_K) - 1$ is an integer at least one, $R^* = hL^{T^*}$, and $\ell_c = R^*/(2L) \to \infty$ as $e \to 0$. Moreover $1/L < \bar{m}h \le 1$, so

$$\bar{m}R^* > \frac{Mr_K}{L^2} \ge r_K \ge v_K, \qquad \exp(-\bar{m}R^*) \le e\sqrt{h}. \tag{11.cxliii}$$

The same upper bound on r_K gives

$$\begin{aligned}e^2 R^* \log(1/e) &< M v_K^r \exp(-2v_K)(v_K + \tfrac{1}{2}\log h)\\ &\le M v_K^r \exp(-v_K)(v_K\exp(-v_K) + \tfrac{1}{2}).\end{aligned}$$

Thus eq. 11.cxlii implies eq. 11.cxxxv. The probe fits every sufficiently small charge and sufficiently fine mesh, since D_β^{buf} is fixed and $\ell_c \to \infty$.

The exponential of the partition defect contributes an inverse power of the charge. The final response exponent is obtained by subtracting this loss from the reference-volume exponent.

Let C_{mat} and C_{ref} be finite constants independent of the charge, mesh and placement. Assume that for $N \geq N_{\mathrm{ref}}(e)$ the mass assignment and reference response satisfy

$$\begin{aligned} D^{\mathrm{mat}} &\leq C_{\mathrm{mat}}, \\ \Theta_4^{(0)} \leq C_{\mathrm{ref}} e^{\mathfrak{e}}, &\qquad (1+|c_2|)\Theta_2^{(0)} \leq C_{\mathrm{ref}} e^{\mathfrak{e}}, \end{aligned} \tag{11.cxliv}$$

where

$$\mathfrak{e} = s(1/4 - 9\varepsilon) - 12\varepsilon, \qquad \sigma = \mathfrak{e} - \frac{2a_L}{N_*} > 0, \qquad 0 < s < 1, \quad 0 < \varepsilon \leq 1/200. \tag{11.cxlv}$$

Here $N_{\mathrm{ref}}(e)$ is a fixed finite mesh threshold. The physical charge ceiling in eq. 13.x is $e_G = \bar{e}_1$ when $h = 1$, and $e_G = \min\{\bar{e}_1/\sqrt{h},\, 1/(\sqrt{h}\log h)\}$ when $h > 1$. For the chamber construction, eq. 8.xxiii and eq. 11.cxxviii give eq. 11.cxliv for $0 < e \leq \min(e_G, e_\sharp(\beta))$. The constants and cutoff thresholds are those fixed in §13.1.

On the reference torus take $c_j = N_0^{(j)}(0)$ from eq. 10.xxi for the same placed probe. The free-jet estimate eq. 11.ciii and $r = e_*^{3\varepsilon}/(4C_b)$ give $|c_2| \leq C_b^2 e_*^{-6\varepsilon}$ and $|c_4| \leq 16 C_b^4 e_*^{-12\varepsilon}$. Since $a_* \geq h/L^2$ and $h \geq 1$, one has $e_* \geq e/L$. Put $J = C_b^2 L^{6\varepsilon}$. Thus

$$|c_2| \leq J e^{-6\varepsilon}, \qquad |c_4| \leq 16 J^2 e^{-12\varepsilon}, \qquad |c_4 - 3c_2^2 - b(f)| \leq (C_\beta^{\mathrm{fr}} + 1) e^{\sigma}. \tag{11.cxlvi}$$

Here C_β^{fr} is independent of e, N, T and placement. Indeed, Prop. 10.2 and eq. 11.cxliii give $|b_{R^*}(f) - b(f)| \leq C_\beta^{\mathrm{fr}} e$. For each fixed $e > 0$, qualitative lattice convergence in that proposition supplies a finite threshold $N_{\mathrm{uv}}(e) \geq N_{\mathrm{ref}}(e)$ so that $|b_{\mathrm{ch}}(\eta, R^*) - b_{R^*}(f)| \leq e^{\sigma}$ for every $N \geq N_{\mathrm{uv}}(e)$. Use $b_{\mathrm{ch}} = c_4 - 3c_2^2$ and $e \leq e^{\sigma}$. All preceding reference estimates hold above this larger threshold. The coefficients are real and independent of lattice placement.

Proposition 11.12. Fix the probe and the scale data above, and assume eq. 11.vi and eq. 11.cxliv. There is a finite constant C_U, independent of e, N, T and placement, such that for all the stated reference charges and $N \geq N_{\mathrm{uv}}(e)$,

$$u_4 \leq C_U e^{\sigma}, \qquad (1+|c_2|)u_2 \leq C_U e^{\sigma}, \qquad |c_4 - 3c_2^2 - b(f)| \leq C_U e^{\sigma}. \tag{11.cxlvii}$$

On every cutoff torus with $T \geq T^* + 1$, every finite set $Y \subset (R^*/L)\mathbb{Z}^3$ of points incongruent modulo $R\mathbb{Z}^3$ satisfies

$$\left| \mathbb{E}_{N,T} \prod_{y \in Y} W_m(F^\sharp(\tau_y f^\eta); v_\eta, c_m) \right| \leq u_m^{|Y|}, \qquad m = 2, 4, \tag{11.cxlviii}$$

where $W_m(x; v, c) = e^{-m} H_m(x; v) - c$.

The tests f^η have one fixed compact support, $f^\eta \to f$ in L^2, and $v_\eta \to \|f\|_2^2$. Thus eq. 11.cxlvii and eq. 11.cxlviii give exactly the cutoff hypotheses of Prop. 13.3. ∎

Proof. Write

$$s_\beta = \|f\|_2 + \|\beta\|_{H^4}, \qquad C_\Gamma^{\mathrm{tr}} = (2\pi)^{a_L/N_*} \exp\left(\frac{K_L + C_{\mathrm{mat}} + 1}{(L-1)^3 N_*} \right).$$

Since $R^* \geq 1$, eq. 11.cxxxix gives

$$\exp(\delta) \leq C_\Gamma^{\mathrm{tr}} e^{-2a_L/N_*}. \tag{11.cxlix}$$

The inverse power here is a power of the charge. The source-curvature identity and Jensen's inequality for the disjoint averaging cubes give $v_\eta^{1/2} \leq \|f^\eta\|_2 \leq s_\beta$. Put

$$B_4 = 600 N_*^2 s_\beta^4 + 16 J^2, \qquad B_2 = 12 N_* s_\beta^2 + J.$$

Because $12\varepsilon < 4$ and $6\varepsilon < 2$, the moment bounds eq. 11.cxxxviii and eq. 11.cxlvi give $M_4 \le B_4 e^{-4}$ and $M_2 \le B_2 e^{-2}$. Consequently

$$\begin{aligned} T_4 &\le C_{\mathrm{ref}} e^{\mathfrak{e}} + 2B_4 e^2 \le (C_{\mathrm{ref}} + 2B_4) e^{\mathfrak{e}}, \\ (1+|c_2|)T_2 &\le C_{\mathrm{ref}} e^{\mathfrak{e}} + 2(1+J)B_2 e^{4-6\varepsilon} \\ &\le [C_{\mathrm{ref}} + 2(1+J)B_2] e^{\mathfrak{e}}. \end{aligned}$$

The last inequalities use $0 < e \le 1$ and $0 < \mathfrak{e} < 1/4 < \min(2, 4-6\varepsilon)$. Take

$$\begin{aligned} C_U = \max\{&C_\Gamma^{\mathrm{tr}}(C_{\mathrm{ref}} + 2B_4), \\ &C_\Gamma^{\mathrm{tr}}[C_{\mathrm{ref}} + 2(1+J)B_2],\ C_\beta^{\mathrm{fr}} + 1\}. \end{aligned} \tag{11.cl}$$

Multiplication by eq. 11.cxlix proves the first two bounds in eq. 11.cxlvii; eq. 11.cxlvi proves the third.

For $T \ge T^* + 1$, $n = 8hL^{T+N} = 8L^{T-T^*} n_*$ is a multiple of p and $n \ge L n_*$. The even indices eq. 11.xi assign distinct cells to incongruent $y \in (R^*/L)\mathbb{Z}^3$. Prop. 11.11 and translation invariance give eq. 11.cxlviii. Any fixed finite set is incongruent once $R > 2\max_{y\in Y} |y|_\infty$.

The compact-support and convergence assertions follow from eq. 11.v and eq. 11.x. □

12. The continuum limit and the Osterwalder–Schrader axioms

We now construct limiting laws and prove their structural axioms. Gaussian estimates give compactness and convergence of moments, and the nine reflection families pass to the limits. The general reflection results in §§A, B, and C supply rotation covariance and clustering. We check their hypotheses for the limiting law and its ergodic components, so that the same components satisfy all the required axioms.

12.1. **Gaussian domination and tensor moments.** The Gaussian bound is used at three different points: to obtain compactness of the curvature laws, to control products of field insertions, and to pass moments to limiting and conditional laws. We record its consequences in a form independent of the regulator. The tensor extension below supplies the stated growth axiom on general tests, rather than only on tensor products.

12.1.1. *Random linear forms.* Let λ be a probability law on the real tempered two-forms, with its cylinder σ-algebra, and suppose

$$\mathbb{E}_\lambda e^{F(f)} \le e^{\|f\|_2^2/2} \qquad (f \in \mathcal{S}(\mathbb{R}^3; \Lambda^2) \text{ real}). \tag{12.i}$$

Set $X(f) = F(f)$ and extend X complex linearly. For $p > 0$ put $c_p = 2\,2^{p/2}\Gamma(1 + p/2)$.

Proposition 12.1. The map X extends uniquely and continuously to $L^2(\mathbb{R}^3; \Lambda^2)$ with values in $L^p(\lambda)$ for every $p \ge 1$. For real f and $u > 0$,

$$\mathbb{E}_\lambda X(f) = 0, \qquad \mathbb{E}_\lambda X(f)^2 \le \|f\|_2^2, \tag{12.ii}$$

$$\lambda(|X(f)| \ge u) \le 2e^{-u^2/(2\|f\|_2^2)}, \qquad \mathbb{E}_\lambda |X(f)|^p \le c_p \|f\|_2^p \quad (p > 0). \tag{12.iii}$$

If $\|f\|_2 = 0$, the tail bound is interpreted as zero for $u > 0$. For real $f_1, \dots, f_n$,

$$\mathbb{E}_\lambda \prod_{j=1}^n |X(f_j)| \le c_n \prod_j \|f_j\|_2, \qquad c_n \le 2\sqrt{2^n n!}. \tag{12.iv}$$

For complex tests the same bound holds with an additional factor $2^{n/2}$. ■

Proof. For a real test, eq. 12.i applied to tf gives $\mathbb{E}_\lambda e^{tX(f)} \le e^{t^2\|f\|_2^2/2}$. Also

$$\mathbb{E}_\lambda e^{t|X(f)|} \le \mathbb{E}_\lambda e^{tX(f)} + \mathbb{E}_\lambda e^{-tX(f)} \le 2e^{t^2\|f\|_2^2/2}.$$

Thus the exponential series can be differentiated at zero. Comparing its linear and quadratic terms with the Gaussian upper bound, for both signs of t, gives eq. 12.ii. Markov's inequality with $t = u/\|f\|_2^2$ gives the upper tail; applying it to $-f$ gives the lower tail. Integrating this bound yields

$$\mathbb{E}_\lambda |X(f)|^p = p\int_0^\infty u^{p-1}\lambda(|X(f)| > u)\,\mathrm{d}u \le 2\,2^{p/2}\Gamma(1+p/2)\|f\|_2^p.$$

Hölder's inequality gives eq. 12.iv; log-convexity of Γ gives $\Gamma(1+n/2) \le \sqrt{n!}$. For integers $r \ge 1$, also $c_{2r} = 2^{r+1}r! \le (2r\exp(1))^r$, since $r! \le r^r$ and $2 \le \exp(r)$. For $f = g + ih$, the triangle inequality in $L^n(\lambda)$ and $\|g\|_2 + \|h\|_2 \le \sqrt{2}\|f\|_2$ give the complex bound. These estimates applied to differences extend X from real Schwartz tests to L^2, consistently for all $p \ge 1$. Fatou along an almost surely convergent approximating subsequence also preserves eq. 12.i on real L^2 tests. □

For fixed n, the moment bound controls differences of products by expanding one factor at a time. In particular, $S_n(f_1,\dots,f_n) = \mathbb{E}_\lambda \prod_j X(f_j)$ is a bounded multilinear form on complex L^2, symmetric in its arguments and real on real tests.

12.1.2. *Extension to tensor tests.* Use the Hermite basis and eigenvalues defined before Thm. A. For a tensor test write

$$\Psi = \sum_{\boldsymbol{\kappa}} c_{\boldsymbol{\kappa}}(\Psi) h_{\kappa_1} \otimes \cdots \otimes h_{\kappa_n}.$$

The Hermite expansion converges in Schwartz space. The estimate we need is

$$\sum_{\boldsymbol{\kappa}} |c_{\boldsymbol{\kappa}}(\Psi)| \le 4^{-n}(34\pi\sqrt{3})^n q_{6n}(\Psi). \tag{12.v}$$

To see the derivative order and the dependence on n, set $Z_4 = \sum_\kappa \lambda_\kappa^{-4} < 1/16$. Weighted Cauchy–Schwarz and Parseval give

$$\sum_{\boldsymbol{\kappa}} |c_{\boldsymbol{\kappa}}(\Psi)| \le Z_4^{n/2}\|(A^2)^{\otimes n}\Psi\|_2.$$

With $M(x) = 1 + |x|^2$,

$$A^2 = M^2 - 2M\Delta - 4x\cdot\nabla - 6 + \Delta^2.$$

The sum of the moduli of the differential coefficients of A^2 is bounded by $34M^2$. Applying this expression in each of the n arguments gives the componentwise bound

$$|(A^2)^{\otimes n}\Psi(x)| \le 34^n q_{6n}(\Psi)\prod_{a=1}^n (1+|x_a|^2)^{-1}.$$

Since $\int_{\mathbb{R}^3}(1+|x|^2)^{-2}\,\mathrm{d}x = \pi^2$ and there are 3^n components, eq. 12.v follows. The bound on Z_4 follows by counting Hermite eigenvalues:

$$Z_4 = \frac{3}{32}\sum_{d\ge 0}\frac{d+1}{(d+2)^3} \le \frac{3}{32}\Big(\frac{\pi^2}{6} - 1\Big) < \frac{1}{16}.$$

Define the random tensor evaluation by the $L^2(\lambda)$ sum

$$Q_\Psi = \sum_{\boldsymbol{\kappa}} c_{\boldsymbol{\kappa}}(\Psi)\prod_{a=1}^n X(h_{\kappa_a}). \tag{12.vi}$$

The product moment bound, now applied to $2n$ real tests of norm one, shows that this sum converges absolutely in $L^2(\lambda)$ and

$$\|Q_\Psi\|_{L^2(\lambda)} \le \sqrt{c_{2n}}\sum_{\boldsymbol{\kappa}} |c_{\boldsymbol{\kappa}}(\Psi)|.$$

For a decomposable test it equals $\prod_a X(f_a)$, by the Hermite expansion of each f_a and the same product bounds. Thus $S_n(\Psi) := \mathbb{E}_\lambda Q_\Psi$ is the unique continuous extension of the moments to $\mathcal{S}_n$. Moreover,

$$\begin{aligned}|S_n(\Psi)| &\le c_n \sum_{\boldsymbol{\kappa}} |c_{\boldsymbol{\kappa}}(\Psi)| \\ &\le 2\sqrt{n!}\left(\frac{34\pi\sqrt{6}}{4}\right)^n q_{6n}(\Psi) \le 2\,66^n\sqrt{n!}\,q_{6n}(\Psi).\end{aligned}$$

This proves (E0′) and supplies the L^2 tensor evaluations used to pass clustering to Schwinger functions.

12.1.3. *The topology of the limits.* Recall the Hilbert scale defined before Thm. A:

$$\|F\|_{-s}^2 = \sum_\kappa \lambda_\kappa^{-2s}|F(h_\kappa)|^2, \qquad H_{-s} = \{F \in \mathcal{S}' : \|F\|_{-s} < \infty\}.$$

A closed bounded ball of H_{-3} is compact in H_{-4}. In Hermite coordinates, the tail of the squared H_{-4} norm on the indices $\lambda_\kappa > M$ is bounded by M^{-2} times the squared H_{-3} norm; the remaining coordinates form a finite-dimensional space. A diagonal subsequence argument proves compactness.

The inequality $|F(f)| \le \|F\|_{-4}\|A^4 f\|_2$ also proves continuity of $H_{-4} \hookrightarrow \mathcal{S}'$ for the strong distribution topology: the second factor is uniformly bounded on every bounded Schwartz set.

12.2. **Volume and continuum limits.** Gaussian domination gives compactness. Identifying each fixed-spacing volume limit as the step-field image of an infinite lattice law will transfer the nine mirrors before removing the spacing.

12.2.1. *The test map and its bounds.* Write $\eta = L^{-N}$, $R = 8hL^T$, and $n = R/\eta$, retaining the reference scale h of §6.1.1. The main theorem uses $h = 1$. A plaquette of the unit covering lattice is $\sigma = (x; \mu\nu)$, with $\mu < \nu$ and center $y_\sigma = x + (e_\mu + e_\nu)/2$. Its averaging cube is

$$Q_\eta(\sigma) = \eta(y_\sigma + [-1/2, 1/2)^3).$$

For each fixed orientation these cubes partition $\mathbb{R}^3$. If u is a real lattice curvature, its physical step field is $\eta^{-3/2}u(\sigma)$ on $Q_\eta(\sigma)$ in component $\mu\nu$. For a Schwartz two-form f, define

$$f_{[\eta]}(\sigma) = \eta^{-3/2}\int_{Q_\eta(\sigma)} f_{\mu\nu}(y)\,\mathrm{d}y, \tag{12.vii}$$

$$f^R(y) = \sum_{k\in\mathbb{Z}^3} f(y + Rk),$$

$$f_\eta([\sigma]) = \eta^{-3}\int_{Q_\eta(\sigma)} f^R_{\mu\nu}(y)\,\mathrm{d}y. \tag{12.viii}$$

Here $[\sigma]$ is the plaquette class modulo $n\mathbb{Z}^3$. The torus cochain norm is $|v|_\eta^2 = \eta^3\sum_{[\sigma]}|v([\sigma])|^2$. The covering cochain norm $\|\cdot\|$ is the unweighted ℓ^2 norm. For the periodic extension $\widetilde{u}$ of u,

$$F^\sharp(f) = \eta^{3/2}\sum_{[\sigma]} f_\eta([\sigma])u(\sigma) = \sum_\sigma f_{[\eta]}(\sigma)\widetilde{u}(\sigma). \tag{12.ix}$$

The first normalization is the cell volume η^3 times the field factor $\eta^{-3/2}$.

The estimates needed below are

$$\|f_{[\eta]}\| \le \|f\|_2, \tag{12.x}$$

$$\sum_\sigma |f_{[\eta]}(\sigma)|(1 + |x(\sigma)|_\infty) \le \eta^{-3/2}\int_{\mathbb{R}^3}\sum_{\mu<\nu}|f_{\mu\nu}(y)|(2 + |y|_\infty/\eta)\,\mathrm{d}y, \tag{12.xi}$$

$$|f_\eta|_\eta \le \|f^R\|_{L^2(\mathbb{R}^3/R\mathbb{Z}^3)} \le p_R(f), \tag{12.xii}$$

where, with $Q_R = [-R/2, R/2)^3$,

$$p_R(f) = \|f\|_2 + 26R^{-2}\||y|^2 f\|_{L^2(\mathbb{R}^3\setminus Q_R)}. \tag{12.xiii}$$

One also has $|f_\eta|_\eta \le \|f\|_2$ if the support of f lies in a translate of an open cube of side R. For $R \ge 1$, $p_R(f) \le p_1(f)$, and $p_R(f) \to \|f\|_2$ as $R \to \infty$. The seminorm p_1 is continuous on Schwartz space.

The first bound follows from Cauchy–Schwarz on each averaging cube. On that cube $1+|x(\sigma)|_\infty \le 2 + |y|_\infty/\eta$, which gives the second. Absolute convergence permits unfolding:

$$\eta^{3/2} f_\eta([\sigma]) = \sum_{k\in\mathbb{Z}^3} f_{[\eta]}(\sigma + nk).$$

This proves eq. 12.ix. Applying the cube Cauchy–Schwarz inequality to f^R gives the first inequality in eq. 12.xii. If the support lies in one open R-cube, its distinct $R\mathbb{Z}^3$ translates are disjoint, so $\|f^R\|_{L^2(\mathbb{R}^3/R\mathbb{Z}^3)} = \|f\|_2$.

For the remaining estimate put $a_k = \|f\|_{L^2(Q_R+Rk)}$. Minkowski's inequality bounds the periodized norm by $\sum_k a_k$. On $Q_R + Rk$, $k \ne 0$, one has $|y| \ge R|k|_\infty/2$; hence

$$\sum_{k\ne 0} a_k \le 4R^{-2}\left(\sum_{k\ne 0} |k|_\infty^{-4}\right)^{1/2} \||y|^2 f\|_{L^2(\mathbb{R}^3\setminus Q_R)}.$$

The number of points with $|k|_\infty = j$ is $24j^2 + 2$, so the sum in parentheses is $4\pi^2 + \pi^4/45 < 41.65$ and its square root times four is less than 26. This proves eq. 12.xii and eq. 12.xiii. The comparison with p_1 and the limit of p_R follow by inclusion of the cubes and integrability of $|y|^4|f(y)|^2$.

For the Hermite two-forms of §12.1,

$$p_1(h_\kappa) \le 31\lambda_\kappa. \tag{12.xiv}$$

Here is the weighted estimate behind this bound. Let $\mathsf{M} = 1 + |y|^2$, $P_j = -i\partial_j$, and $\mathsf{A} = \mathsf{M} - \Delta$. Since $[P_j, [P_j, \mathsf{M}]] = -2$,

$$\|\mathsf{A}v\|_2^2 = \|\mathsf{M}v\|_2^2 + \|\Delta v\|_2^2 + 2\sum_j \langle P_j v, \mathsf{M}P_j v\rangle - 6\|v\|_2^2 \ge \|\mathsf{M}v\|_2^2 - 6\|v\|_2^2.$$

The Hermite eigenvalues satisfy $\lambda_\kappa \ge 4$, and therefore $\|\mathsf{M}v\|_2^2 \le (11/8)\|\mathsf{A}v\|_2^2$. Taking $v = h_\kappa$ gives $p_1(h_\kappa) \le 1 + 26\sqrt{11/8}\lambda_\kappa \le 31\lambda_\kappa$.

The test map respects the affine cubic symmetries. For a signed permutation and lattice translation g of the unit lattice, let $\mathfrak{g}(y) = \eta g(y/\eta)$. With the exterior-square action on two-forms and the cochain pull-back,

$$(\mathfrak{g}\cdot f)_{[\eta]} = (g^{-1})^* f_{[\eta]}, \qquad (\mathfrak{g}\cdot f)_\eta = (g^{-1})^* f_\eta. \tag{12.xv}$$

Indeed, g sends centers to centers and the averaging cubes to averaging cubes up to null boundaries. Its orientation sign is the sign of $\wedge^2 \vec{g}$ on the corresponding plaquette. Changing variables gives the first formula; periodization and the bijection $k \mapsto \vec{g}k$ give the second.

12.2.2. *Compactness and convergence of moments.* Let $P_{N,T}$ be the law of $F^\sharp$ for the all-flux chamber measure. The mass assignment may be any real sequence independent of T. Thm. D and eq. 12.ix give

$$\mathbb{E}_{N,T} e^{F^\sharp(f)} \le e^{|f_\eta|_\eta^2/2} \le e^{p_R(f)^2/2} \qquad (f \text{ real}). \tag{12.xvi}$$

The random-linear-form proof of Prop. 12.1 applies here with the Hilbert seminorm $f \mapsto |f_\eta|_\eta$. In particular its variance bound and eq. 12.xiv give

$$\mathbb{E}_{N,T}\|F^\sharp\|_{-3}^2 \le \sum_\kappa \lambda_\kappa^{-6} p_1(h_\kappa)^2 \le 961 \sum_\kappa \lambda_\kappa^{-4} < 61. \tag{12.xvii}$$

Tonelli's theorem justifies the first sum. Every finite lattice cochain has a step field in H_{-3}: Cauchy–Schwarz in eq. 12.ix bounds its κ-th Hermite coordinate by $31\lambda_\kappa\|u\|_{\ell^2(C(n)^{(2)})}$, with the unweighted norm on the unit-lattice torus plaquettes. The step map from this finite-dimensional cochain space to H_{-3} is consequently continuous.

Markov's inequality gives $P_{N,T}(\|F\|_{-3} > r) \le 61r^{-2}$. The compact H_{-3} balls therefore give uniform tightness on H_{-4} and, by Prokhorov's theorem, nonempty sets $\mathcal{V}_N$ of fixed-spacing volume limits as $T \to \infty$. These balls are closed in H_{-4}, so portmanteau passes their probability lower bounds to every law in $\mathcal{V}_N$. Thus $\bigcup_N \mathcal{V}_N$ is uniformly tight as well and has volume-first limit points: weak limits of $\mu_i \in \mathcal{V}_{N_i}$ with $N_i \to \infty$.

For a joint sequence $N_k, T_k \to \infty$ suppose $P_{N_k,T_k} \to \mu$ weakly on H_{-4}. Truncating $e^{F(f)}$ above by a constant, passing to the weak limit, and then removing the truncation gives

$$\mathbb{E}_\mu e^{F(f)} \le e^{\|f\|_2^2/2}. \tag{12.xviii}$$

The limiting constant follows from $p_{R_k}(f) \to \|f\|_2$. For fixed complex tests $f_1, \dots, f_n$, the product $Y(F) = \prod_j F(f_j)$ is continuous on H_{-4}. The test-map bound and Prop. 12.1 give

$$\sup_k \mathbb{E}_{N_k,T_k}|Y(F^\sharp)|^2 \le 2^n c_{2n} \prod_j p_1(f_j)^2 < \infty.$$

The inequality $|Y|\mathbf{1}_{\{|Y|>M\}} \le |Y|^2/M$ gives uniform integrability from the displayed bound. Continuous truncation and weak convergence give every joint moment limit.

12.2.3. *Closedness and the affine cubic action.* The step fields are closed on the lattice. To pass this identity to distributions, let $\xi \in C_c^\infty(\mathbb{R}^3;\mathbb{R})$ represent the three-form $\xi\,\mathrm{d}y_1 \wedge \mathrm{d}y_2 \wedge \mathrm{d}y_3$. Put

$$\partial_j^\eta \xi(y) = \eta^{-1}(\xi(y + \eta e_j/2) - \xi(y - \eta e_j/2))$$

and define the two-form v^η in the component order $23, 13, 12$ by

$$(v_{23}^\eta, v_{13}^\eta, v_{12}^\eta) = (-\partial_1^\eta \xi, \partial_2^\eta \xi, -\partial_3^\eta \xi).$$

It satisfies

$$F^\sharp(v^\eta) = 0, \qquad \|v^\eta - d^*\xi\|_2 \le \frac{\eta^2}{24} \sum_{j=1}^3 \|\partial_j^3 \xi\|_2. \tag{12.xix}$$

For the exact identity, assign to each unit cube the number $\xi_{[\eta]}(x) = \eta^{-5/2} \int_{\eta(x+[0,1)^3)} \xi(y)\,\mathrm{d}y$. The three plaquette coefficients of $v_{[\eta]}^\eta$ are

$$-(\xi_{[\eta]}(x) - \xi_{[\eta]}(x - e_1)), \quad \xi_{[\eta]}(x) - \xi_{[\eta]}(x - e_2), \quad -(\xi_{[\eta]}(x) - \xi_{[\eta]}(x - e_3)).$$

In the pairing with $\widetilde{u}$ these regroup to $\sum_x \xi_{[\eta]}(x)(d\widetilde{u})(x) = 0$. The weighted summability in eq. 12.xi justifies regrouping; at finite volume $\widetilde{u}$ is bounded. For the error bound, the Fourier multiplier of $\partial_j^\eta - \partial_j$ is $i(2\eta^{-1}\sin(\eta p_j/2) - p_j)$, whose modulus is at most $\eta^2|p_j|^3/24$. Plancherel's theorem gives eq. 12.xix.

For all large k, $d^*\xi - v^{\eta_k}$ is supported in one open R_k-cube. The variance bound and eq. 12.xix then give

$$\mathbb{E}_{N_k,T_k} F^\sharp(d^*\xi)^2 \le \|d^*\xi - v^{\eta_k}\|_2^2 \le \frac{\eta_k^4}{24^2} \left(\sum_j \|\partial_j^3 \xi\|_2 \right)^2.$$

Passing bounded truncations of the square to the weak limit shows $F(d^*\xi) = 0$ almost surely. Choose a countable dense family of compactly supported smooth three-form tests. Continuity of each tempered distribution extends the identity from this family to all tests, so $\mu(dF = 0) = 1$.

By eq. 12.xv, finite-volume lattice translations, signed permutations and sign reversal act on the step-field characteristic functionals. The last two pass to weak limits. Approximate a real translation b by $b_k \in \eta_k\mathbb{Z}^3$ with $|b_k - b|_\infty \le \eta_k/2$; the variance bound gives

$$\left|\mathbb{E}_{N_k,T_k} e^{iF^\sharp(\tau_b f)} - \mathbb{E}_{N_k,T_k} e^{iF^\sharp(\tau_{b_k} f)}\right| \le p_1(\tau_b f - \tau_{b_k} f) \longrightarrow 0.$$

Thus the limiting characteristic functional is invariant under every translation. Characteristic functionals determine the law, so μ has these invariances.

12.2.4. *Identifying a fixed-spacing volume limit.* Fix N, hence η, and take any $\mu^{(N)} \in \mathcal{V}_N$. Choose a defining sequence $T_j \to \infty$ with $P_{N,T_j} \to \mu^{(N)}$ weakly on H_{-4}. Let ν_j be the law on Ω of the periodically extended unit-lattice curvature. We shall identify a product-topology limit of these laws and prove

$$\nu_j \longrightarrow \nu^{(N)}, \qquad \mu^{(N)} = (\sharp_\eta)_* \nu^{(N)}. \tag{12.xx}$$

The step map in this identity is only Borel; the identification requires more than the continuous mapping theorem.

For a closed cochain u put

$$\Omega' = \left\{ u \in \Omega : \sup_\sigma \frac{|u(\sigma)|}{1 + |x(\sigma)|_\infty} < \infty \right\}.$$

On Ω' the componentwise step field $\sharp_\eta u$ is tempered, and $(\sharp_\eta u)(f) = \sum_\sigma f_{[\eta]}(\sigma) u(\sigma)$ is absolutely convergent by eq. 12.xi. Let $\Omega^{(\eta)}$ be the subset on which this distribution has finite H_{-3} norm. Regard $\sharp_\eta$ as a map $\Omega \to H_{-4}$ by setting it to zero outside $\Omega^{(\eta)}$. Both subsets and the map are Borel. Indeed, the growth bound is a countable supremum; each test evaluation is a pointwise limit of finite coordinate sums; and the squared H_{-3} norm is a countable sum of nonnegative evaluations. The Hermite evaluations generate the Borel sigma algebra of H_{-4}.

For each plaquette choose a real smooth bump b^σ, with only its corresponding two-form component nonzero, supported in the interior of $Q_\eta(\sigma)$ and of integral $\eta^{3/2}$. Then $(b^\sigma)_{[\eta]} = \mathbf{1}_\sigma$. The map

$$\mathcal{J}_\eta : H_{-4} \longrightarrow \mathbb{R}^{\{\text{plaquettes}\}}, \qquad \mathcal{J}_\eta(F)(\sigma) = F(b^\sigma) \tag{12.xxi}$$

is continuous for the product topology, and $\mathcal{J}_\eta \sharp_\eta u = u$ on $\Omega^{(\eta)}$. The finite-volume identities are $P_{N,T_j} = (\sharp_\eta)_* \nu_j$ and $\nu_j = (\mathcal{J}_\eta)_* P_{N,T_j}$. Thus the continuous mapping theorem gives $\nu_j \to \nu^{(N)} := (\mathcal{J}_\eta)_* \mu^{(N)}$ in the product space. Closedness of Ω and portmanteau show that $\nu^{(N)}$ is supported on Ω.

This lattice law satisfies Gaussian domination for every finitely supported cochain β. Once n_j is large, the support of β embeds without overlap in the cubic torus, and its periodization has unchanged ℓ^2 norm. Thm. D, weak convergence and bounded truncations of $e^{\langle \beta, u\rangle}$ give

$$\int_\Omega e^{\langle \beta, u\rangle} \, \mathrm{d}\nu^{(N)}(u) \le e^{\|\beta\|^2/2}. \tag{12.xxii}$$

For any law ν on Ω with this bound, the one-coordinate tail estimate is $\nu(|u(\sigma)| \ge t) \le 2e^{-t^2/2}$. There are at most $3(2k+1)^3$ plaquettes with base point of sup norm k, and hence

$$\sum_\sigma \nu(|u(\sigma)| \ge 1 + |x(\sigma)|_\infty) \le 6 \sum_{k \ge 0} (2k+1)^3 e^{-(k+1)^2/2} < \infty.$$

Borel–Cantelli gives $\nu(\Omega') = 1$. Truncate $f_{[\eta]}$ to a finite set of plaquettes. Absolute convergence on Ω' and Fatou's lemma pass eq. 12.xxii to this cochain, giving

$$\int e^{(\sharp_\eta u)(f)} \, \mathrm{d}\nu(u) \le e^{\|f_{[\eta]}\|^2/2} \le e^{\|f\|_2^2/2}.$$

Its variance bound and Tonelli's theorem yield

$$\int \|\sharp_\eta u\|_{-3}^2 \, \mathrm{d}\nu(u) \le \sum_\kappa \lambda_\kappa^{-6}.$$

In particular $\nu(\Omega^{(\eta)}) = 1$.

It remains to recover $\mu^{(N)}$ from $\nu^{(N)}$. Choose a countable Schwartz-dense family $\mathcal{D}_0$ of real compactly supported smooth two-forms. For $f \in \mathcal{D}_0$ set

$$f' = f - \sum_\sigma f_{[\eta]}(\sigma) b^\sigma.$$

The sum is finite and $f'_{[\eta]} = 0$. Consequently $F^\sharp(f') = 0$ at every finite volume. The hyperplane $\{F : F(f') = 0\}$ is closed in H_{-4}, so it has full $\mu^{(N)}$ measure. Intersecting these sets for $f \in \mathcal{D}_0$ with the full-measure set $\mathcal{J}_\eta(F) \in \Omega^{(\eta)}$ gives

$$F(f) = \sum_\sigma f_{[\eta]}(\sigma) F(b^\sigma) = (\sharp_\eta \mathcal{J}_\eta(F))(f) \qquad (f \in \mathcal{D}_0)$$

on one set of full measure. Both sides are tempered distributions, so density extends this equality to every test. This proves eq. 12.xx.

12.2.5. *Nine mirror families in the continuum.* At fixed N the charge $q = e\eta^{1/2}$ and the chamber mass parameter $M = \bar{m}\eta + m_{0,N} + \rho \sum_i w_i/2$ are independent of volume. The sides $n_j = 8hL^{T_j+N}$ satisfy the divisibility condition of Thm. G: every required power of L eventually divides the side, and 8 divides it. That theorem applies to $\nu^{(N)}$. It is invariant under the affine cubic group and change of sign, and reflection positive at the three coordinate and six diagonal mirror families. The identification eq. 12.xx transports its characteristic functional to $\mu^{(N)}$.

Write $\ell(y) = n_\ell \cdot y$ for one of $y_j, y_j - y_k, y_j + y_k$, $j < k$, and let $r_{\ell,c}$ be reflection in the plane $\{y : \ell(y) = c\}$. The admissible unit-lattice heights are $\frac{1}{2}\mathbb{Z}$ for a coordinate family and $\mathbb{Z}$ for a diagonal family. If c is an admissible physical height, the cochain test map intertwines the lattice reflection with $r_{\ell,c}$ by eq. 12.xv. Suppose that real $f_1, \dots, f_s \in C_c^\infty$ have support in $\{\ell \ge c + \delta\}$ with $\delta > 2\eta$. If $f_{a,[\eta]}(\sigma) \ne 0$, its averaging cube meets this support. Every corner ηx of the plaquette is within sup distance η of a point y in that intersection. Since $|n_\ell|_1 \le 2$,

$$\ell(\eta x) \ge \ell(y) - 2\eta > c.$$

Thus $f_{a,[\eta]}$ has finite support strictly on the positive lattice side. Lattice reflection positivity gives

$$\sum_{a,b} \overline{z_a} z_b \mathbb{E}_{\mu^{(N)}} e^{iF(f_b - r_{\ell,c} \cdot f_a)} \ge 0. \tag{12.xxiii}$$

This is the reflected form of the character sum $\sum_b z_b e^{i\langle f_{b,[\eta]}, u\rangle}$.

Proposition 12.2. Every volume-first limit of the chamber cutoff laws is a translation-invariant law of closed tempered two-forms with Gaussian domination. It is invariant and reflection positive at every plane in the nine families $y_j = c$ and $y_j \pm y_k = c$, $j < k$, $c \in \mathbb{R}$. All finite-family moments converge along a joint cutoff sequence realizing that limit. ∎

Proof. Suppose $\mu^{(N_i)} \to \mu$ weakly on H_{-4}, with $N_i \to \infty$ and $\mu^{(N_i)} \in \mathcal{V}_{N_i}$. Fix a metric for weak convergence of probability laws on H_{-4}. For each i, choose a volume index $T_i \ge i$ along the defining volume subsequence such that the distance between P_{N_i,T_i} and $\mu^{(N_i)}$ is less than $1/i$. Then $P_{N_i,T_i} \to \mu$ with both indices tending to infinity. The preceding compactness, moment convergence, closedness and invariance arguments apply to this joint sequence.

Fix ℓ and first take $c \in \mathbb{Z}[L^{-1}]$. Then $c \in \eta_i \mathbb{Z}$ for all large i, and hence it is an admissible height for each lattice mirror family. Any finite family of compact tests in $\{\ell > c\}$ has a common support

margin $\delta > 0$. For all large i, $2\eta_i < \delta$, so eq. 12.xxiii holds for $\mu^{(N_i)}$. The functions in this finite sum are bounded and continuous on H_{-4}; weak convergence passes its nonnegativity to μ.

For an arbitrary height c, choose $c' \in \mathbb{Z}[L^{-1}]$ and $v = (c - c')|n_\ell|^{-2} n_\ell$. Then $r_{\ell,c} = \tau_v r_{\ell,c'} \tau_{-v}$. Translate the test forms by $-v$; they are now supported on the positive side of $\ell = c'$. Translation invariance identifies the reflected form at c with that at c'. Thus eq. 12.xxiii holds at every height. Character sums are dense in the corresponding positive-side L^2 space, and the reflected form is bounded there by Cauchy–Schwarz. Its nonnegativity therefore extends to that space. Finally $r_{\ell,0}$ is a signed permutation, and $r_{\ell,c}$ is its composition with a translation. The invariances already proved give invariance at each mirror. □

12.3. **Passing the hypotheses to ergodic components.** To condition the reflection-positive form, we must be able to insert nonnegative translation-invariant weights into it. The diagonal spectral cone supplies this: invariance of an observable in one coordinate direction implies invariance under all translations, and every invariant observable has a representative in either half-space fixed by reflection. These facts are proved in §A.1.3.

Write λ for the parent law; it satisfies the hypotheses of Prop. C.1. Use $\widehat{g}A(F) = A(g^{-1}F)$, $U(a) = \widehat{\tau}_a$, $\mathcal{R}_j = \widehat{r}_j$ and $\Theta_j A = \overline{\mathcal{R}_j A}$ as in §A.1. Let P and P_j project in $L^2(\lambda)$ onto observables invariant under all translations and under $U(te_j)$, respectively. For an open O, let Σ_O be generated by $F(f)$ with real $f \in C_c^\infty(O;\Lambda^2)$, and put $\mathcal{E}_j^+ = L^2(\Sigma_{\{x_j>0\}}, \lambda)$.

Covariance of a mixture alone does not give covariance of its individual components.

Let $\mathcal{I}$ be the translation-invariant event sigma algebra. The space of tempered distributions, with the sigma algebra generated by its evaluations, is standard Borel. It therefore admits regular conditional laws λ_F given $\mathcal{I}$, with

$$\lambda_F(\Phi) = \mathbb{E}_\lambda[\Phi \mid \mathcal{I}](F), \qquad \lambda = \int \lambda_F \, \mathrm{d}\lambda(F). \tag{12.xxiv}$$

The first equality is understood almost surely for each integrable Φ (and for nonnegative Φ by monotone convergence). The image measure on laws is the translation-ergodic decomposition. We recall why a single full-measure set of ergodic laws can be used.

For an open set O, $\mathcal{S}(\overline{O};\Lambda^2)$ denotes the subspace of $\mathcal{S}(\mathbb{R}^3;\Lambda^2)$ consisting of forms supported in $\overline{O}$, with the induced Schwartz topology. Fix a countable real test space $\mathcal{D}_0 \subset C_c^\infty(\mathbb{R}^3;\Lambda^2)$ over $\mathbb{Q}$, stable under rational translations and cubic isometries, with $\mathcal{D}_0 \cap C_c^\infty(O;\Lambda^2)$ dense in $\mathcal{S}(\overline{O};\Lambda^2)$ for $\mathbb{R}^3$ and the six coordinate half-spaces O. Take the rational span of translated cubic images of countable Schwartz-dense compact-test families in these seven open sets to obtain it. Let $\mathcal{A}_0$ consist of finite sums of $e^{iF(f)}$, $f \in \mathcal{D}_0$, with coefficients in $\mathbb{Q} + i\mathbb{Q}$. This same countable family is dense in L^2 for every law on the evaluation sigma algebra and determines such laws. For density, condition an observable orthogonal to its span on finitely many evaluations. Rational combinations of the tests show that the Fourier transform of the resulting finite complex measure vanishes on rational frequencies. Continuity and Fourier uniqueness make it zero. Conditional expectations on the increasing finite evaluation sigma algebras converge to the observable, proving density. The same finite-dimensional Fourier argument proves determination of laws. Enumerate $\mathcal{A}_0 = \{A_p : p \geq 1\}$ and choose cube-average times $T_k \to \infty$ so that

$$\|M_{T_k} A_p - P A_p\|_2^2 \leq 2^{-k} \quad (p \leq k), \qquad M_T = T^{-3} \int_{[0,T]^3} U(a) \, \mathrm{d}a.$$

eq. 12.xxiv and summability give convergence in $L^2(\lambda_F)$ for every p on one full-measure set of F. Moreover PA_p has zero conditional variance, and thus equals the constant $\lambda_F(A_p)$ in that space. The conditional laws are translation invariant: test the parent invariance against $\mathcal{I}$-measurable bounded weights, first for rational translations and the countable characters, then use continuity and Fourier uniqueness. The mean ergodic theorem for each such law now shows that its invariant projection is the constant projection, first on the dense algebra and then on all $L^2(\lambda_F)$. Thus these

conditional laws are ergodic. This also gives uniqueness of their image measure: the same cube averages converge pointwise along a subsequence and in L^2 under almost every law in any ergodic decomposition, so they determine the same character expectations and hence the same component law.

Fix an axis mirror. A bounded invariant weight h satisfies $\mathcal{R}_j h = h$ by eq. A.iv and has a representative in either half-space. If $h \geq 0$ and $A \in \mathcal{E}_j^+$, reflection positivity applied to $\sqrt{h}A$ gives

$$\mathbb{E}_\lambda[h(\Theta_j A)A] \geq 0. \tag{12.xxv}$$

Reflection invariance similarly gives $\mathbb{E}_\lambda[h\mathcal{R}_j A] = \mathbb{E}_\lambda[hA]$. Taking all bounded $\mathcal{I}$-measurable weights in these identities proves that the corresponding conditional reflection form is nonnegative and the conditional character expectations are reflection invariant. Countably many positive-side character sums suffice; density extends the statements to the full positive-side algebra in each conditional law. Translation invariance moves the mirror to every real height.

For a further mirror with unit normal n, choose $Q \in O(3)$ with $Qe_1 = n$ and apply the axis argument to $(Q^{-1})_*\lambda$. Transport by Q_* preserves translation ergodicity and reflection forms; uniqueness identifies the image decomposition with that of λ. Intersecting the full-measure sets for the three axes and six diagonals transfers all nine mirrors, without assuming $Q_*\lambda = \lambda$.

Suppose now that the parent law satisfies eq. 12.i. To transfer this Gaussian domination with the same constant, fix real $f \in \mathcal{S}$ and put $G = e^{F(f)}$, $c = e^{\|f\|_2^2/2}$, $\phi = PG$. All moments of G are finite. For each integer $n \geq 2$, Tonelli's theorem and the parent bound give

$$\mathbb{E}_\lambda[(M_T G)^n] \leq T^{-3n} \int_{([0,T]^3)^n} \exp\left(\frac{n}{2}\|f\|_2^2 + \frac{1}{2}\sum_{p\neq q}\langle f, \tau_{a_p - a_q} f\rangle\right) \mathrm{d}a. \tag{12.xxvi}$$

The overlaps tend to zero as $|a_p - a_q| \to \infty$ and are bounded by $\|f\|_2^2$. Fix $\epsilon > 0$. Outside the set where one pair is within distance δ, the right side is at most $c^n e^{n(n-1)\epsilon/2}$ once δ is sufficiently large. The exceptional set has relative volume at most $\binom{n}{2}(2\delta)^3 T^{-3}$ and its integrand is at most $e^{n^2\|f\|_2^2/2}$. Since $M_T G \to \phi$ in L^2, and the averages and ϕ are uniformly bounded in L^{2n}, interpolation gives convergence in L^n. Sending $T \to \infty$ and then $\epsilon \to 0$ in eq. 12.xxvi yields $\|\phi\|_n \leq c$ for every n, hence $\phi \leq c$ almost surely. By eq. 12.xxiv,

$$\lambda_F(e^{F(f)}) \leq e^{\|f\|_2^2/2} \quad \text{almost surely.} \tag{12.xxvii}$$

Choose a countable Schwartz-dense real test family and intersect these full-measure sets. For any other test, pointwise convergence of the evaluations under each conditional law and Fatou's lemma extend eq. 12.xxvii to it. Thus all real tests obey the bound on one full-measure set of component laws.

Finally a measurable set of full parent measure has full measure in almost every component by eq. 12.xxiv. In particular closedness passes to the components; its event is measurable by testing $dF = 0$ on a countable dense family of three-form tests. The nine reflections generate the cubic group, so each of these components is also cubically invariant. Prop. C.1 applies to each component and gives clustering. Thm. O applies to the same component laws.

12.4. **Tensor Schwinger functions.** The Hermite expansion controls Q_Ψ in L^2, but its terms do not preserve the spatial support of Ψ. To prove half-space measurability, we use a Fourier expansion with compact cutoffs in that half-space.

12.4.1. *Products and supported tensor tests.* For $\Psi \in \mathcal{S}_k$ and $\Phi \in \mathcal{S}_l$, the tensor evaluation obeys

$$Q_\Psi Q_\Phi = Q_{\Psi\otimes\Phi} \quad \text{in } L^1(\lambda). \tag{12.xxviii}$$

The L^2 bounds, Cauchy–Schwarz and continuity of the Schwartz tensor product make both sides separately continuous into L^1. They agree on elementary tensors, whose density is proved below.

The same argument gives

$$Q_{\overline{\Psi}} = \overline{Q_\Psi}, \qquad Q_{g\cdot\Psi} = \widehat{g} Q_\Psi \quad \text{if } g_*\lambda = \lambda. \tag{12.xxix}$$

Here the second identity uses the exterior-square action, including the tensor indices.

For open $O \subset \mathbb{R}^3$ and $\Psi \in C_c^\infty(O^n)$, choose $\chi_a \in C_c^\infty(O)$ equal to one on the support projections. Periodize Ψ on a cube of side s containing every $\operatorname{supp}\chi_a$ in its interior. For each integer $p \geq 0$, integration by parts with $(1-\Delta)^p$ gives its Fourier coefficients

$$|c_{\mathbf{k}}| \leq C_{p,s,\Psi}(1+|\mathbf{k}|^2)^{-p}, \qquad \mathbf{k} = (k_1, \dots, k_n) \in (\mathbb{Z}^3)^n.$$

Multiplying the periodic series by $\prod_a \chi_a(y_a)$ gives, on all space,

$$\Psi = \sum_{\mathbf{k}} c_{\mathbf{k}} \bigotimes_{a=1}^{n} \left(\chi_a(y_a) e^{2\pi i k_a \cdot y_a / s} \right). \tag{12.xxx}$$

The q_m seminorm of each tensor factor product is $O((1+|\mathbf{k}|^2)^{m/2})$, so $p = m + 3n$ gives absolute Schwartz convergence, componentwise for tensor tests. Growing compact cutoffs prove density for arbitrary Schwartz tests.

Each term has Σ_O-measurable L^2 evaluation. Continuity of Q and closedness of $L^2(\Sigma_O, \lambda)$ therefore give

$$Q_\Psi \in L^2(\Sigma_O, \lambda) \qquad (\Psi \in C_c^\infty(O^n)). \tag{12.xxxi}$$

This extends to tests supported in the closed half-space $H = \{x : v \cdot x \geq c\}$, with $|v| = 1$. Choose a fixed compact cutoff χ equal to one near zero and put

$$\Psi_j(y) = \prod_{a=1}^{n} \chi(y_a/j)\, \Psi(y_1 - v/j, \dots, y_n - v/j).$$

Each $\Psi_j \in C_c^\infty(O^n)$ for $O = \{x : v \cdot x > c\}$, and Leibniz' rule gives

$$q_m(\Psi_j - \Psi) \leq C_{m,n,\chi} j^{-1} q_{m+1}(\Psi).$$

Consequently $Q_{\Psi_j} \to Q_\Psi$ in $L^2(\lambda)$, and

$$Q_\Psi \in L^2(\Sigma_O, \lambda) \qquad (\operatorname{supp}\Psi \subset H^n). \tag{12.xxxii}$$

12.4.2. *The five conditions.*

Proposition 12.3. Let λ be a Gaussian-dominated law of real tempered two-forms. Suppose it is invariant under Euclidean motions and reflection positive at every plane through the origin. Then its tensor Schwinger functions satisfy (E0′)–(E3) of Thm. A. If, for every $A, B \in L^2(\lambda)$,

$$\mathbb{E}_\lambda[\overline{A}\, U(a) B] \longrightarrow \overline{\mathbb{E}_\lambda A}\, \mathbb{E}_\lambda B \quad \text{as } |a| \longrightarrow \infty,$$

they also satisfy (E4). ∎

Proof. The Hermite bound gives (E0′), eq. 12.xxix gives (E1), and commutativity and tensor continuity give (E3).

For (E2), let r be the stated reflection and H the chosen closed half-space. By eq. 12.xxxii, $A = \sum_{k=0}^{M} Q_{\Psi_k}$ lies in its positive-side L^2 space, where $Q_{\Psi_0} = \Psi_0$. Character density and boundedness extend the reflection form to this space. eq. 12.xxviii–eq. 12.xxix give

$$\overline{\mathcal{R}_r Q_{\Psi_k}} = Q_{\Theta_k \Psi_k}, \qquad Q_{\Theta_k\Psi_k} Q_{\Psi_l} = Q_{\Theta_k \Psi_k \otimes \Psi_l} \quad \text{in } L^1.$$

Therefore

$$0 \leq \mathbb{E}_\lambda[(\overline{\mathcal{R}_r A})A] = \sum_{k,l=0}^{M} S_{k+l}(\Theta_k \Psi_k \otimes \Psi_l).$$

Finally, eq. 12.xxviii and translation covariance identify the expression in (E4) with $\mathbb{E}_\lambda[Q_\Psi U(a) Q_\Phi]$. Apply clustering to $A = \overline{Q_\Psi}$ and $B = Q_\Phi \in L^2$ to obtain $\mathbb{E}_\lambda Q_\Psi\, \mathbb{E}_\lambda Q_\Phi$. □

13. An interacting Osterwalder–Schrader theory

We complete the construction by satisfying the parameter conditions simultaneously and selecting a non-Gaussian component of the same limiting law. The structural properties hold on a full-measure set of components; the response estimates will give a positive-measure set with positive fourth cumulant.

13.1. **Choice of the construction constants.** We choose constants before the charge, mesh, volume and twist: local inverses before M, weakened kernels after M, then charge ceilings valid at every smaller charge.

For each fixed probe, put $C_f = 8C_U + 3C_U^2$, where C_U is the constant in eq. 11.cl.

Proposition 13.1. Fix $0 < \rho < 1$ and the standard chamber weights. The constants in the photon and fermion kernel estimates, small-field step, cover comparison and tuned flow can be chosen in the order described below, with positive charge ceilings $\bar{e}_0, \bar{e}_1$ independent of the mesh, volume, twist and reference mass. For every $0 < \bar{m} \le 1$, the physical ceiling e_G in eq. 13.x is positive. The terminal conditions eq. 8.xi hold on the top and reference tori throughout $0 < e \le e_G$. Fix a real probe $\beta \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$ with $b(dd^*\beta) > 0$. There is a positive $e_\beta \le e_G$ for which eq. 11.cxlvii holds with $\sigma > 0$ and $C_f e^\sigma \le b(dd^*\beta)/2$. ∎

13.1.1. *The geometric scales and the positive exponent. Proof.* Set $\bar{\varepsilon} = 1/200$, $c_X = 54$, $\kappa_0 = 54 \log 72$, $K_{\mathrm{pol}} = 72^{54}/36$, $c_{\mathrm{rb}} = 54 K_{\mathrm{pol}}$ and $c_{\mathrm{fl}} = 61 c_X K_{\mathrm{pol}}$. Choose

$$1 - \bar{\varepsilon} < \alpha < 1, \qquad \kappa \ge 7\kappa_0 + 7.$$

An odd integer $L \ge 5$ is required to satisfy

$$\begin{aligned} L^{1/8 - 3\bar{\varepsilon}/2} &\ge 2400, \qquad c_{\mathrm{fl}} L^{-3/8 + 6\bar{\varepsilon}} \le 1/16, \\ L(\kappa - 6\kappa_0 - 4) - 1 &\ge \kappa, \\ c_{\mathrm{rb}} L^3 \exp\bigl(-L(\kappa - \kappa_0 - 1) + \kappa\bigr) &\le 1/2. \end{aligned} \tag{13.i}$$

These are simultaneous lower-bound conditions on L. The powers $1/8 - 3\bar{\varepsilon}/2$ and $3/8 - 6\bar{\varepsilon}$ are positive, as are $\kappa - 6\kappa_0 - 4$ and $\kappa - \kappa_0 - 1$. The last expression tends to zero as $L \to \infty$. Thus every sufficiently large odd L satisfies eq. 13.i.

After L is fixed, the photon and fermion estimates fix a', C_V, γ_w and γ_P, with $a', C_V > 0$ and $\gamma_w, \gamma_P > 0$. Put $N_c = (2L+1)^3$ and $\widehat{\kappa} = N_c\kappa + \log(6c_X) > 0$. The photon and fermion constants have no dependence on M at this stage. Put $\alpha_0 = 1/54$, $\kappa' = \kappa_w + \kappa_0$ and

$$M_F = \max\{4L + 1,\ (108\pi C_V a')^2,\ 108\pi C_V a' \exp(27\kappa')\}.$$

Choose M to be a power of L satisfying

$$\begin{aligned} M &\ge L^2, \qquad M > 4L, \qquad \gamma_w M \ge 16, \\ M^{\alpha_0} &\ge 1 + \exp(\widehat{\kappa}), \qquad M \ge M_F, \qquad 804\alpha_0 \log M \le \gamma_P M. \end{aligned} \tag{13.ii}$$

All these conditions hold for every sufficiently large power of L. For the last one, $\log x \le \sqrt{x}$ on $x \ge 1$ gives the sufficient lower bound $M \ge (804\alpha_0/\gamma_P)^2$. The definition of M_F gives the two fermion conditions in eq. 4.xliv, because $27\alpha_0 = 1/2$. The last inequality gives eq. 3.xlv. The first four give the geometric separation, decay and parameter-radius conditions used in localization.

The exponents can be fixed independently of the charge. Take $r = 2$, $p_0 = 9$, $p = 19$ and $1 - \alpha < \varepsilon \le \bar{\varepsilon}$, and take $s = 1/2$. Then $p_0 > 3r + 2$ and $p > 2p_0$, while

$$\mathfrak{e} = s(1/4 - 9\varepsilon) - 12\varepsilon \ge 17/400.$$

For $N_* = 8L^3$ and $a_L = (L^3 - 1)/(L - 1)^3$,

$$\frac{2a_L}{N_*} < \frac{1}{4(L-1)^3} \le \frac{1}{256} < \frac{17}{400}.$$

It follows that $\sigma = \mathfrak{e} - 2a_L/N_* > 0$. In particular the positive exponent is fixed before the charge is chosen.

13.1.2. *Calibration of the primitive and terminal bounds.* The chamber coercivity, heavy-sector gap and local inverse construction fix a' and $\gamma^F > 0$ before M is chosen; see eq. 4.xxxii and eq. 4.xli. The photon construction likewise fixes C_P^{ph} and $\gamma_P > 0$ before M; its weakened families are considered only after eq. 3.xlv holds. The sums $c(\gamma) = \sum_{z\in\mathbb{Z}^3} \exp(-\gamma|z|_\infty)$ are finite for $\gamma > 0$. Thus the constants used to choose M in eq. 13.ii have already been fixed.

Keep $b = 1$, $b_- = 1 - L^{-1}$ and $b_+ = 1$ from the inverse construction; these are already fixed before M. The coarse resolvent and its auxiliary block matrix give

$$C_{\mathrm{By}} = (1 + 2\exp(1))\sqrt{2}\max(1, b_-^{-1}), \qquad C_\Gamma = \frac{2(1+2\exp(1))}{b_-} + 128\exp(1)(1+2\exp(1))^2 a'.$$

The latter is the constant in eq. 4.lx. Define

$$\begin{aligned} C_1 &= \max\{304\exp(1)\max(1,b)a'M^{27\alpha_0}, C_\Gamma M^{27\alpha_0}, \\ &\qquad 32\exp(1)\max(1,b^2)a'\exp(27\kappa')\}, \\ C^F &= 2\max\{C_1, C_{\mathrm{By}}, 304\exp(1)\max(1,b)a', C_\Gamma, \exp(1)b^2(32a')^2\}. \end{aligned} \tag{13.iii}$$

The factor two converts Euclidean spinor bounds to component bounds. The walk and coarse-family estimates eq. 4.xlv–eq. 4.xlvi and eq. 4.lix–eq. 4.lx are bounded by C^F in this convention. The last entry also bounds the compressed quadratic kernel. At level zero the normalized kernel has the same bound, and $M^{8\alpha_0} \le M^{27\alpha_0}$ bounds its diagonal field multiplier. The unweakened all-pair field bound eq. 11.lx is at most $608\exp(1)\max(1,b)a' \le C^F$; at zero its constant is at most four, also below C^F.

Take the common kernel constants

$$C^P = 116 C_P^{\mathrm{ph}} M^{402\alpha_0}, \qquad C_w^{(1)} = \max\{C^F, C^P, 4\exp(2\gamma_w)M^{27\alpha_0}\}, \qquad C_w = 64 K_{\mathrm{pol}} C_w^{(1)}.$$

These are precisely eq. 5.clii. The top systems and their projections therefore satisfy the same bounds at every required level and side. The determinant, photon-domain and localization constants are given in §§4.2.7, 3.1.8 and 5.3.1. In particular,

$$\begin{aligned} c_C &= 3L^{9/2}, \qquad C_W = 28 C_w^2 c_\gamma, \\ C_d &= 16 c_X C_w \max\{\tfrac{\pi}{2} b\exp(1)(1+2\exp(1)), 2 + 58\exp(1) + b\exp(1)\}. \end{aligned}$$

The kernel-sum constant c_γ is bounded by $2L^3\exp((2L+1)\gamma_w)c(\gamma_w)$; the local-gauge constant C_B depends only on $C_P^{\mathrm{ph}}, \gamma_P, L, M$. All these constants are finite after M is fixed.

For the terminal norms fix

$$\begin{aligned} C_0 &= \max\{C_P^{\mathrm{ph}} c_{\mathrm{bd}}(\gamma_P), C^F, 4b_+ + 16 C^F c(\gamma^F), \\ &\qquad \sqrt{226}\, C^F (24/\gamma^F)^3\}, \qquad c_{\mathrm{bd}}(\gamma) = 3\exp(3\gamma)c(\gamma). \end{aligned} \tag{13.iv}$$

Its last entry ensures $C_0^2 \ge C_K$ and uses only C^F, γ^F, which were fixed before C_0. The photon row sums, both field-map row norms and the terminal Dirac form in §11.3.4 are then bounded by this C_0. In particular $C_0 \ge 1$. The polymer sum has $1 \le c_\kappa \le K_{\mathrm{pol}}$. The differentiated-response constants are now fixed by the terminal inverse and field-map estimates: $\mathcal{S} = C^F(24/\gamma^F)^3$, $C_K = 226\mathcal{S}^2$, $C_K' = 1328\mathcal{S}^2$ and $C_H = 203M(C^F)^2$. The real covariance coefficients g, h_G, g_1, α_G and its rate ϑ_0 are defined with eq. 11.lxxi. The direct coarse twist comparison eq. 8.x now fixes C_Σ, γ_Σ from $C_K', g_1, c_\tau, C_6, \kappa_{(0)}, \mu_-$. They are independent of charge and mesh; the mass lower bound is the fixed $\mu_- = L^{-3}$ on the two reference tori. The insertion and mixed-logarithm coefficients are $C_V^{\mathrm{ins}} = 10^3 C_H^2 (24/\gamma^F)^{12} K_{\mathrm{pol}}$ and C_W^{mix} in eq. 11.cxviii. Their inverse-decay rates are positive and their coefficients finite because $b_- > 0$, $\mu_- = L^{-3} > 0$ and $\gamma^F > 0$. They depend on L, M and the

fixed chamber data; they have no dependence on the later charge or probe. The probe-dependent constants will be chosen last.

13.1.3. *Uniform running-charge bounds.* The preceding choices fix every coefficient of the small-charge conditions. For $n \geq 1$, $A_i, d_i \geq 0$, $\lambda_i > 0$ and $b > 0$, set

$$V(A,d,\lambda;b) = \max_i \left\{ 1, \left(\frac{2d_i}{\lambda_i}\right)^2, \frac{2}{\lambda_i}\log^+\frac{nA_i}{b} \right\}, \tag{13.v}$$

where $\log^+ 0 = 0$. For $v \geq V$, the inequality $\log v \leq \sqrt{v}$ gives $d_i \log v \leq \lambda_i v/2$; the final threshold term therefore gives

$$\sum_i A_i v^{d_i} e^{-\lambda_i v} \leq \sum_i A_i e^{-\lambda_i v/2} \leq b. \tag{13.vi}$$

The kernel domains and the determinant, adjustment and nonlinear step estimates require eq. 3.l, eq. 5.vi, eq. 5.xxxv, eq. 5.xl, eq. 4.lxxviii and eq. 5.c. Their charge conditions are obtained from the following finite families for $v = -\log e_k$. Write $T(A,d,\lambda) = Av^d \exp(-\lambda v)$. The polynomial-exponential conditions are

$$\begin{aligned}
T(1,p_0,\varepsilon/4) &\leq 1, & T(1,0,5\varepsilon/2) &\leq \tfrac{1}{2},\\
T(c_{\mathrm{fl}},0,\varepsilon) &\leq 1, & T(3M,0,\varepsilon) &\leq 1,\\
T(2C_w^2c_\gamma^2c_C,0,\varepsilon/4) &\leq 1, & T(16c_Xc_\gamma C_wC_T^2M^3,0,\varepsilon/2) &\leq 1,\\
T(96M^3,0,1/4) &\leq 1, & T(32C_wC_T,0,3\varepsilon/2) &\leq 1,\\
T(88M^3,0,1/4-6\varepsilon) &\leq 1, & T(112C_w^2c_\gamma,0,2\varepsilon) &\leq 3,\\
T(c_S,0,1/4-5\varepsilon) &\leq 1, & T(c_SL^3,0,\varepsilon) &\leq 1,\\
T(36c_X^2K_{\mathrm{pol}}^2,0,1/4-5\varepsilon) &\leq 1, & T(1,0,\varepsilon) &\leq \tfrac{1}{4},\\
T(3M,0,1/4+\varepsilon)+T(3M,0,2\varepsilon) &\leq 1,\\
T(1,0,1/4)+T(1,0,\varepsilon) &\leq 1,\\
T(160L^{1+3\varepsilon/2}M,0,\varepsilon) &\leq 1, & T(90c_XLM,0,\varepsilon) &\leq 1,\\
T(9ML^\varepsilon,0,2\varepsilon) &\leq 1,\\
T(2M,1,\varepsilon)+T(32M,0,\varepsilon) &\leq 1,\\
T(8C_B,p,3/4-3\varepsilon) &\leq 1, & T(8,0,3/4-7\varepsilon/2) &\leq 1,\\
T(6L,0,\varepsilon) &\leq 1, & T(4C_Va',0,\varepsilon) &\leq 1,\\
T(4,0,1/4+2\varepsilon)+T(2,0,3\varepsilon) &\leq 1,\\
T(8c_{\mathrm{rb}}L^3C_dM^3,0,2\varepsilon) &\leq 1.
\end{aligned} \tag{13.vii}$$

Here $c_S = 10^5 c_X^5 K_{\mathrm{pol}}^5 \exp(6\kappa)$. The inequality with $96M^3$ controls the first step; the one with $C_T^2M^3$ controls the positive-level fluctuation substitution. The conditions with c_S and $36c_X^2K_{\mathrm{pol}}^2$ control the connected logarithm and its scaling. The domain conditions occur with target $1/4$, with rates $(1/4,\varepsilon)$, with coefficients $8C_B, 8$, and with coefficients $4, 2$. The remaining displayed charge hypotheses above are individual inequalities of this inventory; eq. 5.cxxx follows from the condition containing c_S.

Every decay rate in the inventory is positive for $0 < \varepsilon \leq 1/200$. The only additional running-charge restrictions in this group are

$$v^{2p_0}/2 - 7v \geq \log 12, \qquad v^2 \geq 10L, \qquad v \geq 9.$$

They hold respectively for $v \geq 15$, $v \geq \sqrt{10L}$ and $v \geq 9$. Taking the maximum of these thresholds and the finitely many values in eq. 13.v gives a finite V_0. Choose a running-charge ceiling $0 < \bar{e}_0 <$

$\exp(-V_0)$ so that every such condition holds for every $v \geq -\log \bar{e}_0$, equivalently for every running charge in $(0, \bar{e}_0]$. The set of permitted ceilings is downward closed.

The terminal-scale conditions require a second ceiling $\bar{e}_1 \leq \min(\bar{e}_0, 1/3)$. Write $\ell(v) = LMv^r$. They follow from

$$\begin{aligned}
(c_{\mathrm{fl}} + 2)\exp(-\varepsilon v) &\leq \tfrac{1}{16}(1 - L^{-\varepsilon/2}),\\
8C_0^2\ell(v)^6(b_-^{-1} + L\ell(v)) &\leq \exp(\varepsilon v/2),\\
4(2C_0 + c_\kappa + 5c_{\mathrm{fl}} + 6)\ell(v)^3 &\leq \exp(\varepsilon v),\\
\exp(-(1/4 - 9\varepsilon)v) &\leq \tfrac{1}{8},\\
4C_0\ell(v)^2 v^p &< \exp(\varepsilon v),\\
12\ell(v)^3\left(\exp(-v^{p_0}/2) + \exp(-v^{2p}/(2C_0^2))\right) &\leq \exp(-v),\\
C_\Sigma\ell(v)^2\exp(-\gamma_\Sigma M v^r) &\leq \exp(-v),\\
L^2\exp(-2v)\ell(v) &\leq 1,\\
6N_* M v^r\exp(-v)(v\exp(-v) + \tfrac{1}{2}) &\leq \pi^2.
\end{aligned} \tag{13.viii}$$

These are the scalar hypotheses of eq. 6.ii, eq. 8.xi and eq. 11.cxlii, after substituting eq. 6.v. Every polynomial-exponential rate here is positive for $0 < \varepsilon \leq 1/200$. The reference flux condition expands as

$$6N_* M v^{r+1}\exp(-2v) + 3N_* M v^r\exp(-v) \leq \pi^2.$$

The superlinear tails reduce to these bounds: for $v \geq \max(4, 4C_0^2)$,

$$v - \tfrac{1}{2}v^{p_0} \leq -v, \qquad v - \frac{v^{2p}}{2C_0^2} \leq -v.$$

Hence

$$12\ell(v)^3\left[\exp\left(v - \tfrac{1}{2}v^{p_0}\right) + \exp\left(v - \frac{v^{2p}}{2C_0^2}\right)\right] \leq 24(LM)^3 v^{3r}\exp(-v).$$

For the twist tail put $A = \log^+ C_\Sigma + 2\log(LM)$ and $B = 2r + 1$. For $v \geq 1$, $\log v \leq v$ and $v^r \geq v^2$ give

$$\log\left(C_\Sigma(LMv^r)^2 e^{v - \gamma_\Sigma M v^r}\right) \leq A + Bv - \gamma_\Sigma M v^2 \leq 0$$

whenever $v \geq \max\{1, 2B/(\gamma_\Sigma M), \sqrt{2A/(\gamma_\Sigma M)}\}$. Put $\lg = \tfrac{1}{2}\log L$. The scale-ratio condition follows for $v \geq 2r$:

$$\left(1 + \frac{3\lg}{v}\right)^r \leq \exp\left(\frac{3r\lg}{v}\right) \leq L^{3/4} < L. \tag{13.ix}$$

Together with $v \geq 8$, $v \geq 31L + 17$ and $p_0 v^{p_0 - 1} \geq 28$, these bounds give a finite threshold V_1. Its maximum includes $2r$. Let $\mathcal{E}_1$ consist of the numbers $e' \in (0, \min(\bar{e}_0, 1/3)]$ for which the geometric conditions and eq. 13.viii hold for every $v \geq -\log e'$. It contains $\min(\bar{e}_0, 1/3, \exp(-V_1)) > 0$ and is downward closed. Fix

$$\bar{e}_1 = \tfrac{1}{2}\sup \mathcal{E}_1 > 0.$$

There is an element of $\mathcal{E}_1$ larger than $\bar{e}_1$; downward closure therefore gives $\bar{e}_1 \in \mathcal{E}_1$. Thus all terminal-scale conditions hold for every $v \geq -\log \bar{e}_1$.

13.1.4. *Physical and probe-dependent charge intervals.* Fix the reference mass and hence $h \geq 1$. Set

$$e_G = \begin{cases} \bar{e}_1, & h = 1, \\ \min\{\bar{e}_1/\sqrt{h},\, 1/(\sqrt{h}\log h)\}, & h > 1. \end{cases} \tag{13.x}$$

This is positive. For $0 < e \leq e_G$, every running charge up to level K is at most $\bar{e}_1$, and $e\sqrt{h}\log h \leq 1$. The scale proposition therefore applies on the top and reference tori. The terminal inventory gives eq. 8.xi at their respective stops, and eq. 11.cxxxv holds on the reference torus. The common pair of Thm. K is defined for $N \geq N_{\min}(R_{\mathrm{top}}, e)$. At lower cutoffs only its mass shift is assigned zero. For $N \geq N_{\mathrm{ref}}(e)$ the same pair is controlled on the reference torus. Its mass shift is real, independent of the cutoff volume, and has absolute value less than $1/8$.

Fix a compact real probe β and choose its finite constants in eq. 11.li after the differentiated inverse and interaction constants. Prop. 11.10 gives $e_\sharp(\beta) > 0$ satisfying all factor, one-cell, tree, jet and placement conditions for every $0 < e \leq e_\sharp(\beta)$, uniformly in $N \geq N_{\mathrm{ref}}(e)$ and admitted placements.

For $0 < e \leq \min(e_G, e_\sharp(\beta))$, eq. 8.xxiii and eq. 11.cxxviii supply the two inputs in eq. 11.cxliv. The probe placement inequalities imply eq. 11.vi for $\eta \leq 1$. The constants J, B_2, B_4, C_U in the free-coefficient and transfer bounds in §11 are finite and independent of e, N, T and placement. They depend only on the fixed construction and probe. For each fixed charge the coefficients are bounded uniformly in the cutoff; the transfer estimates hold for $N \geq N_{\mathrm{uv}}(e)$.

Suppose now that $b(dd^*\beta) > 0$. With the constant C_f defined before Prop. 13.1, choose

$$0 < e_\beta \leq \min\{e_G, e_\sharp(\beta), 1, (b(dd^*\beta)/(2C_f))^{1/\sigma}\}. \tag{13.xi}$$

The minimum is positive, since $\sigma > 0$ and $C_f > 0$. Prop. 11.12 holds throughout $(0, e_\beta]$ and gives the cutoff hypotheses of Prop. 13.3. In particular $C_f e^\sigma \leq b(dd^*\beta)/2$. The construction constants were chosen before h and the probe; the physical charge interval precedes the probe-dependent restriction. □

13.2. **A non-Gaussian component.** The parent law is a mixture of ergodic laws. Even when each component is a centered Gaussian, variation of its variance can give the mixture a positive fourth cumulant. We therefore control both the fourth-order observable and products of the quadratic observable at separated translates. Averaging the latter bounds the translation-invariant part of the quadratic observable, which measures variation of the conditional second moment. The identity below then gives a positive average of the component fourth cumulants.

Lemma 13.2. Let μ be a probability measure on a standard Borel space with a measure-preserving translation action $U(y)$, and let P be conditional expectation onto the sigma algebra invariant under all translations. Let $X \in L^4(\mu)$ be real-valued and suppose that its conditional mean is zero. For $e > 0$ and real v, c_2, c_4, set

$$G_2 = e^{-2}(X^2 - v) - c_2, \qquad G_4 = e^{-4}(X^4 - 6vX^2 + 3v^2) - c_4.$$

Let $u_2, u_4 \geq 0$. Assume that $|\mu(G_4)| \leq u_4$ and that $|\mu(U(y)G_2\,U(z)G_2)| \leq u_2^2$ for distinct points y, z of an infinite translation lattice. In the conditional-law decomposition $\mu = \int \mu_\xi \,\mathrm{d}\nu(\xi)$,

$$e^{-4}\int \kappa_4^{\mu_\xi}(X)\,\mathrm{d}\nu(\xi) \geq c_4 - 3c_2^2 - u_4 - 6|c_2|u_2 - 3u_2^2. \tag{13.xii}$$

If the right side is positive, a set of components of positive ν-measure has positive fourth cumulant. ■

Proof. Average G_2 over n distinct lattice translates, obtaining Z_n. Conditional expectation commutes with these translations and their invariant part is fixed, so $PZ_n = PG_2$. Conditional Jensen and expansion of the square give

$$\|PG_2\|_2^2 \leq \|Z_n\|_2^2 \leq n^{-1}\|G_2\|_2^2 + (1 - n^{-1})u_2^2.$$

Letting n increase gives $\|PG_2\|_2 \le u_2$, and hence $|\mu(G_2)| \le u_2$. Put $d_m = PG_m$. Centering of the conditional laws and direct expansion give

$$e^{-4}\kappa_4^{\mu_\xi}(X) = c_4 + d_4 - 3(c_2 + d_2)^2. \tag{13.xiii}$$

Here $d_2 \in L^2$ and $d_4 \in L^1$, so every term can be integrated. The bounds just proved and $|\mu(G_4)| \le u_4$ give eq. 13.xii. A positive integral has a positive-measure positive set. □

We apply the lemma to the continuum limit of the transferred estimates. Fix $e > 0$ and the positive probe f constructed in §10.1. Along a joint cutoff sequence converging to μ, write $f_i = f^{\eta_i}$ and use the reference numbers $v_i, c_{2,i}, c_{4,i}, u_{2,i}, u_{4,i}$. The estimates in Prop. 11.12 imply

$$f_i \to f \text{ in } L^2, \qquad v_i \to \|f\|_2^2, \qquad \sup_i(|c_{2,i}| + |c_{4,i}|) < \infty. \tag{13.xiv}$$

Extract a subsequence on which these four bounded reference coefficients $c_{2,i}, c_{4,i}, u_{2,i}, u_{4,i}$ converge, without changing μ. Their limits satisfy

$$u_4 \le Ce^\sigma, \qquad (1 + |c_2|)u_2 \le Ce^\sigma, \qquad |c_4 - 3c_2^2 - b(f)| \le Ce^\sigma. \tag{13.xv}$$

Let $\lambda_0 = R^*(e)/L$, which is independent of the spacing. For $G_m = W_m(F(f); \|f\|_2^2, c_m)$, the transferred estimates give

$$|\mu(G_4)| \le u_4, \qquad |\mu(U(y)G_2\,U(z)G_2)| \le u_2^2 \quad (y \ne z,\ y, z \in \lambda_0\mathbb{Z}^3). \tag{13.xvi}$$

Here is the limit passage. Each fixed pair of translated supports lies in one period cube and occupies incongruent cells for all sufficiently large cutoffs. Expand the corresponding Wick product into monomials. For a monomial of degree $J > 0$, replacing its varying tests by the limiting tests changes expectation by at most $Jc_J B^{J-1}\|f_i - f\|_2$, where $B = \sup_i \max(\|f_i\|_2, \|f\|_2)$. This follows by telescoping and the product-moment bound from Gaussian domination. It tends to zero. Fixed-test moments converge by §12.2, and the polynomial coefficients converge. This proves eq. 13.xvi for every fixed pair and for the one-point fourth-order observable.

Proposition 13.3. Suppose the components of μ satisfy Gaussian domination, and let eq. 13.xv–eq. 13.xvi hold. If $b(f) > 0$ and

$$(8C + 3C^2)e^\sigma \le b(f)/2, \qquad 0 < e \le 1,$$

then a positive-measure set of translation-ergodic components has positive fourth cumulant at f. ■

Proof. Gaussian domination centers every component and supplies all moments. Apply the lemma with $X = F(f)$. Its lower bound is at least

$$b(f) - 8Ce^\sigma - 3C^2e^{2\sigma} \ge b(f)/2 > 0.$$

The stated conclusion follows. A centered Gaussian variable has zero fourth cumulant. □

13.3. **Completion of the construction.**

Proof of Thm. A. Use the fixed values $\rho = 1/2$ and $\bar{m} = 1$. Choose all construction constants in the order established in §13.1; then $h = 1$ and $K = N$. Choose one positive compact probe by Prop. 10.1. The charge interval eq. 13.xi is nonempty. Fix any one charge in it. All subsequent cutoff estimates are used for $N \ge N_{\mathrm{uv}}(e)$; omitting finitely many cutoffs does not change a continuum limit. Thm. K gives the required volume-independent mass shifts, with the bound eq. 6.xix. The terminal partition and response estimates eq. 8.xxiii and eq. 11.cxxviii, together with the free-coefficient estimate eq. 11.cxlvi, verify all hypotheses of Prop. 11.12.

Uniform tightness gives a volume-first parent μ; use the joint cutoff sequence and moment convergence of Prop. 12.2. By that proposition and §12.3, almost every component is centered and translation ergodic, is supported on closed fields, and satisfies Gaussian domination and the nine reflection forms. Thm. O gives their full Euclidean covariance and all-plane reflection positivity. Prop. C.1 gives clustering, and Prop. 12.3 gives precisely (E0′)–(E4).

For this same parent and sequence, §13.2 and Prop. 13.3, with $C = C_U$ and the chosen charge, give a positive-measure set of positive fourth cumulants. Its intersection with the structural full-measure set contains the required λ. □

Appendix A. Consequences of extra reflection positivity

A.1. **Reflection-positive reconstruction and the spectral cone.** The spectral cone gives clustering in all directions. Invariant weights pass reflection positivity to the ergodic components.

A.1.1. *The reflection-positive Hilbert space.* We recall the reflected Hilbert-space construction of [11, §4.1]. Its positive form puts time translations and transverse translations on the same space, where their generators can be compared.

For an open set O, let Σ_O be generated by $F(f)$ with real $f \in C_c^\infty(O;\Lambda^2)$. Let λ be a translation-invariant law satisfying eq. 12.i, and let r_j reflect in $x_j = 0$. Suppose that λ is invariant and reflection positive at this plane. For a Euclidean motion g write $\widehat{g}A(F) = A(g^{-1}F)$ and $U(a) = \widehat{\tau}_a$. For a reflection r put $\mathcal{R}_r = \widehat{r}$. Write $\mathcal{R}_j = \mathcal{R}_{r_j}$ and $\Theta_j A = \overline{\mathcal{R}_j A}$. On $\mathcal{E}_j^+ = L^2(\Sigma_{\{x_j>0\}}, \lambda)$ the form

$$b_j(A,B) = \mathbb{E}_\lambda[(\Theta_j A)B]$$

is positive semidefinite and $b_j(A,A) \le \|A\|^2_{L^2(\lambda)}$. Let $\mathcal{H}_j$ be its quotient completion, $[A]_j$ the quotient map, and $\Omega_j = [1]_j$. Reflection positivity stated on the characters $e^{iF(f)}$, with real $f \in C_c^\infty(\{x_j > 0\};\Lambda^2)$, extends to this form: their linear span is dense in the positive-side L^2 space, and the form is bounded by the L^2 norm.

Proposition A.1. There are a non-negative self-adjoint H_j and a strongly continuous unitary representation V_j of $e_j^\perp$ on $\mathcal{H}_j$ such that

$$e^{-tH_j} V_j(y)[A]_j = [U(te_j + y)A]_j, \qquad t \ge 0, \quad y \perp e_j. \tag{A.i}$$

The two representations commute, and $H_j\Omega_j = 0$. ■

Proof. The construction just recalled uses only translations, reflection invariance and positivity at this axis, and the moment growth implied here by eq. 12.i; no other mirror or rotation covariance is needed. Its form and quotient are exactly b_j and $\mathcal{H}_j$ above, and its positive-time and parallel translations are our $U(te_j + y)$. Its nonnegative generator and commuting transverse unitary groups therefore give eq. A.i, including strong continuity and the fixed constant vector. □

Write $V_j(se_i) = e^{is\mathsf{P}_i}$ for $i \ne j$. The joint spectral measure E_j of H_j and these commuting momenta lies on $[0,\infty) \times e_j^\perp$. For $\psi \in \mathcal{H}_j$ let $\varrho_\psi(A) = \langle \psi, \mathsf{E}_j(A)\psi\rangle$. Then

$$\langle \psi, e^{-tH_j} V_j(y)\psi\rangle = \int e^{-tE + ip\cdot y}\, \mathrm{d}\varrho_\psi(E,p). \tag{A.ii}$$

A.1.2. *A diagonal mirror.*

Proposition A.2. Let $i \ne j$. Assume eq. 12.i and suppose in addition that λ is invariant and reflection positive at $x_j + x_i = 0$. Then

$$\mathsf{E}_j\{(E,p) : E < |p_i|\} = 0. \tag{A.iii}$$

Thus $H_j \ge |\mathsf{P}_i|$ in the joint spectral calculus. ■

Proof. Apply the proof of [11, Prop. 4.1] in the coordinates $(t,x) = (x_j, x_i)$, keeping the third coordinate smeared. That proof uses only translations, reflection invariance and positivity at $x_j = 0$ and $x_j + x_i = 0$, and moment growth. The latter follows from eq. 12.i, which implies the growth hypothesis of that paper with $C = 1/2$. The generators there translate by the same coordinate distances as H_j and P_i here, so its spectral conclusion is exactly eq. A.iii. □

Applying this result to the nine-mirror laws gives $H_j \geq |\mathsf{P}_i|$ for every ordered pair $i \neq j$. This coordinate cone identifies the invariant projections used in clustering.

A.1.3. *Invariant observables and reflection.* Let P_j be the orthogonal projection in $L^2(\lambda)$ onto the observables fixed by all $U(te_j)$, and let P project onto those fixed by all translations. These are the corresponding conditional expectations. The averages $T^{-1}\int_0^T U(te_j)\,\mathrm{d}t$ converge strongly to P_j. The projections commute with one another and with every translation.

Translate compact character approximants of an e_j-invariant observable into either chosen half-space. Their errors are unchanged, so closedness of its L^2 subspace places the observable there.

On this positive-side invariant subspace, reflection positivity makes the self-adjoint involution $\mathcal{R}_j$ positive. Its spectrum is therefore $\{1\}$, giving

$$\mathcal{R}_j A = A \qquad (A \in \operatorname{ran} P_j). \tag{A.iv}$$

Thus the quotient map is an isometry there. Average

$$\langle [A]_j, e^{-tH_j}[B]_j\rangle = \langle \mathcal{R}_j A, U(te_j)B\rangle_{L^2(\lambda)}$$

over $0 \leq t \leq T$. Spectral dominated convergence on the left and the mean ergodic theorem on the right give

$$P_0[B]_j = [P_j B]_j, \qquad P_0 = \mathbf{1}_{\{0\}}(H_j). \tag{A.v}$$

For a law satisfying eq. 12.i and invariant and reflection positive at the nine mirrors, the cone forces the joint spectral measure at zero energy to be concentrated at zero momentum. Hence $V_j(y)$ fixes $\ker H_j$. If $A \in \operatorname{ran} P_j$ and $y \perp e_j$, then

$$[U(y)A - A]_j = 0.$$

The difference is again in $\operatorname{ran} P_j$, so the isometry gives $U(y)A = A$. Consequently

$$P_1 = P_2 = P_3 = P. \tag{A.vi}$$

Appendix B. Rotation invariance from reflection positivity

B.1. **Rotational covariance.** The result of [11, Thm. A] gives covariance when the test supports are pairwise disjoint. To include overlapping supports, we apply the next lemma to the difference between a moment and its rotation. Under the Hodge identification $b = *f = (f_{23}, -f_{13}, f_{12})$, the Gaussian bound makes this difference a bounded multilinear form on vector-valued L^2, and closedness makes it vanish whenever one argument is a gradient. These two properties force the difference to vanish also for overlapping supports.

Proposition B.1. Let $N \geq 1$. For an open $U \subset \mathbb{R}^3$, let $L^2_U = L^2(U;\mathbb{C}^3)$, extended by zero. Let $D : (L^2_U)^N \to \mathbb{C}$ be bounded and multilinear. Suppose D vanishes when its entries are supported in pairwise disjoint compact subsets of U, and whenever one entry is ∇u with $u \in C_c^\infty(U)$. Then $D = 0$. ∎

Proof. The first hypothesis need only be checked on smooth compact fields: disjoint compact supports have disjoint relatively compact open neighborhoods in U. Extend each field by zero and mollify with radius tending to zero inside its neighborhood; convergence in L^2 and boundedness of D give the stated hypothesis, without requiring the compact supports to have interior.

Induct on N, simultaneously for all open U. For $N = 1$, compactly supported fields are dense. For $N = 2$, inner regularity extends locality from disjoint compact sets to disjoint Borel sets. Taking a set A and its complement gives $D(\mathbf{1}_A f, g) = D(f, \mathbf{1}_A g)$. Uniform approximation by simple functions extends this identity to every bounded multiplier h. Hence

$$D(h\nabla u, g) = D(\nabla u, hg) = 0.$$

For compact $K \subset U$, choose $\chi \in C_c^\infty(U)$ equal to one near K and put $u_j(x) = x_j\chi(x)$. If h is bounded and supported in K, then $h\nabla u_j = he_j$. Such fields span a dense subspace of L^2_U, so $D = 0$.

Suppose $N \geq 3$ and the result holds at all smaller orders. First, if one entry is supported in compact $K \subset U$ and all others in $U \setminus K$, fix that entry. The form on the remaining slots satisfies both hypotheses on the open set $U \setminus K$ and is zero. Second, partition the slots into nonempty groups I, J, with all J entries supported in compact $K_J \subset U$ and all I entries in $U \setminus K_J$. Fix the J entries. The resulting form inherits the gradient condition. If its entries have pairwise disjoint compact supports, the first consequence, applied to any one of them, gives zero. Induction on the I slots therefore makes this form zero.

By density place all entries in one compact $K_0 \subset U$. Partition $\mathbb{R}^3$ into half-open cubes Q of side δ and expand by multilinearity; only finitely many cubes meet K_0. A tuple of unequal cubes contributes zero: replace each cube by its concentric closed cube of side $\delta - 2\epsilon$ and apply the second consequence, grouping slots with the same first cube against the others. The fixed group's common support is the compact intersection of K_0 with the union of its closed cubes. Send $\epsilon \downarrow 0$ in L^2. Only equal-cube terms remain, and

$$|D(f_1, \dots, f_N)| \leq \|D\| \max_Q \|\mathbf{1}_Q f_3\|_2 \, \|f_1\|_2 \|f_2\|_2 \prod_{j=4}^{N} \|f_j\|_2.$$

Here the first two cube norms are summed by Cauchy–Schwarz. The maximum tends to zero as $\delta \downarrow 0$, by absolute continuity of the integral of $|f_3|^2$ over sets of volume δ^3. Hence $D = 0$. □

Theorem O. Let μ be a Gaussian-dominated law of closed tempered two-forms. Suppose it is invariant under translations and the cubic group, and reflection positive at the nine planes $x_j = 0$ and $x_j \pm x_i = 0$, $i < j$. Then it is invariant under all Euclidean motions and reflection positive at every plane, on the full L^2 space of either open side. ■□

Proof. We check the hypotheses and conventions of [11, Thm. A]. The inclusion $H_{-4} \hookrightarrow \mathcal{S}'$ is continuous for the weak distribution topology by §12.1, so our law gives the required Borel probability measure. The coefficient space $\Lambda^2\mathbb{R}^3$ is real orthogonal. A rotation in the $(1,2)$ plane fixes $e_1 \wedge e_2$ and rotates the span of $e_1 \wedge e_3, e_2 \wedge e_3$; hence its weights are $0, 1, -1$. Orthogonal conjugation gives the same allowed weights in every plane. The Gaussian bound implies its exponential regularity condition with $C = 1/2$, since

$$\mathbb{E}_\mu e^{F(f)} \leq e^{\|f\|_2^2/2} \leq e^{(\|f\|_1 + \|f\|_2^2)/2} \qquad (f \text{ real Schwartz}).$$

Translations and the nine reflection forms are among our hypotheses. RPM1's fiber pullback $\rho(Q) = \Lambda^2 Q^{\mathsf{T}}$ makes its test pullback our Q^{-1} action; these agree for a reflection. Its characteristic reflection matrix is the transpose of ours, which preserves positive semidefiniteness. Compact L^2 approximation extends these forms to all the Schwartz tests in its statement. Thus RPM1 gives covariance for every moment on smooth tests with pairwise disjoint spatial supports, complexified by multilinearity.

Use the Hodge isometry $*f = (f_{23}, -f_{13}, f_{12})$ and set $\mathbb{B}(b) = F(*^{-1}b)$. The induced action is

$$(Q \cdot b)(x) = \det(Q) Q b(Q^{-1}x), \qquad Q \in O(3).$$

For fixed Q and $N \geq 1$, Prop. 12.1 makes

$$D_Q(b_1, \dots, b_N) = \mathbb{E}_\mu \prod_j \mathbb{B}(Q \cdot b_j) - \mathbb{E}_\mu \prod_j \mathbb{B}(b_j)$$

a bounded multilinear form on complex vector-valued L^2. The just-proved disjoint-support covariance gives its first hypothesis in Prop. B.1 with $U = \mathbb{R}^3$. Closedness gives $\mathbb{B}(\nabla u) = 0$, and $Q \cdot \nabla u = \det(Q) \nabla(u \circ Q^{-1})$, so its gradient hypothesis holds as well. The proposition yields $D_Q = 0$ on all L^2 entries. Density of elementary tensors and Schwartz continuity therefore give covariance on every tensor test, including coincident supports.

For real Schwartz f, $e^{|F(f)|} \in L^1$ justifies termwise expectation of the characteristic Taylor series. Equal moments identify characteristic functionals; finite-dimensional Fourier uniqueness and cylinder generation give $Q_*\mu = \mu$. Together with translations this proves Euclidean invariance. The unitary action of a Euclidean motion carries the axis half-space spaces and reflection to those of any chosen plane and side, transporting its nonnegative form to the full corresponding L^2 space. □

Appendix C. Clustering from reflection positivity

Positive-time translation is e^{-tH_j} in the reflected Hilbert space. For each vector, it approaches the zero-energy part as $t \to \infty$, and commuting transverse unitary translations do not change the norm of the remainder. The diagonal cone identifies the zero-energy classes with conditioning on all translation-invariant observables. The estimate is uniform in the transverse displacement. Choosing the coordinate with largest absolute displacement gives convergence as $|a| \to \infty$. We use the reflection action and projections of §A.1.

C.1. **Correlations in all directions.**

Proposition C.1. Suppose λ satisfies eq. 12.i, is translation invariant, and is invariant and reflection positive at the nine mirrors. For all $A, B \in L^2(\lambda)$,

$$\langle A, U(a)B\rangle \longrightarrow \langle A, PB\rangle \quad \text{as } |a| \longrightarrow \infty. \tag{C.i}$$

If λ is translation ergodic, the limit is $\overline{\mathbb{E}_\lambda A}\,\mathbb{E}_\lambda B$. ■

Proof. First fix j. For $A = \mathcal{R}_j A'$ and $A', B \in \mathcal{E}_j^+$, eq. A.v gives, for $t \geq 0$ and $y \perp e_j$,

$$\begin{aligned} |\langle A, U(te_j + y)(1 - P_j)B\rangle| &= |\langle [A']_j, V_j(y)e^{-tH_j}(1 - P_0)[B]_j\rangle| \\ &\leq \|[A']_j\|\, \|e^{-tH_j}(1 - P_0)[B]_j\|. \end{aligned}$$

The last norm tends to zero by spectral dominated convergence; the bound is uniform in y. Approximate general A, B by local character sums. Translate their supports into opposite half-spaces, at a cost of replacing t by $t - 2r$ for one fixed r. The correlation error is bounded uniformly in t, y by $\|A - A_1\|_2\|B\|_2 + \|A_1\|_2\|B - B_1\|_2$. The adjoint identity gives the same conclusion as $t \to -\infty$. Thus

$$\lim_{|t|\to\infty} \sup_{y \perp e_j} |\langle A, U(te_j + y)(1 - P_j)B\rangle| = 0.$$

Since $P_j = P$, the uniform bound applies along a subsequence with one diverging coordinate, while $U(a)PB = PB$. Every subsequence has such a further subsequence, proving eq. C.i. Under ergodicity, P is the projection onto constants. □

Appendix D. Polymer geometry

Let L be odd, M a power of L, and ℓ a multiple of LM with $\ell \geq 13LM$. The M-cubes have centers in $M\mathbb{Z}^3/\ell\mathbb{Z}^3$. They partition the lattice sites; their closed continuum versions may meet along faces and edges. A polymer is a nonempty union of M-cubes connected through faces. Denote its number of cubes by $|X|_M$. A net in the continuum torus is a connected finite union of images of specified closed straight segments in $\mathbb{R}^3$, a point being allowed as a segment. Its length is the sum of the Euclidean lengths of those segments. Set

$$M d_M(X) = \inf\{\text{length}(N) : N \text{ is a net meeting every closed cube of } X\},$$

where the infimum also extends over the finite segment representations. The same definition is used for finite face-connected unions of M-cubes in $\mathbb{R}^3$, with nets in $\mathbb{R}^3$. Distances are sup distances.

D.1. **Size, diameter, and reblocking.** Put

$$c_X = 54, \qquad \kappa_0 = 54 \log 72, \qquad K_{\text{pol}} = e^{\kappa_0}/36, \qquad c_{\text{rb}} = c_X K_{\text{pol}}.$$

The following estimates are uniform in M and ℓ:

$$d_M(X) \leq |X|_M - 1, \qquad |X|_M \leq c_X(d_M(X) + 1), \tag{D.i}$$

$$d(p, q) \leq M(d_M(X) + 2) \qquad (p, q \text{ in the closed cubes of } X), \tag{D.ii}$$

$$\sum_{X \supset \square} e^{-\kappa_0 d_M(X)} \leq K_{\text{pol}}, \qquad \sum_{X \subset Y} e^{-\kappa_0 d_M(X)} \leq K_{\text{pol}}|Y|_M. \tag{D.iii}$$

Here Y is any union of M-cubes. If ℓ/L satisfies the same side conditions stated above, let $\bar{X}^L$ be the union of LM-cubes containing cubes of X. There is a unique polymer Y on the rescaled torus with $\bar{X}^L = LY$, and

$$Ld_M(Y) \le d_M(X), \tag{D.iv}$$

$$\sum_{X:\bar{X}^L=LY} e^{-\kappa' d_M(X)} \le c_{\mathrm{rb}} L^3 e^{-L(\kappa'-\kappa_0-1)d_M(Y)} \quad (\kappa' \ge \kappa_0). \tag{D.v}$$

To prove the first inequality of eq. D.i, order the cubes along a spanning tree of their adjacency graph. A net meeting one cube can be extended to an adjacent cube by a segment of length at most M perpendicular to their common face. Starting with one point and repeating this construction gives a net of length at most $M(|X|_M - 1)$.

For the reverse inequality, subdivide the segments of any net N at their intersections and discard overlapping pieces. Its total length does not increase. Doubling the edges of the resulting connected graph gives a closed traversal of length at most $2\,\mathrm{length}(N)$. Cut it into at most $2\,\mathrm{length}(N)/M + 1$ arcs of length at most M. Each arc lifts to a curve in $\mathbb{R}^3$ whose range in each coordinate has length at most M. Such an interval meets at most three closed cube intervals, so an arc meets at most 27 closed M-cubes. Every cube of X meets one of these arcs. Taking the infimum over N gives

$$|X|_M \le 27(2d_M(X) + 1) \le c_X(d_M(X) + 1).$$

For eq. D.ii, take a net of length less than $Md_M(X) + \delta$ and points p', q' of the net in the respective cubes of p, q. The path in the net from p' to q' has length at most that of the net. Hence $d(p, q) \le M + (Md_M(X) + \delta) + M$; let $\delta \downarrow 0$. The same argument holds in $\mathbb{R}^3$.

A polymer with n cubes containing a fixed cube determines a closed walk of length $2(n-1)$ starting there: double a spanning tree of its adjacency graph and traverse it. Choose one such walk for each polymer. This assignment is injective, since the visited cubes recover the polymer, and there are at most $6^{2(n-1)} = 36^{n-1}$ possible walks. By the size estimate, $d_M(X) \ge n/c_X - 1$. Summing over n gives

$$\sum_{X \supset \square} e^{-\kappa_0 d_M(X)} \le e^{\kappa_0} \sum_{n \ge 1} 36^{n-1} 72^{-n} = K_{\mathrm{pol}}.$$

Sum this bound over the cubes of Y to obtain the second inequality of eq. D.iii.

Every M-cube belongs to one LM-cube, since L is odd, and adjacent M-cubes belong to the same or to adjacent LM-cubes. Thus $\bar{X}^L$ is connected through faces. Projection by $c \mapsto c/L$ gives its unique rescaled polymer Y. A net for X meets every LM-cube of $\bar{X}^L$; its rescaling has length divided by L. This proves eq. D.iv. For eq. D.v, split the decay into rates κ_0 and $\kappa' - \kappa_0$ and use eq. D.iv. All terms satisfy $X \subset LY$, which has $L^3|Y|_M$ fine cubes, so

$$\sum_{X:\bar{X}^L=LY} e^{-\kappa_0 d_M(X)} \le K_{\mathrm{pol}} L^3 |Y|_M \le c_X K_{\mathrm{pol}} L^3 e^{d_M(Y)}.$$

Combining the two factors proves the stated bound.

D.2. **Nonwrapping lifts.** Let $\mathrm{pr} : \mathbb{R}^3 \to \mathbb{R}^3/\ell\mathbb{Z}^3$ be the projection.

Proposition D.1. If X is a polymer and

$$2M(d_M(X) + 2) < \ell, \tag{D.vi}$$

then $X = \mathrm{pr}(\widetilde{X})$ for a finite face-connected union of M-cubes in $\mathbb{R}^3$. The projection is injective on $\widetilde{X}$, preserves adjacency of its cubes in both directions, and $d_M(\widetilde{X}) = d_M(X)$. The lift is unique up to a translation in $\ell\mathbb{Z}^3$. ∎

Proof. Put $D = M(d_M(X) + 2)$, fix p_0 in a closed cube of X, and fix its lift $\widetilde{p}_0$. The projection is injective on the closed sup ball $\widetilde{B} = \{|p - \widetilde{p}_0|_\infty \le D\}$, since two points there differ in every coordinate by at most $2D < \ell$. It maps this ball onto the torus ball $B = \{p : d(p, p_0) \le D\}$. Its

inverse $\varphi : B \to \widetilde{B}$ is continuous, and is locally isometric. The diameter estimate places every closed cube of X in B. A segment or a closed M-cube lying in B lifts under φ to a segment of the same length or a closed M-cube: the translates $\widetilde{B} + \ell n$ are pairwise disjoint, and a connected lift lies in one of them. The same inverse maps shared faces to shared faces. Thus $\widetilde{X} = \varphi(X)$ has the asserted injectivity, adjacency, and connectedness.

Any other face-connected injective lift can be translated so that it contains the chosen lift of the cube of p_0. There are no two different neighbors of one lifted cube with the same projection: their centers differ by at most $2M < \ell$. Propagate this observation along a chain of adjacent cubes. Every cube of the other lift equals the corresponding cube of $\widetilde{X}$, proving uniqueness.

Projecting a net for $\widetilde{X}$ gives a net for X with no greater length, so $d_M(X) \le d_M(\widetilde{X})$. Conversely, let N be a net for X of length at most $M d_M(X) + M$, and choose its point q_0 in the cube of p_0. Every point of N is joined to q_0 by a curve of length at most that of N, and $d(q_0, p_0) \le M$. Therefore $N \subset B$. Its lift $\varphi(N)$ has the same length and meets every cube of $\widetilde{X}$. Taking an infimizing sequence of these nets proves the reverse inequality. □

For the determinant estimate, suppose in addition that $M(d_M(X) + 4) < \ell$ and $M \ge 2$. The closed cubes of $\widetilde{X}$ have coordinate ranges of length at most $M(d_M(X) + 2)$ by eq. D.ii. Enlarge these intervals by one mesh $\eta_k \le 1$ at each end and intersect with $\eta_k \mathbb{Z}^3$. The resulting box W contains every bond meeting the lifted polymer and has each side of length at most $M(d_M(X) + 3)$. This length, increased by one mesh, is less than ℓ. Thus the projection is injective on W, and every torus bond meeting X has exactly one lift in W.

Appendix E. Notation

E.1. **Scales and coordinates.** The physical and level coordinates have different unit lengths. The following dictionary collects the conventions used in the renormalization estimates.

Physical cutoff	$\eta = L^{-N}$, $R = 8L^T$ and unit-lattice side $n = R/\eta$.
Level j	$a_j = L^{j-N}$, $\eta_j = \eta/a_j$ and $\ell_j = R/a_j$; one level unit has physical length a_j.
Charges	e is the physical charge, $q = e\sqrt{\eta}$ the fine unit charge, and $e_j = e\sqrt{a_j}$ the running charge.
Masses	$\mu_j = \bar{m}a_j$ is the positive reference mass and m_j the mass interaction. The total fine mass is $\boldsymbol{\mu}_N = \mu_0 + m_{0,N}$, which is real.
Dirac operators	$\mathcal{D}_A$ is the finite matrix including the mass; $D^{\mathrm{ch}}[B]$ is the unit-mesh massless matrix with bond phase B; D_A is the massless operator at the level mesh; $D_j(A)$ is the blocked coarse operator.
Curvatures	F^{u} is the unit lattice curvature, $F^{\sharp}$ its rescaled step field, and F the limiting tempered two-form; see §12.2.
Reference cells	R^* is the reference side, $\ell_c = R^*/(2L)$ its cell side, and $N_* = 8L^3$ the number of cells.

E.2. **Notation index.** Symbols are grouped by their role in the construction. Indexed families and their indices are listed together. Reused letters have separate entries for their different roles. References point to defining equations, statements, sections, or pages. Standard notation without a construction-specific definition is explained here in the table.

Symbol or family	Meaning and scope	Defined at
Physical fields, conventions and scales		
L, N, T; η, R, n	Odd block factor, ultraviolet and volume indices; physical mesh and side, and unit-lattice side $n = R/\eta$.	eq. 2.xxiii
e, q; $m_{0,N}$	Physical charge, fine charge $q = e\sqrt{\eta}$, and volume-independent bare mass shift.	eq. 2.xxiii
$\boldsymbol{\mu}_N$; $\bar{m}$	Total fine lattice mass and positive physical reference mass; the main theorem fixes $\bar{m} = 1$.	eq. 2.xxiii
$F^{\sharp}$	Physical centered plaquette step field, with value $\eta^{-3/2}u$ for unit-lattice curvature u.	p. 11
μ, λ	Volume-first parent law and a selected translation-ergodic component in the main theorem.	Thm. A
$F(g)$; S_n, S_0	Evaluation of the limiting random two-form; its nth moment distribution, with $S_0 = 1$.	Thm. A
$\mathcal{S}_n$; q_m	Tensor-valued Schwartz test space and the weighted derivative seminorm used in the OS growth estimate.	eq. 2.xxv
$g \cdot \Psi$; $\tau_a^{\otimes l}$	Euclidean action on tensor tests and translation of the last l arguments.	Thm. A
Θ_k, Θ_0	Antilinear reflected tensor-test action, and conjugation of a scalar.	eq. 2.xxvi
(E0′), (E1)–(E4)	Growth, Euclidean invariance, reflection positivity, symmetry and clustering conditions of the stated theorem.	Thm. A

Continued on next page

Notation (continued)

Symbol or family	Meaning and scope	Defined at
$\beta, f, b(f)$	Compact two-form probe, derived test $f = dd^*\beta$, and positive free fourth coefficient.	eq. 2.xxviii
S_{D}	Mass-one continuum Dirac resolvent.	eq. 2.xxviii
$\kappa_4^{\lambda}(f)$	Fourth cumulant of the field evaluated at f in the selected component.	eq. 2.xxix
$\mathbb{R}, \mathbb{C}, \mathbb{Z}, \mathbb{N}$	Real and complex numbers, integers and natural-number indices.	here
$\mathcal{S}, \mathcal{S}'$; C_c^{∞}	Schwartz tests, tempered distributions and smooth compactly supported tests; coefficient spaces are specified in each use.	here
L^p; $\lVert\cdot\rVert_p, \lVert\cdot\rVert_{\infty}$	Lebesgue or probability-space norm for the stated measure; the operator Schatten notation is distinguished below.	here
$\lvert\cdot\rvert_1, \lvert\cdot\rvert_2, \lvert\cdot\rvert_{\infty}$	Coordinate sum, Euclidean and maximum norms; $\lvert\cdot\rvert_2$ also denotes the Euclidean spinor norm.	here
δ_{ij}	Kronecker delta, equal to one for equal indices and zero otherwise; distinct from the cochain adjoint δ.	here
ε_{ijk}	Completely antisymmetric orientation tensor, normalized by $\varepsilon_{123} = 1$; distinct from the scalar small exponent ε.	here
$\langle\cdot,\cdot\rangle$; $(\cdot,\cdot)$	The stated lattice or continuum pairing; conjugation and lattice weights are fixed in the corresponding definition.	§1.1
O^*, O^T; $O^{\sharp}$	Hermitian adjoint, algebraic transpose for the weighted bilinear pairing, and the displayed reversed-connection transpose.	§4.2
tr, Tr; det	Matrix or operator trace and determinant on the indicated space.	here
Re, Im; $^-$	Real and imaginary parts and complex conjugation; operator real parts use the Hermitian adjoint.	here
$\mathbf{1}_E$; supp	Indicator or support restriction to E, and support of a field, test or kernel.	here
$\mathbb{E}_{\nu}, \nu(G)$	Expectation, or integration of G, against the specified law ν.	here
C, c (unqualified)	Local finite upper-bound and positive lower-bound constants; explicitly named constants are listed in their own scopes.	here
Λ^2; d, d^*	Two-form coefficient space, exterior derivative and Euclidean adjoint, with the three component inner products summed.	§1.1
σ_i, γ_i; γ_v	Pauli matrices, doubled Hermitian Clifford matrices, and contraction of the Clifford triple with a vector.	§2.1.3
$g = (Q, a)$; $\Lambda^2 Q$	Euclidean motion $x \mapsto Qx + a$ and its induced two-form action.	§1.1
$g \cdot f$; gF	Inverse-pullback action on tests and the corresponding dual action on distributions.	§1.1
τ_a; $\det(Q)Q$	Translation convention $\tau_a f(x) = f(x - a)$ and axial-vector action on Hodge duals.	§1.1
a_j, η_j, ℓ_j	Physical length of a level unit, mesh in level units and torus side in level units.	eq. 6.i

Continued on next page

Notation (continued)

Symbol or family	Meaning and scope	Defined at
e_j, μ_j	Running charge and positive reference mass at level j; the interacting mass is indexed separately with the tuned flow.	eq. 6.i
$D^{\mathrm{ch}}[B]$; D_A	Unit-mesh massless chamber matrix with bond phase B; its mesh-normalized covariant version.	eq. 4.iii
F^{u}; $F^{\sharp}, F$	Unit curvature, its physical step-field rescaling, and the limiting distribution; these are different field normalizations.	§12.2
R^*, ℓ_c, N_*	Reference physical side, reference cell side $R^*/(2L)$ and cell count $8L^3$.	§11.1
lattice configurations, chamber graph and fermions		
ρ; $w_0, \dots, w_3$	Chamber Wilson parameter and four positive hopping weights; the construction fixes their standard values.	eq. 2.xi
M (regulator)	Real diagonal chamber-operator coefficient; distinct from the later integer polymer block size.	eq. 2.xiv
$C^k, C^k(\mathbb{Z})$; $C^k(\Gamma)$	Real, integer and periodic lattice cochains with the stated positive-cell orientations.	§2.1.1
d (unit lattice); d^*	Unnormalized lattice coboundary and its counting-inner-product adjoint.	eq. 2.i
Γ, V	Rank-three photon period lattice and its index $V = [\mathbb{Z}^3 : \Gamma]$.	§2.1.1
$\mathrm{Conf}(\Gamma)$	Photon configuration quotient by integer one-cochain shifts and compensating integer curvature shifts.	eq. 2.ii
A, z, F; u_b	Real link potential, closed integer two-cochain, all-flux curvature and representative-independent link phase.	eq. 2.iii
r_q; $\mathrm{d}(A, z)$	Link period $2\pi/q$ and Lebesgue/counting measure on the configuration quotient.	eq. 2.iv
$C_q(\varepsilon), C_q$	Uniform one-dimensional Gaussian lattice sum and its value at $\varepsilon = 1/2$.	eq. 2.v
Ω (lattice)	Polish space of closed real two-cochains on all plaquettes of $\mathbb{Z}^3$, with the product topology.	§2.3.1
$\langle \beta, F \rangle$; $\widetilde{F}$	Finite cochain pairing and periodic extension of a torus curvature.	Prop. 2.3
$\mathcal{L}$; $n_\ell, \mathrm{Adm}(\ell)$	Nine mirror linear forms, their normal vectors and admissible mirror heights.	eq. 2.vii
$\Pi = (\ell, c)$; θ_Π	A lattice mirror and its affine reflection.	eq. 2.viii
W, O_{cub}	Affine lattice reflection group and its 48 signed permutation matrices.	§2.1.2
C_0, s_i, Π_i, n_i	Fundamental chamber, four wall reflections, walls and inward unit normals.	eq. 2.ix
$v_0, \mathcal{X}, x^{(i)}$	Chosen interior chamber site, its reflection orbit and its neighbor of bond type i.	eq. 2.x
$x = k + rv_0$; $\pi(x), y(x)$	Photon-site/cluster-label coordinates and the photon projection and internal displacement of a fermion site.	§2.1.2
d_i	Distance from v_0 to wall Π_i, determining the geometric chamber bond length.	§2.1.2

Continued on next page

Notation (continued)

Symbol or family	Meaning and scope	Defined at
$P_n^{\pm}, r_n$	Clifford eigenspace projections and the linear reflection normal to the unit vector n.	eq. 2.xii
$U_A(x,x')$; $w(x,x')$	Gauge transport and matrix hopping weight along a chamber bond.	eq. 2.xiii
$\mathcal{D}_A$; m	Massive finite chamber operator and its lattice mass $m = M - \rho\sum_i w_i/2$.	eq. 2.xv
χ (boundary); $\mathbb{T}$	Fermion period character and torus data (Γ,χ).	§2.1.3
$\mathrm{Gr}_{\mathbb{T}}$; $\psi,\bar\psi$	Finite Grassmann algebra and its four-component chamber generators.	eq. 2.xvi
$\int_{\mathbb{T}}$; $S_{\mathbb{T}}(A)$	Counting-normalized Berezin integral and the finite fermion quadratic action.	eq. 2.xvi
$Z_{\mathbb{T}}[G]$; $\mathbb{E}_{\mathbb{T}}$	Unnormalized photon–fermion functional and its normalization by $Z_{\mathbb{T}}[1]$.	eq. 2.xvii
$\mu_{\mathbb{T}}$	Normalized curvature pushforward law on the closed-cochain space.	Prop. 2.3
∇_A, H_A, J_A	Covariant chamber difference and the symmetric/antisymmetric hopping sums in the real-part decomposition.	Prop. 2.2
$\check{\mathcal{D}}_A^{\mathbb{T}}$	One two-spin block whose determinant modulus squared is the full determinant.	eq. 2.xviii
$P_{\mathbb{T}}$; C_*	Laurent determinant polynomial and its positive mass-independent extreme Fourier coefficient per photon site.	eq. 2.xxi
$(i_1,\dots,i_6)$	Six successive upward thin-layer bond types in the determinant coefficient argument.	eq. 2.xx
γ_5; $\gamma_n\gamma_5$	Clifford-anticommuting matrix and spin lift of a spatial reflection.	§2.1.4
$C = \sigma_2 \oplus \sigma_2$	Regulator charge-conjugation matrix; the RG matrix $\mathsf{C} = i\gamma_2$ is different.	§2.1.4
Θ_Π; G^g	Antilinear product-reversing Grassmann reflection and induced lattice action on even integrands.	eq. 2.xxx
$g\mathbb{T}$	Torus and boundary character transported by an affine lattice symmetry.	§2.1.4
$e_{n,k}^{\pm}$; ζ_i	Orthonormal Clifford eigenvectors and the four crossing-mode types per projected spin label.	eq. 2.xxxv
ω_i, u_i; ζ_I, ω_I, u_I	Positive crossing weights, phase factors and their ordered subset products.	eq. 2.xxxvi
$S_\Pi^{\times}(J)$; $\mathcal{I}$	Crossing part of the action and its finite mode index set for crossing bonds J.	Prop. 2.4
$\Gamma_\parallel, v, p$	Mirror-parallel period lattice, normal closing vector and normal height period of an adapted torus.	eq. 2.xxxviii
$c', \Pi', U_\pm$	Second periodic mirror and the two open half-period slabs.	§2.1.6
$\mathcal{M}$; $B_\pm, B_0, B_\times$	Boundary mirrors and upper, lower, mirror and cut link sets.	§2.1.7
$P_\pm, P_0, P_\times$; $Q_\pm, Q_\times$	Corresponding plaquette and cube classes at the two mirrors.	§2.1.7

Continued on next page

Notation (continued)

Symbol or family	Meaning and scope	Defined at
d_+, d_-	Upper and lower incidence contributions to the coboundary at cut cells.	eq. 2.xxxix
$A_\pm, A_0, a;\ z_\pm, z_0, z_\times$	Potential and integer-curvature variables partitioned by their mirror-cell class.	§2.1.7
$\alpha_Q, \alpha'_\sigma;\ \beta_\sigma$	Cube Fourier angles, their cut-plaquette incidence sum and dual Poisson frequencies.	eq. 2.xl
$u_{\alpha,k};\ \beta_k$	Upper-side unit-modulus factor and one-dimensional Poisson-summation frequency.	eq. 2.xli
$\mathfrak{q}(I)$	Integer crossing-mode charge on each cut photon link.	eq. 2.xliii
local observables, slab operators and all-flux traces		
$\langle\bar\Psi, O\Psi\rangle_{\mathrm{ch}};\ \int \mathrm{d}\Psi$	Chamber pairing with label weight 1/48 and the compensating weighted Berezin integral.	§2.2
$B^{(j)};\ \mathcal{X}(B)$	Oriented j-cells inside a box and the fermion sites whose complete photon-site cubes lie inside it.	§2.2.1
$\mathcal{G}(B), \mathcal{G}^{\mathrm{ev}}(B)$	Gauge-invariant Borel Grassmann observables with Gaussian-compatible coefficient growth, and their even part.	eq. 2.xliv
$\mathfrak{q}(\Psi)$	Signed photon-site charge of an ordered Grassmann monomial.	eq. 2.xlv
$\widehat{A}, \arg_q$	Borel local potential lift with the correct curvature/link phases and its principal phase representative in $[0, r_q)$.	eq. 2.xlvi
$\mathcal{A}(U)$	Linear span of bounded curvature functions times disjointly supported even local integrands inside U.	eq. 2.xlviii
ε_G	Positive residual Maxwell exponent in the absolute-integrability majorant of a local product.	eq. 2.xlix
$\Xi = (\Gamma_\parallel, \chi_\parallel)$	Rank-two cross-section lattice and its fermion character.	§2.2.2
$\mathcal{M}^j(c), \mathcal{X}^j(c)$	Mirror-contained and mirror-cut j-cells at height c.	§2.2.2
$\mathcal{B}^j(c, c');\ \varpi, \varpi^\circ$	Cells strictly between slab boundaries, a closed-slab configuration and its interior variables.	§2.2.2
$W_{c,c'};\ \mathcal{I}_\Xi[c, c']$	Interior Maxwell/closedness weight and admissible slab-integrand class.	eq. 2.l
C_G, ε_G	Coefficient-growth constants for an integrand in the larger slab class.	eq. 2.li
$\mathcal{J}(c), \mathcal{F}(c), e_I$	Crossing-mode set, its exterior boundary space and its orthonormal occupation basis.	§2.2.2
ω_c, U_{f}	Diagonal positive crossing weights and fermion-parity operator on the boundary Fock space.	§2.2.2
$U_v, s_v(I), r_v(I)$	Mode-translation unitary, reordering/character sign and translated occupation subset.	§2.2.2
$N_{c,c'}[H];\ \mathrm{Ins}_c[\Upsilon]$	Open-slab Grassmann matrix and inverse crossing-mode insertion map.	eq. 2.lii
$d_c, d_c^+;\ \vartheta, k, \varkappa$	Cut incidence maps, cube angles, integer plaquette indices and dual frequencies in the boundary data.	eq. 2.lv
$\mathrm{d}\mu_c^\times, e_c, D_c$	Dual crossing measure, upper-side phase factor and nonnegative Gauss constraint multiplier.	eq. 2.lv

Continued on next page

Notation (continued)

Symbol or family	Meaning and scope	Defined at
$Y_c, \xi, \mathfrak{m}$	Complete mirror-data space, its coordinates and mirror potential/integer variables.	eq. 2.lvii
$\mu_c, \mathcal{H}(c)$	Finite positive boundary measure and boundary Hilbert space $L^2(Y_c, \mu_c)$.	eq. 2.lvii
$S_{(c,c')}$; $\mathfrak{S}[G]$	Interior fermion action and photon–fermion slab kernel.	eq. 2.lviii
$C_{c,c'}$	Fixed slab/cross-section coefficient constant in the pointwise and Hilbert–Schmidt kernel bounds.	eq. 2.lix
$\|\cdot\|_s$; Tr	Schatten norm and Hilbert-space trace in the slab estimates.	eq. 2.lxii
$\mathbb{T}^0, g^0, \Gamma^0, \chi^0$	Straightened torus and its normal closing vector, period lattice and antiperiodic normal character.	eq. 2.lxiii
$\mathrm{fold}_\ell G$	Repeated reflected slab integrand over a full normal period.	eq. 2.lxiv
$Q_\kappa, \lvert Q\rvert, g_\kappa$	Frame cells, their common volume and reflected map from the base cell to cell κ.	eq. 2.lxvi
$G^{[\kappa]}$	Reflected copy of a base-cell integrand assigned to cell κ.	eq. 2.lxvi
$\mathrm{Fold}_{\mathbb{T}} G$; $\|G\|_{\mathbb{T}}$	Full cell fold and its unnormalized chessboard seminorm.	eq. 2.lxvii
$C(a,b,c), C(n)$	Coordinate tori with antiperiodic fermions, and the cubic specialization.	Thm. C
$e_\pm$; $D(a,b,c), D'(a,b,c)$	Diagonal frame directions and adapted tori differing by the character on the e_+ period.	Thm. C
$\Gamma_0, \Gamma_1, \Gamma_2, \Gamma_3$; γ_a	Successive period lattices and normal closing vectors in the three-direction fold/straightening proof.	§2.2.3
$f(n_1,n_2,n_3), f_\infty$	Log partition function per photon site at fixed microscopic parameters and its infinite-volume infimum.	Prop. 2.7
$\Delta(n)$	Nonnegative pressure error between the cubic and final coordinate comparison tori.	eq. 2.lxxiii
$Y = \mathfrak{S}[1]$	Coordinate half-step slab operator used to express pressure by Schatten norms.	§2.2.4
$C_{\mathbf{n}}, \Sigma, v$	Coordinate torus, two-dimensional spatial slice and number of its sites.	§2.2.5
d_s; $\Sigma^{(1)}, \Sigma^{(2)}$	Spatial-slice coboundary and its link and plaquette sets.	§2.2.5
A_j, a_j; $z_j^{\mathrm{el}}, z_j^{\mathrm{mag}}$	Spatial and temporal potentials and electric/magnetic integer curvature slices.	eq. 2.lxxv
$F_j^{\mathrm{el}}, F_j^{\mathrm{mag}}$	Electric and magnetic parts of the lattice curvature.	eq. 2.lxxv
k_j, g, Γ_s	Integer unfolding primitives, winding connection and closed integer slice-shift lattice.	§2.2.5
σ_0, k (slice flux)	Chosen spatial plaquette and the residual integer magnetic flux through the spatial two-torus.	eq. 2.lxxvi
$\mathcal{F}', \mathcal{F}$; ω', ω	Site-plane/link-plane Fock spaces and their crossing-weight matrices.	eq. 2.lxxvii
$N(A), \widehat{N}(A), u(a)$	Open half-slab matrix, its crossing-weighted version and the diagonal mode-charge unitary.	eq. 2.lxxvii
H^s, H^l	Direct sums over magnetic flux of L^2 spaces on real slice connections modulo Γ_s, with Fock fibres for site and link planes.	§2.2.5

Continued on next page

Notation (continued)

Symbol or family	Meaning and scope	Defined at
$E_b, U_G(a), P_G, K$	Electric momentum, gauge unitary, gauge projection and projected electric heat operator.	eq. 2.lxxix
M_b, X_b	Multiplication by the shifted magnetic factor and fermion matrix; X_b then applies $K^{1/2}$.	eq. 2.lxxxi
$g_t, \widetilde{g}_{1/2}$	One-dimensional heat density and the periodized electric half-time Gaussian.	§2.2.5
C_N	Uniform norm of the crossing-weighted half-slab matrix $\widehat{N}(A)$.	eq. 2.lxxviii
$\widehat{g}, C_s$	Gauge-averaged nonnegative heat-kernel majorant and finite spatial Gaussian quotient normalization.	eq. 2.lxxxiii
$Z_{C_{\mathbf{n}}}(\beta)$	Partition function with an external real two-cochain added to the Maxwell curvature.	eq. 2.lxxxiv
X_j, U_j	Magnetic half-step factor and electric-source unitary in the external partition trace.	eq. 2.lxxxv
$A_j^\flat, g_j$	Fundamental-domain representative and closed-shift lift of an unfolded real spatial slice.	§2.2.5
$Z(\beta)$; $\mathbb{E}e^{\sum_i z_i\langle\beta_i,F\rangle}$	Real shifted partition function and complex-source exponential expectation controlled by Gaussian domination.	eq. 2.lxxxviii
$\mathcal{D}$; n_i	Divisibility family and increasing side sequence chosen to accommodate every required cell width.	§2.3
D_n, N_n, Λ_h	Diagonal comparison torus, its number of cells and the lattice of cell translations.	§2.2.7
$\widetilde{O}_n, O_n^{\mathrm{fold}}$	Translated cubic array and reflected diagonal array of a base-cell observable.	eq. 2.xc
$\mu_{D_n}^t$; t (tilt)	Positive law tilted by the reflected observable array and its nonnegative strength.	eq. 2.xci
diagonal comparison, insertion norms and elimination		
$\ell_\pm, c_0, c_+, c_3, h$	Diagonal frame, offsets and even original cell width in the comparison construction.	§2.3.1
D_i, μ_i^t, V_i	Divisible-volume diagonal torus, its tilted functional and reflected tilt array.	§2.3.1
$\lvert O\rvert_1, r_O$	Sum of absolute trigonometric coefficients of O and the radius of a box containing its support.	§2.3.1
Φ_+	Half-period product whose reflected square gives the full bosonic tilt.	§2.3.1
$S, \Lambda, \mathcal{F}_\Lambda, H_S$	Finite spatial link/site supports, their crossing Fock space and local real-connection Hilbert space.	§2.3.2
E_S; $\mathrm{End}_z(\mathcal{F}_\Lambda)$	Local electric momenta and Fock matrices of prescribed integer gauge charge.	§2.3.2
$\mathfrak{A}(S)$	Finite span of bounded Gaussian-exponential electric symbols on subsets of S.	§2.3.2
$z(\delta)$; $\mathfrak{W}(S,\Lambda)$	Integer divergence charge of a local phase and the corresponding gauge-invariant local operator algebra.	eq. 2.xcix
$\alpha_\beta X$	Conjugation of a local operator by a connection character, shifting its electric arguments.	eq. 2.ci

Continued on next page

Notation (continued)

Symbol or family	Meaning and scope	Defined at
ι, X_{Ξ}	Exterior-algebra localization into a cross-section and the quotient realization of a local operator.	§2.3.2
$\|X\|_{\#}$	Supremum over cross-sections and electric-character shifts; equal to the local operator norm.	eq. 2.cii
$G[k]$; $S_c^{\times}(\Lambda)$	Local gauge-invariant insertion of a matrix kernel and the canceled crossing action.	eq. 2.ciii
$B^{\#}(c, S, \Lambda)$	Canonical enlarged support box for a two-slice kernel insertion.	§2.3.3
$k_X^{(s)}$; G_X, G'_X	Kernel of a local operator between two electric heat factors, and its quarter-time and eighth-time insertions.	eq. 2.cvi
$Q_s, \lambda_s, \Upsilon_0$	Regularized Gaussian-symbol quadratic form, shifted linear coefficient and constant Fock matrix in the explicit kernel calculation.	eq. 2.cvii
v, r_I, U (spatial reflection)	Crossing-mode spin unitary, exterior reordering sign and induced Fock reflection unitary.	§2.3.3
J, θ_s (local reflection)	Antiunitary on local connection/Fock wave functions and the reflected spatial-slice map; this J is not the cluster injection.	§2.3.3
φ_β	Curvature character $F \mapsto \exp(i\langle\beta, F\rangle)$.	eq. 2.cix
C (coordinate bound)	Uniform per-insertion constant controlling disjoint coordinate images in every curvature-character trace.	eq. 2.cix
$K^{1/2}_{\beta_j, S_j}, T_j$	Electric heat factor with selected insertion variables and its magnetic half-step product in the coordinate trace proof.	§2.3.3
h'', k (amplification)	Coarse coordinate width and diagonal width multiplier used to separate differently directed insertions.	eq. 2.cxii
$c'_{\pm}, c'_3, Q'_0$	Coarse diagonal offsets and their base cell of width kh.	eq. 2.cxiii
$o_\mu, I_\mu, \kappa(g)$	Coordinate offsets, allowed offset intervals and the injective assignment of diagonal images to coordinate cells.	eq. 2.cxiv
T_i, N'_i, N''_i	Final coordinate comparison torus and the coarse diagonal/coordinate fold cell counts.	§2.3.4
$\Delta_i, \gamma_{t,i}(k)$	Pressure error at side n_i and the amplification factor controlling local tilted expectations.	eq. 2.cxvi
$\mu_i^t(e^{\langle\beta,F\rangle})$	Tilted exponential moments used for volume-uniform tightness of the comparison states.	§2.3.4
$\mathcal{P}$; $\mathcal{P}_1(c)$	Finite gauge-invariant exponential Grassmann polynomials and their two-adjacent-layer subclass.	§2.3.5
$\mathcal{X}_{j,a}$	The six thin fermion layers inside thick photon slice j.	§2.3.5
$\delta_0, \delta_1, \delta_\perp$	Lower spatial, upper spatial and temporal phase exponents of a two-layer monomial.	§2.3.5
Υ_Y; k_Y^0	Finite crossing-mode matrix and continuous heat kernel whose insertion equals the two-layer monomial Y.	eq. 2.cxxii
$U, a(E), b(E)$ (two-layer)	Phase unitary and the two electric quadratic forms defining the regularized middle operators.	§2.3.5
$X'_Y, X_{Y,\epsilon}$	Bounded eighth-heat middle operator and the approximating quarter-heat middle operators.	eq. 2.cxxv

Continued on next page

Notation (continued)

Symbol or family	Meaning and scope	Defined at
$H_\alpha, P_\epsilon, \xi_Y(\epsilon), C_Y$	Electric heat regularizers, the error in approximating the two-layer operator, and its bound proportional to ϵ.	eq. 2.cxxvii
o_i^t; γ (local functional)	Insertion functional $X \mapsto \mu_i^t(G_X)$ and its near-unital norm bound.	§2.3.5
$\mathcal{P}^{\pm}(y)$	Odd exponential polynomials of gauge charge ± 1 at photon site y.	§2.3.6
$\partial_\psi, \partial_{\bar\psi}$	Left Grassmann derivatives used in the exact lattice equations of motion.	eq. 2.cxxxi
$\mathbb{L}, \mathbb{L}^+, \mathbb{L}^{++}$	A thin fermion layer and its next two matching layers.	§2.3.6
$R_{\mathbb{L}}^{\pm}$	Finite recursive elimination maps through upward or downward matching bonds.	eq. 2.cxxxiii
b_a, β_a, Y_a (normal form)	Finite coefficients, curvature characters and two-layer terms obtained by thin-layer elimination.	eq. 2.cxxxiv
C_P, k_P (polynomial bound)	Polynomial-dependent uniform bound and amplification support scale; unrelated to the photon or Poincare constants.	eq. 2.cxxxv
$P^{(u)}, Q_u, V_u$	Imaginary link-shifted polynomial, induced hopping correction and its finite Grassmann exponential.	eq. 2.cxxxvi
$P^{\uparrow}, P^{\flat}$; P_a	Thin-layer-reduced polynomial, outer-plane reduction and its finitely many shifted terms.	eq. 2.cxlvi
$H_\eta, C(\eta)$	Finite change of the bosonic tilt array under imaginary curvature shift and its coefficient bound.	eq. 2.cxliv
$Y_P, C_{P,g}, i_0(P,g)$	Universal two-layer reduction of P, transported reduction-error constant and volume threshold.	Thm. F
ν_{t_k}, ν_0; t_k, i_m	Ordered volume/zero-tilt comparison laws, positive tilt sequence and common volume subsequence.	eq. 2.cxlviii
ν (cubic lattice limit)	Prescribed unperturbed cubic curvature limit dominated, on a chosen observable, by a comparison state.	eq. 2.cxlix
$Q_A = (\Theta_\theta A)A$	Reflected character-sum quadratic product used in the direct reflection-positivity comparison.	§2.3.9
O_A, ν_A	Negative real reflected product and its observable-dependent comparison state.	§2.3.9
$v = ae_1$ (cell placement)	Even integer translation placing an arbitrary local comparison test in an admissible cell.	§2.3.9
photon averages, Gaussian coordinates and domains		
T_k, T_k^0, η_k, ℓ	Mesh-L^{-k} photon torus, its unit lattice, mesh and side in level units.	§3.1.1
$T_k^c, \Omega^p(T)$	The L-spaced coarse torus and real lattice p-cochains.	§3.1.1
$E(\mathscr{A}), Z_0$	Half the fine curvature norm squared and the remainder after fixed-average reconstruction.	eq. 3.vi
$\mathcal{Z}_k$; λ (axial correction)	Zero-average axial unit-lattice fluctuations and the unique zero-block-mean gauge correction.	eq. 3.vii
Δ_z; $\eta^3 \sum_c$	Unit photon blocks and the mesh-weighted cochain pairing.	§3.1.1
$\partial_\mu, \delta, \Delta_{\mathrm{sc}}$	Mesh-normalized forward difference, algebraic cochain adjoint d^T and scalar Laplacian δd.	§3.1.1

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Notation (continued)

Symbol or family	Meaning and scope	Defined at
$\alpha, \delta_\alpha \partial$	Hölder exponent and local Hölder quotient of a first difference.	eq. 3.xi
$N = L^k$ (photon symbol)	Number of fine mesh intervals per unit length; distinct from the physical ultraviolet index N.	eq. 3.i
$\mathbf{Q}_k, \mathbf{Q}_k^{(2)}$	Scalar/bond block averages and the compatible plaquette average.	eq. 3.i
$\mathcal{R}_k$	Scalar orthogonal projection onto $\Delta_{\mathrm{sc}}(\ker \mathbf{Q}_k)$.	eq. 3.iii
$H_k^{\mathrm{ph}}, \Delta_k$	Landau-gauge photon minimizer and its coarse quadratic Maxwell form.	eq. 3.iv
$B(y), \mathbf{Q}, \tau$ (photon)	One-step L-block, bond average and average of six coordinate-order axial path sums.	§3.1.2
$\Gamma_{z,z+Le_\mu}$; $\Gamma^\pi(y,z)$	Straight averaging path and coordinate-order path in the axial functional.	§3.1.2
$p_c, \bar{a}$ (face estimate)	Local curvature maximum and mean face-bond value used in axial coercivity.	eq. 3.ix
$u(t), s(t)$ (photon symbol)	Centered scalar averaging multiplier and mesh sine $2N \sin(t/(2N))$.	§3.1.3
$d_\mu(q), d_\mu^*(q), D(q)$	Analytic difference symbols and scalar Laplacian symbol; the star means argument reversal, $d_\mu^*(q) = d_\mu(-q)$.	§3.1.3
$U(q), V_\mu(q), V_\mu^*(q)$	Product block-average multiplier and directed bond-average multipliers.	§3.1.3
$\rho_\mu(p), \rho_\mu^*(p)$	Unit-lattice difference multipliers, unrelated to the Wilson parameter ρ.	§3.1.3
q_m, u_μ	Aliased momentum $p + 2\pi m$ and the zero-alias component $u(p_\mu)$.	§3.1.3
X_μ, Y, N_μ, N_G	Nonzero-alias sums and regularized denominators in the photon Fourier calculation.	eq. 3.xix
Π (photon symbol)	Positive combination of regularized denominators used to continue the scalar multiplier through zero; distinct from a geometric mirror.	eq. 3.xx
$b, w, \Lambda_{\mu\nu}, h_{\mu\nu}(p,m)$	Fourier Euler and scalar gauge multipliers, coarse energy symbol and reconstruction amplitudes.	eqs. 3.xiv and 3.xvi
κ, B, c, g (symbol bounds)	Real sine lower bound, common complex upper bound, real denominator lower bound and uniform analytic width.	Prop. 3.7
$\langle t \rangle$; S_α	Alias weight $\max(1, \lvert t \rvert)$ and uniform weighted alias-summation bound.	eq. 3.xxii
$K_{\mu\nu}(p;x)$; E_μ, G, r	Periodic summed alias kernel and all-alias denominators in the Fourier reconstruction equations.	§3.1.3
$H_{\mu\nu}^\infty$	Covering-lattice photon minimizer kernel obtained from the Fourier symbol and periodized to finite tori.	§3.1.3
$C_H(\alpha), \gamma_H$	Uniform photon-minimizer regularity constant and exponential kernel-decay rate.	eq. 3.xi
$C_\Delta, C_\Delta^{(2)}, A_1$	Pointwise kernel, L^2 operator and symbol-entry bounds for the coarse photon form.	eq. 3.xii
$\Omega^1(B(y)), \tau_y, e_i^y$	Interior block-bond space, restricted axial functional and translated reflection-eigenvector basis of its kernel.	§3.1.4

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Notation (continued)

Symbol or family	Meaning and scope	Defined at
$V_k, \Upsilon, \delta_b, d_V$	Independent photon coordinate space, its basis vectors, noncentral face coordinates and dimension density.	eq. 3.xxviii
C (photon coordinates)	Real map filling central face bonds so that both coarse average and axial functional vanish.	eq. 3.xxvi
$b_c, n_c(b)$	Central face bond and multiplicity of an interior bond in the face averaging paths.	eq. 3.xxvi
c_C, c_1	Coordinate-supremum and column-ℓ^1 bounds for the photon coordinate map.	eq. 3.xxix
$\mu_I, \mathrm{Box}_k, \widehat{\mu}_k$	Standard independent coordinate Gaussian, its truncation box and normalized restricted Gaussian.	eq. 3.xxxi
$p_{0,k}, \delta\varepsilon_k^0$	Photon-coordinate cutoff and Gaussian-box normalization loss per unit volume.	eq. 3.xxxi
$\widetilde{\Delta}_k, C_k, C_k^{1/2}$	Coordinate fluctuation form, its inverse covariance and positive square root.	eq. 3.xxxii
c_P, C_{Dt}	Spectral enclosure and coordinate-kernel bound for the fluctuation form.	eq. 3.xxxii
$t_0, C_T, g, \phi, S, \theta$ (resolvent)	Spectral gap, decay data, distance-triangle defect, Schur sum and conjugation rate in the generic weighted-resolvent estimate.	§3.1.4
W, E (weighted resolvent)	Exponential diagonal conjugating matrix and its conjugation-error operator.	§3.1.4
$c_{\mathrm{bas}}, S_P, \gamma_P$	Basis-support count, weighted Schur sum and square-root kernel-decay rate.	eq. 3.xxxiv
C_P^{ph}	Common photon minimizer/form/square-root kernel envelope constant.	eq. 3.xxxiv
$A'_L, A_k^{\min}(A')$	Scaled outgoing photon field and axial constrained minimizer of the incoming coarse Maxwell form.	eq. 3.xxxv
Sq_k, ζ	Photon fluctuation embedding $CC_k^{1/2}$ and its standard Gaussian coordinate vector.	eq. 3.xxxvi
$\lambda_L, \omega_k(A')$	Scaled scalar gauge and uniquely normalized gauge relating successive photon backgrounds.	eq. 3.xxxvii
μ (one-step gauge)	Zero-block-mean real scalar gauge used to compare successive constrained minimizers; not a mass or probability law.	§3.1.5
$\square^\dagger, s_d$	Enlarged M-cube $\square^{\sim L}$ and its side $(2L+1)M$.	§3.1.6
$\|B\|_{i,W}, i = 1,2,3$	Local coefficient, first-difference and Hölder-first-difference sup norms.	§3.1.6
C_B	Curvature-to-local-gauge regularity constant for H_k^{ph}.	eq. 3.xxxviii
$a, A^0, A'', \lambda, \nu', \nu$	Mean field, mean-zero and basepoint-gauged fields, and the scalar gauges in the local photon proof.	§3.1.6
$p_A, \mathrm{hol}(v), c_{\mathrm{bd}}(g)$	Curvature maximum, seam holonomy and exponential unit-bond counting bound.	§3.1.6
$\pi_a, \mathrm{Cub}_a(S)$	Overlapping parameter cubes of half-side $a = M$ or LM, and those meeting a site set.	§3.1.7
$\mathrm{Pth}(p, p')$	Shortest ordered rectilinear path with canonical odd-period offsets.	eq. 3.xlii

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Notation (continued)

Symbol or family	Meaning and scope	Defined at
$\vartheta, \varphi, \chi_c$	Piecewise-cubic cutoff, one-dimensional partition factor and product row weights forming a partition of unity.	eq. 3.xliii
$\mathrm{St}(\Upsilon), \mathrm{Pt}(\Upsilon)$	Support set and representative point set of a photon coordinate basis vector.	§3.1.7
$P'(c,b'), P''(\Upsilon,\Upsilon')$	Linked sets of parameter cubes used in the minimizer and square-root kernel products.	eq. 3.xliv
$w(b,b';s)$	Smoothed parameter-product multiplier used to weaken the photon minimizer.	eq. 3.xliv
$H_k^{\mathrm{ph}}(s), C_k^{1/2}(s)$	Weakened photon minimizer and covariance-root families, including the level-zero diagonal family.	eq. 3.xliv
$\alpha_0, n_P, M = L^m$	Weakening-radius exponent, uniform parameter-path count 402, and polymer block size.	eq. 3.xlv
C^P, γ^P	Weakened photon-kernel envelope and decay rate.	Prop. 3.14
Pm	Parameter-product envelope in the discrete product/Hölder-difference proof.	§3.1.7
$\varepsilon, t_{i,k}, R_k$	Small RG exponent, three photon regularity thresholds and the corresponding complex small-field domain.	§3.1.8
$c\widetilde{R}_k, \widetilde{R}_k^-$	Domain enlarged by local real gauges; $\widetilde{R}_k^-$ has the same local bounds as $\widetilde{R}_k$ and smaller bounds on the global remainder.	§3.1.8
Dom_k	Real unit-bond fields with $H_k^{\mathrm{ph}} A \in \frac{1}{4}\widetilde{R}_k$; arbitrary real harmonic parts are allowed.	§3.1.8
C_w, γ_w, c_γ	Common coefficient in the kernel bounds, decay rate and the bond/coordinate exponential row-sum bound.	eq. 3.l
$p_{0,k}, p_k, p_0, p$	Logarithmic coordinate and curvature cutoffs and their fixed powers.	eq. 3.l
$\bar{e}_0$	Fixed running-charge ceiling for the kernel estimates, valid at every smaller charge.	eq. 3.l
$\mathscr{Z}_k(s,\zeta)$	Weakened reconstructed photon fluctuation.	Prop. 3.15
rad_1	Complex fluctuation interpolation radius preserving the open photon domain.	eq. 3.lii
$\mathrm{pj}^*, J, \ell_1, \ell_2$	Periodic photon/coordinate pullback, odd linear cover ratio and the two torus sides.	eq. 3.liii
chamber decomposition, symbols and local inverse construction		
$a, x = (c,g)$, $\mathrm{d}x$	Fermion mesh, photon-site/chamber-label coordinate and normalized chamber measure $a^3/48$.	§4.1.1
J, J^*, P, Q (cluster)	Constant-label injection, cluster mean, light projection and heavy projection $1-P$.	§4.1.1
$M_i(x), R_i(A), \mathcal{L}_A$	Oriented hopping matrices, covariant neighbor maps and weighted chamber graph Laplacian.	eq. 4.ii
e_a, u_A; D_A	Charge at mesh a, photon-bond phase and mesh-normalized massless chamber Dirac matrix.	eq. 4.ii
B_k	Unit-mesh bond phase corresponding to the level connection in $D_A = \eta_k^{-1} D^{\mathrm{ch}}[B_k]$.	eq. 4.iii
$\mathcal{L}_{\mathrm{int}}, \lambda_{\mathrm{int}}, g_{\mathrm{h}}$	Internal cluster Laplacian, its positive gap and resulting heavy Dirac gap.	§4.1.1

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Notation (continued)

Symbol or family	Meaning and scope	Defined at
c_*, c_{lh}	Cluster-to-photon coercivity constant and light–heavy mixing bound.	eq. 4.v
∇_A; $\mathcal{O}_v(c)$	Covariant photon-lattice difference and the eight chamber sites crossing a photon bond in direction v.	eq. 4.v
$V = \ell^2(O_{\mathrm{cub}}; \mathbb{C}^4)$	Single-cluster Fourier-fibre space with label measure 1/48.	§4.1.2
$\mathcal{D}_m(p)$; $D_{\mathrm{ll}}, D_{\mathrm{lh}}, D_{\mathrm{hl}}, D_{\mathrm{hh}}$	Massive chamber symbol and its light/heavy blocks.	§4.1.2
$H_m(p), B_m(p)$	Heavy inverse block and light Schur complement, unrelated to photon minimizers.	§4.1.2
$t(p), E_m(p), K_\rho$	Lattice momentum size, quadratic Schur remainder and its uniform bound.	eq. 4.x
$\theta_0, c_{\mathrm{e}}, \kappa_{\mathrm{strip}}$	Heavy-symbol analytic strip, light ellipticity constant and mass-scaled inverse strip rate.	Prop. 4.2
t_0, c_{l}, K_s	Small-momentum transition, light ellipticity lower bound and analytic Schur-symbol envelope in the proof.	§4.1.2
$G_m, G_m^{\mathrm{l}}, G_m^{\mathrm{h}}$	Free chamber kernel and its label-constant light and improved-decay heavy parts.	Prop. 4.3
C_G; $\widehat{\ }$	Uniform free-kernel envelope and Fourier-transform notation with the displayed Brillouin-zone inverse convention.	Prop. 4.3
$G(x, x')$ (mesh kernel)	Mesh-a free inverse kernel $a^{-2}G_{\mu a}(x/a, x'/a)$.	eq. 4.xix
$d(c, c'), d(b, b')$	Periodic or covering sup distance; bonds use their initial points, and supports use the infimum over pairs of points.	§3.1.1
$d_X(x, \widetilde{x})$	Chamber-pair distance with mesh lower cutoff.	§4.1.3
c_b, P_E, C_P (fermion coercivity)	Dirac coercivity constant, finite-cube mean and chamber Poincare constant.	eq. 4.xxii
$\Delta_z, E_{z'}, \Pi_1, \Pi_L$	Unit and L blocks and their cluster-inclusive averaging projections.	§4.1.4
b, b_L, b_-, b_k	Averaging-penalty normalization, one-step penalty, positive lower bound and accumulated penalty.	§4.1.4
μ (propagator)	Independent positive level reference mass in the inverse estimates.	eq. 4.xxiii
y, α_y, β_y	Real resolvent parameter and its two spin-valued averaging penalties; positive levels also allow $y = \infty$.	eq. 4.xxiii
$F_y(A), \mathrm{Pen}$	Fine resolvent matrix and its block-averaging penalty; F_y is not field strength.	eq. 4.xxiii
C_I, C_5, γ_5'	Uniform zero-field inverse and local Hölder/decay constants.	Prop. 4.5
c_r, a_z, C_{00}	Level-zero real-part gap, allowable exponential-weight rate and resolvent row-bound constant.	§4.1.4
K_1, c_w, a_0	Coercive-estimate coefficient, commutator envelope and allowed exponential weight in the positive-level inverse proof.	§4.1.4
B, B^+, B', c°, r	Nested covering cubes and center/radius in the local free-kernel regularity argument.	§4.1.4
$\tau A(c, c'), \varphi_A(z, c)$	Average coordinate-order path integral and nested-block gauge transport phase.	§4.1.5

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Notation (continued)

Symbol or family	Meaning and scope	Defined at
$Q_k(A), Q_k^T(-A)$	Four-spin fine-to-unit average and its reversed-phase injection.	§4.1.5
$Q(A), Q^T(-A)$	Unit-to-L average and its reversed-phase injection, distinct from the cluster heavy projection Q.	§4.1.5
$\Pi_1(A), \Pi_L(A)$	Gauge-covariant unit and L averaging idempotents on the chamber space.	§4.1.5
$U_A(x, x'), \delta_{\alpha,A}$	Six-path pair transport and its covariant Hölder difference, including intra-cluster pairs.	§4.1.5
$N_\gamma, N_\gamma^{\ker}$	Exponentially weighted unit-block operator row norm and the level-zero normalized-kernel variant.	eq. 4.xxix
$w_k(y), a', \gamma$	Level-dependent resolvent weight, common row envelope and exponential row rate.	eq. 4.xxix
δ, C_V (inverse perturbation)	Small connection size and explicit perturbation constant $2\exp(1)(42+6b)$.	Prop. 4.6
$\widetilde{\mathcal{G}}_k$	Connections with a small global complex part and locally small real-gauge representatives.	§4.1.5
$W_\square, W_\square^-, A^W$	Local inverse neighborhood, its interior and the folded periodic connection on the auxiliary $6M$ torus.	§4.1.5
$\mathrm{ext}, \mathrm{res}, S_\square(A)$	Zero extension, local restriction and the folded-torus local resolvent.	eq. 4.xxxiii
$h_0, h_\square, C_\square$	Scalar patch cutoff, its product partition with $\sum h_\square^2 = 1$, and the commutator $-[F_y, h_\square]$.	§4.1.6
$S^{\mathrm{p}}, R^{\mathrm{p}}, q_0$	Patched parametrix, its right-inverse error and geometric-series norm bound.	eq. 4.xxxviii
$\omega, S_\omega, \lvert\omega\rvert$	Cube walk, its local inverse-product term and its number of commutator steps.	§4.1.6
$q(p)$	Weighted walk-series ratio $54\pi C_V a' p/M$; not the electric charge.	Prop. 4.7
weakened fermions, weighted Grassmann integration and determinants		
O^T (weighted transpose)	Transpose for the stated counting, chamber or coarse bilinear pairing, without complex conjugation.	§4.2
Y_0, Y_k	Resolvent parameter sets $\mathbb{R}$ and $\mathbb{R}\cup\{\infty\}$, with $-\infty=\infty$ at positive levels.	§4.2.1
$\pi_k, \mathcal{P}_k, \lvert X\rvert_M$	Partition into M-cubes, nonempty face-connected polymers and their cube count.	§4.2.1
$d_M(X), X^{\sim r}$	Net-based polymer tree length in units of M and cubical layer enlargement.	eq. 4.xlii
$X_\omega, s_\omega, s_\square$	One-layer-enlarged walk polymer, its weakening monomial and cube parameters.	eq. 4.xlii
$\alpha_0 = 1/54$; κ'	Fermion weakening exponent and enlarged polymer-decay rate.	§4.2.1
$S_y, S_y', S_y^{\mathrm{sy}}, S_y^{\#}$	Right/left weakened resolvents, their transpose-symmetrized average and notation for any of the three choices.	eq. 4.xliii
$S_y^{\#}(X, A), S_{k,y}(X, A)$	Polymer-localized resolvent terms; the second notation retains the level index for determinant integrals.	eq. 4.xliii

Continued on next page

Notation (continued)

Symbol or family	Meaning and scope	Defined at
K_c	Photon-site cube of radius $2L$ whose active parameter cubes are required for a nonzero fermion row or column.	Prop. 4.8
$\mathcal{Q}_k, \mathcal{Q}_k^\sharp, P_L$	Cluster-inclusive fine average, reversed-phase injection and unit-lattice L-block projection.	§4.2.2
$S_k, H_k, \mathfrak{H}_k$	Fine inverse at $y=\infty$, fermion field map and its opposite-direction field map.	eq. 4.l
D_k, M_k	Blocked coarse Dirac matrix and its complementary kernel $b_k - D_k$.	eq. 4.l
$\mathcal{B}_{k,y}$	Explicit inverse of $b_k + b_L P_L + i\gamma_5 y$ on the unit-spinor space.	eq. 4.li
$\Gamma_{k,y}, \Gamma_k$	Coarse resolvent with parameter y and its zero-parameter covariance.	eq. 4.lii
$\delta Z_k(A)$	Fermion block determinant; at level zero it is the determinant of the chamber integration matrix.	eq. 4.liv
F_0, Γ_0	Level-zero penalized chamber matrix and inverse, acting on 192 components per photon site.	eq. 4.liv
$H_k(s,A), \Gamma_k(s,A), M_k(X,A)$	Weakened field map/covariance and polymer-local coarse kernel.	eq. 4.lv
$H_0(s,A), \Gamma_0(s,A), \mathrm{Cub}(c)$	First-step diagonal weakening, symmetrized covariance and all partition cubes meeting K_c.	eq. 4.lvi
$\mathsf{i}(A), C_\Gamma$	Charge-weighted imaginary connection size and the coarse covariance kernel envelope.	eq. 4.lx
$\bar{H}_k(s,A)$	Barred field map $\mathsf{C}^{-1}H_k(s,-A)\mathsf{C}$.	§4.2.3
$\lvert\cdot\rvert_c, \lvert\cdot\rvert_2$ (spin components)	Component maximum and Euclidean spinor norms, with the factor-two conversion used in Grassmann row estimates.	§4.2.3
$\phi_i(I_i), c_F(I), K_n$	Ordered Grassmann monomials, their coefficients and separately antisymmetric integral kernels.	eq. 4.lxi
$\mathsf{w}_i, \nu_i, \lVert F\rVert_{\mathsf{w}}$	Generator weights, index measures and the weighted coefficient ℓ^1 norm.	eq. 4.lxi
$W, \bar{W}$, Top, $\mathrm{d}W$	Additional fluctuation generators, their ordered top product and weighted Berezin extraction.	eq. 4.lxiii
$\mu_I(W), \mu_C(W)$	Grassmann Gaussians with identity and invertible covariance, distinguished from the real photon Gaussian by their arguments.	eq. 4.lxiii
w_W, w_W	Fluctuation-generator integration weight $\max \nu^{-1/2}$; 1 at positive levels and $\sqrt{48}$ at level zero.	eq. 4.lxviii
Pf	Pfaffian of the antisymmetric contraction matrix in the weighted Wick formula.	eq. 4.lxvii
$O_{ij}(t,s), \mathsf{w}'_j$	Linear Grassmann substitution kernels and target-generator weights entering their coefficient row bounds.	Prop. 4.11
$\mathcal{D}_k, \mathcal{A}_k, \mathcal{A}_k^\sharp$	Common notation for the integration matrix, next block average and reversed-connection transpose, including level zero.	§4.2.5
K_k, D_{k+1}^0	Penalized integration matrix and the unscaled outgoing quadratic Dirac operator.	eq. 4.lxix

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Notation (continued)

Symbol or family	Meaning and scope	Defined at
$\Psi^c, \bar{\Psi}^c$	Independent coarse-field-induced incoming Grassmann means in the square completion.	eq. 4.lxix
n_k, c_k^d	Unit-site count and explicit free penalty contribution per site in the determinant integral.	eq. 4.lxxi
λ_j (determinant proof)	Real eigenvalues of the Hermitian matrix $K_k\gamma_5$, counted with multiplicity.	§4.2.6
$\gamma_{\mathrm{w}}, \kappa_{\mathrm{w}}, C_w$	Spatial decay, polymer decay and common component-norm envelope for local determinant propagators.	§4.2.7
$\widetilde{E}_k, E_k^d, E_k^{\mathrm{det}}$	Raw determinant logarithm, charge/reality-symmetrized local term and its zero-field subtraction.	eq. 4.lxxiii
c_X, C_d	Polymer-volume counting constant and explicit local determinant envelope.	eq. 4.lxxiv
$N_y(X), \mathcal{L}(A)$ (determinant proof)	Block-compressed polymer resolvent and sum of raw local determinant logarithms.	§4.2.7
$R_k, \widetilde{R}_k$ (determinant domain)	Complex fields satisfying three norm bounds, and their enlargement by local real gauges.	§4.2.8
$\mathfrak{X}_k$	External connection domain with its two charge-dependent coefficient/derivative thresholds.	§4.2.8
v_k, r_k	Logarithmic charge size $-\log e_k$ and the later logarithmic stopping-radius scale.	eq. 4.lxxviii
$W, \Gamma(o,c), \lambda, B$ (gain proof)	Local lifted neighborhood, ordered path, real gauge and globally small representative used for determinant Cauchy gain.	§4.2.8
activity spaces, extraction, reconstruction and contraction		
$\Sigma_3, \mathsf{C}, \varkappa$	RG spin matrices $\sigma_3 \otimes 1_2$, $i\gamma_2$ and $\gamma_5\mathsf{C} = \sigma_2 \otimes \sigma_2$; $\mathsf{C} = i\Sigma_3 C$ relates the regulator convention.	§5.1
$\mathrm{Vol}(X)$; small X	Polymer volume $M^3\lvert X\rvert_M$; smallness means $d_M(X) \le L$.	§5.1.1
$\psi, \bar{\psi}, \chi, \bar{\chi}$ (RG)	Independent ordinary and ordered-pair Grassmann legs; the pair fields are Hölder legs, not boundary characters.	eq. 5.i
ξ, ζ, ω (leg indices)	Ordinary and pair indices and the bar-status index $\omega = 0, 1$; ζ here is not the real photon fluctuation.	§5.1.1
$\nu(\xi), \nu(\zeta)$	Ordinary and pair index measures $\eta_k^3/48$ and $(\eta_k^3/48)^2$.	§5.1.1
$\mathscr{E}, \mathscr{E}_{nm}$	Balanced activity family and its separately antisymmetric ordinary/pair kernels.	eq. 5.i
$h_k = (h_{k,1}, h_{k,2})$	Ordinary/pair generator weights $e_k^{-1/4}$ and $e_k^{-1/4+\varepsilon}$.	eq. 5.ii
$\lVert\cdot\rVert_{h_k}$; $\lVert\mathscr{E}\rVert_k^x$	Weighted Grassmann coefficient norm and connection-uniform exponentially weighted polymer norm.	eq. 5.iii
$K_k^{x,\mathbb{C}}, K_k^x$	Two-connection complex and real activity spaces with support, holomorphy, balance and covariance requirements.	eq. 5.iv
$K_k^{\mathbb{C}}, K_k$; $\mathscr{E}\lvert_0$	One-connection activity spaces and restriction to $\mathscr{C} = 0$.	§5.1.1
$\widetilde{g} = (S, a), r(S), a_{\widetilde{g}}$	Lifted scheme symmetry, its coordinate-sign action and induced Grassmann automorphism.	eq. 5.iv
$G_\Lambda, \mathrm{Cg}, \mathbf{K}$	Local-gauge, charge-conjugation and antilinear-reality automorphisms of the activity generators.	§5.1.1

Continued on next page

Notation (continued)

Symbol or family	Meaning and scope	Defined at
$w_k^{\mathrm{m}}, w_k^{\mathrm{E}}$; $\mathrm{Ball}_k^x(c)$	Mass/activity windows and their scaled product ball; level zero permits only zero incoming activity.	eq. 5.v
$\mathscr{T}$; $\mathrm{ev}_{\mathscr{T}}$	Total connection $\mathscr{A}+\mathscr{C}$ and evaluation of independent legs at reconstructed fields.	§5.1.1
$\widetilde{G}_k$	Domain $\widetilde{\mathcal{G}}_k$ for the inverse estimates: a real field made small by local real gauges, plus a globally small complex remainder.	§5.1.2
$\mathfrak{i}(B), q_d(B)$	Charge-weighted imaginary connection and curvature sup norms.	eq. 5.viii
$\mathscr{E}^{[n,m]}$; K_X	Bidegree component of an activity and its ordinary balanced quadratic matrix kernel.	eq. 5.ix
$a_0(\mathscr{E}, X), a_2(\mathscr{E}, X)$	Volume-normalized zero-connection vacuum and integrated quadratic extractions on a small polymer.	eq. 5.x
$\varepsilon_{\mathrm{v}}(\mathscr{E}, \square), M(\mathscr{E}, \square)$	Local relevant vacuum and matrix mass counterterms obtained by summing small polymers through a cube.	eq. 5.xi
$\varepsilon_{\mathrm{v}}(\mathscr{E}), m(\mathscr{E})$	Translation-independent vacuum extraction and scalar mass extraction.	Prop. 5.1
$\kappa_0, K_{\mathrm{pol}}, c_X$	Fixed polymer counting exponent, summability constant and tree-length-to-volume constant.	§D
$R\mathscr{E}$ (normalization)	Activity remainder after subtracting the two relevant local terms; R here is not a torus side.	eq. 5.xv
$T_k(A)$	Local left inverse of the fermion field map $H_k(A)$.	eq. 5.xviii
$c_j(c), \tau_k A$	Centers of nested blocks and sum of averaged path transports from a fine point to its unit-block center.	eq. 5.xx
$T_Y^{(1)}, T_Y^{(2)}$	Rows that reconstruct a coarse field over Y using interior ordinary generators and outward boundary pairs.	eq. 5.xxiii
$\|\cdot\|_{\mathrm{su}}, C_T$	Operator norm for the Euclidean spinor sup norm, and a common bound for the coefficient row sums in reconstruction.	eq. 5.xxvi
E_1, E_2, E_3 (row proof)	Interior, same-polymer external-block and outward-boundary site classes in the reconstruction row estimate.	§5.1.4
$j(c, a), v(c, a)$	First common nested-block scale and phase defect for neighboring photon sites.	eq. 5.xxx
$\bar{T}_Y^{(i)}, \lambda^{\mathrm{u}}, \sigma(S)$	Barred reconstruction rows, gauge restricted to unit sites and canceling spin-transpose sign.	§5.1.4
$\mathrm{Mn}, p_i, U_i^{\pm}$	Mean over shortest cube chains, their connecting chamber points and partial forward/inverse phase products.	eq. 5.xxxvi
$\Theta_{X,B}$ (adjustment)	Ordinary-leg homomorphism into transported mean legs and admissible pair differences; distinct from reflection.	eq. 5.xxxvi
$\mathscr{E}^{\mathrm{nat}}$	Quadratically adjusted activity, with unchanged value after covariant field evaluation.	eq. 5.xxxvii
ψ_{av}; $E = (R\mathscr{E})^{\mathrm{nat}}$	Mean ordinary leg and the normalized adjusted family; its vacuum and ordinary quadratic kernels vanish at zero connection on small polymers.	eq. 5.xxxviii
L_ψ, L_χ, s_K	Ordinary/pair pieces of the adjustment row and the absolute integrated quadratic-kernel coefficient.	§5.1.5

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Notation (continued)

Symbol or family	Meaning and scope	Defined at
$\bar{\varepsilon}, \theta_i, d_i$ (thresholds)	Reference exponent ceiling, three small-field norm thresholds and two external-connection norm thresholds.	§5.1.6
A_L, λ_L	Photon-connection and scalar-gauge scaling with powers $L^{-1/2}$ and $L^{1/2}$.	eq. 5.xli
S_L; $p(n,m)$	Grassmann scaling homomorphism and its bidegree contraction exponent.	eq. 5.xlvii
$\bar{X}^L, \mathcal{L}$ (reblocking)	Smallest containing union of LM-cubes and scaled sum over its fine-polymer preimages.	eq. 5.xlix
c_{rb}	Uniform polymer reblocking summation constant $c_X K_{\mathrm{pol}}$.	§D
$\lambda_*, \mathscr{K}_1, \mathscr{K}_2$	Real local gauge and two constant connection pieces on a lifted small polymer.	eq. 5.l
$\mathscr{A}^{\mathrm{b}}, \mathscr{C}^{\mathrm{b}}$	Globally small representative fields admitting the analytic disk $\lvert t\rvert \le L^{7/4}$.	eq. 5.li
b_1 (constant-field bound)	Allowed norm bound for the first extracted constant connection piece.	eq. 5.lii
K_{loc}, c_*, r (local gauge proof)	Nonwrapping local box, chosen base point and cutoff radius $2LM$.	§5.1.7
$f(t), \mathrm{cut}_r$	Piecewise-quadratic cutoff and its product action on bond fields before zero extension.	§5.1.7
$V[b], \mathrm{Lin}[b], \mathrm{rem}[b]$	Basepoint derivative matrix, linear Taylor field and first-order remainder of a connection.	§5.1.7
V^0, V^{s}, F (local Taylor proof)	Derivative matrix, its symmetric part and antisymmetric curvature remainder; F here is a local matrix.	§5.1.7
$\mathscr{K}_j^{\mathrm{R}}, \mathscr{K}_j^{\mathrm{I}}$	Real and imaginary parts of the constant fields continued through the local gauge identity.	Prop. 5.4
$\Omega, A(z), C(z), \Lambda_z, H(z)$ (continuation)	Connected complex parameter domain, continued connection pair, affine scalar gauge and gauge-compensated kernel in constant-field removal.	§5.1.7
c_{fl}	Fixed envelope for normalized activity contraction and relevant-counterterm flow bounds.	eq. 5.lxii
$N(X), \rho = L^{7/4}$ (contraction proof)	Connection-uniform adjusted polymer norm and analytic disk radius; this ρ is not the Wilson parameter.	§5.1.8
exact density, normalization and the first vertex		
$\mathcal{S}(a,\ell), \mathcal{S}^{\mathrm{ch}}(a,\ell)$	Finite antiperiodic unit-spinor and chamber-spinor spaces, not continuum Schwartz spaces.	§5.2.1
$\mathscr{V}_k, \mathcal{D}_k(B)$	Incoming integration space and its quadratic Dirac operator, with the separate chamber interpretation at $k = 0$.	§5.2.1
$\mathcal{Q}_k^+, \mathcal{Q}_k^{+,\sharp}, \bar{b}_k$	Fine-to-L average, reversed-phase embedding and outgoing accumulated penalty divided by L.	eq. 5.lxxi
F_k^+, Σ_k^+	Fine matrix with outgoing averaging penalty and its inverse.	eq. 5.lxxi
$\mathcal{B}_k$	Explicit $y = 0$ block inverse $(b_k + b_L P_L)^{-1}$ in the Schur calculation.	eq. 5.lxxii
R_L	Linear scaling bijection on actual chamber/unit spinor fields, distinct from the Grassmann homomorphism S_L.	eq. 5.lxxv

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Notation (continued)

Symbol or family	Meaning and scope	Defined at
$\Psi^c, \bar{\Psi}^c, \psi', \bar{\psi}'$	Incoming Gaussian means and outgoing reconstructed ordinary fields.	eq. 5.lxxvii
n_k, n_{k+1}, c_k	Incoming/outgoing site counts and exact Berezin block normalization $b^{-4n_{k+1}}$.	eq. 5.lxxx
$\mathscr{A}', \mathscr{C}', \mathscr{T}'$	Outgoing reconstructed photon background, external connection and their total.	§5.2.3
$A_k(\zeta), \mathscr{Z}(\zeta), \mathscr{B}_k$	Incoming fluctuating coarse photon field, reconstructed fluctuation and gauge-related incoming background.	eq. 5.lxxxi
I_k^x, Act_k^x	Evaluated interaction action and full quadratic-plus-interaction action with external connection $\mathscr{C}$.	eq. 5.lxxxiii
$\mathfrak{P}_k(X, C)$	Arbitrary finite external-only scalar family retained with its exact normalization.	eq. 5.lxxxiv
$\varepsilon^{\mathrm{v}}, m, \mathscr{E}$	Incoming vacuum density, mass interaction and balanced activity in the exact RG density.	eq. 5.lxxxiv
$\mathscr{F}_k^x[\varepsilon^{\mathrm{v}}, m, \mathscr{E}]$	Normalized Grassmann density with vacuum, mass, activity and external scalar factors.	eq. 5.lxxxiv
$\mathbf{T}_k^x, \mathcal{Q}_{\mathrm{bl}}$	Exact photon/Berezin block transform and its gauge-adjusted quadratic blocking exponent.	eq. 5.lxxxv
$m', m^+, \varepsilon^{\mathrm{v},+}, \mathscr{E}^r$	Extracted mass, scaled mass/vacuum outputs and normalized incoming remainder.	Prop. 5.8
$\psi^{0,\#}, \Psi^f, \bar{\Psi}^f$	Scaled outgoing ordinary/pair fields and coarse means plus independent fermion fluctuations.	Prop. 5.8
V_k^x, Ξ_k^x	Exact fluctuation exponent and its normalized truncated photon/Grassmann integral.	eq. 5.lxxxvii
$\mathcal{R}(\Psi)$ (square-completion proof)	Residual action separated from the background quadratic density.	§5.2.3
$\mathrm{sub}_{Y,B}$	Local substitution of independent unit generators by interior ordinary and boundary pair reconstruction rows.	eq. 5.lxxxix
w_1, w_2, w_3 (substitution)	Ordinary-leg, pair-leg and auxiliary unit-generator norm weights in the reconstruction substitution.	eq. 5.xci
$\mathcal{V}_0(Z, \square)$	Explicit local first-step hopping vertex before fluctuation integration.	eq. 5.xciii
V_0^x	First-level fluctuation action with no incoming activity and identity field maps.	eq. 5.xcii
δ_x, δ'_x	Opposite oriented hopping-phase increments in the first-vertex coefficient estimate.	§5.2.5
nonlinear contours, localization and connected logarithm		
e, h (nonlinear step)	Local abbreviations for the running charge e_k and weight pair h_k, not the physical charge or a cell width.	§5.3.1
Pd_k, s', s''	Joint weakening polydisc and its photon and fermion parameter families.	§5.3.1
N_c, c_S	Local parameter-cube count and fixed cluster-smallness envelope.	§5.3.1
$Z(s'), \lvert Z\rvert_*$	Weakened reconstructed photon fluctuation and maximum of its three regularity norms.	eq. 5.ci

Continued on next page

Notation (continued)

Symbol or family	Meaning and scope	Defined at
r_1, r_2, r_4	Closed complex contour radii for connection, activity/mass and fermion-fluctuation differences.	eq. 5.cii
$\psi^{\#}, H_k^{\#}$	Four-tuple of ordinary/pair/barred legs and corresponding reconstructed field-map tuple.	§5.3.1
$J_k^{\#}, J_k, \bar{J}_k$	Connection-increment field-map difference and its two components without Hölder quotients.	§5.3.1
$\widetilde{\Gamma}_k, W_k^{\#}$	Covariance substitution on the independent fluctuation pair and resulting reconstructed fluctuation tuple.	§5.3.1
$\mathrm{Wc}, \overline{\mathrm{Wc}}$	Ordinary unbarred/barred components of $W_k^{\#}$.	§5.3.1
$P_0(X), Y'(X,P)$	Parameter cubes contained in X and the partition-cube support generated by X and selected outside parameters.	eq. 5.ciii
$\partial_P, \mathcal{T}_P$	Mixed parameter derivative and its finite unit-interval interpolation integral.	eq. 5.ciii
$\mathrm{Loc}\, G$	Sum of the interpolation terms $\mathcal{T}_P G(X)$ grouped by the polymer $Y'(X,P) = Y$.	eq. 5.civ
B_0, κ' (resummation)	Uniform analytic-family envelope and input polymer-decay exponent in the finite localization bound.	Lem. 5.11
$R = M^{\alpha_0}$ (resummation)	Weakening polydisc radius in the interpolation/Cauchy proof, unrelated to a physical torus side.	Lem. 5.11
$\mathcal{C}_r$	Contour difference functional with kernel $1/[t(t-1)]$, equal to $f(1) - f(0)$.	§5.3.2
G_1, G_2, G_3	First finite localization inputs: kinetic difference, normalized activity difference and mass difference.	eq. 5.cviii
$E_i^*, E_i, E_{\le 3}$	First localized terms, their local mean-field substitutions and their sum.	§5.3.2
$G_{\le 3}^W, G_4, G_5$	Second localization inputs: first terms with fermion fluctuations and additional activity/mass fluctuation differences.	eq. 5.cxi
$F(Y)$ (finite activity)	Full second-localization output before photon/Grassmann integration; not a field strength.	eq. 5.cxiv
Ps, Y_1'	Active parameter set and the component of $Y'(X,P)$ containing X.	§5.3.2
κ_w, a_1, a_2, C_W	Input decay exponent, first/second localization decay rates and fermion fluctuation row envelope.	§5.3.3
H_{cl}	Scalar smallness envelope for the connected-polymer logarithm, unrelated to any Hilbert space or field map.	eq. 5.cxxii
$\lvert\cdot\rvert_{1,\infty}$; w_3 (finite bounds)	Absolute component row norm and temporary weight of auxiliary unit-spinor generators.	§5.3.3
$u^T(X_1, \dots, X_n)$	Connected incompatibility-graph coefficient, with the one-polymer value equal to one.	eq. 5.cxxiii
$\mathbf{G}, z_X$	Tuple incompatibility graph and formal commuting polymer variables used before Banach-algebra substitution.	eq. 5.cxxiii
$T(G), R(\tau)$ (tree bound)	Rooted least-parent tree of a connected graph and its interval closure in the incompatibility graph.	eq. 5.cxxiv

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Notation (continued)

Symbol or family	Meaning and scope	Defined at
$V(X)$ (cluster input)	Even Grassmann family satisfying the finite-activity norm bound, with fluctuation indices in X and dependence on ζ only through coordinates supported in X.	Prop. 5.15
$K(Y), H(Y)$ (cluster)	Mayer activity on a cube union and its photon/Grassmann integrated activity.	eq. 5.cxxv
$E^{\#}(Y), E_k^{\#}$	Connected local logarithm for a general input and its nonlinear-step specialization.	eq. 5.cxxvii
a_3, c'_S	Output logarithm decay exponent and explicit logarithm envelope constant.	eq. 5.cxxix
a_4, a_5, a_6, H_1, z_S	Intermediate decay margins and scalar polymer/tree majorants in the connected-series proof.	§5.3.4
$\mu(Y), S_n(Y)$ (cluster proof)	Scalar integrated-activity majorant and its n-polymer connected graph sum.	§5.3.4
$p_N(V), E_N^{\#}, D$ (approximation)	Truncated exponential, uniform polynomial logarithm approximant and number of available polymers.	§5.3.4
δ_N (logarithm approximation)	Uniform relative integrated-activity error controlling convergence of the polynomial approximants.	§5.3.4
$\mathbb{G}_k^L$	Chamber scheme symmetry group with translations in $LM\mathbb{Z}^3$, preserving all parameter partitions.	§5.3.5
$\phi^{\#}, \mathrm{ad}, \mathcal{V}(X)$	Independent leg copies, their addition homomorphism and the auxiliary source algebra used to check Wick symmetries.	§5.3.5
$\phi_\pm, \bar\phi_\pm, \phi', \bar\phi'$	Reconstructed fluctuation fields at opposite connections and their charge-conjugate substituted versions.	§5.3.5
$E_k^*(m, \mathscr{E}), E_k^*(0,0)$	Reblocked scaled nonlinear correction and its charge-source term at zero interaction data.	eq. 5.cxxxix
D, r, B (coupling Cauchy)	Normalized data separation, contour radius $3/D$ and common correction bound in the direct coupling-difference estimate.	§5.3.6
$\delta E_k^{\mathrm{det}}$	Difference of local determinant terms between total and external-only connections.	§5.3.7
$\mathscr{E}^+, m^+, \varepsilon^{\mathrm{v},+}$	Full one-step activity, mass and vacuum outputs.	eq. 5.cxli
$\mathfrak{P}_{k+1}$	Reblocked external scalar family including the determinant contribution at the external connection.	eq. 5.cxliv
$\mathbf{T}_k^x \mathscr{F}_k^x$	Exact incoming density after the block transform, identified with the fully normalized outgoing density.	eq. 5.cxlvi
covers, projected primitives and common local algebras		
$C^F, \gamma^F, C^P, \gamma^P$	Constants and decay rates in the fermion and photon kernel bounds used in the cover construction.	eq. 5.clii
$C_w^{(1)}, C_w, \gamma_w$	Top-system envelope, common projected-system envelope and common spatial-decay rate.	eq. 5.clii
$T_i = \mathbb{L}_k/\ell_i\mathbb{Z}^3$; J_c, p, p^*	Two level tori, odd linear cover ratio, quotient map and its periodic/antiperiodic pullback.	eq. 5.cliii
$\chi(w)$ (cover character)	Periodic photon character or simultaneous antiperiodic fermion/pair character $(-1)^{w_1+w_2+w_3}$.	eq. 5.cliii

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Notation (continued)

Symbol or family	Meaning and scope	Defined at
$\widetilde{O}(s,A),\widetilde{O}(X,A),O_1$	Pulled-back top family, polymer-preimage sum and restriction to the smaller character subspace.	eq. 5.cliv
$d=2L+1,N(\widetilde{X})$	Cover comparison buffer scale and six-buffer enlargement of the smallest rectangular cube union.	p. 135
diam $\widetilde{X}$; flat polymer	Largest rectangular side of a covering cube union; a flat polymer has a connected admissible lift.	p. 135
admissible $\widetilde{X}$	Covering cube union with diameter at most ℓ_1-14dM, so its comparison neighborhood embeds.	p. 135
common local algebra	Identification of covering ordinary/pair/fluctuation generators, preserving bar status, ordering and coefficient norms at matching connections.	Lem. 5.19
$B_i(t)=A_i+\mathscr{C}_i+tZ_i(s')$	Paired contour connections whose kernel rows are identified on the common local algebra.	Thm. I
$\mathscr{E}_i,\mathfrak{P}_k$	Incoming activity and external scalar data compared on flat supports; output equality uses the explicit local agreement hypotheses.	Thm. I
scales, tuned trajectories and terminal coordinates		
h,K,a_j,e_j,μ_j	Physical mass scale, top level, level length, running charge and positive reference mass; m_j is a separate interaction.	eq. 6.i
$v_j,r_j,p_j,p_{0,j}$	Logarithmic charge, rounded localization range and large-/small-field cutoffs.	§6.1.1
$R_{\mathrm{top}},R^*,T^*(e)$	Physical top/reference sides and reference volume exponent.	eq. 6.iii
$\mathcal{T}(e),\ell_j(R),T_j(R),\mathfrak{n}_j,\mathrm{Cl}$	Reference-side family, level sides/tori, depth and admissible unit-side class.	§6.1.1
$\widehat{v}_s,\widehat{r}_s,\widehat{n}_s(R),s_*(R,e)$	Depth-indexed scales and their last-zero stopping depth.	Prop. 6.1
$k_*(R),k_{\mathrm{top}}$	Stopping level on the chosen side and on the top torus.	eq. 6.vi
$N_{\min}(R,e),N_{\mathrm{ref}}(e)$	Individual and common reference mesh thresholds.	eq. 6.vii
$a_*,e_*,\mu_*,\ell_*,r_*,v_*,p_*,p_{0,*}$	Starred running data are evaluated at the chosen torus's stop k_*.	eq. 6.v
$w_j^{\mathrm{m}},w_j^{\mathrm{E}}$	Running mass/activity normalizing weights.	§6.1.2
$m_{\mathrm{in}},\varepsilon_{\mathrm{in}}^{\mathrm{v}},m_j,\varepsilon_j^{\mathrm{v}}$	Incoming real mass/vacuum pair and running scalar couplings.	eq. 6.ix
$E_j,\mathscr{E}_j,\mathfrak{P}_j$	Activity at $\mathscr{C}=0$, activity with external connection $\mathscr{C}$, and background scalar family along the trajectory.	eq. 6.xi
Φ_j,Φ_j^x,d_m,d_E	Activity step maps and normalized input distances.	eq. 6.xii
$\|\mathbf{z}\|_{\mathfrak{fl}},\mathcal{B}$	Weighted finite-trajectory norm and closed unit set with $E_0=0$.	§6.1.3
$\widehat{m}_j,\widehat{E}_j$	Backward-mass/forward-activity fixed-point map.	eq. 6.xvii
b,B_j,V_j (trajectory proof)	Vacuum-bound exponent and normalized mass/vacuum sizes; not the averaging penalty or fluctuation space.	§6.1.3
$P_N(e)=(m_{0,N},\varepsilon_{0,N}^{\mathrm{v}})$	Common microscopic mass/vacuum pair constructed on the top torus.	§6.1.4
$\boldsymbol{\mu}_N$	Total fine lattice mass $\bar{m}\eta+m_{0,N}(e)$.	Thm. K
$A_j^{\downarrow},\zeta_j$	Downward photon fields and their fluctuation coordinates.	eq. 6.xxi

Continued on next page

Notation (continued)

Symbol or family	Meaning and scope	Defined at
$\mathbf{Q}^{(j)},B^{[j]}$	Iterated photon average and connection rescaling in the coordinate construction.	§6.2.1
$\mathrm{Ax}_k,\mathrm{Sl}_k$	Multiscale axial slice and its mean-zero coclosed terminal slice.	§6.2.1
$V_{\mathrm{t}},F_N,\gamma_N$	Transverse potentials, exact fine-curvature space and standard Gaussian; F_N here is a space.	§6.2.1
$\mathcal{J}_{\mathrm{ph}}$	Curvature-to-terminal/fluctuation-coordinate bijection.	eq. 6.xxiii
m_*,μ_I	Terminal photon Gaussian and identity-covariance fluctuation Gaussians.	eq. 6.xxv
ϕ_j,Υ,ϕ_p	Curvature vectors representing fluctuation and terminal-plaquette coordinates.	eq. 6.xxvi
$\mathscr{C}_*,\mathscr{A},B$	Terminal external connection, photon background and total connection.	eq. 6.xxix
$Y_*^x(A;\mathscr{C}_*),Y_*(A)$	Normalized terminal Grassmann density, with external connection or at zero.	eq. 6.xxix
$w_N(B),B_N(a)$	Fine determinant/vacuum density and fixed-cutoff complex-source majorant.	eq. 6.xxxi
$\nu^{\theta,a}$	External-source complex curvature measure at fixed harmonic twist.	§6.2.2
$\mathfrak{G},\chi_*$	Multiscale small-field event and terminal-curvature indicator.	eq. 6.xxxii
zero-flux Gaussian bounds and tails		
$\bar{\mu},\bar{\nu},\nu^\theta$	Normalized average zero-flux law, unnormalized average and fixed-twist zero-source measure.	§7.2
P_k (flux),S_2,S_3,b_ξ,K_j^ξ	Magnetic-flux projector, slice cycles, Bloch shift and electric kernels; not invariant-observable projections.	eq. 7.i
$Z^{(0)}[\,\cdot\,]$	Partition functional restricted to zero flux in all directions.	eq. 7.i
$\varepsilon_3,\varepsilon_4$	Excluded-field and terminal-curvature tail errors.	eq. 7.iii
free pressure, terminal errors and partition defects		
$\mathfrak{d}_m(p),p_{\mathrm{ch}}^0(m)$	Determinant of the chamber symbol and infinite-volume free pressure.	eq. 8.i
$\kappa_{(0)},\varepsilon_5(x),C_6$	Free-pressure strip rate, finite-size error and its coefficient.	eq. 8.ii
$z_{r,n}^{\mathrm{free}}$	Harmonic average of the positive-mass free determinant.	eq. 8.iv
$P_{n,\theta}$	Twisted antiperiodic finite momentum grid for the chamber determinant product.	eq. 8.v
f_H,f_S	Heavy- and Schur-block logarithms in the free-pressure proof.	eq. 8.vi
$r_m(p'),C_{\mathrm{lH}},C_{\mathrm{lS}},c_{\mathrm{F}}$	Massive momentum radius and heavy/light derivative/integration constants.	eq. 8.vii
$c_\theta,R_\theta^{\mathrm{tw}},C_\Sigma,\gamma_\Sigma$	Terminal harmonic connection, blocked twist ratio and comparison coefficient/rate.	eq. 8.x
$F(\theta),\widehat{F}(m)$ (twist proof)	Blocked log determinant and its Fourier coefficients.	Prop. 8.2
c_κ	Anchored exponentially weighted polymer sum.	§8.2

Continued on next page

Notation (continued)

Symbol or family	Meaning and scope	Defined at
$\varepsilon_1,\varepsilon_2$	Terminal interaction and blocked-twist errors.	eq. 8.xii
$A^g,\mathscr{A}^0,\mathscr{A}$	Gauge representative, photon lift and lift plus harmonic twist.	eq. 8.xiii
V_C,K_C	Terminal perturbation and coarse Gaussian covariance at external connection C.	eq. 8.xiv
$r_*(\theta,$ $A)$ (density remainder)	Relative unmarked density error; not the localization range r_*.	eq. 8.xvii
$D_{\mathrm{mat}},C_{\mathrm{mat}}$	Reference-torus matter defect and its uniform bound.	eq. 8.xxiii
Q_n,E_n (partition proof)	Interacting/free pressure remainders after bulk subtraction.	Thm. M
$\mathcal{L}(n_1,n_2,n_3),\mathcal{L}(n)$	Scalar Laplacian log determinant omitting zero mode; not RG reblocking.	eq. 11.xx
G_j,K_j (photon defect)	Explicit photon log-determinant comparison constants.	eq. 8.xxiv
$\omega(k'),g(z),M(p),$ c_j (photon determinant)	Transverse dispersion, logarithmic tail, lattice symbol and comparison coefficient.	eq. 8.xxv
$Z_n^{(w)}[1],P_n(w),\varsigma_n$	Flux-sector partitions, sector probabilities and nonzero-flux Gaussian tail.	eq. 8.xxvii
$D_j^{\mathrm{ph}},D_j^{\mathrm{mat}},D_j^{\mathrm{fl}}$	Photon, matter and flux pieces of the two-volume defect.	eq. 8.xxxi
reflection transfer		
$Q_c^\circ,C(ws),O^{[c]}$	Reflection cells, rectangular torus and reflected one-cell observable.	eq. 9.i
$O_{ws}^\sharp,\mathcal{Z}_s(O)$	Fully folded observable and its partition functional.	eq. 9.i
s_0,n_r,N_r,D_j,δ_j	Reference cell count/side, total cells, partition defect and normalized cost.	eq. 9.ii
$\|O\|_*$	Normalized folded reference seminorm; not a terminal activity norm.	eq. 9.iii
positive probe and continuum/free quartic response		
$V_a,S_M,q_{4,M}(a)$	Clifford multiplication, continuum massive Dirac inverse and quartic coefficient; M here is mass.	§10.1
$S_M(z),\widehat{S}_M(p)$	Position/Fourier kernels of the continuum Dirac inverse.	eq. 10.ii
$G = S_M V_a S_M$, $u(p)$	Hilbert–Schmidt factor and momentum majorant in the trace bound.	eq. 10.iv
$s(z),s^{*j}(z)$	Scalar propagator majorant and its convolution powers.	eq. 10.v
$R_a^{(j)}(x,y),r_4(x;M,a)$	Resolvent expansion kernels and diagonal fourth-kernel trace.	eq. 10.viii
$\lambda_x,a^x,A_F,\Psi(F)$	Local Taylor gauge, gauged potential, constant-curvature potential and local quartic coefficient.	eq. 10.xii
$B,A,D_u,h_u,W_1,$ W_2 (constant-field proof)	Axial curvature, symmetric-gauge potential, covariant derivative, squared Pauli operator and perturbations.	eq. 10.xiii
$k_u(t;x,y),\tau,x_p,x\wedge y$	Mehler kernel, magnetic heat parameter, planar coordinates and oriented area.	eq. 10.xiv
$\check{h},\alpha,\phi$ (Mehler proof)	Planar magnetic operator and Gaussian exponent parameters.	eq. 10.xiv

Continued on next page

Notation (continued)

Symbol or family	Meaning and scope	Defined at
$k^{(n)}$,$K^{(n)}$,$m_{j,\alpha}$, $\mu_j(s)$ (heat proof)	Heat/resolvent coefficients and scalar regularity integrals; not running mass μ_j.	eq. 10.xvi
φ,$\beta^{(\ell)}$,a_ℓ,f_ℓ,U_ℓ,I	Compact bump, dilated probe/potential/test, unitary dilation and positive curvature integral.	eq. 10.xvii
D_β,$a = d^*\beta$,$f = dd^*\beta$	Support-cube side and potential/curvature test for response.	§10.2.1
J_η,$\delta_{\eta,\mu}$	Cube average and centered differences of smooth physical tests.	§10.2.1
$a^0_{\eta,\nu}$,$A_{\eta,\nu}$,B_t	Smooth/periodic sampled probe and its phase source family.	eq. 10.xix
$N_0(t)$,c_j,$b_{\mathrm{ch}}(\eta, R)$	Free determinant ratio, Taylor derivatives and fourth logarithmic derivative.	eq. 10.xx
S,S^R,a^R,$b(f)$,$b_R(f)$	Free space/torus inverses, periodized potential and continuum fourth coefficients.	eq. 10.xxii
V,J,$\mathcal{H}_R$,$\mathcal{P}_R$,$\mathcal{B}_\eta$, P_η (free response)	Label space/injection, common Fourier space, antiperiodic momenta, window and projection.	p. 164
R_η,R_0	Physical lattice inverse extended by zero and continuum light-sector limit.	eq. 10.xxv
$T^\pm_\nu$,$U^\pm_\nu$,D_j,$V_{j,\eta}$, V_0 (free response)	Directional chamber/site shifts, source derivatives and rescaled vertices.	eq. 10.xxvi
cell probes and exact external response		
Q_β,z_β,D^{buf}_β	Buffered support cube, center and side $D_\beta + 4$.	§11.1
a'_η,f^η	Smooth finite-difference potential and compact approximate curvature test.	eq. 11.i
b_η,a^o_η,d_η,d^*_η	Midpoint plaquette samples, coexact potential and mesh differential/adjoint.	eq. 11.ii
$\hat{\mathfrak{a}}^o_\eta$,$\hat{\mathfrak{a}}_\eta$	Covering unit-lattice source and torus periodization.	eq. 11.iii
p,w,ℓ_c,Q°_c,N_* (cells)	Odd cell period, half-period width, physical side, open cells and total cell count $8L^3$.	eq. 11.vi
v (placement),B_0,r_0,B_c	Placement and strictly interior support boxes of reflected sources.	eq. 11.viii
g_c,$\hat{\mathfrak{a}}^o_c$,$\hat{\mathfrak{a}}_c$,X_c	Cell wall map, covering/periodic sources and local curvature observable.	eq. 11.ix
v_η	Common normal-ordering variance $\|d\hat{\mathfrak{a}}^o_\eta\|^2$.	eq. 11.ix
$c(y)$,$X_{c(y)}$	Even cell index for a pure translation and its translated observable.	eq. 11.xi
μ_{lat},M_{ch},D_B,$\det_n$	Lattice/shifted chamber mass, massless operator with phases e^{iqB} and finite determinant.	§11.2.1
$V = \|n\|$,B_n, $\mathcal{L}_n$ (external response)	Rectangular volume, site representatives and oriented link orbits.	§11.2.1
$K_n(s)$	Complex-strip determinant majorant, uniform in the real connection at a fixed volume.	eq. 11.xiii
H_θ,$\mathbf{1}_\mu$	Harmonic twist and constant coordinate bond cochains.	eq. 11.xiv
$k^{(\mu)}$,$\lambda^{(\mu)}$	Integer seam cochains and scalar gauges implementing twist periods.	eq. 11.xv

Continued on next page

Notation (continued)

Symbol or family	Meaning and scope	Defined at
$V_{\mathrm{g}},V_{\mathrm{h}},V_{\mathrm{t}},\mathrm{Harm}^2$	Exact/harmonic/transverse one-cochains and constant two-cochains in Hodge decomposition.	eq. 11.xvi
$\mathcal{P}_n^{\mathrm{per}},e_p,\xi_\mu(p),\iota_{\bar{\xi}}$	Periodic momenta, normalized Fourier characters, coboundary symbol and contraction.	§11.2.2
$\Delta_0,\det{}'\Delta_0,\mathcal{L}(n)$	Scalar Laplacian, determinant omitting the constant mode and log determinant.	eq. 11.xx
$\mathcal{G}_n,\mathrm{d}\sigma(A)$	Unnormalized transverse Gaussian integral and Euclidean measure on its slice.	eq. 11.xxii
$\mathcal{N}_n$	Gauge/harmonic factor in zero-flux unfolding.	eq. 11.xxv
$\mathcal{Z}^1_{\mathbb{Z}},\mathcal{R},\mathcal{D}_1,\mathcal{D}_2,\mathcal{F}_{\mathrm{g}},\mathcal{F}_{\mathrm{h}}$	Closed integer cochains, quotient representatives and unfolding domains.	p. 172
$P(A)$ (unfolding)	Curvature Gaussian times observable and matter insertion; not a projection.	p. 172
$F_n,\gamma_n,d_{\mathrm{t}},A(u)$	Exact curvatures, standard Gaussian, transverse coboundary and inverse potential.	§11.2.4
$Z_n^{\mathrm{ph}},z_{r,n}$	Photon Gaussian factor and normalized harmonic-average matter factor.	eq. 11.xxxi
$\mathrm{dens}^{\theta,a},\nu^{\theta,a},\bar{\nu}^a$	External-source density, its complex measure and harmonic-twist average.	eq. 11.xxxiii
$\mathfrak{F}(n),\mathbf{Z}^{\mathrm{x}}_n(a;\Phi)$	Exponentially bounded Borel observables and external partition with optional insertion.	eq. 11.xxxvi
$\mathbb{T}$ (Cauchy contour)	Product-oriented contour surrounding a complex source polydisc.	eq. 11.xxxvii
$\mathbb{E}_n^{\mathrm{av}},X_j$	Harmonic-average zero-flux expectation and linear probes in the generating identity.	eq. 11.xli
$\mathcal{C}_*,\widehat{a}_c,a(t),\bar{Z}^{\mathrm{ext}}(t),\mathring{\Phi}(t)$	Reference cells/sources, fine marked connection, external partition and normalized generating function.	eq. 11.xliii
$H_j(x;v),H_2,H_4$	Hermite/Wick polynomials with variance v; not block reconstruction maps $H_j(A)$.	p. 175
terminal operators, marked supports and covariance		
$\eta_*,T_*,T_*^0,b_-,b_*,\mu_-$	Terminal mesh and fine/coarse tori, penalty bounds and fixed positive mass lower bound.	§11.3.2
$a^{[j]},\widehat{c}_c,\mathscr{C}(t)$	Rescaled connection, terminal cell source and full terminal marked connection.	eq. 11.xlix
$\lvert a\rvert_{\mathrm{ext}},\lvert\delta_\alpha\partial A\rvert_\infty$	External sup/derivative/Hölder norm and its derivative seminorm.	p. 176
$\lvert\beta\rvert_{C^j},U_{\tau'}\beta,g_\nu,G_\nu,W_\nu$	Component derivative norm and reflected/rescaled profiles for uniform probe bounds.	p. 176
$C_\beta,\mathbf{D}$	Explicit external-probe norm and energy constants independent of charge and cutoff.	eq. 11.l
$P_c,P_c^+,d_{\mathrm{sep}},p_\beta$	Marked support centers/neighborhoods, separation and support-count majorant.	eq. 11.liv
p_β^+	Uniform majorant of the stop-scale support count.	eq. 11.cxxxii
$F_*(A),S_*(A),H_*(A),D_*(A)$	Penalized fine operator, its inverse, reconstruction and coarse Dirac operator; F_* is not curvature.	eq. 11.lv

Continued on next page

Notation (continued)

Symbol or family	Meaning and scope	Defined at
$\varphi_A,\phi_A,U_A,\upsilon_A$	Averaging phases, pair transport and their real linear phase functions.	eq. 11.lvi
P_B,P_B^+,ϵ_B	Unit-block support/neighborhood and scaled size of direction B.	eq. 11.lvii
$\lvert O\rvert(y,y'),\lVert O\rVert_\gamma$	Unit-block majorant and exponentially weighted maximum row/column norm.	eq. 11.lvii
$E(y,y'),C^F,\gamma^F$	Terminal fermion decay kernel, calibrated coefficient and rate.	eq. 11.lix
$\mathcal{W},\mathfrak{i}$	Domain of a real background plus a complex external connection, and the scaled size of the connection's imaginary part.	eq. 11.lx
$\mathcal{S},h_{\mathrm{L}},C_H,C_K$	Summed fermion majorant, leg weight and local insertion coefficients.	Prop. 11.3
$\mathrm{Leg}_A(u;y',\beta'),\mathrm{pos}(u)$	Coarse-generator coefficient of a substituted fine/pair generator and its one/two block positions.	eq. 11.lxiii
$c_\tau,C'_K,\mathcal{W}^+$	Complex-enlargement radius, enlarged derivative constant and enlarged connection domain.	§11.3.3
R,Q,C,Y (inverse proof)	Fine massive operator, compressed average/inverse and finite-rank Schur correction.	eq. 11.lxix
$g,h_G,g_1,\theta_{\mathrm{CT}},\vartheta_0,\alpha_G$	Coarse Gaussian bounds, Grassmann weight and inverse/tree decay coefficients.	eq. 11.lxxi
$G=D_*(A_r)^{-1},K_C$	Coarse covariance at a real connection and at its complex external perturbation.	eq. 11.lxxii
$a_1,a_2,h_*;\ \lvert\cdot\rvert_{a_i},\lVert\cdot\rVert_{h_*}$	Terminal coarse/fine Grassmann weights and coefficient norms.	§11.3.4
$c(\gamma)$	Infinite-lattice exponential sum used in row/column estimates.	§11.3.4
source logarithms, connected trees and fourth jets		
$C_b,t_x,t_x^+,r,\mathbb{D}(t_x^+)$	Mark radii and source polydisc; r here is a Cauchy radius.	eq. 11.lxxvii
$E_j^{\mathrm{det}},\mathfrak{P}_j$	Subtracted determinant activity and accumulated external scalar family.	eq. 11.lxxix
$Z_{\mathrm{in}}(A,C),Z_{\mathrm{tm}}(t),f_{\mathrm{int}}(t)$	Two-field terminal integral, marked specialization and interaction factor.	eq. 11.lxxxiv
$R_{\mathrm{tm}}(t;\theta,A),\mathcal{G}(t;\theta,A)$	Nonzero density ratio and its logarithm normalized at zero.	eq. 11.xci
$g_D^\theta(t),\ell_0(t),w_c(t)$	Free coarse log, full one-cell free log and one-cell interacting error.	eq. 11.xcv
$C_{\mathrm{fr}},B_{\mathrm{wr}},B_1,B_0,\tau_s$	Free-log gain/wrapping/terminal errors, their sum and twist strip width.	eq. 11.xciii
$B_j,\mathcal{Q}_j$ (probe gain)	Level cell connection and M-cubes meeting its support.	eq. 11.c
$\phi_0(t)$	Fourth-degree Taylor polynomial of N_0.	eq. 11.ciii
$\rho_G(F;w),N_G(F;w),F_p$	Weighted Grassmann kernel sums: ρ_G takes a supremum over one argument, while N_G sums all arguments; F_p are antisymmetric kernels.	eq. 11.civ
w_δ,w_P	Exponential argument-diameter and distance-to-support weights.	§11.4.3

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Notation (continued)

Symbol or family	Meaning and scope	Defined at
$E^a(X;t),E^a_{nm},w^{\mathrm{E}}_*,a_w,q_x$	Mass-absorbed activity, fine kernels, activity weight and Cauchy factor $2/t_x$.	eq. 11.cv
$V(t),K(t),V_0(t),V_+(t)$	Real-covariance interaction, coarse source difference and scalar/positive-degree parts.	eq. 11.cvi
$\bar r,s_F,C_V^{\mathrm{ins}}$	Small tree-norm radius, exponential leg sum and insertion constant.	eq. 11.cviii
$X^+,d_0,d_1,d_2,\mathcal{D}(X)$	Enlarged activity support, marked-set distances and mixed-derivative decay weight.	Prop. 11.6
$\Delta^{ij},\Delta_{\{i,j\}},\Delta[s],\epsilon_0,s^T_{ij}(\tau)$	Replicated contractions, constant-term extraction and forest minimum-path matrix.	eq. 11.cxii
$T_n(F_1,\dots,F_n),W(t)$	Connected tree form and its summed logarithm.	Prop. 11.7
$t^T,f_S,\mathcal{G}_S$	Coordinate restriction and variable-set Möbius components of a source function/logarithm.	eq. 11.cxvi
$C_W^{\mathrm{mix}},\varpi_{\mathrm{sep}},\varpi^+_{\mathrm{sep}}$	Mixed-log coefficient and errors before/after geometric separation.	eq. 11.cxviii
$\mathfrak{G}^-,\varpi_0,t_2,\varpi_L,M_L,\lambda_L$	Lowered small-field event, tail/domain majorants and complement fourth-jet error.	eq. 11.cxix
$\bar\kappa_1,v_{\mathrm{sf}},L(t)$	Singleton error scale, positive small-field measure and normalized entire complement.	Prop. 11.9
$\omega(z)$ (two constants)	Semicircle harmonic measure propagating real-source smallness into a complex disc.	p. 195
$L^{[4]},\mathcal{J},\|P\|_r$	Fourth jet, finite polynomial quotient and weighted coefficient norm (degree at most four per cell).	§11.4.5
$k_S,\ell_c,\ell,\mathcal{Y},a(\mathcal{Y})$	Error jets, recentered derivative functionals and covering families/cost; ℓ_c here is not cell side.	Thm. N
$\mathfrak{e},\sigma,s,r_{\mathrm{loc}}$	Terminal/final charge gain exponents, interpolation fraction and localization power.	eq. 11.cxxxi
$e_\sharp(\beta)$	Probe-dependent charge threshold for all response conditions.	Prop. 11.10
A_8,V,e_V	Complement-jet majorant and sufficient half-line/charge thresholds.	eq. 11.cxxxiv
arbitrary-volume Wick transfer		
$H_m(x;v),O_m^{[c]},W_m(x;v,c_m),e^{-m}$	Wick polynomial and recentered observables; e^{-m} is an inverse physical-charge power.	eq. 11.cxxxvi
$\Theta_m^{(0)},m=2,4$	Full-cell recentered derivative error after its N_*-th root.	eq. 11.cxxxvii
M_2,M_4,T_m,u_m	Moment majorants, combined reference errors and transferred errors.	eq. 11.cxxxviii
$a_L,D^{\mathrm{mat}},\delta$	Geometric loss coefficient, matter defect and specialized normalized partition cost.	eq. 11.cxxxix
$C_{\mathrm{ref}},C_{\mathrm{mat}}$	Uniform reference-response and matter-defect input constants.	eq. 11.cxliv
$J,C_\beta^{\mathrm{fr}},N_{\mathrm{uv}}(e)$	Free-jet majorant, free finite-size coefficient and enlarged cutoff threshold.	eq. 11.cxlvi
$s_\beta,C_\Gamma^{\mathrm{tr}},B_2,B_4,C_U$	Probe energy bound, partition-loss coefficient, flux-moment majorants and final error constant.	eq. 11.cl

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Notation (continued)

Symbol or family	Meaning and scope	Defined at
random fields, tensor moments and limit spaces		
$X(f)$,c_p	Extended random linear field and explicit Gaussian moment constants.	eq. 12.iii
$A = 1 + \lvert x\rvert^2 - \Delta$,$h_\kappa$,$\lambda_\kappa$, $c_{\boldsymbol{\kappa}}(\Psi)$,$Z_4$	Hermite oscillator/basis/eigenvalues, tensor coefficients and summability constant.	eq. 12.v
$M(x)$,M,A, P_j (Hermite proof)	Quadratic multiplication, oscillator and differential momenta in the test-seminorm estimate; not invariant projections.	eq. 12.xiv
Q_Ψ,$S_n(\Psi)$	Random tensor evaluation and its expected Schwinger function.	eq. 12.vi
$\mathcal{S}_n$,q_m	Tensor Schwartz space and explicit growth seminorm.	Thm. A
$\lVert F\rVert_{-s}$,H_{-s}	Negative Hermite Hilbert norm and distribution space; limits are laws on H_{-4}.	p. 203
$Q_\eta(\sigma)$,y_σ	Physical averaging cube and unit plaquette center.	eq. 12.vii
$f_{[\eta]}$,f^R,f_η,$\lvert\cdot\rvert_\eta$	Covering/periodized/torus test maps and mesh-weighted cochain norm.	eq. 12.viii
$F^\sharp$,$\widetilde{u}$	Physical finite-volume step field and periodic curvature extension.	eq. 12.ix
Q_R,$p_R(f)$,$p_1(f)$	Centered period cube and periodization-control seminorms.	eq. 12.xiii
$\mathfrak{g}(y)$	Physical affine symmetry induced by a unit-lattice cubic isometry.	eq. 12.xv
$P_{N,T}$,$\mathcal{V}_N$,$\mu^{(N)}$,μ	Cutoff law, fixed-spacing volume-limit class, a class member and continuum parent.	p. 204
$\partial_j^\eta \xi$,v^η	Centered derivatives and signed two-form test for discrete Bianchi.	eq. 12.xix
$\xi_{[\eta]}$	Cube-averaged three-form test with the scaling used in the exact discrete Bianchi pairing.	eq. 12.xix
Ω',$\Omega^{(\eta)}$,$\sharp_\eta$	Closed cochains with linear growth; those whose step field also has finite H_{-3} norm; and the measurable step map on the infinite lattice.	eq. 12.xx
b^σ,$\mathcal{J}_\eta$,$\nu^{(N)}$	Plaquette-extracting bump, inverse step coordinate map and identified lattice law.	eq. 12.xxi
ℓ,n_ℓ,$r_{\ell,c}$	Height form, its normal and continuum mirror reflection; ℓ here is not torus side.	§12.2.5
$\mathcal{I}$,λ_F	Translation-invariant event sigma algebra and regular conditional ergodic law.	eq. 12.xxiv
$\mathcal{S}(\overline{O};\Lambda^2)$,$\mathcal{D}_0$,$\mathcal{A}_0$	Supported Schwartz subspace, countable rational tests and determining character algebra.	§12.3
M_T,T_k,A_p	Cube-average operator, averaging subsequence and enumerated determining characters in disintegration.	§12.3
$G = e^{F(f)}$,$\phi =$ PG (component bound)	Exponential field observable and its invariant conditional expectation in the component Gaussian proof.	eq. 12.xxvi
$c_{\mathbf{k}}$,χ_a,s (tensor support)	Fourier coefficients, cutoffs and period-cube side in supported tensor approximation.	eq. 12.xxx

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Notation (continued)

Symbol or family	Meaning and scope	Defined at
Ψ_j (half-space approximation)	Translated compact cutoffs of a closed-half-space tensor test, supported strictly in its open side.	eq. 12.xxxii
$\Theta_k\Psi_k,\Theta_0 c,\tau_a^{\otimes l}\Phi$	Reflected tensor test and simultaneous argument translation in the OS axioms.	eq. 2.xxvi
ordered constants and non-Gaussian components		
$\bar{\varepsilon},\alpha,\kappa,c_{\mathrm{fl}}$	Epsilon ceiling, Hölder exponent, polymer decay and flow constant in the ordered construction.	eq. 13.i
$N_c,\widehat{\kappa},M_F,\alpha_0,\kappa'$	Block count, enlarged decay, fermion size threshold and weakening exponents.	eq. 13.ii
r,p_0,p,s,ε (chosen powers)	Localization, field-cutoff, interpolation and RG exponents fixed in the final construction.	p. 211
$C_{\mathrm{By}},C_\Gamma,C_1,C^F$	Explicit coarse-resolvent and common fermion calibration coefficients.	eq. 13.iii
$b,b_-,$ b_+ (penalty calibration)	Fixed averaging penalty and its lower/upper geometric bounds.	eq. 13.iii
$C^P,C_w^{(1)},C_w$	Common photon and cover constants after fermion/photon calibration.	eq. 5.clii
c_C,C_W,C_d,c_γ,C_B	Determinant, fluctuation/localization and local-gauge calibration constants.	p. 212
$C_0,c_{\mathrm{bd}}(\gamma)$	Common terminal row/Grassmann bound and photon boundary summation coefficient.	eq. 13.iv
$V(A,d,\lambda;b),T(A,d,\lambda),$ $\log^+$	Explicit tail threshold, polynomial-exponential term and positive logarithm ($\log^+ 0 = 0$).	eq. 13.v
$\ell(v),V_0,V_1,\mathcal{E}_1,\mathrm{lg}$	Terminal-side majorant, charge thresholds, downward-closed ceiling set and $\frac{1}{2}\log L$.	eq. 13.viii
$\bar{e}_0,\bar{e}_1$	Running and terminal charge ceilings for which the estimates hold at every smaller charge.	p. 214
e_G	Physical charge ceiling before probe restrictions.	eq. 13.x
e_β,C_f	Final positive-probe charge ceiling and cumulant-error coefficient.	eq. 13.xi
$G_2,G_4,d_m = PG_m,Z_n$	Recentered Wick observables, invariant conditional parts and translated empirical average.	eq. 13.xiii
$\mu_\xi,\nu(\,\mathrm{d}\xi),\kappa_4^{\mu_\xi}(X)$	Conditional laws, mixing measure and centered fourth cumulant.	eq. 13.xii
λ_0	Fixed physical translation-grid spacing $R^*(e)/L$ for continuum Wick bounds.	eq. 13.xvi
reflection Hilbert spaces and the spectral cone		
$\Sigma_O,\mathcal{E}_j^+,b_j,\mathcal{H}_j,[A]_j,\Omega_j$	Local sigma algebra, positive-side L^2 space, reflected form, quotient space/class and constant vector.	§A.1
$r_j,\widehat{g},U(a),\mathcal{R}_r,\mathcal{R}_j,\Theta_j$	Coordinate reflection, observable Euclidean action, translations and linear/antilinear reflection actions.	§A.1
H_j,V_j,P_i	Positive-time generator, transverse translations and momenta; V_j here is not a fluctuation space.	eq. A.i
$\mathsf{E}_j,\varrho_\psi$	Joint energy–momentum spectral measure and scalar vector-state measure.	eq. A.ii

Continued on next page

Notation (continued)

Symbol or family	Meaning and scope	Defined at
P_j,P, P_0 (invariant projections)	Axis/all-translation conditional expectations and zero-energy projection; not flux projectors.	eq. A.v
rotation and contact terms		
L^2_U,D	Zero-extended vector L^2 space and contact multilinear form.	Prop. B.1
D_Q	Multilinear rotation-covariance difference of axial-field moments.	Thm. O
$\rho(Q)$,$*$,$\mathbb{B}(b)$,$Q \cdot b$	Two-form fiber pullback, Hodge isometry, axial-dual random field and axial orthogonal action.	Thm. O
clustering		
$\langle A, U(a)B\rangle$,$\langle A, PB\rangle$	Ordinary L^2 translated correlation and its invariant conditional limit.	eq. C.i
polymer geometry		
$\lvert X\rvert_M$,$d_M(X)$,length(N)	Cube count, normalized minimal net length and Euclidean length of a connected segment net.	§D
c_X,κ_0,K_{pol},c_{rb}	Explicit geometric counting, entropy and reblocking constants.	eq. D.iii
Y,$\bar{X}^L = LY$	Rescaled coarse M-polymer and its dilation, equal to the union of LM blocks containing X.	eq. D.iv
pr,$\widetilde{X}$	Torus covering projection and nonwrapping lift, unique up to periods.	§D.2
$\varphi : B \longrightarrow \widetilde{B}$ (covering lift)	Inverse torus projection on an injectively covered local box in the nonwrapping-lift proof.	Prop. D.1

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